$\mathbb{Z}$ is Diophantine in $\mathbb{Q}$

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Abstract

We construct an integer-coefficient polynomial in 292 890 231 905 variables whose rational zero projection is exactly $\mathbb{Z}$. Its degree is less than $2 \uparrow\uparrow 7$, and the absolute value of every coefficient is less than $2 \uparrow\uparrow 8$.

We show the minimum degree of a Diophantine definition for $\mathbb{Z}$ in $\mathbb{Q}$ is 4 and specify two smaller quartics. The first has 540 variables, 35 213 nonzero monomials, and largest absolute coefficient $3^{2048} + 1\,005\,072\,314\,631\,333 < 2^{3247}$; its complete decimal coefficient table occupies 708 848 bytes. The second has 523 variables, 35 163 nonzero monomials, and largest absolute coefficient $64 \cdot 3^{2^{27}}$. Its table writes coefficients relative to the fixed integer $S_0 = 3^{2^{26}}$. Both rational zero projections are exactly $\mathbb{Z}$.

Well-known consequences of $\mathbb{Z}$ being Diophantine in $\mathbb{Q}$ include a refutation of Mazur's conjecture that the topological closure of the rational points of a variety over $\mathbb{Q}$ has finitely many components, and existence of a universal rational Diophantine polynomial. Moreover, the set $\mathrm{HTP}(\mathbb{Q})$ of integer polynomials with a rational zero is many-one equivalent to the halting set $0'$.

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1 The main theorem and proof structure

This construction builds on OpenAI's preprint *Hilbert's tenth problem over the rational numbers* [78].

The strategy of using rank-one elliptic curves and existential local-divisibility conditions to approach undecidability over $\mathbb{Q}$ goes back to Pheidas [94]. His work also developed partial valuation predicates anticipating the pole conditions used here.

Earlier definability results provide the logical context. Robinson gave a first-order definition of $\mathbb{Z}$ in $\mathbb{Q}$ [71, Theorem 3.1]. Poonen gave a definition with universal quantifiers followed by existential quantifiers [72, Theorems 3.1 and 4.1]. Koenigsmann gave a universal definition and proved that $\mathbb{Q} \setminus \mathbb{Z}$ is Diophantine in $\mathbb{Q}$ [83, Theorem 1 and Corollary 2]. These are historical context, not additional premises of our construction.

We use $\mathbb{N} = \{0, 1, 2, \ldots\}$. For a reduced rational number a/b, with $b > 0$, put $h(a/b) = \log \max(|a|, b)$. Define $2 \uparrow\uparrow 1 = 2$ and $2 \uparrow\uparrow (r+1) = 2^{2 \uparrow\uparrow r}$; every tower is associated to the right.

Theorem 1.1 (A polynomial definition of the integers)**.** *There exists one polynomial*

$$P \in \mathbb{Z}[X_0, X_1, \ldots, X_{292890231904}]$$

such that

$$\begin{aligned}
&\deg P < 2 \uparrow\uparrow 7, \\
&|[X^\alpha]P| < 2 \uparrow\uparrow 8 \qquad \textit{for every multi-index } \alpha, \\
&q \in \mathbb{Z} \iff \exists (w_1, \ldots, w_{292890231904}) \in \mathbb{Q}^{292890231904}, \quad P(q, w_1, \ldots, w_{292890231904}) = 0 \quad (q \in \mathbb{Q}).
\end{aligned}$$

All conclusions concern the same polynomial. Here $q \in \mathbb{Z}$ means the ordinary embedded integers. The ambient variable count does not assert that every variable occurs essentially.

The file `ordinary-integer-definition.json` specifies the polynomial P_*. The argument first represents integer indices by points on an elliptic curve over $\mathbb{Q}(\sqrt{2})$. A separate value relation identifies the ordinary rational value of an index. The point coordinates and the integer index have distinct roles throughout.

Theorem 1.1 refutes Mazur's conjecture that the closure of the rational points of a variety in its real locus has finitely many connected components. Mazur proposed this in 1992 [70]; his observation that it obstructs a Diophantine definition of $\mathbb{Z}$ is recalled in [76, p. 166]. It also gives a universal rational Diophantine polynomial, with $\mathrm{HTP}(\mathbb{Q})$ many-one equivalent to $0'$. Section 13.1 proves these three consequences.

Version 10 clarifies determinant specialization, the published prime-two Euler-system input, the local valuation deduction, and the identification of the saved templates, following the review in repository Issue 2. It also corrects and expands the historical attribution. The polynomial specifications and numerical bounds are unchanged. We thank `manvirjaswal` for the detailed review [1].

Version 9 incorporated Gabriel Conant's odd-index product-height argument and asymmetric formal-depth refinement, developed with ChatGPT. Section J specifies the resulting 523- and 540-variable quartics and proves their connection to the arithmetic inputs established in this manuscript. The earlier quartics below are retained as proof and source baselines; they are not the current variable records.

Section 15 gives the explicit 4 973-variable quartic. Its arithmetic source revises the base residual compiler of Section 14, with a separate proof of soundness and completeness. The complete coefficient table and exact algebraic conversion certificate are included with this manuscript. A third polynomial, in 962 variables, is constructed in Section 17. Its complete monomial table records the large integer coefficients using exact powers. The 4 973-variable polynomial is retained with its ordinary, fully expanded integer coefficients. Section 18 proves that a single polynomial defining $\mathbb{Z}$ in $\mathbb{Q}$ must have total degree at least four, following the agentic notes cited there. Thus the degree of a quartic definition is optimal.

The final composition uses the following precise statements.

[1] `https://github.com/elliotglazer/diophantine-z-q/issues/2`

G — Canonical point codes form exactly the infinite cyclic group $2E(\mathbb{Q}(\sqrt{2})) = \mathbb{Z}P_0$. The membership polynomial types arbitrary inputs; raw addition describes the group law on canonical codes.

M — The actual guarded multiplication template has rational witnesses exactly for triples with indices a, b, ab.

H — The six-contact height theorem holds with the constants used by the precision computation and, separately, by multiplication.

PC — The explicit pole polynomial is soluble for every input in Theorem 5.2.

C — The actual five-input natural polynomial and its rational translation have the exact computation and substitution semantics of Section 10, including the power template.

B — The templates, fixed constructors and private witness substitutions specify the same saved expression P_*. The quantitative syntax calculation applies to that expression.

Lemma 1.2 (Assembly)**.** *Assume* G, M, H, PC, C, B*. Then P_* satisfies Theorem 1.1.*

Proof. Section 11 proves the exact ordinary-value relation using one shared shift and disjoint private witness blocks. The literal substitutions in B identify its existential polynomial with P_*. Section 12 proves the stated bounds for that same expression. □

Proof of Theorem 1.1. Section 3, including Proposition 3.1, proves G with its canonical membership and addition formulas. Theorem 4.1 proves H for each of the two sets of constants used below. Sections 7–9, with Appendices E–G, prove the restricted converse in Theorem 5.3. The simultaneous descent in Section 5 then proves PC. Using these results, Section 6 proves M on all rational input codes, including the finite false-triple guard.

Section 10 proves C for the fixed five-input natural polynomial and its rational model translation. Its witness variables remain natural before that translation. Sections 2 and 12 and Appendix C prove B, including the fixed scalars and private witness substitutions. All six statements therefore hold, and Lemma 1.2 applies to the specified P_*. □

2 A finite specification of the polynomial

The delivered specification is `construction/ordinary-integer-definition.json`, a 23 179 783-byte JSON expression with SHA256

`4d639d4c44e69ea360d36d4591b8fdc82269d68b9363af0157d1e5c1174c7af6.`

It contains 23 library templates and seven named recipes. The root recipe is `integers`; its only parameter is the rational variable at index zero. Every other variable is existentially quantified over $\mathbb{Q}$. Sources S01–S03 fix the construction and its semantics. Their historical specification had 23 180 023 bytes and SHA256

`a86790bbf1095d6b1cea9c66067ad8432dd35adc320254eecde21f9f98574714.`

The delivered expression replaces two diagnostic `sourceFile` strings and the path key in `sourceHashes` by portable relative paths. Direct comparison shows that every polynomial-defining field is unchanged. Since these three provenance fields contribute no equations, both byte representations denote exactly the same integer polynomial, with the same variables and bounds. The shipped `construction/SOURCE_MANIFEST.json` records both identities and the normalization; the companion `SOURCE_MAP.json` retains the historical source references.

2.1 Literal operations and witness scope

A scalar node is a variable, integer constant, sum, product, or a call to an earlier template. A call substitutes its parameter expressions and a private consecutive witness block into the callee's polynomial. Shared nodes represent repeated subexpressions, not new equations. A recipe sums squares of constraint polynomials or multiplies branch polynomials. Over $\mathbb{Q}$,

$$\sum_i f_i^2 = 0 \iff \bigwedge_i f_i = 0, \qquad \prod_i f_i = 0 \iff \bigvee_i f_i = 0.$$

Private witness blocks can be moved to the front because the rational domain is nonempty. A whole inactive product-call block may receive arbitrary values. This does not discard fixed equations inside an active callee. A rational constant p/q, $q \neq 0$, is represented by a fresh c and the globally conjoined equation $qc = p$. In particular the cached coordinates of P_0 satisfy

$$8g_0 = 1641, \qquad 4g_1 = -15465, \qquad 32g_2 = 87483.$$

These equations remain active outside internal disjunctions in the multiplication, signed-encoding and exception templates. Inverses are equations $uv = 1$, never partial evaluation operations.

Strict positivity has the polynomial expression

$$s > 0 \iff \exists z_1, z_2, z_3, z_4 \in \mathbb{Q}, \quad s \sum_i z_i^2 = 1.$$

Indeed, if $1/s = a/b > 0$, express ab as four integer squares and divide their coordinates by b. The reverse direction follows from the order on $\mathbb{Q}$. Absolute value has the exact expression

$$a = |x| \quad \iff \quad \exists z_1, \ldots, z_4 \in \mathbb{Q}, \quad a = \sum_i z_i^2, \quad a^2 = x^2.$$

2.2 Fixed finite constructors

Three node types abbreviate finite polynomials. A `fixed_integer` inserts a fixed integer expression; `psi_component` inserts one coordinate of a division polynomial; `finite_guard_product` inserts a product over a fixed box of integers. Their bounds never depend on the free rational input.

For example, on $y^2 = x^3 + Ax$, write $\psi_n = f_n$ for odd n and $\psi_n = 2yf_n$ for even n. The bases are

$$f_0 = 0, \qquad f_1 = f_2 = 1,$$
$$f_3 = 3x^4 + 6Ax^2 - A^2, \qquad f_4 = 2(x^6 + 5Ax^4 - 5A^2x^2 - A^3).$$

For $m \geq 3$, $f_{2m} = f_m(f_{m+2}f_{m-1}^2 - f_{m-2}f_{m+1}^2)$. For $m \geq 2$, put $L = f_{m+2}f_m^3$, $R = f_{m-1}f_{m+1}^3$, and $D_0 = 16(x^3 + Ax)^2$. Then $f_{2m+1} = D_0L - R$ for even m and $L - D_0R$ for odd m. Negative indices satisfy $\psi_{-n} = -\psi_n$. This is a terminating recursion. These are the short-Weierstrass normalizations of [47, §§5.5–5.6, Lemmas 5.20 and 5.24, Theorem 5.21]. In its Jacobian-coordinate convention, $x([n]P) = \phi_n(P)/\psi_n(P)^2$ and $y([n]P) = \omega_n(P)/\psi_n(P)^3$. The numerator formula has a minus sign:

$$\omega_n = \frac{\psi_{n+2}\psi_{n-1}^2 - \psi_{n-2}\psi_{n+1}^2}{4y}, \qquad \psi_{2n} = 2\psi_n\omega_n.$$

Thus the fixed-scalar ordinate equation is $2y([n]P)\psi_n(P)^4 = \psi_{2n}(P)$. For $n \neq 0$ and a nonidentity point in the torsion-free carrier, $\psi_n(P) \neq 0$, so clearing these denominators is reversible.

The fixed prime catalogue consists of the first 22 primes above 324 in each unit residue class modulo $W = 4\prod_{p\le 324} p$. Dirichlet's theorem ensures termination. Put $N_s = (\prod_{p\le 324} p)H_{\mathrm{cat}}$, where H_{cat} is the catalogue product. Assign inverse abscissa zero at the identity. For the normalized representation $1/x([m]P_0) = (a+bs)/c$ with $a, b, c \in \mathbb{Z}$, $c > 0$, and $\gcd(a, b, c) = 1$, define M_0 to be the least $m \ge 1$ such that $N_s^4 \mid a, b$ and $\gcd(c, N_s) = 1$ (S31). This is a fixed integer, with a successful index bounded in Section 12. The fixed point coordinates in the guard have nonzero denominators by the infinite order of P_0. Thus finite expansion gives one integer polynomial without requiring a monomial expansion. Its arithmetic zero set remains a separate issue. In particular, bounding the integer T does not prove that it excludes all false multiplication triples.

3 The elliptic model of integer indices

The rank-one model and its generator calculation adapt [78, Proposition 2.1 and its proof].

Let

$$K_0 = \mathbb{Q}(s), \quad s^2 = 2, \quad d = 75 - 53s, \quad E : y^2 = x^3 - d^2x,$$

and fix

$$P_0 = \left(\frac{1641 - 1160s}{8}, \frac{-123720 + 87483s}{32}\right).$$

The identity has code $(1, 0, 0, 0, 0)$. An affine point has code $(0, x_0, x_1, y_0, y_1)$ with $x = x_0 + x_1 s$, $y = y_0 + y_1 s$. Linear independence of $1, s$ makes this unique. Write $\iota(n)$ for the code of $[n]P_0$.

Proposition 3.1 (Generator certificate)**.** *The point P_0 has infinite order and $2E(K_0) = \mathbb{Z}P_0$.*

Proof. We give the argument of S05–S07, making its finite certificate explicit. The ring of integers $\mathbb{Z}[s]$ has class number one because its Minkowski bound is $\sqrt{2} < 2$. It has four unit squareclasses. The sign vectors of $-1, 1+s$ are independent, so every totally positive unit is a square.

The x-squareclass map in the two-isogeny descent on E has image contained in the eight classes generated by $-1, 1+s, d$. Away from (d), odd $v(x)$ makes the minimum of $3v(x)$ and $v(x) + 2v(d)$ odd, impossible for y^2. The torsion points give $1, -1, d, -d$. Put $\epsilon = 1 + s$, $\delta = 3 - s$. The change $(x, y) \mapsto (\epsilon^4 x, \epsilon^6 y)$ takes E to $C : y^2 = x^3 - \delta^2 x$, and $R = (1 + s, 2s)$ provides the remaining unit class. Thus the first image has order eight.

On $E' : y^2 = x^3 + 4d^2x$, a nonzero x is positive in both real embeddings. Its valuations are even. At (s) the two terms have valuations $3m$ and $m + 4$, with equality only at $m = 2$. At (d) the only odd case would require $u^2 = -4$ in $\mathbb{F}_7$. Thus the second image is trivial. The exact two-isogeny sequence gives $[E(K_0) : 2E(K_0)] = 8$; its possible extra kernel term vanishes since a nonkernel rational two-torsion point maps to the dual kernel.

The good reductions at $(17, s - 6)$ and $(23, s - 5)$ have 20 and 24 points. Prime-to-residue-characteristic torsion injects, so the torsion group has order at most four, hence is exactly the displayed full two-torsion. Mordell–Weil now gives rank one. The x-class of R has opposite real signs, whereas all torsion classes have equal signs. Its free index is therefore odd.

The duplication forms on C are

$$F_2(a, b) = (a^2 + \delta^2 b^2)^2, \qquad G_2(a, b) = 4ab(a^2 - \delta^2 b^2).$$

At either real embedding their maximum lies between $\max(|a|, |b|)^4$ and $676 \max(|a|, |b|)^4$. For primitive a, b their common ideal has valuations at most six at (s) and four at (δ), and none elsewhere. For the latter bound use that -1 is nonsquare modulo seven. For the former, a local

unit is $A + Bs$ with A odd, and $v_s((A + Bs)^2 + 1)$ is two or three according to the parity of B. Hence

$$4h_x(T) - \log 392 \le h_x(2T) \le 4h_x(T) + \log 676.$$

With $2\widehat{h}(T) = \lim_m 4^{-m} h_x(2^m T)$ this gives $h_x(T) \le 2\widehat{h}(T) + \log 392/3$. Direct doubling gives $x(2R) = (57 - 28s)/8$, with height $\log 41$. If $R = mS + T$ for torsion T and $|m| \ge 3$, it follows that

$$h_x(S) \le \frac{\log(41^3 \cdot 676)}{108} + \frac{\log 392}{3} < \log 9.$$

The strict inequality is the exact integer comparison $41^3 \cdot 676 \cdot 392^{36} < 9^{108}$.

A nonrational $x \in K_0$ in this height range has primitive polynomial aX^2+bX+c with $1 \le a \le 81$, $|b| \le 162$, $|c| \le 81$, and discriminant $2k^2$, $1 \le k \le 162$. Both roots are included. Rational possibilities have coprime numerator of absolute value at most nine and denominator from one to nine. The exact finite square test of $x^3 - \delta^2 x$ gives precisely

$$0, \quad \delta, \quad -\delta, \quad 1 + s, \quad 23 - 17s, \quad 2 + 4s, \quad (5 - 4s)/2.$$

The certificate covers 7,004 primitive quadratics, their 14,008 roots, and 111 rational candidates. To test whether $u + vs$ is a square, take both signs of a rational square root t of $u^2 - 2v^2$, test $y_0^2 = (u + t)/2$, then set $y_1 = v/(2y_0)$ if $y_0 \ne 0$. The $y_0 = 0$ case requires $v = 0$ and $y_1^2 = u/2$. Substitution verifies each result. This is a necessary and sufficient test, including zero. The last four listed coordinates are those of $R + T$ for $T \in C[2]$. Every surviving nontorsion point therefore has height $\widehat{h}(R)$, contradicting $\widehat{h}(R) = m^2\widehat{h}(S)$. This finite computation is an explicit part of the proof, with a recorded independent replay; it is not replaced by a rank oracle.

Thus $2C(K_0) = \mathbb{Z}(2R)$. The inverse coordinate change sends $2R$ to P_0, as asserted. □

3.1 Polynomial membership and addition

Apart from the identity branch, membership in $2E(K_0)$ is expressed by choosing $X, Y, H \in K_0$, $u \in \mathbb{Q}$, and imposing

$$Y^2 = X^3 - d^2X, \quad u\,\mathrm{N}(Y) = 1, \quad 2YH = 3X^2 - d^2, \quad x = H^2 - 2X, \quad y = H(X - x) - Y.$$

Every nonidentity double has a preimage with nonzero ordinate. Each field equation is two rational equations. Membership is an all-input statement. Raw addition has the narrower exact contract

$$\mathrm{Add}_{\mathrm{raw}}(\iota(a), \iota(b), R) \quad \Longleftrightarrow \quad R = \iota(a + b).$$

Its branches cover both identities, opposite points, distinct abscissas and doubling with nonzero ordinate. Every later use supplies canonical input codes. Multiplication first imposes membership of all three roots and is sound on arbitrary rational tuples. In the natural predicate, four initially untyped tuples enter multiplication calls, so are typed before their squared values are added. Lagrange's theorem then gives exactly $\iota(\mathbb{N})$. This uses multiplication completeness; the generator proposition alone does not supply the natural predicate.

4 The six-contact height estimate

The contact and odd-support framework follows [78, Section 4]. The effective geometric estimates below give the height input used in this manuscript; the preprint's height assertion is not taken as an additional premise.

Set

$$\mathcal{C} = \{0, -15, -20, -24, -30, -60\}, \qquad F(u) = u(u+15)(u+20)(u+24)(u+30)(u+60),$$

and $G(u) = 7u^2 + 300u + 3600$. For a finite prime set S containing all primes at most 324 and $u \notin \mathcal{C}$, define

$$M_S(u) = \prod_{\substack{p \notin S:\ v_p(u-c)>0 \text{ odd} \\ \text{for some } c \in \mathcal{C}}} p.$$

Negative valuations of u do not enter the product. Put

$$\begin{aligned} F_0 &= 324!, \qquad D = F_0\, 3^{256F_0^4}, \qquad G_0 = 116064000000, \\ E_0 &= 10(DG_0)^2, \qquad Q_R = R^{RE_0}, \\ B_{\mathrm{cm}} &= D^2\, 3^{21000D^2} + 1, \\ C_{\mathrm{cm}} &= 2B_{\mathrm{cm}} + 4096(1028 + 5D + 84000D^2), \\ H_6(S;R) &= 2 \cdot 10^{13} \max(1, C_{\mathrm{cm}}, Q_R^{17}), \\ R &\geq \max(D, \max S, 10^{12}). \end{aligned}$$

When the second argument is omitted, we use $H_6(S) = H_6(S; \max(D, \max S, 10^{12}))$. The geometric construction is followed by a direct modularity argument at coefficient prime five and a separate effective CM argument. The constants in the displayed definition of $H_6(S;R)$ are unchanged.

Theorem 4.1 (Six-contact estimate)**.** *For every such S, R and every $u \in \mathbb{Q} \setminus \mathcal{C}$,*

$$h(u) \leq H_6(S;R) M_S(u)^{24}.$$

Proof. Take $B = (2, -3)$, the maximal order $\mathcal{O} = \mathbb{Z}[i, (1+j)/2]$, and $\mu = 2j + ij$. With its positive sign choice, the form $\mathrm{Trd}(\mu^{-1} x \overline{y})$ has matrix

$$\begin{pmatrix} 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & -2 \\ 1 & -1 & 0 & 0 \\ -1 & 2 & 0 & 0 \end{pmatrix}$$

and determinant one, in the ordered basis $(1, i, (1+j)/2, (i+ij)/2)$. The Riemann form is the convention of [21, equation (3) and Lemma 3]; its Rosati involution is $a \mapsto \mu^{-1} \overline{a} \mu$. This is the principal-polarization datum whose normalizer quotient is $\mathbf{P}^1_j$, with branch values $0, \infty, -16/27$ and indices $4, 6, 2$ [19, 21]. Put $K_0 = \widehat{\mathcal{O}}^*$ and $K_3 = \ker(K_0 \longrightarrow (\mathcal{O}/3\mathcal{O})^*)$. Let N be the finite-adelic normalizer of $\widehat{\mathcal{O}}$. It normalizes K_3. At a split prime the local normalizer is the center times the maximal-order units; at each of the two division primes its quotient by these groups has order two. Thus $|N/(\mathbb{A}_f^* K_0)| = 4$. A central finite idele acts trivially on the level-three curve: multiply it by a rational scalar to remove all finite valuations, and choose the scalar's sign so that the residue at three is one. The remaining idele lies in K_3, and the rational central scalar acts trivially on the symmetric domain. Consequently the entire level-three curve Y_3, with all its determinant components, admits the action of $N/(\mathbb{A}_f^* K_3)$ with quotient $\mathbf{P}^1_j$, and

$$\deg(Y_3 \longrightarrow \mathbf{P}^1_j) \leq 4|(\mathcal{O}/3\mathcal{O})^*| \leq 4 \cdot 81 = 324.$$

We construct its integral model without changing this degree. Apply [20, Theorem 2.1 and Corollary 2.3] to $V_1(5)$ over $\mathbb{Z}[1/30]$. Its determinant image is full, its level is maximal at the

division primes, and $5 \geq 4$ is prime to six. First add a full five-marking refining the $V_1(5)$-marking. This gives the full-five moduli scheme, including every determinant component, as a finite étale cover. Next add the finite étale sheaf

$$\mathrm{Isom}_{\mathcal{O}}((\mathcal{O}/3\mathcal{O}), A[3]).$$

This sheaf is nonempty on every geometric fiber. Indeed B_3 is a division algebra and the rational Tate module is a rank-one left B_3-module. Every $\mathcal{O}_3$-stable lattice in that module is principal: choose a nonzero lattice element of least division valuation and divide every other element by it on the right. Thus the Tate lattice is free of rank one over $\mathcal{O}_3$. One can also check this in characteristic zero and extend it to all fibers by local constancy of the finite étale $A[3]$ with its $\mathcal{O}$-action.

The resulting smooth projective scheme X_{35} carries full markings at three and five. The group $G_5 = (\mathcal{O}/5\mathcal{O})^*$ changes only the five-marking. Its action is free: a stabilizer would give a polarized automorphism preserving the full three-marking, hence a finite-order element of $1+3M_4(\mathbb{Z}_3)$. That group is torsion-free, as follows by expanding a hypothetical prime-order element at its first nonzero 3-adic term. The automorphism is therefore the identity, and the full five-marking forces the group element to be the identity. The quotient X_{35}/G_5 is smooth projective, its covering map is finite étale, and the universal abelian scheme, its canonical principal polarization, its $\mathcal{O}$-action and its three-marking descend. This quotient represents exactly the full-three moduli problem, so its generic fiber is Y_3. The auxiliary five-marking has contributed no degree factor to $Y_3 \to \mathbf{P}^1_j$.

Each geometric component of Y_3 is a torsion-free cover of the norm-one orbifold of Euler characteristic $-1/3$, and hence has genus at least two. The normalizer action extends to its smooth proper model by uniqueness of the stable model, including permutations of components. Fix $p > 324$ and work after strict henselization. The action is tame. On each component use its effective action. A finite-order automorphism of order prime to p cannot specialize to the identity, by considering its first nonzero term modulo a power of the uniformizer. At a fixed point the same power-series argument shows that the tangent character of the stabilizer is faithful. Thus the stabilizer is cyclic.

For a nonidentity generator the tangent eigenvalue minus one is a unit. Its fixed-point equation therefore has a unique lift by Hensel's lemma. All powers of the generator fix that lift. It follows that no new branch orbit appears and that distinct branch orbits cannot collide: their stabilizers at a purported common specialization would belong to one cyclic group, whose nontrivial elements have the same unique fixed lift. Averaging a parameter gives a local invariant ring $R[[z^e]]$. Hence the quotient is smooth and retains the three disjoint branch sections and their indices $4, 6, 2$.

Its genus-zero model with these three ordered sections is isomorphic to $\mathbf{P}^1_{\mathbb{Z}_p}$ carrying the ordered triple $(0, \infty, -16/27)$, which is disjoint since 2 and 3 are units. The isomorphism agrees with j on the generic fiber, by uniqueness of a projective coordinate on three points. Thus it is the integral model in the stated j-coordinate; no further denominator prime has been discarded.

Pull back by $J(u) = -57600F(u)^2/G(u)^6$. If

$$H(u) = (u^2 - 60u - 1800)(13u^2 + 480u + 3600)(11u^2 + 600u + 7200),$$

direct calculation gives $G^6 - 97200F^2 = H^2$. The pullback indices are $2, 6, 2$. Tame normalization replaces e by $e/\gcd(e, a)$ and leaves $2, 1, 1$. The discriminants and resultants have prime support at most 13, so this persists at $p > 324$. The normalized cover of $\mathbf{P}^1_u$ has degree at most 324, not $12 \cdot 324$, and is branched only at the six contacts. Locally it has equation $u - c = z^2$ after unramified extension. Positive even contact valuation is unramified; negative valuation specializes to the unbranched point at infinity. This is exactly the support specified by $M_S(u)$.

Let K be the splitting field of the fiber over u of the normalized cover. The fiber is finite étale of degree at most 324, so $K/\mathbb{Q}$ is normal with $[K : \mathbb{Q}] \leq F_0 = 324!$. The preceding local calculation

makes it unramified outside $T = S \cup \operatorname{supp} M_S(u)$. Choose a fiber A/K. The proper model and its universal abelian scheme give good reduction outside the places over S.

All conjugates of A have the same geometric polarized moduli point. Indeed Baba–Granath's polarized equivalence relation gives $V_{\mathbb{Z}} = V/W$; for discriminant six its two twisting involutions generate the whole Atkin–Lehner group W. On the finite nonzero j-locus the polarized point is equivalently specified by

$$[J_2, J_4, J_6, J_{10}] = [12(j+1), 6(j^2+j+1), 4(j^3-2j^2+1), j^3].$$

Here $j = J(u)$ is rational, finite and nonzero: $F(u) \neq 0$ and $\operatorname{disc}(G) = -10800 < 0$.

To define the Hom field without choosing a Galois action on a labelled collection, form $W_0 = \operatorname{Res}_{K/\mathbb{Q}} A$. Its geometric endomorphism lattice is the direct sum of the ordered Hom groups among the conjugates. Its rank r is at most $16F_0^2$, by its faithful integral-homology realization. Its Galois image is finite, and every finite subgroup of $\mathrm{GL}_r(\mathbb{Z})$ injects modulo three. If E is the kernel field of this action and $L = EK$, then $L/\mathbb{Q}$ is normal and

$$[L : \mathbb{Q}] \leq F_0\, 3^{r^2} \leq F_0\, 3^{256F_0^4} = D.$$

At a prime outside T, the field K is unramified and every conjugate has good reduction. Choose a different Tate prime. Inertia acts trivially on the Tate module of W_0, and therefore on its endomorphism lattice. Thus L is unramified outside T. It defines every geometric endomorphism and every ordered conjugacy homomorphism of A.

By [18, Part I, Theorems 2.8–2.9], polarizable complex tori are abelian varieties and their analytic homomorphisms are algebraic. After clearing lattice denominators, the same assertion identifies rational Hodge endomorphisms with algebraic quasi-endomorphisms. These endomorphisms are defined over $\overline{\mathbb{Q}}$: every automorphism of $\mathbb{C}/\overline{\mathbb{Q}}$ fixes their action on all torsion points, and torsion is Zariski dense. On rational homology the complex structure is right multiplication by an element $J \in B \otimes \mathbb{R}$ with $J^2 = -1$. In $\operatorname{End}_{\mathbb{Q}}(B) = B \otimes B^{\mathrm{op}}$, commutation with J requires each right coefficient to commute with J. Its rational centralizer is either $\mathbb{Q}$ or an imaginary quadratic field k. Hence the geometric endomorphism algebra is either B or $M_2(k)$. We treat these two cases separately. Throughout, put $M = M_S(u)$, $d = [L : \mathbb{Q}] \leq D$, and $Q = Q_R$.

CM fibers. Suppose $\operatorname{End}^0(A_{\overline{\mathbb{Q}}}) = M_2(k)$. All its elements are defined over L. Multiples of matrix units therefore give an elliptic curve E/L with complex multiplication and an L-isogeny $E^2 \to A$. Let $\Delta < 0$ be the discriminant of the full geometric endomorphism order of E. The degree of $j(E)$ is its order class number, so $h(\Delta) \leq d \leq D$. The effective bound of [29, Proposition 2.7] gives

$$\sqrt{|\Delta|} \leq 2D^2 \exp(21000D^2).$$

This bound includes every imaginary quadratic order and has no exceptional-field exclusion.

Here is the height comparison with its normalization explicit. After a finite extension E has everywhere good reduction, since its j-invariant is integral. This extension does not change its stable Faltings height. For a reduced CM period $\tau = x + iy$ we have $\sqrt{3}/2 \leq y \leq \sqrt{|\Delta|}/2$. The archimedean summand in the original Faltings normalization is

$$-\log(2\pi) + \frac{\pi y}{6} - \frac{1}{2}\log y - 2\sum_{n \geq 1} \log|1 - e^{2\pi i n\tau}|,$$

by the discriminant formula [30, Eq. (6.2)]. For $\rho = e^{-2\pi y} < 1/100$, the final sum is at most $2\rho/(1-\rho)^2 < 200/9801$, and $-\frac{1}{2}\log y < 1/10$. Averaging and discarding the negative $-\log(2\pi)$ gives

$$h_F(E) \le \frac{\pi\sqrt{|\Delta|}}{12} + 1 \le D^2 3^{21000D^2} + 1 = B_{\mathrm{cm}}.$$

Apply the published isogeny estimate [28, Theorem 1.4] over the original field L, with source E^2. It gives an isogeny to A of degree at most

$$\bigl(28^{256} d \max(h_F(E^2), \log d, 1)^2\bigr)^{8192}.$$

The maximum is at most $2B_{\mathrm{cm}}$. Height variation under isogeny [33] and additivity give

$$\begin{aligned} h_F(A) &\le 2B_{\mathrm{cm}} + 4096\bigl(256\log 28 + \log d + 2\log(2B_{\mathrm{cm}})\bigr) \\ &\le 2B_{\mathrm{cm}} + 4096(1028 + 5D + 84000D^2) = C_{\mathrm{cm}}. \end{aligned}$$

For the last line use $\log 28 < 4$, $\log d \le D$, and $\log(2B_{\mathrm{cm}}) \le 2 + 2D + 42000D^2$. No bound for the degree of the auxiliary good-reduction extension enters the isogeny estimate.

Non-CM fibers and scalar descent. Now suppose $\mathrm{End}^0(A_{\overline{\mathbb{Q}}}) = B$. In particular A is absolutely simple. For each $\sigma \in \mathrm{Gal}(L/\mathbb{Q})$ choose a geometric isomorphism $\psi_\sigma : {}^\sigma A \to A$. The automorphism $b \mapsto \psi_\sigma {}^\sigma b \psi_\sigma^{-1}$ of B is inner. Correcting it by Skolem–Noether and clearing denominators gives a B-linear isogeny $\mu_\sigma : {}^\sigma A \to A$, defined over L because all the relevant Hom groups are defined there. Take $\mu_1 = 1$ and inflate this finite choice to $G_{\mathbb{Q}}$. The discrepancy

$$c(\sigma,\tau) = \mu_\sigma {}^\sigma\mu_\tau \mu_{\sigma\tau}^{-1}$$

commutes with B, hence belongs to $\mathbb{Q}^\times$. Set $a_\sigma = (\deg \mu_\sigma)^{-1/4} > 0$ and $\nu_\sigma = a_\sigma \mu_\sigma$ in the category with algebraic coefficients. Throughout these coefficient extensions Galois acts on the varieties and morphisms and acts trivially on the added scalars; the induced homology maps are extended linearly. In particular, the a_σ are constant coefficients in the products below. Multiplicativity of degree and $\deg[c] = c^4$ show that

$$\nu_\sigma {}^\sigma\nu_\tau = \varepsilon(\sigma,\tau)\nu_{\sigma\tau}, \qquad \varepsilon(\sigma,\tau) \in \{1,-1\}.$$

Thus ε is a normalized sign cocycle inflated from $\mathrm{Gal}(L/\mathbb{Q})$. Only finitely many algebraic coefficients a_σ occur.

We need a splitting with controlled ramification, not merely existence of a projective lift. Identify $H^2(G_{\mathbb{Q}}, \{1,-1\})$ with $\mathrm{Br}(\mathbb{Q})[2]$. At a finite prime outside T the class of ε is zero: its restriction is unramified and the unramified local quotient has cohomological dimension one. Let T_- be the finite primes where its invariant is $1/2$. For each odd $p \in T_-$ choose an odd character of $(\mathbb{Z}/p\mathbb{Z})^\times$; at $2 \in T_-$ use the nontrivial character modulo four. Their product is a Dirichlet character χ, unramified outside T, whose order m satisfies

$$m \le 2 \prod_{\substack{p \in T \\ p \ne 2}} (p-1) \le 2R^R M.$$

Here the product over an empty set is one.

To check that χ has the required square-root obstruction, use the exact sequence of constant discrete Galois modules $1 \to \{\pm 1\} \to \mu_\infty \xrightarrow{z \mapsto z^2} \mu_\infty \to 1$. By local reciprocity, a finite-order

character of $G_{\mathbb{Q}_p}$ has a character square root exactly when its value on $-1 \in \mathbb{Q}_p^\times$ is 1. Indeed this is the only obstruction on the torsion part of $\mathbb{Q}_p^\times$; the valuation factor and the procyclic principal-unit factors admit such roots. Hence the boundary of χ and $[\varepsilon]$ agree at every finite place. Brauer reciprocity makes them agree at the real place, and the injective global-to-local Brauer map makes them equal globally [42, Chapter I]. Choosing a continuous pointwise square root of χ and correcting it by a sign cochain consequently gives a continuous cochain $\beta : G_{\mathbb{Q}} \to \mu_{2m}$ such that

$$\beta^2 = \chi, \qquad \beta(\sigma)\beta(\tau)\beta(\sigma\tau)^{-1} = \varepsilon(\sigma, \tau).$$

Its restriction to G_L is a character invariant under $G_{\mathbb{Q}}$. Outside T, its inertia restrictions are quadratic and nontrivial at only finitely many primes. Twisting by the quadratic character of the product of those odd primes kills all these restrictions: at an odd prime there is a unique nontrivial quadratic character on inertia. This introduces ramification only at those primes and possibly at $2 \in T$, and changes neither displayed identity. We henceforth use this choice of β.

Let L' be the kernel field of $\beta|_{G_L}$. Conjugacy invariance shows that $L'/\mathbb{Q}$ is Galois, and

$$[L' : \mathbb{Q}] \le 2md \le 4DR^R M \le QM.$$

The last inequality follows from $R \ge \max(D, 10^{12})$ and $E_0 = 10(DG_0)^2$.

Modularity and a uniform conductor bound. Choose one number field C containing all a_σ, the values of β, and a splitting field of B. For each coefficient prime ℓ choose an embedding of C into $\overline{\mathbb{Q}}_\ell$ and extend scalars on $V_\ell(A)$ to a finite extension containing its image. With the constant-coefficient convention above, define

$$\rho_\ell(\sigma) = \beta(\sigma)^{-1} a_\sigma \mu_\sigma \circ \sigma.$$

The cocycle identities make this a representation of $G_{\mathbb{Q}}$. It commutes with B, so matrix units give $\rho_\ell \simeq r_\ell \oplus r_\ell$ with $\dim r_\ell = 2$. On $G_{L'}$ it is the actual Tate action of A. Faltings' semisimplicity and endomorphism theorems [33] show that $r_\ell|_{G_{L'}}$ is absolutely irreducible: after splitting B, its commutant is the scalars. Lemma H.1 below shows that the representation is de Rham at ℓ, by proper-smooth comparison and finite descent [15, 16], and that its two Hodge–Tate weights are $0, -1$ in the Tate convention. Berger's p-adic monodromy theorem [22] makes this de Rham representation potentially semistable, as required by Pan's theorem. It is odd. To see this without an assumption on a residual representation, compare with Betti homology at a complex embedding. Take the map on rational Betti homology induced by antiholomorphic conjugation followed by the holomorphic quasi-isogeny μ_c, and extend it $\mathbb{C}$-linearly. It exchanges the two Hodge summands. Each summand is a simple module for $B \otimes \mathbb{C} = M_2(\mathbb{C})$. Consequently its action on the two-dimensional multiplicity space is not scalar; since $r_\ell(c)^2 = 1$, its eigenvalues are $1, -1$. It is unramified outside $T \cup \{\ell\}$, since there L is unramified, A has good reduction, and β is unramified.

Use the Tate convention in which the cyclotomic character has Hodge–Tate weight -1. Apply [23, Theorem 1.0.4] at $\ell = 5$; there is no residual-image proviso at this prime. It gives a cuspidal eigenform up to character twist. The twisting character is a de Rham subrepresentation of the Hom representation and is unramified outside finitely many primes. Local algebraicity [24, Appendix B, pp. 32–33] and class field theory over $\mathbb{Q}$ make it a finite character times an integral power of the cyclotomic character: removing its integral local Hodge–Tate weight leaves finite inertia at five and at every other prime, hence a finite ray-class character. Comparing $\{0, -1\}$ with the modular weights $\{0, 1-k\}$ after that power forces $k = 2$ and power zero. Absorb the finite Dirichlet character into the eigenform and take its primitive newform f. Li's Fricke formula [25, p. 296, before Lemma

6] makes the span of its coefficient conjugates Fricke-stable. Shimura's quotient construction and Euler-factor identity [26, Theorem 1] therefore show that r_5 is a constituent of the Tate module of the modular abelian variety A_f, or equivalently $r_5^\vee$ is its cohomological constituent. This duality preserves the conductor.

The representations at five and three belong to this same system. Enlarge C to contain the coefficients of f, retaining the chosen embedding at five. For $p \notin T \cup \{3,5\}$, choose a place v of L and a lift $\sigma \in G_{\mathbb{Q}_p}$ of the residue field map $x \mapsto x^p$ in its decomposition group. Since $L_v/\mathbb{Q}_p$ is unramified and A has good reduction, relative p-Frobenius maps the reduction $\overline{A}_v$ to the reduction of ${}^\sigma A$ at v. Composing with $\overline{\mu}_\sigma$ gives an endomorphism of $\overline{A}_v$. The characteristic polynomial of an abelian-variety endomorphism on rational Tate modules is independent of the coefficient prime. Thus half its trace, multiplied by $\beta(\sigma)^{-1}a_\sigma$, is one element of C whose images are the traces of every r_ℓ. Away also from the finite set of primes dividing the level of f, the equality with its modular trace at five is an equality of algebraic numbers. Transporting it through an embedding at three, Chebotarev and semisimplicity identify r_3 with the corresponding modular Tate representation. We may therefore use r_3 to compute the conductor at five and r_5 at all other primes.

Here are uniform local bounds, including the small primes. For a finite extension of $\mathbb{Q}_p$ of degree at most a, its normal closure has degree at most $a!$. For a finite Galois local extension with ramification index e and different exponent δ, let f be the Eisenstein polynomial of a uniformizer π over the maximal unramified subfield. Then $\delta \le e - 1 + ev_p(e)$. The derivative terms have distinct valuations modulo e, so there is no cancellation at the smallest valuation. The identity $f'(\pi) = \prod_{\sigma \neq 1}(\pi - \sigma\pi)$ over the inertia conjugates gives $\delta = \sum_{i \ge 0}(|G_i| - 1)$. Changing variables by the Herbrand function yields $\delta/e = 1 - 1/e + \int_0^\infty (1 - |G^u|^{-1})\,du$. Before the last positive upper break the integrand is at least $1/2$, so that break is bounded by $2\delta/e \le 2(1 + \log_2(a!)) \le 4a^2$ [32, Sections 4.1–4.3].

At p, take $F = L_v$ and adjoin $A[3]$, using $A[5]$ if $p = 3$. This extension has degree at most DG_0 over $\mathbb{Q}_p$. Raynaud's criterion [31, Theorem 3.5] gives semistable reduction there. Its wild inertia acts trivially on each $V_\ell(A)$ with $\ell \neq p$. The preceding break bound and the subgroup rule for upper numbering therefore show that all absolute ramification groups above $U_0 = 4(DG_0)^2$ act trivially on that Tate module. On these groups $a_\sigma \mu_\sigma = 1$ because they lie in G_L. For odd p, the character $\beta|_{G_F}$ is trivial on wild inertia: its square is the tame character $\chi|_{G_F}$, and a pro-p group has no quotient of order two. At $p = 2$, local reciprocity and the norm map give $\beta^4 = 1$ on the units of F, since the 2-primary local factor of χ has conductor at most four. If $e = v_F(2) \le D$, the logarithm for $\widehat{\mathbf{G}}_m$ identifies U_F^{e+1} with $\mathfrak{m}_F^{e+1}$ since $e + 1 > e$ [48, IV.6.4, p. 132]. Multiplication by four gives $(U_F^{e+1})^4 = U_F^{3e+1}$. Thus the relative upper break of β is at most $3e$. To make the change of numbering explicit, the same normal-closure estimate bounds the last upper break of $F/\mathbb{Q}_2$ by $4D^2$. Above it, the absolute upper group $G^u_{\mathbb{Q}_2}$ lies in G_F, and the subgroup formula is

$$G^u_{\mathbb{Q}_2} \cap G_F = G_F^{\psi_{F/\mathbb{Q}_2}(u)}, \qquad \psi_{F/\mathbb{Q}_2}(u) \ge u \quad (u \ge 0).$$

Consequently β is trivial on every absolute upper group with $u > \max(4D^2, 3e)$, which is at most U_0. This proves the asserted absolute bound without a factor involving $[L' : \mathbb{Q}]$. The Artin conductor exponent of r_ℓ is consequently at most $2 + 2U_0 \le E_0$.

For $p > R$, the extension $L_v/\mathbb{Q}_p$ is tame, since its degree is at most $D < p$. The surface has good reduction, and the same argument kills the wild part of β. Hence the conductor exponent is at most two there. Outside T it is zero. Local–global compatibility for the weight-two modular representation therefore bounds the primitive level N of f by

$$N \le \Big(\prod_{p \le R} p^{E_0}\Big) M^2 \le R^{RE_0} M^2 = Q M^2.$$

Transfer from a modular Jacobian. Over L' the Tate modules of A and $J = J_1(N)$ have a common constituent. Invariants in their finite-dimensional Hom space commute with extension of the coefficient field. Faltings' Hom theorem thus gives a nonzero L'-homomorphism $A \to J$. Absolute simplicity and Poincaré reducibility give an L'-isogeny $A \times Z \to J$ for an abelian variety Z/L'. In particular $g = \dim J \geq 2$. The index of $\Gamma_1(N)$ is at most N^2, so both the genus and the degree of its standard Belyi map are at most N^2. By [27, Theorem 1.1.1 and Section 2.3],

$$h_F(J) \leq 13000000N^{12} \leq Q^{13}M^{24}, \qquad g \leq Q^3M^4, \qquad d' = [L' : \mathbb{Q}] \leq QM.$$

The normalization in that theorem is the original stable Faltings height of the Jacobian.

Apply [28, Theorem 1.4] with source J and target $A \times Z$, both over L'. There is such an isogeny φ with

$$\log \deg \varphi \leq 1024g^3\bigl(64g^2 \log(14g) + \log d' + 2 \log \max(h_F(J), \log d', 1)\bigr).$$

For $Q \geq 10^{12}$ and $M \geq 1$, the three logarithms in order are at most $5QM^4$, $2QM^4$, and $19QM^4$. These follow from $\log Q \leq Q$ and $\log M \leq M^4/4$. It follows that

$$\log \deg \varphi \leq 368640Q^{16}M^{24} \leq Q^{17}M^{24}.$$

The general lower bound $h_F(Z) \geq -\frac{3}{2} \dim Z \geq -2g$ is recorded in [28, Section 9]. Additivity and height variation now give

$$h_F(A) \leq Q^{13}M^{24} + \tfrac{1}{2}Q^{17}M^{24} + 2Q^3M^4 \leq Q^{17}M^{24}.$$

Thus both cases prove the exact estimate needed below:

$$\max(1, h_F(A)) \leq \max(1, C_{\mathrm{cm}}, Q_R^{17}) M_S(u)^{24}.$$

Use the Hauptmodul normalized by the values $0, \infty, -16/27$ at the CM points of discriminants $-4, -3, -24$. In a common weighted-projective representative, [21, equations (2) and (14), pp. 3 and 17] gives

$$J_2 = 12(j+1), \quad J_4 = 6(j^2 + j + 1), \quad J_{10} = j^3, \qquad \psi_4 = J_2^2 - 24J_4 = 144j, \quad \chi_{10} = -j^3/4.$$

The conventions of [35, §2.2, p. 4] are $\psi_4 = 2^{-2} \sum_{m \text{ even}} z_m^8$ and $\chi_{10} = -2^{-14} \prod_{m \text{ even}} z_m^2$. The quotient χ_{10}^2/ψ_4^5 is unchanged by weighted rescaling. Since our j is finite and nonzero, its denominator is nonzero, and substitution gives

$$j = 16 \cdot 144^5 \frac{\chi_{10}^2}{\psi_4^5} = 3779136 \frac{\bigl(\prod_{m \text{ even}} z_m\bigr)^4}{\bigl(\sum_{m \text{ even}} z_m^8\bigr)^5}.$$

The two homogeneous polynomials have degree 40 and coefficient ℓ^1 norms 3779136 and 10^5. Hence $h(j) \leq 40h([z_m]) + 22$. The product and sum use the ten even theta nulls of the same principally polarized surface. Adjoining the six zero odd coordinates does not change their projective height.

In [36], the level $r = 2$ theta coordinates are the sixteen half-characteristic values used here; the six odd values vanish. The permitted changes of theta structure permute coordinates and multiply them by roots of unity. His Euclidean projective height dominates our supremum projective height. Put $X = \max(1, h_F(A))$ and $X_D = \max(1, h_F(A) + \log \pi) \leq 3X$. His Corollary 1.3(2), in the Deligne metric, gives

$$h_\Theta(A) \leq \tfrac{1}{2}X_D + C_2 \log(X_D + 2), \qquad C_2 = 1000 \cdot 2^4 (\log(2^4))^5 \leq 1000 \cdot 2^{24}.$$

Using $\log(X_D+2)\le 2X_D$ proves $h([z_m])\le 10000\cdot 2^{24}X$.

Write $j=a/b$ in lowest terms. The nonzero integer polynomial $-57600bF(X)^2-aG(X)^6$ has u as a root and coefficient norm at most $2\cdot 4000^6\max(|a|,|b|)$. Remove any initial power of X and use the actual highest nonzero coefficient. The rational-root theorem bounds both numerator and denominator of u, giving $h(u)\le h(j)+\log(2\cdot 4000^6)<h(j)+74$.

A field extension for the theta structure changes neither absolute height. Since $40\cdot 10000\cdot 2^{24}+96<10^{13}$, the preceding inequalities give

$$h(u)\le 10^{13}\max(1,h_F(A))\le 10^{13}\max(1,C_{\mathrm{cm}},Q_R^{17})M_S(u)^{24}=\tfrac{1}{2}H_6(S;R)M_S(u)^{24}.$$

Every estimate used only the displayed lower bound on R, so the conclusion holds for every permitted larger R as well. □

The final value argument uses $S_{\mathrm{ht}}=\{p:p\le 324\}$. Multiplication uses $S_{\mathrm{mul}}=S_{\mathrm{ht}}\cup\mathcal{H}\cup\mathrm{supp}(c_*)$, where $\mathcal{H}$ is the fixed catalogue. A common numerical R may bound both sets. It does not identify them.

5 The pole predicate and its converse input

The pole construction and simultaneous-descent strategy adapt [78, Section 3].

Let $S_0=\{2,3,5,7,13\}$. With input B and eleven witnesses $b_0,\dots,b_5,e,z,r_1,r_2,w$, define

$$\begin{aligned}
m&=(1-2b_0)(1+b_1)(1+2b_2)(1+4b_3)(1+6b_4)(1+12b_5),\\
A&=mB,\qquad H=A(A-B^2),\\
T_1&=A(A-e)(A+3e),\qquad T_2=A(Az^2-eH)(Az^2+3eH),\\
Q_{\mathrm{pole}}&=B\Big(\sum_{i=0}^{5}(b_i^2-b_i)^2+(eHzw-1)^2\\
&\qquad\qquad+(r_1^2-T_1)^2+(r_2^2-T_2)^2\Big).
\end{aligned}$$

Write $\Phi(B)$ for rational solubility. For $B\neq 0$, m is a signed divisor of 2730, and the equations give points $(A,r_1)\in E_e$ and $(Az^2,zr_2)\in E_{eH}$ with $eHz\neq 0$, where $E_\ell: y^2=x(x-\ell)(x+3\ell)$. The polynomial has total degree 75.

Lemma 5.1 (One-sided obstruction)**.** *If $B\neq 0$ and $\Phi(B)$, then $v_p(B)$ is not negative and odd for any $p\notin S_0$.*

Proof. If $k=v_p(B)<0$ were odd, then $v_p(A)=k$, $v_p(H)=3k$. Exactly one of e,eH has even valuation. Square scaling makes its parameter a unit, preserving the parity of the abscissa's valuation. For a unit parameter the cubic has valuation $v_p(x)$ if $v_p(x)>0$ and $3v_p(x)$ if $v_p(x)<0$. Its being a square forces even valuation. But both points have abscissa in the squareclass A, a contradiction. □

Theorem 5.2 (Pole completeness, PC)**.** *For every rational B with square reduced denominator, $0<B<1/2$ and $v_p(B)\ge 4$ for every $p\in S_0$ imply $\Phi(B)$.*

The following statement is the $E=E_\ell$, $s_2(E)=1$ case of [74, Theorem 1.1], and also follows as a special case of [75, Theorem 1.1]. We give the restricted argument here; these preprints are cited for attribution, not assumed as additional converse inputs.

Theorem 5.3 (Restricted converse, RC)**.** *For every positive squarefree* $\ell \equiv 11 \pmod{12}$*,*

$$\operatorname{corank}_{\mathbb{Z}_2} \mathrm{Sel}_{2^\infty}(E_\ell/\mathbb{Q}) = 1 \quad \Longrightarrow \quad \operatorname{ord}_{s=1} L(E_\ell, s) = 1.$$

Here $c_4 = 208\ell^2$, $\Delta = 2304\ell^6$, and $j = 35152/9$. The nonintegral rational j shows these curves are non-CM. They are different from the CM generator curve. Theorem 5.3 is proved in Sections 7–9, with its coefficient and normalization arguments in Appendices E–G. We now prove Theorem 5.2 from this restricted converse.

5.1 The exact descent reduction

The signed divisors of 2730 realize all real, 2-adic and 3-adic squareclass signatures. After $-1, 2, 3$ fix sign and valuation parities, the remaining unit vectors of $5, 7, 13$ are $(0,1,1), (1,0,0), (0,1,0)$ and form a basis. Choose positive m making $A = mB$ square at two and a nonsquare unit squareclass at three. Let β, θ be the positive squarefree representatives of A, H. Then $\beta \equiv 17 \pmod{24}$, $\theta \equiv 1 \pmod{24}$, with disjoint supports, $\beta > 1$, and $\theta = 1$ permitted. Each is a local square at the support of the other. At a prime of β, the relevant odd valuation of B is positive, so $1 - B/m$ is a square unit. At a prime of θ, $v(B)$ must be zero, since otherwise $v(H)$ is even; there $B/m \equiv 1$, so A is a square.

The simultaneous construction in S11–S12 gives $n, n\theta$ positive squarefree and congruent to eleven modulo twelve, both with two-Selmer dimension three and with extra Kummer class $(\beta, 1)$. Its matrix conditions can be stated explicitly. For support primes q_i, set $s_i = [-1/q_i]$, $\lambda_i = [3/q_i]$, and

$$(Kv)_i = \sum_{j \neq i} [q_j/q_i](v_j + v_i).$$

One arranges $\ker K = \langle \mathbf{1}, \mathbf{1}_{\operatorname{supp}\beta} \rangle$ and $\lambda \notin \operatorname{im} K$ on both supports. Add $d - 1$ prime slots of type $(s, \lambda) = (0, 0)$ and one slot q_0 of type $(1, 1)$, where $d = \omega(\beta)$. The cross-block has columns $e_i + e_d$. After the rank-one correction $K^\sharp = K + \lambda a^t$, deleting q_0 leaves

$$\begin{pmatrix} D_1 & C_1 \\ C_1^t & 0 \end{pmatrix}, \qquad \mathbf{1}^t D_1 \mathbf{1} = 1.$$

This is nonsingular: $C_1^t x = 0$ makes x constant, and the first equation forces that constant and then y to vanish. Adding the support of θ preserves this block. Free q_0-symbols toggle the new diagonal entries; the determinant's full-product coefficient is one. Thus a single Boolean assignment works on both supports. Reciprocity, CRT and Dirichlet realize it by distinct auxiliary primes.

In Kummer coordinates $(x + 3\ell, x - \ell)$, a positive-second-coordinate Selmer representative is $(3^\delta \prod q_i^{u_i}, \prod q_i^{v_i})$. The necessary local equations are

$$Ku + \operatorname{diag}(\lambda)v = \delta\lambda, \quad (K + \operatorname{diag}(s))v = 0, \quad (\lambda + s)^t v = 0.$$

The final equation is the condition at three. The matrix properties force $v = 0$, $\delta = 0$, $u \in \ker K$, so there are at most eight classes in total. The extra class is actually local at every place, including two, and independent of the two torsion classes, giving dimension three. This sufficiency check for the particular class is essential; arbitrary parity conditions at two would not suffice.

Lemma 5.4 (Dimension three gives corank one)**.** *For an elliptic curve with full rational two-torsion,* $\dim_{\mathbb{F}_2} \mathrm{Sel}_2 = 3$ *implies* $\operatorname{corank}_{\mathbb{Z}_2} \mathrm{Sel}_{2^\infty} = 1$.

Proof. The finite Kummer sequence gives $r + \dim \mathrm{Sha}[2] = 1$. Write $\mathrm{Sha}[2^\infty] = (\mathbb{Q}_2/\mathbb{Z}_2)^t \oplus F$ with F finite. The Cassels–Tate pairing on the quotient by the maximal divisible subgroup is nondegenerate and alternating for an elliptic curve [37]. Its finite primary cyclic factors are paired, so $\dim F[2]$ is even. Thus $F = 0$ and $r + t = 1$. The restriction of the pairing to $F[2]$ need not be nondegenerate and is not used. □

Lemma 5.5 (Forward analytic rank zero and one)**.** *Let $A/\mathbb{Q}$ be an elliptic curve. If $\mathrm{ord}_{s=1} L(A, s) = r \le 1$, then $\mathrm{rank}\, A(\mathbb{Q}) = r$ and $\mathrm{Sha}(A/\mathbb{Q})$ is finite. Consequently $\mathrm{corank}_{\mathbb{Z}_2} \mathrm{Sel}_{2^\infty}(A/\mathbb{Q}) = r$.*

Proof. Modularity follows from [50, Theorem A]. Apply [54, Theorem 1, pp. 4–5], whose statement includes both $r = 0$ and $r = 1$. Its Theorems 3–5 provide the auxiliary imaginary quadratic field, the nonzero Heegner height, and finiteness over that field; the restriction kernel for Sha is contained in the finite group $\mathrm{Sha}(A/\mathbb{Q})[2]$. The ordinary Kummer sequence

$$0 \longrightarrow A(\mathbb{Q}) \otimes \mathbb{Q}_2/\mathbb{Z}_2 \longrightarrow \mathrm{Sel}_{2^\infty}(A/\mathbb{Q}) \longrightarrow \mathrm{Sha}(A/\mathbb{Q})[2^\infty] \longrightarrow 0$$

gives the corank assertion. Independently, the $r = 0$ finiteness assertion follows directly from [55, Corollary 14.3], applied to the modular quotient $J_1(N) \twoheadrightarrow A$ over $\mathbb{Q}$ and the trivial character. This result includes the prime two and imposes no residual irreducibility hypothesis. □

Applying Theorem 5.3 to both n and $n\theta$, Lemma 5.5 gives rank one and finite Sha, hence $\mathrm{Sha}[2] = 0$. The Kummer class is then represented by actual rational points. Their abscissas have squareclass β, by the cubic equation. Write $H = \theta c^2$ and rescale the second point to E_{nH}. If the first abscissa is Aq^2, divide both parameters by q^2 and both coordinates by q^2, q^3. The abscissas become A, Az^2, giving precisely the eleven witnesses, with $w = (eHz)^{-1}$. If $\theta = 1$, one can reuse a point on the same curve. This proves Theorem 5.2, with both restricted-converse applications in their stated domain.

6 Multiplication and the finite false-triple guard

The local-test construction and prime-pattern strategy adapt [78, Section 2 and Lemma 6.1].

The actual multiplication template first forces all three input codes to be members of $2E(K_0)$. These membership calls are conjuncts inside the height certificate; they are not assumed caller preconditions. Thus even untyped rational inputs, if accepted, have unique indices. The template has 35,134 rational witnesses. Its construction is S03 and its mathematical specification is S13.

We use the following local conventions for the displayed integral Weierstrass equation. Put $z = -x/y$, and put $w = 1/x$ with $w(O) = 0$. This w differs from the auxiliary variable $-1/y$ used in [48, IV.1]. The formal expansions give $w = z^2$ times an integral unit, so $v(w) = 2v(z)$. For a complete discrete valuation with $e = v(p)$, the formal logarithm converges on $v(z) > 0$ and is an isometry with inverse exponential on $v(z) > e/(p-1)$ [48, IV.5.5, IV.6.3–6.4, pp. 129–132]. Indeed its degree-r coefficient has denominator dividing r, so each relative error has valuation at least $(r-1)v(z) - ev_p(r) > 0$ in this range. Thus $\log([n]Q) = n \log(Q)$ and $v(z([n]Q)) = v(z(Q)) + v(n)$ for nonzero Q and n there. These statements concern the formal neighborhood of our integral equation. When identifying it with the full reduction kernel of a minimal model, we use [48, VII.2.1–2.2, pp. 188–191].

For this particular equation the invariant differential is

$$\frac{dx}{2y} = \frac{dz}{\sqrt{1 + 4d^2 z^4}}, \qquad \log(z) = \sum_{j \ge 0} \frac{(-1)^j \binom{2j}{j} d^{2j}}{4j+1} z^{4j+1}.$$

This proves the displayed coefficients of L below. At a contact prime $p > 324$, with $v(z) \geq 1$ and $v(p) = 1$, every omitted term relative to z has valuation at least $r - 1 - v_p(r) \geq 28$ for $r \geq 29$. Consequently $L(z)/16575 = \log(z)(1 + \epsilon_z)$ with $v(\epsilon_z) \geq 28$.

For indices n_1, n_2, n_3, each test group chooses $H = [h]P_0$, $U_i = [z_i]P_0$, and sets $V_0 = H - [c_*]P_0$, $V_i = U_i - [c_* n_i]P_0$, where $c_* = 4M_0$ is fixed. Use

$$\begin{aligned} L(z) = 16575z &- 6630d^2z^5 + 11050d^4z^9 - 25500d^6z^{13} \\ &+ 68250d^8z^{17} - 198900d^{10}z^{21} + 612612d^{12}z^{25}. \end{aligned}$$

This is 16575 times the degree-25 truncation of the normalized formal logarithm. The twelve tests are four inverse abscissas at H, U_i, their four conjugate counterparts at V_0, V_i, the three differences $A_i - \sigma(A_i')$, and $A_1A_2 - A_3$, where

$$A_i = \frac{L(z(U_i))}{L(z(H))}, \qquad A_i' = \frac{L(z(V_i))}{L(z(V_0))} \quad (1 \leq i \leq 3),$$

and σ is the nontrivial automorphism of $K/\mathbb{Q}$. In particular the primed ratio is conjugated before the difference is taken. Identity inverse abscissas are zero; all denominators are guarded.

For a slope $t \notin \mathcal{C} \cup \{1\}$, put

$$R_0 = \eta F(t)/(t-1)^8, \quad \eta^2 = 1, \quad a = 1 - R_0q^2 \neq 0, \quad R = \ell_0^2 a^2 R_0.$$

Denote the twelve unscaled test values in group g by D_{gj}. Their normalized values are

$$C_{gj} = \frac{D_{gj}}{R_0^{k_j}(t-1)^{k_j+1}}, \qquad k_j = \begin{cases} 1, & 1 \leq j \leq 8, \\ 25, & 9 \leq j \leq 12. \end{cases}$$

For six independent groups set $W_g = \prod_j (1 + RC_{gj}^2)$ and $\Theta = \operatorname{Tr} \sum_g W_g^{-1}$, require them nonzero, and impose $\Phi(R/\Theta^2)$. The pole input is nonzero, so the prefactor B in Q_{pole} cannot make these constraints vacuous.

Here is the valuation deduction at a positive odd contact prime $p \notin S_{\text{mul}}$. Normalize every $v \mid p$ by $v(p) = 1$ and put $\delta = v_p(R_0) > 0$. The contact roots are distinct modulo p, and $t - 1$ is a unit, so δ is the positive odd contact order. The fixed ℓ_0 is supported on S_{mul}. Since $\delta + 2v_p(q)$ is odd, the two summands in $a = 1 - R_0q^2$ have unequal valuations. Thus $v_p(a) \leq 0$ and

$$\alpha = v_p(R) = \delta + 2v_p(a) \leq \delta, \qquad \alpha \text{ is odd}.$$

Every factor has valuation $v(1 + RC_{gj}^2) = \min\{0, \alpha + 2v(C_{gj})\} \leq 0$. If test j fails its threshold $v(D_{gj}) \geq k_j$ at v, then

$$v(1 + RC_{gj}^2) \leq \alpha + 2(k_j - 1) - 2k_j\delta \leq \alpha - 2\delta < 0.$$

Consequently $v(W_g^{-1}) \geq 2\delta - \alpha$ for that group and place. If every group fails at every place above p, the trace of their sum satisfies

$$v_p(\Theta) \geq 2\delta - \alpha, \qquad v_p(R/\Theta^2) \leq 3\alpha - 4\delta \leq -\delta < 0.$$

The latter valuation is odd, contradicting Lemma 5.1. Therefore one common group and place has its first eight tests of valuation at least one and its last four of valuation at least 25. This argument does not require all factors in that group to be units. The first tests put the points in the formal group. At least one of $h, h - c_*$ is a unit at the prime, so its full-log ratio is integral. The truncated

logarithm is the full logarithm times a unit congruent to one in valuation at least 28. Its ratio is therefore integral. The difference test transfers integrality to the other truncated ratio, then to the other full-log ratio. Only now do the relative errors give absolute congruences modulo p^{28}. Write $A_i = L(z(U_i))/L(z(H))$. The difference test and the full-log ratios give

$$z_i - hA_i \equiv 0, \qquad z_i - c_* n_i - (h - c_*)A_i \equiv 0 \pmod{p^{25}}.$$

Their difference is $c_*(n_i - A_i) \equiv 0$. Since $p \notin S_{\mathrm{mul}}$ implies $p \nmid c_*$, this yields $n_i \equiv A_i \pmod{p^{25}}$. The product test then implies $p^{25} \mid n_1 n_2 - n_3$.

To specify the height certificate, write the abscissa of each input as $x_{i0} + x_{i1} s$, using $x_{i0} = x_{i1} = 0$ at the identity. Choose $a_{i0}, a_{i1}, b_i \in \mathbb{Q}$ with $b_i \neq 0$ and $b_i x_{ij} = a_{ij}$. Express each of these nine rational quantities as a sum of three slopes. Impose the preceding slope predicate on all 27 slopes, with private witnesses, and impose membership on the three input codes. This conjunction is the height certificate. The multiplication predicate conjoins it with the fixed scalar graphs and the inverse of the finite guard below.

Proposition 6.1 (Finite-box deduction)**.** *Assuming these local test identities and the height estimate for S_{mul} with constant A_{ht}, every false triple accepted by the unguarded certificate has $\max_i |n_i| < T = (48A_{\mathrm{ht}})^{13}$.*

Proof. Put $f = n_1 n_2 - n_3 \neq 0$. For each of the 27 slopes, $M_{S_{\mathrm{mul}}}(t)^{25} \mid f$, so $h(t) \leq A_{\mathrm{ht}} |f|^{24/25}$. The nine rational homogeneous quantities for the three abscissas are sums of three slopes. The sum and quotient height bounds give $h(x_j) \leq 6A_{\mathrm{ht}} |f|^{24/25} + 2\log 3$ for each rational component. The fixed coordinate change and the duplication bounds yield

$$2n_i^2 \widehat{h}(P_0) \leq 12A_{\mathrm{ht}} |f|^{24/25} + B_0,$$

where $2\widehat{h}(P_0) > 1$ and $B_0 = 4\log 3 + \frac{3}{2}\log 2 + 2\log(1+\sqrt{2}) + \frac{1}{3}\log 676 < 10$. For $N = \max |n_i|$, $|f| \leq 2N^2$, so $N^2 < 24A_{\mathrm{ht}} N^{48/25} + 10$. If $N \geq 5$, this implies $N^{2/25} < 48A_{\mathrm{ht}}$, hence $N < (48A_{\mathrm{ht}})^{25/2} \leq T$. The case $N \leq 4$ is immediate. □

For each false triple $(a, b, c) \in [-T, T]^3$, write its fifteen code coordinates as A_j/B_j with $B_j > 0$, put $D_{abc} = \prod B_j$, and form

$$g_{abc}(X) = \sum_{j=1}^{15} (D_{abc} X_j - (D_{abc}/B_j) A_j)^2.$$

The guard requires the product of all these factors to have an inverse. A false accepted triple makes its own factor zero. At a true triple every factor is positive by injectivity of codes, and the inverse exists. Thus the sound direction uses the generator and corrected all-S height theorem but does not use pole completeness.

Proposition 6.2 (The fixed twenty-two-prime pattern)**.** *Put*

$$\mathcal{B} = \{0, -15, -20, -24, -30, -60, 1\}, \qquad P_s = \prod_{p \leq 324} p, \qquad W = 4P_s.$$

Let $\mathcal{H}$ consist of the first twenty-two primes greater than 324 in each unit residue class modulo W, as in the fixed constant prescription, and set

$$\mathcal{K} = \{dh : d \mid P_s^4,\ d > 0,\ h \in \mathcal{H}\}.$$

For every $A \in \mathbb{Z}$ and every real B_0 there are integers u_1, u_2, v, with $v \neq 0$, such that, on putting $u_3 = Av - u_1 - u_2$, each of the twenty-two values

$$v, \qquad u_j - bv \quad (1 \leq j \leq 3,\ b \in \mathcal{B})$$

equals ϵkp, where $\epsilon \in \{1, -1\}$, $k \in \mathcal{K}$, and $p > B_0$ is prime. Every such p splits in $\mathbb{Q}(\sqrt{2})$, the two reductions of

$$E : y^2 = x^3 - (75 - 53\sqrt{2})^2 x$$

are good, and their orders d_1, d_2 satisfy

$$\gcd(d_1, d_2) = 4, \qquad p \nmid d_1 d_2.$$

The prime values need not be distinct. The assertion concerns the two orders at the same prime only.

Proof. Write $T = (u_1, u_2, v)$ and denote the twenty-two homogeneous forms by F_ν. Their coefficient vectors are

$$(0, 0, 1), \quad (1, 0, -b), \quad (0, 1, -b), \quad (-1, -1, A - b).$$

Each form has a coefficient equal to 1 or -1. Within each of the three seven-form families, a nonzero two-by-two minor is a contact difference, of absolute value at most 61. Between two different families the first two coordinates give a minor 1 or -1; a pair involving v also has such a minor. Thus every pair is independent over $\mathbb{Q}$ and over $\mathbb{F}_q$ for every prime $q > 324$, for every integer A. No condition on three-form dependencies is required below.

Choose a bounded open rectangular box $\Omega \subset \mathbb{R}^3$ of positive volume whose closure avoids all the zero planes. For each ν choose ϵ_ν such that $\epsilon_\nu F_\nu > 0$ on $\overline{\Omega}$. These signs allow all integers A, including zero and negative integers.

The fixed multipliers and initial progression. For each prime $q \leq 324$, the proportion of T modulo q^5 for which at least one $F_\nu(T)$ vanishes is at most $22/q^5 < 1$. Choose a residue outside this union and lift it to a residue modulo q^8. On this residue each F_ν has a fixed valuation $e_{\nu,q} \in \{0, 1, 2, 3, 4\}$. Set

$$d_\nu = \prod_{q \leq 324} q^{e_{\nu,q}}.$$

The unit $\epsilon_\nu F_\nu / d_\nu$ is now fixed modulo 8 at 2 and modulo q at each odd $q \leq 324$. Prescribe the prime residues

$$c_2 = 1 \pmod 8, \qquad c_7 = 3 \pmod 7, \qquad c_q = 1 \pmod q \quad (q \leq 324,\ q \neq 2, 7).$$

Choose $h_\nu \in \mathcal{H}$ in the unit class specified by $h_\nu \equiv (\epsilon_\nu F_\nu / d_\nu) c_q^{-1}$ at these moduli. There are at most twenty-two demands in any class, so the h_ν can be chosen pairwise distinct. Put $k_\nu = d_\nu h_\nu$. Dirichlet's theorem proves that this is the same finite catalogue as the prescribed one; no extra catalogue prime is introduced.

At $h = h_\nu$, impose $F_\nu(T) = 0$. This plane has h^2 points. For every $\mu \neq \nu$, the restriction of F_μ is nonconstant, so excluding

$$F_\mu(T) = 0, \qquad \epsilon_\mu F_\mu(T) = -k_\mu$$

removes at most $42h < h^2$ points. Choose a remaining point. Among its lifts modulo h^2, the unit $\epsilon_\nu F_\nu / (d_\nu h)$ runs through every residue modulo h: changing a coordinate with coefficient 1 or -1

by h proves surjectivity. Choose this residue different from 0 and -1. The restrictions for $\mu \neq \nu$ remain valid.

The Chinese remainder theorem now gives one vector residue r_0 modulo

$$W_0 = \prod_{q \le 324} q^8 \prod_{\nu} h_\nu^2.$$

On $r_0 + W_0\mathbb{Z}^3$ the functions $\Psi_\nu = \epsilon_\nu F_\nu / k_\nu$ are integral. At every prime dividing W_0 they are nonzero, and at every odd such prime they are different from -1. In the affine coordinates on this progression every variable coefficient is divisible by each prime dividing W_0. Indeed, q^8/d_ν is divisible by q even when $e_{\nu,q} = 4$, and h^2/k_ν is divisible by h.

All these choices depend on A and the fixed catalogue, but not on B_0 or the cutoff L used next. Fix also

$$Q_* = \max(324, \max \mathcal{H}).$$

In both upper sieves below the excluded fixed primes are exactly those at most Q_*; in particular this exclusion does not grow with L. There are constants $0 < \alpha_A < \beta_A$ such that

$$\alpha_A X \le \Psi_\nu(T) \le \beta_A X \quad (T \in X\Omega,\ 1 \le \nu \le 22).$$

The lower count with its cutoff dependence. Take $L > \max(Q_*, B_0, 3)$. At each prime $q \le L$ not dividing W_0, exclude the forty-four affine planes

$$F_\nu(T) = 0, \qquad \epsilon_\nu F_\nu(T) = -k_\nu.$$

Each has q^2 points and all these primes exceed 324. With

$$W_L = W_0 \prod_{\substack{q \le L \\ q \nmid W_0}} q,$$

let $\mathcal{R}_L$ be the resulting vector residues modulo W_L that reduce to r_0 modulo W_0. Then

$$\frac{|\mathcal{R}_L|}{W_L^3} \ge W_0^{-3} \prod_{\substack{q \le L \\ q \nmid W_0}} (1 - 44/q) \ge c_A (\log L)^{-44}. \tag{1}$$

Here and below a positive constant with subscript A may depend on the choices already made, including the fixed catalogue, but never on L, X, or a varying fiber prime. The last inequality follows from $\log(1 - 44/q) = -44/q + O(q^{-2})$ and Mertens' theorem [81, Theorem 1, with modulus 2].

Fix L. For $r \in \mathcal{R}_L$ apply the finite-complexity prime asymptotic of [38, Main Theorem, p. 1763], with its hypotheses supplied by [39, Theorem 1.1] and [40, Theorem 1.3], to

$$n \longmapsto \frac{\epsilon_\nu F_\nu(r + W_L n)}{k_\nu}, \qquad r + W_L n \in X\Omega.$$

These are integral affine forms with pairwise independent linear parts. Singleton classes in [38, Definition 1.5] give complexity at most 20, so the inputs GI(20) and MN(20) suffice. The inverse theorem is read in arXiv version 5 together with the separate erratum, in particular its Proposition 3.1 and modified deduction of Proposition 7.3. All coefficient bounds and moduli are fixed before

X tends to infinity. At $q \mid W_L$ the local factor is $(1-1/q)^{-22}$. At $q \nmid W_L$, scaling and changing variables preserve the twenty-two homogeneous zero planes, so

$$1 - 22/q \le \mathbb{P}(F_\nu \neq 0 \text{ for all } \nu) \le 1 - 22/q + \binom{22}{2} q^{-2}.$$

Consequently the local factor is $1 + O(q^{-2})$, and

$$\log \beta_q \ge \log(1 - 22/q) - 22\log(1 - 1/q) \ge -484/q^2.$$

The singular product is therefore bounded below by an absolute positive constant, uniformly in L and r. The real factor is $\mathrm{vol}(\Omega)X^3/W_L^3$.

There are finitely many r for fixed L, so their asymptotic errors can be made smaller than half their main terms with one threshold for X, depending on A, L. This uses no uniform prime-asymptotic error as L varies. Summing over r and using (1) gives a weighted lower bound $c_A(\log L)^{-44}X^3$. Proper prime powers contribute at most $O_A(X^{5/2}(\log X)^{23})$: there are $O_A(X^{1/2}\log X)$ possible proper powers in each form's range, each equation fixing a form value has $O_A(X^2)$ integer solutions in the box, and each weight is $O_A((\log X)^{22})$. Thus, for all sufficiently large X depending on A, L,

$$\#\{T \in X\Omega :\ T \bmod W_L \in \mathcal{R}_L,\ \Psi_\nu(T) \text{ all prime}\} \ge a_A \frac{X^3}{(\log L)^{44}(\log X)^{22}}, \qquad a_A > 0. \tag{2}$$

An upper sieve with uniform constants. We give the estimates needed for both upper bounds. Consider an axis-parallel square of side $N \ge 1$ in $\mathbb{R}^2$. For each sieving prime q let $\Omega_q \subset \mathbb{F}_q^2$ have size $\rho(q)$, where $0 < \rho(q) \le 21q$ and $q > Q_*$. Put

$$g(q) = \rho(q)/q^2, \qquad h(q) = g(q)/(1 - g(q)),$$

and extend both multiplicatively on squarefree integers supported on the sieving primes. For such d, CRT and the elementary count of lattice points in each residue square give

$$A_d = N^2 g(d) + R_d, \qquad |R_d| \ll \rho(d)(N/d + 1) \le Nd + d^2. \tag{3}$$

The implied absolute constant is uniform in the square's location and in the sets Ω_q. The last inequality uses $\rho(q) \le q^2$.

For $Z \ge 2$ define

$$G(Z) = \sum_{d \le Z} h(d), \qquad \lambda_d = \frac{\mu(d)}{G(Z)\prod_{q \mid d}(1 - g(q))} \sum_{\substack{m \le Z/d \\ (m,d)=1}} h(m) \quad (d \le Z),$$

and set $\lambda_d = 0$ otherwise. All sums here are over the stated squarefree support, with $g(1) = h(1) = 1$. These are Selberg's weights. They satisfy $\lambda_1 = 1$, $|\lambda_d| \le d$, and

$$\sum_{e,f \le Z} \lambda_e \lambda_f g([e,f]) = G(Z)^{-1}.$$

For completeness, set $y_r = \sum_{r \mid d} \lambda_d g(d)$. The displayed quadratic form equals $\sum_r y_r^2/h(r)$, because $1/g(m) = \sum_{r \mid m} 1/h(r)$; the stated weights give $y_r = \mu(r)h(r)/G(Z)$. The bound on the weights follows from the inner sum being at most $G(Z)$ and $(1 - g(q))^{-1} \le q$ for $q > 324$.

Squaring the divisor sum majorizes the indicator of avoiding all Ω_q for $q \le Z$. Inserting (3), and using $[e,f] \le Z^2$, there are at most Z^2 pairs with $|\lambda_e \lambda_f| \le Z^2$. It follows that

$$S(N,Z) \ll \frac{N^2}{G(Z)} + NZ^6 + Z^8. \tag{4}$$

Here are the two denominator estimates, including truncation. For $z = Z^{1/100}$ put

$$H(z) = \prod_{\substack{q \le z \\ q \text{ sieved}}} (1 + h(q)).$$

Give each squarefree divisor d of this prime product probability $h(d)/H(z)$. The events $q \mid d$ are independent and have probability $g(q)$. Hence

$$\mathbb{E} \log d = \sum_{\substack{q \le z \\ q \text{ sieved}}} g(q) \log q \le 21 \sum_{q \le z} \frac{\log q}{q} \le 42 \log z = 0.42 \log Z$$

for sufficiently large z, using $\sum_{q \le z} (\log q)/q = \log z + O(\log \log z)$. Indeed, [81, Theorem 1, with modulus 2] gives $S(z) := \sum_{q \le z} q^{-1} = \log \log z + C + O(1/\log z)$, including the prime 2. Partial summation in $S(z) \log z - \int_2^z S(t)\, dt/t$ gives the asserted bound. Markov's inequality gives probability at least 0.58, and thus at least 1/2, that $d \le Z$. Every such divisor is included in $G(Z)$, so

$$G(Z) \ge \tfrac{1}{2} H(Z^{1/100}). \tag{5}$$

This proves a lower bound for the truncated sum, rather than merely for an untruncated Euler product.

In the first application the sieved primes are $q > Q_*$ with $q \equiv 1 \pmod 4$, with at most one prime omitted, and $g(q) = 2/q - 1/q^2$. Mertens' theorem in the fixed progression 1 (mod 4) [81, Theorem 1] gives

$$\log H(z) = \log \log z + O_{Q_*}(1).$$

Omitting a prime ℓ changes $H(z)$ by $1 - g(\ell) \ge 1/2$ when $\ell \le z$, and has no effect otherwise. Therefore $G(Z) \gg_{Q_*} \log Z$ uniformly in the omitted prime. In the second application all primes $q > Q_*$ are sieved and

$$\frac{21}{q} - \frac{210}{q^2} \le g(q) \le \frac{21}{q}.$$

Then $\log H(z) = 21 \log \log z + O_{Q_*}(1)$, with an error independent of the individual sets Ω_q, and hence $G(Z) \gg_{Q_*} (\log Z)^{21}$. These statements and (4) establish every uniform upper sieve estimate used below.

The conjugate orders. Every prime counted by (2) satisfies $p \equiv 1 \pmod 8$ and $p \equiv 3 \pmod 7$. Thus it splits in $\mathbb{Q}(\sqrt{2})$ and $(7/p) = -1$. The two reductions of $d = 75 - 53\sqrt{2}$ have product 7 and opposite square classes. The two reduced curves are the quadratic twists of $y^2 = x^3 - x$ by these two classes. Their traces are opposite. The discriminant $64d^6$ is a unit at these primes, so both reductions are good.

For $p \equiv 1 \pmod 4$, choose $i \in \mathbb{F}_p$ with $i^2 = -1$. The automorphism $\iota : (x,y) \mapsto (-x, iy)$ satisfies $\iota^2 = [-1]$ and commutes with the arithmetic p-power Frobenius φ, directly from its coordinate formula. The injection

$$\mathrm{End}(\overline{E}) \otimes \mathbb{Q}_3 \hookrightarrow \mathrm{End}_{\mathbb{Q}_3}(V_3 \overline{E})$$

is [48, III.7.4, pp. 89–91]; here $p > 324$. Since $T^2 + 1$ is irreducible over $\mathbb{Q}_3$, the centralizer of ι on this two-dimensional Tate space is $\mathbb{Q}_3(i)$. The rational centralizer therefore has dimension at most two and already contains $\mathbb{Q}(i)$, so it is exactly $\mathbb{Q}(i)$. The Frobenius endomorphism is integral over $\mathbb{Z}$ by [48, V.2.3.1(b), p. 142]. Hence it lies in $\mathbb{Z}[i]$; write it as $a + bi$ with $a, b \in \mathbb{Z}$. The trace and determinant formulas [48, III.8.6, p. 99] give

$$a^2 + b^2 = p, \qquad d_1 = p + 1 - 2a, \qquad d_2 = p + 1 + 2a.$$

Here d_1, d_2 are the cardinalities of the two reduced elliptic curve groups; the opposite twist changes the Frobenius trace's sign. In particular $a \neq 0$, since p is not a square. Both orders are divisible by 4 because all the two-torsion is rational. Their sum is $2(p+1)$, of 2-adic valuation exactly 2, so the 2-part of their gcd is exactly 4. Hasse's bound [48, V.1.1, p. 138] gives $0 < d_i < 2p$ for $p > 324$, and $d_i \neq p$ since $4 \mid d_i$. Thus $p \nmid d_1 d_2$.

If an odd prime ℓ divides both orders, it divides a and $p + 1$, and $b^2 \equiv -1 \pmod{\ell}$. The progression conditions already rule out every odd $\ell \leq L$. It remains to discard prime values with such an $\ell > L$.

The number of bad prime values. For a fixed form its prime value lies between $\alpha_A X$ and $\beta_A X$. Representations $p = a^2 + b^2$ lie in $[-C_A\sqrt{X}, C_A\sqrt{X}]^2$. Overcount them to bound the number of bad primes. If $L < \ell \leq X^{1/8}$, there are at most two residues b_0 with $b_0^2 \equiv -1 \pmod{\ell}$. Write

$$a = \ell U, \qquad b = b_0 + \ell V.$$

The required (U, V) lie in a square of side $N \asymp_A \sqrt{X}/\ell \gg_A X^{3/8}$. For every $q > Q_*$ with $q \equiv 1 \pmod 4$ and $q \neq \ell$, the condition $a^2 + b^2 = 0 \pmod q$ is the union of two distinct lines through the origin, of size $2q - 1$. The affine substitution is bijective modulo q. Take $Z = X^{1/100}$. Since the prime norm is at least $\alpha_A X > Z$ for large X, every required pair avoids these residues. Equations (4) and (5) give

$$\#(U, V) \ll_A \frac{N^2}{\log X} + NZ^6 + Z^8 \ll_A \frac{X}{\ell^2 \log X}.$$

The absorption is uniform: $Z^6/N \ll_A X^{0.06-3/8}$ and $Z^8/N^2 \ll_A X^{0.08-3/4}$, both tending to zero faster than $1/\log X$. Summing over $\ell > L$ gives $O_A(X/(L \log X))$.

For $\ell > X^{1/8}$ use the lattice count directly. Since $a \neq 0$, there are $O_A(\sqrt{X}/\ell)$ possible multiples a, with no additive constant. For each of the at most two root classes there are $O_A(\sqrt{X}/\ell + 1)$ choices of b. Also $\ell \leq |a| \leq C_A\sqrt{X}$. Bounding the sum over primes by the corresponding sum over positive integers yields

$$\sum_{X^{1/8} < \ell \leq C_A\sqrt{X}} O_A\left(\frac{X}{\ell^2} + \frac{\sqrt{X}}{\ell}\right) \ll_A X^{7/8} + \sqrt{X} \log X.$$

We have therefore proved the bound, uniformly in L,

$$\#\{\text{bad prime values for a fixed form}\} \ll_A \frac{X}{L \log X} + X^{7/8} + \sqrt{X} \log X. \tag{6}$$

The number of patterns on each nonzero fiber. Fix ν and $p \in [\alpha_A X, \beta_A X]$, and put $c = \epsilon_\nu k_\nu p$. The integral solutions of $F_\nu(T) = c$ are parametrized by two original coordinates: solve for a coordinate whose coefficient is 1 or -1. For example, for $F_\nu = -u_1 - u_2 + (A - b)v$, use (u_1, v)

and set $u_2 = (A-b)v - c - u_1$. This parametrization is valid over every $\mathbb{F}_q$, and the points from $X\Omega$ lie in a square of side $O_A(X)$ in these coordinates.

For $Q_* < q \le Z = X^{1/100}$, one has $q \nmid k_\nu$ and $q < p$ when X is large. Thus $c \neq 0$ modulo q. Each remaining form restricts to a nonconstant affine form. Their twenty-one zero lines are distinct: if the restrictions of F_j, F_k had the same zero line, then for some $\lambda \neq 0, \mu$,

$$F_j - \lambda F_k = \mu(F_\nu - c)$$

as affine forms over $\mathbb{F}_q$. Comparing constants gives $\mu c = 0$, hence $\mu = 0$, contradicting pairwise independence of F_j, F_k. Parallel lines and multiple intersections cause no difficulty. The union consequently has between $21q - \binom{21}{2}$ and $21q$ points.

Every target pattern on the fiber avoids this union: its other prime values are also at least $\alpha_A X > Z$, and every k_j is a unit at $q > Q_*$. Discarding the CRT progression conditions only increases the count. The second denominator estimate in (5) gives

$$\#\{\text{target patterns on } F_\nu = c\} \ll_A \frac{X^2}{(\log X)^{21}} + X^{1.06} + X^{0.08} \ll_A \frac{X^2}{(\log X)^{21}}. \tag{7}$$

This is uniform in the fiber prime p and independent of L. In particular $q = p$ is outside the sieving range; no varying exceptional-prime product has been suppressed.

The order of choices. Multiply (6) by (7) and sum over the twenty-two forms. The number of bad patterns is at most

$$b_A \frac{X^3}{L(\log X)^{22}} + O_A\left(\frac{X^{23/8}}{(\log X)^{21}} + \frac{X^{5/2}}{(\log X)^{20}}\right),$$

where b_A is independent of L. First choose $L > \max(Q_*, B_0, 3)$ so large that

$$\frac{b_A}{L} < \frac{a_A}{2(\log L)^{44}}.$$

Such an L exists because $(\log L)^{44}/L \to 0$. Then keep A, L and every preceding choice fixed, and take X large enough for (2), the upper bounds, and the remaining error to be less than $a_A X^3/(2(\log L)^{44}(\log X)^{22})$. A pattern survives. Its signed prime values tend to infinity with X, $v \neq 0$, and the preceding order calculation proves all the assertions. □

Apply this construction to the nine integral homogeneous quantities of a true triple. Each slope has at most six odd-contact primes. At one such prime with reduced group cardinalities d_1, d_2, which annihilate the reduced point, solve

$$h \equiv 0 \pmod{\operatorname{lcm}(M_0, d_1)}, \qquad h \equiv c_* \pmod{d_2}.$$

Compatibility follows from

$$\gcd(\operatorname{lcm}(M_0, d_1), d_2) \mid M_0 \gcd(d_1, d_2) = 4M_0 = c_*.$$

Choose a solution larger than c_*. Each contact prime uses its own private group within its own slope. No common shift across different primes is imposed, and selected prime values need not be distinct. The saved library allocates six private groups per slope and twenty-seven private slope calls. Set $U_i = [n_i]H$ and assign one group to each prime. Fixed local clearing at the old primes

can be checked as follows. Put $S_b = S_{\mathrm{ht}} \cup \mathcal{H}$, $B_p = 4$ on S_{ht} and $B_p = 1$ on the remaining catalogue primes, $C_p = 14B_p$, and $D_p = 51C_p$. The selected integral forms give $|v_p(R_0)|, |v_p(t-1)| \le C_p$, while the small formal subgroup in the definition of M_0 gives $v_p(D_{gj}) \ge 0$ at S_b. Consequently $v_p(C_{gj}) \ge -D_p$. The fixed normalization has $v_p(\ell_0) = E_p = 52C_p + v_p(12) + 4$.

Weak approximation chooses q small at S_b, integral at the selected primes, and arbitrarily close to a real root of $1 - R_0 q^2$, without equality; choose η to make $R_0 > 0$. Then a is a unit at the specified finite primes. At $p \in S_b$,

$$v_p(RC_{gj}^2) \ge 2E_p - C_p - 2D_p > v_p(12), \qquad v_p(\Theta - 12) > v_p(12),$$

so $v_p(R/\Theta^2) \ge 103C_p + 8 \ge 4$. At a selected odd-contact prime, $v_p(R) = 1$. Each of the twelve summands of the trace of the six W_g^{-1} has residue zero or one; the assigned group and place contributes one. Since $p > 324$, their sum is a unit and the pole input has valuation one. At every other prime outside S_b, its valuation is even. Thus its reduced denominator is square. In both real embeddings each W_g is positive; as $a \to 0$, $W_g \to 1$, $\Theta \to 12$, and $R/\Theta^2 \to 0$ through positive values. This verifies exactly PC's hypotheses, including all nonzero guards. Negative indices, zero indices and coincident points are handled by the group-law branches. Therefore Theorem 5.2 supplies the required pole witnesses. This proves the full multiplication statement M, including completeness for every integer triple $(n_1, n_2, n_1 n_2)$ and soundness on arbitrary rational input codes.

7 The restricted converse and its dependency order

The organization of the restricted-converse argument follows [74], particularly its ring-class interpolation, determinant transfer, coefficient systems and fixed-curve assembly. The broader converse is stated in [75]. Our specialization uses the same full Selmer-corank convention, including any divisible part of Sha, as those sources.

For a positive squarefree $\ell \equiv 11 \pmod{12}$ put

$$E_\ell : y^2 = x(x - \ell)(x + 3\ell).$$

Its invariants are $c_4 = 208\ell^2$, $\Delta = 2304\ell^6$ and $j = 35152/9$. Thus it has full rational two-torsion and is not a CM curve, since a rational CM j-invariant is an integer. The curve in this section is different from the fixed CM curve used to represent integer indices.

The assertion needed in Theorem 5.3 is

$$\operatorname{corank}_{\mathbb{Z}_2} \mathrm{Sel}_{2^\infty}(E_\ell/\mathbb{Q}) = 1 \quad \Longrightarrow \quad \operatorname{ord}_{s=1} L(E_\ell, s) = 1. \tag{RC}$$

We prove this assertion in this section and the next two sections. Appendices E–G give the fixed-form, theta, nonvanishing and normalization constructions used in this deduction. By [50, Theorem A and Introduction, conditions (1)–(6)], $E = E_\ell$ is modular. Here N denotes its conductor. The modular realization gives a rational weight-two newform f of level N and an elliptic quotient $q_f : J_0(N) \twoheadrightarrow A_f$ in the rational isogeny class of E; see [51, Theorem 11.19 and Remark 11.18]. Choose a $\mathbb{Q}$-isogeny $\alpha : A_f \to E$ and put $\varphi = \alpha q_f$ and $\pi(x) = \varphi([x] - [\infty])$. Thus $\pi(\infty) = 0$ and, for $q \nmid N$, $\varphi T_q = [a_q(f)]\varphi$. Fix the integral Tate lattice and the local squareclasses allowed at the primes of $2N$. Constants depending on this data are chosen before any conductor, companion field or binary dimension varies. Quotienting by a rational point of order two gives a rational two-isogeny. The Selmer parity theorem [49, Theorem 1.8] therefore gives root number -1 under the left side of (RC). This application does not replace Selmer corank by Mordell–Weil rank and does not assume finite Sha.

Remark 7.1 (Why this family is the input to the converse). The congruence $\ell \equiv 11 \pmod{12}$ comes from the pole-completeness construction. In the proof of Theorem 5.2, the two parameters $n, n\theta$ are positive squarefree numbers in this congruence class; their two-Selmer groups have dimension three. Lemma 5.4 gives full two-primary Selmer corank one, so these are exactly the two applications required of Theorem 5.3.

In the converse argument we fix one such ℓ. In particular ℓ is odd, is prime to three, and is greater than one, hence is not a square. We use the displayed full rational two-torsion and non-CM properties, the resulting rational two-isogeny, and the explicit roots $0, \ell, -3\ell$. The last roots enter the K-split seed argument below. The constants may depend on this fixed curve; there is no bound on $\omega(\ell)$ and no assertion that they are uniform as ℓ varies. The statements about coefficient graphs below are the rational, full-two-torsion case used by E_ℓ and its quadratic twists.

Here is the dependency order of the proof. A fixed-form Euler-system calculation gives the even coefficient bound. A varying-companion arithmetic estimate, proved by the two limits below, then gives the genus-point lower bound. Only after that lower bound do we normalize the odd coefficients. Their finite-stage forms and the even forms give one common binary family. Finally a fixed-field determinant argument excludes a zero derivative at the base. The even estimate does not use that final determinant argument on Heegner classes.

7.1 The even coefficient calculation

Use normalized central values $L_*(h)$ with the real and imaginary periods fixed by π and the integral lattice. Only their valuations are used; every subsequent comparison uses this same normalization. For every twist whose two-primary Tate–Shafarevich group is finite, let $s(h) = \operatorname{length}_{\mathbb{Z}_2} \operatorname{Sha}(E^{(h)}/\mathbb{Q})[2^\infty]$.

Proposition 7.2 (The fixed-form even scalar)**.** *Use the fixed E, π, integral lattice and allowed old local types specified above. Put $\Lambda = \mathbb{Z}_2[[t]]$. For a finite elementary abelian two-group G, let $\chi_x = \chi_0(x \circ \varepsilon)$ $(x \in \widehat{G})$ be a family of quadratic characters with a common real sign and the same restrictions at every place dividing $2N$. There is one scalar $u^{\mathrm{ev}} \in \Lambda[1/2][G]$ satisfying (3.2). Its clearing exponent and the bounded error in that formula depend only on the fixed data, not on G or the good-prime conductor support. The central identity is used only at corank-zero constituents with nonzero central value; at a corank-zero constituent the same central coordinate is zero exactly when the central value is zero.*

Proof. The following construction is expanded in Propositions E.1 and E.2, with its central calculation in Proposition E.3. The fixed-form construction uses one integral symbol combination with nonzero projections to both real signs. Kato's integral maps, with j_* cohomology taken modulo each power of two before passage to the limit, give classes whose inertia restrictions vanish integrally. Indeed the low-degree Leray edge map gives this vanishing before rationalization. Restriction of transfer is the Mackey sum of the corresponding upper inertia restrictions. Hence unnormalized transfer preserves the vanishing even when the intervening extension is ramified. No character projector divided by a growing group order is used.

For a finite binary character group, let F be the least common multiple of the odd parts of all its character conductors, including the base character. Write Z_F for the compatible Iwasawa class obtained by pushing the retained-smoothed Kato classes at levels $F2^n$ through the integral coefficient map. Here and throughout the common-scalar construction, the binary family is required to have one real sign and fixed local squareclasses at $2N$. Writing its characters as $\chi_x = \chi_0(x \circ \varepsilon)$, this means $\varepsilon(c) = 1$ for complex conjugation c, and $\varepsilon|_{G_{\mathbb{Q}_v}} = 1$ at every $v \mid 2N$. Indeed every character of G evaluates these elements to one, and the characters separate G. The lower bound

uses $G = 1$; the upper-transfer cubes lie in the fixed signed unit filter of their coefficient source. The final cubes have the same property. At a character χ its unused-prime factors are

$$E_q(\chi) = 1 - \frac{a_q \chi(q)}{q} \sigma_q^{-1} + \frac{1}{q} \sigma_q^{-2}. \tag{3.1}$$

They are the actual determinants of the unused good local terms. Active ramified good primes have rational inertia -1, so have no such rational local term. The fixed bad and dyadic corrections form a factor F_E with nonzero central value.

The rational divisibility input is Kato's Theorem 13.4(2), together with the nonzero-class input in 13.5(2) [55]. At two we justify the divisibility through [73, Theorem 0.8]; Appendix E checks its stronger abelian-extension hypothesis and compares the two cohomology conventions. We use the height-one primes not containing two. Rational irreducibility is preserved on a quadratic subgroup of the non-CM open image. That subgroup contains a nontrivial unipotent, which supplies the required one-dimensional fixed space. Oddness supplies the two real eigenspaces, and purity supplies the common Frobenius weight. This application does not use residual irreducibility at two or the stronger hypotheses of Theorem 13.4(3).

The smoothing constants c, d_s can be chosen to satisfy

$$c, d_s \equiv 1 \pmod{3N_{\mathrm{odd}}F}, \qquad c, d_s \equiv 1 + 2^e \pmod{2^{e+1}},$$

where $e \geq \max\{3, v_2(N)\}$ is fixed by the original dyadic data. In weight two and cohomological degree one the two smoothing exponents are one. Kato's explicit reciprocity formula therefore contributes valuation $2e$, independent of F. For the central normalization, first descend to the primitive character conductor and then use the invariant basis $\sqrt{h}\, e_{\chi_h}$ of the quadratic character line. The resulting varying scalar $1/h$ is a two-adic unit. The complete calculation in Appendix E includes discriminant $4h$; a Gauss basis there has the extra factor $\sqrt{\Delta_h/h}$, which must also be included in its change of basis. The numerical sizes of c, d_s may grow, while their relevant valuation remains fixed.

For completeness, the integral calculation at two uses the positive complex. Finite continuous cochains are filtered unions of finite free modules and hence flat. Splitting the residual complex into cohomology and disks, then using the nilpotent perturbation calculation of Section 9, gives finite free models with their maps. In the fixed real cyclotomic tower the ordinary two-primary class group vanishes. One proves this inductively: only the prime above two ramifies in each quadratic step; the abelianization of a possible Hilbert two-class tower is generated by that inertia of order two. A finite two-group with cyclic abelianization is cyclic, and inertia excludes a larger cyclic extension. The unit-signature calculation is separate. A norm-minus-one unit in positive layers, and -1 at the base, has signature vector a unit in the characteristic-two cyclic group algebra. For a scalar coefficient the resulting residual dimensions are

$$\dim H^1 = 2^n + s_n - 1, \qquad \dim H^2 = s_n - 1.$$

For each fixed support s_n is bounded. Universal coefficients give generic residual rank one in degree one; the two trivial elliptic factors give rank two. Unit cancellations thus identify the positive complex over $\mathcal{O}[G]$ with $\mathcal{O}[G]^2[-1]$. This last rank is independent of support, although preliminary models need not have a uniform size.

A generator of the fixed real invariant lattice gives a connecting class a. Its central image need not be saturated. Twice an ordinary class lifts because the relevant real H^1 is killed by two. Wedge a with a lift of $2F_E Z_F$ in the original determinant line. This produces a common scalar u^{ev}. The

first assertion below holds for the full family. The central valuation identity is asserted only at a constituent of Selmer corank zero with $L_*(h) \neq 0$:

$$2^{C_{\mathrm{ev}}} u^{\mathrm{ev}} \in \Lambda[G], \qquad v_2 u_h^{\mathrm{ev}}(0) = v_2 L_*(h) - 2\omega(h) - s(h) + O_E(1). \tag{3.2}$$

Here $\Lambda = \mathbb{Z}_2[[t]]$ and $\mathcal{O} = \widehat{\Lambda_{(2)}}$. The error and C_{ev} are fixed before the binary group and support. To justify membership in $\Lambda[G]$, first use rational characterwise divisibility and Fourier inversion for membership in $\Lambda[1/2][G]$, without estimating the Fourier denominator. Intersect coefficientwise with the integral $\mathcal{O}[G]$ result. The explicit coefficient construction also covers field discriminant $4h$, with its dyadic character retained before the cofinal norm limit, as in Appendix E. At a corank-zero constituent the same coordinate vanishes centrally if and only if the central value vanishes. At such a nonzero corank-zero constituent, keeping the real saturation and torsion terms gives the local formula

$$v_2 u(0) = v_2 \exp^* z - s - \sum_v g_v + 2g + 1 - r_1 - e_2.$$

At an active good odd prime $g_v = 2$, since $A^{I_v} = E[2]$. Unused terms cancel (3.1); all other terms belong to the fixed local list. This proves (3.2) with the indicated active-prime coefficient. □

7.2 Nonvanishing with all support restrictions

Proposition 7.3 (Nonvanishing in a prescribed unit filter)**.** *The curve E is the fixed curve of this section. The following assertion allows any nonempty signed unit filter and any additional finite avoidance set; its bound on the number of prime factors does not depend on either set. A signed unit filter prescribes a real sign and unit squareclasses at a finite set S containing $2, 3$ and the bad primes. Let B be any additional finite set of forbidden primes. If its root number is even, there is a squarefree parameter in this filter, coprime to B, with at most five prime factors and nonzero central value. In an odd filter there is a simple-zero parameter, with no asserted prime-count bound.*

Proof. Here is the reduction to the precise classical statements. Choose a signed prime q_0 in the prescribed unit classes by Dirichlet's theorem, avoiding $S \cup B$. Apply Hoffstein–Luo [57] in the even case, or Bump–Friedberg–Hoffstein [56] in the odd case, to the actual curve $E^{(q_0)}$. Require their positive discriminant d to split at $S \cup B \cup \{|q_0|\}$ and at the conductor primes. Then $q_0 d$ has the prescribed original classes. Hoffstein–Luo gives at most four prime factors of d, so at most five in total. The BFH clause used in the odd case is the simple-zero clause.

The root-number check includes ramified dyadic quadratic characters. At the primes in S equal squareclasses give isomorphic local curves. At a good odd prime newly ramified in the parameter, the root factor is $(-1/p)$ [58, Proposition 3.3]. Its product outside S is fixed by the real sign and the dyadic unit class. Thus unit classes 3 or 7 modulo eight are allowed; the field discriminant may be four times the parameter. We do not invoke an odd-discriminant root formula on those parameters. See also Proposition G.1 for the same five-prime bound and its subsequent use in the companion construction. □

7.3 Coefficient symbols and the finite address lemmas

We state the finite interface before using it for the even upper bound. The graph language records quadratic residue data of the coefficient indices. It does not replace a Selmer group by an abstract graph. Throughout this subsection all vertices are in the rational full-two-torsion case. Thus there is one independent residue bit per pair; the split/simple distinction needed for other two-torsion fields does not occur here.

Definition 7.4 (Compatible typed graphs)**.** Fix the finite support S of a coefficient source, containing two, and the finite quadratic characters used to read its coefficients. The *type* of a prime $p \notin S$ records its values under these characters and its unit squareclasses at S, including its class modulo eight. We write the resulting finite elementary two-group additively as $\mathcal{L}$. We retain all relations among these coordinates; a tuple of arbitrary character values need not be a type. Additional fixed splitting conditions restrict us to their actual realizable types. For an odd theta stage, its reduction prime is part of this fixed support during that stage. Let S_{lab} also contain the ramification of any additional stationary label tests. Enlarging S_{lab} records labels and prime exclusions; it does not change the coefficient source or add relaxed places to a Selmer complex.

A *compatible typed graph* consists of a finite ordered vertex set V, an individually realizable type $\ell_v \in \mathcal{L}$ for each vertex, and quadratic residue bits. For prime representatives, write $z_{vw} = 0$ or 1 according as $(p_v/p_w) = 1$ or -1. For each unordered pair just one direction is freely specified; the other satisfies

$$z_{vw} + z_{wv} = \epsilon_v \epsilon_w, \qquad \epsilon_v = (p_v - 1)/2 \pmod 2,$$

where ϵ_v is part of the type. There are no self-residue bits. Only these compatible, realizable data are used below. A fixed filter prescribes the total type $\sum_v \ell_v$ needed for the coefficient index, including the signed parameter convention of the source. If that convention is $h = \sigma D$, then $D = \prod_v p_v > 0$ is the Fourier index and σ is fixed by the source. The signed factors $p_v^* = (-1/p_v)p_v$ will be used when the same graph defines a family of quadratic characters.

Contracting two vertices replaces their type by its sum and every incidence to an external vertex by the sum of the two incidences. Equivalently its template is the product $p_v p_w$; it is represented by a fresh prime with those data. Reverse incidences are determined by reciprocity. No bit internal to the contracted cluster is retained. A vertex is *primary* if its generator is a square unit at the fixed support and its type is neutral in every retained quadratic and stationary coordinate. It is *neutral relative to a graph* if in addition all its incidences to the other vertices are zero.

Two symbols may read projections of one common type group. The projections retain the relevant coordinates, preserve primary types, and commute with the product and external-incidence rules just given. We require each indicated projection to be surjective onto its own realizable alphabet; we do not require the map to their product to be surjective.

Definition 7.5 (Coefficient symbols)**.** A *symbol* is the function A with values in $\mathbb{F}_2$ obtained from one fixed coefficient source and one fixed normalized depth δ. On an allowed positive squarefree index D it is

$$A(D) = \begin{cases} 1, & v_2(a(D)) = \delta + \omega(D), \\ 0, & v_2(a(D)) > \delta + \omega(D), \end{cases}$$

and it is zero off that source's filter. For the even source, exact phase removal makes the relevant coefficient rational. For the odd source, (3.8) gives the required phase removal modulo the retained power 2^{M_θ}; use its integer representative at that precision. A coefficient zero modulo 2^{M_θ} is invisible. Whenever a unit is tested, the precision exceeds $\delta + \omega(D)$, so the test is independent of the representative. No exact rationality of an arbitrary lifted odd theta coefficient is asserted. The lower bound making these alternatives exhaustive is part of the coefficient construction, not part of this definition. For the even source it follows from (3.2) and the coefficient-square formula of Proposition E.4.

For an odd source this definition at one finite stage is used only through a specified finite squarefree-weight cutoff and at a stage where the lower bound holds through that cutoff. The chosen cutoff and all required divisions must be included in the operator budget. The fixed minimum and the compatible finite tables are constructed in Propositions G.3 and G.5. The resulting limit symbol

is a function on finite compatible typed graphs. It is not a valuation rule at all weights of one finite form. The same source and depth are kept when a vertex is omitted.

Lemma 7.6 (The coefficient rules)**.** *After the stated coefficient lower bounds have been proved, the even symbol and the stabilized odd symbol satisfy the following rules on compatible typed graphs. At an odd stage a rule is asserted only when all its inputs, outputs and coefficient divisions lie in the chosen finite budget. Configurations in these rules, and outputs of contractions, are nonempty.*

1. Graph dependence and whole-prime omission. *For a new vertex q adjoining a list U, its reading factors through the elementary quadratic extension unramified outside $S \cup U$. It is determined by its type and its incidences with U. The reading after omission of $p \in U$ uses the same original form, phase characters and depth, with varying support $U \setminus \{p\}$. That extension is unramified at the whole rational prime p. Changing inertia at the omitted prime has no effect on this reading.*

2. Contraction. *For the independent bit of a pair v, w, write $\Delta_{vw}A = A|_{z_{vw}=0} + A|_{z_{vw}=1}$, changing the reverse bit as required by reciprocity. Then*

$$\Delta_{vw}A(\Gamma) = A(\Gamma/(v, w)).$$

   The graph on the right has the summed type and every *summed external incidence from Definition 7.4. Its fresh prime representative is read using the omitted-prime source in the first rule.*

3. Primary isolation. *If a configuration contains a primary vertex neutral relative to all its other vertices, its symbol is zero.*

These are identities of the actual coefficient sources, rather than axioms imposed on an arbitrary binary function.

Proof. Proposition F.5 proves the rules from the integral shell recurrence, the unary subtraction and the weight-two trace and inertia identities. Divisibility by the integer two precedes reduction modulo the residue field. In the present full-two-torsion case every new vertex is split in $\mathbb{Q}(E[2]) = \mathbb{Q}$; the contraction coefficient is one. The omitted form is constructed from the same original source, so its unramifiedness removes the entire prime and does not choose a new minimum. The final fresh-copy comparison in that proof shows dependence on the whole typed graph, including overlapping prime realizations. The same argument proves the rules for the even source; no odd normalization is needed in that application. □

Lemma 7.7 (Finite realization)**.** *Keep finitely many old prime vertices and all their mutual data fixed. Given individually realizable types for finitely many new vertices, every prescription of the independent forward bits involving a new vertex can be realized by distinct fresh primes outside any prescribed finite avoidance set. This includes primary vertices and extra fixed splitting conditions. A product template can likewise be replaced by a fresh prime with the same type and all external incidences whenever that output type is one of the permitted realizable types.*

Proof. For rational quadratic characters the fixed type is a reduced residue condition at a modulus supported on S_{lab}, including eight. Choose the new primes successively. A desired residue bit at an old prime is a nonzero residue class there. The Chinese remainder theorem combines these classes with the fixed type, and Dirichlet's theorem supplies infinitely many primes in the resulting reduced residue class. At each step avoid all previously chosen primes and the finite avoidance set. Reciprocity gives the required reverse bits. Apply the same construction to the external residues

and support type of a product template. The graph dependence in Lemma 7.6 makes replacement by this fresh representative legitimate for the symbol. □

Definition 7.8 (Unit seeds and partitionable lists)**.** A *unit seed* is a finite compatible typed graph with symbol one. A *minimum nonempty seed* has the least positive vertex count among such seeds. This terminology does not exclude the possibility that the empty configuration is also a unit. A nonempty prime list is *partitionable* if its vertices admit a partition into nonempty clusters, with a contraction tree in each cluster, whose contracted graph is a unit seed. These trees and the resulting types and incidences constitute a contraction certificate. This condition makes no assertion about a Selmer group.

Lemma 7.9 (Minimum seeds and contraction certificates)**.** *At the least positive vertex count of a unit seed, its value is independent of all internal pair bits and depends only on its vertex types. A contraction certificate for a nonempty list makes its polynomial in the independent pair bits nonzero. In particular, a list with the same total type as a minimum nonempty seed is partitionable if it contains anchors for the first $r-1$ seed types and a nonempty remainder, where r is the seed's size.*

Proof. If $r \geq 2$, a nonzero derivative in an internal bit would, by contraction, give a unit on $r-1$ nonempty vertices. This contradicts minimality. If $r = 1$ there is no internal bit. Iterating the contraction identity along a forest gives its contracted symbol; an output equal to one therefore proves that the original polynomial is nonzero. For the last assertion keep the $r-1$ anchors and merge the whole remainder into the last cluster. Equality of total types forces the last seed type. The internal bits of the minimum seed do not matter. All contractions have nonempty output. □

Definition 7.10 (Addresses and terminal polynomials)**.** For an integer $b \geq 1$, an *address* is $x \in \mathbb{F}_2^b$. A vertex with activation $\lambda \in (\mathbb{F}_2^b)^* \setminus \{0\}$ is present at x exactly when $\lambda(x) = 1$. Stationary vertices are present at every address. At a fixed address, retain all vertex types and any fixed external incidences, and regard a symbol as a function of the independent pair bits between the active terminals. Its unique multilinear polynomial over $\mathbb{F}_2$ is the *terminal polynomial*. Its variables describe possible changes of those bits; the initial prime realization already has fixed bit values. Finite differences in these variables are formal derivatives.

Lemma 7.11 (One symbol with a stationary base)**.** *Fix a coefficient symbol satisfying Lemma 7.6, a nonempty stationary terminal list, its types and any stationary external data. Suppose its terminal polynomial is nonzero and its total data lie in the source's filter. For every $b \geq 1$ and every fixed assignment of the list's pair bits, finitely many new primary vertices with nonzero linear activations can be adjoined, with only their new incidences chosen, so that the symbol is one at every $x \neq 0$. The original vertices and all their mutual data are unchanged. No new vertex is present at zero. Fixed extra splitting and neutral external-incidence requirements may be included in the primary types. For a fixed even source the original prime representatives can be kept throughout. For a stabilized odd symbol, the fixed data in this statement are the abstract types and bits; its actual theta stage is chosen before its prime representatives, as specified in the proof.*

Lemma 7.12 (Two symbols on one address family)**.** *Fix one or two coefficient symbols satisfying Lemma 7.6, reading compatible projections of one common realizable alphabet. Fix $b \geq 1$ and terminals with nonzero linear activations, including at least one terminal for each nonzero linear form. Stationary external data may also be fixed. Assume that at every $x \neq 0$ the active configuration is nonempty, its total data satisfy every required filter, and each symbol's terminal polynomial is separately nonzero. Then, starting with any fixed assignment of these compatible types and bits,*

finitely many new primary vertices with nonzero linear activations can be adjoined so that all the symbols are one at every $x \neq 0$. Only incidences involving the new vertices are chosen. All original data and the zero address are preserved, and the same fixed splitting and neutral external-incidence conditions may be imposed as above. If an odd limit symbol is present, this is first a statement about the finite typed graphs. It does not fix rational prime representatives before the eventual theta stage.

Proof of the two address lemmas. The forest and network calculation in Proposition G.6 proves both statements. For the stationary case its grounded-network columns attach to a fixed terminal, so no terminal activation anchors are required. For variable terminals the stated anchors give the row-sum identity on every active restriction. Singular networks isolate prescribed derivatives at one nonzero address; varying networks supply the corresponding difference when two symbols share a variable. The two-symbol proof takes the product of the top forms in the ordinary polynomial ring before Boolean reduction. Its nonzero coefficient forces a single assignment that works at all nonzero addresses. It does not require the two symbols to have a common nonzero evaluation before the auxiliary vertices are added.

Each construction uses a finite number of vertices and coefficient readings once b and the terminal polynomials are fixed. For an odd symbol, include all of them in one finite weight and operator budget, then use Proposition G.5 to choose one actual theta stage carrying those tables and divisions. Only afterward use Lemma 7.7 to choose its primes. Any old prime representatives already chosen at that stage may be retained when they have its full prescribed types and bits, by the same realization lemma. In the even stationary application all original primes can be retained from the outset. The number of auxiliaries and the required theta precision may depend on b and the terminal data. The source's depth and the valuation constants attached to a unit do not change. No common finite stage for all values of b is asserted. □

The extra upper-bound argument. Nonvanishing alone does not bound the valuation of L_*, and (3.2) by itself gives a lower bound. The needed upper bound uses a separate even-symbol selection and transfer argument, which we now give.

Proposition 7.13 (The partitionable even upper estimate)**.** *Fix one even coefficient source, its signed unit filter and its depth. If a parameter d in that filter has a nonempty partitionable prime list and $E^{(d)}$ has full two-primary Selmer corank zero, then (3.3b) holds. Its constant depends only on the fixed source, not on the number or sizes of the primes of d or on later finite exclusions.*

Proof. For the even coefficient system described below, a unit symbol at a parameter d gives

$$L_*(d) \neq 0, \qquad v_2 L_*(d) \leq 2\omega(d) + C_0. \tag{3.3a}$$

This follows from its same-filter coefficient-square formula and its fixed normalization. That even normalization uses the lower bound from (3.2), not the as-yet undefined odd minimum. Partitionability is the nonempty contraction-certificate condition in Definition 7.8. By Lemma 7.9, its iterated finite difference is nonzero and hence so is the terminal polynomial. An empty unit, when present, satisfies its coefficient bound directly and does not require the stationary address lemma.

For a partitionable parameter d already known to have Selmer corank zero, keep all its prime factors stationary. Lemma 7.11 gives arbitrarily large binary cubes based at d, with no old prime removed, whose other addresses satisfy (3.3a). The fixed-form common scalar in (3.2) has one uniformly cleared denominator on every such cube. Choose a two-adic precision larger than the nonzero-address scalar valuation bound and its fixed clearing exponent. Its sign-evaluation coefficient is a Boolean function with the bounded bit degrees proved in Section 9. A cube dimension

larger than their sum therefore supplies a nonzero address congruent to the base at this precision. If the base scalar were zero or had larger valuation than the bound, so would that nonzero address, a contradiction. The corank-zero central comparison in (3.2) consequently gives

$$L_*(d) \neq 0, \qquad v_2 L_*(d) - 2\omega(d) - s(d) \leq C_1. \tag{3.3b}$$

The constants are fixed by the even system. The dimension needed may depend on d; the valuation bound does not. This is the cyclotomic missing-address argument on the fixed-form Kato scalar, not the later Heegner transfer. □

Proposition 7.14 (A bounded companion in the prepared filter)**.** *For (3.3), restrict h to positive squarefree parameters in the prepared odd-sign unit filter containing one, prime to its fixed support. Thus $h \equiv 1 \pmod 8$ and $w(E^{(h)}) = -1$. The simple-zero parameters used in (3.4)–(3.6) lie in this filter. There are constants A, C_E fixed before h varies such that, outside any prescribed additional finite avoidance set, one can choose a negative odd fundamental discriminant k' which is a square at every place dividing $2Nh$, avoids the extra-unit fields, and has $L_*(hk') \neq 0$. It satisfies (3.3), in particular $\omega(k') \leq A$. The valuation estimate uses the fixed even sources of the original E, although the analytic nonvanishing theorem will be applied to an auxiliary twist of E.*

Proof.

Here is why the actual rough companion is partitionable. Prepare an even symbol for every possible total type at the finite fixed support. There are finitely many. In our full rational two-torsion case the alphabet is all split. A least nonempty unit seed has no dependence on internal edge bits: an edge derivative would be a smaller nonempty unit configuration. Its individual labels determine it. The negative-discriminant parity at two makes the total label nonzero, so this seed is nonempty. A list with the same total label and enough copies of the seed labels contracts to it: keep the first $r - 1$ seed anchors and merge all remaining vertices into one nonempty cluster. Its label is forced to be the last seed label. Thus the list is partitionable, regardless of the number or labels of the remaining vertices.

Before h varies, fix a bounded multiset of dummy labels supplying the required anchors and reserves for every prepared total type. For each h, realize these labels by distinct primes avoiding the fixed support, the support of h, and all already fixed exclusions. Put in B the entire fixed support, the support of h, the chosen dummy primes, and every other already fixed forbidden prime. Include any already fixed reduction prime in B as well. Prescribe the total k' to be a negative odd fundamental discriminant, square at every place of $2Nh$ and outside the extra-unit cases. Absorb h and the dummy product into the curve and use the preceding nonvanishing theorem in its even filter. It adds at most five new primes. For the actual resulting total type, the already chosen dummy supply makes hk' partitionable. Its analytic nonvanishing gives Selmer corank zero by the forward rank-zero theorem. Apply (3.3b), obtaining

$$\omega(k') \leq A, \qquad v_2 L_*(hk') \leq 2\omega(h) + 2\omega(k') + s(hk') + C_E. \tag{3.3}$$

Here A is the fixed dummy count plus five. This is the argument of the partitionable-even and rough-companion lemmas, with their stationary primes and order of choices retained. Neither prime sizes nor a uniform analytic threshold are asserted. Excluding a later coefficient prime does not enlarge the old Selmer support. The finite source preparation and the distinction between fixed dummy labels and their later prime realizations are also detailed in Proposition G.2. □

7.4 The genus lower bound precedes normalization

Fix a negative genus partner h_* with $L_*(h_*) \neq 0$. Here h_* is chosen odd fundamental with $|h_*| > 4$ and in the same squareclasses at $2N$ as the prepared positive filter containing one. Throughout (3.4)–(3.6), h belongs to that filter and is prime to the fixed support, including h_*; the companion k' is supplied by the preceding construction with the primes of h_* also in its avoidance set. For a known simple-zero h, let P_h^{gen} be the unaveraged conductor-one genus sum over $\mathbb{Q}(\sqrt{hh_*})$. Let $P_{h,k'}^{\mathrm{rc}}$ be the unaveraged conductor-$|h|$ ring-class sum over $\mathbb{Q}(\sqrt{k'})$, weighted by χ_h. They are different points in different native lattices. Write $j_{\mathrm{gen}}(h)$ and $j_{\mathrm{rc}}(h,k')$ for their index valuations.

Proposition 7.15 (The genus bound before odd normalization)**.** *Use h_*, h, k' and the two native index valuations defined in the preceding paragraph, with h a known simple-zero parameter and k' as in Proposition 7.14. Then (3.4)–(3.6) hold. In particular their constant in (3.6) is independent of h. The arithmetic lower estimate (3.5) holds uniformly for the admissible companions with $\omega(k') \leq A$; it is only asserted here at these known simple-zero parameters.*

Proof. The explicit height formula [59, Theorem 1.1] applies twice. All level primes split, conductors are prime to N, extra-unit fields are excluded, and the parametrization is the same. Its conductor factor is $c\sqrt{|\operatorname{disc} K|}$ and its sum is unaveraged. The two Rankin products have the common factor $L(E^{(h)}, s)$. Each rational free line has index at most two in its native rank-one lattice: conjugation fixes the free class, and the norm makes twice the class rational. Thus

$$2j_{\mathrm{rc}}(h,k') - 2j_{\mathrm{gen}}(h) = v_2 L_*(hk') - v_2 L_*(h_*) + \epsilon, \qquad |\epsilon| \leq 2, \tag{3.4}$$

up to the one fixed initial period convention.

The arithmetic estimate of Section 9, for each fixed A, is

$$2j_{\mathrm{rc}}(h,k') \geq 4\omega(h) + s(h) + s(hk') - C_R(E,A), \qquad \omega(k') \leq A. \tag{3.5}$$

Combining (3.3)–(3.5) cancels $s(hk')$ and gives

$$2j_{\mathrm{gen}}(h) \geq 2\omega(h) + s(h) - C. \tag{3.6}$$

This calculation uses no odd normalization. For the forward reference in (3.5), combine Proposition 9.6 with the same-class central identity in Proposition 8.10. Those arguments use the known nonzero point in the present setup; they do not use an odd coefficient minimum or the final family. □

7.5 A fixed nonzero witness and actual odd forms

Choose one positive squarefree D_w in the odd filter containing one, after adjoining the primes of h_* and their compatible unit tests. Section 7.2 supplies $L'(E^{(D_w)}, 1) \neq 0$. The explicit height formula makes $P_{D_w}^{\mathrm{gen}}$ nontorsion. Its unwanted h_* component has rank zero. One fixed multiplier L kills the bounded-degree torsion and the projection denominator, so $Q = LP_{D_w}^{\mathrm{gen}}$ is a rational nontorsion point on $E^{(D_w)}$.

Let C be complex conjugation on its Tate module and choose a nonzero integral row b with $bC = b$. The Kummer translation image of Q is a full lattice. Indeed, if it vanished, its rational cocycle would descend to the Tate image. A nonidentity central scalar z gives

$$(z-1)c(g) = (g-1)c(z),$$

so that descended cocycle is a coboundary, contrary to Kummer injectivity. Irreducibility makes a nonzero translation span the Tate space, and intersection with any fixed finite kernel preserves full rank. Twisting preserves open image by restriction to an index-at-most-two subgroup.

Choose one such translation v with $2b(v) \neq 0$. Multiplying conjugation by this translation gives square translation $(1+C)v$, detected by $2b(v)$ with one fixed valuation c_w. The element is a single actual Galois element on the compositum. It is trivial on $\sqrt{D_w}$ and nontrivial on $\sqrt{h_*}$ because $D_w > 0 > h_*$. Its restrictions therefore already agree on intersections. Finite Chebotarev, with a place chosen above the resulting prime, gives arbitrarily advanced stages (M_θ, ℓ_0) such that

$$\rho(\mathrm{Fr}_{\ell_0}) \equiv C \pmod{2^{M_\theta}}, \quad (D_w/\ell_0) = 1, \quad (h_*/\ell_0) = -1, \quad v_2\lambda_{M_\theta}(\overline{Q}) = c_w < M_\theta. \tag{3.7}$$

Here $\lambda_{M_\theta} = b\kappa_{M_\theta}$ is the actual finite-field Kummer map on $E(\mathbb{F}_{\ell_0^2})$. It has Frobenius sign $+1$. All reductions use that same place. This is a fixed-witness statement, not a bound for arbitrary points or arbitrary reduction primes.

The weighted ternary theta construction gives actual holomorphic forms at these stages. At original level primes the root weight uses the summand selected by the cyclic level structure. At two, the integral spectral projectors of $(1+x)/2$ specify that summand; the operator itself is not in general idempotent. The nilpotent weight at h_* and the division-algebra residue-line weight at ℓ_0 have the prescribed conjugation and uniformizer signs. Their product is even. The root-character field is fixed before ℓ_0 and M_θ.

Appendix F gives these functions, their exact transformation rules, and the direct extraction using the form Q/m. Poisson summation for the resulting compactly supported local functions gives the scalar transformation law and holomorphy at every cusp. A fixed stabilizer multiple, for example one divisible by $|\,\mathrm{GL}_4(\mathbb{F}_3)|$, clears the finite automorphism groups. For the Hecke recurrence, a split or irreducible nonzero reduction has $1 + (-n/p)$ invariant lines, a nonzero nilpotent has one, and zero has $p+1$. The unique integral primitive nilpotent neighbor supplies the remaining square term. Tracking the scaling characters gives the recurrence with the same character convention as the coefficient comparison.

For an admissible D with $(D/\ell_0) = 1$, the oriented embedding count [62, Theorem 3.1], with changed set $\{\ell_0\}$, gives

$$a_{M_\theta}(D) = c_0\, 2^{\omega(N)}\zeta_{M_\theta}(D)\lambda_{M_\theta}(\overline{LP_D^{\mathrm{gen}}}) \pmod{2^{M_\theta}}, \tag{3.8}$$

where $c_0 \neq 0$ is fixed and ζ is a permitted root-of-unity phase. For admissible D with $(D/\ell_0) \neq 1$, including $\ell_0 \mid D$, the local Schwartz support instead gives $a_{M_\theta}(D) = 0$. Formula (3.8) is asserted only on the first support. There is no division by a class number. Formula (3.7) makes D_w visible with fixed upper valuation in (3.8). It remains on every stage's support, including the extra test $(D_w/\ell_0) = 1$.

After phase removal, the coefficients used for valuation are integers. Equation (3.6) gives a uniform lower bound for $v_2 a_{M_\theta}(D) - \omega(D)$ on visible bounded-weight sequences. Take the finite integer values constant on a set belonging to the chosen nonprincipal ultrafilter. This set of values is nonempty by D_w, is bounded below, and hence has an attained minimum δ. If the resulting lower inequality failed for a fixed weight bound on an ultrafilter-large set, choosing a violating index and then one of finitely many smaller integer values would contradict minimality. For any fixed finite operator budget, the coarse lower bound handles the large weights and stabilization handles the bounded weights. Consequently every required division is valid at one sufficiently advanced actual stage. No infinite-precision modular form or arbitrary-ring base-change assertion is needed.

7.6 One common family and the final order of choices

The even source is a fixed algebraic vector in the full good-Hecke eigenspace. Waldspurger's Corollary 2, including its same-dyadic-squareclass removal of the auxiliary condition, compares the actual squarefree Fourier functionals [63]. Local upper-unipotent projections retain a nonzero initial functional; smoothness supplies lower-unipotent invariance. The algebraic linear conditions choose a vector over a number field. Galois conjugation followed by the matching finite character twist shows that each normalized coefficient ratio is a rational number times a two-power root of unity. This supplies the valuation and nonvanishing comparison on the whole filter. Only flat field extension of the algebraic model is used. The comparison is proved at the original support S_0 before any extra coprimality sieve B. The sieve is applied last, and every later good recurrence prime avoids the full support $S_0 \cup B$.

For both coefficient systems the integral operators have the required omission and contraction rules. At a new prime V_p raises the level exponent only to one, with ramified quadratic character. The recurrence makes the relevant differences divisible by the integer two before residue reduction. Multiplying by ordinary theta, then taking a fresh trace, removes the integral unary corrections. Two fresh traces vanish. The matrix trace identity gives $t(gu^2) = t(g)$, hence elementary quadratic labels. At a new odd level prime the weight-two local representation has one unramified and one ramified line. The normalized-induction and Whittaker calculation below identifies the Fourier U_p eigenvalue with arithmetic Frobenius on the unramified line. The diagonal-matrix calculation gives the two inertia differences. These identities prove omission and primary isolation for the same original form and depth, without choosing another minimum.

If the intrinsic odd limit symbol is a unit at the empty configuration, choose an actual stage carrying that value with $M_\theta > \delta_o$. The index $D = 1$ lies in every moving support and then satisfies $v_2 a_{M_\theta}(1) = \delta_o < M_\theta$. The visibility implication (O.8) gives $\operatorname{ord}_{s=1} L(E, s) = 1$, which is exactly (RC). No final companion or determinant bound is needed in this case. For the rest of the final-family and missing-address argument, assume that the empty configuration is not a unit. The bounded-weight witness then supplies a least nonempty odd seed, as in Appendix G.

Fixing the final companion. Use a separate even symbol on negative discriminants square at $2N$, excluding the fixed odd support, the ramification support of $E[8]$, and the exceptional imaginary fields. Choose a unit k of this symbol. Then $L(E^{(k)}, 1) \neq 0$. Fix k and $K = \mathbb{Q}(\sqrt{k})$ before choosing the binary dimension. At least one prime ramified in K is outside all the old fixed label extensions and torsion ramification. Choose every later moving theta prime ℓ_0 away from k. The intrinsic odd symbol continues to omit the primes of k.

At any such stage, the normal closure of the old odd label extension is unramified outside its old support and ℓ_0. The fresh ramified prime of K lies outside this set, so its intersection with K is $\mathbb{Q}$. Thus each old odd label still has a lift split in K. The added stationary tests at the primes of k may have relations; include those relations in the common alphabet, rather than requiring independent arbitrary prescriptions. Stabilize its separate projection maps on the same ultrafilter. This preserves the old odd minimum. Fresh ramification also makes K independent of $\mathbb{Q}(E[8])$ and of the old fields $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{\ell})$.

The forward theorem at this known nonzero central value gives $s_2(E^{(k)}/\mathbb{Q}) = 0$. The sum-and-difference isogeny over $\mathbb{Q}$ gives

$$s_2(E/K) = s_2(E/\mathbb{Q}) + s_2(E^{(k)}/\mathbb{Q}) = 1.$$

Indeed its kernel and cokernel on the relevant restriction and corestriction maps are killed by a fixed power of two. Tensoring the compact Selmer modules, equivalently the Pontryagin duals of

the discrete Selmer groups, with $\mathbb{Q}_2$ gives the direct sum and hence this identity of coranks. No integral direct-sum assertion is used.

Lemma 7.16 (The nonempty seed for the present curve)**.** *For $E = E_\ell$ and the k, K just chosen, the positive even source for $E^{(k)}$, normalized at index one, has a nonempty unit seed consisting of K-split primes.*

Proof. This is the specialization to the roots $0, \ell, -3\ell$ of the coefficient calculation in Proposition G.4. If no such nonempty unit existed, its integral p^3 coefficient and a fresh trace would force $a_p(E^{(k)}) \equiv p + 1 \pmod 8$ at every fresh good K-split prime outside the support, whether or not it lies in the coefficient filter. Indeed, off the filter $a(p) = 0$ in (G.4a); the second fresh prime p' matches the old labels so that pp' lies in the filter, as used in (G.4d). The selected p below need not lie in the filter. The cited calculation uses positive Fourier indices and the positive twist character convention; it derives this congruence inside the assumed absence of nonempty units.

Choose such a prime with $p \equiv 1 \pmod 4$, $(\ell/p) = -1$ and $(k/p) = 1$, avoiding the finite support. In fact this p is outside the square filter: quadratic reciprocity and $p \equiv 1 \pmod 4$ give $\prod_{r|\ell}(p/r) = (\ell/p) = -1$, so p is a nonsquare at some odd $r \mid \ell$, and $r \mid N$. This is why the off-filter scope of the preceding calculation is needed. The fresh ramification of K permits these simultaneous conditions. Because k is a square modulo p, the reductions of $E^{(k)}$ and E_ℓ are isomorphic over $\mathbb{F}_p$. None of the three nonzero rational points of order two can be doubled: their necessary square differences include, respectively, $-\ell, \ell, -4\ell$. Each is a nonsquare under the chosen conditions. The two-primary reduction group therefore has order exactly four, contradicting the asserted congruence. This proof uses the particular curve E_ℓ; no nonempty-seed theorem for an arbitrary elliptic curve is being asserted. □

We record the finite combinatorial realization used here. Each label space is a finite elementary two-group; the common alphabet projects onto each separately. Product surjectivity is unnecessary. Take sufficient repeated copies of every label, with total label zero. Keeping the seed anchors, contract the unused copies into the last anchor. Minimal positive seed size makes the internal bits irrelevant, so each of the two terminal polynomials is nonzero. Grounded Laplacian minors express the constant and pair-contraction operators, including singular matrices. A cyclic singular network isolates its target radical and a path network isolates its selected pair coefficient. In the product of the two top forms, taken in the ordinary polynomial ring before Boolean reduction, a variable of exponent two uses one singular program in each factor; exponent one uses one varying program. The total degree kills the other contractions and leaves coefficient one. Thus one choice works for both forms. CRT and Dirichlet realize the prescribed forward residue bits by distinct primes; quadratic reciprocity supplies the reverse bits.

Proposition 7.17 (One family for the two coefficient tests)**.** *Keep the fixed E, h_*, normalized odd symbol and final k, K of the preceding paragraphs. Assume, as in the remaining branch above, that the intrinsic odd empty configuration is not a unit. In this remaining branch, for every sufficiently large b there is one finite set Q of distinct good odd primes split in K and outside the fixed support, and nonzero linear forms $\lambda_q : \mathbb{F}_2^b \to \mathbb{F}_2$, such that the family*

$$h_x = \prod_{q \in Q} (q^*)^{\lambda_q(x)} \quad (x \in \mathbb{F}_2^b), \qquad q^* = (-1)^{(q-1)/2} q,$$

has $h_0 = 1$, all fixed local types unchanged, and, for $x \neq 0$,

$$\operatorname{ord}_{s=1} L(E^{(h_x)}, s) = 1, \quad L_*(h_x k) \neq 0, \quad j_{\mathrm{gen}}(h_x) \leq \omega(h_x) + C_o, \quad v_2 L_*(h_x k) \leq 2\omega(h_x) + C_e. \tag{3.9}$$

The constants are independent of b. By (3.4), now with fixed k, $2j_{\mathrm{rc}}(h_x, k) \leq 4\omega(h_x) + C_J$.

Proof. The packet argument just given supplies separate nonzero terminal polynomials in one common alphabet. Apply Lemma 7.12. The full proof in Proposition G.6 and Proposition G.7 preserves the projected total labels, all stationary tests and the original two normalizations.

The order is fixed data and D_w, then the genus lower bound and odd minimum, then the final $K = \mathbb{Q}(\sqrt{k})$ and common alphabet, then determinant precision M, then b and its finite operator budget, then a sufficiently advanced actual theta stage M_θ and its prime realization. In particular M_θ is not M. The finite graph construction, its local conditions and the coefficient tests all occur at that same final stage. Appendix G spells out the absolute five-prime completion, normalization constants, and the final use of the one corrected coordinate. □

8 Actual classes, local conditions and central comparison

Let $K = \mathbb{Q}(\sqrt{k})$ be imaginary, split at $2N$, and different from $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$. Active primes are good odd primes split in K. Derivative primes are good inert primes outside all current supports. The actual finite coefficients are

$$A_i = (\mathbb{Z}/2^i\mathbb{Z})[t]/t^i, \quad M_i = (T_2E \otimes A_i)(\chi_h\Psi_i).$$

For a fixed binary group G, use $A_i[G]$ and the universal quadratic character. The character Ψ_i is finite, trivial at $2N$, ramified at a moving conductor prime r_i, and has prescribed odd scalar exponents at the finite active list. In a binary family let Q be the union of its prime supports. Apply Proposition 8.5 to an orientation above every $q \in Q$, prescribing $\xi_i(\mathrm{Fr}_q) = 1$ at each finite stage. This prescription includes the primes unused by an individual character, including every graph prime at augmentation. The same Ψ_i is used for all characters of the family. The finite set Q is fixed before choosing the sequence of moving primes r_i; the trace bound c_E does not depend on its size. The prime r_i varies with precision. There is no assertion that one fixed odd prime supports an infinitely ramified $\mathbb{Z}_2$ character.

Definition 8.1 (Coefficient and local-condition conventions)**.** In Sections 8 and 9, retain the fixed non-CM curve E, parametrization π, Tate lattice, and old local types of Section 7. Put $\Lambda = \mathbb{Z}_2[[t]]$ and $\mathcal{O} = \widehat{\Lambda_{(2)}}$. A binary family uses one finite elementary abelian two-group G, one real sign and common local squareclasses at $2N$. Its universal varying quadratic character is therefore trivial at the real place and at $2N$. The active primes are good odd primes split in the indicated imaginary quadratic field K; the moving character is the finite character specified above, trivial at $2N$. The primitive and relaxed complexes use the actual derived Kummer, unramified, full and orthogonal local conditions specified below, with their maps and homotopies. At derivative primes the four local conditions are those of Lemma 9.5. Every determinant coordinate uses the retained integral determinant basis.

Lemma 8.2 (Conjugate coefficients and local duality)**.** *For the finite coefficient module M_i and coefficient ring $R_i = A_i$ or $A_i[G]$ above, let τ be an actual complex conjugation. Pullback of the Galois action gives an R_i-linear identification $\tau^* M_i \simeq M_i^*(1)$ through $(v, w) \mapsto e_2(v, \rho_T(\tau)w)$; it fixes t and the group-ring basis. With the conjugate places, actual orthogonal local conditions and their isotropy homotopies, finite Selmer duality gives*

$$C_U(M_i) \simeq R\,\mathrm{Hom}_{R_i}(C_{U^\perp}(M_i^*(1)), R_i)[-3].$$

The local Kummer and derivative comparisons used here are proved below and in Lemma 9.5.

Proof.

The dual coefficient system. Fix an actual complex conjugation $\tau \in G_{\mathbf{Q}}$, so that $\tau^2 = 1$. The moving character is constructed on the ring-class group:

$$\xi_i : \mathrm{Gal}(H_{r_i}/K) \longrightarrow \mathbf{Z}/2^{n_i}\mathbf{Z}, \qquad \Psi_i = (1+t)^{\xi_i},$$

where n_i is large enough that $(1+t)^{2^{n_i}} = 1$ in A_i. Conjugation acts on this ring-class group by inversion. Consequently

$$\Psi_i(\tau g \tau^{-1}) = \Psi_i(g)^{-1} \qquad (g \in G_K).$$

Let $R_i = A_i$ in the scalar construction and $R_i = A_i[G]$ in the binary construction. Write $\alpha_i = \chi_h \Psi_i$, respectively $\alpha_i = \chi_0 \varepsilon \Psi_i$, where $\varepsilon : G_K \to G$ is the restriction of the universal rational quadratic character. Then $\varepsilon^\tau = \varepsilon = \varepsilon^{-1}$ and $\alpha_i^\tau = \alpha_i^{-1}$. Here $\tau^* M_i$ denotes pullback of the Galois action; it does not act on R_i.

The Weil pairing gives the following R_i-linear perfect pairing:

$$M_i \otimes_{R_i} \tau^* M_i \longrightarrow R_i(1), \qquad (v, w) \longmapsto e_2(v, \rho_T(\tau) w).$$

Indeed $\rho_T(\tau)^2 = 1$, the two scalar characters multiply to one, and e_2 transforms by the cyclotomic character. Thus $\tau^* M_i \simeq M_i^*(1)$. We do not identify M_i with its ordinary dual when Ψ_i is non-quadratic. In this convention t and every element of the group-ring basis are fixed. Inverting $1+t$ is already accounted for by pullback of the character.

At a split place v, the dual local condition is transported from the conjugate place τv. At a fixed place it is transported by the actual local representative of conjugation. Kummer conditions at $2N$ are preserved by this transport. A full condition is paired with the zero condition; the full local complex is replaced by zero only at the localizations where its displayed contraction exists. At a derivative prime the finite and singular planes are checked with this transported pairing. In the chosen coordinates this is the pairing $\beta(x, y) = \epsilon\langle x, Jy\rangle$. The J is therefore part of the duality convention, not an extra identification of an arbitrary twist with its dual.

Use the coefficient-valued pairing in the finite Selmer diagrams, with the conjugate local places and their isotropy homotopies. The resulting duality is

$$C_U(M_i) \simeq R\,\mathrm{Hom}_{R_i}(C_{U^\perp}(M_i^*(1)), R_i)[-3].$$

These are the maps and homotopies retained in both limits. □

8.1 Finite Heegner descent and the derivative identities at two

Fix the non-CM elliptic curve $E/\mathbf{Q}$, its conductor N, its Tate module $T = T_2 E$, and a modular parametrization $\pi : X_0(N) \longrightarrow E$. Normalize $\pi(\infty) = 0$. Let K be imaginary quadratic, split at every prime dividing $2N$, and different from $\mathbf{Q}(i)$ and $\mathbf{Q}(\sqrt{-3})$. The conductor f below is prime to $2ND_K$; it may contain all the active primes in a finite binary family and the moving prime. Write H_f for the ring-class field of the order $\mathcal{O}_f = \mathbf{Z} + f\mathcal{O}_K$.

Let I be any finite set of good inert primes outside $6fND_K$, and put $n_I = \prod_{\ell \in I} \ell$ and $M_I = H_{fn_I}$. At the end, a finite character θ of $\Delta_f = \mathrm{Gal}(H_f/K)$ will take values in $A^\times$, where $A = (\mathbf{Z}/2^m\mathbf{Z})[t]/t^b$, or in $A[G]^\times$ for a finite binary group G. It has the form $\chi_h \Psi$, or the corresponding universal quadratic character times Ψ. At $2N$, Ψ is trivial and the universal quadratic character has the fixed local type already specified in the manuscript. These are hypotheses on the actual finite character. Its construction is separate from the descent proved here.

Proposition 8.3 (Finite Heegner descent with one multiplier)**.** *With E, π, K, f, I, M_I, A and the finite character θ as just specified, there are integers $e \geq 0$ and $L > 0$ depending only on the fixed curve, parametrization and allowed old local types, and on any other prescribed fixed multiplier requirements, with the following properties. Form the derivative over the complete extension H_{fn_I}/H_f and descend it before applying the character. This gives finite classes $P_{I,m}$ with the old Kummer conditions, the stated good unramified conditions and full active and moving conditions. For $I = \varnothing$ the class is the Kummer class of the L-multiplied unaveraged weighted point.*

The unaveraged norms are (4.1). If $a \geq m + e$ and every new derivative prime has $\rho_T(\mathrm{Fr}_\ell) \equiv \rho_T(\tau) \pmod{2^{a+1}}$, the classes projected to precision m satisfy (4.2). These assertions hold over $A[G]$ before character evaluation. They are realized by finite continuous cocycles, local-condition representatives and boundary witnesses, with the weighted coefficient map applied to all this data. There is no multiplier depending on K, f, I, m, the truncation length, the number of active primes or $|G|$.

Proof.

1. One uniform torsion annihilator. By the non-CM open-image theorem [69], choose $n \geq 2$ such that $1 + 2^n M_2(\mathbf{Z}_2) \subset \rho_T(G_{\mathbf{Q}})$, and choose $g \in G_{\mathbf{Q}}$ with $\rho_T(g) = uI$, $u = 1 + 2^n$. For an actual complex conjugation τ set

$$z = g^2 \tau g^2 \tau^{-1}, \qquad e = v_2(u^4 - 1) = n + 2. \tag{H1}$$

The element g^2 fixes every quadratic field. Its image in the abelian group $\mathrm{Gal}(H_d/K)$ is inverted by τ, for every d and every allowed K. Thus z fixes every such H_d, while $\rho_T(z) = u^4 I$. If $P \in E(H_d)[2^\infty]$, then $(u^4 - 1)P = 0$, so

$$2^e E(H_d)[2^\infty] = 0. \tag{H2}$$

This bounds the whole invariant torsion group, with no restriction on the number of conductor primes. It also holds for a compositum of the ring-class fields in a finite construction, since z fixes each of them.

For the matrix-span statement used elsewhere, square preimages of $I + 2^n E_{11}$, $I + 2^n E_{12}$ and $I + 2^n E_{21}$. The squares belong to every quadratic subgroup, and their images, together with I, are

$$I, \quad I + (2^{n+1} + 2^{2n})E_{11}, \quad I + 2^{n+1}E_{12}, \quad I + 2^{n+1}E_{21}.$$

Their $\mathbf{Z}_2$-span contains $2^{n+1}M_2(\mathbf{Z}_2)$: subtract I, use that $1 + 2^{n-1}$ is odd, and obtain E_{22} from $I - E_{11}$. This bound also depends only on E.

2. Point relations with the full old conductor. Choose the CM points $x_d \in X_0(N)(H_d)$ by compatible proper ideal lattices and the fixed orientations at N, and put $y_d = \pi(x_d)$. The split Heegner hypothesis identifies the curve $X_K(N)$ in Cai–Chen–Liu, Theorem 2.15 [60], with $X_0(N)$: $N_1 = 1$, and the condition $(*)$ reduces to $(d, N) = 1$. Take their old conductor to be any d currently in use and the new prime outside $6dND_K$. Their factor u_m is one because $\mathcal{O}_K^\times = \{\pm 1\}$. Applying the elliptic quotient to that theorem gives

$$\mathrm{Tr}_{H_{d\ell}/H_d}\, y_{d\ell} = a_\ell y_d \quad (\ell \text{ inert}), \qquad \mathrm{Tr}_{H_{dq}/H_d}\, y_{dq} = (a_q - \sigma_q - \sigma_q^{-1}) y_d \quad (q \text{ split}). \tag{4.1}$$

The inverse Frobenius pair at a split prime follows because their product is the Artin symbol of the rational principal ideal (q). These are unaveraged sums. On degree-zero divisor classes,

$\varphi T_q = [a_q]\varphi$. Since $q \nmid N$, $T_q[\infty] = (q+1)[\infty]$, so the displayed norm relations for $y_d = \pi(x_d)$ have no cusp correction. For other changes of CM convention, let $\mathcal{C}_N$ be the subgroup of $J_0(N)(\overline{\mathbb{Q}})$ generated by $[c]-[\infty]$ for all cusps c. This is a finite group by Manin–Drinfeld [52, p. 59, Theorem]. Fix $C_\pi \geq 1$ annihilating $\varphi(\mathcal{C}_N)$. This single integer depends only on N, π. For an Atkin–Lehner involution W acting on f by w, the exact formula is $\pi(Wx) = [w]\pi(x) + \pi(W\infty)$; the final constant lies in $\varphi(\mathcal{C}_N)$ and vanishes after multiplication by C_π.

At an inert new prime ℓ, a compatible place gives

$$\overline{y}_{d\ell} = F_\ell \overline{y}_d. \tag{H4}$$

Here F_ℓ is rational arithmetic Frobenius on the reduced elliptic curve. This is the usual CM congruence, and it does not require the old conductor to be an inert squarefree product. Indeed $(d, \ell) = 1$ makes the local CM order the maximal unramified quadratic order. The CM elliptic curve has supersingular reduction at this inert prime. Each of its $\ell + 1$ cyclic raising isogenies specializes to the quotient by the unique subgroup of order ℓ, namely the relative Frobenius kernel. The prime-to-ℓ cyclic N-subgroup specializes by the same Frobenius map. The resulting equality of reduced modular points passes through π, which extends over the good integral models at ℓ. This also proves (H4) after every old ideal-class translate, with its place transported. Howard, Section 1.7, formula (13) [79], records this congruence in his conductor subfamily; his odd-prime descent and residual-surjectivity theorem are not used here.

3. The relative group and invariant derivative class. The idelic description $\mathrm{Pic}(\mathcal{O}_d) = K^\times \backslash \mathbf{A}_{K,f}^\times / \widehat{\mathcal{O}}_d^\times$ identifies the kernel of $\mathrm{Pic}(\mathcal{O}_{fn_I}) \to \mathrm{Pic}(\mathcal{O}_f)$ with

$$\prod_{\ell \in I} \frac{(\mathcal{O}_K/\ell\mathcal{O}_K)^\times}{(\mathbf{Z}/\ell\mathbf{Z})^\times} \simeq \prod_{\ell \in I} C_{\ell+1}. \tag{H5}$$

There is no additional quotient by units: the only global units are ± 1 and already belong to the denominator at every factor. This proves (H5) without choosing a splitting of the full class group. Let σ_ℓ be a generator of its ℓ-factor, with the local Artin orientation fixed. Set

$$D_\ell = \sum_{j=1}^{\ell} j\sigma_\ell^j, \qquad D_I = \prod_{\ell \in I} D_\ell, \qquad (\sigma_\ell - 1)D_\ell = (\ell+1) - \mathrm{Tr}_\ell\,. \tag{H6}$$

If 2^a divides a_ℓ and $\ell + 1$ for all $\ell \in I$, (4.1) and (H6) show that

$$c_{I,a} = \delta_a(C_\pi D_I y_{fn_I}) \in H^1(M_I, E[2^a])^{\mathrm{Gal}(M_I/H_f)}. \tag{H7}$$

For example its difference under σ_ℓ is the Kummer class of $C_\pi D_{I\setminus\{\ell\}}((\ell+1)y_{fn_I} - a_\ell y_{fn_I/\ell})$, an actual 2^a-multiple of an M_I-point. The factors in (H5) commute.

4. Descend once, before twisting. Put $B = H_f$, $M = M_I$, $\Gamma = \mathrm{Gal}(M/B)$ and $V_a = E[2^a]$. The five-term sequence for finite discrete coefficients [44, formula (4.1.4.1), p. 96] gives

$$0 \longrightarrow H^1(\Gamma, V_a^{G_M}) \longrightarrow H^1(B, V_a) \xrightarrow{\mathrm{res}} H^1(M, V_a)^\Gamma \longrightarrow H^2(\Gamma, V_a^{G_M}). \tag{H8}$$

By (H2), multiplication by 2^e is zero on the coefficient module $V_a^{G_M}$ and hence on every cochain and cohomology group of Γ with those coefficients. For an invariant class c, choose b with $\mathrm{res}(b) = 2^e c$ and define

$$d_a(c) = 2^e b. \tag{H9}$$

Any two choices of b differ by the first group in (H8), so (H9) is independent of the choice. It is additive, natural for coefficient reduction, and equivariant for transport of the whole field diagram. Its restriction is exactly $2^{2e}c$. This is one application to M_I/H_f, not one application for each factor of (H5). The coefficient module in (H8) is untwisted $E[2^a]$; the scalar and binary coefficient push comes only later. Thus no invariant module containing a varying nonquadratic scalar character has been silently substituted in this argument. For $I = \varnothing$, the relative group is trivial and this same definition gives $d_a(c) = 2^{2e}c$. In particular it uses the same normalizing multiplier for the primitive class and its derivatives.

5. Integral local descent at the old places. We record the local fact needed to put $d_a(c_{I,a})$ in the old Kummer conditions. If F'/F is a finite unramified extension of nonarchimedean local fields and B/F is an elliptic curve, then

$$H^1(\mathrm{Gal}(F'/F), B_0(F')) = 0, \qquad c_B H^1(\mathrm{Gal}(F'/F), B(F')) = 0, \tag{H10}$$

where $c_B = \#\Phi_B(\overline{k}_F)$ is the geometric component group order. To prove the first assertion, reduce a cocycle to the connected smooth special fiber. Lang's theorem [80, Theorem 1 and its corollary] makes its H^1 over the finite residue field zero; inflation–restriction then makes its H^1 for the finite residue extension zero. Lift the resulting coboundary through the smooth reduction map. The remaining cocycle takes values in the formal group. For a uniformizer of F, fixed by the unramified Galois group, each successive formal quotient is an additive copy of the residue field of F' [48, IV.3.2(a), p. 124]. A normal basis makes this an induced module, so its positive-degree cohomology vanishes. Correct the coboundary successively in these quotients. The corrections converge in $B_1(F')$, and the group is finite, so their limit solves all the cocycle equations. The component exact sequence proves the second assertion of (H10). No division by $[F' : F]$ was used.

If a class $v \in H^1(F, B[2^a])$ becomes Kummer over F', its image in $H^1(F, B)[2^a]$ belongs to the kernel of restriction. That kernel is the image of $H^1(\mathrm{Gal}(F'/F), B(F'))$. Thus $c_B v$ is Kummer over F, by the Kummer exact sequence and (H10).

All ring-class extensions in use are unramified at $2N$, since the conductors are prime to $2N$. The base fields there are unramified extensions of $K_v = \mathbf{Q}_p$. Geometric component groups are unchanged under unramified extension. Let C_{bad} be a common multiple of their orders for the finitely many allowed local quadratic types at $2N$. The same integer works for every conductor, every place above $2N$ and every K. Choose a fixed positive integer m_0, allowing the other fixed multiplier requirements in the argument to be included, and define

$$L = m_0 C_\pi C_{\mathrm{bad}} 2^{2e}, \qquad b_{I,a} = m_0 C_{\mathrm{bad}} d_a(c_{I,a}) \in H^1(H_f, E[2^a]). \tag{H11}$$

Thus any common fixed multiple of this descent multiplier and the detection multiplier is permitted; every class is scaled by the same m_0. Then $b_{I,a}$ is Kummer at $2N$ and restricts to $\delta_a(LD_I y_{fn_I})$. At a good place away from $2fn_I$, the same assertion has component order one. It gives the unramified local condition, since at good odd places the finite Kummer image equals the unramified image. At active and moving primes the required condition is full. The integer L is now fixed, independently of K, f, I, m, b and G.

6. Exact finite precision for one derivative prime. Let the retained torsion precision be m, choose $a \geq m + e$, and require, for every retained derivative prime,

$$\rho_T(\mathrm{Fr}_\ell) \equiv J \pmod{2^{a+1}}, \qquad J = \rho_T(\tau), \quad J^2 = I. \tag{H12}$$

Then 2^{a+1} divides a_ℓ and $\ell+1$, since $\operatorname{tr} J = 0$ and $\det J = -1$. Choose $\ell \notin I$, and perform the construction for $I \cup \{\ell\}$. Apply the other derivatives first, writing

$$X = D_I y_{fn_I}, \qquad Y = D_I y_{fn_I \ell}.$$

The K-prime (ℓ) splits completely in H_{fn_I}: it is prime to the conductor and is a rational principal ideal in the ring-class Artin map. Thus its residue field is $\mathbf{F}_{\ell^2}$, and

$$F^2 \overline{X} = \overline{X}, \qquad \operatorname{Tr} Y = a_\ell X, \qquad \overline{Y} = F\overline{X}, \qquad F^2 - a_\ell F + \ell = 0. \tag{H13}$$

The same splitting makes the old class $b_{I,a}$ at this place exactly the Kummer class of LX, by its restriction identity in (H11); there is no local descent ambiguity for this old class. For $2^a Q = LX$ its finite coordinate is

$$f_{\text{old}} = (F^2 - 1)\overline{Q} = (a_\ell F - (\ell+1))\overline{Q}. \tag{H14}$$

Although the two integer coefficients are highly divisible, Q is a division point, so (H14) does not assert $f_{\text{old}} = 0$.

Choose a continuous cocycle representing $b_{I\cup\{\ell\},a}$ and choose Z with $2^a Z = LD_\ell Y$. Its restriction to $G_{H_{fn_I\ell}}$ is cohomologous to $g \mapsto (g-1)Z$. Their difference is the coboundary of a single element of $E[2^a]$; subtract the same global coboundary to make the restriction exactly this Kummer cocycle. Denote the adjusted representative by b. Restrict it to $G_{H_{fn_I}}$ and choose a local inertia lift of σ_ℓ. Put

$$U = (\sigma_\ell - 1)Z - b(\sigma_\ell).$$

For $g \in G_{H_{fn_I\ell}}$, compare $b(g\sigma_\ell)$ with $b(\sigma_\ell(\sigma_\ell^{-1} g \sigma_\ell))$. Using the exact restricted Kummer representative gives $gU = U$. Consequently U is an actual point of $E(H_{fn_I\ell})$, and

$$2^a U = L((\ell+1)Y - a_\ell X).$$

The point

$$U_0 = L\frac{\ell+1}{2^a}Y - L\frac{a_\ell}{2^a}X \tag{H15}$$

also belongs to this field, with the same 2^a-multiple. Its coefficients are integers. Hence $U - U_0$ is upper-field 2-primary torsion and is killed by 2^e, by (H2). This comparison with an integer quotient point is necessary; comparison with an arbitrary division point would give no bound.

Inertia acts trivially on the reduction of every algebraic point at a good prime. Reduction is injective on 2-primary torsion. Thus the singular coordinate $b(\sigma_\ell)$ is

$$s_{\text{new}} = (a_\ell - (\ell+1)F)\overline{Q} + \epsilon = (F - a_\ell) f_{\text{old}} + \epsilon, \qquad 2^e \epsilon = 0. \tag{H16}$$

The second equality uses the exact endomorphism identity

$$(F - a_\ell)(a_\ell F - (\ell+1)) = a_\ell - (\ell+1)F. \tag{H17}$$

Reduction $T/2^a T \to T/2^m T$ is multiplication by 2^{a-m} on geometric torsion points. It kills ϵ. Conditions (H12) then turn (H16) into $s_{\text{new}} = J f_{\text{old}}$.

For the new finite coordinate, all inertia translates of Y have the same reduction, so

$$\overline{LD_\ell Y} = L\frac{\ell(\ell+1)}{2}F\overline{X}. \tag{H18}$$

This is 2^a times an F^2-fixed reduced point, because $2^{a+1} \mid \ell+1$. The local extension obtained by adjoining the conductor ℓ is totally ramified, of degree $\ell+1$. Its residue field is still $\mathbf{F}_{\ell^2}$. Lift

the indicated 2^a-division point in the residue field; the discrepancy lies in the formal group, where multiplication by 2^a has an integral formal inverse because ℓ is odd [48, IV.2.3 and IV.3.2, pp. 122–124]. This produces a local 2^a-division point of $LD_\ell Y$ in the upper local field itself. The restricted Kummer class is therefore zero. Choose a Frobenius lift in that upper local Galois group. It acts trivially on $E[2^a]$ by (H12), so its cocycle value is zero, independently of a coboundary adjustment. Hence

$$f_{\mathrm{new}} = 0, \qquad s_{\mathrm{new}} = J f_{\mathrm{old}} \quad \text{at precision } 2^m. \tag{H19}$$

The generator σ_ℓ fixes the tame orientation. A different generator changes the corresponding singular coordinate by the matching unit. Frobenius lifts and places in (H19) are part of the retained local diagram. The extra bit in (H12) pays exactly for the division by two in (H18). Repeat this calculation with each prime of $I \cup \{\ell\}$ singled out: all other lower conductors split at that prime by the same principal-ideal argument. Thus the previously imposed singular conditions are retained. The maximum of the required finite precisions suffices for the whole finite set; neither (H11) nor its exponent is repeated.

7. The actual coefficient push and its local conditions. Reduce $b_{I,a}$ to $E[2^m]$, extend coefficients freely to A, and only now use the character θ of Δ_f. Under Shapiro, followed by the equivariant coefficient map,

$$E[2^m] \otimes A[\Delta_f] \longrightarrow (E[2^m] \otimes A)(\theta), \qquad v \otimes [\delta] \longmapsto \theta(\delta) v, \tag{H20}$$

obtain the class $P_{I,m}$ over K. Here the induced module is written with the diagonal Tate action and the left regular action on its class-group basis. Formula (H20) is equivariant since left translation multiplies the value by $\theta(g)$. It remains equivariant with values in $A[G]$, before evaluation at any sign character.

For an explicit degree-one cochain formula, choose representatives r_δ of G_K/G_{H_f} using the normal quotient Δ_f, and write $g r_\delta = r_{g\delta} h(g,\delta)$ with $h(g,\delta) \in G_{H_f}$. The transferred cocycle is

$$(\mathcal{T}_\theta b)(g) = \sum_{\delta \in \Delta_f} \theta(r_{g\delta}) \rho_T(r_{g\delta}) b(h(g,\delta)). \tag{H21}$$

The identity $h(gk,\delta) = h(g,k\delta)h(k,\delta)$ proves the cocycle identity with target action $\theta\rho_T$. In degree zero the formula is the weighted sum $\sum_\delta \theta(r_\delta)\rho_T(r_\delta)Q$; applying its boundary gives (H21) applied to the boundary of Q. More generally, put $R(g) = \theta(g)\rho_T(g)$ and, for $g_1, \dots, g_n$, set $\delta_i = g_{i+1} \cdots g_n \delta$. The degree-$n$ map is

$$(\mathcal{T}_\theta c)(g_1, \dots, g_n) = \sum_{\delta \in \Delta_f} R(r_{g_1 \cdots g_n \delta}) c\big(h(g_1,\delta_1), \dots, h(g_n,\delta_n)\big).$$

With the ordinary inhomogeneous differential, the relation for h above gives $d\mathcal{T}_\theta = \mathcal{T}_\theta d$. These are the integral Shapiro and transfer maps of [44, Sections 8.1.1–8.1.6, pp. 189–194], written with left cosets. Changing sections gives the integral homotopy in Section 8.1.4; compatible sections give strictly transitive transfer. No averaging factor occurs.

Localizing (H21) separates the cosets into decomposition-group orbits: this is the local Mackey formula, a sum of the local corestrictions from the transported places. At $2N$, the deformation character Ψ is trivial locally; the fixed quadratic character need not be trivial. Under the fixed quadratic-twist identification, the local map is the ordinary integral norm on Kummer classes. One can see this without a projector: the sum $\sum_\delta \chi(\delta)\delta Q$ is a point of the quadratic twist over the lower field, and Kummer of this sum equals the cohomological transfer. The extra A coefficients take

finite linear combinations of such maps. Thus the old Kummer conditions proved in paragraph 5 survive (H20), also in the universal family with its common local quadratic type. Local Kummer corestriction is natural even for a ramified field extension; in the present old-place descent the extensions used in (H10) are unramified.

At a derivative prime, H_f/K splits completely and θ is trivial on the local Galois group. Each Mackey term uses a transported place and a transported Frobenius. Transporting the Tate basis transports F as well, so (H19) is equivariant, not an assertion that an arbitrary change-of-place matrix commutes with a fixed untransported J. The same weights occur on the upper and lower class. Summing gives, in the common target coordinates,

$$f(P_{I\cup\{\ell\},m}) = 0, \qquad s(P_{I\cup\{\ell\},m}) = Jf(P_{I,m}). \tag{4.2}$$

Active and moving conditions are full; all other good conditions are unramified and preserved by the same maps.

The split norm formula in (4.1) also survives this construction. For a new conductor prime the class-field squares obtained by retaining the derivative factors are Cartesian, by (H5) and the same local-unit calculation at that prime. Restriction and corestriction commute in these squares. If b lifts $2^e c$ in (H9), its corestriction lifts 2^e times the corresponding corestricted class; uniqueness after the second multiplication by 2^e proves the required compatibility of d_a. The weighted sums are transitive finite sums. Thus the two horizontal Artin terms in (4.1) become $\theta(\sigma_q)$ and $\theta(\sigma_q)^{-1}$, giving $a_q - \theta(\sigma_q) - \theta(\sigma_q)^{-1}$ on the class. For $I = \varnothing$, the class is precisely the Kummer image of the L-multiplied weighted point, so the same statement gives the central split-norm relation with that exact L.

The local diagrams underlying the coefficient push. At a retained derivative place put $q = \ell^2$ and choose tame and unramified generators τ_v, ϕ_v with $\phi_v \tau_v \phi_v^{-1} = \tau_v^q$. The local coefficient character is trivial. The two-generator tame resolution is

$$[M \xrightarrow{u\mapsto((\Phi_v-1)u,0)} M \oplus M \xrightarrow{(x,y)\mapsto(\Phi_v-q)y} M], \qquad \Phi_v = \rho(\phi_v).$$

At the retained exponent $m \le a$, (H12) makes this the zero-differential model $D^0 \oplus (F_v \oplus S_v)[-1] \oplus D^2[-2]$. Moreover $q \equiv 1 \pmod{2^{a+2}}$. Represent the singular generator by a uniformizer adjusted by a unit so its Kummer value on the chosen ϕ_v is zero. Residue-unit Kummer characters allow this adjustment. The latter congruence makes its self-cup zero: -1 is a 2^m-th power, so the adjusted uniformizer Kummer class has square zero. The finite self-cup is zero by unramified cohomological dimension one. Retain the transported mixed pairing and the D^0, D^2 pairing.

In the derivative diagram the old ring-class fields split locally, and the new conductor extension is totally ramified. For the latter choose the lower Frobenius in the upper subgroup and take $\tau_{\text{upper}} = \tau_{\text{lower}}^e$; here $f = 1$. For split terms $e = f = 1$. Transport these choices with the Tate basis. With these compatible generators the local matrices, in the order D^0, F_v, S_v, D^2, are

$$\text{res} = \text{diag}(1, f, e, ef), \qquad \text{cor} = \text{diag}(ef, e, f, 1).$$

The possible transfer shear is divisible by $(q-1)/2$ and vanishes at retained precision. These matrices preserve each of the four conditions retaining D^0, and satisfy the local cup-product adjunctions. Split Mackey terms have $e = f = 1$; coefficient weights and reduction commute with these matrices. An arbitrary new Frobenius lift is not used without transporting the resulting finite/singular decomposition.

Compare the small free resolution with the continuous bar resolution and choose augmentation-preserving comparison maps and their composite homotopies. Continuous Hom into M retains these identities. Since the small model has zero differential, the composite on it is the identity; normalize the other homotopy to obtain a contraction. Here are its compatibility maps. For this retained contraction $1 - i_v p_v = dh_v + h_v d$ and a full local cochain arrow F, take $F_L = p_w F i_v$ and restrict it to U_v. The inclusion square has homotopy $h_w F i_v|_{U_v}$. For composable arrows the transferred composition homotopy is $p_z G h_w F i_v$, together with the transferred original bar homotopy when present. Every such degree-minus-one map on U_v lands in D^0, which is retained in every condition; thus these homotopies restrict too. For three composable maps, write $K(G, F) = pGhFi$ and $L(H, G, F) = pHhGhFi$, with the intermediate subscripts understood. Direct expansion gives

$$K(H, GF) - K(HG, F) + H_L K(G, F) - K(H, G)F_L = Ld - dL.$$

The degree-minus-two map L vanishes on every derivative condition. Transfer the original bar comparison homotopies and cubes as well; unrelated classwise boundaries are not substituted for them. For the isotropy diagrams, $U_v \otimes U_v'$ is free in degrees $0, 1, 2$, whereas the trace target is $A[-2]$. Its negative Hom groups vanish, so the restricted zero pairing has compatible null-homotopies and the requisite 2-homotopies. This constructs the local data required in [44, 8.6.2.1–8.6.2.4] before using 8.6.4. At old Kummer places use the natural restriction/norm diagrams of the multiplication-by-2^m exact sequences, including local torsion, and their derived scalar extensions; at good unramified, full and zero places use the corresponding standard subcomplexes. The witnesses in (H22) are chosen in these same diagrams.

8. Actual finite Selmer-cone witnesses. Represent $P_{I,m}$ by a continuous global cocycle p. At each constrained place, membership in the indicated local condition gives a local cycle u_v and a degree-zero cochain q_v with

$$\mathrm{loc}_v\, p - \iota_v u_v = dq_v. \tag{H22}$$

These entries give a cycle in $\mathrm{Cone}(C^\bullet(G_{K,S}, M) \oplus \bigoplus U_v \to \bigoplus C^\bullet(K_v, M))[-1]$. For (H19), choose in the same way a degree-zero cochain whose boundary is the difference of the two representatives being compared. At the derivative primes use the already specified D^0-retaining finite/singular local complexes, not full/zero conditions. Their coordinate equality is the finite local cohomology equality just proved. All coefficient groups and coset sets at this stage are finite; the cocycles and the finite list of witnesses factor through finite quotients.

Apply the coefficient push to these witnesses at the same time as to the cocycle, including the natural Kummer comparison homotopies. If (i, p, h) is a retained contraction to a finite target model, a relation $z - z' = dq$ becomes $pz - pz' = d(pq)$. For a comparison F to another model with projection p', the correction between $p'Fz$ and $p'Fipz$ is the boundary of $p'Fhz$ (with the consistent contraction sign). Thus the finite arithmetic provides actual cycles, boundaries and comparison entries before compression. Adding a derivative prime extends this same finite diagram. Existence and size bounds for those target contractions, and the subsequent two limits, are separate inputs; this proof makes no inference from abstract isomorphism of cohomology groups to those bounds.

The resulting finite arithmetic uses the single multiplier (H11) and the precision rule $a \geq m + e$, (H12). No factor depending on $[H_f : K]$, $|G|$, $|I|$ or the number of active primes occurs. □

8.2 The derived Kummer condition

Lemma 8.4 (The actual derived Kummer condition)**.** *For the actual local twist $B = E^{(h)}$ over $F = K_v$, the condition $U_F = B(F)^\wedge_2[-1] \to R\Gamma(F, T_2 B)$ includes the finite two-primary torsion.*

Its reduction modulo 2^a has $B(F)[2^a]$ in degree zero and $B(F)/2^aB(F)$ in degree one. Its Kummer image is its exact local annihilator, and the comparison has integral isotropy homotopies. At $v \mid 2N$, triviality of the moving character and the common old quadratic type identify the scalar and binary versions by derived coefficient extension. These are the local maps used in the Selmer-cone duality above.

Proof. For the actual local twist $B = E^{(h)}$ over $F = K_v$, put

$$U_F = B(F)^{\wedge}_2[-1] \longrightarrow R\Gamma(F, T_2B).$$

The completion includes finite two-primary torsion. Derived reduction modulo 2^a has $B(F)[2^a]$ in degree zero and $B(F)/2^aB(F)$ in degree one. At $v \mid 2N$, Ψ_i is trivial, so derived coefficient extension defines the scalar A_i condition for this actual twist. In the universal binary family every h_x has the same local squareclasses at $2N$ as $h_0 = 1$. Hence every sign character evaluates the universal character ε trivially on G_{K_v}, and therefore $\varepsilon|_{G_{K_v}} = 1$ in G, before reduction modulo two. The $A_i[G]$ Kummer condition and its null-homotopy are consequently derived scalar extensions of the fixed local data. Omitting the scalar twist, the universal local-squareclass condition, or degree-zero torsion changes this argument.

The local finite dualities [42, I.2.3, I.3.2–I.3.5] make the Kummer map adjoint to $H^1(F, B[2^a]) \to H^1(F, B)[2^a]$. The latter has Kummer image as kernel, so the Kummer image is its exact annihilator. To obtain the integral chain statement, write $M = B(F)^{\wedge}_2$. The restricted cup map $M[-1] \otimes^L M[-1] \to \mathbb{Z}_2[-2]$ has homotopy class determined by $\mathrm{Hom}(M \otimes M, \mathbb{Z}_2)$. Finite isotropy makes every value divisible by every power of two, hence zero. A projective resolution therefore supplies a null-homotopy. The possible ambiguity group in degree minus one vanishes. Reduction modulo two, including the torsion in degree zero, gives the perfect quotient-to-dual comparison; derived Nakayama lifts it. The global Selmer duality theorem [44, Sections 5.1, 5.2.4, 5.4.3 and 6.3.4] is applied to these actual local maps and homotopies.

For clarity the Selmer complex is the cone

$$C_U = \mathrm{Cone}\left(R\Gamma(G_{K,S}, M_i) \oplus \bigoplus_{v \in S} U_v \longrightarrow \bigoplus_{v \in S} R\Gamma(K_v, M_i) \right)[-1]. \tag{4.3}$$

The compatible trace in (4.3) is coefficient-valued. A top-coefficient functional used to test a finite Frobenius ring's perfection is not used as a coefficient-reduction-compatible arithmetic trace. □

8.3 Actual support comparisons

At a tame split prime the local free model has

$$d^0(v) = (Av, Cv), \quad d^1(v, w) = -Cv + Dw, \quad A = \Phi - 1,\ C = g - 1,\ D = \Phi - N_q(g),$$

$$N_q(g) = \frac{q+1}{2} + \frac{q-1}{2}g, \qquad DC = CA. \tag{4.4}$$

At augmentation $g = 1$, this is the direct sum of the unramified complex $[\Phi - 1]$ in degrees zero and one and the singular quotient $[\Phi - q]$ in degrees one and two.

The support-change theorem [44, Proposition 7.8.8(ii)] retains arbitrary old local conditions and puts the unramified condition at new places. Its finite-module proof applies at two. Use it only when the actual coefficient representation is unramified at the removed place. There are five uses here: universal augmentation at Q; an unused scalar prime $q \nmid h$; characteristic-two residual removal after $\chi_h = 1$; the moving prime after $t = 0$; and a new good inert derivative prime outside

h, r_i and the old support. In the last case the coefficient module is unramified and the chosen finite local plane is the unramified condition. It is not applied to an active ramified scalar prime or to r_i before central specialization. Replacing the unramified condition by the full condition then gives the singular quotient by a cone of Selmer cones. In particular the old complex is an actual subcomplex of an upper triangular model, not just a complex with the same Euler characteristic.

Proposition 8.5 (One moving character for the complete active list)**.** *Fix an allowed K and any finite list of oriented active primes. For every $n, B \geq 1$ and finite avoidance set there are infinitely many split rational primes $r \equiv 1 \pmod{2^B}$ and characters $\xi : \mathrm{Gal}(H_r/K) \to \mathbb{Z}/2^n\mathbb{Z}$ that are surjective on inertia at r, trivial on the local decomposition groups at $2N$, and take value one at every prescribed active orientation. They can be chosen with*

$$v_2(2 \pm a_r) \leq c_E, \qquad v_2 \det(1 \pm \mathrm{Fr}_r) \leq c_E,$$

where c_E depends only on the fixed Tate image, not on K, n, B or the active list. One such character serves the entire binary family before evaluation.

Proof.

The simultaneous moving character. We give the construction for the non-CM curve used here. For every $n, B \geq 1$ and every finite avoidance set, there are infinitely many split rational primes $r \equiv 1 \pmod{2^B}$ and characters

$$\xi : \mathrm{Gal}(H_r/K) \longrightarrow \mathbb{Z}/2^n\mathbb{Z}$$

which are surjective on inertia at r, trivial at $2N$, and equal to one at each prescribed oriented active prime. The trace bounds below have one constant c_E, independent of n, B, K and the active list. The degrees of the finite fields and the sizes of the primes chosen in the construction need not be uniformly bounded.

Choose prime ideals $\mathfrak{p}_1, \ldots, \mathfrak{p}_s$ in distinct conjugate pairs, including the active orientations, one prime over each prime dividing $2N$, and enough additional split prime ideals to generate $\mathrm{Cl}(K)$. The latter exist outside any finite set by Chebotarev in the Hilbert class field. Set $w_j = 1$ at the active primes and $w_j = 0$ at the other chosen primes. Put

$$\mathcal{R} = \ker(\mathbb{Z}^s \longrightarrow \mathrm{Cl}(K)), \qquad (\beta_a) = \prod_j \mathfrak{p}_j^{m_j^{(a)}}, \qquad \alpha_a = \beta_a/\overline{\beta}_a,$$

where the $m^{(a)}$ form a $\mathbb{Z}$-basis of $\mathcal{R}$. The generators β_a may be fractional. Disjointness of the conjugate pairs gives $v_{\mathfrak{p}_j}(\alpha_a) = m_j^{(a)}$.

Let $F_\infty = K(\mu_{2^\infty})$ and adjoin the 2^n-th roots of all the α_a. Prescribe the multiplier

$$\zeta_{2^n}^{-\sum_j m_j^{(a)} w_j} \quad \text{on } \alpha_a^{1/2^n}. \tag{4.4a}$$

These prescriptions respect every Kummer relation. Indeed, if $\prod_a \alpha_a^{b_a}$ is a 2^n-th power in F_∞, its root lies in a finite cyclotomic layer. At every nonzero-weight prime that layer is unramified over K, since the prime is odd. Consequently 2^n divides $\sum_a b_a m_j^{(a)}$ at each such prime, and hence divides $\sum_{a,j} b_a m_j^{(a)} w_j$. The dyadic primes have weight zero. This tests all multiplicative relations, including those involving units; it does not assume independence of the radicands. Kummer theory therefore gives the specified automorphism of the finite abelian extension

$$L_\infty = F_\infty(\alpha_a^{1/2^n} : a)/F_\infty.$$

Here is why these are precisely the extension conditions for ξ. For a split r avoiding the chosen primes, the ring-class sequence is

$$1 \longrightarrow \mathbf{F}_r^\times \longrightarrow \mathrm{Pic}(\mathbb{Z} + r\mathcal{O}_K) \longrightarrow \mathrm{Cl}(K) \longrightarrow 1.$$

The kernel is the ratio of the two residue factors in $(\mathcal{O}_K/r\mathcal{O}_K)^\times$ modulo diagonal rational residues. There is no additional unit quotient because $\mathcal{O}_K^\times = \{\pm 1\}$. Write q_j for the ideal-class lift of $\mathfrak{p}_j$, and orient the connecting map δ so that its principal relations are

$$\sum_j m_j^{(a)} q_j + \delta(\mathrm{res}\,\alpha_a) = 0.$$

The kernel and the q_j generate the ring-class group. Any relation between them projects to $\mathcal{R}$; subtracting its basis relations leaves a relation within the kernel. Thus these are a complete presentation, not merely necessary relations. For $r \equiv 1 \pmod{2^n}$ the power-residue character $\kappa_r : \mathbf{F}_r^\times \to \mathbb{Z}/2^n\mathbb{Z}$ is surjective. The assignments $\xi(q_j) = w_j$ and $\xi\delta = \kappa_r$ extend exactly when

$$\kappa_r(\mathrm{res}\,\alpha_a) = -\sum_j m_j^{(a)} w_j. \tag{4.4b}$$

Below we choose r so that its radical Frobenius is (4.4a), which gives (4.4b). Reversing the connecting-map convention reverses the power-residue convention as well. No class number is divided out.

We next preserve all these conditions while controlling the Tate trace. Let $J = \rho_T(G_{\mathbb{Q}(\mu_{2^\infty})})$. The determinant is the cyclotomic character, so J is the determinant-one part of the fixed Tate image. Non-CM open image [69] gives a fixed $d \geq 2$ with

$$\Gamma_d := \ker\bigl(\mathrm{SL}_2(\mathbb{Z}_2) \to \mathrm{SL}_2(\mathbb{Z}/2^d\mathbb{Z})\bigr) \subset J.$$

The subgroup $J_K = \rho_T(G_{F_\infty})$ has index at most two in J, and therefore contains its squares. Write $U(a)$ and $V(b)$ for the upper and lower unipotent matrices. Then J_K contains $U(2^{d+1}\mathbb{Z}_2)$ and $V(2^{d+1}\mathbb{Z}_2)$. The identity

$$U(a)V(b)U\bigl(-a/(1+ab)\bigr)V\bigl(-b(1+ab)\bigr) = \mathrm{diag}(1+ab, (1+ab)^{-1})$$

and Gaussian decomposition show that a subgroup containing $U(2^k\mathbb{Z}_2), V(2^k\mathbb{Z}_2)$ contains Γ_{2k}. Hence $\Gamma_m \subset J_K$ for $m = 2d+2$. Commuting $\mathrm{diag}(1+2^m, (1+2^m)^{-1})$ with the upper and lower unipotents of level m gives all upper and lower unipotents of level $2m+1$, since $v_2((1+2^m)^2 - 1) = m+1$. Applying the same decomposition gives

$$\Gamma_{4m+2} \subset \overline{[J_K, J_K]}.$$

Every matrix commutator lifts to a commutator in G_{F_∞}. Such lifts fix L_∞, whose Galois group over F_∞ is abelian. The image of the compact group G_{L_∞} is closed. Thus, for the fixed integer $\nu = 4m + 2 = 8d + 10$,

$$\Gamma_\nu \subset \rho_T(G_{L_\infty}). \tag{4.4c}$$

This is a subgroup in the image of the actual radical-fixing kernel. It requires no disjointness of the radical and Tate fields, and is independent of the number of radicands and of K.

For the trace estimate one can take $c_E = 2\nu + 1 = 16d + 21$. We verify the elementary bound used here. Put $\delta = 2^\nu$ and $D(v) = \mathrm{diag}(v, v^{-1})$. For any $A = \left(\begin{smallmatrix} a & b \\ c & f \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z}_2)$, one of

$$A, \quad AU(\delta), \quad AD(1+\delta), \quad AD(1+2\delta) \tag{4.4d}$$

has both trace distances to 2 and -2 of valuation at most $2\nu+1$. If A itself does not, its trace is within $2^{2\nu+2}\mathbb{Z}_2$ of one of these two values, while its distance to the other has valuation two. Its diagonal entries have the same parity. If both are even, c is odd by $af-bc=1$, and $AU(\delta)$ changes the trace by $c\delta$, of valuation ν. If both are odd, the two diagonal multipliers change the trace by

$$\delta\frac{(a-f)+a\delta}{1+\delta}, \qquad 2\delta\frac{(a-f)+2a\delta}{1+2\delta}.$$

If $v_2(a-f)<\nu$, the first change has valuation at most $2\nu-1$. Otherwise write $a-f=\delta z$. For even z the first change has valuation 2ν; for odd z the second has valuation $2\nu+1$. All the chosen changes have valuation at least three. They therefore leave the distance to the other forbidden value of valuation two, and give the claimed bound for the nearer one.

Lift the prescribed Kummer automorphism to $\sigma_0 \in G_{F_\infty}$. By (4.4c), every right multiplier in (4.4d) lifts to an element of G_{L_∞}. Choose one giving the trace bound and multiply σ_0 by that lift. The resulting actual Galois element σ fixes F_∞, preserves every prescribed radical multiplier, and satisfies $v_2(2\pm\operatorname{tr}\rho_T(\sigma))\le c_E$.

Choose $B'\ge\max(B,n)$ with $B'>c_E$, and take a finite normal extension of $\mathbb{Q}$ containing $K,\mu_{2^{B'}},E[2^{B'}]$ and all the chosen radical roots. Restrict σ to this extension. Its restrictions on all intersections agree because it is already an actual Galois element. Finite Chebotarev supplies infinitely many rational primes in its Frobenius class. Choose a place realizing this representative and avoid all the finitely many excluded primes. Then r splits in K and $r\equiv 1 \pmod{2^{B'}}$. The Frobenius formula on $\alpha_a^{1/2^n}$ gives (4.4b), and hence the character ξ. Each local unit factor at r maps onto the residue-ratio kernel, so local reciprocity proves that ξ is surjective on inertia there. At $2N$ it is unramified with Frobenius value zero, and is therefore trivial on the whole local decomposition group. Conjugation negates its values because a prime ideal times its conjugate is rational principal.

Finally $a_r\equiv\operatorname{tr}\rho_T(\sigma) \pmod{2^{B'}}$ and $\det(\mathrm{Fr}_r)=r$. Since $B'>c_E$, we obtain

$$v_2(2\pm a_r)\le c_E, \qquad v_2\det(1\pm\mathrm{Fr}_r)=v_2(1\pm a_r+r)\le c_E.$$

The same bound controls $a_r-2\chi_h(r)$ for every constituent unramified at r. Repeat with increasing n,B and avoidance sets to obtain distinct r_i. Taking $n_i\ge 2i$ makes $(1+t)^{2^{n_i}}=1$ in A_i: for $1\le j<i$, $v_2\binom{2^{n_i}}{j}\ge n_i-v_2(j)\ge i$. Thus $\Psi_i=(1+t)^{\xi_i}$ is the required finite character. Its inertia generator can be oriented to have value $1+t$, and its active oriented Frobenius values have odd exponent. One character works for the entire fixed active list before any binary evaluation. □

8.4 The same functional at every localization

Proposition 8.6 (The paired functional and its integral representatives)**.** *Use the retained inner-limit diagrams of Lemma 9.1, with the local conditions of Definition 8.1, and let z be the actual retained class. Transport z by conjugation, lift it through the orthogonal comparison at the generic point, and pair by the retained duality. This defines one generic closed degree-two functional λ_z. It has an integral representative after multiplication by 2^a over $\mathcal{O}[G]$, for a fixed old-support exponent a independent of the active list and G. For a scalar constituent with its primitive local conditions, its paired functional has an integral representative at every height-one prime $\mathfrak{p}\neq(2)$ of Λ, including (t) and $(t+2)$. These representatives have the same generic value for that primitive constituent. The universal relaxed functional is asserted integral after the fixed power of two over $\mathcal{O}[G]$; scalar regularity of its corrected coordinate is obtained instead from the actual primitive comparison of Lemma 8.11. All these assertions concern the inner limit and the indicated localizations, not inverses of t in a finite Artin ring.*

Proof. After the inner limit, put $\mathcal{O} = \widehat{\mathbb{Z}_2[[t]]_{(2)}}$. In (4.4), the residual active Frobenius is $(1+t)^a I$ with a odd. Thus A, D are invertible over $\mathcal{O}[G]$. The formulas

$$h^1(v,w) = A^{-1}v, \qquad h^2(u) = (0, D^{-1}u)$$

contract the complex; the identities follow from $DC = CA$.

Which functional is paired with the class. Let z denote the retained class and transport it by τ and the preceding Weil identification to the conjugate coefficient system. At a localization where the full local complexes at active and moving primes are acyclic, the comparison from the orthogonal condition to the transported condition is a quasi-isomorphism. Apply its inverse in the derived category to the transported class, and call the resulting class $z^{\perp}$. Define

$$\lambda_z(w) = \mathrm{PD}(w, z^{\perp}), \qquad w \in H^2(C_U).$$

Thus λ_z is a closed degree-two functional, with a cochain representative obtained using the displayed local contractions and the retained global duality map. The inverse is unique in the derived category, so representatives built at distinct localizations define the same generic functional. Integrality still requires the stated integral contractions and fixed old-support correction; it does not follow from abstract duality alone. The coordinate definition below uses this same λ_z.

Here the representatives must be constructed separately. Over $\mathcal{O}[G]$, the active Frobenius operators A, D are units, and moving-prime inertia minus one is t times a unit; the displayed homotopies are integral. For the height-one assertion take the scalar primitive constituent. Its active primes are precisely the primes ramified in that constituent, so inertia minus one is -2, a unit at $\mathfrak{p} \neq (2)$. At the moving prime inertia minus one is invertible unless $\mathfrak{p} = (t)$; at (t) the nonzero central Frobenius determinants give integral adjugate contractions. In particular $(t+2)$ is treated by the inertia contraction. Unused good primes in the primitive constituent retain the unramified condition and its orthogonal comparison; no inverse of a full local Frobenius complex at such a prime is required. This proves the scalar assertion only for the primitive functional. In the universal relaxed family an unused prime can specialize to trivial inertia, so the -2 contraction is not available there. Its correction is the determinant identity of Lemma 8.11. The common generic value alone would not prove any of these integrality statements.

In the inertia contraction the maps are

$$h^1(v,w) = C^{-1}w, \qquad h^2(u) = (-C^{-1}u, 0).$$

The fixed old-support lattice and the exponent. Here is the integral content of the old-support comparison. For each $p \mid 2N$ and each of the fixed allowed local quadratic types δ, choose once a bounded free $\mathbf{Z}_2$-model of the diagram

$$U_{p,\delta} \longrightarrow L_{p,\delta}, \qquad U_{p,\delta} = (E^{(\delta)}(\mathbf{Q}_p)_2^{\wedge})[-1], \qquad L_{p,\delta} = R\Gamma(\mathbf{Q}_p, T_2E^{(\delta)}),$$

together with its Kummer map, cup pairing, isotropy homotopy, and quotient-to-dual comparison. Finite torsion is retained in $U_{p,\delta}$; it is represented by a free resolution, not removed. The local duality and Kummer exact sequence identify that comparison integrally:

$$\mathrm{Cone}(U_{p,\delta} \to L_{p,\delta}) \simeq R\,\mathrm{Hom}_{\mathbf{Z}_2}(U_{p,\delta}, \mathbf{Z}_2)[-2].$$

This is the chain-level comparison of Lemma 8.4, with its specified isotropy homotopy. In particular it is not a rational isomorphism of free parts of cohomology. A quasi-isomorphism between these

bounded free models has an integral homotopy inverse: its bounded acyclic free cone is contractible, by splitting from its last nonzero term and proceeding downwards. Consequently neither torsion lengths nor inverse determinants of torsion differentials are denominators in this comparison.

Every allowed K splits at $2N$. Its two completions above p are therefore $\mathbf{Q}_p$, with precisely two copies of this same diagram, interchanged by conjugation. The moving character and the universal varying quadratic character are trivial on their decomposition groups. All old-place diagrams over $A_i[G]$ are thus derived scalar extensions of this finite fixed list. Tensoring the chosen free models and their homotopy identities preserves all these identities. The remaining good unramified conditions use the integral unramified comparison, and the derivative conditions use their integral orthogonal planes. Neither requires a new lattice comparison when a prime is added.

For completeness, there is also no hidden group-order denominator in duality. The coefficient pairing is the $A_i[G]$-valued Weil pairing. If one checks its perfection over A_i, the auxiliary pairing on the group algebra is

$$(a, b) \longmapsto [1](ab), \qquad ([1](gh))_{g,h\in G} = (\mathbf{1}_{h=g^{-1}})_{g,h\in G}.$$

Its matrix is a permutation matrix, with integral inverse. It is not the normalized trace $|G|^{-1}\,\mathrm{Tr}$. The global cup-product and trace maps are integral as well. The hypotheses of Selmer duality here are a perfect coefficient pairing and the actual orthogonal local diagrams with their isotropy homotopies [44, Sections 6.2.5–6.2.8 and Theorem 6.3.4]; the old Kummer diagrams just described supply those hypotheses. Since K is imaginary there is no real-place Tate correction in this argument.

Over $S = \mathcal{O}[G]$ the only remaining differences between the transported conditions and their orthogonal conditions occur at the active and moving places. Their full local complexes have the integral contractions displayed above. Their direct sum, of any finite size, is still an integral contraction. The resulting comparison $j : C_{U^\perp} \to \tau^* C_U$ is an integral homotopy equivalence over S. Write j^{-1} for such a homotopy inverse and c for conjugate transport. On the retained bounded free models,

$$z^\perp = j^{-1}c(z), \qquad \lambda_z = \mathrm{PD}(-, j^{-1}c(z))$$

are integral cochains, and the second is closed. Every map in this formula is integral; no inverse on a torsion cohomology quotient is taken. The finite-model contractions and their retained limits preserve these identities. In detail, represent the comparison on the bounded free models and contract its acyclic cone over S. The cone contraction yields a chain map j^{-1} and homotopies $jj^{-1} \simeq 1$ and $j^{-1}j \simeq 1$ over S. Thus $d(j^{-1}c(z)) = j^{-1}c(dz) = 0$ as a cochain identity, and its image differs from $c(z)$ by an integral boundary. Applying the integral duality chain map gives a closed element of $\mathrm{Hom}_S(C_U^2, S)$, with the claimed generic value. This proves the existence of an actual integral closed lift, rather than only a lift after rationalization.

With the conventions of this paper the actual class z already contains the fixed multiplier L. Thus the preceding construction permits $a = 0$. One may keep any larger fixed nonnegative a in the subsequent inequalities. This does not change z, divide out L, or change the determinant basis. It proves the existence and independence of the exponent from G, the active graph, and the allowed varying K; a numerical computation of a new denominator is unnecessary because no such denominator occurs in these integral diagrams. This conclusion concerns S. The separate primitive height-one argument above, and the corrected relaxed coordinate comparison, retain exactly their stated scopes. □

Definition 8.7 (The class-functional coordinate)**.** For a constituent whose generic cohomology has dimensions $(1, 1)$ in degrees one and two and vanishes otherwise, use the determinant line

$$\mathcal{D} = \det{}^{-1} C \simeq H^1(C) \otimes H^2(C)^*.$$

The class-functional coordinate u is the coordinate of $[z] \otimes [\lambda] \in \mathcal{D}$ in the retained integral determinant basis. Here z is the actual class and λ is the closed functional just constructed. For a generically acyclic constituent set $u = 0$. The primitive and relaxed subscripts specify the corresponding local conditions on C.

8.5 Central identity, torsion and the primitive coordinate

Lemma 8.8 (Central specialization preserves the same class)**.** *The finite classes of Proposition 8.3 satisfy the retained boundary (4.5), with the same fixed L. For the functional of Proposition 8.6, the actual central Kummer comparison i satisfies*

$$z_0 = i(LmP), \qquad (\lambda_z)_0(ix) = \mathrm{PD}_K(x, c_K(LmP)).$$

Here P is the unaveraged ring-class point and c_K is its conjugate-dual transport. On a rank-one Mordell–Weil free lattice, both free-line coefficients consequently have valuation $j_{\mathrm{rc}}(h,k) + v_2(Lm)$. The statement identifies the same point and functional; their integral torsion factors are still computed in the integral Kummer complex and local quotient triangles.

Proof. The split norm relation gives an actual finite-stage boundary

$$z_i(0) - Lm_i\, j_i(b_h) = dv_i, \qquad m_i = a_{r_i} - 2\chi_h(r_i). \tag{4.5}$$

The equality in global cohomology lifts to the Selmer cone because the local degree-zero maps are onto and the degree-one maps inject. Retain v_i. The uniform trace bound bounds $v_2(m_i)$ and the local determinants by c_E; their adjugate identities multiplied by a fixed power of two survive the limit.

Specialization of the same paired functional. Put $A = \Lambda_{(t)}$, with residue field $\mathbf{Q}_2$. The representative of λ_z over A is the one obtained from the actual orthogonal comparison and the retained cup pairing. At an active prime its contraction uses $C = -2 \in A^{\times}$. At the moving prime the central Frobenius determinants are nonzero; their adjugate inverses therefore belong to A. Both contractions commute with $A \to \mathbf{Q}_2$: their defining identities are matrix identities with invertible denominators in A. In particular the central full local complexes are rationally acyclic. The Kummer local objects at these changed odd places are also rationally zero: local points have finite two-primary completion, and the retained central Frobenius matrices have nonzero determinants. Thus the full/Kummer quotients are rationally acyclic as well. At the fixed old places the Kummer maps and their duality homotopies are retained unchanged.

Let j denote the resulting orthogonal comparison over A and let $c(z)$ be the conjugate-dual transport of the actual class. The definition is $z^{\perp} = j^{-1}c(z)$ and $\lambda_z = \mathrm{PD}(-, z^{\perp})$. Derived base change preserves the specified chain maps, cup products, and homotopies. It also preserves j^{-1}, since the base change of the two inverse identities is again the two inverse identities. Consequently

$$(\lambda_z)_0 = \mathrm{PD}_0(-, j_0^{-1}c(z_0)).$$

To make the central identification explicit, write $R_0 = C_U(M)_0$, $R_0^c = C_{cU}(M^*(1))_0$, and $S_0 = C_{U^{\perp}}(M^*(1))_0$. Let K_0 be the ordinary rational Kummer complex and K_0^c its conjugate-dual transport. The actual comparison factors as

$$i : K_0 \longrightarrow R_0, \qquad s : S_0 \longrightarrow K_0^c, \qquad q : K_0^c \longrightarrow R_0^c, \qquad j_0 = qs.$$

All three maps are quasi-isomorphisms, by the preceding local vanishing and the unchanged old conditions. The retained duality homotopies give

$$\mathrm{PD}_R(ix, y) = \mathrm{PD}_K(x, sy), \qquad c(iP) = qc_K(P).$$

The boundary (4.5) gives $z_0 = i(LmP)$. Therefore

$$(\lambda_z)_0(ix) = \mathrm{PD}_K(x, s(qs)^{-1}qc_K(LmP)) = \mathrm{PD}_K(x, c_K(LmP)).$$

Thus this is the Kummer-dual functional of the same transported point. On the integral rank-one Mordell–Weil free lattice, the central conjugation transport is an involutive automorphism (including the quadratic sign), hence acts by $+1$ or -1. It preserves the point index; it need not fix the point. The two free-line coefficients both have valuation $j_{\mathrm{rc}}(h, k) + v_2(Lm)$. An orientation sign is a fixed unit; there is no independently chosen scalar multiplying the functional. The integral torsion correction is still computed from the ordinary integral Kummer complex and the integral local quotient triangles, not from their rationally acyclic reductions. This is the compatibility needed to apply the arithmetic length calculation in (4.6). □

Lemma 8.9 (Determinant specialization with no dimension jump)**.** *Let A be a discrete valuation ring with uniformizer t, fraction field F, and residue field k. Let C be a bounded finite free complex over A. If*

$$\dim_F H^i(C \otimes_A F) = \dim_k H^i(C \otimes_A k) \quad \text{for every } i,$$

then its cohomology is free and commutes with reduction modulo t. In particular, when the only nonzero dimensions are $(1, 1)$ in degrees one and two, the coordinate of an actual cycle and an actual closed functional in $\det^{-1} C$ is regular at $t = 0$ and specializes to the coordinate of their central tensor in the specialized retained determinant basis.

Proof. Smith reduction splits the bounded free complex into free summands with zero differential and two-term summands $[A \xrightarrow{t^r u} A]$, where u is a unit and $r \geq 0$. A summand with $r > 0$ contributes one extra residue-field cohomology dimension in each of two adjacent degrees and contributes none over F. The dimension equalities therefore exclude every such summand. The remaining two-term summands are contractible over A. On the zero-differential summands both cohomology and the class-functional tensor commute with base change. The determinant identifications include the same contractible summands before and after reduction, with their unit differentials; there is no independent rescaling of the retained basis. □

Proposition 8.10 (Arithmetic calibration at known simple zeros)**.** *Let $K = \mathbb{Q}(\sqrt{k})$ and h satisfy the ring-class hypotheses above, with every prime dividing h split in K. Suppose $L(E^{(h)}, s)$ has a simple zero at 1 and $L(E^{(hk)}, 1) \neq 0$. Then the specialization at $t = 0$ of the generic coordinate in Definition 8.7 is regular and satisfies*

$$v_2 u_{\mathrm{prim}}(0) = 2j_{\mathrm{rc}}(h, k) - 4\omega(h) - s(h) - s(hk) - \sum_{p \mid k} t_p + O_{E,\pi}(1).$$

The error depends on the fixed old local types and fixed multiplier, but not on h or k. Here $0 \leq t_p \leq 2$, and $t_p = 2$ in the present full rational two-torsion setting. Only after a bound on $\omega(k)$ is imposed may the companion sum be absorbed. The finite-Sha assertion in this calibration is supplied by the two stated analytic hypotheses.

Proof. First we justify the specialization of the generic coordinate. The forward theorem, Lemma 5.5, gives rank one and finite Sha for the first twist and rank zero and finite Sha for its companion. Hence the central rational primitive complex has cohomology dimensions $(1,1)$ in degrees one and two and zero elsewhere. Over $A = \Lambda_{(t)}$, cancel the unit disks outside these two degrees; Nakayama's lemma permits these cancellations because the other central cohomology groups vanish. The retained cycle has nonzero central class by (4.5), the height formula at the known simple zero, and its specified nonzero multipliers. In the remaining two-term free complex this forces a nonzero generic kernel. Semicontinuity bounds its dimension by one. The Euler characteristic is zero, so the generic dimensions are also $(1,1)$. Lemma 8.9 now identifies the specialization of the generic coordinate with the central tensor computed in Lemma 8.8, in the same retained determinant basis. This step uses neither the numerical calibration (4.6) nor the later uniform determinant bound.

For the ordinary compact Kummer complex C_K, universal coefficients and the local degree-zero identifications give

$$H^1(C_K)_{\mathrm{tors}} = E^{(h)}(K)[2^\infty], \qquad 0 \to H^1(C_K) \otimes \mathbb{Q}_2/\mathbb{Z}_2 \to \mathrm{Sel}_{2^\infty} \to H^2(C_K)_{\mathrm{tors}} \to 0.$$

The last finite group is Selmer modulo its maximal divisible subgroup. Self-duality identifies the degree-three torsion with the degree-one torsion and gives a unimodular pairing of the free degree-one and degree-two parts. Thus their rational dimensions equal full Selmer corank, including a possible divisible part of Sha.

At known simple zeros, Lemma 5.5 gives rank one and finite Sha. For restriction of scalars and the product of the two twists, the sum/difference isogenies are dual and compose to $[2]$. Their free-lattice indices satisfy $IJ = 2$, with regulator ratio $2/I^2$. At odd primes the differential lattices agree; at two the pullback determinants have bounded total valuation. Milne's isogeny ratio [42, proof of I.7.3, (7.3.1)] follows from duality without assuming the BSD leading-term equality. It applies here with finite Sha, canonical heights and pullback measures.

At every active $p \mid h$, K splits, so the isogeny local factors cancel exactly. At either split place the full/Kummer quotient has length $\dim_{\mathbb{F}_2} E[2]^{\mathrm{Fr}_p} = 2$, since all of $E[2]$ is rational. The two places therefore contribute exactly -4. At a companion prime $p \mid k$, the rational component number of the ramified twist is $\#E[2]^{\mathrm{Fr}_p}$, whereas the restriction of scalars has connected special fiber. Thus $t_p = \dim_{\mathbb{F}_2} E[2]^{\mathrm{Fr}_p} \le 2$; in the present full rational two-torsion case it is exactly two. The result is

$$v_2 u_{\mathrm{prim}}(0) = 2j_{\mathrm{rc}}(h,k) - 4\omega(h) - s(h) - s(hk) - \sum_{p \mid k} t_p + O_{E,\pi}(1). \tag{4.6}$$

Only the companion sum can be absorbed when $\omega(k)$ is bounded. No class number or unbounded degree-two torsion length is absorbed.

Here is the determinant and isogeny bookkeeping in more detail. This calculation is used only at the known simple-zero constituents, where the forward theorems give finite Sha. Put $\mathcal{A} = \mathrm{Res}_{K/\mathbb{Q}}(E_K^{(h)})$ and $\mathcal{B} = E^{(h)} \times E^{(hk)}$, and let $g_{\mathcal{A}}$ and $s_{\mathcal{A}}$ be the lengths of their two-primary rational torsion and Tate–Shafarevich group, respectively, with analogous notation for $\mathcal{B}$. The ordinary Kummer complex has torsion lengths $g_{\mathcal{A}}, s_{\mathcal{A}}, g_{\mathcal{A}}$ in degrees one, two and three. A torsion disk with cokernel of length e in degree i contributes $(-1)^{i+1}e$ to the inverse determinant coordinate. The central class $Lm_i P^{\mathrm{rc}}_{h,k}$ and its paired functional each have free-coordinate valuation $j_{\mathrm{rc}}(h,k) + v_2(Lm_i)$. Thus their ordinary Kummer coordinate has valuation $2j_{\mathrm{rc}} + 2v_2(Lm_i) + 2g_{\mathcal{A}} - s_{\mathcal{A}}$.

At either place above the split moving prime r_i, the central full/Kummer quotient has degree-two length

$$v_2(N_i), \qquad N_i = r_i + 1 - \chi_h(r_i) a_{r_i} = (r_i - 1) - \chi_h(r_i) m_i.$$

Use the moving-character construction with $r_i \equiv 1 \pmod{2^{B_i}}$, $B_i > c_E$ tending to infinity. Since $v_2(m_i) \le c_E$, both local lengths equal $v_2(m_i)$. Their total negative contribution cancels the two occurrences of $v_2(m_i)$ in the class-functional tensor. This equality persists in the common limit. At each active split place the corresponding quotient has only degree-two cohomology $E[2]$, of length two. The two places above each active prime therefore contribute -4.

For clarity, orient the isogeny indices as

$$I = [\mathcal{B}(\mathbb{Q})_{\mathrm{free}} : \phi\mathcal{A}(\mathbb{Q})_{\mathrm{free}}], \qquad J = [\mathcal{A}(\mathbb{Q})_{\mathrm{free}} : \psi\mathcal{B}(\mathbb{Q})_{\mathrm{free}}].$$

Then $IJ = 2$ and $\mathrm{Reg}_{\mathcal{B}} / \mathrm{Reg}_{\mathcal{A}} = 2/I^2$. Choose global generators $\omega_{\mathcal{A}}$ and $\omega_{\mathcal{B}}$ of the top cotangent lattices of their Néron models, and use their period measures over $\mathbb{Q}$ throughout. The integral pullbacks $\phi^*\omega_{\mathcal{B}} = a\omega_{\mathcal{A}}$ and $\psi^*\omega_{\mathcal{A}} = b\omega_{\mathcal{B}}$ satisfy $ab = 4$. Thus $a, b \in \{\pm 1, \pm 2, \pm 4\}$. Real change of variables gives

$$|a|\Omega_{\mathcal{A}} = \frac{\#\ker \phi_{\mathbf{R}}}{\#\operatorname{coker} \phi_{\mathbf{R}}}\Omega_{\mathcal{B}},$$

where both finite groups have order at most four. Hence the period ratio has uniformly bounded two-adic valuation, without a separate discriminant normalization. With these differential choices the finite measure factors are the products $C_{\mathcal{A}}, C_{\mathcal{B}}$ of component numbers. The arithmetic isogeny identity gives

$$s_{\mathcal{A}} - s_{\mathcal{B}} = 2g_{\mathcal{A}} - 2g_{\mathcal{B}} + v_2\frac{\mathrm{Reg}_{\mathcal{B}}}{\mathrm{Reg}_{\mathcal{A}}} + v_2\frac{\Omega_{\mathcal{B}}}{\Omega_{\mathcal{A}}} + v_2\frac{C_{\mathcal{B}}}{C_{\mathcal{A}}}.$$

The $2g_{\mathcal{A}}$ terms cancel on substitution into the determinant formula. The remaining rational torsion is uniformly bounded by the fixed local types at two. Component factors cancel at every split prime. At $p \mid k$, put $E_0 = E^{(h)}$; it has good reduction, and $K_p/\mathbb{Q}_p$ is a tame ramified quadratic extension. By [46, Theorem 5.3 and Remark 5.4.1, pp. 299–300], the component group of the ramified twist is the fixed subgroup of the good special fiber under its descent action $[-1]$:

$$\Phi(E_0^{(k)}) \simeq \widetilde{E}_0[2].$$

This is an isomorphism of finite étale residue-field group schemes, so its rational component number is $\#E[2]^{\mathrm{Fr}_p}$. The Weil restriction of the good model has special fiber an extension of $\widetilde{E}_0$ by a vector group, and is connected. Thus its component number is one. All other component factors cancel at split primes or equal one at the remaining good primes. Consequently

$$v_2(C_{\mathcal{B}}/C_{\mathcal{A}}) = \sum_{p \mid k} t_p, \qquad t_p = \dim_{\mathbb{F}_2} E[2]^{\mathrm{Fr}_p}.$$

Substitution into the coordinate containing $-s_{\mathcal{A}}$ gives the negative sign in (4.6). In the full rational two-torsion case each $t_p = 2$. □

Lemma 8.11 (Unused-prime comparison and one corrected coordinate)**.** *At an unused good split prime, use the actual primitive-to-relaxed maps and the retained class and functional above. Write z for the scalar Frobenius value. With $e_q = a_q - z - z^{-1}$ and l_q the two-place quotient factor, they satisfy (4.7) with $\varepsilon_q \in \Lambda^{\times}$. For a finite binary family the universal correction η lies in $\mathcal{O}[G]$, and its corrected coordinate $\widetilde{u} = \eta u_{\mathrm{rel}}$ satisfies $\widetilde{u}_x = \varepsilon_x u_{\mathrm{prim},x}$ at every character, with $\varepsilon_x \in \Lambda^{\times}$. At any character where the primitive coordinate is regular at $t = 0$, this identity preserves its central valuation and vanishing. Only the product is specialized; η_x itself may have a pole.*

Proof. At an unused prime the exact point factor is $e_q = a_q - z - z^{-1}$. The second factor comes from the same closed functional, as follows. Let $i : C_P \to C_R$ be the primitive-to-relaxed map, and let $j : C_S \to C_{P^\perp}$ be the paired strict-to-primitive map. All old orthogonal conditions and their homotopies are retained. At q these are the maps from the unramified condition to the full condition and from zero to the unramified condition. Naturality of the cup pairing [44, Sections 6.4.1–6.4.3] gives

$$\beta_R(v, iw) = \beta_P(jv, w).$$

The actual norm identity is $z_R = e_q i(z_P)$, with its retained boundary. The generic paired strict lift satisfies $jz_S = e_q^c z_{P^\perp}$. Conjugation sends z to z^{-1}, so $e_q^c = e_q$. Consequently

$$\lambda_R \circ i = e_q \lambda_P.$$

Thus the class and its paired functional each contribute one e_q; no integral self-duality of the primitive condition is assumed. The two quotient determinants satisfy

$$\delta_q = \det(z \operatorname{Fr}_q - q) \det(z^{-1} \operatorname{Fr}_q - q) = q^4 l_q.$$

In the original integral determinant bases, the support triangle therefore gives

$$l_q u_{\mathrm{rel}} = e_q^2 \varepsilon_q u_{\mathrm{prim}}, \qquad \varepsilon_q \in \Lambda^\times. \tag{4.7}$$

To see the unit assertion, retain the finite mapping-cone bases. Their comparison is a sign; replacing them by the retained homotopy equivalent free models contributes only determinants of unit disks. These are A_i-units. Their limits and inverse limits are Λ-units, and q^{-4} is one also. An arbitrary $\mathcal{O}$-unit would not suffice.

We check the integrality needed for the universal correction. The graph-wide prescription at the start of this section applies also to this unused prime. Modulo the maximal ideal $(2, I_G)$ of $\mathcal{O}[G]$, every group element and fixed quadratic sign becomes one, and the scalar z becomes $1 + t$. Since $E[2]$ is rational, a_q is even at every good odd prime. Thus

$$\overline{e}_q = (1 + t) + (1 + t)^{-1} = \frac{t^2}{1 + t} \neq 0 \quad \text{in } \mathbf{F}_2((t)).$$

Consequently e_q is a unit in $\mathcal{O}[G]$. This assertion uses the prescribed exponent at every $q \in Q$, not merely at the primes ramified in a particular constituent. With

$$l_q = \det(1 - q^{-1} z \operatorname{Fr}_q) \det(1 - q^{-1} z^{-1} \operatorname{Fr}_q),$$

the trace and determinant of Fr_q are a_q and q. Expanding these two degree-two polynomials gives the exact identity

$$q^2 l_q = e_q^2 + (q - 1)(q - 1 - (z + z^{-1}) e_q).$$

Because q is odd and e_q is a unit, this identity implies $l_q / e_q^2 - 1 \in 2\mathcal{O}[G]$. Set $v_q = (l_q / e_q^2 - 1)/2 \in \mathcal{O}[G]$. Then $1 + v_q(1 + g_q)$ selects l_q / e_q^2 at an inactive character and one at an active character. Their finite product is therefore one integral $\eta \in \mathcal{O}[G]$, without any division by the group order. Put $\widetilde{u} = \eta u_{\mathrm{rel}}$. It already is one common family; no interpolation of the individual ε_x is required. Formula (4.7) makes $\widetilde{u}_x = \varepsilon_x u_{\mathrm{prim},x}$. The central zero and valuations therefore transfer unchanged. The multiplier η_x may itself have a pole at $t = 0$. Only the product $\widetilde{u}_x$, identified with the regular primitive coordinate times a Λ-unit, is specialized there.

Simultaneous prescriptions and local lifts. We spell out why the preceding calculation applies to the whole graph. First choose its actual finite set Q by Proposition G.7. Its primes are good, odd, split in K, and outside $2ND_K$. The theta prime has already been chosen and is an exclusion. Choose one prime of K above each $q \in Q$. In Proposition 8.5, "active list" can be this entire list of orientations: the construction uses only these splitting and avoidance properties, not ramification in any particular quadratic constituent.

Here is the compatibility check in its class-group presentation. Adjoin orientations above $2N$ and auxiliary split primes generating $\mathrm{Cl}(K)$, all in distinct conjugate pairs. Assign weight one to every member of Q and weight zero to the other primes. For the relation basis $m^{(a)}$ and $\alpha_a = \beta_a/\overline{\beta}_a$ used there, a relation $\prod_a \alpha_a^{b_a} \in K(\mu_{2^\infty})^{\times 2^n}$ implies

$$2^n \mid \sum_a b_a m_j^{(a)} \qquad (j \text{ corresponding to } Q).$$

Indeed, a root belongs to a finite cyclotomic layer, which is unramified at each of these odd primes, and its valuation is integral. Multiplying by the weights and summing proves that the prescribed Kummer multipliers respect every relation. At a resulting split prime r, their power-residue values therefore satisfy every relation in the presentation of $\mathrm{Pic}(\mathbf{Z} + r\mathcal{O}_K)$. The resulting character has value one at every chosen orientation in Q, value zero at $2N$, and is surjective on the residue-ratio kernel, hence on inertia at r. The conjugate orientation has the negative value because the product ideal is rational principal. The radical-fixing subgroup (4.4c) preserves all these prescriptions while imposing the two trace bounds. Its level depends only on the Tate image. Enlarging Q consequently changes the finite Chebotarev field, but not c_E.

Perform this construction separately at each coefficient stage, excluding Q, the theta prime, the old support and the previously chosen moving primes. Taking $n_i \geq 2i$ gives $(1+t)^{2^{n_i}} = 1$ in A_i, so these are actual finite characters. They need not be restrictions of one character of G_K: the retained inner limit is the limit of the finite diagrams of Lemma 9.1. At each fixed graph orientation their values are already exactly $1+t$, so that identity survives every coefficient reduction and that limit. The norm identities (4.1), their coefficient maps, and their boundary witnesses hold before character evaluation. This is the same construction for all constituents. At $t = 0$ the moving character is trivial, so its added prescriptions do not alter the central quadratic family.

There is a local-lift issue only for the universal quadratic character. Write $g_q \in G$ for its inertia generator at q; thus $x(g_q) = (-1)^{\lambda_q(x)}$. Fix a lift of arithmetic Frobenius at the chosen orientation and define z_q using that lift. The moving character is unramified there, and its value is $1+t$. The fixed quadratic character is unramified there also. Every other graph prime contributes an unramified quadratic character at q. Changing the Frobenius lift can therefore only replace z_q by $g_q z_q$. For an inactive character x, $x(g_q) = 1$, and $x(z_q)$ is the actual unramified Frobenius scalar. For an active character its value is immaterial to the correction, which is one. Thus we never assign an unramified Frobenius eigenvalue to a ramified constituent.

This independence also holds inside $R = \mathcal{O}[G]$, before evaluation. Both choices of z_q have reduction $1+t$ modulo $(2, I_G)$, so both corresponding e_q are units. Put $r_q = l_q/e_q^2$ and $r'_q = l'_q/(e'_q)^2$. They agree in $R/(g_q - 1)$, hence $(1+g_q)(r'_q - r_q) = 0$. Since R is two-torsion-free and $r_q = 1 + 2v_q$, $r'_q = 1 + 2v'_q$, this gives

$$(1+g_q)(v'_q - v_q) = 0.$$

The correction $1 + (1+g_q)v_q$ is consequently independent of the lift. Reversing the orientation replaces z_q by z_q^{-1} and leaves both e_q and l_q unchanged.

For clarity, write $s_q = z_q + z_q^{-1}$. Direct expansion gives

$$q^2 l_q = (q-1)^2 + a_q^2 - a_q(q+1)s_q + q s_q^2, \qquad v_q = \frac{q-1}{2q^2 e_q^2}(-(q+1)e_q^2 + (q-1) - s_q e_q).$$

Every term in the second expression is integral in R because q is odd and e_q is a unit. These inverses are taken only after the inner limit and localization at (2), never in the finite ring $A_i[G]$. Reduction modulo $(2, I_G)$ gives $\overline{e}_q = t^2/(1+t)$ and $\overline{l}_q = t^4/(1+t)^2$. The latter identity also holds at each sign evaluation modulo two. At augmentation the local two-place determinant is $q^4 l_q$, so its residual t-order is exactly four. Multiplying the correction factors over the finite set Q proves the asserted universal integrality and all the characterwise identities simultaneously. □

Lemma 8.12 (The unknown base has zero coordinate if its derivative vanishes)**.** *Use the base $h_0 = 1$ in the final fixed-K construction, with $L(E^{(k)}, 1) \neq 0$ and full two-primary Selmer corank one for E/K. If $L'(E/K, 1) = 0$, then $u_{\mathrm{prim},0}(0) = \widetilde{u}_0(0) = 0$ for the same coordinates as above. The primitive coordinate is regular at (t) by the local argument below, and its corrected relaxed coordinate has the same property by Lemma 8.11. No finite* Sha *assertion at the base is required.*

Proof. Put $A = \Lambda_{(t)}$, with residue field $\mathbf{Q}_2$ and fraction field F. Use the actual finite free primitive complex and its retained class and functional over A. The latter is integral over A by Proposition 8.6. The conductor-one height formula makes the specified point torsion when $L'(E/K, 1) = 0$. Thus its rational Kummer class vanishes, and the retained boundary (4.5) makes the specialization of the actual cycle z zero. No new normalization of L is made here.

Full Selmer corank one gives central dimensions $(1, 1)$ in degrees one and two and zero in every other degree. This uses the full Selmer group, including a possible divisible part of Sha. Over the DVR A, a bounded finite free complex decomposes, by successive Smith reductions, into free zero-differential summands and two-term disks $[A \xrightarrow{t^r} A]$, up to units. A disk with $r > 0$ contributes one additional residue-field cohomology class in each of its two adjacent degrees. Semicontinuity and Euler characteristic therefore leave only two generic possibilities: the complex is acyclic, or its dimensions are $(1, 1)$ in degrees one and two.

In the acyclic case the specified coordinate is zero by definition. In the second case the generic and central dimensions agree in every degree. Lemma 8.9 applies: no disk with $r > 0$ occurs. All cancellable disks have unit differential, so integral determinant base change identifies the coordinate of $[z] \otimes [\lambda_z]$ with the coordinate of its central tensor, in the retained determinant basis. That tensor is zero because $[z]_0 = 0$. This proves regularity at (t) and $u_{\mathrm{prim},0}(0) = 0$ directly. Finally $\widetilde{u}_0 = \varepsilon_0 u_{\mathrm{prim},0}$ with $\varepsilon_0 \in \Lambda^\times$ by Lemma 8.11, so the corrected coordinate is also regular and zero at the center. This proof uses neither the numerical lower bound of Proposition 9.6 nor finite Sha at the unknown base. □

9 The two limits and determinant transfer

This section uses the actual diagrams of Section 8. Their cohomology ranks alone would not suffice: maps to original cochains, local comparisons, cup products, classes and boundary witnesses are retained together. The same retained data include the integral duality maps, their homotopy inverses, and the homotopies expressing compatibility of duality with localization.

9.1 Finite free models and the inner limit

Lemma 9.1 (Finite free models and retained inner-limit diagrams)**.** *Let A be an Artin local ring with nilpotent maximal ideal, and let a complex of flat A-modules have finite bounded residual cohomology. This includes the actual continuous cochains with finite free coefficient modules used here. It admits a bounded finite free model with actual inclusion, projection and contraction homotopy.*

Every fixed finite diagram of maps and homotopies, including the duality and local-condition data used here, transports to these models.

For one fixed active support and binary family, retain a tuple whose model ranks are bounded over the finite coefficient stages. One fixed nonprincipal ultrafilter gives a diagram over Λ at all coefficient precisions, preserving its matrix identities and retained homotopy inverses. This assertion imposes no rank bound uniform over all active supports.

Proof. Let A be an Artin local ring with nilpotent maximal ideal $\mathfrak{m}$. A flat A-module is free, including in infinite rank: lift a basis modulo $\mathfrak{m}$, use nilpotence for surjectivity, and flatness and nilpotence for vanishing of the kernel. A continuous finite-coefficient cochain module is a filtered union of modules of functions on finite clopen partitions, hence is flat.

Suppose the residual cohomology of a complex of such modules is finite and bounded. Split the residual complex into cohomology and disks, and lift graded bases. Let d_0 be the lifted disk differential and $d = d_0 + \epsilon$, where ϵ has entries in $\mathfrak{m}$. Use the graded decomposition into cohomology, disk sources and disk targets. Let i_0, p_0 include and project the cohomology summand, and let h_0 invert d_0 from disk targets to disk sources, zero on the other summands. Then

$$p_0 i_0 = 1, \quad h_0^2 = h_0 i_0 = p_0 h_0 = 0, \quad 1 - i_0 p_0 = d_0 h_0 + h_0 d_0.$$

With this normalized contraction, the finite geometric series give

$$\begin{aligned} d_C &= p_0 \epsilon (1 + h_0 \epsilon)^{-1} i_0, \quad & i &= (1 + h_0 \epsilon)^{-1} i_0, \\ p &= p_0 (1 + \epsilon h_0)^{-1}, & h &= h_0 (1 + \epsilon h_0)^{-1}. \end{aligned}$$

Substitution using $d^2 = 0$ gives $pi = 1$ and $1 - ip = dh + hd$. The resulting bounded finite free model carries actual maps and homotopies to the original cochains. Transport any fixed finite collection of arithmetic maps by $f \mapsto p' f i$. The identity $1 - ip = dh + hd$ transports their compatibility homotopies as well, including the second homotopies in the cup-product diagrams.

Fix one nonprincipal ultrafilter on coefficient stages. Bounded ranks can be fixed on a set in that ultrafilter. Each finite tuple of matrix entries then has a compatible limit at every coefficient precision, giving matrices over Λ. Every polynomial identity between the matrices survives. Retained homotopy inverses remain homotopy inverses. Mere stagewise quasi-isomorphisms without those maps would not justify a nonflat coefficient change. □

Lemma 9.2 (The reduced generic residual rank bound)**.** *Fix E, an integer $A \geq 0$, and a number j of derivative labels. For the ring instances with $\omega(k) \leq A$, reduction to characteristic two followed by the stated support comparison and inversion of t gives a generic residual model of rank at most*

$$N(E, A, j) \leq 56\omega(2N) + 8A + 56j + 70.$$

This bound is independent of the number and sizes of active primes and of $v_2(r_i - 1)$. It bounds the reduced generic model, not the whole retained support-comparison tuple before t is inverted.

Proof. All ranks in this proof are sums of the ranks in all degrees of one complex. They are not the size of the diagram retaining all comparison maps. Put $v = \omega(2N)$ and $m = v + 1 + j$. Let S_0 be the finite places of K above $2N$, the moving prime, and the derivative primes. Let T be the set of rational primes below S_0, together with the primes ramified in $K/\mathbb{Q}$, and put $s = |T|$. Since $2 \mid 2N$,

$$s \leq v + A + 1 + j. \tag{5.1a}$$

Primes added to T solely because they ramify in K are used for the global estimate. They do not impose additional Selmer conditions.

We first give the global estimate at the residue field $B_i/(t) = \mathbb{F}_2$, where $B_i = \mathbb{F}_2[t]/(t^i)$. The residual coefficient module is the trivial module $W = E[2] = \mathbb{F}_2^2$. Write S_T for all finite places of K above T, and let r be the number of finite rational primes ramified in K. Kummer theory gives the exact sequence

$$0 \longrightarrow \mathcal{O}_{K,S_T}^{\times}/(\mathcal{O}_{K,S_T}^{\times})^2 \longrightarrow H^1(G_{K,S_T}, \mathbb{F}_2) \longrightarrow \mathrm{Cl}(\mathcal{O}_{K,S_T})[2] \longrightarrow 0.$$

Here $2 \in T$ and $\mu_2 = \mathbb{F}_2$ as a Galois module. The unit term has dimension $1 + |S_T|$: the unit rank is $|S_T|$ and the roots of unity contribute one dimension. Moreover

$$|S_T| \le 2s - r, \qquad \dim_{\mathbb{F}_2} \mathrm{Cl}(\mathcal{O}_{K,S_T})[2] \le r - 1. \tag{5.1b}$$

For completeness, the class-group inequality follows directly from the imaginary quadratic genus argument. Conjugation acts by inversion on the ideal class group. If $[\mathfrak{a}]$ has order dividing two, write $\overline{\mathfrak{a}} = (\alpha)\mathfrak{a}$. Taking ideal norms gives $N_{K/\mathbb{Q}}\alpha = 1$, since this norm is positive. Hilbert 90 gives $\alpha = \beta/\overline{\beta}$ (take $\beta = 1 + \alpha$ unless $\alpha = -1$, and take $\beta = \sqrt{k}$ in that case), so $\beta\mathfrak{a}$ is a conjugation-invariant fractional ideal. Modulo rational principal ideals such ideals are generated by the ramified prime ideals. Their classes have exponent two, and the principal ideal of $\sqrt{d}$, where d is the squarefree radicand of K, supplies a nonempty relation among them. Thus their span has dimension at most $r - 1$. The excluded field $\mathbb{Q}(i)$ would also satisfy the inequality, since its class group is trivial. Finally an S-class group is a quotient of the ordinary class group, and the two-rank of a quotient of a finite abelian group cannot increase. This proves (5.1b), including the possible ramification at two. The bound on $|S_T|$ counts one prime over each ramified rational prime and at most two over each other prime.

Inflation from G_{K,S_0} into G_{K,S_T} is injective on H^1 with trivial coefficients. Consequently

$$h^0(G_{K,S_0}, W) = 2, \qquad h^1(G_{K,S_0}, W) \le 4s.$$

The global Euler characteristic and cohomological-dimension statements [42, I.4.10 and I.5.1], applied over the imaginary quadratic field K, give

$$h^0 - h^1 + h^2 = -2, \quad H^{>2} = 0, \quad h^2 = h^1 - 4.$$

Thus the total residual cohomology dimension of the global complex is at most $8s - 2$, and in particular at most $8s + 2$. The latter weaker bound can also be obtained without genus theory: the maximal pro-two group over $\mathbb{Q}$ unramified outside T and infinity has $s + 1$ generators, by the squareclasses of -1 and the primes in T; its subgroup for K has at most $2s + 1$ generators by the index-two Schreier bound. The maximal pro-two group for G_{K,S_0} is a quotient of this subgroup. Hence $h^1(G_{K,S_0}, W) \le 4s + 2$, giving total dimension at most $8s + 2$. In either computation the H^2 estimate is supplied by the global Euler characteristic, not by a generator estimate alone.

We next bound the local terms in the Selmer cone (4.3). For a finite place w, put $d_w = [K_w : \mathbb{Q}_2]$ if $w \mid 2$ and $d_w = 0$ otherwise. Local duality and the local Euler characteristic [42, I.2.3 and I.2.8] give, for the trivial module W,

$$(h^0, h^1, h^2)(K_w, W) = (2, 4 + 2d_w, 2).$$

At an old Kummer place, the derived condition has residual dimensions $(2, 2 + d_w)$ in degrees zero and one. Indeed $E^{(h)}(K_w)_2^{\wedge}$ is the direct sum of a free $\mathbb{Z}_2$-module of rank d_w and finite two-primary

torsion; that torsion has two generators because $E^{(h)}[2]$ is rational. Derived reduction retains its two-dimensional degree-zero torsion. For the other conditions at the odd moving or derivative places, the following table records the sum of the dimensions of the local condition and of the full local complex. Cone shifts do not change this sum.

condition U_w	$\sum h^a(U_w)$	$\sum h^a(U_w) + \sum h^a(K_w, W)$
old Kummer	$4 + d_w$	$12 + 3d_w$
unramified, $w \nmid 2$	4	12
0, $w \nmid 2$	0	8
full, $w \nmid 2$	8	16
derivative strict	2	10
derivative finite or singular	4	12
derivative relaxed	6	14

The last three rows use exactly the conditions retaining D^0 in Lemma 9.5; the relaxed condition does not include D^2. There are at most two places over each rational prime, and $\sum_{w|2} d_w = 2$. The old prime two therefore costs at most 30, any other old rational prime at most 24, the moving prime at most 32, and a derivative prime at most 28 (in fact it is inert and costs at most 14). Each is bounded by 48. Thus all finite local terms cost at most $48m$. No prime dividing k alone is charged a local condition.

There are no real places of K. Its one complex place has full cohomology W in degree zero only. With the usual modified archimedean convention its local complex is zero; even retaining both an ordinary full term and a full local condition costs at most four additional generators. Complex conjugation used to transport duality is not an additional real local condition over K. In particular no infinite two-periodic real resolution is being included in this count.

At each finite coefficient stage, apply Lemma 9.1 separately to these global and local complexes and transport the actual local maps. The mapping cone then has a finite free model whose total rank is bounded by the sum just computed. Further cancellation can only decrease it. The estimate applies before inverting t to the smaller-support model, uniformly in the moving prime and in the order of its finite character: reduction modulo t kills that character. Allowing the weaker global estimate and the archimedean allowance still gives

$$\begin{aligned} \operatorname{rank}_{\mathrm{tot}} C_{S_0} &\le (8s+2) + 48m + 4 \\ &\le 8(s+1) + 6 + 48(v+1+j) \\ &\le 56v + 8A + 56j + 70. \end{aligned} \tag{5.1}$$

It remains to transfer this estimate to the original support. First consider a scalar ring instance. Reduction modulo two kills χ_h, so the coefficient system is unramified at every active prime. The actual support comparison of [44, Proposition 7.8.8(ii)] therefore identifies the smaller-support complex with the complex having unramified conditions at those primes. At an oriented active place, reduction of (4.4) has

$$C = 0, \qquad A = D = \big((1+t)^e - 1\big) I_2,$$

where e is odd; at the conjugate place the exponent is $-e$. Both $((1+t)^e - 1)/t$ and $((1+t)^{-e} - 1)/t$ are units of $\mathbb{F}_2[[t]]$. Thus the full, unramified and singular comparison blocks all become contractible after inverting t. For each fixed finite active support retain the comparison maps and homotopies, take the inner limit, and only then invert t. Every active comparison cone is then contractible, so the original complex has the same generic residual cohomology as the bounded smaller-support model. No count of active primes remains.

For clarity, a universal binary character is not trivial merely after reduction modulo two. In the binary setting make the preceding calculation after augmentation $\mathbb{F}_2[G] \to \mathbb{F}_2$. After the inner limit and inversion of t, the ring $\mathbb{F}_2((t))[G]$ is local with nilpotent augmentation ideal. A finite free model over this ring has a minimal model obtained by cancelling unit entries; its ranks are the dimensions of the cohomology after augmentation, and hence obey (5.1). These are ranks over $\mathbb{F}_2((t))[G]$, not dimensions over $\mathbb{F}_2((t))$. For scalar instances they are precisely the generic residual ranks needed in the outer limit. The whole retained comparison diagram can still grow with G or with the active support; no bound for that diagram is asserted. Finally the minimum ranks over $\mathcal{O}$ are the dimensions after reduction to $\mathbb{F}_2((t))$, so the same estimate bounds the scalar $\mathcal{O}$-minimal models used below. □

9.2 Evaluation and the outer valuation ring

Lemma 9.3 (Evaluation, coefficient change and the outer DVR)**.** *For the retained global models and for the primitive and derivative Selmer models, evaluations at sequences of Galois elements satisfy (5.2). A cycle defines an abstract crossed homomorphism of the indicated product group. The induced map on H^1 is injective, and the finite detecting-list comparison remains injective after field extension. Direct evaluation of other Selmer models is asserted only when their local maps induce isomorphisms on residual H^0 and injections on residual H^1. For orthogonal zero/full comparisons, the assertion may instead be transported at a test ring where the actual local comparison complexes are contractible.*

For a sequence of ring instances with $\omega(k_j) \le A$ and a fixed number of derivative labels as in Lemma 9.2, the quotient $\mathcal{O}_\infty$ in (5.3) is a DVR with uniformizer two. The bounded minimal-model matrices, evaluations, duality maps and their compatibility homotopies pass to it. Sequences whose differentials tend to zero give actual crossed homomorphisms there. No continuity assertion on a single original Galois group, or completeness of $\mathcal{O}_\infty$, is required.

Proof. For fixed K let $\Gamma_K = \prod_i G_K$ and include one simultaneous complex conjugation. Evaluation of retained cochain maps at arbitrary sequences $g = (g_i), h = (h_i)$ gives

$$e^1_g d = (\rho(g) - 1)e^0, \qquad e^2_{g,h} d = \rho(g)e^1_h - e^1_{gh} + e^1_g. \tag{5.2}$$

One fixed ultrafilter supplies these identities for all sequences. A cycle therefore defines an abstract crossed homomorphism of the product group. We do not assert that its restriction to one original Galois group is a continuous pointwise limit.

Evaluation remains injective after field extension. At each finite stage take representatives V for H^1 and coefficient space W. Evaluate (c, w) by $b_c(g) - (\rho(g) - 1)w$. The intersection of all kernels is $(0, W^\Gamma)$. A finite list of at most $\dim V + \dim W$ evaluations already has that intersection. Choose and pad these lists at the finite stages before taking their common limit. The comparison to $[W \to W^d]$ is an isomorphism on H^0 and injective on H^1. Its cone has no residual cohomology in degrees at most zero. Cancel unit disks there so that the free cone starts in degree one; the same is true after every coefficient extension. The long exact sequence gives the claimed injection. For the primitive and derivative Selmer complexes the required residual comparison with global cochains follows from their actual local conditions. At an old Kummer place, derived reduction retains $B(F)[2]$ in degree zero, isomorphically onto the local invariants, and has the Kummer injection in degree one. The unramified condition at a good place has the same degree-zero invariants as the full local complex and injects in degree one. A full condition has the identity map. Every derivative condition retains D^0 and includes its stated degree-one subspace. The Selmer-cone exact sequence therefore gives an isomorphism on global H^0 and an injection on global H^1.

This does not apply directly to the zero conditions in the retained orthogonal complexes: at $(2,t)$ their local degree-zero invariants need not vanish. Indeed, two zero local conditions with residual invariant space W can contribute $(W \oplus W)/\operatorname{diag}(W)$ to Selmer H^1, and global cochain evaluation kills this contribution. Such orthogonal complexes are retained for duality, without asserting residual evaluation injectivity. If an actual zero/full comparison is contractible at a chosen test ring, its retained contraction identifies the two Selmer models there and transports the evaluation injection. This use requires that contraction at the chosen ring; it is not an assertion for arbitrary unused relaxed primes at every scalar height-one localization.

Now allow K_j, h_j to vary with $\omega(k_j) \le A$. For copies $\mathcal{O}_j$ of $\mathcal{O}$, form

$$\mathcal{O}_\infty = \left(\prod_{\mathcal{U}} \mathcal{O}_j\right) / \{(x_j) : v_2(x_j) \to_{\mathcal{U}} \infty\}. \tag{5.3}$$

A nonzero element has valuation one fixed nonnegative integer on a set in $\mathcal{U}$ and is that power of two times a unit. Every nonzero ideal has an element of least valuation and is principal. Thus (5.3) is a DVR with uniformizer two and characteristic-zero fraction field. Completeness is unnecessary.

Integral minimal-model matrices and (5.2) pass to (5.3). So do the retained duality maps, inverse homotopies and duality-to-localization compatibility homotopies. After minimal cancellation an integral homotopy equivalence is degreewise invertible: its residual matrices are invertible, and Nakayama applies. These inverse matrices and all compatibility identities pass to the same outer quotient. Thus perfect duality and exact orthogonality are preserved there; they are not inferred merely from isotropy. Approximate cycles suffice: if $v_2(dw_j) \to \infty$, the degree-two term in (5.2) tends to zero, so w_∞ is an actual crossed homomorphism. The finite detecting lists have bounded length by (5.1); retain their cone contractions to preserve injection. One must not replace approximate cycles by unexplained exact cycles. □

9.3 One new prime detects two classes

Lemma 9.4 (The common kernel and finite realization of a rank-two test)**.** *Use the exact kernel of sequences acting trivially on the full Tate module, the relevant ring-class fields and quadratic characters. On the characteristic-zero tests used below, restriction of the evaluated cohomology to this kernel is injective. When the moving scalar character is nonquadratic, two independent classes have a rank-two localization at a single actual kernel element. Finite Chebotarev realizes the required conjugation coset and basis-cochain evaluations, with exact inner-limit matrix identity (5.4) before any $\mathcal{O}$-minimal-model denominators are applied.*

At the residual test only scalar-inertia vanishing, bounded cohomology and finite detecting lists are asserted. The quadratic character exception at characteristic-zero height-one tests is treated in Proposition 9.6.

Proof. The exact kernel consists of sequences trivial on the full Tate module, all relevant ring-class fields and quadratic characters. The scalar element of Section 8 is central modulo this kernel, though not in the kernel itself. For a cocycle zero on the kernel,

$$(u^4 - 1)b(g) = (g - 1)b(z)$$

makes it a coboundary over characteristic zero. The same fixed scalar survives the outer quotient. To eliminate residual invariants, use moving-prime inertia: its scalar difference is t, invertible in $\mathbb{F}_2((t))$. These are two separate arguments.

On the characteristic-zero tests used for rank reduction, the fixed finite matrix-span witness gives absolute irreducibility of each rank-two summand. When the scalar character is not quadratic, moving inertia has distinct characters on the two summands and simultaneous conjugation exchanges them. The induced rank-four representation is therefore absolutely irreducible with scalar endomorphisms. The outer fraction field is one such characteristic-zero test; the quadratic character exception at an inner test is treated separately below.

For two independent cohomology classes on these tests, the span of their joint values on the exact kernel projects onto both copies of this absolutely irreducible representation. A proper submodule with both projections onto would be the graph of a scalar isomorphism, contradicting independence. Thus the span is the full sum. If the determinant of their two finite localizations vanished on every actual kernel element, additivity and polarization would make it vanish on the entire span, impossible. Hence one actual element supplies a rank-two localization on these characteristic-zero tests.

The polarization identity itself is also valid in characteristic two, but no residual irreducibility or residual rank-two localization is asserted. Here $E[2]$ is trivial of rank two, so its scalar twist is isotypic and the induced residual representation has multiplicity two. At the residual test we use only scalar-inertia vanishing of H^0, the bounded residual cohomology dimensions, and finite detecting lists from intersections of linear evaluation kernels. Those lists need not consist of one rank-two localization. The rank-two selection for the uniform valuation bound occurs only after passage to the characteristic-zero fraction field of the outer DVR.

Chebotarev is applied at finite stages, not to the product group. Each of the finitely many original basis cochains in degrees zero through two factors through a finite quotient. Include their quotient, the required Tate precision, ring-class and quadratic fields and old diagrams in one compositum. The element just found, multiplied by conjugation, is a single actual automorphism of that compositum. Its restrictions agree on intersections. Choose the new prime and a place above it to realize that representative.

This also resolves the precision issue caused by localization. Choose the new prime to match all original basis-cochain evaluations at a common retained precision $n(i) \to \infty$. Taking the same inner ultrafilter gives an exact equality of Λ-matrices

$$L_j = E_{\sigma_j^2}. \tag{5.4}$$

Only afterward compose with the inclusion I_j from an $\mathcal{O}$-minimal model: $L_j I_j = E_{\sigma_j^2} I_j$. The denominators in I_j may depend on j; equality (5.4) makes this irrelevant. Long t-jets are never treated as approximations in the two-adic valuation of $\mathcal{O}$, where t is a unit. □

9.4 Switches and the varying-companion estimate

Lemma 9.5 (Derivative local conditions and determinant-preserving switch)**.** *At a derivative prime use $L_v = D^0 \oplus (F_v \oplus S_v)[-1] \oplus D^2[-2]$ with the transported pairing, and retain D^0 in all four conditions. Take $U_{\mathrm{str}} = D^0$, $U_{\mathrm{rel}} = D^0 \oplus (F_v \oplus S_v)[-1]$, $U_F = D^0 \oplus F_v[-1]$, and $U_S = D^0 \oplus S_v[-1]$. Then strict and relaxed are orthogonal, the two plane conditions are self-orthogonal, and their cone triangles are the ones below.*

Over the stated fraction or test field, suppose the actual old class spans generic H^1 and has nonzero finite localization v. Its derivative from Proposition 8.3 has finite coordinate zero and singular coordinate Jv. The strict generic H^1 is zero, the new generic H^1 is spanned by this derivative, and the two class-functional determinant coordinates have equal valuations, even when v is not primitive in the local integral lattice.

Proof.

The strict and relaxed conditions at a derivative prime. The words strict and relaxed have a different local meaning at a derivative prime from zero and full at an active or unused split prime. At a derivative prime use the actual split local model

$$L_v = D^0 \oplus (F_v \oplus S_v)[-1] \oplus D^2[-2]$$

with zero differential at the retained precision, and retain its pairing and isotropy homotopies. Here D^0 and D^2 are dual, and F_v, S_v are the finite and singular isotropic planes. Put

$$\begin{aligned} U_{\mathrm{str}} &= D^0, & U_{\mathrm{rel}} &= D^0 \oplus (F_v \oplus S_v)[-1], \\ U_F &= D^0 \oplus F_v[-1], & U_S &= D^0 \oplus S_v[-1]. \end{aligned}$$

Thus $U_{\mathrm{str}}^{\perp} = U_{\mathrm{rel}}$ and the two plane conditions are self-orthogonal under the transported duality. All four maps retain local degree-zero cohomology. In particular, the relaxed condition here omits $D^2[-2]$ and is not the full local complex. The actual cone triangles are consequently

$$C_{\mathrm{str}} \longrightarrow C_F \longrightarrow F_v[-1] \longrightarrow C_{\mathrm{str}}[1], \qquad C_{\mathrm{str}} \longrightarrow C_S \longrightarrow S_v[-1] \longrightarrow C_{\mathrm{str}}[1].$$

These are the triangles used in the determinant switch. Zero/full conditions at the other primes retain their previously stated meaning. Exact annihilator and dimension assertions in the switch argument are made after passage to the stated fraction or test field, using these integral diagrams and their retained dualities. At a new inert prime the finite and singular planes are free of rank two, with cross pairing $\beta(x, y) = \varepsilon\langle x, Jy\rangle$. Suppose the old actual class A spans generic H^1 and has $f(A) = v \neq 0$. Its derivative B satisfies $f(B) = 0$, $s(B) = Jv$. The strict H^1 vanishes. The transported duality and localization identities identify the relaxed localization image with the kernel of its dual localization map, hence with its exact annihilator [44, Sections 6.3.3–6.3.4]. This image contains $(v, 0)$ and $(0, Jv)$, so these span it. Hence the new generic H^1 is spanned by B.

The determinant comparison is exact even if v is divisible. In the fraction field choose x with $v \wedge x$ of unit volume, and similarly y with $Jv \wedge y$ of unit volume. The finite and singular determinant exact sequences have the same strict factor, so their class-functional coordinates have equal valuations. All strict torsion remains in that common factor. Requiring x integral would be wrong when v is divisible. □

Proposition 9.6 (Uniform ring lower bound and height-one regularity)**.** *Fix E, π, the old local types and a bound A. For every ring instance satisfying the hypotheses of Proposition 8.10 and $\omega(k) \leq A$, the actual primitive coordinate satisfies*

$$v_2 u_{\mathrm{prim}} \geq -C(E, A), \qquad u_{\mathrm{prim}} \in \Lambda[1/2],$$

where the first valuation is that of $\mathcal{O}$ and the constant is independent of the active support and companion field. Consequently its central specialization and (4.6) imply (3.5).

The regularity assertion also holds at the unknown full-corank-one base used in Lemma 8.12. The uniform numerical lower bound here is asserted only for the known simple-zero ring instances; it is not asserted at that unknown base.

Proof. The starting generic kernel has rank one. Indeed, at a known simple zero the ordinary class is nonzero. Equation (4.5), with its bounded nonzero multiplier, makes the retained central class nonzero. Over $\Lambda_{(t)}$ cancel the degree-zero and degree-three disks. If the generic kernel of the resulting two-term free complex vanished, the actual lifted cycle would vanish, contradicting its central value. Semicontinuity bounds that kernel by the central dimension one.

Suppose no uniform lower bound exists. Choose instances with $v_2 u_j \to -\infty$, $\omega(k_j) \le A$, and write their minimal completed models as $[\mathcal{O}_j^n \xrightarrow{D_j} \mathcal{O}_j^n]$, fixing the bounded n on an outer-large set. In each generic one-dimensional kernel choose a primitive vector v_j. Its outer image is nonzero, so the outer kernel dimension s is at least one. If $s > 1$, one new derivative prime gives rank-two localization including that vector. The relaxed localization image is self-annihilating and contains the whole finite plane, hence equals that plane. The strict kernel has dimension $s - 2$ and the singular switch adds no localized vector, so the new outer dimension is $s - 2$.

The actual inner class is a nonzero scalar multiple of v_j. Even if this scalar tends to zero, its generic localization is nonzero on an outer-large set. Exact switch preservation therefore keeps each inner determinant valuation. The new inner kernels again have primitive vectors. Thus their outer dimension remains at least one; a proposed drop to zero is already a contradiction. There can be only finitely many drops. For each fixed number of them (5.1) bounds the models; no bound uniform in an unbounded number of switches is needed.

At terminal outer dimension one a maximal boundary minor Δ is nonzero. Its inner minors have one finite valuation b on an outer-large set. Determinant-one Schur operations in a square two-term rank-one-kernel complex give $u = \pm\alpha\beta/\Delta$, where α is an integral surviving cycle coordinate and β has valuation at least $-a$. Consequently $v_2 u_j \ge -a - b$, the required contradiction. We obtain

$$v_2 u_{\mathrm{prim}} \ge -C(E, A). \tag{5.5}$$

Here (5.5) is asserted for the known simple-zero ring instances used to obtain the varying-companion estimate. Their nonzero actual central class supplied the rank-one start. It is not a uniform valuation assertion at the unknown corank-one base. Bounded matrix size alone would not prove it: $[2^j]$ is a counterexample to that inference.

Height-one regularity, including quadratic specializations. Fix a height-one prime $\mathfrak{p} \ne (2)$, put $R = \Lambda_{\mathfrak{p}}$ and $k = \mathrm{Frac}(\Lambda/\mathfrak{p})$, and use the actual primitive scalar Selmer complex over R. Here R is a DVR and k has characteristic zero. All subsequent switches use the derivative conditions of Lemma 9.5, retaining D^0. The primitive old local conditions and these derivative conditions have the evaluation injection of Lemma 9.3. Cancel the degree-zero and degree-three disks and then the unit disks. Write $C = [R^s \xrightarrow{d} R^s]$ in degrees one and two, with $d \otimes k = 0$. Its generic kernel has dimension one. Choose a primitive vector $v \in \ker d$. Such a vector exists because $\ker d$ is saturated in R^s: divide a nonzero kernel vector by its largest common power of a uniformizer. Thus $\bar{v} \ne 0$ in $V = H^1(C \otimes_R k)$ and $\dim_k V = s \ge 1$. The actual class z is a nonzero fraction-field multiple of v. We will only make switches whose finite localization is nonzero on $\bar{v}$. Consequently their generic localization of z is nonzero, which is the hypothesis needed for the actual-class determinant switch.

If the scalar character specialized to k is nonquadratic, the rank-two detection of Lemma 9.4 applies to $\bar{v}$ and an independent vector whenever $s > 1$. The finite localization then has rank two. Exact duality makes the relaxed localization image equal to the finite plane, so switching to the singular plane lowers s by two. We give the quadratic case in detail.

Let α be the specialized character on the index-two group. Since the moving inertia generator has value $1 + t$, quadraticity implies $t(t + 2) = 0$ in k. The only possible primes are therefore (t) and $(t+2)$. The equality $\alpha^\tau = \alpha^{-1} = \alpha$ uses the actual conjugation τ; it involves no automorphism of R or substitution for t. The action

$$c^\tau(g) = Jc(\tau^{-1} g \tau), \qquad J = \rho_T(\tau),$$

on cocycles gives a k-linear involution on cohomology. Extend α to the index-two overgroup by choosing its value on τ to be $+1$. The other extension differs by the character of K. The induced

rank-four representation is consequently $W_+ \oplus W_-$, with each $W_\pm$ of rank two. The matrix-span witness in (H1)–(H2) and the paragraph following them makes their restriction to the index-two group absolutely irreducible. Twisting its matrices by nonzero scalar character values does not change their linear span. The summands are nonisomorphic: an intertwiner restricts to a scalar, whereas their actions of τ are J and $-J$. A scalar intertwining these two actions is zero since 2 is invertible. We write $V = V_+ \oplus V_-$ for the resulting cohomology signs.

At a new inert derivative prime write F, S for the finite and singular planes. The action on F is J, and that on S is $-J$. Indeed, the local rational Frobenius acts on the Tate module by J; it fixes its square, the finite Frobenius generator, and conjugates tame inertia by its ℓ-th power. The retained congruences give $\ell = -1$ in the limit, so pullback on the tame coordinate contributes $\ell^{-1} = -1$. These assertions hold at every previously chosen derivative prime as well. A common scalar sign would simply interchange the names + and −. The transported cross pairing is exactly

$$\beta(x, y) = \varepsilon e_2(x, Jy), \qquad \varepsilon \in k^\times,$$

from Lemma 9.5. The finite and singular planes are isotropic. These are the retained local cup-product identities, including the extra finite precision in (H12), not a residual alternatingness assertion. Since $\det J = -1$ and $J^2 = 1$, $e_2(x, Jy) = e_2(y, Jx)$. Choose nonzero J-eigenvectors e_+, e_-. They form a basis and $e_2(e_+, e_-) \neq 0$. The two local cohomology signs are therefore

$$H_+ = F(e_+) \oplus S(e_-), \qquad H_- = F(e_-) \oplus S(e_+).$$

They are orthogonal nondegenerate symmetric hyperbolic planes. In either plane the self-pairing of $aF(e_\pm) + bS(e_\mp)$ is $2ab$ times a nonzero constant. Its only isotropic lines are thus its finite and singular axes.

All old conditions are stable under this involution. Conjugate old Kummer conditions are transported into each other; primitive ramified full conditions occur at the conjugate pair; unused good conditions are unramified; and both F and S at a previous derivative prime are stable under the actions just computed. At the new prime let C_{str} and C_{rel} have the actual truncated conditions D^0 and $D^0 \oplus (F \oplus S)[-1]$. Put

$$U = \mathrm{im}\big(H^1(C_{\mathrm{rel}} \otimes k) \longrightarrow F \oplus S\big), \qquad K_0 = H^1(C_{\mathrm{str}} \otimes k).$$

The retained localization-duality diagrams give $U = U^\perp$. Specifically, localization's connecting map is its dual under the strict/relaxed duality, so its kernel is both U and $U^\perp$; this is the exactness in [44, 6.3.3–6.3.4] used in Lemma 9.5. Isotropy alone would not suffice. The cone triangles, with D^0 retained, give exact sequences

$$0 \longrightarrow K_0 \longrightarrow V \longrightarrow U \cap F \longrightarrow 0, \qquad 0 \longrightarrow K_0 \longrightarrow V' \longrightarrow U \cap S \longrightarrow 0, \tag{5.5a}$$

where V' is the switched fiber cohomology. There is no preceding local degree-zero term, since the quotient conditions are $F[-1]$ and $S[-1]$. The space U is stable under signs. Each $U_\pm \subset H_\pm$ is its own annihilator in $H_\pm$, hence is one of that plane's two isotropic axes.

We also need one prime detecting prescribed classes of opposite signs. Choose nonzero $c_+ \in V_+$ and $c_- \in V_-$. Their restrictions to the exact kernel H of Lemma 9.4 are nonzero additive equivariant maps. Their joint k-linear evaluation span in $W_+ \oplus W_-$ surjects to each summand. Simplicity and nonisomorphism force this span to be the full direct sum. At an element $h \in H$, evaluation at $(\tau h)^2$ on a sign-δ cocycle is

$$c_\delta((\tau h)^2) = (1 + \delta J)c_\delta(h) \qquad (\delta = +1, -1). \tag{5.5b}$$

This follows from additivity on H and the sign relation; a coboundary is zero on H. Thus the two finite-coordinate maps on the joint span surject respectively onto ke_+ and ke_-. There is an actual h for which both coordinates are nonzero. For completeness, if their product vanished at every actual h, additivity and the identity $Q(x+y)-Q(x)-Q(y)$ for that product would make its polar form zero on all pairs of actual evaluations. It would then vanish on their linear span, contradicting surjectivity onto the two axes. Finite Chebotarev and the exact matrix identity (5.4) realize this choice by a new inert prime, avoiding every previously used prime. All character, Tate, cochain and old-diagram constraints are included in the same finite quotient as in that lemma. No residual irreducibility or infinite-level Chebotarev assertion is used. The same argument for one nonzero sign class gives nonzero localization in its finite axis.

Suppose first that both V_+ and V_- are nonzero. Choose the class in one sign to be a nonzero component of $\bar{v}$ and choose any nonzero class in the other. Detection as above puts both finite axes in U, so $U = F$ and $U \cap S = 0$. Equation (5.5a) gives $\dim V' = s-2$. The localization of $\bar{v}$ is nonzero: its two sign components lie on different finite axes, so cannot cancel.

Suppose instead that only one sign occurs, say $V = V_+$, with $s > 1$. Detect $\bar{v}$ in its finite axis. Then $U_+ = F(e_+)$. Since $V_- = 0$, (5.5a) gives $U_- \cap F(e_-) = 0$. The isotropic-axis calculation forces $U_- = S(e_+)$. Consequently K_0 has sign dimensions $(s-1, 0)$, while V' has sign dimensions $(s-1, 1)$. This proves that the first switch introduces the other sign and preserves total dimension. Since $s - 1 \geq 1$, the next switch uses the preceding mixed-sign case and lowers the total dimension by two. After the first switch choose the primitive vector in the new integral generic kernel; include one of its nonzero sign components in the second detection. Thus the actual-class nonvanishing requirement holds at both steps. The case $V = V_-$ is identical with signs interchanged.

At each step the new class is the actual derivative $P_{I\cup\{\ell\}}$ of Proposition 8.3, with the same fixed multiplier L. Equation (4.2) gives finite coordinate zero and singular coordinate $Jf(P_I)$, which is nonzero generically. The construction retains the old Kummer conditions and all earlier singular conditions, by the final assertion following (H19). Lemma 9.5 therefore makes this class span the new generic kernel and preserves the valuation of its class-functional determinant coordinate. In particular we do not replace the actual class by the chosen primitive vector. That vector is used only to select the prime. The paired functional is the one transported from the same actual class. The determinant comparison uses the common strict determinant factor; fractional complements to $f(P_I)$ and $Jf(P_I)$ have unit exterior volume. Consequently a divisible localization creates no extra valuation, and all strict torsion factors remain present on both sides.

Each new generic kernel has rank one and has a primitive integral vector, so its fiber dimension is at least one. A proposed drop from dimension two to zero is therefore a contradiction; that configuration cannot occur. Whenever the dimension exceeds one, one or two switches give a strict drop by two. Thus finitely many switches leave dimension one. After cancellation the terminal complex has one free generator in each of degrees one and two; its differential is zero because its generic kernel has rank one. The actual class is integral. To apply Proposition 8.6 after these switches, we check its orthogonal comparison on the added local conditions. In the integral derivative model of Lemma 9.5, D^0 pairs with D^2 and S pairs perfectly with F, while the pairing on S is zero with its retained isotropy homotopy. Hence $(D^0 \oplus S[-1])^\perp = D^0 \oplus S[-1]$ as the specified local condition, with the retained integral comparison maps. Conjugate transport preserves this condition as well. At a derivative prime the comparison from the orthogonal condition to the conjugate condition is therefore the identity under those integral maps. It requires no inverse of a localization vector or of a local Frobenius operator. Using this identity in the Selmer-cone comparison, together with the same old-place comparisons as before, constructs the integral orthogonal lift of the new transported class. Composing with the retained integral global duality gives a closed integral degree-two functional. Generically this is exactly the functional of

Proposition 8.6: both use the inverse of the same orthogonal comparison, which is unique in the derived category. Thus its primitive integrality proof extends to every finite list of these derivative conditions. At the active ramified primes this uses the unit -2; at the moving prime it uses t except at (t), where the central Frobenius determinants are nonzero and hence units in R. Unused good primes retain their unramified condition. No inverse at an unused relaxed prime is used. The terminal coordinate is the product of the two integral coefficients, up to a determinant-basis unit, so belongs to R. Valuation preservation transports this conclusion through every switch. This proves regularity at every $\mathfrak{p} \neq (2)$, including both quadratic specializations. Finally the normal-domain intersection formula for $\Lambda[1/2]$ gives $u_{\mathrm{prim}} \in \Lambda[1/2]$.

The same proof applies at the unknown full-corank-one base. If its actual generic class is zero, its coordinate is zero. Otherwise its nonzero generic cycle and the central dimension bound give the rank-one start used above. This requires no uniform outer valuation bound at that base. Combining (5.5), for the known simple-zero instances, with (4.6) and $\sum_{p|k} t_p \leq 2A$ proves (3.5). No fixed-field class number is absorbed into the varying-field bound.

□

9.5 Actual augmentation and the fixed-field count

Lemma 9.7 (Augmentation and the fixed-field rank budget)**.** *Fix the final field K and the family of Section 7, in the branch where the odd empty configuration is not a unit. There are no derivative labels. At augmentation all graph primes are unramified. The actual old-support subcomplex gives (5.6). With $d_K = \dim_{\mathbb{F}_2} \mathrm{Sel}_2(E/K)$, $t_K = \mathrm{length}\, E(K)[2^\infty]$ and $h_K = \mathrm{length}\, H^2(C_K)_{\mathrm{tors}}$, it has the rank, Smith-disk and torsion bounds (5.7). These are fixed-K bounds, independent of the binary dimension and graph size.*

At every sign character, the inverse new local blocks and the resulting old-rank Schur complex have no Laurent terms below $-4M$ modulo 2^M, uniformly in the number of new blocks. This is a bound on sign evaluations, not on coefficients in the group basis. The finite number h_K does not assume finiteness of the whole Sha *at the base.*

Proof. We now use the final K in the remaining branch of Section 7, where the intrinsic odd empty configuration is not a unit. There are no derivative labels in this last binary family. At augmentation every graph prime is unramified, and the actual support triangle gives

$$C_{\mathrm{old}} \longrightarrow (C_b)_0 \longrightarrow \bigoplus_{q \in Q} Q_q \longrightarrow C_{\mathrm{old}}[1], \tag{5.6}$$

with C_{old} an actual subcomplex of an upper triangular model. The graph-wide prescription gives scalar $1+t$ at each chosen orientation, even though the quadratic character is trivial at augmentation. The Euler-factor calculation in Lemma 8.11 therefore gives

$$\overline{l_{q,0}} = \overline{e}_{q,0}^{\,2} = \frac{t^4}{(1+t)^2}.$$

The two-place local determinant differs only by the unit q^4. Its residual t-order is consequently four, uniformly in q and the number of graph primes.

Put $T = T_2E$ and let C_K be the ordinary compact Kummer Selmer complex over the fixed imaginary quadratic field K, with the derived Kummer condition, including its degree-zero torsion, at every finite place. Lengths in this paragraph are lengths over $\mathbb{Z}_2$. Write

$$d_K = \dim_{\mathbb{F}_2} \mathrm{Sel}_2(E/K), \qquad t_K = \mathrm{length}\, E(K)[2^\infty], \qquad h_K = \mathrm{length}\, H^2(C_K)_{\mathrm{tors}}.$$

The finite Selmer group defines d_K, and perfection of C_K makes its cohomology finitely generated, so h_K is finite. This does not assert that the full two-primary Tate–Shafarevich group is finite. The already fixed companion has $L(E^{(k)},1)\neq 0$, so the sum-and-difference isogeny and the forward corank-zero theorem give full Selmer corank one for E/K. This includes any divisible part of the Tate–Shafarevich group.

Here are the cohomology calculations underlying the budget. At finite coefficients, the local degree-zero maps are isomorphisms onto local invariants and the local degree-one maps are the Kummer injections. The Selmer cone therefore has $H^0 = E(K)[2]$ and $H^1 = \mathrm{Sel}_2(E/K)$. The Weil pairing and the exact orthogonality of the Kummer conditions give the finite self-duality with shift three [44, 5.1.1–5.1.6 and 6.3.4]. Since K has no real places and $E[2]$ is rational of rank two, this gives

$$\dim_{\mathbb{F}_2} H^i(C_K \otimes^{\mathbf{L}} \mathbb{F}_2) = (2, d_K, d_K, 2) \quad (i = 0,1,2,3), \tag{5.7a}$$

with zero cohomology in other degrees. In particular a minimal finite free $\mathbb{Z}_2$ model of C_K has total rank $2d_K + 4$.

For the integral torsion calculation, $H^0(C_K) = T^{G_K} = 0$: a compatible system of rational points of unbounded two-power order cannot lie in the finite group $E(K)_{\mathrm{tors}}$. The exact sequence for $0 \to T \to T \otimes \mathbb{Q}_2 \to E[2^\infty] \to 0$, and the local Kummer conditions, identify

$$H^1(C_K)_{\mathrm{tors}} = E(K)[2^\infty].$$

Indeed this finite group injects as torsion global cohomology, all of its elements satisfy the Kummer conditions, and a torsion class maps to zero with rational coefficients, so is in this image. Integral Kummer self-duality $C_K \simeq R\mathrm{Hom}_{\mathbb{Z}_2}(C_K, \mathbb{Z}_2)[-3]$ now gives

$$H^3(C_K) \simeq \mathrm{Ext}^1_{\mathbb{Z}_2}(H^1(C_K), \mathbb{Z}_2), \qquad \mathrm{length}\, H^3(C_K) = t_K. \tag{5.7b}$$

It also identifies the free ranks in degrees one and two. They both equal one: universal coefficients with $\mathbb{Q}_2/\mathbb{Z}_2$ give

$$0 \longrightarrow H^1(C_K) \otimes \mathbb{Q}_2/\mathbb{Z}_2 \longrightarrow \mathrm{Sel}_{2^\infty}(E/K) \longrightarrow H^2(C_K)_{\mathrm{tors}} \longrightarrow 0,$$

and the middle group has full corank one. Thus the total integral torsion length of C_K is $h_K + 2t_K$.

At $t = 0$ the moving character is trivial. At either place over the good split moving prime r_i, the unramified condition is the ordinary Kummer condition. Replacing it by the full condition adds the singular complex

$$Q_{r_i,w} = [\, \mathbb{Z}_2^2 \xrightarrow{\Phi_{r_i} - r_i} \mathbb{Z}_2^2 \,] \quad \text{in degrees one and two.}$$

Here Φ_{r_i} is arithmetic Frobenius on T, with trace a_{r_i} and determinant r_i. In particular

$$\det(\Phi_{r_i} - r_i) = r_i(r_i + 1 - a_{r_i}) = r_i \det(1 - \Phi_{r_i}),$$

so this determinant is nonzero and has valuation at most c_E. The same bound holds at the conjugate place. The singular complexes have zero degree-one cohomology and finite degree-two cokernels of total length at most $2c_E$.

This comparison is retained before the inner limit. The field K and C_K are fixed, while the two moving Frobenius matrices and the comparison maps may vary with i and with the graph. Pass to their limit at every coefficient precision using the same retained finite diagram. A matrix whose

determinant has valuation at most c_E cannot acquire a zero determinant in this limit: the property is determined modulo 2^{c_E+1}. Consequently, if $D_c = C_{\mathrm{old}} \otimes^{\mathbf{L}}_{\Lambda} \mathbb{Z}_2$, the resulting triangle is

$$C_K \longrightarrow D_c \longrightarrow Q_r \longrightarrow C_K[1], \tag{5.7c}$$

where Q_r is the sum of two rank-two singular disks and $H^2(Q_r) = J_r$ has length at most $2c_E$. The notation r in this limiting complex does not designate a fixed rational prime supporting characters of unbounded order.

Reducing (5.7c) modulo two, the two singular disks contribute at most eight to total cohomology dimension. Equation (5.7a) thus gives a finite free model of C_{old} over Λ of total rank at most $2d_K + 12$, by cancelling unit entries at its maximal ideal $(2, t)$. The integral long exact sequence of (5.7c) gives

$$H^1(D_c) \simeq H^1(C_K), \qquad 0 \to H^2(C_K) \to H^2(D_c) \to J_r \to H^3(C_K) \to H^3(D_c) \to 0. \tag{5.7d}$$

In particular $H^3(D_c)$ is a quotient of $H^3(C_K)$. The image of the torsion of $H^2(D_c)$ in J_r has length at most length J_r, and its kernel is $H^2(C_K)_{\mathrm{tors}}$. Hence, defining the symbol used below by

$$h_1 := \sum_i \operatorname{length} H^i(D_c)_{\mathrm{tors}},$$

we have $h_1 \le h_K + 2c_E + 2t_K$.

Finally, a finite free complex over $\mathbb{Z}_2$ decomposes into free cohomology generators with zero differential and Smith disks $[\mathbb{Z}_2 \xrightarrow{2^e u} \mathbb{Z}_2]$, $u \in \mathbb{Z}_2^{\times}$, $e \ge 0$. Its only free cohomology generators here are one in degree one and one in degree two. If m counts all its Smith disks, including those with $e = 0$, then $\operatorname{rank}_{\mathrm{total}} C_{\mathrm{old}} = 2 + 2m$. Every disk contributes e to h_1, so the sum of the valuations of all its nonzero Smith entries equals h_1. We have proved

$$\operatorname{rank}_{\mathrm{total}} C_{\mathrm{old}} \le 2d_K + 12, \qquad m \le d_K + 5, \qquad h_1 \le h_K + 2c_E + 2t_K. \tag{5.7}$$

The model and its matrices may vary with the moving characters, but these bounds depend only on the final fixed K and fixed old data. They do not depend on the binary dimension or graph size. The varying-field estimate (5.1) is not used with $j = b$, and h_K is not treated as a varying-field constant.

The lift is made over $R = \Lambda[G]$ before passing to $\mathcal{O}[G]$. Let C be the retained finite free family model. Its augmentation C_0 is the actual scalar cochain model, by the retained coefficient-change maps. After augmentation the representation is unramified at Q, so the support-change triangle (5.6) gives a finite free model D of C_0 whose underlying graded module is $C_{\mathrm{old}} \oplus \bigoplus_q Q_q$ and whose differential is upper triangular. This follows by representing the connecting map between these bounded free complexes by a chain map and forming its cone.

Cancel unit disks in C_0. The same pivots and elementary operations lift to C over R, since the augmentation ideal is contained in the Jacobson radical of R. Let C' be the resulting family model, so that C'_0 is minimal. Decompose D over Λ into a minimal complex $D_{\min}$ and unit disks E. The actual comparison gives a homotopy equivalence $C'_0 \simeq D_{\min}$. A homotopy equivalence between minimal bounded free complexes over a local ring is degreewise invertible, by reduction modulo its maximal ideal and Nakayama. Add to C' the same unit disks E_R and lift the resulting graded isomorphism $C'_0 \oplus E \simeq D$ to R. Its matrices are invertible because their augmented determinants are units. Conjugating the family differential by these lifted matrices makes its augmentation exactly the differential of D. All matrices and their inverses at this step lie in $\Lambda[G]$; thus their sign

evaluations are power series. The added disks are precisely those needed for D, so this procedure does not enlarge its stated old-support rank. The block containing the local quotients has

$$B_0 = A = \operatorname{diag}(A_q), \qquad T_0 = 0$$

in a differential $\left(\begin{smallmatrix} P & Q \\ T & B \end{smallmatrix}\right)$. Over $\mathcal{O}[G]$, cancel B. The surviving Schur differential $P - QB^{-1}T$ augments to P_0.

Write $\det A_q = t^{d_q} v_q + 2w_q$, $d_q \le 4$, with v_q a power-series unit, and put $C_q = \operatorname{adj}(A_q)/(t^{d_q} v_q)$. For a character x, $B_x C = 1 + 2F$ with both C and F having no Laurent terms below degree -4. Therefore

$$B_x^{-1} = C \sum_{j=0}^{M-1} (-2F)^j \pmod{2^M} \tag{5.8}$$

has pole at most $4M$, independent of the number of blocks. For completeness the adjacent-differential assertion is an exact block identity. In the two degrees containing the new local block, write the differential as $d = \left(\begin{smallmatrix} P & Q \\ T & B \end{smallmatrix}\right)$, with B invertible. Then

$$\begin{pmatrix} 1 & -QB^{-1} \\ 0 & 1 \end{pmatrix} d \begin{pmatrix} 1 & 0 \\ -B^{-1}T & 1 \end{pmatrix} = \begin{pmatrix} P - QB^{-1}T & 0 \\ 0 & B \end{pmatrix}.$$

If the preceding differential has components $(X, Y)^{\mathsf{T}}$, then $d^2 = 0$ gives $TX + BY = 0$. In the new source coordinates its components are $(X, B^{-1}TX + Y)^{\mathsf{T}} = (X, 0)^{\mathsf{T}}$. If the following differential has components (U, V), then $UQ + VB = 0$, and in the new target coordinates they are $(U, UQB^{-1} + V) = (U, 0)$. Thus the surviving entries of both adjacent differentials are literally unchanged; no second inverse is introduced there.

Here A_q is the direct sum of the two rank-two singular differentials at the conjugate graph places. Modulo two their scalar entries are t and $-t/(1+t)$, since their Frobenius scalars are $1+t$ and $(1+t)^{-1}$ and $E[2]$ is trivial. Consequently

$$\det \overline{A}_q = \frac{t^4}{(1+t)^2}.$$

This independently gives the residual order four in (5.6). The factor q^4 in the Euler-factor convention is a unit and does not affect it. Moreover every sign evaluation of a matrix over $\Lambda[G]$ has the same reduction modulo two as its augmentation, because each sign is congruent to one modulo two. Thus $B_x = A + 2V_x$ with V_x a power-series matrix. For the block-diagonal matrix C in (5.8), the identities

$$A_q C_q = 1 + \frac{2w_q}{t^{d_q} v_q}, \qquad B_x C = 1 + 2F_x$$

show that every entry of C and F_x has Laurent order at least -4. Formula (5.8) therefore has Laurent order at least $-4M$, because its jth summand has order at least $-4(j+1)$. The estimate counts the largest pole of an entry, not the number of summands or blocks. The remaining Schur differential has the same bound since P_x, Q_x, T_x are power-series matrices, and the adjacent surviving entries have no poles at all. These arguments take place after localization where t is invertible; they do not invert t in a finite Artin coefficient ring. □

9.6 Determinant clearing and the missing-address contradiction

Proposition 9.8 (A common denominator for the corrected coordinate)**.** *Under Lemma 9.7, use the paired functional exponent a of Proposition 8.6 and the universal correction η of Lemma 8.11.*

There are common $\Delta, Q \in \mathcal{O}[G]$ such that, wherever $\Delta_x \neq 0$, $\widetilde{u}_x = 2^{-a}\eta_x Q_x/\Delta_x^3$. Putting $f = 2^a\Delta^4$ and $d = \eta\Delta Q$ gives $f_x\widetilde{u}_x = d_x$ at every character, including $\Delta_x = 0$, and $f_0 \in \Lambda$. Moreover

$$v_2 f_0(0) \leq a + 4(h_K + 2c_E + 2t_K) = c_{\det}, \qquad L_T(M) = 16(d_K + 5)M$$

bounds the negative Laurent support of every f_x modulo 2^M, uniformly in the sign character x. The elements f, d themselves belong to $\mathcal{O}[G]$ in the original group basis. These bounds concern the same corrected coordinate throughout; no pole bound for d_x or for the group coefficients of f is required.

Proof. Put $S = \mathcal{O}[G]$ and first perform the simultaneous new-local cancellation of Lemma 9.7. Its determinant comparison factor κ is an S-unit: the canceled block is invertible over S, so its determinant and inverse are integral over S. Retain this factor, with the determinant-line sign prescribed by the original basis. Transport the actual cycle z and the closed functional $\mu = 2^a\lambda_z$ to the remaining complex D. Both transported cochains are integral over S. The augmentation of D is the old complex over Λ, and each sign-evaluated differential entry has pole at most $4M$ modulo 2^M. No bound on the entries of the transported z or μ is needed below.

Write $D_c = D_0 \otimes_\Lambda \mathbf{Z}_2$. Its integral Smith decomposition gives graded decompositions

$$D^i = B_i \oplus H_i \oplus E_i,$$

lifted by constant $\mathbf{Z}_2$ basis changes, where centrally the only nonzero differentials are the Smith disks $E_i \to B_{i+1}$. Here H_1, H_2 have rank one and all other H_i vanish. Let $m_i = \operatorname{rank} E_i$, $m = \sum_i m_i$, and let A_i be the actual differential submatrix from E_i to B_{i+1}. Set

$$\delta_i = \det A_i, \qquad \Delta = \prod_i \delta_i.$$

An empty determinant is one. The entries in this definition are the original entries of D, before any of these old-disk cancellations. Thus Δ has degree m in those entries, $\Delta_0 \in \Lambda$, and

$$v_2\Delta_0(0) = \sum_{\text{central Smith disks}} v_2(\text{disk}) \leq h_1 \leq h_K + 2c_E + 2t_K.$$

Here is the full denominator calculation. Work temporarily in $S[1/\Delta]$. Group all sources and targets for a given degree, and cancel the whole block A_i once. For that differential the two elementary matrices are

$$\begin{pmatrix} 1 & 0 \\ -CA_i^{-1} & 1 \end{pmatrix} \begin{pmatrix} A_i & B \\ C & D' \end{pmatrix} \begin{pmatrix} 1 & -A_i^{-1}B \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} A_i & 0 \\ 0 & D' - CA_i^{-1}B \end{pmatrix}.$$

Both elementary matrices have determinant one. In the adjacent differentials, the equation $d^2 = 0$ makes the components into and out of this canceled pair zero after these changes; the other entries of those adjacent differentials are unchanged. Hence a cancellation does not modify any other surviving A_j. Every inverse used is $\operatorname{adj}(A_i)/\delta_i$, and each grouped block is used only once.

More explicitly, in degree k the projection associated to the incoming cancellation is

$$(b, v) \longmapsto v - CA_{k-1}^{-1}b,$$

and the inclusion associated to the outgoing cancellation is

$$v \longmapsto (-A_k^{-1}Bv, v).$$

The other map in each of these two degrees is the corresponding coordinate inclusion or projection. Compose these maps in the chosen cancellation order. At each step each entry is linear in the one inverse A_i^{-1} at that step, and no A_i is encountered again. Products of denominators therefore divide $\prod_i \delta_i$, not a power depending on the number of individual Smith disks. In particular, if $p : D \to H$ and $\iota : H \to D$ are the final projection and inclusion, the two scalars

$$\alpha = (pz)^1 \in H_1, \qquad \beta = (\mu\iota)^2 \in H_2^*$$

satisfy $Z := \Delta\alpha \in S$ and $W := \Delta\beta \in S$. The final complex H has just the map $H_1 \to H_2$.

For the inverse determinant line, a disk in degrees $i, i+1$ contributes $\delta_i^{(-1)^i}$ to the coordinate of a cohomology tensor. This can be checked on the based disk $e \mapsto \delta_i b$: replace the boundary basis b by $\delta_i b$ in $\bigotimes_k \det(D^k)^{(-1)^{k+1}}$. Thus the remaining comparison factor is

$$T = \epsilon \prod_i \delta_i^{(-1)^i}, \qquad P := \Delta T = \epsilon \prod_{i \text{ even}} \delta_i^2 \in S,$$

where ϵ is the integral unit from the fixed basis ordering and the constant Smith basis changes. Including the previously retained new-local factor, put

$$Q = \kappa ZWP \in S.$$

This definition takes place over S, before sign evaluation. It uses no interpolation and no division by $|G|$.

If $\Delta_x \neq 0$ and the constituent has the rank-one cohomology pattern, its coordinate in the original determinant basis is exactly

$$\widetilde{u}_x = \eta_x \kappa_x \alpha_x (2^{-a}\beta_x) T_x = 2^{-a}\eta_x Q_x / \Delta_x^3.$$

This is the same z, functional, basis and multiplier η as in the statement. If the constituent is generically acyclic, the remaining map between the two lines is nonzero. Its cycle α_x is then zero over the fraction field, so $Q_x = 0$ and the formula agrees with the defined coordinate zero. These are the only generic patterns in the present family, by central specialization and the central corank-one bound.

Now define $f = 2^a\Delta^4$ and $d = \eta\Delta Q$ in S. The preceding formula proves $f_x\widetilde{u}_x = d_x$ whenever $\Delta_x \neq 0$. When $\Delta_x = 0$, both f_x and d_x are zero; the already defined coordinate $\widetilde{u}_x$ is finite in its fraction field, and the same equality follows directly. No rational expression with a zero denominator is specialized. In particular $f_0 \in \Lambda$ and

$$c = v_2 f_0(0) \leq c_{\det} := a + 4(h_K + 2c_E + 2t_K), \qquad L_T(M) = 16(d_K + 5)M. \tag{5.9}$$

Indeed f is a polynomial of degree $4m$ in the differential entries of D, with constant integral coefficients. Each such entry, at a sign character, has pole at most $4M$ modulo 2^M. A product of $4m$ entries has pole at most $16mM$, and sums cannot increase this bound. Finally $m \leq d_K + 5$.

Only the last pole estimate is characterwise. The actual elements f, d belong to the original group ring S. For $i < -L_T(M)$ all sign values of $[t^i]f$ are divisible by 2^M; the group-ring coefficient itself need not be divisible by 2^M. No Laurent support bound for d or for the group coefficients of f has been used. The constants d_K, t_K, h_K are fixed only after the final K is fixed; a and c_E precede that choice. None depends on the graph or binary dimension. A varying-K estimate cannot be deduced by treating h_K as an absolute constant. □

Proposition 9.9 (Finite coefficient transfer and the final contradiction)**.** *Use the fixed field, common family, regular coordinates and integral identity of Proposition 9.8. Let B be the fixed upper bound on $v_2\widetilde{u}_x(0)$ at every nonzero address supplied by (3.9), (3.4), (4.6) and Lemma 8.11. Choose M, b as in (5.10), before the finite graph and its actual theta stage are realized. For some nonzero address x,*

$$v_2(\widetilde{u}_x(0) - \widetilde{u}_0(0)) \geq M - 2c > B, \qquad c = v_2 f_0(0) \leq c_{\mathrm{det}}.$$

The convolution includes every negative Laurent coefficient of f_x. If $L'(E/K, 1) = 0$, Lemma 8.12 makes $\widetilde{u}_0(0) = 0$, contradicting the upper bound. Since the fixed complementary value $L(E^{(k)}, 1)$ is nonzero, this proves $\operatorname{ord}_{s=1} L(E, s) = 1$.

Proof. For a coefficient of any element of $\mathcal{O}[G]$, its sign-value function on $\mathbb{F}_2^b$ expands as

$$A(x) = a_\varnothing + \sum_{I \neq \varnothing} 2^{|I|} a_I \prod_{i \in I} x_i.$$

Its low bit is constant and its bit of index $j \geq 1$ has Boolean degree at most 2^{j-1}. One can see this by extracting the Y^{2^j} coefficient of $(1+Y)^{A(x)}$ modulo two: the first possible contribution of x_I has degree at least $2^{|I|}$ in Y, and $|I| \leq 2^{|I|-1}$. Thus q coefficient equalities modulo 2^M have sum of Boolean degrees at most $q(2^{M-1} - 1)$.

If that sum is less than b, multiply the complementary indicators of the equations. Its degree is less than b, so its sum over the Boolean cube is zero: every monomial misses a variable. The support of this indicator is the simultaneous solution set. It has even cardinality and contains zero, hence another address.

Apply this to the coefficients of f from $-L_T(M)$ through zero and the single constant coefficient of d. It suffices to choose

$$M > \max(c_{\mathrm{det}}, B + 2c_{\mathrm{det}}), \qquad b > (L_T(M) + 2)(2^M - 1), \tag{5.10}$$

where B bounds the nonzero-address central valuations. These choices precede the graph and its theta precision. At the resulting nonzero address, negative coefficients of f_x vanish modulo 2^M, and its constant coefficient agrees with $f_0(0)$ and has valuation c. Positive coefficients may have smaller valuation. Nevertheless $v_{\mathcal{O}}(f_x) \leq c$, so integrality of $d_x = f_x \widetilde{u}_x$ gives $v_{\mathcal{O}}(\widetilde{u}_x) \geq -c$. This estimate comes from the displayed product at the selected nonzero address; it does not apply (5.5) to the unknown base. Both $\widetilde{u}_x$ and $\widetilde{u}_0$ lie in $\Lambda[1/2]$ by the height-one argument above, so neither has negative powers of t. In particular, since $f_0 \in \Lambda$, $[t^0]d_0 = f_0(0)\widetilde{u}_0(0)$. Put $F_0 = [t^0]f_0$ and $F_x = [t^0]f_x$. Since d_0 is integral and $[t^0]d_0 = F_0\widetilde{u}_0(0)$, we also have $v_2\widetilde{u}_0(0) \geq -c$, without using (5.5) at the base. The convergent convolution has the form

$$[t^0]d_x = F_x\widetilde{u}_x(0) + E_x, \qquad E_x = \sum_{i<0} [t^i]f_x[t^{-i}]\widetilde{u}_x, \qquad v_2(E_x) \geq M - c.$$

The untested coefficients below the Laurent cutoff have the same divisibility by the pole bound, so this sum includes every negative coefficient. Subtracting the two constant terms gives

$$F_x(\widetilde{u}_x(0) - \widetilde{u}_0(0)) = ([t^0]d_x - [t^0]d_0) - (F_x - F_0)\widetilde{u}_0(0) - E_x.$$

Every term on the right has valuation at least $M - c$ and $v_2(F_x) = c$. Consequently

$$v_2(\widetilde{u}_x(0) - \widetilde{u}_0(0)) \geq M - 2c > B. \tag{5.11}$$

By (3.9), (3.4), (4.6) and the unit comparison (4.7), every nonzero address has $v_2\widetilde{u}_x(0) \leq B$ for one B fixed before b; the finite Sha terms have negative sign and may be discarded for this

upper bound. If the base derivative vanished, Section 8 would give $\widetilde{u}_0(0) = 0$, contradicting (5.11). Hence $L'(E/K, 1) \neq 0$. The complementary value $L(E^{(k)}, 1)$ is nonzero, and $L(E/K, s) = L(E, s)L(E^{(k)}, s)$, so E has analytic rank one. This proves (RC), and hence Theorem 5.3, using the classical applications and finite constructions specified above. □

10 Exact precision computation and polynomial compilation

Encode a signed integer by

$$\operatorname{enc}(n) = \begin{cases} 2n, & n \geq 0, \\ -2n - 1, & n < 0. \end{cases}$$

The five natural inputs decode in the fixed order (n, κ, a, l, j). The decision predicate rejects unless $n \notin \mathcal{C}$, $\kappa \neq 0$, $a, l \geq 1$, and $j = na$. For nonzero multiples of P_0, $z = -x/y$ is well-defined. Set

$$z([l]P_0) = u_l + v_l s, \qquad \epsilon_l = |u_l| + |v_l|, \qquad r(a, j) = \tfrac{1}{2} \operatorname{Tr}_{K_0/\mathbb{Q}} \frac{z([j]P_0)}{z([a]P_0)}.$$

With the fixed constant $A_{\mathrm{ht}}^{(0)} = H_6(S_{\mathrm{ht}})$, its final tests are

$$|r(a, j) - n| < \epsilon_l, \qquad 4\epsilon_l < 4^{-A_{\mathrm{ht}}^{(0)} |\kappa F(n)|^{24}}.$$

These are strict tests on exactly computable rationals and integers.

The reference program S21 uses exact quadratic-field arithmetic and capped integer operations. The ideal register intermediate representation S22 computes the same predicate on every five-natural tuple. Its independent all-input argument covers normalized signed arithmetic, origin and zero-ordinate cases, equal-abscissa addition, and every early rejection. Nine exact polynomial identities verify the nonexceptional projective formulas and preservation of $Y^2 = X^3 + AXZ^4$. Scalar loops decrease by halving; the precision loop divides by four; factorial iteration is bounded. This is a proof about the ideal intermediate representation, rather than an assertion that a budgeted simulator always finishes.

For example, if $\epsilon = U/V > 0$ with natural U, V and integer $B \geq 0$,

$$4\epsilon < 4^{-B} \quad \Longleftrightarrow \quad 4^B \leq \left\lfloor \frac{V - 1}{4U} \right\rfloor.$$

This permits a finite cap. For positive L, $\sigma(x) = \min(x, L + 1)$ commutes with saturated addition and multiplication. Powers and factorials can be computed with this saturation. Capping an exponent is safe for base at least two because an exponent above L already makes the value exceed L. Base zero, base one and exponent zero are separate cases. This proves equality with the fixed-height expression without requiring its full evaluation when the available precision is too small.

10.1 The natural polynomial and the power interface

The saved precision polynomial has 1 305 811 natural variables, the first five being inputs. Its SHA256 is

`f3e163f5c60b972c98b48647274f956dd7d78428e27ddc0543574a5f18120b8c.`

Source S23 specializes a fixed universal natural halting polynomial at the code of a wrapper. The wrapper inverts four Cantor pairs and then runs the unchanged precision machine. Its pairing equations are

$$2t_i = (c_i + t_{i+1})(c_i + t_{i+1} + 1) + 2t_{i+1} \quad (0 \le i < 4), \qquad t_4 = c_4.$$

Their triangular intervals prove unique natural unpairing. The new polynomial sums the square of the old halting polynomial and the four squared differences above.

The universal compiler packs a finite run of a fixed m-state increment/decrement/branch program into base-B integers. Acceptance means reaching the unique final halt with all registers zero; the fixed interpreter includes a finite cleanup stage. Choose

$$c > 0, \quad K = 2^c, \quad B = 4K > m, \quad T = B^q, \quad F_q = 1 + B + \cdots + B^q, \quad D = (K-1)F_q,$$

and require every input to be below K. For natural integers, say X is masked by Y when every set binary bit of X occurs in Y. Require each register history R_j to be masked by D and satisfy $R_j < T$. Each state history S_i is masked by F_q, and

$$\sum_{i=0}^{m-1} S_i = F_q, \qquad S_{\text{halt}} = T.$$

Thus register digits are below K, terminal register digits are zero, and state digits are zero or one. Since $B > m$, the state sum has no carries and selects exactly one state at each time.

The nonzero mask N_j is determined by

$$KN_j = (R_j + D) \text{ AND } (KF_q).$$

Indeed each digit $r + K - 1 < 2K < B$ has its K-bit set exactly when $r > 0$. For a decrement/branch instruction at state i testing register j, impose

$$V_i = S_i \text{ AND } N_j, \qquad S_i = V_i + W_i.$$

These are the nonzero and zero branch histories. Form the packed increment indicator A_j from states incrementing j, the decrement indicator C_j from the appropriate V_i, and each incoming-state indicator I_i from the states or branches directed to i. They have zero-one digits, since there is just one active instruction and branch at each time. The transition equations are

$$R_j + BC_j = \text{initial}_j + BR_j + BA_j, \qquad S_i = \mathbf{1}_{i=0} + BI_i.$$

Every register coefficient on either side is at most $K < B$; uniqueness of base-B digits gives the initial conditions and each exact transition. The possible extra final slot vanishes because halt has no outgoing instruction and the terminal registers are zero. For $q = 0$ the same equations require the initial state already to be halt and the inputs to vanish. Conversely, any finite accepting run fits after choosing c sufficiently large, and supplies all these masks and equations.

10.2 The literal alpha and power formulas

The alpha system below is `Exp_Matrices.alpha_equations`, with equivalence `alpha_equivalence`; the power system is `Exponential_Relation.exp_repr` and its polynomial expansion `exp_dioph`

in the pinned AFP source [45]. Their variables range over natural numbers. The alpha recurrence itself is integer-valued; the bounds below justify its natural-valued uses. Define $U_n = \alpha_b(n)$ by

$$U_0 = 0, \qquad U_1 = 1, \qquad U_{n+2} = bU_{n+1} - U_n \quad (n \geq 0).$$

For $b \geq 3$ this recurrence gives $0 \leq U_n < U_{n+1}$ and $U_n \geq n$. For $n \geq 1$, the matrix

$$M = \begin{pmatrix} b & -1 \\ 1 & 0 \end{pmatrix}, \qquad M^n = \begin{pmatrix} U_{n+1} & -U_n \\ U_n & -U_{n-1} \end{pmatrix}$$

has determinant one. Thus adjacent terms are coprime and $U_n^2 - bU_nU_{n-1} + U_{n-1}^2 = 1$. Parameter congruence follows directly from the recurrence: $b \equiv b' \pmod{d}$ implies $\alpha_b(n) \equiv \alpha_{b'}(n) \pmod{d}$. In particular $\alpha_2(n) = n$. This is parameter congruence, not merely congruence between indices.

Every natural solution of $x^2 + y^2 = 1 + bxy$, $b > 2$, is an unordered adjacent pair. To prove this, order it as $X > Y$. If $Y = 0$, $X = 1$. Otherwise $Z = bY - X$ satisfies $XZ = Y^2 - 1$, hence $0 \leq Z < Y$, and (Y, Z) satisfies the same equation. Descent reaches $(1, 0)$; reversing the steps generates exactly the adjacent pairs. No ordering assumption on the original pair is needed.

For $k > 0$, reduction of M^k modulo U_k is the invertible scalar $U_{k+1}I$. Dividing an index by k and using strict growth proves $U_k \mid U_m$ if and only if $k \mid m$. More precisely, for $m > 0$,

$$U_k^2 \mid U_m \quad \Longleftrightarrow \quad kU_k \mid m. \tag{C.1}$$

For $k = 0$ both sides are false. For $k > 0$ write $m = kq$ and expand $M^{kq} = (U_kM - U_{k-1}I)^q$ modulo U_k^2. Its lower-left entry is $qU_k(-U_{k-1})^{q-1}$, so coprimality proves (C.1), including $U_k = 1$.

For $m > 0$ put $v = U_{m+1} - U_{m-1} = \operatorname{tr}(M^m)$. Cayley–Hamilton gives $M^{2m} \equiv -I \pmod{v}$. Extend the recurrence by $U_{-j} = -U_j$. Every $n \geq 0$ has $n = 2lm + \varepsilon j$ with $l \geq 0$, $0 \leq j \leq m$ and $\varepsilon \in \{1, -1\}$; then $U_n \equiv (-1)^l \varepsilon U_j \pmod{v}$. The value sign need not equal the index sign. If $b \geq 4$, $v = bU_m - 2U_{m-1} > 2U_m$.

The actual alpha formula has natural witnesses r, s, t, u, v, w, x, y and the following fifteen conditions:

$$\begin{gathered} b > 3, \quad u^2 + t^2 = 1 + but, \quad s^2 + r^2 = 1 + bsr, \quad r < s, \quad u^2 \mid s, \quad v + 2r = bs, \\ w \equiv b \pmod{v}, \quad w \equiv 2 \pmod{u}, \quad w > 2, \quad x^2 + y^2 = 1 + wxy, \\ 2X < u, \quad 2X < v, \quad x \equiv X \pmod{v}, \quad 2N < u, \quad x \equiv N \pmod{u}. \end{gathered} \tag{C.2}$$

A strict inequality $a < b$ is $b = a+1+d$; divisibility is $b = ad$; congruence is $A+Mu = B+Mv$, with fresh natural witnesses. At zero modulus the last equation says equality, so there is no exception hidden in this replacement.

For soundness, the unordered Pell result gives $u = U_k$, $k > 0$, and $x = \alpha_w(n)$. The ordered pair gives $s = U_m, r = U_{m-1}$, $m > 0$. By (C.1), $u \mid m$. Parameter congruence gives $x \equiv U_n \pmod{v}$ and $x \equiv n \pmod{u}$. Reflect n to $j \in [0, m]$. Since $2X < v$ and $2U_j < v$, the congruence $X \equiv \pm U_j \pmod{v}$ forces $X = U_j$, with the negative case forcing both zero. Then $j \leq X$, $2j < u$ and $2N < u$ force $N = j$ from $N \equiv \pm j \pmod{u}$. Thus $X = U_N$.

Conversely choose an odd $u = U_k > 2U_N$. Such terms are unbounded, since adjacent terms are coprime and the sequence grows. Set $t = U_{k+1}$, $m = ku$, $s = U_m$, $r = U_{m-1}$ and $v = bs - 2r$. Equation (C.1) gives $u^2 \mid s$, and the required strict bounds hold. The moduli u, v are coprime: a common divisor is odd, divides s and $2r$, hence divides both adjacent terms. CRT gives $w \equiv 2 \pmod{u}$, $w \equiv b \pmod{v}$ with $w > 2$. Take $x = \alpha_w(N)$, $y = \alpha_w(N + 1)$. Parameter congruence verifies every remaining condition. This includes $N = X = 0$.

The literal power formula separates $Q = 0, R = 0, P = 1$ and $Q = 0, R > 0, P = 0$. For $Q \geq 1$ it imposes

$$a_0 = \alpha_{Q+4}(R+1), \quad b = a_0 + Q^2 + 2, \quad m + Q^2 + 1 = bQ, \quad P < m,$$
$$a_1 = \alpha_b(R), \quad a_2 = \alpha_b(R+1), \quad P + ba_1 \equiv Qa_1 + a_2 \pmod{m}. \tag{C.3}$$

Induction gives $a_0 \geq (Q+3)^R$, so $m > Q^R$ and $m > 0$. Let $E_R = QU_R - bU_R + U_{R+1}$. The recurrence and $bQ \equiv Q^2 + 1 \pmod{m}$ give $E_0 = 1$ and $E_{R+1} \equiv QE_R \pmod{m}$. Thus (C.3) says $P \equiv Q^R \pmod{m}$. Both values lie in $[0, m)$, forcing equality. Conversely equality supplies every witness. This includes $R = 0$, $Q = 1$ and $0^0 = 1$.

These formulas make the binomial and bit operations polynomial. For $B = 2^{n+1}$, the equation

$$(1+B)^n = B^k(Bq + C) + r, \quad C < B, \quad r < B^k$$

selects exactly $\binom{n}{k}$, since all its base-B coefficients are below B, including zero for $k > n$. The identity $(1+t)^n = \prod_{i:\,\mathrm{bit}_i(n)=1}(1+t^{2^i})$ over $\mathbb{F}_2$ makes odd binomial coefficients equivalent to bit inclusion. Bit disjointness of a, b is equivalent to inclusion of a in $a + b$: at the least shared bit there is no incoming carry and that bit is cleared. Finally $x = z + a, y = z + b$ with z, a, b pairwise bit-disjoint says exactly $z = x$ AND y.

10.3 Interpreter, lowering and five-input specialization

The universal program numbering is $e + 1 = 2^w(2D + 1)$. At $w = 0$ it halts. Otherwise the four base-2^w digits at positions $4p, \ldots, 4p + 3$ give the opcode, register and two targets at state p. Opcode one increments, opcode two decrements or takes its zero branch; every other opcode halts. Missing digits are zero, so decoding is total.

The interpreter stores a finite tape as 0 or $\mathrm{cons}(h, t) = 2^h(2t + 1)$. Repeated halving decodes a nonempty head and tail; the tail is smaller than the code. To update a register, save the finite reversed prefix, replace the entry and restore the prefix. Absent entries are zero. The reserved jump register remains zero. Copies restore their source; aliased addition snapshots its operand before overwriting it. Division maintains $x = qy + r$ and subtracts only when $r \geq y$; the interpreter's divisors are at least two. These invariants prove that one interpreter loop executes exactly one decoded instruction, with finite work. A simulated halt enters cleanup and clears all registers. An infinite simulation never enters cleanup. Thus its all-zero halt is equivalent to virtual halting, without requiring the virtual final tape itself to be zero.

The precision IR uses normalized signed pairs for integer arithmetic and pairs over the basis $1, \sqrt{2}$ for field arithmetic. Norm zero implies the field element is zero: after clearing denominators, a primitive solution of $a^2 = 2b^2$ would have both entries even. Every division is guarded by that exact test. The Jacobian point formulas satisfy the curve equation and the affine chord or tangent formulas after multiplying denominators. Equal abscissas split into doubling and opposite-point branches; zero ordinate doubles to the identity. The fixed projective point is the same affine P_0. These are polynomial identities, independent of the generator theorem.

The scalar loop halves its remaining count. The precision loop divides its remainder by four. The capped arithmetic computes exactly $\min(x, L + 1)$; zero bases, unit bases and zero exponents are separate. For bases at least two an exponent exceeding L already exceeds the cap. Thus the ideal IR terminates and agrees with the stated rational predicate on all five natural inputs, including every early rejection.

Each primitive lowering uses fresh auxiliary registers, snapshots overlapping sources, and preserves the live-register map at the end of each finite block. Saturating decrement gives natural

subtraction, and repeated addition gives multiplication. All variable divisions are guarded. Induction on completed loop iterations proves the same claim for loops. Acceptance enters cleanup; rejection enters a permanent self-loop. Hence the primitive machine halts exactly on acceptance, although the ideal decision program is total.

For $\mathrm{pair}(a, b) = (a + b)(a + b + 1)/2 + b$, unpairing maintains remainder = input $- s(s + 1)/2$ and subtracts $s + 1$ while $s <$ remainder. At exit $0 \leq$ remainder $\leq s$, so $b =$ remainder, $a = s - b$. Four repetitions put the five codes into registers zero through four. Fresh scratch lies above the old register range and is cleared; old work registers remain zero. Relocating every target by the prefix length therefore preserves the precision machine.

The polynomial specialization fixes the universal program input to this wrapper code, maps the old packed input to variable five, and maps every old variable $i \geq 2$ and private-block start by $i \mapsto i + 7$. The four pairing variables occupy five through eight, disjoint from the five inputs. The appended four squared equations force their unique Cantor values. Thus, for the exact saved polynomial C,

$$\forall \mathbf{c} \in \mathbb{N}^5, \qquad \exists \mathbf{z} \in \mathbb{N}^{1305806}\ C(\mathbf{c}, \mathbf{z}) = 0 \quad \Longleftrightarrow \quad \text{accepts_naturals}(\mathbf{c}) = \text{True}. \tag{C.4}$$

The natural compiler proof has no height, restricted-converse or rational-model premise. Its use of the height expression concerns only computation of that fixed integer expression.

10.4 Translation to point codes

The typing order is independent of the natural predicate. First, `member` forces a canonical code in D, and raw addition on two canonical elliptic inputs forces their canonical sum, including the origin and two-torsion cases. The fixed scalar graph gives $[m]Q$ on D. In `multiply`, the three root membership calls are forced conjuncts inside `height_certificate`. Thus the four scalar shifts are in D before they are passed, unchanged, to the 27 slope calls and the six test-group calls within each slope. This supplies the hypotheses of every raw addition in those groups. Only after globally typed multiplication is established do the four square products and three successive additions justify `natural`. Their outputs are canonical by the raw-addition postcondition, so no separate output membership assumption is needed. The signed-encoding and exceptional branches start their addition chains at this natural code, the literal origin, or the fixed signed generator. An inactive product branch is not assumed true. The fixed generator equations are conjuncts outside the internal products in all three templates that use them. These observations give the base cases for the source-DAG induction below. Apply (C.4) using the exact signed index model. Induction on the actual source DAG gives each node the code of its signed integer value. Natural guards constrain source variables, not intermediate sums or products. Raw addition receives canonical codes; globally typed multiplication justifies the natural guards. Fixed generator equations and scalar templates supply constants.

The power-value recipe has three signed code inputs, seventy-nine private natural witnesses, and an output code for the value of the power-template polynomial. That value is not independently required to vanish. Its saved root equality is $[17751579, 15]$. The precision parent has 54056 explicit variables and 15845 disjoint private blocks of seventy-nine variables. No parent variable node accesses a private coordinate. Replacing each block by its private power-value call therefore preserves the full existential assignment. Only the final parent root is equated to the identity.

At each primitive formula, equality is a difference, conjunction is a sum of squares, and disjunction is a product. A topological induction, including the globally conjoined fixed equations, proves equality of polynomial values under substitution, not merely equisatisfiability of an unrelated formula. This is the required link to the saved AST.

11 A variable-shift relation for ordinary values

We prove the ordinary-value deduction from the exact index model, the compiled precision relation, the height theorem and PC. These are supplied by Sections 3, 10, 4 and Theorem 5.2, respectively. Capital letters below denote five-coordinate point codes; their lower-case indices are ordinary integers. The external variable t is any rational number.

11.1 Approximation for every fixed shift

Define $\mathrm{Close}(N, \mathcal{K}, t)$ by choosing A, L, J, linking all five signed codes to precision inputs in the order (n, κ, a, l, j), and imposing $|t - r(a, j)| < \epsilon_l$ on those same coordinates. Put

$$B(n, \kappa) = H_6(S_{\mathrm{ht}})|\kappa F(n)|^{24}.$$

The precision contract gives

$$\mathrm{Close}(\iota(n), \iota(\kappa), t) \Longrightarrow |t - n| < e^{-B(n,\kappa)}, \tag{8}$$

$$\mathrm{Close}(\iota(n), \iota(\kappa), n) \quad \text{for every } n \notin \mathcal{C},\ \kappa \neq 0. \tag{9}$$

The first follows from $|t - n| < 2\epsilon_l < 4^{-B}/2 < e^{-B}$. For the second, positive multiples of P_0 approach the identity simultaneously in the compact group $E(\mathbf{R}) \times E(\mathbf{R})$. This follows, for instance, by partitioning the compact closure of the cyclic subgroup into arbitrarily small neighborhoods and subtracting two sufficiently close multiples. If the resulting positive differences were bounded, a fixed nonzero multiple would be the identity; infinite order excludes this. Choose l first to make the positive ϵ_l small enough. Since the derivative of $[n]$ at the identity is n [48, IV.2.3(a), p. 122], $r(a, na) \to n$ along positive multiples a approaching it. Choose such an a and put $j = na$. This works for every already fixed nonzero shift κ, regardless of its size.

11.2 The six local tests

For $N = \iota(n)$, $\mathcal{K} = \iota(\kappa)$, form the code for κn, choose $H = \iota(h)$, $U = \iota(s_0)$, and put $V_0 = H - \mathcal{K}$, $V_1 = U - \iota(\kappa n)$. Require $\kappa \neq 0$, $H, V_0 \neq O$. Define $w(Q) = 1/x(Q)$ for $Q \neq O$, $w(O) = 0$. For $t \notin \mathcal{C}$ use

$$D_1 = w(H), \qquad D_2 = w(U), \qquad D_3 = \sigma w(V_0), \qquad D_4 = \sigma w(V_1),$$
$$D_5 = z(U)/z(H) - t, \qquad D_6 = \sigma(z(V_1)/z(V_0)) - t.$$

Here σ is conjugation in K_0. Choose $\eta, q \in \mathbb{Q}$ and impose

$$\eta^2 = 1, \qquad R_0 = \eta 2730^6 F(t), \qquad a_0 = 1 - R_0 q^2 \neq 0,$$
$$R = R_0 a_0^2, \qquad C_j = D_j/F(t), \qquad W = \prod_{j=1}^{6} (1 + RC_j^2) \neq 0,$$
$$\Theta = \mathrm{Tr}_{K_0/\mathbb{Q}}(W^{-1}) \neq 0, \qquad B_{\mathrm{pole}} = R/\Theta^2, \qquad \Phi(B_{\mathrm{pole}}).$$

Call this relation $\mathrm{Test}(N, \mathcal{K}, t)$. All quotients are guarded polynomially. There is one common denominator $F(t)$, one product W, one trace and one pole instance.

Lemma 11.1 (Divisibility)**.** *If* $\mathrm{Test}(\iota(n), \iota(\kappa), t)$ *holds, then* $M_{S_{\mathrm{ht}}}(t) \mid \kappa F(n)$.

Proof. Let $p \mid M_{S_{\mathrm{ht}}}(t)$. Exactly one $c \in \mathcal{C}$ has $\delta = v_p(t-c) > 0$ odd. The contact differences are units, so $v_p(F(t)) = \delta$; the field is unramified at p. Put $b = v_p(a_0)$. If $q = 0$, then $b = 0$. Otherwise $\delta + 2v_p(q)$ is odd and nonzero, so $b = \min(0, \delta + 2v_p(q)) \le 0$. In either case $\alpha = v_p(R) = \delta + 2b$ is odd. At each place above p,

$$v(1 + RC_j^2) = \min(0, \alpha + 2(v(D_j) - \delta)).$$

Let $\lambda = \max_{v|p} v(W) \le 0$. Then $v_p(\Theta) \ge -\lambda \ge 0$. The pole input has odd valuation $\alpha - 2v_p(\Theta)$, which is positive by Lemma 5.1. Therefore $\alpha > 0$ and $\lambda > -\alpha/2$.

At a place attaining λ, each nonpositive factor valuation is at least λ. Consequently

$$v(D_j) \ge \delta - 3\alpha/4 = \delta/4 - 3b/2 > 0 \quad (1 \le j \le 6).$$

Integral valuations are at least one, at the same place for all tests. The first four put H, U in one formal group and V_0, V_1 in the conjugate formal group. At the identity we set both $z(O) = 0$ and $\log(O) = 0$; thus the cases $U = O$ and $V_1 = O$ obey the same ratio identities below. Their denominators are nonzero by the guards on H and V_0. For a nonzero point in either formal group,

$$\log(Q) = z(Q)(1 + e_Q), \qquad v(e_Q) \ge 1.$$

Here $p > 324$ and $v(p) = 1$. By [48, IV.5.5 and IV.6.4, pp. 129–132], the logarithm is an injective isometry on $v(z) \ge 1$. The integral invariant differential gives in relative degree $r-1$ the bound $(r-1)v(z) - v_p(r) \ge 1$ for every $r \ge 2$. Thus the guarded nonzero points H, V_0 have nonzero logarithms. The fifth test first makes $z(U)/z(H)$ integral and congruent to t; multiplication by the units $1+e_Q$ preserves that congruence. The relation $[s_0]H = [h]U$ then gives $\log(U)/\log(H) = s_0/h$. Likewise the sixth gives $(s_0 - \kappa n)/(h - \kappa) \equiv t$. Multiplying by the integer denominators and subtracting yields $v_p(\kappa(n-t)) \ge 1$. Since $t \equiv c$, we get $p \mid \kappa(n-c)$, hence $p \mid \kappa F(n)$. The squarefree product divides as asserted. No unit assumption on κ was used. □

Lemma 11.2 (One suitable shift)**.** *For every $n \in \mathbb{Z} \setminus \mathcal{C}$ there is $\kappa \ne 0$ with* $\mathrm{Test}(\iota(n), \iota(\kappa), n)$.

Proof. Choose the sign so $R_0 = \eta 2730^6 F(n) > 0$ and let S' be its finite prime support. Each sufficiently small local formal neighborhood has finite index in the compact local group [48, VII.6.3, p. 200]. A positive common multiple of the quotient orders gives κ with $[\kappa]P_0$ in every chosen neighborhood. Take

$$H = [2\kappa]P_0, \quad U = [2\kappa n]P_0, \quad V_0 = [\kappa]P_0, \quad V_1 = [\kappa n]P_0.$$

The formal parameter and the derivative of $[n]$ make each C_j integral above S' after sufficiently shrinking the neighborhoods. Infinite order preserves the nonzero guards.

Choose rational q integral at S' and arbitrarily close to $1/\sqrt{R_0}$ in $\mathbf{R}$, without equality. Denominators $R_0 m + 1$ with nearest integer numerators give such choices, adjusting by one if equality occurs. At S', a_0 is a unit and $W \equiv 1$, $\Theta \equiv 2$ modulo $p^{v_p(R_0)}$. Integral trace justifies this also at the ramified place two. The pole input has valuation at least four at two and at least six at the other primes of S_0. Outside S', its valuation is $2v_p(a_0) - 2v_p(\Theta)$ and is even, so its reduced denominator is a square. In both real embeddings, R tends to zero positively, $W \to 1$ and $\Theta \to 2$. All guards hold and $0 < B_{\mathrm{pole}} < 1/2$ for a sufficiently close choice. Theorem 5.2 supplies the pole witnesses. □

11.3 Exact ordinary value

Define $\mathrm{Val}(N,t)$ by membership of N and the disjunction

$$\bigvee_{c\in\mathcal{C}}(N=\iota(c)\wedge t=c)\quad\vee\quad\exists\mathcal{K}\,(\mathrm{Close}(N,\mathcal{K},t)\wedge\mathrm{Test}(N,\mathcal{K},t)).$$

The same five coordinates $\mathcal{K}$ are used in both calls. In an accepting main branch $n\notin\mathcal{C}$, $\kappa\neq0$, and Lemma 11.1 and the height estimate imply

$$h(t)\le H_6(S_{\mathrm{ht}})|\kappa F(n)|^{24}=B(n,\kappa).$$

For $t=a/b\neq n$ in reduced form, $|a-nb|\ge1$, hence

$$|t-n|\ge1/b\ge e^{-h(t)}\ge e^{-B(n,\kappa)},$$

contradicting (8). Thus $t=n$. Conversely choose κ by Lemma 11.2, then use (9) for that same shift. Private witness blocks are disjoint, so the two calls can be satisfied simultaneously. The six remaining integers have exactly their displayed branches.

Therefore $\mathrm{Val}(\iota(n),t)$ is exactly the graph $t=n$. Existentially quantifying N gives ordinary $\mathbb{Z}$ in $\mathbb{Q}$. S01 and S16 multiply the exception and main-branch polynomials and conjoin membership by a sum of squares. The literal substitutions in Section 2 and Appendix C identify this polynomial with the saved root P_*. This proves the zero-set part of Lemma 1.2.

12 The cutoff and polynomial size

The literal constants in S03 define the catalogue product H_{cat}, its maximum $H_{\max}$, $N_s=P_sH_{\mathrm{cat}}$ with $P_s=\prod_{p\le324}p$, the least formal multiple M_0, and $c_*=4M_0$. They take

$$R=\max(D,H_{\max},c_*,10^{12}),\quad A_{\mathrm{ht}}=2\cdot10^{13}\max(1,C_{\mathrm{cm}},Q_R^{17}),\quad T=(48A_{\mathrm{ht}})^{13}.$$

The pole clearing constant is $\ell_0=12P_s^{2916}H_{\mathrm{cat}}^{732}$. The symbol $F_0=324!$ is distinct from the six-contact polynomial $F(u)$.

Proposition 12.1 (Literal cutoff)**.** *The fixed prescription satisfies*

$$R=D,\qquad T=(96\cdot10^{13})^{13}D^{2210G_0^2D^3}<2^{2^{2^{9013}}}.$$

This is a statement about the prescribed integer, independently of its semantic role in excluding false multiplication triples.

Proof. We give the bounds of S25. Exact small calculations give $10^5<W<2^{512}$ and $2^{2000}<F_0<2^{2^{250}}$. For $q\ge10^5$, the unconditional progression estimate [41, Theorem 1.2, (1.10)–(1.11), Corollary 6.17 with $A=1$] applies for $x\ge\exp(0.03\sqrt{q}(\log q)^3)$, including the possible exceptional-zero case. At $X=2^{W^3}$ its error is at most $X/(160\log X)$ and the hypothesis follows from $0.03\sqrt{W}(\log W)^3<(\log2)W^3$. Hence $\vartheta(X;W,a)>X/(2W)$, giving more than $X/(2W^4)>22$ primes in each unit class. No prime at most 324 belongs to such a class. Thus

$$H_{\max}<X,\quad H_{\mathrm{cat}}\le X^{22W},\quad\log_2N_s<2^{2054}.$$

For every odd prime dividing N_s, take the actual reduction order of P_0 at each place above it, and let their product be m_0. All preceding positive multiples stay in the dyadic formal subgroup,

since P_0 starts there and the subgroup is closed under addition. A nonidentity point there has negative abscissa valuation, so its reciprocal is defined; the constructor accepts the identity directly. Thus every earlier search step is defined without a nontorsion premise. At the bad place $s \equiv 3 \pmod 7$, the point reduces to $(2, 6)$ on $y^2 = x^3$ and is nonsingular because $2y \not\equiv 0$. The other place $s \equiv 4 \pmod 7$ has good reduction. Take the actual order of the reduced point in the nonsingular reduction group, using [48, VII.2.1, p. 188]. At a good place the residue degree is at most two, so Hasse's bound [48, V.1.1, p. 138] gives order at most $(p+1)^2 < p^3$ for odd p. The singular smooth group over $\mathbb{F}_7$ has order seven. There are at most two places over each rational prime, so $m_0 \le N_s^6$. If $[N_s^4 m_0]P_0$ is the identity, the search has already succeeded. Otherwise its reciprocal coordinate is defined and the following estimates apply. At each odd place, $[m_0]P_0$ is formal, and multiplication by N_s^4 gives

$$v(1/x([N_s^4 m_0]P_0)) \ge 2 + 8v_p(N_s).$$

At two normalize $v_s(s) = 1$, so $v_s(2) = 2$. Initially $v_s(x) = -6$, $v_s(y) = -9$, and $v_s(z) = 3 > 2$. The log isometry therefore gives $v_s(z([2]Q)) = v_s(z(Q)) + 2$ on this subgroup, and hence

$$v_s(1/x([N_s^4 m_0]P_0)) \ge 6 + 16v_2(N_s).$$

Writing this reciprocal as $a + bs$, extraction of a and b uses division by 2 and $2s$ and loses at most three units of v_s. Thus both rational coefficients lie in $N_s^4 \mathbb{Z}_{(N_s)}$, where denominators in $\mathbb{Z}_{(N_s)}$ are prime to N_s. This proves the required divisibility and guard. Therefore the literal search succeeds by $N_s^4 m_0$ and $M_0 \le N_s^{10}$. This uses actual finite orders, not numerical upper bounds incorrectly treated as multiples of those orders.

Consequently $c_* < 2^{2^{2059}}$, whereas $2^{2^{8008}} < D < 2^{2^{9010}}$, so $R = D$. Next $C_{\mathrm{cm}} < 2^{50000D^2}$ and

$$\log_2 Q_D^{17} = 170 G_0^2 D^3 \log_2 D > 50000 D^2.$$

The maximum is therefore Q_D^{17}, giving the identity for T. Finally $G_0 < 2^{37}$, $E_0 < D^3$, $\log_2 Q_D < D^5$, $\log_2 A_{\mathrm{ht}} < D^7$, and $\log_2 T < D^8 < 2^{2^{9013}}$. □

12.1 Quantitative syntax bounds

Use coefficient ℓ^1 norm for integer polynomials and the weighted pair norm $\|a+bs\| = \|a\|_1 + 2\|b\|_1$. They are submultiplicative. The division-polynomial recurrences give

$$\deg f_n \le (n+1)^3, \qquad \|f_n\| \le 2^{100(n+1)^3}.$$

For the even recurrence the exponent slack is $4m^3 - 24m - 27 \ge 9$ for $m \ge 3$; for the odd recurrence it is at least ten for $m \ge 2$. This absorbs the fixed factors and addition.

The following facts have been checked for the actual saved expression: the absolute values of all literal fixed constants and all occurring division-polynomial indices are at most T; the specified syntax-height traversal has root height at most 256; each model recipe has at most 2^{20} constraints and each composition recipe has at most 2^{20} terms. The check accounts for all 23 templates, seven recipes, private witness blocks, 173 scalar constant nodes and 50 finite constructors. Its exact artifact identity and independent replay are recorded in the source map. Together with the constructor bounds below, this discharges the quantitative part of B. The all-input arithmetic interpretation of the same constituents is the separate substitution induction of Section 10.

Put $B_* = T + 1$. Coordinates of $\iota(n)$ for $|n| \le T$ have absolute numerator and positive denominator at most 2^{K_*}, with $K_* = 2^{30}(T+1)^3$. Evaluate the division polynomials at P_0 and use the fixed coordinate formulas; at most eight quadratic-field operations are needed, each taking

an available height bound H to at most $3H^4$. There are at most $8(T+1)^3$ guard factors. The resulting bounds for each finite constructor are

$$\deg \le B_*^4, \qquad \log_2 \max(1, \|\cdot\|_1) \le B_*^8.$$

Give variables syntax height zero, constants height one, and add one for each arithmetic operation or finite constructor. A template call adds one plus the callee's intrinsic height and the maximum argument height. The substitution inequalities

$$\deg P(Q) \le \deg P \max(1, \max_i \deg Q_i), \qquad \|P(Q)\|_1 \le \|P\|_1 \max(1, \max_i \|Q_i\|_1)^{\deg P}$$

give degree and logarithmic norm at most B_*^{8h} at positive height h, with the bounded sums in the actual recipes treated as in S26. This general call rule applies to polynomial arguments. In each outer composition recipe the arguments are coordinate variables, so the call is only a renaming, even when coordinates repeat. For any such map σ,

$$\deg P(Y_{\sigma(i)}) \le \deg P, \qquad \|P(Y_{\sigma(i)})\|_1 = \sum_\beta \left| \sum_{\alpha \mapsto \beta} c_\alpha \right| \le \sum_\alpha |c_\alpha| = \|P\|_1.$$

Thus these calls retain the callee's majorant height before charging the sum-of-squares or product connective. The seven recipe heights are $228, 233, 235, 251, 252, 254, 256$. A rule charging an extra unit for each of these mere renamings would describe a looser syntax-depth bound, not this proved degree-and-norm majorant. For the model wrappers use the safe height bound four for the generator equations and root differences; the callee contribution dominates this bound. The literal height-two claim for those equations in S26 is not used. This height is a majorant for the compact constructor syntax, not for an expanded monomial tree. The height-256 bound yields

$$\deg P_* \le (T+1)^{2048}, \qquad |[X^\alpha]P_*| \le 2^{(T+1)^{2048}}.$$

Since T is an integer, $T+1 \le U = 2^{2^{2^{9013}}}$. Hence

$$(T+1)^{2048} \le U^{2048} = 2^{2^{2^{9013}+11}} < 2^{2^{2^{9014}}}.$$

As $9014 < 65536 = 2 \uparrow\uparrow 4$, the degree is less than $2 \uparrow\uparrow 7$ and the coefficient magnitude less than $2 \uparrow\uparrow 8$. The root recipe declares $292\,890\,231\,904$ witnesses. Together with the preceding estimates, these complete the bounds for the same P_* in Theorem 1.1.

13 Consequences of the integer definition

The proof of Theorem 1.1 combines the following results.

Result	Mathematical inputs	Proof location
Index group	Two-isogeny descent, reductions, finite height search	Section 3
Six-contact bound	Quaternionic model, tame pullback, compatible-isogeny height theorem, theta comparison	Section 4
Pole completeness	Simultaneous descent, restricted converse, forward rank-one theorem	Section 5
Multiplication	Index group, height bound, selected primes, pole completeness, finite guard	Section 6
Precision relation	Decision program, lowering, power and packed histories	Section 10
Ordinary value	Shared-shift Close and Test, height and rational separation	Section 11
Bounded P_*	Literal substitutions and quantitative syntax	Sections 2, 12; Appendix C

13.1 Three corollaries

We apply the exact equality between the rational zero projection of P_* and ordinary $\mathbb{Z}$ proved in Theorem 1.1.

Corollary 13.1 (Negation of Mazur's finite-component conjecture)**.** *Mazur's finite-connected-components conjecture for the real-topological closure of rational points, Conjecture 3 of [70, 76], is false. Its stronger Conjecture 2 is false as well.*

Proof. We recall Mazur's projection argument, originating in [70] and repeated in [76, p. 166]. Conjecture 3 in [76, p. 165] asserts that the closure of the rational points of any variety over $\mathbb{Q}$ in its real locus has finitely many connected components. In that source a variety means a reduced scheme of finite type; irreducibility is not required. Put $r = 292\,890\,231\,904$ and

$$V = \left(\{P_*(t, w_1, \ldots, w_r) = 0\} \subset \mathbf{A}^{r+1}_{\mathbb{Q}}\right)_{\mathrm{red}}, \qquad C = \overline{V(\mathbb{Q})} \subset V(\mathbf{R}).$$

Reduction changes neither the rational nor the real points. Let $\pi : V(\mathbf{R}) \to \mathbf{R}$ be projection to t. Theorem 1.1 gives $\pi(V(\mathbb{Q})) = \mathbb{Z}$. Since $\mathbb{Z}$ is closed in $\mathbf{R}$ and π is continuous, $C \subset \pi^{-1}(\mathbb{Z})$, so $\pi(C) \subset \mathbb{Z}$. Every integer already occurs on $V(\mathbb{Q})$, hence $\pi(C) = \mathbb{Z}$. The image of any connected component of C is a connected subset of $\mathbb{Z}$, therefore a singleton. Points projecting to distinct integers belong to distinct components. Thus C has infinitely many connected components, contradicting the stated conjecture. Conjecture 2 in the same source would give the homotopy type of a finite simplicial complex and hence only finitely many connected components, so it implies Conjecture 3 and is contradicted too. □

Corollary 13.2 (A universal rational Diophantine polynomial)**.** *There exist $m \geq 1$ and a fixed polynomial $U \in \mathbb{Z}[e, n, x_1, \ldots, x_m]$ such that, for every $e, n \in \mathbb{N}$,*

$$\exists \mathbf{x} \in \mathbb{Q}^m \; U(e, n, \mathbf{x}) = 0 \quad \Longleftrightarrow \quad \textit{machine } e \textit{ halts on input } n.$$

Here e and n are ordinary integer parameters, with $\mathbb{N}$ embedded in $\mathbb{Q}$, and are not elliptic-curve coordinates. In particular, every computably enumerable subset of $\mathbb{N}$ is represented by a suitable fixed value of e in this one polynomial.

Proof. Fix an effective enumeration of machines. The Davis–Putnam–Robinson–Matiyasevich theorem, [77, Basic Theorem, p. 3], applied to the universal halting relation gives a fixed $D \in \mathbb{Z}[e, n, z_1, \ldots, z_s]$, with $s \geq 1$, such that, for $e, n \in \mathbb{N}$,

$$\text{machine } e \text{ halts on } n \quad \Longleftrightarrow \quad \exists \mathbf{z} \in \mathbb{Z}^s \; D(e, n, \mathbf{z}) = 0.$$

To use integer witnesses in this statement, replace each nonnegative witness by a sum of four integer squares. In the positive-integer convention of the cited source, apply the theorem to the predicate that machine $a - 1$ halts on $b - 1$, substitute $a = e + 1$, $b = n + 1$, and use one plus that sum for each witness; see also [77, p. 2]. A dummy witness ensures $s \geq 1$ if needed. For each z_i take a separate block $\mathbf{w}_i$ of r rational variables, where r is the witness count of P_*. Define

$$U(e, n, \mathbf{z}, \mathbf{w}_1, \ldots, \mathbf{w}_s) = D(e, n, \mathbf{z})^2 + \sum_{i=1}^{s} P_*(z_i, \mathbf{w}_i)^2.$$

A rational zero makes every summand zero. Theorem 1.1 then forces every z_i to be an integer and yields the displayed halting condition. Conversely, integer witnesses for D have separate rational witness blocks for P_*, giving a zero of U. All blocks are private and the polynomial is fixed independently of e, n. No variable, degree or coefficient bound proved for P_* is asserted for this new polynomial U. □

OpenAI already obtained Turing degree $0'$ for this set in [78, Corollary 7.1, pp. 56–57]; that paper explicitly does not assert many-one completeness. The stronger direction below is $0' \leq_m \mathrm{HTP}(\mathbb{Q})$, obtained from the integer definition through the universal polynomial. The reverse many-one reduction is the standard enumeration of rational solutions.

Corollary 13.3 (The many-one degree of $\mathrm{HTP}(\mathbb{Q})$)**.**

$$\mathrm{HTP}(\mathbb{Q}) \equiv_m 0', \qquad 0' = \{e \in \mathbb{N} : \textit{machine } e \textit{ halts on input } e\}.$$

Here $\mathrm{HTP}(\mathbb{Q})$ *is the set of codes of integer-coefficient polynomials, of arbitrary finite arity, having a rational zero, under a fixed effective coding.*

Proof. Given e, specialize the fixed polynomial of Corollary 13.2 at its ordinary integer parameters to obtain $U(e, e, \mathbf{x})$. Computing its coefficient list and code is effective, and

$$e \in 0' \quad \Longleftrightarrow \quad \ulcorner U(e, e, \mathbf{x}) \urcorner \in \mathrm{HTP}(\mathbb{Q}).$$

This proves $0' \leq_m \mathrm{HTP}(\mathbb{Q})$.

Conversely, from the code of a polynomial f construct a machine that ignores its input and enumerates all rational tuples of the appropriate arity. It evaluates f using exact rational arithmetic and halts as soon as it finds a zero. Its index $j(f)$ is computable from the finite code of f, and

$$f \in \mathrm{HTP}(\mathbb{Q}) \quad \Longleftrightarrow \quad j(f) \in 0',$$

where the left side uses the code of f. Thus $\mathrm{HTP}(\mathbb{Q}) \leq_m 0'$. Invalid polynomial codes, if present in the coding convention, are sent to a fixed nonhalting index. Using rational-coefficient polynomial codes gives the same result, since clearing their finitely many denominators is effective and preserves rational zeros. The two reductions prove the assertion. □

More generally, an integer Diophantine problem translates to a rational one by adjoining P_* for every integer witness, with private witness blocks. Natural witnesses can first be expressed as sums of four integer squares. Bounds for the resulting polynomial require their own composition calculation; the bounds for P_* do not automatically transfer.

For example, a machine that enumerates ZFC proofs and halts upon finding a contradiction has a rational Diophantine halting condition by Corollary 13.2. This equivalence does not assert that a contradiction exists, and no additional explicit polynomial is specified here.

14 The base residual compiler

This section records the base residual compiler used as a construction stage for the quartic in Section 15. Its degree-eight sum-of-squares encoding fixes the arithmetic interfaces and conventions; it is not a third final polynomial specification. The construction uses a joint comparison of four elliptic indices, a Lucas equation, and the five-contact height estimate proved below. The archived counts describe this base compiler before its reciprocal revision and quadratic lift.

Proposition 14.1 (Finite specification of the base compiler)**.** *The prescription below specifies a polynomial*

$$F_8 \in \mathbb{Z}[X, V_1, \dots, V_{15671}]$$

of degree exactly eight, with exactly 129497 *nonzero monomials and maximum absolute coefficient*

$$685783949153203534233601 < 2^{80}.$$

These statements are finite identities about the displayed construction and its integer coefficient table.

Theorem 14.2 (Arithmetic interpretation)**.** *For every $x \in \mathbb{Q}$,*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists (v_1, \dots, v_{15671}) \in \mathbb{Q}^{15671}, \qquad F_8(x, v_1, \dots, v_{15671}) = 0.$$

The required height estimate is proved below; Theorem 4.1 uses a different contact set. The seven-variable pole lemma below is proved from Theorem 5.3; it is not inferred from Theorem 5.2, which has different variables and a restricted completeness domain. The finite coefficient calculation does not establish the height bound. The supplied unpublished Freitag archive is the provenance of this construction and of the original coefficient table. It is not cited here as an arithmetic theorem. The published inputs already used in Sections 3 and 6 remain available, together with the quaternion trace theorem specified below.

14.1 The height input and the exact pole predicate

Put

$$S_* = \{p : p \le 76176\}, \qquad \mathcal{D} = \{0, 1, 2401, -343/32\},$$

$$H_* = 2(4608^{165888} + 6^{100}) \left(\prod_{p \le 76176} p \right)^{24} + 128.$$

For $a \in \mathbb{Q} \setminus \mathcal{D}$, let $M_5(a)$ be the product of the primes outside S_* at which the point $(a : 1)$ has positive odd contact order with one of $\infty, 0, 1, 2401, -343/32$. Explicitly, choose local homogeneous coordinates $(u : v)$ and subtract $\min(v_p(u), v_p(v))$ from the valuations of v and of $u - \delta v$ for $\delta \in \mathcal{D}$.

Theorem 14.3 (The five-contact height estimate)**.** *For every $a \in \mathbb{Q} \setminus \mathcal{D}$,*

$$h(a) \le H_* M_5(a)^{24}.$$

The displayed constant satisfies $\log_2 H_* < 4783227$*.*

Proof. Write $s = a$, $M(s) = M_5(a)$, $H = H_*$, $B_* = \prod_{p \le 76176} p$, and $\mathcal{C} = \{\infty, 0, 1, 2401, -343/32\}$. We prove the geometric comparison, the non-CM estimate through the modular level, and the CM estimate over the same fiber field. All heights use the original stable Faltings normalization.

Recovering the contact line from published quotients Let $B/\mathbb{Q}$ be the indefinite quaternion algebra of discriminant 210, $\mathcal{O}$ a maximal order, X its maximal-level Shimura curve, and W its Atkin–Lehner group. The standard genus formula gives

$$g(X) = 1 + \frac{\varphi(210)}{12} = 5,$$

because the elliptic-point terms vanish: 5 splits in $\mathbb{Q}(i)$ and 7 splits in $\mathbb{Q}(\sqrt{-3})$, so neither quadratic field embeds in B. The group W is $(\mathbb{Z}/2\mathbb{Z})^4$ with involutions w_m indexed by square classes of divisors of 210.

The published models of [84, Theorem 4.2 and Table 2, pp. 938–939] have the following coefficient rows (a, b, c, d, e) for $y^2 = ax^4 + bx^3 + cx^2 + dx + e$:

m	a	b	c	d	e
30	−43	−686	−2915	−1372	−172
42	−43	600	−986	−600	−43
70	−43	256	−130	−768	−387
105	−43	2	505	12	−1548

The cited theorem identifies each with $X/\langle w_m \rangle$. The genus formula and the Atkin–Lehner group law used here also appear in the same published paper, Proposition 5.2 and the paragraph following Proposition 5.3, pp. 940–941.

Here is a deduction of all the required branch information, rather than an additional assumption about it. Each of these four genus-one quotients has eight fixed points upstairs, by Riemann–Hurwitz. A stabilizer for the W action on a characteristic-zero smooth curve is cyclic, and therefore has order at most two. Each nonempty fixed set of an involution is a union of W-orbits of size eight. Riemann–Hurwitz for $X \to X/W$ now gives

$$8 = 16(2g(X/W) - 2) + \sum_{1 \ne w \in W} \#\operatorname{Fix}(w).$$

The four known fixed sets already contribute 32. Thus $g(X/W) = 0$, the total is 40, and there is exactly one further orbit of eight fixed points. Inertia generators around the branch points have product one. The four classes $30, 42, 70, 105$ are independent in W, and their product is the class 210. Hence there are exactly five branch points, each of index two, with distinct labels $30, 42, 70, 105, 210$. Since the involutions are defined over $\mathbb{Q}$ and each label occurs only once, each branch point is rational. Three of them trivialize the genus-zero quotient.

Normalize $b_{210} = \infty$, $b_{105} = 0$, $b_{70} = 1$, and write $b_{42} = \lambda$, $b_{30} = \mu$. On a genus-one quotient the residual group $(\mathbb{Z}/2)^3$ has translation subgroup of order four, necessarily the full two-torsion group after choosing an origin. Its quotient has the same Jacobian j-invariant: it is multiplication

by two over the algebraic closure. The remaining double cover of X/W branches at the four points other than b_m. Therefore the four quartic invariants determine the four Legendre invariants of these four-point subsets.

For a quartic put

$$I = 12ae - 3bd + c^2, \quad J_q = 72ace + 9bcd - 27ad^2 - 27b^2e - 2c^3, \quad j = \frac{6912I^3}{4I^3 - J_q^2}.$$

The exact computation gives the Legendre parameters $1/2401, 32/375, 1/225, 5/1029$, in row order. Enumerating the six cross-ratio values of each of the first two parameters and testing the last two invariants leaves exactly

$$\lambda = 2401, \qquad \mu = -343/32.$$

The script `check_height.py` reproduces this rational computation. Every nonzero determinant of the five primitive contact forms has prime support in $\{2, 3, 5, 7\}$.

Integral geometry and the degree 576 field Choose $\mu_0 \in \mathcal{O}$ with $\mu_0^2 = -210$. The embedding criterion allows this choice in a maximal order; maximal orders are conjugate here. Use the maximal moduli stack $\mathcal{X}$ and the entire principal-level three scheme Y, including its geometric components. Yuan's integral moduli theorem [85, Theorem 1.1 and Sections 2.1–2.2] makes $\mathcal{X}$ regular, proper and semistable over $\mathbb{Z}$, with a universal abelian surface, and smooth away from 210. Its generic stabilizer is precisely $\{\pm 1\}$. Principal level three is fine and gives a universal full three-torsion marking.

Work first over $R_0 = \mathbb{Z}[1/210]$. The effective level-change group has order 36, prime to every residue characteristic of R_0. Its generic action is free. For a finite tame group acting on a smooth relative curve, the fixed locus of an element is smooth locally along its fixed points; an isolated special fixed point consequently lifts. Thus no new fixed point occurs here, and $Y_{R_0} \to X_{R_0}$ is finite etale. The W action extends to the smooth stable model of X. Its stabilizers remain cyclic, of order at most two. Distinct involution fixed loci cannot intersect; each fixed locus is etale over the base. Consequently X_{R_0}/W is a smooth genus-zero curve with five disjoint branch sections of index two. The three rational sections already chosen identify it with $\mathbb{P}^1_{R_0}$. Their values are the five elements of $\mathcal{C}$.

Set $K = \widehat{\mathcal{O}}^\times$, $U = U(3)$, $Z = \mathbb{A}_f^\times$, and $N = N_{B(\mathbb{A}_f)^\times}(\widehat{\mathcal{O}})$. Normalization preserves $3\widehat{\mathcal{O}}$, so U is normal in N. The canonical adelic action is defined over the reflex field $\mathbb{Q}$ [86, 2.2.5 and 2.3.1]; this refers to the entire canonical model, not an individually selected complex component. Every central finite idele is a rational scalar times a unit idele. At three its scalar residue is 1 or -1, which a rational sign removes. Thus $Z \subset \mathbb{Q}^\times U$ acts trivially on Y.

At three the quaternion algebra is ramified. Its maximal order modulo three has 81 elements, radical of size 9, and hence 72 units. It follows that

$$1 \longrightarrow (\mathcal{O}/3\mathcal{O})^\times/\{\pm 1\} \longrightarrow \Gamma = N/(ZU) \longrightarrow W \longrightarrow 1, \qquad |\Gamma| = 36 \cdot 16 = 576.$$

Above an unbranched point the actions of W and of the effective level-change kernel are free. The fiber therefore is a torsor of size 576 for this constant group. The Galois permutations commute with its regular action, so belong to its regular centralizer, also of order 576. The normal splitting field L consequently satisfies $[L : \mathbb{Q}] \mid 576$: the Galois group is a subgroup of that regular centralizer. In particular $[L : \mathbb{Q}] \leq 576$.

All fiber surfaces and their three-torsion markings are defined over L. For two such surfaces, a homomorphism annihilating the three-torsion factors through [3]. Hence reduction embeds

$$\operatorname{Hom}(A, B)/3\operatorname{Hom}(A, B) \hookrightarrow \operatorname{Hom}(A[3], B[3]).$$

The finite Galois image on the integral Hom lattice is trivial modulo three. The principal congruence subgroup modulo three in an integral general linear group is torsion-free, so this image is trivial. Thus every geometric homomorphism between fiber surfaces is defined over L.

At a prime outside 210, tame local coordinates for the quotient are $w = z^2$ up to a unit at a branch section, and etale elsewhere. A base point with even contact gives no ramification after an unramified extension. The universal abelian scheme extends at the resulting integral point of Y_{R_0}. Therefore

$$\operatorname{rad}(D_L N_{\mathrm{bad}}) \mid B_* M(s). \tag{R}$$

This assertion includes contact with infinity. It is not a statement obtained merely from the numerical prime support of pairwise determinants.

The normalizer correspondences give isogenies between the conjugate surfaces. In the non-CM case the geometric endomorphism algebra is B; Skolem–Noether adjusts a conjugate isogeny to commute with its chosen quadratic subfield. All such maps lie over L by the Hom argument. These facts establish the usual GL_2-type and compatibility hypotheses; they do *not* establish the separate bounded descent to a GL_2-type variety over $\mathbb{Q}$.

The metric comparison, with its exact additive budget Put $t = 32s$, $(a_1, a_2, a_3, a_4) = (0, 32, 76832, -343)$ and $P(t) = \prod_i (t - a_i)$. The sections

$$q_0 = (dt)^2/P(t), \qquad q_1 = tq_0$$

are holomorphic bicanonical sections on X, with no common zero. At a finite branch point use $t - a_i = z^2$; at infinity use $t = z^{-2}$. At infinity q_0 vanishes to order two and q_1 does not vanish.

For each prime let $\delta_p = \max_{i<j} v_p(a_i - a_j)$, so

$$J = \prod_p p^{\delta_p} = 2^{10}3^2 5^3 7^4 = 2765952000.$$

Both $J^2 q_i$ extend across the maximal regular stack. Indeed, at the generic point of a vertical component semistability makes p a uniformizer, and the relative differential satisfies $v(du) \geq v(u)$. Writing $r_i = v(t - a_i)$ gives $v(dt) \geq \max r_i$. If $v(t) < 0$, all r_i equal $v(t)$ and both sections have nonnegative valuation. Otherwise all $r_i \geq 0$ and at most one exceeds δ_p. In either case $2\max r_i - \sum r_i \geq -2\delta_p$. This proves the required vertical integrality; regularity handles codimension two.

For the archimedean estimate write $r_i = |t - a_i|$, $U_0 = 1 + \sum_i r_i$ and

$$ds^2_\rho = \frac{U_0}{\prod_i r_i}|dt|^2, \qquad C_{\mathrm{met}} = 16(12 \cdot 76832 + 9)^3 = 12540153541944538512.$$

Its pullback is smooth and positive at every branch point and infinity. The curvature formula, $\Delta U_0 = \sum_i r_i^{-1}$ and $|\nabla U_0|^2 \leq 16$ give

$$-K_\rho \geq \frac{\sum_i \prod_{j \neq i} r_j}{2U_0^3}.$$

When $|t| \leq 2 \cdot 76832 + 2$, at most one $r_i < 1/2$, and $U_0 \leq 12 \cdot 76832 + 9$. The displayed bound is at least C_{met}^{-1}. Outside this disk, $|t|/2 \leq r_i \leq 3|t|/2$ and $U_0 \leq 7|t|$, giving the stronger bound $1/1372$. The maximum principle against the curvature -1 metric implies $ds^2_\rho / C_{\mathrm{met}} \leq ds^2_{\mathrm{hyp}}$. Yuan's Petersson norm is $\|d\tau\| = 2\operatorname{Im}\tau$. Consequently

$$\|q_0\|_{\mathrm{Pet}} \leq 4C_{\mathrm{met}}/U_0, \qquad \|q_1\|_{\mathrm{Pet}} \leq 4C_{\mathrm{met}}|t|/U_0 \leq 4C_{\mathrm{met}}.$$

Yuan's isometry [85, Theorem 1.1 and metric definitions, pp. 2–3] identifies the square of the Hodge bundle with $210\omega_{\mathcal{X}}^{\otimes 2}$ with the induced metric. The Hodge height in his normalization equals the original stable Faltings height plus $\log \pi$ in dimension two. Thus

$$h_{\bar{\omega}^{\otimes 2}}(y) = 2h_F(A_y) + 2\log \pi + \log 210.$$

The sign of $\log 210$ is positive: the sublattice $210\omega^{\otimes 2}$ has height $h_{\bar{\omega}^{\otimes 2}} - \log 210$. Evaluate on an extension where the point extends to the proper maximal stack; its universal abelian scheme computes stable height. The integral sections and the supremum estimate yield

$$h(s) \le 2h_F(A_y) + 2\log \pi + \log 210 + 2\log J + \log(4C_{\mathrm{met}}) + \log 32.$$

The integer comparison $210J^2(4C_{\mathrm{met}})32 \cdot 16 < 2^{160}$, together with $\pi < 4$ and $\log 2 < 3/4$, bounds the additive term by $120 < 128$. Therefore

$$h(s) \le 2\max(0, h_F(A_y)) + 128. \tag{G}$$

A level bound for the non-CM fibers This argument allows the scalar character order and coefficient field to vary. It does not assert bounded same-field descent to a rational GL_2-type variety. Let s be a rational point outside the five contacts, let $L/\mathbb{Q}$ be the normal fiber field constructed above, and let A/L be a non-CM fiber surface. Thus $[L : \mathbb{Q}] \mid 576$, all geometric homomorphisms between conjugate fibers are defined over L, and $\mathrm{End}^0(A_{\overline{\mathbb{Q}}}) = B$, the quaternion division algebra of discriminant 210. Put

$$R = B_* M(s), \qquad T = \{p : p \mid R\}.$$

The geometric argument gives that L is unramified outside T and A has good reduction outside the primes above 210. All conjugate QM structures lie in the same normalizer orbit.

Good reduction after full torsion. First a QM surface with the indicated division action has potentially good reduction at every finite place. Indeed, pass to an extension with semistable reduction, using [31, Theorem 3.4, p. 406]. Every endomorphism extends uniquely to the Néron model by its mapping property. On the identity component of the geometric special fiber it preserves the maximal torus $\mathcal{T}$: this torus is the maximal connected affine subgroup, and every homomorphism from a torus to an abelian variety is zero. If its rank t is positive, restriction gives a unital homomorphism

$$B \longrightarrow \mathrm{End}^0(\mathcal{T}) = M_t(\mathbb{Q}), \qquad 1 \le t \le 2.$$

The kernel is a two-sided ideal of the simple algebra B, and the map is unital, so it is injective. For $t = 1$ dimensions prohibit this; for $t = 2$ it would be an isomorphism from a division algebra to $M_2(\mathbb{Q})$, also impossible. Hence $t = 0$. The semistable identity component is abelian, which is good reduction. These Néron-model and reduction conventions are those of [31, §3, pp. 405–406].

Set $E = L(A[11])$. We next justify that $E/\mathbb{Q}$ is Galois. For each σ, the normalizer relation supplies a compatible isogeny ${}^{\sigma}A \to A$ of degree prime to 11. Here is the prime-to-11 detail. In the analytic description $A_z = \mathbb{C}^2/(\mathcal{O}(z,1)^t)$ of the published 2021 edition of Voight [89, equations (43.1.2)–(43.1.3)], a positive normalizer element $b = \left(\begin{smallmatrix} a & b' \\ c & d \end{smallmatrix}\right)$ gives the scalar compatible map $(cz+d)I : A_{bz} \to A_z$. When $\mathcal{O}b \subset \mathcal{O}$, its degree is $[\mathcal{O} : \mathcal{O}b] = \mathrm{Nrd}(b)^2$. At 11 the algebra is split and the maximal-order normalizer is $\mathbb{Q}_{11}^* \mathrm{GL}_2(\mathbb{Z}_{11})$. Rescale b rationally to make it an 11-adic unit, then clear the remaining denominators by an integer prime to 11. This produces the claimed isogeny. Changes of level marking use isomorphisms. All these geometric homomorphisms are L-defined by

construction. Consequently the 11-torsion representations of A and its conjugates over L have the same kernel. Thus every conjugate of E equals E.

The faithful action on $A[11]$ now gives

$$[E:\mathbb{Q}] \le 576|\operatorname{GL}_4(\mathbb{F}_{11})| < 2^{10}11^{16} < 2^{66}. \tag{LR1}$$

For residue characteristic different from 11, full rational 11-torsion gives semistable reduction over E by [31, Theorem 3.5, p. 406]. The preceding torus argument makes this good reduction. In particular it applies at each of $2, 3, 5, 7$.

The scalar lift with controlled support. Choose B-compatible isogenies $\mu_\sigma : {}^\sigma A \to A$ over L, with $\mu_1 = 1$, and inflate this finite choice to $G_\mathbb{Q}$. Their discrepancies $c(\sigma,\tau)$ commute with B, so lie in $\mathbb{Q}^*$. Put $a_\sigma = (\deg \mu_\sigma)^{-1/4} > 0$ and $\nu_\sigma = a_\sigma \mu_\sigma$, with trivial Galois action on the added algebraic coefficients. Since a scalar c has degree c^4, the discrepancies of the ν_σ form a sign cocycle $\varepsilon \in Z^2(G_\mathbb{Q}, \{\pm 1\})$, inflated from $\operatorname{Gal}(L/\mathbb{Q})$.

The sign class is unramified, hence zero, locally outside T. At each odd prime where its Brauer invariant is $1/2$, choose an odd Dirichlet character modulo p; at 2 use the odd character modulo 4 if required. Their product χ is tame at every odd prime, has local conductor at most 4 at 2, and has order

$$m \le 2 \prod_{\substack{p \in T \\ p \ne 2}} (p-1) \le 2R.$$

For constant coefficient groups, the boundary in $1 \to \{\pm 1\} \to \mu_\infty \xrightarrow{2} \mu_\infty \to 1$ is the obstruction to a character square root. Local reciprocity identifies this obstruction with the value at -1: the cyclic torsion factor has precisely this obstruction, while the valuation and principal-unit factors admit character square roots. Thus the boundary of χ equals $[\varepsilon]$ at every finite place. Brauer reciprocity and the global-to-local injection give equality globally. A pointwise square root, corrected by a sign cochain, gives

$$b : G_\mathbb{Q} \longrightarrow \mu_{2m}, \qquad b^2 = \chi, \qquad \delta b = \varepsilon. \tag{LR2}$$

Only this *sign* cochain is asserted to take root-of-unity values; the full scalar splitting also involves the a_σ.

On G_L, b is a conjugation-invariant character. Outside T its inertia characters are quadratic. Only finitely many are nontrivial; twist by the quadratic character of the product of those odd primes to kill them. At each odd prime there is a unique nontrivial quadratic inertia character. This changes no square or coboundary in (LR2), and introduces possible ramification only at $2 \in T$. The kernel field L' of $b|G_L$ is normal over $\mathbb{Q}$ and

$$[L':\mathbb{Q}] \le 2m[L:\mathbb{Q}] \le 2304R. \tag{LR3}$$

The compatible modular representations. Choose a number field containing all normalization scalars, the values of b, and a splitting field of B. For each coefficient prime ℓ and an embedding into $\overline{\mathbb{Q}}_\ell$, set

$$\rho_\ell(\sigma) = b(\sigma)^{-1} a_\sigma \mu_\sigma \circ \sigma.$$

The cocycle identity makes this a representation commuting with B; after splitting B, matrix units give $\rho_\ell = r_\ell \oplus r_\ell$ with $\dim r_\ell = 2$. On $G_{L'}$, ρ_ℓ is the Tate action of A. Faltings' semisimplicity and endomorphism theorems and $\operatorname{End}^0(A) = B$ show that the multiplicity representation is absolutely irreducible. Proper-smooth comparison and finite descent, as in Lemma H.1, make r_ℓ de Rham with Tate Hodge–Tate weights $\{0, -1\}$; Berger's monodromy theorem makes it potentially semistable.

It is odd. On Betti homology complex conjugation followed by the holomorphic isogeny μ_c exchanges the two Hodge summands, each a simple $B\otimes\mathbb{C}$-module. Its action on the multiplicity space is therefore nonscalar. After the scalar correction its square is one, so its eigenvalues are $1,-1$. The representation is unramified outside $T\cup\{\ell\}$, by (LR2) and the reduction and field properties. Pan's regular two-dimensional modularity theorem applies at $\ell=5$. We use its published formulation in [90, Section 7, p. 624], with the definition of a geometric representation on pp. 611–612, and the detailed author proof in [23, Theorem 1.0.4]. It requires no additional residual-image restriction at this prime and identifies r_5 with a cuspidal modular representation up to a character twist. The published formulation is used as stated; no byte or text identity between Pan's author version and the journal version is asserted.

The twist can be taken finite and the weight is two, without invoking a separate local-algebraicity theorem for characters. The twisting character is a de Rham subrepresentation of the relevant Hom representation. Comparing its integer weight with $\{0,-1\}$ and the modular weights $\{0,1-k\}$ forces $k=2$ and twist weight zero. On $G_{L'}$, the identity $(\det r_5)^2=\det V_5(A)=\chi_{\mathrm{cyc}}^2$ shows that $\det r_5/\chi_{\mathrm{cyc}}$ has finite image, since $G_{L'}$ has finite index in $G_{\mathbb{Q}}$. The modular determinant divided by χ_{cyc} also has finite image. The square of the twisting character consequently has finite image, so the character itself has finite image. Absorb it in the modular form and take the primitive weight-two newform f of level N.

The conductor at 5 is not omitted. For each prime $p\notin T\cup\{3,5\}$, reduction of μ_σ composed with relative p-Frobenius is a quasi-endomorphism of the good reduction of A at a chosen place. After clearing denominators its characteristic polynomial is independent of the Tate prime. Half its trace, multiplied by $b(\sigma)^{-1}a_\sigma$, is therefore one algebraic number giving the trace of every r_ℓ. Enlarge the coefficient field to contain the coefficients of f. The equality with the trace of f at five is an equality of algebraic numbers; transporting it to an embedding at three and applying Chebotarev and semisimplicity identifies r_3 with the three-adic realization of f. Thus use r_3 to compute the conductor at 5 and r_5 at every other prime. The modular level is the conductor of these realizations by [66, §§0.3–0.7, Theorem A, pp. 409–411] and newform theory [65, Theorems 1 and 4]. Here the totally real field is $\mathbb{Q}$, of odd degree one, so Carayol's extra finite-place hypothesis for even degree does not apply. His unramified normalization twist and contragredient preserve the conductor; his theorem allows arbitrary nebentypus. Li's Fricke relation and Shimura's Jacobian construction, as in the baseline modularity argument, also place r_5 in the Tate module of $J_1(N)$.

The primitive level equals the local conductor. Let $f\in S_2(\Gamma_0(N),\chi)$ be the primitive newform obtained after absorbing the finite character twist, and let Π_f be its unitary cuspidal automorphic representation. The character χ is arbitrary; its conductor need only divide N. For every prime p, write $a(\Pi_{f,p})$ for the local conductor exponent. Then

$$a(\Pi_{f,p})=v_p(N). \tag{CL1}$$

Here is the exact published newform input. In [65, Theorem 1, p. 302], the local conductor ideal is the largest ideal $\mathfrak{c}$ for which the appropriate central character eigenspace under $\Gamma_0(\mathfrak{c})$ is nonzero; at that ideal the eigenspace is one-dimensional. The remark on pp. 306–307 identifies this ideal with the exponent in the local epsilon factor. Theorem 4 on p. 313 gives the global level spaces: they vanish unless $\mathfrak{c}(\Pi)\mid M$, and at $M=\mathfrak{c}(\Pi)\prod_p p^{b_p}$ their dimension is $\prod_p(b_p+1)$. The paragraph immediately following that theorem identifies all primitive newforms with the minimal level $M=\mathfrak{c}(\Pi)$. This proves (CL1).

This statement includes nebentypus. Casselman's classical setup on p. 309 allows every character ϵ modulo M and uses $f|_k\gamma=\epsilon(a)^{-1}f$ for $\gamma=\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\in\Gamma_0(M)$. Since $ad\equiv 1\pmod M$, this is exactly

$\epsilon(d)f$. Thus the usual nebentypus convention is already covered. No squarefree-level hypothesis occurs.

For a coefficient characteristic $\ell \neq p$, Carayol's published Theorem A gives the full local Weil–Deligne compatibility, not only an equality at unramified primes [66, §§0.3–0.7, pp. 409–411]. Its base field here is $\mathbb{Q}$, its real weight is two, and its extra finite discrete-series-place condition applies only to even degree totally real fields, so imposes no additional hypothesis here. Theorem A includes the dyadic cases; the weaker Theorem B is not the result being used.

Carayol normalizes reciprocity by sending geometric Frobenius to a uniformizer. His Hecke correspondence in §0.5 differs from local Langlands by a contragredient and an unramified norm twist. It is characterized there by equality of local L- and epsilon-factors, including character twists. The conductor is unchanged by either duality or an unramified twist. In particular, the passage to the Tate-module convention used for r_ℓ changes no conductor exponent. Equality of good Frobenius traces identifies the global representation supplied by Carayol with the already constructed irreducible modular representation, in this convention. This uses Chebotarev and Brauer–Nesbitt; it does not use the unpublished irreducibility observation in Carayol's remark after Theorem A. Since the matching representation is irreducible, its matching semisimplification also forces the original representation to be irreducible.

The epsilon-factor conductor in (CL1) therefore equals the Artin conductor of $r_\ell|_{G_{\mathbb{Q}_p}}$:

$$a_p(r_\ell) = a(\Pi_{f,p}) = v_p(N), \qquad p \neq \ell. \tag{CL2}$$

The Artin conductor here includes its tame term and Swan term; equivalently one uses the full Weil–Deligne conductor, including the monodromy contribution. One must not discard monodromy by replacing the local representation with its semisimplification on inertia. Carayol's compatibility supplies precisely the required full local object. His epsilon-factor normalization and Casselman's conductor-zero additive character give the same exponent; changing an additive character only introduces the usual common dimension-dependent shift, so no such change is needed.

Apply (CL2) with $\ell = 5$ at every $p \neq 5$ and with $\ell = 3$ at $p = 5$. The companion proof identifies both r_5 and r_3 with the realizations of this same f. Thus the asserted conductor bounds control every prime divisor of the primitive level, even when $p^2 \mid N$. No compatibility theorem at $p = \ell$ is needed.

The conductor bound. Let $F = E\mathbb{Q}(\chi)$, where $\mathbb{Q}(\chi)$ means the kernel field of the character, not its value field. On G_F, b is a conjugation-invariant quadratic character. Its kernel gives a Galois extension $F'/\mathbb{Q}$ of degree at most two over F. For $p = 2, 3, 5, 7$, let $e_E, e_{F'}$ denote local ramification indices over $\mathbb{Q}_p$. From (LR1), $v_p(e_E) \leq 65$. At odd p both added extensions are tame, whereas at 2 each adds ramification index at most two. Consequently

$$v_p(e_{F'}) \leq 65 \ (p = 3, 5, 7), \qquad v_2(e_{F'}) \leq 67. \tag{LR4}$$

The possibly unbounded tame ramification index causes no problem. For a finite Galois extension of $\mathbb{Q}_p$ with ramification index e and different exponent δ, an Eisenstein polynomial over the maximal unramified subfield gives

$$\delta \leq e - 1 + ev_p(e).$$

Indeed the derivative terms have distinct valuations modulo e, and its leading derivative term has the stated valuation. In upper numbering the different formula is

$$\delta/e = 1 - 1/e + \int_0^\infty (1 - |G^u|^{-1})\, du.$$

Before the last positive break the integrand is at least $1/2$. Thus (LR4) bounds this break for $F'/\mathbb{Q}_p$ by 136. These ramification and conductor conventions are those of [32, §§4.1–4.3, pp. 155–159]. Over F' the scalar correction is trivial and A has good reduction. Above that break, therefore, the two-dimensional representation is trivial on upper wild groups. Its Swan conductor is at most 272, and its tame conductor is at most two. Hence

$$a_p(r) \le 274 < 300 \qquad (p = 2, 3, 5, 7).$$

For $p > 7$, $L/\mathbb{Q}$ is tame because its degree divides 576, A/L has good reduction, and b is trivial on wild inertia: on that pro-p group its square is tame and its remaining image has order at most two. Thus $a_p(r) \le 2$ there; it is zero outside T. Using the coefficient-prime comparison above proves

$$N \le 210^{300} R^2. \tag{LR5}$$

The original constant is preserved. Apply the accompanying modular-Jacobian transfer calculation with $C = 210^{300}$ and (LR3)–(LR5). It gives

$$h_F(A) < 2^{24} 210^{3600} R^{24} < 2^{28824} R^{24} < 4608^{165888} R^{24}. \tag{LR6}$$

The last comparison uses $210 < 2^8$, $4608 > 2^{12}$, and $28824 < 1990656 = 12 \cdot 165888$. Together with the separately proved CM bound and geometric height comparison, (LR6) yields exactly

$$h(s) \le (2(4608^{165888} + 6^{100}) B_*^{24} + 128) M(s)^{24}.$$

No contact, coefficient constant, or radical exponent has been changed.

Transfer from the level, allowing unbounded coefficient degree. Let A/L be the non-CM quaternionic surface. Write $R \ge 1$ for the squarefree product of the prescribed rational bad primes, including 2, and assume $[L : \mathbb{Q}] \le 576$. The modularity construction provides a weight-two newform of level N and a field L' over which the Tate modules of A and $J = J_1(N)$ share an irreducible constituent. In the notation of the sign-splitting construction, the auxiliary Dirichlet character has order

$$m \le 2 \prod_{\substack{p \mid R \\ p \ne 2}} (p - 1) \le 2R.$$

The sign-splitting cochain has values in μ_{2m}. If one instead uses the full scalar splitting map, it also includes the square root of the degree map; that factor equals one on G_L. Thus in either notation the restriction to G_L is a character valued in μ_{2m}, and its kernel field satisfies

$$d' = [L' : \mathbb{Q}] \le 2m[L : \mathbb{Q}] \le 2304R. \tag{LT1}$$

The quadratic twist used to remove extra ramification does not increase this bound, since 2 divides $2m$. There is no bound here on m independent of R.

Faltings' Hom theorem, followed by scalar extension in the Hom space, gives a nonzero L'-homomorphism $A \to J$. Absolute simplicity of A and Poincaré reducibility therefore give an L'-isogeny $A \times Z \to J$ for an abelian variety Z/L'. Put $g = \dim J$; then $g \ge 2$. The modular genus estimate [88, (5.9), p. 13] gives $g \le N^2/24$. The standard Belyi map has degree at most N^2 and [27, Theorem 1.1.1 and §2.3] gives

$$h_F(J) \le 13000000 N^{12}. \tag{LT2}$$

All heights in this argument use the original stable Faltings metric.

Suppose the companion local argument has proved $N \le CR^2$, where $C \ge 1$. Then

$$g \le C^2R^4/24, \qquad h_F(J) \le 13000000C^{12}R^{24}.$$

Set $l = \log C$ and $r = \log R$. The published Gaudron–Rémond Theorem 1.9(1) [87, pp. 7–8], combined with half-logarithmic isogeny-height variation, gives

$$h_F(A) + h_F(Z) \le h_F(J) + V, \quad V = 2g^2\left(8g^2\log(7g) + \log d' + \log\max(1, \log d', h_F(J))\right). \quad \text{(LT3)}$$

Here the source variety for the isogeny estimate is $J_{L'}$. The three logarithms in (LT3) satisfy respectively

$$\log(7g) \le 2l + 4r, \quad \log d' < 8 + r, \quad \log\max(1, \log d', h_F(J)) < 17 + 12l + 24r.$$

For the last assertion, the bound in (LT2) also dominates 1 and $\log d'$; use $8 + r \le 9R^{24}$. It follows that

$$V \le \frac{C^8R^{16}}{24^4}(32l + 64r) + \frac{C^4R^8}{288}(25 + 12l + 25r). \quad \text{(LT4)}$$

Using $l \le C^4/4$, $r \le R^8/8$, and also $r \le R^{16}/16$ in the second summand gives

$$\frac{V}{C^{12}R^{24}} \le \frac{16}{24^4} + \frac{25 + 3 + 25/16}{288} < 1. \quad \text{(LT5)}$$

The Bost lower bound in [88, (2.2), p. 5] gives $h_F(Z) \ge -\frac{1}{2}\dim Z\log(2\pi^2) > -\frac{3}{2}g$. Consequently

$$h_F(A) < (13000000 + 1 + 1)C^{12}R^{24} < 2^{24}C^{12}R^{24}. \quad \text{(LT6)}$$

This calculation needs no uniform bound on the degree of the newform's coefficient field and no assertion that all its geometric factors are isogenous to A.

For the companion conductor estimate one may take $C = 210^{300}$. Since $210 < 2^8$ and $4608 > 2^{12}$,

$$2^{24}C^{12} = 2^{24}210^{3600} < 2^{28824} < 2^{1990656} < 4608^{165888}. \quad \text{(LT7)}$$

Thus the non-CM height inequality is strictly within the original coefficient budget, once the stated modularity and level estimate have been established. In the five-contact application $R = B_*M(s)$, so the radical exponent is exactly 24, as required.

The CM height estimate over the same field Let A/L be an abelian surface of CM type, let B be a quaternion division algebra over $\mathbb{Q}$ embedded in $\mathrm{End}^0_L(A)$, and suppose that all geometric endomorphisms of A are defined over L. Assume

$$d = [L : \mathbb{Q}] \le 576, \qquad \mathrm{rad}(|D_L|) \le R, \qquad R \ge 1.$$

All heights below are stable Faltings heights in Faltings's original normalization. We prove

$$h_{\mathrm{F}}(A) < 854R < 6^{100}R^{24}. \quad (10)$$

No assertion about the existence or degree of L is included in this lemma.

First, $A_{\overline{L}}$ is isogenous to E^2 for a CM elliptic curve E. Here is a dimension argument which makes the alternative cases explicit. If A were geometrically simple, its division endomorphism algebra would act faithfully on the four-dimensional rational homology. Its dimension therefore

divides four. As it contains B, it would equal B; but B has no commutative semisimple subalgebra of dimension four, contrary to CM type. If the two elliptic factors were not isogenous, their endomorphism algebra would be a product of commutative fields and could not contain B. Finally, if E had no CM, the endomorphism algebra would be $M_2(\mathbb{Q})$, and the embedding of the four-dimensional division algebra B would be an impossible isomorphism.

The elliptic factors and the isogeny may be taken over L. Indeed, the rational idempotents in $\mathrm{End}^0(A)$ are defined over L, and the image of an integral multiple of such an idempotent is an elliptic subvariety over L. Endomorphisms and homomorphisms of the resulting factors extend to rational endomorphisms of A by the inclusions and rational projections; these are fixed by $\mathrm{Gal}(\overline{L}/L)$. Thus the factor isogenies, and all endomorphisms of E, are defined over L.

Put $F = \mathrm{End}^0(E)$, an imaginary quadratic field, and $\mathcal{O} = \mathrm{End}(E)$. Differentiation gives an injective unital map $F \longrightarrow \mathrm{End}_L(\mathrm{Lie}\, E) = L$: it is nonzero, and every nonzero elliptic endomorphism in characteristic zero is separable. Consequently $F \subset L$.

We next pass to the maximal order without extending L. Choose $f \geq 1$ with $f\mathcal{O}_F \subset \mathcal{O}$, choose an integral basis α_1, α_2 of $\mathcal{O}_F$, and form the finite subgroup scheme

$$K = \sum_{i=1}^{2} (f\alpha_i)(E[f]) \subset E.$$

Each $f\alpha_i$ is defined over L, so K and the quotient $E' = E/K$ are defined over L. Under a complex uniformization $E(\mathbb{C}) = \mathbb{C}/\Lambda$, this subgroup is $\mathcal{O}_F\Lambda/\Lambda$; thus $E'(\mathbb{C})$ has lattice $\mathcal{O}_F\Lambda$ and endomorphism ring $\mathcal{O}_F$. The ring cannot be larger, since the rational endomorphism algebra remains F. This is also a direct special case of the same-field maximal-order theorem of Rémond, but the construction just given is the proof used here. It follows that $A' = (E')^2$ is L-isogenous to A.

Write Δ_F for the fundamental discriminant of F. The discriminant tower formula implies that every rational prime dividing Δ_F divides D_L. A quadratic fundamental discriminant has odd prime exponents one and exponent at two at most three. Therefore

$$|\Delta_F| \leq 4\,\mathrm{rad}(|D_L|) \leq 4R.$$

The multiplier four includes the possible factor $2^3/2$.

For completeness we give an elementary analytic bound on $h_{\mathrm{F}}(E')$. CM elliptic curves have potentially good reduction. After a finite extension with everywhere good reduction, the elliptic discriminant formula of Silverman, reproduced as equation (6.2) in [30], gives

$$h_{\mathrm{F}}(E') = \frac{1}{[L' : \mathbb{Q}]} \sum_{\sigma : L' \hookrightarrow \mathbb{C}} \left(-\log(2\pi) - \log|\eta(\tau_\sigma)|^2 - \tfrac{1}{2}\log \mathrm{Im}\,\tau_\sigma \right).$$

Here L' is used only to evaluate the stable height; it is not the field used in the isogeny estimate below. The discriminant term is zero because reduction is good. We use the individual-embedding formula (6.2), not the class-average assertion of Proposition 6.1, so no hypothesis on Weierstrass coefficients lying in $\mathbb{Q}(j)$ is needed.

For each embedding choose a reduced quadratic parameter of discriminant Δ_F. If $y = \mathrm{Im}\,\tau$, then

$$\frac{\sqrt{3}}{2} \leq y = \frac{\sqrt{|\Delta_F|}}{2a} \leq \frac{\sqrt{|\Delta_F|}}{2}, \qquad a \in \mathbb{Z}_{\geq 1}.$$

Put $r = |e^{2\pi i\tau}| < 1/16$. The product defining η gives

$$-\log|\eta(\tau)|^2 \leq \frac{\pi y}{6} + 2\sum_{n\geq 1} \frac{r^n}{1-r^n} \leq \frac{\pi y}{6} + \frac{2r}{(1-r)^2} < \frac{\pi\sqrt{|\Delta_F|}}{12} + \frac{32}{225}.$$

Also $-\frac{1}{2}\log y \le \frac{1}{4}\log(4/3) < 1/12$. Discarding the negative term $-\log(2\pi)$ and using $\pi < 4$ shows

$$h_{\mathrm{F}}(E') < \sqrt{|\Delta_F|} + 1 \le 2\sqrt{R} + 1 \le 3R.$$

The height is additive on products, hence $h_{\mathrm{F}}(A') < 6R$.

Apply [87, Theorem 1.9(1)] over L, with source A' and target A and with no extension of the base field. For a surface, its quantity $\Xi(A')$ is

$$\Xi(A') = \left(14^{32} d \max(1, \log d, h_{\mathrm{F}}(A'))\right)^8.$$

The general assertion supplies an isogeny $A' \to A$ of degree at most $\Xi(A')^2$; no identification of CM type with that paper's separate "MM" condition is made or needed. The isogeny-height inequality then gives

$$h_{\mathrm{F}}(A) \le 6R + 8\log\left(14^{32}\, 576 \max(1, \log 576, 6R)\right).$$

One can see the last inequality directly from invariant differentials: after semistable base change, pullback on determinant differentials has integral cokernel, while its archimedean squared norm is multiplied by the isogeny degree. Thus the height difference is half the logarithm of that degree minus a nonnegative normalized cokernel length.

Finally, $\log 14 < 3$, $\log 576 < 7$, and $\max(1, \log 576, 6R) \le 7R$. Since $\log 7 < 2$ and $\log R \le R$,

$$h_{\mathrm{F}}(A) < 6R + 8(96 + 7 + 2 + \log R) \le 854R.$$

As $6^4 = 1296 > 854$ and $R \ge 1$, this proves (10). In particular the original $6^{100}R^{24}$ allowance absorbs this CM estimate with the field degree still exactly bounded by 576.

The unconditional numerical certificate and scalar detector The specified integer H satisfies

$$\log_2 H < 4783227.$$

Indeed the exact sieve enumerates 7499 primes in S_*, with largest prime 76163, and their product has bit length 109445. Hence $B_* < 2^{109445}$. With $A_0 = 4608^{165888}$, we have $6^{100} < A_0$ and $128 < A_0$, whence $H < 8A_0B_*^{24}$. Since $4608 < 2^{13}$,

$$\log_2 H < 3 + 165888 \cdot 13 + 24 \cdot 109445 = 4783227.$$

This is a check of the number H, not a proof that H is a height constant.

For completeness, the scalar product used by the original polynomial is

$$r(\beta) = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1).$$

At a prime $p > 76176$, if $v_p(\beta) < 0$, its valuation is $5v_p(\beta)$ and so detects odd contact with infinity. If β is p-integral and has odd contact with a finite member of $\mathcal{C}$, only that factor has positive valuation; every other factor is a unit. The extra root $1/2$ introduces only exceptional collision primes (notably factors of 4801 and 718). Thus every prime of $M(\beta)$ has odd valuation in $r(\beta)$. If a separate comparison proves $M(\beta)^{48} \mid s \ne 0$, the five-contact estimate just proved gives $h(\beta) \le H\sqrt{|s|}$.

Combining the non-CM and CM estimates with (R) and (G) gives

$$h(s) \le 2(4608^{165888} + 6^{100})(B_* M(s))^{24} + 128 \le H_* M(s)^{24},$$

because $M(s) \ge 1$. The numerical certificate above proves the remaining assertion about H_*. □

Define $\Psi(b)$ by seven rational witnesses t, U, e, p, q, s, h and four equations, with $B = 17b$:

$$\begin{aligned} U(1+bt^2) &= b(1-bt^2), & p^2 &= B(1-e)(1+3e), \\ h &= e(B-U^2), & Bq^2 &= (s^2-Bh)(s^2+3Bh). \end{aligned} \tag{11}$$

Lemma 14.4 (Exact seven-variable pole statement)**.** *If $b \neq 0$ and $\Psi(b)$, then $v_p(b)$ is not a negative odd integer for every $p \notin \{2, 3, 17\}$. Conversely, $\Psi(b)$ holds whenever $b > 0$, the denominator of b in lowest terms is a square, and b is a square in each of $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$.*

Proof. These are Lemmas 14.5 and 14.7. Their proof gives the simultaneous two-twist construction explicitly, and applies the existing Theorem 5.3 and Lemma 5.4. It introduces no new simultaneous-realization assumption. □

The lemma includes the degenerate values of the seven witnesses permitted by (11). No inverse condition on e, h, s, or U may be inserted into this particular polynomial. For $b \neq 0$, the first equation itself implies $1 + bt^2 \neq 0$. The predicate is soluble at $b = 0$; the norm argument below accounts for that case where necessary.

14.1.1 The seven-witness parity predicate and its finite correction

Put $B = 17b$. The predicate used here is exactly the existence of $t, U, e, p, q, s, h \in \mathbb{Q}$ satisfying

$$U(1+bt^2) = b(1-bt^2), \tag{12}$$

$$p^2 = B(1-e)(1+3e), \tag{13}$$

$$h = e(B-U^2), \tag{14}$$

$$Bq^2 = (s^2-Bh)(s^2+3Bh). \tag{15}$$

Denote it by $\Psi(b)$. No nonzero conditions are added to this formula.

Lemma 14.5 (The exact one-sided assertion)**.** *If $b \neq 0$ and $\Psi(b)$, then $v_\ell(b)$ is not a negative odd integer for any prime $\ell \notin \{2, 3, 17\}$.*

Proof. Suppose $k = v_\ell(b) < 0$ is odd. Equation (12) forces $1 + bt^2 \neq 0$: if it vanished, its right side would be $2b \neq 0$. If $t = 0$, then $U = b$. Otherwise $v_\ell(bt^2)$ is odd and nonzero, so $(1 - bt^2)/(1 + bt^2)$ is a unit. Thus $v_\ell(U) = k$ in all cases, and $v_\ell(B - U^2) = 2k$.

If $e = 0$, equation (13) gives $p^2 = B$, impossible. Write $j = v_\ell(e)$. If $j < 0$, the right side of (13) has valuation $k + 2j$, which is odd. If $j > 0$, it has valuation k, again odd. Consequently $j = 0$. This includes the possibilities $p = 0$, $e = 1$, and $e = -1/3$. Now $h \neq 0$ and $v_\ell(Bh) = 3k$ is odd. If $s \neq 0$, its square has even valuation, different from $3k$, so the two factors on the right of (15) have the same finite valuation. Their product has even valuation and is nonzero. The same conclusion holds when $s = 0$, since the product is $-3(Bh)^2$. The left side is either zero or has odd valuation, a contradiction. □

The omitted guards therefore cause no loss of the asserted soundness. At $b = 0$, the equations require $U = p = h = s = 0$ and permit arbitrary t, e, q. For $b \neq 0$, the six-witness source formula is recovered exactly by $e_{\text{old}} = Be$, $p_{\text{old}} = Bp$, $q_{\text{old}} = Bq$, with s, t, U unchanged. Its first curve residual is B^2 times (13)'s residual and its second is B times (15)'s residual after (14). Division by B reverses this transformation.

Lemma 14.6 (Simultaneous realization from the existing restricted converse)**.** *Use Theorem 5.3, the restricted converse* RC *for $E_l : y^2 = x(x-l)(x+3l)$ with $l > 0$ squarefree and $l \equiv 11 \pmod{12}$. Let $\beta, \theta > 0$ be squarefree, with disjoint supports avoiding $2, 3$, such that $\beta \equiv 17 \pmod{24}$, $\theta \equiv 1 \pmod{24}$, and each is a local square at every prime dividing the other. Then there is a positive squarefree n, prime to 6θ, for which both E_n and $E_{n\theta}$ have rational points with nonzero abscissa in $\beta\mathbb{Q}^{\times 2}$. The case $\theta = 1$ is allowed.*

Proof. Here are the descent steps, so that no separate realization theorem is being assumed. Write $B_0 = \operatorname{supp}(\beta)$, $J = \operatorname{supp}(\theta)$, and $d = |B_0| \geq 1$. For a support prime q_i, put $s_i = [-1/q_i]$, $\lambda_i = [3/q_i]$, and define over $\mathbb{F}_2$

$$(Kv)_i = \sum_{j \neq i} [q_j/q_i](v_j + v_i).$$

Add $d-1$ prime slots I of type $(s, \lambda) = (0, 0)$ and one slot q_0 of type $(1, 1)$. Let the B_0-by-I symbol matrix have columns $e_i + e_d$ for $1 \leq i < d$. Set all other symbols incident with I to zero, prescribe $[q_0/q_b]$ as the corresponding row sum, and obtain reverse symbols by reciprocity. The q_0–J symbols remain free. Mutual squareness gives zero row sums for both directed B_0–J blocks; hence $K\mathbf{1}_{B_0} = 0$ on both supports.

Choose a with ones at one fixed member of B_0 and at q_0. The matrix $K^\sharp = K + \lambda a^t$ kills $\mathbf{1}$ on both sides. After deleting q_0, its smaller-support matrix has the form

$$M_0 = \begin{pmatrix} D & C \\ C^t & 0 \end{pmatrix}, \qquad \mathbf{1}^t D \mathbf{1} = 1.$$

The columns of C span the even-sum hyperplane. If $M_0(x, y)^t = 0$, then $x = c\mathbf{1}$; multiplication of the first equation by $\mathbf{1}^t$ gives $c = 0$, and injectivity of C gives $y = 0$. This also proves the case $d = 1$, where $M_0 = (1)$. Adding J preserves M_0. Each free q_0–J bit toggles just one new diagonal entry. The enlarged determinant is multilinear in these bits, with full-product coefficient $\det M_0 = 1$, so some Boolean choice makes it nonzero. CRT and Dirichlet's theorem realize the prescribed symbols by distinct fresh primes, with types imposed modulo 12. Thus $n = \prod_{B_0 \cup I \cup \{q_0\}} q$ and $n\theta$ are squarefree and congruent to 11 modulo 12.

The projection $T = 1 - \mathbf{1}_{B_0} a^t$ satisfies $K = K^\sharp T$. Nonsingularity of the deleted matrix gives $\ker K^\sharp = \langle \mathbf{1} \rangle$, and therefore

$$\ker K = \langle \mathbf{1}, \mathbf{1}_{B_0} \rangle, \qquad \lambda \notin \operatorname{im} K$$

on both supports. For the latter assertion, $Kx = \lambda$ would make $Tx - \mathbf{1}_{B_0}$ constant, contradicted by applying a^t.

For clarity, the local descent needed here is the manuscript's full two-descent in coordinates $(x + 3l, x - l)$. The two torsion classes are $(l, 1)$ and $(3, -l)$. At an odd good prime, the local image consists of the four pairs with even valuations; at a prime dividing l, the two torsion classes span it by their independent valuation vectors. At 3 it is the set of all first-coordinate classes with square second coordinate. At 2 both coordinate valuations are even: the only problematic case $x - l = 2u$, u odd, would give $(y/2)^2 \equiv l \pmod 8$, impossible for $l \equiv 3 \pmod 4$. When both differences are divisible by 4, their difference $4l$ and the cubic equation give even valuations for both. At infinity the first coordinate is positive, and the positive-second-coordinate subgroup has index two. Its possible representatives consequently are $(3^\delta \prod q_i^{u_i}, \prod q_i^{v_i})$ with necessary equations

$$Ku + \operatorname{diag}(\lambda)v = \delta\lambda, \quad (K + \operatorname{diag}(s))v = 0, \quad (\lambda + s)^t v = 0.$$

Reciprocity gives $K + K^t = ss^t + \operatorname{diag}(s)$. Hence $\ker(K + \operatorname{diag}(s)) = \ker K^t \cap s^\perp$ is a line. The functional λ^t is nonzero on this line, because $\lambda \notin \operatorname{im} K$ and $\lambda^t \mathbf{1} = 0$, whereas $s^t \mathbf{1} = 1$. The displayed

equations force $v = 0$, $\delta = 0$, and $u \in \ker K$. There are at most eight Selmer classes. The class $(\beta, 1)$ is actually local everywhere: it is the identity at 2 and infinity, the first torsion class at 3, satisfies the torsion comparison at support primes by $K\mathbf{1}_{B_0} = 0$, and is unramified at the other odd primes. It is independent of the two torsion classes since B_0 is nonempty and $q_0 \notin B_0$. Thus each two-Selmer group has dimension exactly three.

Lemma 5.4 gives full two-primary Selmer corank one. Explicitly, the Kummer sequence gives $r + \dim \mathrm{Sha}[2] = 1$; writing $\mathrm{Sha}[2^\infty] = (\mathbb{Q}_2/\mathbb{Z}_2)^t \oplus F$, the alternating Cassels–Tate pairing on the finite quotient makes $\dim F[2]$ even, so $F = 0$ and $r + t = 1$. Apply RC to n and $n\theta$, and the manuscript's published forward analytic-rank-one theorem. Both curves have rank one and finite Sha, hence $\mathrm{Sha}[2] = 0$. Kummer exactness now represents $(\beta, 1)$ by actual rational points. These are not two-torsion, so their abscissas are nonzero of class β. When $\theta = 1$ the same point can be used twice. □

Lemma 14.7 (The completeness scope actually used)**.** *Assume only the existing* RC *and the published descent and forward-rank facts used above. If $b > 0$ has square reduced denominator and is a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, then $\Psi(b)$ holds.*

Proof. Choose $t \in \mathbb{Q}$ close to a solution of $bt^2 = 1$ at infinity and at $2, 3, 17$. In (12) this makes U arbitrarily small at those places. We can and do require $V = B(B - U^2) > 0$ and V locally square at all three primes. Explicit sufficient conditions are $U^2 < B$ in $\mathbb{R}$, $v_2(U^2/B) \geq 3$, and $v_3(U^2/B), v_{17}(U^2/B) \geq 1$. The denominator tends locally to $2 \neq 0$ and is positive globally.

Let β, θ be the positive squarefree representatives of B, V. The number $B = 17b$ is square at 2, a nonsquare unit class at 3, and has odd valuation at 17. Thus $\beta \equiv 17 \pmod{24}$ and $\beta > 1$; $\theta \equiv 1 \pmod{24}$ and $17 \nmid \theta$. At any prime $\ell \mid \beta$ outside $2, 3, 17$, the valuation of b is odd and therefore positive, since its reduced denominator is square. The same quotient calculation as in soundness gives $v_\ell(U) = v_\ell(b)$; hence $1 - U^2/B$ is a square unit. This proves that V, and hence θ, is square at every prime of β, including 17 by construction. The supports are disjoint. Conversely, if $\ell \mid \theta$, then $\ell \notin \{2, 3, 17\}$ and $v_\ell(B)$ is even. Odd valuation of $B(B - U^2)$ forces $v_\ell(B) = 2v_\ell(U)$ and strict cancellation, so $B/U^2 \in 1 + \ell\mathbb{Z}_\ell$ is square. This proves the other mutual-square condition. If $U = 0$, then $\theta = 1$ and this last argument is unnecessary.

Apply the preceding lemma. Write $V = \theta c^2$ and rescale the second point by $(x, y) \mapsto (c^2 x, c^3 y)$ to put it on E_{nV}. If the first point has abscissa Ba^2, divide both curve parameters by a^2 and the point coordinates by a^2, a^3. The resulting points have abscissas B, Bs^2 with $s \neq 0$, on E_{e_0} and $E_{e_0 V}$ respectively. Write their ordinates as y_1, y_2. Set

$$e = e_0/B, \qquad p = y_1/B, \qquad q = y_2/(B^2 s), \qquad h = e(B - U^2).$$

Direct substitution verifies all four equations. Every division made in choosing these witnesses is by a nonzero number; none is an additional equation or guard in the original polynomial. □

Finite correction and the unguarded norm root. At 2 use valuation parity followed by unit coordinates $[-1], [5]$; at 3 and 17 use valuation parity followed by the nonsquare-unit bit. The columns for $2, 3, 5, 7, 11, 13, 17$ are exactly

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 \end{pmatrix}.$$

The accompanying exact computation verifies rank seven and all 128 distinct subset signatures. Rational solutions of $e^2 - e = 0$ are exactly $0, 1$. Therefore the original eight bits and two products

$$A = (1 + e_2)(1 + 2e_3)(1 + 12e_{13})(1 + 16e_{17}), \quad D = (1 + 4e_5)(1 + 6e_7)(1 + 10e_{11})(1 - 2\eta)$$

give exactly $AD = (-1)^\eta \prod p^{e_p}$, always nonzero. For nonzero integral r, choose $\delta = \mathrm{sign}(r)$ and the unique subset product μ having the same local signature as δr. Taking $AD = \delta\mu$ makes ADr positive and locally square at $2, 3, 17$.

Choose J_0 sufficiently deep at all places over primes dividing μr, then S in a sufficiently divisible subgroup satisfying the anchor. Since $f = z^2$ times a formal unit, the coefficients of $f(J_0), f(S), z(S)/z(J_0)$ can attain any prescribed finite lower valuation bounds. This remains true at ramified and bad primes by working in sufficiently deep formal subgroups. Thus at $\ell \mid r$ all normalized leaves are integral, and at $\ell \mid \mu$ with $\ell \nmid r$ all nonconstant leaves can be made divisible by ℓ. The norm tree $N_r(a, b) = a^2 + rb^2$ has its constant leaf 1 on the all-left path; its root W is a unit at these primes. In the actual application $r = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1)$ with $\beta = X^2 + 2$ and $X \in \mathbb{Z}$. It is nonzero and $|r| > 1$ since $X^2 \neq 2399$ and $\beta \geq 2$. Consequently $W \neq 0$, and $b = ADrW^2$ is positive, locally square at $2, 3, 17$, and has square reduced denominator. Completeness applies.

For soundness no global guard $W \neq 0$ is needed. At any prime where $v_\ell(r)$ is odd, the two nonzero summands of $a^2 + rb^2$ have different valuation parity. Such a norm vanishes only if both inputs vanish. Recursing to the constant leaf 1 proves $W \neq 0$ at every prime relevant to an odd-contact argument. Outside the fixed exceptional set, AD is a unit and $v_\ell(ADrW^2) = v_\ell(r) + 2v_\ell(W)$. The exact original norm argument therefore applies unchanged. If r has no relevant odd valuation, there is no local conclusion to prove. No guard or witness has been inserted, and the original polynomial and coefficient bound are unchanged.

14.2 Elliptic charts and the anchor

Use the normalized curve and point

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2 = 11 - 6\sqrt{2}, \quad E : y^2 = x^3 - Dx,$$

$$P_* = (57/8 - 7\sqrt{2}/2, -165/8 + 417\sqrt{2}/32), \qquad G = 2E(K) = \mathbb{Z}P_*.$$

The identification of G follows from Proposition 3.1 by the unit change of variables $\lambda = 3 + 2\sqrt{2}$. Write $f = 1/x$ and $z = -x/y$. A nonzero point of G is represented by eight field variables u, v, w, z, f, h, x, y satisfying

$$f = u^2, \quad v^2 + df = 1, \quad w^2 + v^2 = 2, \quad h = vw, \quad zh + u = 0, \quad xf = 1, \quad yz + x = 0. \tag{16}$$

Each field equality means its two rational coefficient equalities.

Lemma 14.8 (The sparse point chart)**.** *The equations* (16) *parametrize precisely the nonzero points of G, and their reciprocal functions are defined.*

Proof. It gives $x = u^{-2}$, $y = vw/u^3$ and the three square root differences $x = (1/u)^2$, $x - d = (v/u)^2$, $x + d = (w/u)^2$. Here is a direct verification that these squares give precisely the required doubles. Write $e_1 = 0, e_2 = d, e_3 = -d$ and choose $r_i^2 = x - e_i$ with $r_1 r_2 r_3 = -y$. Put

$$B = r_1 r_2 + r_1 r_3 + r_2 r_3, \qquad X' = x + B, \qquad Y' = -(r_1 + r_2)(r_1 + r_3)(r_2 + r_3).$$

Since $X' - e_i = (r_i + r_j)(r_i + r_k)$, the point (X', Y') lies on E. None of the pair sums is zero, because the e_i are distinct. Its tangent slope is $-(r_1 + r_2 + r_3)$, and the duplication equations give $[2](X', Y') = (x, y)$. Conversely, if $(x, y) = [2](a, b)$ is nonzero, then $b \neq 0$ and direct duplication gives

$$x - e_i = \left(\frac{(a - e_i)^2 - (e_i - e_j)(e_i - e_k)}{2b} \right)^2 .$$

Choose the signs of these square roots to have product y, and recover u, v, w, h, z, f from (16). The group G is torsion-free, so its nonzero points have $x, y \neq 0$. Thus the sparse chart and the membership equations in Section 3 have the same nonzero image. The inverse relation $xf = 1$ excludes zero denominators. In particular u, h, z, y are nonzero. □

Subtraction uses the equations

$$\ell x - y = j, \quad \ell a + b = j, \quad x' = \ell^2 - x - a, \quad \ell x' + y' = j, \quad f'x' = 1, \quad z'y' + x' = 0$$

for $(x, y) - (a, b)$, with an inverse for the rational norm of $x - a$. Doubling uses the usual tangent formula with its nonzero denominators.

The baseline duplication estimates give

$$4h(x(Q)) - \log 392 \leq h(x(2Q)) \leq 4h(x(Q)) + \log 676.$$

Since $h(x(P_*)) = \log 41$, passage to the canonical height gives $2\widehat{h}(P_*) \geq \log 41 - \frac{1}{3} \log 392 > 1$, and hence

$$h(x(nP_*)) \geq n^2 - 6 \qquad (n \neq 0). \tag{17}$$

The deliberately weaker constant six suffices throughout this section.

Let $\mathcal{T}(a)$ be the seven-variable formula

$$u_0^2 - 2u_1^2 - 3u_2^2 + 6u_3^2 = 1, \qquad (a - 2u_0)^2 - 8v_1^2 - 12v_2^2 + 24v_3^2 = 4. \tag{18}$$

Then $\mathcal{T}(\mathbb{Q}) = \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}$. Indeed the quaternion algebra $(2, 3)$ is ramified exactly at $2, 3$ and split at infinity. Koenigsmann's trace-sum theorem, Proposition 6 of [83], identifies the sum of two norm-one trace sets with the intersection of these two local rings. Eliminating the scalar coordinate of the second norm-one element gives exactly (18), with no translation of a.

Introduce twenty-five rational variables

$$c_0 = 3, \qquad c_{i+1} = c_i^2 \quad (0 \leq i < 24), \qquad C = c_{24}. \tag{19}$$

Thus $C = 3^{2^{24}}$, but this large integer is not a coefficient in F_8. For $S = sP_* \neq O$ and $f(S) = f_0 + f_1\sqrt{2}$, impose $Ca_i = f_i$ and $\mathcal{T}(a_i)$, for $i = 0, 1$. At the prime above 3, P_* reduces to the nonidentity point $(\sqrt{2}, 0)$ of order two, and $v_3(z(2P_*)) = 1$. The formal group therefore gives $v_3(f(sP_*)) = 2 + 2v_3(s)$ whenever this valuation is positive. The anchor forces

$$|s| \geq 3^{2^{23}-1} > (20H_*)^2. \tag{20}$$

For the last inequality use Theorem 14.3, $\log_2 3 > 3/2$, and $3(2^{23}-1) = 25165821 > 4(5+4783227) = 19132928$. The allowed indices contain a nonzero subgroup: sufficiently divisible indices make $f(sP_*)/C$ integral at both 2 and 3. The anchor can therefore be imposed together with any finite list of formal divisibilities.

14.2.1 The literal seven-witness semilocal predicate

For $t \in \mathbb{Q}$, let $\mathcal{T}(t)$ mean that there are $u_0, u_1, u_2, u_3, v_1, v_2, v_3 \in \mathbb{Q}$ with

$$u_0^2 - 2u_1^2 - 3u_2^2 + 6u_3^2 = 1, \qquad (t - 2u_0)^2 - 8v_1^2 - 12v_2^2 + 24v_3^2 = 4.$$

Then

$$\mathcal{T}(t) \quad \Longleftrightarrow \quad t \in \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}.$$

Indeed, use the quaternion convention $i^2 = 2$, $j^2 = 3$ and $ij = -ji$. Its reduced norm is $x_0^2 - 2x_1^2 - 3x_2^2 + 6x_3^2$ and its reduced trace is $2x_0$. Put $v_0 = (t - 2u_0)/2$. The two displayed equations say exactly that t is a sum of two reduced traces of norm-one elements. No eighth variable is needed, since v_0 is recovered by this rational expression with fixed nonzero denominator 2.

In the notation of [83, Definition 4 and Proposition 6], this is $T_{2,3} = S_{2,3} + S_{2,3}$. That proposition identifies $T_{a,b}$ with the intersection of the local integer rings at the ramified places, with a separate real interval only when infinity is ramified. Here the Hilbert symbols are -1 at 2 and 3, and 1 elsewhere. At 3 this follows from $(2/3) = -1$; at every other odd prime both parameters are units. At 2, writing $2 = 2^1 \cdot 1$ and $3 = 2^0 \cdot 3$, the usual exponent is $0 + 1 \cdot (3^2 - 1)/8 + 0 = 1$. The algebra splits over $\mathbb{R}$ because both parameters are positive. Thus the ramification set is precisely $\{2, 3\}$, which proves the claimed equivalence in both directions. This uses Koenigsmann's sum of two trace sets, not a predicate with an additional finite offset.

14.2.2 The original anchor and its exact index subgroup

Write $K = \mathbb{Q}(\rho)$, $\rho^2 = 2$, and use the literal normalized curve and generator

$$E : y^2 = x^3 - (11 - 6\rho)x, \qquad P = \left(\frac{57}{8} - \frac{7}{2}\rho, -\frac{165}{8} + \frac{417}{32}\rho\right).$$

As in the original construction, use Proposition 3.1, transported by the unit change of variables, to obtain $2E(K) = \mathbb{Z}P$. For a nonzero $S = nP$, put $f(S) = 1/x(S) = f_0 + f_1\rho$. Retain the exact constant chain

$$c_0 = 3, \qquad c_{j+1} = c_j^2 \quad (0 \le j < 24), \qquad C = c_{24}.$$

It forces $C = 3^{2^{24}}$, with 25 shared rational variables and 25 quadratic-or-linear equations. The anchor introduces two variables a_i, imposes $Ca_i = f_i$, and imposes $\mathcal{T}(a_i)$ for each $i \in \{0, 1\}$. It has 16 private witnesses and six quadratic equations, exactly as in the saved module.

Set $N = 2^{24}$ and $m = N/2 = 2^{23}$. We claim that, for nonzero n,

$$\text{the anchor holds on } nP \quad \Longleftrightarrow \quad n \in 2 \cdot 3^{m-1}\mathbb{Z}.$$

First consider the place above 3. It is unramified quadratic, with residue field $\mathbb{F}_9$ and normalized valuation $v(3) = 1$. For rational a, b one has $v(a + b\rho) = \min(v_3(a), v_3(b))$: after dividing by the smaller power of 3, linear independence of $1, \bar{\rho}$ over $\mathbb{F}_3$ precludes cancellation. The curve has good reduction there. Direct calculation gives

$$\bar{P} = (\bar{\rho}, 0), \qquad z(2P) = \frac{3396356159088 - 4135204317756\rho}{3287811625151}, \qquad z = -x/y.$$

The reduction of P has order two. The denominator in this expression is a 3-unit and both numerator coefficients have valuation one. Consequently nP is in the formal kernel exactly when n is even, and $v(z(2P)) = 1$.

For completeness, the valuation of every such multiple can be computed without an unproved uniformity assertion. The integral formal group in the parameter z has inverse $-z$, so its multiplication series are odd. Their linear coefficient is the corresponding integer. Thus, for a 3-unit integer u, $[u]z = uz + O(z^3)$ preserves positive valuation, whereas $[3]z = 3z + O(z^3)$ increases valuation by exactly one at depth at least one. The higher terms have valuation at least $3v(z) > v(z) + 1$. Apply this repeatedly to $[n/2](2P)$, allowing negative integers by negation. It gives

$$v(z(nP)) = 1 + v_3(n), \qquad v(f(nP)) = 2 + 2v_3(n) \quad (2 \mid n).$$

The second equality follows from $f = z^2(1-(11-6\rho)f^2)$ on the formal branch. The integral formal-group construction and its linear multiplication coefficient are given in [48, IV§1, Example IV.2.2.3 and Proposition IV.2.3(a)]. No minimal-model assertion at the bad prime 2 is used below.

At the place above 2, normalize $v(2) = 1$, so $v(\rho) = 1/2$. The two summands of a rational pair have valuations of opposite integer parity after multiplication by 2, and hence cannot cancel. The displayed coordinates give

$$v(x(P)) = -3, \qquad v(y(P)) = -9/2, \qquad v(z(P)) = 3/2.$$

Thus P lies on the formal branch at infinity, even though reduction at 2 is bad. Every integral multiple lies in this formal subgroup. One can see the integral branch directly by setting $w = -1/y$: the equation becomes $w = z^3 - (11 - 6\rho)zw^2$, which has a unique solution $w \in z^3\mathcal{O}[[z]]$; its group law is the integral formal group just cited. In particular $f(nP)$ is integral at 2. Since

$$v(a + b\rho) = \min(v_2(a), v_2(b) + 1/2),$$

its rational coefficients are both 2-integral. As C is a 2-unit, the two anchor conditions at 2 impose no further restriction on n.

At 3, the two anchor conditions are equivalent to $v(f(nP)) \geq N$. Positive valuation of f implies negative valuation of x, hence membership in the formal kernel, so n is even. The preceding exact formula makes the remaining condition $v_3(n) \geq m - 1$. Conversely these two conditions give both required coordinate divisibilities at 3, and the condition at 2 already holds. The proved characterization of $\mathcal{T}$ supplies its rational witnesses, proving the subgroup claim.

In particular, anchor indices contain a nonzero finite-index subgroup and have positive representatives. They can be combined with any finite collection of requirements that indices be divisible by fixed integers: take a common multiple with $2 \cdot 3^{m-1}$. More generally, any finitely many formal-neighborhood conditions whose preimages are finite-index subgroups can be imposed in the same way. This statement does not claim compatibility with arbitrary nonzero residue conditions.

The old and new reciprocal coordinates are equivalent for this anchor. With $\lambda = 3 + 2\rho$, one has $f_{\mathrm{old}} = \lambda^2 f$ and

$$\begin{pmatrix} f_{\mathrm{old},0} \\ f_{\mathrm{old},1} \end{pmatrix} = \begin{pmatrix} 17 & 24 \\ 12 & 17 \end{pmatrix} \begin{pmatrix} f_0 \\ f_1 \end{pmatrix}.$$

This matrix has determinant one and integral inverse, so it preserves coordinatewise membership in $C(\mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)})^2$ in both directions.

14.2.3 The unchanged numerical cutoff

In this calculation $H = H_*$ is the constant in Theorem 14.3.

The anchor gives $|n| \geq 3^{2^{23}-1}$. Conditional only on the separately audited original height bound $\log_2 H < 4783227$, it follows that $|n| > (20H)^2$. Indeed $\log_2 3 > 3/2$ and $\log_2 20 < 5$, and

$$3(2^{23} - 1) = 25165821 > 19132928 = 4(5 + 4783227).$$

Thus the original 24 squarings suffice whenever that exact height input is established. This calculation does not replace the required five-contact height theorem by a six-contact theorem or by a larger constant. No part of this proof changes the saved polynomial.

14.3 The exact norm tests

For a nonzero rational contact r and rational tests X_i of positive orders L_i, apply the binary operation $N_r(a,b) = a^2 + rb^2$ to the ordered leaves $1, X_1/r^{L_1}, \dots, X_k/r^{L_k}$ in the following fixed order. First pair 1 with X_1/r, where $L_1 = 1$. Group the remaining tests by increasing order, retaining their original order within each group. Pair successive tests within each group and carry any last unpaired test. Then pair successive resulting nodes at each level and carry any last node until only a root W remains. The polynomial equations for the bottom pairs are

$$rw - r - X_1^2 = 0, \qquad r^{2L}w - X_i^2 - rX_j^2 = 0;$$

an unpaired normalized leaf satisfies $r^L w - X_i = 0$. Higher nodes satisfy $w - a^2 - rb^2 = 0$. Powers of r are computed by binary squaring. Add an inverse for r, a variable for W^2, and the predicate $\Psi(rW^2)$.

At a prime with $R = v_p(r)$ odd, the two summands in every norm node have distinct valuation parity. There is no cancellation. The constant leaf is on the all-left path, so $v_p(W) \le 0$ if $R < 0$. Consequently $v_p(rW^2) > 0$ first implies $R > 0$. It also gives $v_p(W) > -R/2$. Descending any norm node, a bound $v_p(N_r(a,b)) > -cR$ with $c < 1$ gives $v_p(a) > -cR/2$ and $v_p(b) > -(c+1)R/2$. Every normalized leaf therefore has valuation greater than $-R$. Thus $v_p(X_i) > (L_i - 1)R \ge L_i - 1$, or $v_p(X_i) \ge L_i$. The root cannot vanish at such a prime. Indeed $N_r(a,b) = 0$ forces $a = b = 0$ over $\mathbb{Q}_p$, and descent would make its constant leaf zero. Thus Lemma 14.4 applies even though the polynomial has no separately imposed inverse for W.

For completeness at a positive prime $r = p$, tests of orders L_i can be made so deep that every nonconstant normalized leaf is p-integral and the root W is a p-adic unit. Then rW^2 has square reduced denominator. If r is a local square at $2, 3, 17$, the completeness half of Lemma 14.4 applies.

14.4 Forty-one contacts and one common comparison point

Set

$$R = 47027 = 31 \cdot 37 \cdot 41, \quad W_0 = 25878772920, \quad c = 18484837801, \quad V_0 = 32W_0.$$

For nine rational targets a_i, put $A_i = Ra_i + 3$ and $w_i = A_i v - u$. Use the forty-one nonzero contacts

$$v/V_0, \qquad u - \delta v \ (\delta \in \mathcal{D}), \qquad -(w_i - \delta v) \ (\delta \in \mathcal{D},\ 1 \le i \le 9). \tag{21}$$

For integer a_i, the prime-pattern argument of Section 6 applies on $v = 32W_0(c + W_0 y)$, $u = c + W_0 x$ and gives positive ordinary prime values for every distinct contact, all congruent to c modulo W_0. Repeated targets use the same prime values. These are local squares at $2, 3, 17$. The precise application uses at most forty-one distinct affine forms and their successor exclusions. At $31, 37, 41$ there are at most eight slopes since every A_i reduces to 3; at $p \ge 43$ there are at most forty. Choosing two slopes outside that set proves local admissibility. The small-prime conditions are built into c. The full uniform fiber sieve and same-prime order argument are given in Subsection 14.4.1 below. No conditions relating orders at different selected primes are needed.

An odd normalized contact of u/v or w_i/v outside S_* forces some raw contact in (21) to have odd valuation. To see this, let m be the minimum of the two coordinate valuations. For a finite contact of odd depth, if m is even its distinguished linear form has odd valuation; if m is odd

another finite contact form has valuation m. At an odd contact with infinity the same argument uses the denominator if m is even and a finite form if m is odd. All differences of distinct finite contact points are units outside S_*. The signs and V_0 do not affect parity there.

For nonzero points $Q_i = n_i P_*$, $1 \le i \le 4$, take $Q_0 = P_*$, one common $J = jP_* \neq O$, and four rational proxies b_i. At each raw contact introduce $X_0 = hP_*$ and $X_i = m_i P_*$ and the subtraction charts $Y_i = X_i - 4Q_i$, $0 \le i \le 4$. The points $4Q_i$ are computed by two doublings. Put

$$L(Z) = \sum_{j=0}^{15} \frac{(-1)^j \binom{2j}{j} D^j}{4j+1} Z^{4j+1}.$$

This polynomial has degree sixty-one; the first omitted logarithm term has degree sixty-five. Test order one for

$$f(X_i)\ (0 \le i \le 4), \quad f(J^\sigma), \quad f(Y_i^\sigma)\ (0 \le i \le 4),$$

and order sixty-four for

$$L(z(X_i))/L(z(X_0)) - b_i\ (1 \le i \le 4), \qquad z(Y_i^\sigma)/z(J^\sigma)\ (0 \le i \le 4).$$

The inverses in these formulas are imposed explicitly. For each field test $a + b\sqrt{2}$ of order L_i, use the rational tests $(\theta^2 - 2)b$ and $a + \theta b$ of that order, together with $\theta^2 - 2$ of order one. There is one common rational θ for all forty-one comparisons. This gives twenty-three rational tests of order one and eighteen of order sixty-four.

At an odd raw contact the norm tests select a place above p, and the conjugate parameter ratios imply $h \equiv 4$ and $m_i \equiv 4n_i \pmod{p^{64}}$. In particular h is a unit. The logarithm expansion and its degree sixty-five remainder give

$$b_i \in \mathbb{Z}_p, \qquad b_i \equiv n_i \pmod{p^{64}}. \tag{22}$$

This uses the same formal-group estimates as Section 6.

For completeness choose a consistent orientation at each distinct selected prime. Let $d_{1,p}, d_{2,p}$ be the two reduction orders. The ordinary-prime conditions give $\gcd(d_{1,p}, d_{2,p}) = 4$ and $p \nmid d_{1,p} d_{2,p}$. Choose $j = \operatorname{lcm}_p d_{2,p}$. For each occurrence at p choose

$$h \equiv 0 \pmod{d_{1,p}}, \qquad h \equiv 4 \pmod{d_{2,p}}, \qquad h \equiv 4 \pmod{p^{64+v_p(j)}}.$$

These congruences are compatible. Avoid $h = 0, 4, -4$, and set $X_i = hn_i P_*$. They give all the required formal conditions and parameter ratios relative to the single point J. Increase the last precision if necessary and choose θ last by the Chinese remainder theorem. Repeated occurrences of a prime use the same orientation. Every subtraction denominator is nonzero, and the formal-logarithm unit calculation at p proves $L(z(X_0)) \neq 0$.

14.4.1 The original forty-one-prime pattern

Let

$$\mathcal{D} = \{0, 1, 2401, -343/32\}, \quad R = 47027 = 31 \cdot 37 \cdot 41, \quad W = 25878772920 = 4 \prod_{q \le 29} q,$$

$$c = 18484837801, \qquad V_0 = 32W = 828120733440.$$

For integers $a_1, \dots, a_k$, $k \le 9$, put $A_i = Ra_i + 3$, $w_i = A_i v - u$. There are $5 + 4k$ displayed positions

$$v/V_0, \qquad u - \delta v, \qquad u - (A_i - \delta)v \quad (\delta \in \mathcal{D},\ 1 \le i \le k). \tag{23}$$

For every lower bound these positions can simultaneously be positive primes beyond that bound, congruent to c modulo W, with the following property. For the curve $E : y^2 = x^3 - (3-\sqrt{2})^2 x$, both reductions above each prime are good, and their cardinalities d_1, d_2 satisfy

$$\gcd(d_1, d_2) = 4, \qquad p \nmid d_1 d_2.$$

This is a same-prime statement and asserts no cross-prime gcd condition.

Proof. Use $u = c + Wx$, $v = V_0(c + Wy)$. Put

$$\mathcal{S} = \mathcal{D} \cup \bigcup_i (A_i - \mathcal{D}), \quad K_s = V_0 s \in W\mathbb{Z}, \quad \Psi_0 = c + Wy, \quad \Psi_s = c + Wx - K_s(c + Wy).$$

Remove duplicate targets. The remaining slopes are all distinct. Indeed, the nonzero integral differences in $\mathcal{D} - \mathcal{D}$ are $\pm 1, \pm 2400, \pm 2401$, none divisible by 31. The integral sums in $\mathcal{D} + \mathcal{D}$ are $0, 1, 2, 2401, 2402, 4802$, with residues $0, 1, 2, 14, 15, 28$ modulo 31, none equal to $A_i \equiv 3$. These facts exclude intersections between secondary families and between primary and secondary families. The $m = 1 + |\mathcal{S}| \le 41$ forms have pairwise independent linear parts. Repeated targets may be restored afterwards with repeated values.

At $q \mid W$, the forms equal c; direct reduction gives $c \equiv 1 \pmod 8$, $c \equiv 3 \pmod 7$, and $c \equiv 1 \pmod q$ at all other odd primes $q \le 29$. Thus the forms and their successors are units at every odd $q \mid W$. For $q > 29$, the affine map from (x, y) to (u, v) is invertible. At $q = 31, 37, 41$, all $A_i = 3$ modulo q, so $|\mathcal{S} \bmod q| \le 8$. For $q \ge 43$ this cardinality is at most 40. In either case choose distinct residues b, e outside the slopes. Put $\gamma = V_0^{-1}$ modulo q and, reversing b, e if necessary, require $e - b \ne -\gamma$. Then

$$v = (e - b)^{-1}, \quad u = bv$$

satisfies

$$u - sv = v(b - s) \ne 0, \quad u - sv + 1 = v(e - s) \ne 0, \quad \gamma v \ne 0, \quad \gamma v + 1 \ne 0.$$

This proves all the value and successor local conditions. No condition on a successor is imposed at 2.

Add an unused integer coordinate z, and choose a fixed rectangular box Ω inside the cone $v > 0$, $u/v > \max(0, |A_1|, \dots, |A_k|) + 2413$ in the homogeneous leading coordinates. All forms on $X\Omega$ lie between αX and βX for positive fixed constants. Fix $Q > 82$ containing every prime divisor of W and every nonzero $K_s - K_t$. Impose one good residue at every prime up to Q. For a cutoff $L > Q$, exclude $\Psi_\nu = 0, -1$ at every additional prime up to L. The surviving proportion is

$$\gg_{\mathbf{a}} \prod_{Q < q \le L} (1 - 2m/q) \gg_{\mathbf{a}} (\log L)^{-2m},$$

by [81, Theorem 1, modulus 2].

For fixed L, apply [38, Main Theorem, p. 1763] to every surviving lattice residue in the convex box. The singleton partition in [38, Definition 1.5 and Lemma 1.6] bounds complexity by $m - 2 \le 39$. The inputs MN(39) and GI(39) follow from [39, Theorem 1.1] and [40, Theorem 1.3]. At imposed primes the local factor is $(1 - 1/q)^{-m}$; at the other primes it is $1 + O(m^2/q^2)$. The singular product therefore has a positive lower bound independent of L. All coefficient and modulus bounds are fixed before X tends to infinity. Summing the finitely many asymptotics and removing proper prime powers gives

$$\#\{\text{prime patterns}\} \ge c_{\mathbf{a}} \frac{X^3}{(\log L)^{2m} (\log X)^m}. \tag{24}$$

The constant $c_{\mathbf{a}} > 0$ is independent of L; the threshold for X may depend on it. Proper powers contribute at most $O_{\mathbf{a}}(X^{5/2}(\log X)^{m+1})$ before removing von Mangoldt weights, since each fixed value has $O_{\mathbf{a}}(X^2)$ fibers.

We give the uniform upper estimates needed to impose the order condition. For forbidden sets of cardinality $\rho(q)$ in a two-dimensional box of side N, put $g(q) = \rho(q)/q^2$, $h(q) = g(q)/(1-g(q))$, and extend multiplicatively on squarefree support. CRT gives

$$A_d = N^2 g(d) + R_d, \qquad |R_d| \ll Nd + d^2.$$

With $G(Z) = \sum_{d \le Z} h(d)$, take

$$\lambda_d = \frac{\mu(d)}{G(Z)\prod_{q \mid d}(1-g(q))} \sum_{\substack{e \le Z/d \\ (e,d)=1}} h(e) \quad (d \le Z).$$

Then $\lambda_1 = 1$, $|\lambda_d| \le d$, and the Selberg quadratic form equals $G(Z)^{-1}$. To verify the last assertion, use $y_r = \sum_{r \mid d} \lambda_d g(d) = \mu(r)h(r)/G(Z)$ and $1/g(n) = \sum_{r \mid n} 1/h(r)$. The number of sifted points is thus

$$\ll N^2/G(Z) + NZ^6 + Z^8. \tag{25}$$

For $\rho(q) = 2q - 1$, supported on $q \equiv 1 \pmod 4$ with at most one prime omitted, $G(Z) \gg_Q \log Z$, uniformly in that omission. For $\rho(q) = (m-1)q$ at all $q > Q$, $G(Z) \gg_{Q,m} (\log Z)^{m-1}$. These follow from [81, Theorem 1, moduli 4 and 2]. Truncation is justified by using $z = Z^{1/512}$ and the probability distribution $h(d)/\prod_{q \le z}(1+h(q))$: its expected $\log d$ is at most $40\sum_{q \le z}(\log q)/q < \frac{1}{2}\log Z$ for large Z. Thus at least half the mass lies below Z. Removing one norm-sieve prime changes its product by a factor at least $1/2$.

For a selected prime, $(7/p) = -1$ and $p \equiv 1 \pmod 8$. The two reductions of $3 - \sqrt{2}$ have product 7 and opposite square classes, so their traces are opposite. Their discriminants $64d^6$ are units. The automorphism $(x, y) \mapsto (-x, iy)$ is rational over $\mathbb{F}_p$. By the Tate-module injection [48, III.7.4], its rational centralizer is $\mathbb{Q}(i)$: at the auxiliary prime 3 the centralizer has dimension two because $T^2 + 1$ is irreducible. Frobenius is integral by [48, V.2.3.1(b)]; hence it is $a + bi \in \mathbb{Z}[i]$. The trace and determinant formulas [48, III.8.6] give

$$a^2 + b^2 = p, \qquad d_1 = p + 1 - 2a, \qquad d_2 = p + 1 + 2a.$$

Full rational two-torsion gives $4 \mid d_1, d_2$, and $v_2(d_1 + d_2) = 2$, so their gcd has 2-part exactly 4. Also $0 < d_i < 2p$ and $d_i \ne p$, so $p \nmid d_1 d_2$. An odd common divisor ℓ divides a and $p + 1$, and forces $b^2 \equiv -1 \pmod \ell$. The successor conditions exclude $\ell \le L$.

The number of bad prime values of any fixed form is

$$\ll_{\mathbf{a}} X/(L \log X) + X^{7/8} + \sqrt{X} \log X. \tag{26}$$

Indeed, for $L < \ell \le X^{1/8}$, write $a = \ell r$, $b = b_0 + \ell s$, with at most two b_0 modulo ℓ. The box has side $N \asymp \sqrt{X}/\ell \gg X^{3/8}$. Use the norm case of (25) with $Z = X^{1/100}$; its error is uniformly $o(N^2/\log X)$. Sum $O(X/(\ell^2 \log X))$. For $\ell > X^{1/8}$, $a \ne 0$ gives $O(\sqrt{X}/\ell)$ choices with no additive constant, and the b count is $O(\sqrt{X}/\ell + 1)$. Summation proves the remaining terms.

On a fiber $\Psi_0 = p$, use (x, z); the other forms are $c + Wx - K_s p$. On a fiber $\Psi_{s_0} = p$, use (y, z) and

$$x = K_{s_0} y + \frac{p - c(1 - K_{s_0})}{W}, \qquad \Psi_s = p + (K_{s_0} - K_s)\Psi_0.$$

The quotient is integral for an actual fiber. For $Q < q \le X^{1/100} < p$, the remaining zero lines are distinct and parallel, with total cardinality $(m-1)q$. The second case of (25) bounds each fiber by $O_{\mathbf{a}}(X^2/(\log X)^{m-1})$, independently of p, L. Multiply by (26) and sum over m positions. The bad patterns are at most

$$C_{\mathbf{a}} \frac{X^3}{L(\log X)^m} + O_{\mathbf{a}}\left(X^{23/8}/(\log X)^{m-1} + X^{5/2}/(\log X)^{m-2}\right).$$

Choose L so large that $C_{\mathbf{a}}/L$ is less than half $c_{\mathbf{a}}/(\log L)^{2m}$, then choose X large. A good pattern survives, proving the assertion. □

14.4.2 A shared auxiliary point with private shifts

Let $\mathcal{P}$ be the finite set of distinct selected primes, all larger than 76176. At each p fix an orientation of its two places, and let $d_{p,1}, d_{p,2}$ be the corresponding group cardinalities. Set

$$j = \operatorname{lcm}_{p\in\mathcal{P}} d_{p,2} > 0, \qquad J = [j]P_*.$$

For each p, independently choose an integer h_p satisfying

$$h_p \equiv 0 \pmod{d_{p,1}}, \qquad h_p \equiv 4 \pmod{d_{p,2}}, \qquad h_p \equiv 4 \pmod{p^{64+v_p(j)}}. \tag{27}$$

The first two congruences are compatible because their modulus gcd is 4. The last is compatible because $p \nmid d_{p,1}d_{p,2}$. The resulting arithmetic progression contains an h_p outside $\{0, 4, -4\}$. Put

$$X_0 = [h_p]P_*, \qquad X_i = [h_p n_i]P_*, \qquad Y_i = [(h_p-4)n_i]P_*, \quad n_0 = 1.$$

These are private for each contact occurrence, whereas J is common. Use the same orientation at repeated occurrences of p.

At the primary place all X_i are formal; at the opposite place J and all Y_i are formal. The formal logarithm is a homomorphism and an isometry on $p\mathbb{Z}_p$ by [48, IV.6.4 and its proof; VII.2.1–2.2]. The hypotheses hold because p is odd, split, and of good reduction. For any integers k, j whose multiples of P_* lie in the formal group,

$$\frac{\log([k]P_*)}{\log([j]P_*)} = \frac{k}{j}.$$

To see this without defining a logarithm of P_* itself, apply the homomorphism law to $[j]([k]P_*) = [k]([j]P_*)$. Nonzero formal multiples have nonzero logarithm. Therefore

$$v_p\left(\frac{z(Y_i)}{z(J)}\right) = v_p\left(\frac{(h_p-4)n_i}{j}\right) \ge 64, \qquad \frac{\log(X_i)}{\log(X_0)} = n_i.$$

This proves the required local comparison conditions. Although other $d_{q,2}$ may be divisible by p, their effect is exactly the term $v_p(j)$ in (27); no cross-prime gcd assertion is needed.

For the original truncation

$$L(z) = z\sum_{r=0}^{15} \frac{(-1)^r\binom{2r}{r}}{4r+1}(Dz^4)^r,$$

the full logarithm has the same series extended over all $r \ge 0$. This follows by integrating $(1 + 4Dz^4)^{-1/2}$, the invariant differential in the parameter $z = -x/y$. If $v_p(z) \ge 1$ and $p > 76176$, every

omitted term has valuation at least $65v_p(z)$: its integer numerator can only increase valuation, and for $n = 4r + 1 \geq 65$, $nv_p(z) - v_p(n) \geq 65v_p(z)$. For $n < p$ this is immediate; for $n \geq p$ use $n - 65 \geq v_p(n)$. Thus $L(z)/\log(z) \in 1 + p^{64}\mathbb{Z}_p$. The required primary truncated-log ratios are congruent to the integers n_i modulo p^{64}.

Conversely, in an accepted comparison at any $p > 76176$, its conjugate parameter-ratio tests imply

$$m_i - 4n_i \equiv 0 \pmod{p^{64}}, \qquad h - 4 \equiv 0 \pmod{p^{64}},$$

since j is a nonzero integer. Hence h is a p-unit. The same logarithm identity and truncation estimate give $L(X_i)/L(X_0) \equiv m_i/h \equiv n_i \pmod{p^{64}}$ and show that this ratio is integral. The rational proxy test then gives the claimed integrality and congruence for every b_i.

14.4.3 The common root approximation and parity boundary

For each distinct selected p, retain the chosen square root ρ_p of 2. After constructing the private points, each field test $A + B\sqrt{2}$ of order k has $v_p(A + B\rho_p) \geq k$. Choose a common integer θ sufficiently close to ρ_p at every selected prime that $v_p(\theta - \rho_p) \geq k - v_p(B)$ for every nonzero coefficient B, and at least one. There are finitely many requirements; CRT supplies such an integer. Since $\theta + \rho_p$ is a unit,

$$v_p((\theta^2 - 2)B) \geq k, \qquad v_p(A + \theta B) \geq k.$$

Repeated primes require the same root and impose compatible precision bounds. Conversely, the root test and these two rational tests imply the field test at the selected root. This verifies the circuit's one common θ, not a separate root variable per contact.

Every raw contact parameter in the complete assignment is a positive prime congruent to c modulo W. It is therefore a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$. The binary norm batch has unit root at its contact prime, because normalized leaves are integral and its leftmost constant leaf is 1. Its root is positive and nonzero in the real embedding. The argument $r\mathcal{W}^2$ consequently is positive, has square reduced denominator, and is locally square at $2, 3, 17$. These are the exact hypotheses of the original specialized parity-completeness input. This verifies its consumption but does not constitute a new proof of that input, which is supplied here by Lemma 14.7. In particular no integrality of the comparison points at these three small primes is required here.

14.5 Scalar height and the defining relation

Put $\beta = X^2 + 2$ and

$$r(\beta) = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1).$$

This is nonzero for every rational X, since $X^2 = 2399$ has no rational solution. Introduce $J_0 \neq O$ and apply a scalar norm batch to the two coefficients of each of $f(J_0), f(S), z(S)/z(J_0)$, of orders $1, 1, 1, 1, 48, 48$. In this batch replace the pole argument by $\delta\mu rW^2$, where $\delta \in \{1, -1\}$ and μ is a squarefree product of $2, 3, 5, 7, 11, 13, 17$. Precisely, eight Boolean variables produce

$$A = (1 + e_2)(1 + 2e_3)(1 + 12e_{13})(1 + 16e_{17}), \quad B = (1 + 4e_5)(1 + 6e_7)(1 + 10e_{11})(1 - 2\eta),$$

and the pole argument is $ABrW^2$. These factors have no valuation outside S_*. Every odd five-contact of β gives an odd valuation of $r(\beta)$ there. At infinity its valuation is five times that of β; at a finite contact all other factors are units. Thus the scalar tests force $M_5(\beta)^{48} \mid s$, and Theorem 14.3 gives

$$h(\beta) \leq H_* \sqrt{|s|}. \tag{28}$$

For integral X, the scalar r is a nonzero integer. The square classes of $2,3,5,7,11,13,17$ form a basis of $\mathbb{Q}_2^*/\mathbb{Q}_2^{*2}\times\mathbb{Q}_3^*/\mathbb{Q}_3^{*2}\times\mathbb{Q}_{17}^*/\mathbb{Q}_{17}^{*2}$. Choose δ and μ so that $\delta\mu r$ is positive and a square at these three places. Choose J_0 in the required formal subgroups and then choose S sufficiently divisible relative to J_0 at all primes dividing $r\mu$, together with the anchor conditions. All normalized nonconstant leaves are then integral at primes dividing r, and vanish at primes dividing μ but not r. The root is a unit at these primes. Consequently $\delta\mu rW^2$ has square reduced denominator, and Lemma 14.4 supplies the scalar pole witnesses.

Take $A=nP_*, S=sP_*, T=tP_*, Z=zP_*\neq O$, and use (β,ξ,τ,ζ) as the four proxies. Impose

$$\tau^2-\xi(\xi-2)\tau\zeta+(\xi-2)^2\zeta^2=1. \tag{29}$$

The nine rational targets in (21) are a nonzero common denominator e and the eight coefficients of $ex(A), ex(S), ex(T), ex(Z)$. Require its forty-one comparison batches, the scalar batch, and the anchor just described.

If every associated five-contact radical has its Kth power dividing a nonzero integer N, Theorem 14.3 and $Ra_i+3=u/v+w_i/v$ give

$$h(a_i)<2H_*|N|^{24/K}+14, \qquad h(x(Q))<7H_*|N|^{24/K}$$

for each of the four points. The second bound follows by applying the first to e and both coefficients of $ex(Q)$ and absorbing the constant smaller than fifty into H_*.

Lemma 14.9 (Lucas growth)**.** *If $s,t,z\in\mathbb{Z}$, $|s|=a\geq 7$, $z\neq 0$, and*

$$t^2-s(s-2)tz+(s-2)^2z^2=1,$$

then $|t|>\exp(a-3)$. Conversely each such s admits positive integers t,z satisfying this equation.

Proof. Put $w=\operatorname{sgn}(s)(s-2)z$ and $\rho=(a+\sqrt{a^2-4})/2$. The equation becomes $t^2-atw+w^2=1$. Every positive norm-one unit of $\mathbb{Z}[\rho]$ is a power of ρ: multiply it by a power into $[1,\rho)$, write the result as $v=A+B\rho$, and use $B=(v-v^{-1})/(\rho-\rho^{-1})\in[0,1)\cap\mathbb{Z}$. Thus $v=1$. If $U_0=0$, $U_1=1$ and $U_{k+1}=aU_k-U_{k-1}$, the identities for $\rho^{\pm k}$ give $|w|=U_k$ and $|t|\in\{U_{k-1},U_{k+1}\}$ for some $k\geq 1$. Modulo $a-2$ one has $U_k\equiv k$, and modulo $a+2$ one has $U_k\equiv(-1)^{k-1}k$. The divisibility of w by $|s-2|$ therefore gives $k\geq a-2$ when $s>0$ and $k\geq a+2$ when $s<0$. Induction gives $U_k\geq(a-1)^{k-1}$, whence $|t|\geq(a-1)^{a-4}>e^{a-3}$ for $a\geq 7$. For existence take $k=a-2$, $z=U_k/(a-2)$ if $s>0$, and $k=a+2$, $z=U_k/(a+2)$ if $s<0$, in both cases with $t=U_{k+1}$. □

Proof of Theorem 14.2. Suppose the relation holds, and let $B=\max(|n|,|s|,|t|,|z|)$ and $\Delta=t^2-s(s-2)tz+(s-2)^2z^2-1$. If $\Delta\neq 0$, the congruences (22) imply that every coordinate contact radical to the sixty-fourth power divides Δ. Since $|\Delta|<5B^4$, the index-height and shared-height bounds give

$$B^2-6<7H_*|\Delta|^{3/8}<14H_*B^{3/2}.$$

Hence $B<(14H_*+6)^2\leq(20H_*)^2$, contradicting the anchor. Thus $\Delta=0$. Lemma 14.9 gives $|t|>\exp(|s|-3)$ because $z\neq 0$.

If $\beta\neq n$, put $N=|\operatorname{num}(\beta-n)|$ in lowest terms. The same congruences give

$$B^2<7H_*N^{3/8}+6, \qquad N\leq 2e^{h(\beta)}B,$$

whence $\log B<\frac{8}{13}\log(16H_*)+\frac{3}{13}h(\beta)$. Writing $a=|s|>(20H_*)^2$, (28) implies $h(\beta)<a/20$ and $\log(16H_*)<a/20$. Therefore $\log B<11a/260<a-3$, another contradiction. We obtain $\beta=n\in\mathbb{Z}$, and $X\in\mathbb{Q}$, $X^2=\beta-2\in\mathbb{Z}$ imply $X\in\mathbb{Z}$ by reducing its denominator.

Conversely let $X \in \mathbb{Z}$ and set $A = \beta P_*$. Choose the scalar height data and then an anchored S with all required divisibilities. Lemma 14.9 gives positive integers t, z satisfying the index equation; set $T = tP_*$ and $Z = zP_*$. Choose an integer e clearing their abscissa coefficients. The ordinary prime pattern provides (21). Set the proxies equal to their integer indices, choose J and the private points as above, and finally choose the common θ. The norm and pole completeness statements supply all remaining witnesses. No subsequent choice changes the target points. □

14.6 The base compiler prescription and finite verification

Every equation above is implemented with named arithmetic intermediates so that each rational residual has degree at most four. A field product uses $(a+b\sqrt{2})(c+d\sqrt{2}) = (ac+2bd)+(ad+bc)\sqrt{2}$. Scalar inverses are equations $uv = 1$. Field inverses use the two coefficient equations of the field product being one; subtraction has the separate scalar inverse of its abscissa-difference norm. The logarithm uses Horner evaluation in $(Dz^4)^2$ with block length two. The two doublings of the fixed point P_* also use arithmetic intermediates; inserting its large precomputed fourth multiple as a coefficient would give a different polynomial. The chart is exactly (16), including the intermediate $h = vw$.

For each nonzero rational residual, multiply by the least common multiple of its coefficient denominators, then divide all coefficients by their positive greatest common divisor. Denote the resulting integral rows by $R_1, \dots, R_{15107}$. In the variable order specified by `degree8-original/build.py` and its residual table, define

$$F_8 = \sum_{i=1}^{15107} R_i^2. \tag{30}$$

This equation is an exact finite prescription for every coefficient. The residual table, its full collected expansion, and the construction certificate accompany the manuscript. No huge constant is treated as an additional free parameter. In particular, all twenty-five variables in (19) belong to the stated witness count.

The recursive module counts are

module	new variables	residuals
pole	7	4
anchor	16	6
scalar batch, including pole	34	31
forty-one-test batch, including pole	58	55
one joint comparison, including batch	373	360
root, including all its calls	15646	15082
shared anchor chain	25	25

The root has 303 direct variables and 285 direct residuals; its calls are one anchor, one scalar batch, and forty-one comparisons. Thus $303+16+34+41\cdot373 = 15646$ and $285+6+31+41\cdot360 = 15082$. Expansion of (30) gives the finite assertions of Proposition 14.1. Over $\mathbb{Q}$, a sum of squares vanishes exactly when every summand vanishes, so the residual interpretation proves the theorem. This computation checks the polynomial and its compilation. The proof of the height estimate is the arithmetic argument given above.

15 An explicit quartic defining the integers

This is the second final polynomial of the manuscript. The original huge polynomial remains unchanged. The base residual compiler in Section 14 supplies construction machinery and inherited

arithmetic inputs; it is not a third final polynomial.

Theorem 15.1 (An explicit quartic definition)**.** *With precisely the inherited arithmetic hypotheses and qualifications of Section 14 and Appendix B, the coefficient file below defines one integer polynomial Φ of total degree 4, with 72 925 nonzero monomials in 4 973 variables. For every $q \in \mathbb{Q}$,*

$$q \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists (w_1, \dots, w_{4972}) \in \mathbb{Q}^{4972}, \qquad \Phi(q, w_1, \dots, w_{4972}) = 0.$$

There are 4 972 existential witnesses besides the original input. The coefficient bound is

$$\|\Phi\|_\infty = 685783949153203534233890 < 2^{80}.$$

15.1 Literal specification and coefficient file

The complete coefficient table is `quartics/quartic-4973/polynomial.expanded.txt`. After one comment line, each line gives a nonzero signed integer coefficient followed by a nondecreasing list of variable indices. Repeated indices give their multiplicities as exponents; a coefficient alone is the constant. Thus `-6 0 2 2` denotes $-6X_0X_2^2$. The original input is X_0. Like terms are collected, and the rows are in lexicographic index-list order. The occurrence-frequency relabeling is recorded in `quartics/quartic-4973/variable-map.json` and fixes index zero. The gzip companion has exactly the same uncompressed bytes. The coefficient table has 1 622 053 bytes, including its comment and line endings, and SHA-256

```
094f6a7ed17e7cc8d5d17a2f809127bbaffd0fe343cb2a32d18ac97e61279876
```

This specifies every coefficient without an auxiliary choice of witnesses.

15.2 Arithmetic source and retained foundations

The selected source uses the following retained height, contact and reciprocal-logarithm proofs before the new source equivalence is established. An element of K is represented by two rational coefficients, with

$$(a_0, a_1)(b_0, b_1) = (a_0b_0 + 2a_1b_1, a_0b_1 + a_1b_0), \qquad a^{-1} = \frac{(a_0, -a_1)}{a_0^2 - 2a_1^2} \quad (a \neq 0).$$

The inverse denominator is nonzero by the irrationality of $\sqrt{2}$.

Theorem 15.2 (A sharper five-contact height estimate)**.** *Keep S_*, $\mathcal{D}$ and M_5 from Theorem 14.3, and put*

$$B_* = \prod_{p \le 76176} p, \qquad A_{19} = 2^{109}210^{1800}, \qquad H_{19} = 2A_{19}B_*^{19} + 128.$$

For every $a \in \mathbb{Q} \setminus \mathcal{D}$,

$$h(a) \le H_{19}M_5(a)^{19}, \qquad \log_2 H_{19} < 2093966.$$

Here h is the absolute logarithmic Weil height.

Proof. We use the geometric construction and compatible representations in the proof of Theorem 14.3, not merely its final exponent-24 inequality. Write $R = B_*M_5(a)$ and $C = 210^{300}$. The geometric comparison there is

$$h(a) \le 2\max(0, h_F(A)) + 128.$$

All h_F in this proof use the original stable Faltings metric. The CM estimate in that proof is $h_F(A) < 854R$.

In the non-CM case the same construction gives an absolutely simple quaternionic surface A/L, a normal field $L'/\mathbb{Q}$, and the final primitive weight-two newform f, after the finite twist has been absorbed, with

$$[L:\mathbb{Q}] \le 576, \quad d' = [L':\mathbb{Q}] \le 2304R, \quad N \le CR^2, \quad \operatorname{rad}(N) \mid R.$$

Its two-dimensional compatible representation r_ℓ satisfies

$$V_\ell(A)|_{G_{L'}} \simeq (r_\ell \oplus r_\ell)|_{G_{L'}}, \qquad (\det r_\ell)^2|_{G_{L'}} = \chi_{\mathrm{cyc}}^2|_{G_{L'}}.$$

These are (LR3)–(LR5) and the determinant calculation in the existing modularity argument. There is no bounded-degree hypothesis on the coefficient field of f.

Let ϵ be the nebentypus of this final form. In the Tate-module convention, with arithmetic Frobenius, $\det r_\ell = \epsilon\chi_{\mathrm{cyc}}$; see [91, §12.5, (12.5.1)–(12.5.2) and Remark 12.5.1, pp. 119–120]. Equivalently, at a good prime its characteristic polynomial is $X^2 - a_pX + p\epsilon(p)$. A consistent contragredient convention replaces the finite character by its inverse, with the same kernel and order. The character ϵ^2 is trivial on $G_{L'}$, so its image order divides d'. Consequently

$$t = \operatorname{ord}(\epsilon) \le 2d' \le 4608R.$$

This bounds the character order, not the coefficient-field degree.

Put $H = \ker\epsilon \subset (\mathbb{Z}/N\mathbb{Z})^*$ and

$$\Gamma_H = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N) : d \bmod N \in H \right\}.$$

Since $\epsilon(-1) = 1$, this group contains $-I$. Let $X_H = X_1(N)/H$ for the diamond action and $J_H = \mathrm{Jac}(X_H)$. The curve, diamond action and quotient are defined over $\mathbb{Q}$. The rational model here parametrizes a point of exact order N, up to H, and is the model specified by [93, §2.1, (2.1.8)]; it introduces no cyclotomic field-degree factor.

Write $q : J_1(N) \to A_f$ for the newform quotient. Its pulled-back differentials are spanned by all Galois conjugates of f; see [26, Theorem 1(i),(iv), p. 526] and [92, §3.1]. Every $h \in H$ fixes those differentials, since its eigenvalue on each conjugate is a conjugate of $\epsilon(h) = 1$; the diamond convention is [92, §3.2.1]. Homomorphisms of abelian varieties in characteristic zero are determined by their differentials. Hence $q \circ \langle h \rangle = q$. For

$$e_H = \frac{1}{|H|} \sum_{h \in H} \langle h \rangle \in \mathrm{End}^0_{\mathbb{Q}}(J_1(N))$$

we therefore have $qe_H = q$.

Let H_{eff} be the image of H in $\mathrm{Aut}(X_1(N))$, and let $\pi : X_1(N) \to X_H$ be the quotient. On Jacobians,

$$\pi^*\pi_* = \sum_{h \in H_{\mathrm{eff}}} h, \qquad \pi_*\pi^* = [|H_{\mathrm{eff}}|].$$

The averages over H and H_{eff} agree. In particular the trivial action of -1 is not counted twice in the first identity. The image of e_H is thus $\mathbb{Q}$-isogenous to J_H, and its restriction of q is surjective. Poincare reducibility makes A_f a $\mathbb{Q}$-isogeny factor of J_H.

The accepted realization of r_ℓ in A_f puts it in $V_\ell(J_H)$. Over L', Faltings' Hom theorem and extension of scalars give a nonzero homomorphism $A \to J_H$. Absolute simplicity and Poincare reducibility give an L'-isogeny $A \times Z \to J_H$. In particular $g = \dim J_H \geq 2$. The Jacobian has its canonical principal polarization; no bound on a polarization degree of A or Z is being assumed.

The map $X_H \to X(1)$ has degree

$$d_H = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_H] = Nt \prod_{p|N} (1 + 1/p).$$

There is no factor of two here, because $-I \in \Gamma_H$. It is branched only over the cusp and the two elliptic points, so $\deg_B(X_H) \leq d_H$. The congruence-curve genus bound gives $g \leq 1 + d_H/12 \leq d_H$, using $g \geq 2$. These degree and genus conventions, and the resulting height inequality, are recorded in [88, §4, (4.1)–(4.2), p. 8]. For every prime $p \geq 5$, $(1 + 1/p)^6 \leq p$, while the possible factors at two and three have product at most two. It follows that

$$d_H \leq 9216 C R^{19/6} < T, \qquad T = 2^{14} C R^{19/6}, \qquad g < T.$$

By [27, Theorem 1.1.1 and §2.3],

$$h_F(J_H) \leq 13000000\, d_H^5 g < 2^{24} T^6.$$

This is the generic Belyi-curve estimate; no squarefree-level restriction or trivial-nebentypus assumption enters it.

Apply [87, Theorem 1.9(1), pp. 7–8] with source $(J_H)_{L'}$ and target $A \times Z$. Combining its degree bound with half-logarithmic isogeny-height variation gives

$$h_F(A) + h_F(Z) \leq h_F(J_H) + V, \quad V = 2g^2 \left(8g^2 \log(7g) + \log d' + \log \max(1, \log d', h_F(J_H)) \right).$$

The Bost lower bound in [88, (2.2), p. 5] gives $h_F(Z) > -3g/2$. Put $u = \log T$. Then

$$V \leq 16 T^4 (2 + u) + 2T^2 (25 + 7u) \leq T^6.$$

Indeed $\log d' < 8 + \log R \leq 8 + u$, and the last logarithm in V is less than $17 + 6u$. For $T \geq 16$ the inequalities $2 + \log T \leq T^2/32$ and $25 + 7 \log T \leq T^4/4$ follow at 16 and remain true by differentiation. Thus

$$h_F(A) < (2^{24} + 1) T^6 + \tfrac{3}{2} T < 2^{25} T^6 = A_{19} R^{19}.$$

The earlier CM estimate is also less than this bound. The geometric comparison and $M_5(a) \geq 1$ prove the assertion. Finally $B_* < 2^{109445}$, $210 < 2^8$ and $A_{19} B_*^{19} > 64$ give

$$H_{19} < 4 A_{19} B_*^{19} < 2^{\,2 + 109 + 1800 \cdot 8 + 19 \cdot 109445} = 2^{2093966}.$$

□

Lemma 15.3 (Thirty-seven contacts for rational coefficients)**.** *Keep $K = \mathbb{Q}(\sqrt{2})$, $d = 3 - \sqrt{2}$, $D = d^2$, and $E : y^2 = x^3 - Dx$ from Section 14. Let A, S, T, Z be nontorsion points of $E(K)$. Order the eight rational coefficients of $x(A), x(S), x(T), x(Z)$ as $a_1, \ldots, a_8$. Put*

$$R_{\mathrm{c}} = 47027, \quad W_0 = 25878772920, \quad c = 18484837801, \quad V_0 = 32 W_0,$$

$$A_i = R_{\mathrm{c}} a_i + 3, \quad w_i = A_i v - u.$$

Use the following thirty-seven nonzero raw contacts:

$$r_0 = v/V_0, \qquad u - \delta v \quad (\delta \in \mathcal{D}), \qquad -(w_i - \delta v) \quad (1 \le i \le 8,\ \delta \in \mathcal{D}). \tag{31}$$

Here and below the order within $\mathcal{D}$ is $(0, 1, 2401, -343/32)$. The common five positions are numbered $0, \dots, 4$; the other positions are numbered $5, \dots, 36$ in the displayed order.

Every normalized odd contact of u/v or w_i/v outside S_* is detected by an odd valuation at a raw position belonging to that slope. If the radicals for both slopes $u/v, w_i/v$ have their eth powers dividing an integer $N \neq 0$, then, with $H = H_{19}$,

$$h(a_i) < 2H|N|^{19/e} + 14.$$

Consequently the two coefficients of any one of the four abscissas give

$$h(x(Q)) < 4H|N|^{19/e} + 30 < 7H|N|^{19/e} \quad (Q \in \{A, S, T, Z\}) \tag{32}$$

whenever all its required positions have those divisibilities.

Conversely, for any fixed eight rational coefficients, the inherited ordinary-prime construction supplies these raw contacts with every finite contact a positive selected prime and r_0 a positive rational square times a selected prime. Repeated positions may have equal values. These contacts have the positivity and local squareclasses used by Lemmas 15.10 and 15.11.

Proof. Nonzero raw contacts imply $v \neq 0$ and exclude the four finite branch values for every slope. At a prime outside S_* all differences of distinct members of $\mathcal{D}$ are units. Write m for the minimum of the valuations of the two coordinates of one slope. If an odd finite normalized contact has m even, its distinguished raw linear form has odd valuation. If m is odd, another finite form has valuation m. For an odd contact at infinity use the denominator in the first case and a finite form in the second. The unit V_0 and the signs do not change parity. This proof uses rational coordinates and also covers negative valuations; it does not presuppose that a contact is prime.

Apply Theorem 15.2 to the two slopes. The identity $R_{\mathrm{c}} a_i + 3 = u/v + w_i/v$ and the elementary logarithmic height inequalities give the first bound, since $\log(6R_{\mathrm{c}}) < 14$. For $x(Q) = a + b\sqrt{2}$ use $h(x(Q)) \le h(a) + h(b) + h(\sqrt{2}) + \log 2$. This proves (32), since $H > 128$ and $|N| \ge 1$.

For the converse choose a positive integer D_0 with $D_0 a_i \in \mathbb{Z}$ for every i, after fixing the eight coefficients. This integer is a choice in the existence proof, not a variable of the source. Set

$$T_0 = V_0 D_0^2, \qquad p_0 = c + W_0 y, \qquad u = c + W_0 x, \qquad v = T_0 p_0.$$

For each distinct slope $b \in \mathcal{D} \cup \bigcup_i (A_i - \mathcal{D})$ put $K_b = T_0 b \in W_0\mathbb{Z}$ and $\Psi_b = u - K_b p_0$. Together with p_0, these are at most thirty-seven integer affine forms, each congruent to c modulo W_0. Distinct slopes give independent linear parts. Exact duplicate slopes are removed only while applying the prime theorem and restored at all their original positions afterward.

The value and successor admissibility in Section 14.4.1 remains valid. At primes dividing W_0 use the same residue c. If a prime $\ell > 29$ divides D_0, all K_b vanish modulo ℓ, and $u = p_0 = 1$ avoids zero and minus one. If $\ell > 29$ does not divide D_0, the change from (u, p_0) to (u, v) is invertible. For $\ell = 31, 37, 41$ there are at most eight slope residues, since $R_{\mathrm{c}} a_i \equiv 0$; for $\ell \ge 43$ there are at most thirty-six. Choose distinct missing residues b_0, e_0 and order them so that $e_0 - b_0 \neq -T_0^{-1}$. Then

$$v = (e_0 - b_0)^{-1}, \qquad u = b_0 v$$

make every Ψ_b different from zero and minus one, and make $p_0 = v/T_0$ different from zero and minus one. Interchanging the two missing residues permits the required ordering, because ℓ is odd.

Choose a fixed positive cone $u/v > \max_i |A_i| + 2413$, $v > 0$. Enlarge the fixed initial cutoff past 82, every prime dividing D_0, and the nonzero determinants of the distinct forms. The inherited prime-pattern proof then applies with its same unused third coordinate and with complexity at most 35. For its fiber upper estimates, fixing $\Psi_{b_0} = p$ gives the same one-variable forms $\Psi_b = p + (K_{b_0} - K_b)p_0$. All coefficients, the cone, the denominator D_0 and the cutoff are fixed before the box grows. Thus the proof uses precisely the inherited fixed-system prime and successor estimates, not a uniform estimate in a growing denominator. It gives the positive ordinary primes and the same-prime elliptic-order conditions used there.

Now $r_0 = D_0^2 p_0$. At a prime $\ell \mid D_0$ outside S_*, both u and p_0 are units, $v_\ell(v) = 2v_\ell(D_0)$, and $v_\ell(A_i) \geq -v_\ell(D_0)$. Hence $w_i = A_i v - u$ is a unit. Both slopes have negative valuation $-2v_\ell(D_0)$, and their normalized infinite depths are even. No new normalized odd contact is created at such a denominator prime. At p_0, the odd valuation of r_0 is one.

For each occurrence choose its private comparison witnesses sufficiently deep that its norm root is a unit at its assigned prime. The remaining denominator valuations in the pole argument occur in the square of that root. The raw contacts are positive, their reduced denominators are squares, and they are squares in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$ by the retained residue c and the square factor D_0^2. Lemmas 15.10 and 15.11 supply the selected ordinary-pole witnesses. Equal prime occurrences use the same orientation and may use separate private points. No condition on reduction orders at different primes has been added. □

Lemma 15.4 (Reciprocal logarithm and its error)**.** *Put $f = 1/x$, $z = -x/y$, and define*

$$a_0 = 1, \qquad a_1 = -\frac{2}{5}, \qquad a_j = \frac{4j-5}{4j+1} a_{j-1} \quad (j \geq 2), \qquad G(w) = \sum_{j \geq 0} a_j w^j.$$

On the formal branch at the identity,

$$\log_E(z) = zG(Df^2).$$

For every $J \geq 0$ put $G_J(w) = \sum_{j=0}^{J} a_j w^j$. For every integer $10 \leq e \leq 64$, set $m = \lceil e/4 \rceil$ and

$$G_{m-1}(w) = \sum_{j=0}^{m-1} a_j w^j, \qquad L_e(z, f) = zG_{m-1}(Df^2).$$

Let $p > 76176$ be split and choose either place above it. If the point is nonidentity and lies in the formal group at that place, so $r = v_p(z) \geq 1$ and $v_p(f) > 0$, then

$$v_p\big(\log_E(z) - L_e(z, f)\big) \geq (4m+1)r, \qquad \frac{L_e(z, f)}{\log_E(z)} \in 1 + p^e \mathbb{Z}_p.$$

Proof. The curve equation gives

$$f = z^2(1 - Df^2), \qquad w = Df^2, \qquad Dz^4 = \frac{w}{(1-w)^2}.$$

Coefficient comparison in the defining recurrence gives

$$4w(1-w)G'(w) + (1+w)G(w) = 1 - w.$$

Indeed the coefficient of w is $5a_1 + a_0 = -1$, and each later coefficient gives $(4j+1)a_j = (4j-5)a_{j-1}$. Differentiating the curve identity yields $z\,dw/dz = 4w(1-w)/(1+w)$. Consequently

$$\frac{d}{dz}\{zG(w)\} = \frac{1-w}{1+w} = (1 + 4Dz^4)^{-1/2}.$$

The square root has constant term one. This is the normalized invariant differential for z, and $zG(w)$ has zero constant term, which proves the formal identity.

At the stated place, D is a unit and the formal branch gives $v_p(f) = 2r$, hence $v_p(w) = 4r$. The denominator of a_j, for $j \geq 1$, divides $5\prod_{k=2}^{j}(4k+1)$ and therefore divides $(4j+1)!$. Thus

$$v_p(a_j) \geq -v_p((4j+1)!) \geq -\frac{4j+1}{p-1}.$$

For $j \geq m$ with $4j+1 < p$, the coefficient is p-integral and $v_p(a_j w^j) \geq 4mr$. Otherwise $j \geq 19044$, so $3j \geq 4m$ and $(4j+1)/(p-1) \leq j$. Therefore

$$v_p(a_j w^j) \geq 4jr - j \geq 4mr.$$

These lower bounds tend to infinity. They control the entire infinite tail, including terms whose denominators are divisible by p. Multiplication by z proves the first estimate. Formal logarithm isometry gives $v_p(\log_E(z)) = r$, and proves the relative estimate. □

Lemma 15.5 (Global nonvanishing)**.** *For every $f \in K$ and every $J \geq 2$,*

$$G_J(Df^2) \neq 0.$$

In particular $L_e(z, f) \neq 0$ whenever $z \neq 0$ and $10 \leq e \leq 64$.

Proof. Let ν_2 be the rational exponent-of-two valuation and put

$$v(a + b\sqrt{2}) = \tfrac{1}{2}\nu_2(a^2 - 2b^2) = \min\{\nu_2(a), \nu_2(b) + \tfrac{1}{2}\}.$$

The two terms in the norm have valuations of opposite parity when finite. Norm multiplicativity and the displayed minimum prove the valuation identities. Its finite values lie in $\frac{1}{2}\mathbb{Z}$, and $v(D) = 0$. Every a_j with $j \geq 1$ has valuation one, because $a_1 = -2/5$ and all subsequent recurrence factors have odd numerator and denominator. If $f = 0$, the assertion is immediate. If $t = v(f) \geq 0$, the constant term of $G_J(Df^2)$ is uniquely least in valuation. If $t < 0$, then $t \leq -1/2$ and the nonconstant valuations $1+2jt$ strictly decrease; the last is at most $1-J < 0$, below the constant term as well. In either case a uniquely least term proves nonvanishing. No good-reduction assumption at two is used. □

15.3 Conventions and inherited hypotheses

Retain the field and curve of Section 14:

$$K = \mathbb{Q}(\sqrt{2}), \qquad d = 3 - \sqrt{2}, \qquad D = d^2, \qquad E:\ y^2 = x^3 - Dx.$$

For an affine point with nonzero abscissa write $f(P) = 1/x(P)$ and $p(P) = y(P)/x(P)$. The rational function $z = -x/y$ extends with value zero at O and $T_0 = (0, 0)$; its other poles are $(d, 0)$ and $(-d, 0)$. The letter J denotes a common comparison point. We write $\jmath$ for $1/z(J)$, to distinguish that field element from a point or an integer index. The implementation conjugates the complete alternate-comparison identity: its boundary value is $1/\overline{z(J)}$, and its primary and target coefficients in that identity are conjugated as well.

The following arguments retain all inherited qualifications of Section 14 and Appendix B. In particular, the restricted converse, rank and original arithmetic assertions retain their stated proof status. The prime-pattern input retains its corrected author-version and erratum conventions,

including the stated version of arXiv:1009.3998v5. The Pan interface retains the inspected author theorem and the cited Calegari formulation, without claiming comparison with inaccessible journal text. The Yuan input retains its author-attested translation qualification. The stable Faltings metric, arithmetic-Frobenius, diamond-quotient and final-twist conventions are unchanged. None of the finite source or polynomial checks strengthens these assumptions.

In particular we use, in exactly their previous scopes, the generator and torsion statements $2E(K) = \mathbb{Z}P_*$ and $E(K)_{\mathrm{tors}} = E[2](K)$; the formal-group logarithm and valuation statements; the fixed-system prime-pattern theorem with arbitrary prescribed finite exclusions; the height, scalar, anchor and Lucas lemmas; and the restricted-converse and simultaneous-realization inputs used for the pole predicate. The one additional explicit trace application below is identified in Lemma 15.12.

Put

$$B_* = \prod_{p \le 76176} p, \qquad H = 2(2^{109}210^{1800})B_*^{19} + 128, \qquad \log_2 H < 2093966.$$

This is H_{19} of Theorem 15.2. The source parameters remain high order 40, beta order 12, scalar orders $(1, 1, 1, 1, 32, 32)$ and anchor constant $C = 3^{2^{25}}$.

We also retain Lemma 15.4. Explicitly, write

$$\sqrt{1-w} \sum_{j \ge 0} \frac{\binom{2j}{j}}{4^j(4j+1)} w^j = \sum_{j \ge 0} a_j w^j, \qquad G_m(w) = \sum_{j=0}^{m} a_j w^j.$$

The polynomials used here are G_9 and G_2. They are nonzero at every argument Df^2 with $f \in K$, by the previously proved dyadic argument. For a nonidentity formal-group point at either place above a split prime $p > 76176$, the approximation $-G_9(Df^2)/p(P)$ has relative order at least 40, and $-G_2(Df^2)/p(P)$ has relative order at least 12. Actual formal membership is required: small z alone is not the premise of this lemma.

15.3.1 The retained scalar, anchor and beta interfaces

The scalar input is still $\beta = X^2 + 2$. The scalar batch tests the two rational coefficients each of $f(J_0), f(S), z(S)/z(J_0)$, with orders $(1, 1, 1, 1, 32, 32)$ in that order. Its original pole predicate is retained. The anchor constant is computed by

$$c_0 = 3, \qquad c_{i+1} = c_i^2 \quad (0 \le i < 25), \qquad C = c_{25}.$$

Thus the integer $C = 3^{2^{25}}$ is a witness value, not a polynomial coefficient. The four-witness anchor below consumes this same terminal value. The height targets are the eight rational abscissa coefficients of Lemma 15.3.

For every beta-bearing comparison, the common and beta quotient equations are explicitly

$$\Lambda p_H = G_H, \qquad B_A = C_A p_A, \qquad B_A \Lambda = G_A.$$

The global nonvanishing in Lemma 15.5 forces $\Lambda, p_H, B_A, p_A \neq 0$. Hence C_A is the ratio of the beta and common approximate logarithms, with their two minus signs cancelled. The beta residual is $C_A - 2\beta$. In full comparisons G_H has order 40 and G_A order 12; in beta-only comparisons both have order 12. The Lucas numerators use the normalized equations below, whose permitted zero-ordinate branches must not be excluded by this beta argument.

15.4 Selective comparisons and the local elliptic chart

Lemma 15.6 (Three-mode coverage)**.** *Number the unchanged raw contacts from* 0 *to* 36*. Retain full comparisons at* $0, \dots, 4$ *and* $21, \dots, 28$*, beta-only comparisons at* $5, \dots, 12$*, and Lucas-only comparisons at* $13, \dots, 20$ *and* $29, \dots, 36$*. The contacts carrying Lucas information cover the three points* S, T, Z*, and those carrying beta information cover* A, T*.*

Proof. The first five positions are shared. Each of A, S, T, Z has its original block of eight finite positions, in that order. Thus the Lucas set contains the five shared positions and all three finite blocks of S, T, Z; the beta set contains the five shared positions and the two finite blocks of A, T. The unchanged two-slope rational-height argument applies separately to these sets. No position is deleted when two slopes, raw values, or assigned primes coincide. The two maxima used below are consequently $B_L = \max(|s|, |t|, |z|)$ and $B_{AT} = \max(|n|, |t|)$. □

Lemma 15.7 (Doubled formal comparisons)**.** *At a split good prime above the cutoff, if* $v(z(R)) > 0$*, then either* R *or* $R + T_0$ *is in the formal group. Whenever the relevant rational function is defined,*

$$v(\log_E([2]R)) = v(z(R)),$$

with infinite valuation for $R = O$ *or* T_0*.*

Proof. For an affine nontorsion point, $z(R)^2 = x(R)/(x(R)^2 - D)$. If $v(x) = 0$, the right side has nonpositive valuation. If $v(x) < 0$, the point is formal. If $v(x) > 0$, translation by T_0 gives $x' = -D/x$ and $z' = -z$, so the translated point is formal with the same parameter valuation. Doubling kills T_0 and is an isometry on the formal group at an odd good prime. The asserted equality follows from the retained logarithm isometry. The two other nonzero two-torsion points are poles of z and are not in the domain. □

Private primary points may therefore lie in $E(K)$ rather than $2E(K)$. If $[2]P = [2\alpha(P)]P_*$, their coefficients $\alpha(P)$ may be half-integral. They are integral at each consumed odd prime. One compares logarithms only after writing an actual integer-multiple identity between formal points; torsion is not discarded by equating these half-indices. The primary f tests still put the primary points themselves in the formal group. An alternate test of $z(P - Q)/z(J)$, together with the opposite-place $f(J)$ test, implies $v(z(P - Q)) > 0$, and the lemma supplies the needed doubled logarithm.

Lemma 15.8 (Joint primary and difference graph)**.** *Let* $Q = (a, aq)$ *be nontorsion and* $\jmath \neq 0$*. Set*

$$F_Q = 1/a, \quad c = a + DF_Q, \quad r = \jmath/q, \quad s_Q = c/q, \quad t_Q = 1/q^2, \quad u = rs_Q.$$

The three field equations

$$k = (A - r)p, \tag{33}$$

$$kp + up - 2Drf - 4Dt_QA = 0, \tag{34}$$

$$2D(A + r)f + s_Qk + 4Dt_QA + \jmath q = 0 \tag{35}$$

give the exact graph $f = 1/x(P)$*,* $p = y(P)/x(P)$*,* $A = \jmath z(P - Q)$ *for affine primary points* P *with* $x(P) \neq 0$ *and finite* $z(P - Q)$*. In particular this primary chart excludes* O *and* T_0*. The equations permit* $A = 0$ *and do not require inverses of* A, k *or of a difference parameter.*

Proof. Set $x = p^2 + Df$. The target identity is $c^2 - q^4 = 4D$. Reversible linear combinations of the displayed equations give

$$AU = \jmath H_2, \qquad AV = \jmath H_1,$$

where

$$U = q(x + Df) + cp, \quad V = q^2(x - Df) - 4D, \quad H_1 = q(x + Df) - cp, \quad H_2 = x - Df - q^2.$$

Direct expansion yields

$$H_2V - H_1U = -4Dq^2(fp^2 + Df^2 - 1).$$

Thus $xf = 1$ and $P = (x, xp)$ is on E. If $U = V = 0$, then for $v \in \{d, -d\}$ one has $p = -2v/q$ and $x = v(a + v)/(a - v)$, precisely $P = Q + (v, 0)$. At either point $H_2 = (4D - q^4)/q^2 \neq 0$: equality $q^4 = 4D$ would imply $N_{K/\mathbb{Q}}(q)^2 = 28$, impossible for a rational norm. These genuine poles cannot satisfy the graph equations. Else U or V is nonzero and $A/\jmath$ is uniquely determined. The chord identities

$$HU = NH_2, \qquad HV = NH_1, \quad H = (p - q)(1 + DfF_Q), \quad N = pq(1 - DfF_Q) + 2D(f + F_Q)$$

identify it with $z(P - Q)$ on the chord domain and hence at every regular point. At $P = Q$ and $Q + T_0$ the value is zero. Opposition is covered by $V \neq 0$. At $p = 0$, the curve has $x = \pm d$, $U = 2qx \neq 0$, and the graph still gives its unique regular value. Conversely substituting any point and its regular difference value reverses these identities and recovers all three equations with $k = (A - r)p$. □

The last $p = 0$ cases are needed below. They are excluded at consumed places by the positive primary-f valuation, but need not be excluded globally for every normalized numerator.

15.5 Private conics, packed field tests and the ordinary pole

Each contact r has its own rational variables θ, s_c, S_c with

$$S_c = s_c^2, \qquad \theta^2 - rS_c = 2.$$

In particular $r, s_c \neq 0$. For a field value $F_i = a_i + b_i\sqrt{2}$ of required order e_i, define

$$Y_i = a_i + \theta b_i, \qquad T_i = 2a_iY_i - a_i^2 + 2b_i^2 = Y_i^2 - (\theta^2 - 2)b_i^2, \qquad r^{2e_i}W_i = T_i.$$

The upper tree uses successive pairs $N_r(u, v) = u^2 + rv^2$, starting with the literal leaf 1 followed by the W_i in stable order, carrying an unpaired final child unchanged. Denote its root by W.

Lemma 15.9 (Local conic and tree deductions)**.** *Assume the ordinary pole gives no negative odd valuation to nonzero rW^2 outside its fixed exceptional primes. At any $p > 76176$ with odd $R = v_p(r)$, the equations above imply $R > 0$ and choose one root $\rho \in \mathbb{Q}_p$ of 2 such that $v_p(a_i + \rho b_i) \geq e_i$ for every retained field.*

Proof. The conic makes $\delta = v_p(\theta^2 - 2) = R + 2v_p(s_c)$ odd. It cannot be negative, since then $v_p(\theta^2)$ would be odd; it cannot be zero. Thus $\delta > 0$, θ is a unit, and one root ρ satisfies $v_p(\theta - \rho) = \delta$ while the other factor is a unit. The summands in $N_r(u, v)$ have opposite valuation parity and cannot cancel. The literal leaf proves $W \neq 0$. If $R < 0$, ascent along that leaf gives $v_p(W) \leq 0$, contradicting the pole conclusion. Hence $R > 0$. The pole conclusion gives $v_p(W) > -R/2$. Descent through the tree uses

$$v_p(N_r(u, v)) > -cR \implies v_p(u) > -cR/2, \quad v_p(v) > -(c + 1)R/2,$$

with $0 \le c < 1$, and gives $v_p(W_i) > -R$. Consequently $v_p(T_i) > (2e_i-1)R \ge 2e_i-1$, so $v_p(T_i) \ge 2e_i$. The two summands of T_i also have opposite parity, yielding $v_p(Y_i) \ge e_i$ and $\delta + 2v_p(b_i) \ge 2e_i$. It follows that $v_p((\theta-\rho)b_i) \ge e_i$ and hence the claimed field order. Zero summands are treated with infinite valuation. No integrality of a_i, b_i and no global squareclass of F_i was assumed. □

The proof uses $v_p(T_i) \ge 2e_i$, not the generally unjustified stronger bound $2e_iR$. It is independent of the elliptic meaning of the fields.

Lemma 15.10 (Conic completeness)**.** *In the retained integral-input construction, each contact $r = d_0^2 p_0 > 0$ admits the needed private conic and arbitrarily deep approximation to its chosen split branch, independently of the other contact occurrences.*

Proof. Here $p_0 \equiv 1 \pmod 8$ and $d_0 \neq 0$ is a p_0-unit. The ring $\mathbb{Z}[\sqrt{2}]$ is norm Euclidean: subtracting nearest integers from two coefficients leaves $|a|, |b| \le 1/2$ and $|a^2 - 2b^2| \le 1/2 < 1$. The split prime ideal therefore gives integers a, b with $a^2 - 2b^2 = p_0$, after multiplying by the norm-minus-one unit if needed. Since $b \neq 0$, $(\theta_0, s_0) = (a/b, 1/(d_0 b))$ is a rational conic point. The parametrization

$$\Delta = \frac{2(rs_0 - \theta_0 t)}{t^2 - r}, \qquad s_c = s_0 + \Delta, \quad \theta = \theta_0 + t\Delta$$

and rational approximation give simultaneous approximation at the finite set of required places. There is no rational pole $t^2 = r$. Near the selected root, $\delta = 1 + 2v_{p_0}(s_c)$ can be made arbitrarily large. After all field coefficients are fixed, choose it so that $\delta + 2v_{p_0}(b_i) \ge 2e_i$ for every i. The original field orders then make all W_i integral; the constant path makes W a p_0-unit. At the real place the tree is positive, even if some W_i are negative. The extra approximations used for the following pole may be imposed at the same time. No conic variable is shared between occurrences. □

Proposition 15.11 (Linear-unit ordinary pole)**.** *Set $g = 3 - 2\theta$. Replace only the ordinary pole module by*

$$\ell = gW, \quad z_\ell = \ell^2, \quad c_r = re, \qquad p_r^2 - 17r - 34c_r + 51ec_r = 0,$$
$$w = 17e - c_r z_\ell, \quad j_s = s_r^2, \quad k_r = rq_r,$$
$$j_s^2 + 34 j_s w - 867 w^2 - 17 q_r k_r = 0.$$

It has the one-sided pole conclusion used in Lemma 15.9. It has integral-input completeness under the same inherited realization and rank hypotheses. The scalar-height pole module is unchanged.

Proof. At an odd prime with odd $v_p(r)$, the conic argument makes θ integral, and $(3-2\theta)(3+2\theta) = 1 - 4rs_c^2$ is a unit. Thus g is a unit. Suppose $b = rW^2 \neq 0$ has negative odd valuation outside $2, 3, 17$. The first conic $p_r^2 = 17r(1-e)(1+3e)$ forces e to be a unit: positive, negative or infinite valuation of e would make the right side have odd valuation. Since $v_p(\ell) = v_p(W)$, one has $v_p(w) = v_p(b) < 0$, an odd integer. In

$$(s_r^2 - 17w)(s_r^2 + 51w) = 17 r q_r^2$$

the left side is nonzero with even valuation, since the valuations of s_r^2 and w differ in parity. The right side is zero or has odd valuation. This contradiction proves the asserted exclusion, including zero ordinate branches of the first conic. If $W = 0$, the conclusion has no nonzero pole input; explicit witnesses are $\ell = z_\ell = p_r = q_r = k_r = 0$, $e = 1$, $c_r = r$, $w = 17$, $s_r = 17$, $j_s = 289$.

For completeness retain the generalized fixed-U realization argument under its original hypotheses. For $b > 0$ with square reduced denominator and square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, it applies to a prescribed U_0 when

$$U_0^2/(17b) < 1 \text{ at infinity}, \quad v_2(U_0^2/(17b)) \ge 3, \quad v_3(U_0^2/(17b)), v_{17}(U_0^2/(17b)) \ge 1,$$

and $v_p(U_0) = v_p(b)$ at every other odd-support prime of b. Indeed, put $B = 17b$ and $V = B(B-U_0^2)$, and let $\beta_{\mathrm{sf}}, \eta_{\mathrm{sf}}$ be their positive squarefree representatives. The displayed conditions make V positive and a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$. Consequently

$$\beta_{\mathrm{sf}} > 1, \quad \beta_{\mathrm{sf}} \equiv 17 \pmod{24}, \quad \eta_{\mathrm{sf}} \equiv 1 \pmod{24}, \quad 17 \nmid \eta_{\mathrm{sf}}.$$

At an odd-support prime of B outside $2, 3, 17$, the square reduced denominator makes $v_p(b)$ positive, and the hypothesis on U_0 gives $v_p(U_0^2/B) > 0$. Thus $1-U_0^2/B$ is a square unit, so η_{sf} is a square in $\mathbb{Q}_p$ and $p \nmid \eta_{\mathrm{sf}}$. The same conclusion at 17 follows from the displayed smallness condition. Conversely, at a prime dividing η_{sf} the necessary strict cancellation in $B - U_0^2$ gives $v_p(B) = 2v_p(U_0)$ and makes B/U_0^2 a square unit. Thus β_{sf} is a square in $\mathbb{Q}_p$. In particular $\gcd(\beta_{\mathrm{sf}}, \eta_{\mathrm{sf}}) = 1$, and both mutual-square conditions hold. If $U_0 = 0$, then $V = B^2$ and $\eta_{\mathrm{sf}} = 1$, so the converse condition is empty. These are precisely the hypotheses consumed by the inherited simultaneous-realization proof; no new rank or converse statement is substituted.

Now set $b = rW^2$ and $U_0 = rW^2 g$. In the complete construction $r = d_0^2 p_0$ and W is a p_0-unit, so $v_{p_0}(b) = 1$ and all other valuations of b are even. Hence its reduced denominator is a square. The retained local squares of r at $2, 3, 17$ also make b a square at those places. After the fields are fixed, approximate the private conic also near $(\theta, s_c) = (3/2, 1/(2\sqrt{r}))$ at infinity, $2, 3, 17$. These local points exist by the original contact construction. The continuous function $(r/17)W(\theta)^2 g(\theta)^2$ vanishes there, so it satisfies the displayed smallness conditions while the selected prime still makes W a unit. The local-unit argument gives the required odd-support valuations of U_0. The resulting old three-conic witnesses satisfy

$$p_o^2 = 17b(1-e)(1+3e), \quad h_o = e(17b - U_0^2), \quad (s_o^2 - 17bh_o)(s_o^2 + 51bh_o) = 17bq_o^2.$$

Here r, W, b are nonzero. Set

$$p_r = p_o/W, \quad w = h_o/b, \quad s_r = s_o/b, \quad q_r = q_o/(r^2W^3).$$

These substitutions give $w = e(17 - r\ell^2)$ and the final conic in the displayed normalized system; the other coordinates are its definitions. Thus the witness conversion uses only stated nonzero quantities. Equivalently the unnormalized form has $L = r\ell$, $z_L = r^2 z_\ell$, $w_L = r^2 w$, $s_L = r s_r$, $q_L = r^2 q_r$, $j_L = r^2 j_s$, $k_L = r^2 k_r$, with e, p_r, c_r unchanged. All divisions in the reverse scaling are powers of the conic-proved nonzero r. This also proves that normalization loses no zero branch. □

Lemma 15.12 (Four-witness anchor)**.** *For rational a, b, both belong to $\mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}$ if and only if $9 + 3(a^2 + b^2)$ is a norm-one trace for the quaternion algebra $(2, 3)$. Consequently the original two-coefficient anchor is equivalent to*

$$CH - 9C^2 - 3f_0^2 - 3f_1^2 = 0, \qquad H^2 - 8U^2 - 12V^2 + 24W_a^2 - 4C^2 = 0.$$

Proof. By Definition 4, Observation 5 and Facts (a), (b), (c), (e) in the proof of Proposition 6 of [83], the algebra is ramified exactly at $2, 3$, the trace set is unrestricted at the other places, and it is integral at its two ramified places $2, 3$, contains integral traces with residues 1 at 2 and 0 at 3, and is determined by these local trace conditions. These are the explicit trace facts used here, in addition to the older sum-of-traces criterion. If $m = \min(v_p(a), v_p(b)) < 0$, then at 3 the sum of squares has valuation $2m$, while at 2 it has valuation $2m$ for unequal valuations and $2m + 1$ for equal finite valuations. Thus $9 + 3(a^2 + b^2)$ would be nonintegral, proving one implication, with zeros covered by infinite valuation. Conversely integral a, b give a trace congruent to zero at 3. At 2 it is odd or belongs to $4\mathbb{Z}_2$. For the latter case put $x_0 = h/2 \in 2\mathbb{Z}_2$, where h is this trace. If

$x_0 \equiv 2 \pmod 4$, then $(x_0^2-1)/3 \equiv 1 \pmod 8$ is a square, giving $x_0^2 - 3v^2 = 1$. If $x_0 \equiv 0 \pmod 4$, then $1 - x_0^2/6$ is a square, and $u = v = 1$, $w^2 = 1 - x_0^2/6$ gives $x_0^2 - 2u^2 - 3v^2 + 6w^2 = 1$. The local trace criterion supplies rational witnesses. Finally divide the two homogeneous equations by C^2, which is allowed because the retained constant C is nonzero. This is an equality of existential projections, not a rational formula choosing quaternion witnesses. □

15.6 Local and normalized Lucas tests

Write

$$\mathcal{F}(s,t,z) = t^2 - s(s-2)tz + (s-2)^2z^2 - 1.$$

Lemma 15.13 (Local Lucas residual)**.** *The primary and alternate logarithm comparisons, before any proxy equality is used, give local integral quotients C_s, C_t, C_z congruent to the integer root indices modulo p^{40}. An order-40 test of $\mathcal{F}(C_s, C_t, C_z)$ implies $p^{40} \mid \mathcal{F}(s,t,z)$.*

Proof. The implication follows by substitution into the polynomial with integer coefficients. One may evaluate the field residual using

$$V = (C_s - 2)C_z, \quad D_L = C_sC_t - V, \quad E_L = C_t^2 - VD_L - 1.$$

These are quadratic equations, and $E_L = \mathcal{F}(C_s, C_t, C_z)$. They replace the three proxy comparisons and the global proxy equation. They do not impose $E_L = 0$ as a field equality. Removal of the proxy coordinates requires their exclusive incidence in that equation and those comparison arguments. Integer completeness and the reverse input implication use the two height contradictions below; no conversion of arbitrary old proxy witnesses is asserted. □

Lemma 15.14 (Normalized local Lucas evaluation)**.** *Let $\Lambda p_H = G_H$ be the common negative-logarithm equation. For the three Lucas numerators replace their quotient blocks and the preceding evaluation by*

$$\begin{aligned} B_i\Lambda = G_i \quad (i = s,t,z), \qquad PB_s = p_s, \\ (U+V)p_t = PB_t, \qquad K_Lp_t = (1-2P)B_t, \\ Vp_z = (1-2P)B_z, \qquad E_L = U^2 - K_LV - P^2. \end{aligned}$$

An order-40 test of E_L has the same required integer consequence. Every integer input still admits the changed witnesses.

Proof. The global nonvanishing of the retained G_i implies $\Lambda, p_H, B_s, B_t, B_z \neq 0$. Also $p_t \neq 0$: otherwise the two equations with left factor p_t would force both $P = 0$ and $1 - 2P = 0$. At a consumed place all primary points are formal, so their p_i are nonzero. Put $C_i = B_i/p_i$. Then $P = 1/C_s$ and direct substitution gives

$$E_L = P^2\mathcal{F}(C_s, C_t, C_z).$$

Since C_s is locally integral, the implication $v(E_L) \geq 40 \Rightarrow v(\mathcal{F}(C_s, C_t, C_z)) \geq 40$ is valid even when C_s is not a unit. No local denominator has been discarded.

For completeness the integer $s \neq 0$ and all root points are fixed before the prime-pattern theorem is used. Its prescribed finite-exclusion version allows every chosen prime to avoid the divisors of s. Then C_s is a unit at the consumed place. The old order-40 residual implies the new one, and the displayed equations assign the new witnesses. This imposes no compatibility between group orders at different primes.

Globally p_s or p_z may vanish at an occurrence not being consumed in the local argument. Such points have $x = \pm d$ and are covered by the regular extension in Lemma 15.8; they must not be excluded by reusing an earlier blanket nonzero assertion. At a consumed place $C_s = 2$ gives $P = 1/2$ and $V = K_L = 0$, since both p_t and p_z are nonzero there. No division by $1-2P$ is needed. Globally the branch $P = 1/2$, $p_z = 0$ forces $K_L = 0$ but need not force $V = 0$; the latter value may be chosen in the completeness construction. □

The retained field lists can now be stated without proxy coordinates. In a full occurrence there are the five primary f values, the common $\overline{f(J)}$, the beta residual, E_L, and five alternates. In a beta-only occurrence there are two primary f values, the common field, the beta residual and two alternates. In a Lucas-only occurrence there are four primary f values, the common field, E_L and four alternates. These lists have lengths $13, 6, 10$. The formal fields all have order one. The beta residual and beta alternate have order 12; the Lucas residual and Lucas alternates have order 40. The denominator alternate has order $e_0 = 40$ in full and Lucas-only occurrences, and $e_0 = 12$ in beta-only occurrences. Its G precision is the same e_0.

Theorem 15.15 (The two height contradictions)**.** *Retain the inherited scalar and contact-height conclusions, and the integer root indices n, s, t, z of the chosen carrier, with $s, z \neq 0$. Let $\beta = X^2+2$ and $\beta' = 2\beta$. Suppose the local tests above give $C_i \equiv n_i \pmod{p^{e_i}}$ at all consumed contacts, with beta residual $C_A - \beta'$. Put $H_* = 5H$. Suppose*

$$n_Q^2 - 6 < 7H_*N^{19/e}, \qquad h(\beta') < a/40, \qquad \log(40H_*) < a/40, \qquad a = |s| > (24H_*)^{10},$$

where the first inequality is the contact-height bound whenever the appropriate radical powers divide N. Then the changed arithmetic source has the same integer-input conclusion as Version 3.

Proof. If $\Delta = \mathcal{F}(s,t,z) \neq 0$, every relevant order-40 radical power divides $|\Delta|$. By three-mode coverage, all three heights in $B_L = \max(|s|,|t|,|z|)$ satisfy the assumed inequality. Since $|\Delta| < 5B_L^4$, the same elementary estimate as in Version 3 gives

$$B_L^{2-76/40} < 24H_*.$$

This contradicts $B_L \geq a > (24H_*)^{10}$. Therefore $\Delta = 0$. The displayed lower bound on a gives $|s| \geq 7$, and $z \neq 0$ was retained. The Lucas lemma therefore gives the stronger individual conclusion $|t| > \exp(a-3)$, not merely a bound on $\max(|t|,|z|)$.

If $\beta' \neq n$, put $N = |\operatorname{num}(\beta' - n)|$. The beta contacts cover A, T, so with $B = \max(|n|,|t|)$ one has

$$B^2 < 7H_*N^{19/12} + 6, \qquad N \leq 2e^{h(\beta')}B.$$

Thus $B^{2-19/12} < 40H_*e^{(19/12)h(\beta')}$ and

$$\log B < \frac{12+19}{40(24-19)}a = \frac{31}{200}a < a - 3,$$

contrary to $B \geq |t|$. Hence $\beta' = n$ is an integer. $\beta' = 2\beta = 2(X^2+2)$; writing $X = u/v$ in lowest terms gives $v^2 \mid 2$, hence again $v = 1$. □

15.7 General root carrier

Lemma 15.16 (Integer doubled root indices)**.** *Let $R = (1+\sqrt{2}, 2\sqrt{2})$, so direct duplication gives $2R = P_*$. Every $Q \in E(K)$ has a unique integer n_Q with $2Q = n_QP_*$. It is nonzero exactly when Q is nontorsion. For such a point,*

$$h(x(Q)) > n_Q^2/4 - 4.$$

Proof. The inherited generator assertion gives $2Q = n_Q P_*$. The torsion assertion makes its kernel exactly $E[2](K)$, so equivalently $Q = n_Q R + \tau$, $\tau \in E[2](K)$. Uniqueness follows from the infinite order of P_*. The unchanged height bounds give

$$n_Q^2 - 6 \le h(x(2Q)) \le 4h(x(Q)) + \log 676.$$

Since $(6 + \log 676)/4 < 4$, the stated inequality follows. □

The root points A, S, T, Z, J_0, J may now be any nontorsion points of $E(K)$. Their indices always mean the integer doubled indices of the lemma. They are distinct from the half-integer coefficients used for private points in the earlier carrier convention.

Lemma 15.17 (Small general-root charts)**.** *For A, S, T, Z retain x, y and the actual first-doubling square $q = x^2$, and impose $y^2 = xq - Dx$. For S also retain $xf = 1$, $yz + x = 0$. Keep the nonzero tangent denominator, or the direct tangent equation of Lemma 15.19. For each of J_0, J use f, z, k with*

$$k = fz, \qquad f = z^2 - Dk^2,$$

with $z \neq 0$ enforced by its retained inverse. These charts reconstruct exactly the required nontorsion root points.

Proof. The first equations are the curve equation with a reused square. The nonzero tangent denominator excludes all points with $y = 0$, hence all torsion. The two extra S equations preserve its reciprocal and parameter. In the other chart, $z \neq 0$ implies $f \neq 0$: $f = 0$ would force $k = 0$ and then $z = 0$. Set $x = 1/f$, $p = -1/z$, $y = xp$. Dividing the second equation by z^2 gives $fp^2 = 1 - Df^2$, so $P = (x, y)$ lies on the curve and $xy \neq 0$. Conversely every nontorsion point gives these witnesses. These are rational maps on the stated nonzero domains. The new equations cannot justify deleting an unrelated guard without its incidence proof. □

For clarity, the inherited scalar argument still applies to the new indices. At each odd contact prime of β outside S_*, the constant-left scalar norm tree excludes negative odd raw valuation and gives all six assigned coefficient orders. The first four put S, J_0 in the formal group at every place above that prime. Writing $2S = sP_*$ and $2J_0 = j_0 P_*$, with nonzero integer j_0, the last two give $v_p(s/j_0) \ge 32$, hence $v_p(s) \ge 32$. The logarithm comparison uses the exact identity $[2j_0]S = [2s]J_0$. It is valid also at inert primes: both rational coefficients are tested and the extension is unramified outside two. The unchanged scalar raw polynomial and its small-prime correction detect every prime of $M_5(\beta)$, including an odd infinite contact. Consequently

$$M_5(\beta)^{32} \mid s, \qquad h(\beta) \le H M_5(\beta)^{19} \le H|s|^{19/32}.$$

The height theorem's domain condition holds. The value $\beta = X^2 + 2$ is none of $0, 1, -343/32$, and $\beta = 2401$ would require the nonsquare integer 2399 to be a rational square. Doubling both indices therefore changes neither the scalar ratio nor its divisibility argument.

Take $H_* = 5H$ and $\beta' = 2\beta$. The unchanged contact-height bound $h(x(Q)) < 7HN^{19/e}$ and the preceding lemma give

$$n_Q^2 - 6 < 28HN^{19/e} + 7 < 35HN^{19/e} = 7H_* N^{19/e}.$$

The scalar ratio is still s/j_0, since doubling both indices cancels the factor two. Its order-32 tests give $h(\beta) \le H|s|^{19/32}$. The full anchor at both 2 and 3 is retained. Its 3-adic consequence, using $2S = sP_*$, is

$$s \in 2 \cdot 3^{2^{24}-1}\mathbb{Z} \setminus \{0\}, \qquad \log_2 |s| > 25165822.5.$$

Here formal doubling at 3 is an isometry and the retained calculation is $v_3(f(sP_*)) = 2 + 2v_3(s)$. The 2-adic part of the anchor is not replaced by this consequence. Since $\log_2 H_* < 2093969$, one has

$$|s| > (24H_*)^{10}, \qquad |s| > (40H_*)^{32/13}, \qquad h(2\beta) < |s|/40, \quad \log(40H_*) < |s|/40.$$

For example the first two required binary cutoffs are respectively $10 \cdot 2093974$ and $(32/13) \cdot 2093975$, both below the displayed anchor lower bound. Thus Theorem 15.15 applies with unchanged orders. Both beta-bearing comparators must now test $C_A - 2\beta$; the scalar definition $\beta = X^2 + 2$ and its raw scalar polynomial remain unchanged.

15.7.1 Doubled targets

Lemma 15.18 (Targets $2Q_i$ and P_*)**.** *The general-root construction may use targets $2Q_i = n_i P_*$ and fixed denominator target $P_* = 2R$, requiring only one doubling of each variable root. It preserves the local integer congruences and integer completeness.*

Proof. For soundness write $2X_0 = hP_*$, $2X_i = m_i P_*$ and $2J = j_d P_*$. The root J is nontorsion, so $j_d \in \mathbb{Z} \setminus \{0\}$. The doubled differences have indices $h - 2$ and $m_i - 2n_i$. Lemma 15.7 and the alternate tests give

$$v_p((h-2)/j_d) \geq e_0, \qquad v_p((m_i - 2n_i)/j_d) \geq e_i,$$

where $e_0 = 40$ in full/Lucas mode and $e_0 = 12$ in beta-only mode. Since j_d is integral, $h \equiv 2$ and $m_i \equiv 2n_i$ at those orders. In particular h is a unit at the odd prime. The exact identity $[2h]X_i = [2m_i]X_0$ permits logarithm comparison on the primary formal group, giving ratio m_i/h. The retained relative errors, with $e_i \leq e_0$, give $C_i \equiv n_i \pmod{p^{e_i}}$.

For completeness let $N_{p,1}, N_{p,2}$ be the two full finite group cardinalities supplied by the inherited ordinary-prime pattern. It gives $\gcd(N_{p,1}, N_{p,2}) = 4$ and $p \nmid N_{p,1} N_{p,2}$. Each reduction has exactly four rational two-torsion points, so its doubling image has order $M_{p,i} = N_{p,i}/4$. Consequently $M_{p,1}$ and $M_{p,2}$ are coprime and prime to p. Every reduction of a doubled global point is killed by the corresponding $M_{p,i}$; no global reduction-surjectivity or cyclicity claim is needed. Put $j_c = \operatorname{lcm}_p M_{p,2}$ and $J = j_c P_*$, whose doubled index is $j_d = 2j_c$. Separately for each contact choose

$$k \equiv 0 \pmod{M_{p,1}}, \quad k \equiv 1 \pmod{M_{p,2}}, \quad k \equiv 1 \pmod{p^{40+v_p(j_c)}}, \qquad k \notin \{0, 1, -1\}.$$

The three moduli are compatible. Set $X_0 = kP_*$ and $X_i = kn_i P_*$. The primary points are formal at the first place. Their differences from the chosen targets, and J, are formal at the second. The alternate logarithm ratios are $(k-1)/j_c$ and $(k-1)n_i/j_c$, while the primary ratios are n_i. This supplies every required order. The finite exclusions avoid zero points and the old open-chart exceptional cases. There is no simultaneous CRT condition on different contact occurrences. □

The fixed target can be specified directly by

	1	$\sqrt{2}$
$x(P_*)$	$57/8$	$-7/2$
$q = y(P_*)/x(P_*)$	$-87/41$	$129/164$
$F_Q = 1/x(P_*)$	$456/1681$	$224/1681$
$s_Q = (x(P_*) + DF_Q)/q$	$-55123/14637$	$19301/58548$
$t_Q = 1/q^2$	$7624/14161$	$39904/127449$

Direct substitution verifies the curve and target identities. The source stores x, q, F_Q and the conjugates of s_Q, t_Q. Its remaining coefficients are affine in $\jmath$:

$$r = \jmath\left(-\frac{232}{357} + \frac{86}{357}\sqrt{2}\right), \qquad u = \jmath\left(\frac{291673}{127449} - \frac{29440}{42483}\sqrt{2}\right),$$

when $\jmath = 1/\overline{z(J)}$. The ten constant bindings and four affine bindings replace the fixed derivation, subject to full incidence checks. Their isolated coefficient calculation does not bound the complete sum of squares.

15.8 Exact implementation refinements

Lemma 15.19 (Direct tangent equation)**.** *For a curve point with $q = x^2$, replace the inverse and slope equations $(2y)h = 1$, $\lambda = (3q - D)h$ by $2y\lambda = 3q - D$. This preserves all retained rational coordinates.*

Proof. The new equation and the curve force $y \neq 0$. If $y = 0$, then $x = 0$ or $x^2 = D$, and $3q - D$ is respectively $-D$ or $2D$, a contradiction. The unique inverse is restored by $h = (2y)^{-1}$. There is also the polynomial restoration

$$h = \frac{2\lambda(3q - 2D) - 9xy}{4D^2},$$

because $(3q - D)(3q - 2D) - 9x(x^3 - Dx) = 2D^2$. For a retained second doubling its input is on the curve: with $t = X - x$, $Y_0 = -y - \lambda t$, the identity

$$\begin{aligned} Y^2 - X^3 + DX = y^2 - x^3 + Dx + t(2y\lambda - 3x^2 + D) \\ + t^2(\lambda^2 - X - 2x) + (Y - Y_0)(Y + Y_0) \end{aligned}$$

proves curve preservation from the first doubling equations. Target-two sources have only the first four variable-root doublings; no second set of savings or equations is present in that branch. □

Lemma 15.20 (Shared field-square evaluation)**.** *For the common field $F = a + b\sqrt{2} = \overline{f(J)}$, bind once $c = a^2 + 2b^2$, $d_s = 2ab$. In each designated order-one packed field replace*

$$Y = a + \theta b, \qquad PW = 2aY - a^2 + 2b^2$$

by $PW = c + \theta d_s$, with the same $P = r^2$. This is an exact projection preserving W, every higher norm node and the pole witnesses.

Proof. Substitution gives $2a(a + \theta b) - a^2 + 2b^2 = c + \theta d_s$. The inverse map restores $Y = a + \theta b$, without division. All zero cases are included. The two coefficients describe F^2, but their packed use still enforces the original order-one condition on F; they must not be relabelled as a new order-one test of F^2. The same identity permits reuse of an already available primary-f square in its own packed numerator, after checking that square's defining rows. □

Lemma 15.21 (First packed denominator and norm fusion)**.** *For the first order-one primary field only, replace its virtual denominator $W_0 = T_0/r^2$ by $W_0 = T_0/r$, and its first norm $N = 1 + rW_0^2$ by*

$$T_0 = 2a_0Y_0 - a_0^2 + 2b_0^2, \qquad rN - r - T_0^2 = 0.$$

Keep every other denominator, the conic, all higher tree nodes and the ordinary pole. The arithmetic input projection is unchanged.

Proof. The earlier tree proof still gives $v_p(W_0) > -R$, so $v_p(T_0) > 0$. More generally $v_p(T) \geq 2e - 1$ suffices for an order-e field test: opposite parity gives $v_p(Y) \geq e$ and $\delta + 2v_p(b) \geq 2e - 1$, hence $\delta + v_p(b) \geq e + (\delta - 1)/2 \geq e$. Thus the first field still has order one, and every other deduction is unchanged. For completeness the old depth-two choice has $v_{p_0}(T_0) \geq 2$ and $R = 1$, so the new

virtual leaf is integral and the root remains a unit. For fixed nonzero r, all new tree values remain rational polynomials in θ, allowing the same exceptional-place pole approximation.

After that arithmetic change, the displayed fusion is exact: set $T_0 = rW_0$ forward and $W_0 = T_0/r$ backward. The retained conic proves $r \neq 0$. These maps preserve N and all later nodes, including $N = 0$ or $T_0 = 0$ and negative rational r. The unguarded case $r = 0$ would be invalid and is not admitted. The old r^2 coordinate remains for other leaves. The changed field is primary $f(H)$, not the separately shared common field $\overline{f(J)}$. □

15.9 Composition, completeness and finite realization

For integral input first set $\beta = X^2 + 2$ and take $A = \beta P_*$, whose doubled index is 2β. Choose J_0, S together in the old subgroup with the required scalar and full anchor divisibilities. Their doubled-index ratio is unchanged. The Lucas existence lemma then gives t, z for the now fixed s, and take $T = tR$, $Z = zR$. Fix the eight abscissa coefficients and denominator data before using the prime-pattern theorem. Exclude the divisors of the fixed nonzero s as well as every earlier finite exceptional set.

Next choose one common comparison point J using the stated least common multiple, then choose the private points separately for each occurrence, using precisely the target-two CRT rule. Construct all joint-graph, reciprocal-G, normalized-ratio and shared-square fields. Only then choose each private conic point and its norm/pole witnesses by simultaneous approximation. Repeated primes retain their common orientation but separate conic and private-point witnesses. The constructions above give every required equation. In the reverse direction the conic and pole decode field orders at every odd-contact prime, without an ordinariness assumption. The graph and doubled logarithm arguments give integer-index congruences, and the two height contradictions give the integer input. The exact fusions preserve the specified retained coordinates; arithmetic source replacements use the full soundness and completeness argument rather than a claim about arbitrary old private witness tuples.

Finally each selected source must be compiled afresh into rational quadratic residual equations with the original input coordinate retained. For real, hence rational, values the all-weights-one sum of their squares vanishes exactly when every residual vanishes. Clearing nonzero rational row denominators preserves those zero sets and gives an integral polynomial of degree at most four. This elementary last step does not prove the inherited arithmetic inputs. The exact source, complete flattening map, allowed algebraic transformations, and full coefficient collection must be checked for the single selected endpoint. None of the local identities, allocation counts or isolated coefficient bounds in this appendix is an assertion of that final check.

Theorem 15.22 (The selected arithmetic source)**.** *With precisely the inherited arithmetic hypotheses and source-version qualifications stated above, the source consisting of the selected equations has rational input projection $\mathbb{Z}$. Its input coordinate is the original rational number X.*

Proof. First suppose that rational witnesses satisfy the source. At every prime above the cutoff with an odd raw-contact valuation, the private conic, linear-unit pole and constant-left norm tree give the required field orders, by Lemmas 15.9 and 15.21. This argument starts at arbitrary odd-contact primes and does not assume that they are ordinary. The joint graph supplies actual primary points and their regular difference parameters. Positive primary-f valuations put the primary points in the formal group; the common conjugate-f test and each alternate test allow Lemma 15.7 to be applied to the doubled difference. Lemma 15.18, with denominator order 12 in beta-only mode and 40 otherwise, gives the integer-index congruences for each retained numerator. The beta quotient equations and Lemma 15.14 therefore give the precise local tests used in Theorem 15.15.

The retained scalar system gives its order-32 divisibility, while the four-witness anchor is equivalent to the original two-place condition. The general-root height and anchor estimates establish every cutoff of that theorem with $H_* = 5H$. The three-mode position sets cover exactly the two maxima needed in its proof. It follows first that $\mathcal{F}(s,t,z) = 0$, then that $2\beta = n \in \mathbb{Z}$. Write $X = u/v$ in lowest terms with $v > 0$. Since $2X^2 = 2\beta - 4$ is integral, v^2 divides 2; hence $v = 1$ and $X \in \mathbb{Z}$.

Conversely fix any $X \in \mathbb{Z}$. Set $\beta = X^2 + 2$ and $A = \beta P_*$. Choose the scalar pair S, J_0 in the retained subgroup using the inherited scalar and full anchor construction, and choose t, z by the inherited Lucas existence lemma for the resulting nonzero integer doubled index s. Set $T = tR$, $Z = zR$. All four roots and the two common roots are nontorsion and admit the general-root charts. Fix their rational abscissa coefficients and denominator data before applying Lemma 15.3. Its fixed-system prime theorem permits the finite exclusion of divisors of s, as required by the normalized Lucas construction.

Choose one common J by the least common multiple in Lemma 15.18; at each occurrence choose its private points using that lemma's three compatible congruences. No congruences between different occurrences are imposed. The joint graph, exact reciprocal evaluators, beta quotients and normalized Lucas equations then have the stated witnesses and local orders. The explicit order of choices in Section 15.9 now chooses each private conic after its field coefficients are fixed. Conic approximation supplies the integral packed leaves and a unit norm root; the fixed-parameter pole realization supplies its remaining witnesses. This remains true with the first denominator changed to r. The shared square and direct tangent equations are exact identities with the reconstruction maps proved above. Thus every equation of the selected source has a rational solution at this input. This proves both implications without identifying arbitrary old and new private witness tuples. □

The finite evaluations used here can themselves be written quadratically. For precision 40 bind four field pairs

$$w = Df^2, \qquad w_2 = w^2, \qquad w_3 = w_2 w, \qquad U_6,$$

with the fourth pair defined by the quadratic equation

$$U_6 = \frac{a_3 + a_4 w + a_5 w_2 + w_3(a_6 + a_7 w + a_8 w_2 + a_9 w_3)}{a_6}.$$

The quadratic output expression is

$$G_9(w) = a_0 + a_1 w + a_2 w_2 + a_6 w_3 U_6, \qquad a_6 = -\frac{418}{16575} \neq 0.$$

For precision 12 bind only $w = Df^2$ and use $G_2(w) = 1 + a_1 w + a_2 w^2$. Substitute these expressions directly into $\Lambda p_H = G_H$ and the beta or normalized-Lucas equations $B_i \Lambda = G_i$. Every product is quadratic in the bound rational coordinates. Expansion gives each required a_j coefficient exactly. The scale is one; no finite-error estimate is used as a substitute for these polynomial identities.

15.10 Exact source specification and audit boundary

The source just proved has 4 935 rational coordinates, including input zero, and 4 513 residual equations. Its distinct name in `quartics/quartic-4973/source-entry.json` is

`target2-tangent-first-norm-normalized-linear-pole-k40-q12-a25-h19-s32-4a82d9a8dc1b`

The compressed source `quartics/quartic-4973/source-residuals.jsonl.gz` has SHA-256

94f8a1972cbfc0cf6493d418e628a18482307dec5c427db72e86d3eacb1a6f5d

Its uncompressed bytes have SHA-256

4a82d9a8dc1b5c92eb40a49751c0a6aae073c8f765ca5e276a74802fd56b84be

The source contains 11 modules. Its complete construction map is `quartics/quartic-4973/construction-map.json`. The independent source audit checks the full module equations, exact ordered expression lists and contact positions, root and scalar interfaces, all retained guards, the anchor terminal coordinate and complete flattening. It is recorded in `quartics/quartic-4973/source-audit.json`. The source-auditor SHA-256 is

58bf1da4007247bc4558eacce6bdbbc83d2394ac2a47f128390f20f26567d50a

Those checks establish finite source identities. The source's rational input projection is proved in Theorem 15.22, with the inherited qualifications unchanged. A source digest is not an arithmetic proof, and the historical base compiler remains construction machinery rather than another final polynomial.

15.11 Quadratic reductions preserving the rational input projection

We describe the algebraic reductions applied to the new, fixed source. All scalar coordinates range over $\mathbb{Q}$, and coordinate zero is the original input X. Each field equation denotes its two rational coefficient equations. These reductions prove equality of rational input projections to that source; its arithmetic interpretation is Theorem 15.22.

Definitions, row operations, and interfaces. An equation $cv - F = 0$, with $c \in \mathbb{Q}^*$ and F independent of v, permits substitution $v = F/c$ in every other equation and deletion of v. Restriction is the forward solution map, and adjoining $v = F/c$ is its inverse. Each use below retains degree at most two after substitution. The initial quadratic lift uses the reverse operation: introduce targets for products of two available coordinates until the source residuals are quadratic. Evaluating these product definitions and forgetting their targets give the two solution maps. A previously defined product can be reused only with its retained defining equation.

An operation replacing R_i by $aR_i + bR_j$, while retaining R_j, has inverse $R_i = (R_i' - bR_j)/a$ when $a \neq 0$ and $i \neq j$. Thus every recorded linear row operation preserves the simultaneous zero set. A shear replaces an old witness by $v = v' + cw$, with fixed rational c and another witness w retained. Its forward and reverse coordinate maps are $v' = v - cw$ and $v = v' + cw$. This is an invertible rational coordinate change and preserves quadratic degree.

Every removed internal coordinate must have no occurrence outside the replaced rows, unless the stated substitution is made in *every* such row. Retained coordinates keep their identities and values. All new coordinates are fresh, and input coordinate zero is never a target of deletion or redefinition. These conditions apply to the current system after all preceding transformations.

Elimination modulo retained equations. For a recorded relation elimination, a current defining row has the form $x - N(y) = 0$ up to nonzero rational scale, with N quadratic and independent of the deleted witness x. For each other affected row $F_j(x, y)$ the certificate gives an exact polynomial identity

$$F_j(N(y), y) = G_j(y) + \sum_k M_{jk}(y) H_k(y),$$

up to the recorded nonzero row scales. Every H_k is an unchanged, x-free row, and every new row G_j is at most quadratic. Forgetting x takes an old solution to a new one, since the H_k

vanish. Conversely adjoin $x = N(y)$: the same identities recover every affected $F_j = 0$. Complete occurrence checks ensure that no other row uses x. This is an identity over $\mathbb{Q}$, not a division by a possibly zero field value. The retained support rows are never replaced in this operation.

The recorded contact-power circuits. The circuit replacements preserve the boundary values r^2, r^{24}, r^{80}. The replacement definitions are

$$r_2 = r^2, \quad r_4 = r_2^2, \quad r_8 = r_4^2, \quad r_{16} = r_8^2, \quad r_{24} = r_{16}r_8, \quad r_{40} = r_{24}r_{16}, \quad r_{80} = r_{40}^2.$$

Their exact outputs agree with the old acyclic product definitions. For either direction retain r and the boundary outputs, and assign each internal target its displayed power of r. These maps work also for $r = 0$ and use no inverse. Each deleted internal target has no use outside the replaced definitions; every new internal target is fresh. The certificate checks those conditions and the exact output identities in the current system. Only circuit events with these retained powers are included in this description.

A quadratic presentation of the parity block. This replacement concerns only its rational equations, with unchanged input b. Its old private variables t, U, e, p, q, s, h satisfy

$$U(1 + bt^2) = b(1 - bt^2), \qquad p^2 = 17b(1 + 2e - 3e^2),$$
$$h = e(17b - U^2), \qquad 17bq^2 = (s^2 - 17bh)(s^2 + 51bh).$$

Use instead twelve private rational variables $u, v, a, z, e, p, c, w, s, q, j, k$ and nine quadratic equations

$$z = u^2, \quad a = bv, \quad z + va = b^2, \quad c = be,$$
$$p^2 - 17b - 34c + 51ec = 0, \qquad w - 17bc + cz = 0,$$
$$j = s^2, \qquad k = bq, \qquad j^2 + 34jw - 867w^2 - 17qk = 0.$$

For the forward map put $\delta = 1 + bt^2$. It is nonzero: otherwise $bt^2 = -1$ and the first old equation gives $0 = 2b$, a contradiction. Keep e, p, s, q and set

$$u = U, \quad v = 2bt/\delta, \quad a = bv, \quad z = U^2, \quad c = be, \quad w = bh, \quad j = s^2, \quad k = bq.$$

The identity $[b(1-bt^2)]^2 + b(2bt)^2 = b^2(1+bt^2)^2$ proves the first conic; substitution proves the other equations.

Conversely $u^2 + bv^2 = b^2$. Set $h = e(17b - z)$ and keep e, p, s, q. If $b = 0$, take $U = t = 0$. If $b \neq 0$ and $u + b \neq 0$, take $U = u$ and $t = v/(u + b)$. In this case the first old residual, multiplied by $(u + b)^2$, is

$$(u - b)(u + b)^2 + b(u + b)v^2 = (u + b)(u^2 + bv^2 - b^2) = 0.$$

If $b \neq 0$ and $u = -b$, then $v = 0$; take $U = b$ and $t = 0$. In every case $U^2 = z$, and the remaining old equations follow from $c = be$, $w = bh$, $j = s^2$, and $k = bq$. The possible sign change of U is harmless because all old witnesses and deleted lifting coordinates are private. These maps prove equality of existential projections; they are not asserted to be inverse on the private witness tuples. This rule is applied only at the certificate's height-parity blocks; the ordinary comparison pole is the linear-unit source predicate already proved above. The actual multiplicity is given in the following census.

15.12 Finite realization and proof of the quartic theorem

The fresh quadratic lift introduces only defining product equations. Their evaluation and forgetting maps preserve rational solutions while fixing the original input. The certificate then applies exactly the operations below, under the current-row and complete-incidence hypotheses just proved. The resulting file `quartics/quartic-4973/residuals.json.gz` contains 4 552 nonzero integer residuals of degree at most two. Every weight in `quartics/quartic-4973/weights.json` is one, and

$$\Phi(X_0, \ldots, X_{4972}) = \sum_{i=1}^{4552} R_i^2.$$

At rational values each square is nonnegative, so this sum vanishes if and only if all residuals vanish. Nonzero rational row normalization preserves each row's zero set before this integral sum is formed.

The certificate `quartics/quartic-4973/certificate.jsonl.gz` records 551 transformations:

height-parity replacements	1
contact-power circuit replacements	13
invertible linear row operations	266
rational definition eliminations	121
invertible rational shears	2
eliminations modulo retained equations	148

Its SHA-256 is

`81ca768459a13b76dbdf4bfe81e94fbab9ab99f14a5731f6b1a1b03e28c1df2b`

The independent verifier `quartics/quartic-4973/verify_certificate.py` has SHA-256

`b7180b01029b3d9364d94914401a4f1f66b190762f10b99a38e9fd4f49437ffe`

It checks the named source binding, fresh lift, every recorded polynomial identity and reconstruction map, defining order, current-row supports, every occurrence of each removed coordinate, final quadratic degrees, and the bijective variable map fixing input zero. It then compares the complete collected integer expansion, including every cross term. The endpoint verification receipt is `quartics/quartic-4973/evidence/final-verification.json`.

Proof of Theorem 15.1. Theorem 15.22 proves both rational-input implications for the selected arithmetic source. For any fixed rational input, every defining lift and every recorded transformation has the forward and reverse solution maps proved above. Their composition preserves the existence of rational witnesses and keeps that input fixed. Hence the final residual system has rational input projection exactly $\mathbb{Z}$. The all-one sum of its squares vanishes precisely on its simultaneous zero set, so the same equivalence holds for Φ.

The literal coefficient table is the verified complete integral expansion of that sum. Its nonzero degree-four term makes its degree exactly four; the collected coefficients give the stated monomial count and maximum. The checked variable map gives the stated ambient coordinates, one of which is the input. Thus all numerical claims and the input-projection equivalence concern this one polynomial. □

The finite checks do not replace the inherited rank, height, prime-pattern, formal-group, norm or pole arguments. Their qualifications remain those stated in the source proof and Appendix B.

16 Arithmetic interfaces for the smaller quartic

Lemma 16.1 (The original scalar certificate at order thirty-two)**.** *Let $K = \mathbb{Q}(\sqrt{2})$, $d = 3 - \sqrt{2}$, $D = d^2$, and $E : y^2 = x^3 - Dx$, with the point P_* of Section 14. Put*

$$\beta = X^2 + 2, \qquad r = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1), \qquad X \in \mathbb{Q}.$$

For $h_0, a \in \mathbb{Z} \setminus \{0\}$ and points $J_0 = h_0 P_*$ and $S = aP_*$, take the two rational coefficients, in the basis $1, \sqrt{2}$, of $f(J_0), f(S), z(S)/z(J_0)$ in that order. Denote them by $X_1, \dots, X_6$, and put

$$(m_1, \dots, m_6) = (1, 1, 1, 1, 32, 32), \qquad l_i = X_i / r^{m_i}.$$

Use the same ordered norm tree as the original scalar construction:

$$\begin{aligned} N_r(u, v) &= u^2 + rv^2, \\ n_0 &= N_r(1, l_1), \quad n_1 = N_r(l_2, l_3), \quad n_3 = N_r(l_5, l_6), \\ W &= N_r(N_r(n_0, n_1), N_r(l_4, n_3)). \end{aligned}$$

Let

$$c \in \left\{ \pm \prod_{p \in I} p : I \subseteq \{2, 3, 5, 7, 11, 13, 17\} \right\}.$$

If $b = crW^2 \neq 0$ satisfies the original pole predicate $\Psi(b)$, then

$$M_5(\beta)^{32} \mid a, \qquad h(\beta) \leq H_* |a|^{3/4}.$$

Here M_5 and the exact constant H_* are those of Theorem 14.3; in particular $1 \leq H_* < 2^{4783227}$.

For every integral X, the scalar certificate is complete with successively chosen positive indices h_0, a. One may require h_0 to be divisible by any prescribed positive integer and a to be arbitrarily divisible and arbitrarily large. After S has been chosen, the normalized-leaf conditions and the nonvanishing of W persist when h_0 is replaced by every integer in one suitable congruence class $h \equiv h_0 \pmod{L}$. The modulus L may also be required to be divisible by any prescribed positive integer.

Proof. The value r is nonzero: $\beta \geq 2$, and $\beta = 2401$ would give $X^2 = 2399$, which has no rational solution. At a prime $p > 76176$, the multiplier c is a unit. Every prime in $M_5(\beta)$ has odd valuation in r, by the contact computation in the proof of Theorem 14.3. One can also observe that a negative valuation of $\beta = X^2 + 2$ is even, so only its finite contacts contribute here.

Write $R = v_p(r)$. The pole obstruction in Lemma 14.4 and $b \neq 0$ make the odd valuation $R + 2v_p(W)$ positive. In a norm node $u^2 + rv^2$ the two nonzero summands have opposite valuation parity, so there is no cancellation. The constant leaf on the all-left path first excludes $R < 0$. Then $v_p(W) > -R/2$. Descending the tree, the implication

$$v_p(N_r(u, v)) > -tR, \quad t < 1 \quad \Longrightarrow \quad v_p(u) > -tR/2, \quad v_p(v) > -(t + 1)R/2$$

shows that every leaf l_i has valuation greater than $-R$. Thus $v_p(X_i) > (m_i - 1)R \geq m_i - 1$, and integrality of the valuation gives $v_p(X_i) \geq m_i$. This is the original norm argument with the two assigned orders changed to thirty-two.

The order-one tests put J_0 and S in the formal group at each place above p. Such a completion is unramified, and the formal logarithm is an isometry on that group. Since $[h_0]S = [a]J_0$, it follows that

$$v_{\mathfrak{p}}\left(\frac{z(S)}{z(J_0)} \right) = v_p(a/h_0).$$

Both rational coefficients of this ratio have valuation at least thirty-two, so $v_p(a/h_0) \geq 32$. As h_0 is a nonzero integer, $v_p(a) \geq 32$. This proves $M_5(\beta)^{32} \mid a$. Theorem 14.3 now gives $h(\beta) \leq H_* M_5(\beta)^{24} \leq H_* |a|^{3/4}$. No exponent-nineteen height estimate is used.

For completeness, let $X \in \mathbb{Z}$. The integer r is nonzero and $|r| > 1$. The original seven-prime squareclass basis, proved in the finite-correction argument following Lemma 14.7, permits a choice of c such that $cr > 0$ and cr is square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$. Let $\mathcal{S}$ be the finite prime support of the integer cr.

At the places above $\mathcal{S}$, choose h_0 sufficiently divisible that J_0 is in arbitrarily deep formal subgroups. This remains possible after requiring any fixed divisibility of h_0, because these subgroups have finite index in the local elliptic groups. Having fixed this nonzero J_0, choose a sufficiently divisible that $S = aP_*$ is still deeper. The coefficients of $f(J_0), f(S)$ and $z(S)/z(J_0)$ then attain every finite lower valuation bound needed here. This includes ramified and bad primes, by working in sufficiently deep formal subgroups. We arrange that all normalized leaves are integral when $p \mid r$ and divisible by p when $p \mid c$ but $p \nmid r$. The index a may simultaneously exceed any fixed bound.

At $p \mid r$, the root of the displayed tree is congruent to one modulo p, because $r \equiv 0$ and the all-left leaf is one. At $p \mid c$, $p \nmid r$, all other leaves vanish modulo p, so the root is again congruent to one. Thus W is a unit on $\mathcal{S}$, and in particular is nonzero. At these primes $v_p(crW^2) = v_p(cr) \geq 0$; outside $\mathcal{S}$ its valuation is even. Hence b has square reduced denominator, is positive, and has the same local squareclasses as cr at two, three, and seventeen. Lemma 14.7 supplies the scalar pole witnesses.

Finally, with S fixed, these scalar tests are rational functions of J_0 defined at the chosen point. Their coefficient maps are continuous in the product of the completions above each prime in $\mathcal{S}$. The finite lower valuation bounds persist in sufficiently small neighborhoods. A common finite-index subgroup of the integer orbit gives a congruence $h \equiv h_0 \pmod{L}$ contained in all those neighborhoods. Enlarging L preserves that property. The same unit calculation proves the asserted stability for the final shared point. □

Lemma 16.2 (The fixed multiplier seventeen with preparation zero)**.** *Let $b \in \mathbb{Q}^\times$, put $B = 17b$, and define $\Psi_0(b)$ by the existence of $e, p, q, s \in \mathbb{Q}$ satisfying*

$$p^2 = B(1-e)(1+3e), \qquad h = e(B - b^2), \qquad Bq^2 = (s^2 - Bh)(s^2 + 3Bh),$$

where h is the displayed expression, not a further free choice. Then $\Psi_0(b)$ excludes negative odd valuations of b at every prime outside $\{2, 3, 17\}$. Conversely, it holds whenever b has square reduced denominator, is positive and square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, and

$$b < 17, \qquad v_2(b) \geq 3, \qquad v_3(b) \geq 1, \qquad v_{17}(b) \geq 2.$$

No additional nonzero condition on e, p, q, s, h is imposed.

Proof. Taking the preparation witnesses $t = 0$ and $U = b$ in (11) gives exactly the displayed equations. The obstruction therefore follows from Lemma 14.5, including its degenerate witness cases.

For completeness put

$$V = B(B - b^2) = B^2(1 - b/17).$$

The real inequality makes V positive. The valuation bounds make $1 - b/17$ a square at two, three, and seventeen: its difference from one has valuation at least three at two and at least one at the two odd primes. Thus V is square at these three places.

Let μ and θ be the positive squarefree representatives of B and V. The prescribed squareclasses of b give $\mu \equiv 17 \pmod{24}$; in particular $\mu > 1$ and its support contains seventeen but neither two nor three. Also $\theta \equiv 1 \pmod{24}$, and seventeen is absent from its support. At a prime $\ell \mid \mu$ outside the three exceptions, $v_\ell(b)$ is positive and odd, since the reduced denominator of b is square. Hence $1 - b/17$ is a square unit there, so V and θ are local squares. The same assertion at seventeen was already arranged. These facts make the supports disjoint.

Conversely, if $\ell \mid \theta$, then $\ell \notin \{2, 3, 17\}$. If $v_\ell(b) > 0$, the valuation of V is $2v_\ell(b)$; if $v_\ell(b) < 0$, it is $3v_\ell(b)$, again even because the denominator is square. Thus an odd valuation of V forces $v_\ell(b) = 0$ and $1 - b/17 \equiv 0 \pmod{\ell}$. It follows that $B = 17b \equiv 17^2 \pmod{\ell}$ is a nonzero square unit. This proves the other mutual-squareness condition.

Lemma 14.6 now gives the two rational points in the squareclass of B. To make the conversion explicit, write $V = \theta c^2$ and scale the second point to the curve with parameter nV. If the first abscissa is Ba^2, scale both points and parameters by a so that their abscissas are B and Bs^2, with $s \neq 0$, and their curve parameters are e_0 and $e_0 V$. For their ordinates y_1, y_2, set

$$e = e_0/B, \qquad p = y_1/B, \qquad q = y_2/(B^2 s), \qquad h = e(B - b^2).$$

The first curve equation gives $p^2 = B(1 - e)(1 + 3e)$. Since $e_0 V = B^2 h$, the second gives $Bq^2 = (s^2 - Bh)(s^2 + 3Bh)$. These are the required witnesses. The only arithmetic existence theorem used in this specialization is the existing simultaneous-realization lemma, with its stated restricted-converse and descent dependencies. □

Lemma 16.3 (Real bounds independent of later progression moduli)**.** *Let G be the closure of the integer orbit*

$$\{([n]P_*, [n]P_*^\sigma) : n \in \mathbb{Z}\} \subset E(\mathbb{R}) \times E^\sigma(\mathbb{R}),$$

and let G^0 be its identity component. Fix a finite collection of rational functions formed from nonzero integral multiples and fixed translates of either elliptic coordinate, with all denominators nonzero as rational functions on the relevant curve. Suppose their pole set is contained in a finite union of inverse images of finite sets under these nonzero multiplication-and-translation maps on either factor. There is a bound C such that every nonconstant integer progression contains arbitrarily large positive indices at which all these functions are defined and their real absolute values are at most C. The bound is chosen from the functions before the progression modulus is specified.

If $m = |G/G^0|$, $m \mid L$, and $m \mid h_0$, the progression $h \equiv h_0 \pmod{L}$ is dense in G^0, and its positive tail meets every nonempty open subset of G^0 infinitely often.

Proof. The real elliptic groups are compact Lie groups. Their closed subgroup G is a compact abelian Lie group with finitely many components, and G^0 is a torus. The projection of G^0 onto each elliptic identity circle is surjective. Indeed the projected integer orbit is infinite, since P_* has infinite order and each real embedding is injective on the group of algebraic points. If the connected projection were trivial, the projected G would be finite; a nontrivial connected closed subgroup of a circle is the whole circle.

Each nonzero multiplication map on a circle has finite fibers. Consequently each of the prescribed inverse images of a finite set has Haar measure zero in every component of G. The finite union of pole sets cannot cover a component. Choose in each component a nonempty open neighborhood whose compact closure avoids the poles. Continuity bounds all the fixed rational functions on these finitely many compact sets by one constant C.

For a progression $h = r + kn$, its closure is $[r]P_* + [k]G$. Multiplication by the positive integer k maps G^0 onto itself, so this closure contains an entire component of G. The positive tail of the progression is dense in its closure: compactness gives unbounded positive multiples tending to the

identity, so each negative multiple is also a limit of positive multiples. It therefore meets the chosen neighborhood in that component at arbitrarily large positive indices.

If $m \mid k$, then $[k]G = G^0$. If also $m \mid r$, the coset is G^0 itself. Apply this with $k = L$ and $r = h_0$ for the final assertion. □

17 The shared-point quartic in 962 variables

This section gives a new complete quadratic compilation of the shared-point construction in the public agentic notes [6]. The shared comparison points, reciprocal-trace detectors, fixed height scales, direct Lucas test, and affine anchor change are arithmetic and algebraic improvements recorded in those notes. The simultaneous prime theorem used to choose the shared points is Sun's Theorem 4.1, in the precise form of Theorem I.1 below. Our contribution here is an independently specified gate program, its source verification, and further changes of coordinates and circuit identities leading to a 962-variable polynomial after 32 certified eliminations. The definition does not depend on an unavailable coefficient table. The relevant passages in the public notes are shared-point construction, fixed height scales, and quadratic compilation and anchor translation.

We retain the arithmetic conclusions and conventions of Section 14, including the restricted-converse input to Lemma 14.4. In this section the height estimate is Theorem 14.3, with exponent 24 and its original constant H_*; it is not the exponent-19 estimate used for the preceding quartic. Thus

$$1 \le H_* < 2^{4783227}.$$

The exact finite compilation and coefficient assertions below require no arithmetic conjecture. Identifying the rational input projection with $\mathbb{Z}$ uses the stated arithmetic results and the simultaneous prime theorem.

The point, pole, and correction variables below specify the arithmetic source predicate. The final compilation subsection gives explicit maps to its implementation coordinates.

17.1 Points, local parameters, and quotient domains

Use the normalized curve and generator already established in Section 14:

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2, \quad E : y^2 = x^3 - Dx,$$

$$P_* = (57/8 - 7\sqrt{2}/2, -165/8 + 417\sqrt{2}/32), \qquad 2E(K) = \mathbb{Z}P_*.$$

The last assertion follows from Proposition 3.1 by the unit change of variables recorded there. Conjugation of $K/\mathbb{Q}$ is denoted by σ. All field equations below mean both rational component equations in the basis $1, \sqrt{2}$.

The source uses nine nonidentity points

$$A, S, T, Z, H, U_A, U_S, U_T, U_Z \in 2E(K).$$

Each is represented as the double of an affine halfpoint $(\alpha, \eta) \in E(K)$, using four rational halfpoint coordinates. The equations for a double are

$$\eta^2 = \alpha^3 - D\alpha, \qquad 2\eta m = 3\alpha^2 - D, \qquad x = m^2 - 2\alpha, \qquad y = m(\alpha - x) - \eta. \tag{36}$$

Products of degree greater than two in this display are evaluated by the quadratic gates specified below. Set

$$Q_0 = P_*, \quad (Q_1, Q_2, Q_3, Q_4) = (A, S, T, Z), \quad (U_0, U_1, U_2, U_3, U_4) = (H, U_A, U_S, U_T, U_Z).$$

Compute $[4]Q_i$ by two doublings, or by fixed rational arithmetic for Q_0, and compute

$$V_i = U_i - [4]Q_i \quad (0 \le i \le 4).$$

For the subtraction of (x', y') from (x, y) we retain an inverse of $x - x'$ in K and use

$$m = (y + y')/(x - x'), \quad x'' = m^2 - x - x', \quad y'' = m(x - x'') - y.$$

Lemma 17.1 (Domains of the point quotients)**.** *The equations* (36) *force $\eta \neq 0$ and describe a finite nonidentity point of $2E(K)$. Every such point has nonzero abscissa and ordinate. The guarded differences V_i have the same properties. Consequently $f(Q) = 1/x(Q)$ and $z(Q) = -x(Q)/y(Q)$ are defined and nonzero at every point where they are used here.*

Proof. If $\eta = 0$, the curve equation gives $\alpha = 0$ or $\alpha^2 = D$. The right side of the slope equation is then $-D$ or $2D$, respectively, a contradiction. The ordinary duplication formulas therefore apply. They give

$$x([2](\alpha, \eta)) = \left(\frac{\alpha^2 + D}{2\eta}\right)^2 \neq 0,$$

because $D = d^2$ and the numerator is positive at either real embedding. An ordinate-zero point has abscissa $0, d$, or $-d$. The latter two are not squares in K: their rational norm is 7, whereas the norm of a square is a rational square. Thus the doubled ordinate is nonzero. A guarded affine difference is finite and is again in $2E(K)$, so the same argument applies. Conversely any nonidentity point of $2E(K)$ has an affine halfpoint of nonzero ordinate and is represented by these formulas. □

Write

$$a_0 = 1, \quad a_1 = -2/5, \quad a_j = \frac{4j - 5}{4j + 1} a_{j-1} \ (j \ge 2), \qquad G_J(w) = \sum_{j=0}^{J} a_j w^j,$$

and, for $e = 16, 52$, put

$$\ell_e(Q) = z(Q) G_{e/4-1}(Df(Q)^2). \tag{37}$$

Lemma 15.5 applies to both truncation degrees 3 and 12. Thus all these values are nonzero globally.

Lemma 17.2 (The logarithm estimate at unramified places)**.** *Let $p > 76176$ be a rational prime and $\mathfrak{p} \mid p$ a place of K. If a nonidentity point Q is in the formal group at $\mathfrak{p}$, normalize $v(p) = 1$ and put $r = v(z(Q))$. Then $r \ge 1$, and for $e \in \{16, 52, 64\}$,*

$$v(\log_E Q) = r, \qquad v(\log_E Q - \ell_e(Q)) \ge (e + 1)r, \qquad \ell_e(Q)/\log_E Q \in 1 + p^e \mathcal{O}_{K_\mathfrak{p}}.$$

Here ℓ_{64} is defined by the same formula with G_{15}. Ratios with numerator precision e and denominator precision e' in this set have the valuation of their full logarithm ratio and relative error of valuation at least $\min(e, e')$.

Proof. The local extension is unramified, including when p is inert. The curve has good reduction and D is a unit. On the formal branch,

$$f = z^2(1 - Df^2), \qquad w = Df^2, \qquad v(w) = 4r.$$

The recurrence gives $4w(1-w)G'(w) + (1+w)G(w) = 1 - w$ and hence the same identity $\log_E Q = zG(w)$ as in Lemma 15.4. For $j \ge 1$ the denominator of a_j divides $(4j + 1)!$, so

$$v(a_j) \ge -\frac{4j + 1}{p - 1}.$$

Set $m = e/4 \le 16$. If $j \ge m$ and $4j + 1 < p$, then $v(a_j w^j) \ge 4mr$. Otherwise $j \ge 19044$ and

$$v(a_j w^j) \ge 4jr - j \ge 3jr \ge 4mr.$$

The lower bounds tend to infinity, proving the estimate for the entire tail. Every nonconstant term of $G(w)$ has positive valuation, so the leading term gives $v(\log_E Q) = r$. Division proves the relative estimate, and division of two relative estimates proves the ratio assertion. No step uses residue degree one. □

17.2 The complete rational source

The ordinary input is X, and $\beta = X^2 + 2$. Define

$$(k_1, k_2, k_3, k_4) = (16, 52, 52, 52), \qquad L_i = \frac{\ell_{k_i}(U_i)}{\ell_{52}(H)}, \qquad L_i' = \sigma\Big(\frac{\ell_{k_i}(V_i)}{\ell_{52}(V_0)}\Big).$$

The sixteen field tests D_j, with their orders m_j, are exactly

$$\begin{array}{c|c} f(U_0), \dots, f(U_4),\ \sigma f(V_0), \dots, \sigma f(V_4) & 1 \quad (10) \\ L_i - L_i' \ (1 \le i \le 4) & 16, 52, 52, 52 \\ L_1 - \beta & 16 \\ F(L_2, L_3, L_4) & 52, \end{array} \tag{38}$$

where

$$F(U, V, W) = V^2 - U(U-2)VW + (U-2)^2 W^2 - 1. \tag{39}$$

There is no rational proxy for an integer point label.

For $j = 1, \dots, 4$ write $x(Q_j) = x_{j,0} + x_{j,1}\sqrt{2}$ and set

$$\mathcal{A}_0 = 1, \quad \mathcal{A}_{2j-1} = x_{j,0}^2, \quad \mathcal{A}_{2j} = (1 + x_{j,1}^2)^2.$$

These are positive nonzero rational squares. In fact $x(Q_j)$ is positive at both real embeddings by Lemma 17.1, so $x_{j,0} > 0$. Retain one rational coordinate λ and put

$$\Theta = \{0, -1/2, -1, 1/2399, -32/407\}, \qquad F_5(T) = \prod_{\theta \in \Theta} (T - \theta), \qquad t_r = \mathcal{A}_r \lambda, \quad \delta_r = t_r - 1.$$

For every $0 \le r \le 8$, require $t_r \notin \Theta \cup \{1\}$ and define

$$R_r = \mathcal{A}_r^{-5} F_5(t_r)/\delta_r^6, \qquad \widetilde{D}_{r,j} = D_j/\delta_r^{m_j+1}, \tag{40}$$

$$W_r = \prod_{j=1}^{16} \Big(1 + R_r(\widetilde{D}_{r,j}/R_r^{m_j})^2\Big), \qquad T_r = \mathrm{Tr}_{K/\mathbb{Q}}(W_r^{-1}), \qquad B_r = R_r/T_r^2. \tag{41}$$

Require $W_r, T_r \ne 0$. For $B = B_r$ retain four rational witnesses e_r, s_r, p_r, q_r and impose

$$\begin{aligned} C_r &= 17B_r, \qquad h_r = e_r(C_r - B_r^2), \\ p_r^2 &= C_r(1 - e_r)(1 + 3e_r), \\ C_r q_r^2 &= (s_r^2 - C_r h_r)(s_r^2 + 3C_r h_r). \end{aligned} \tag{42}$$

The first two expressions are arithmetic definitions. The last two are the preparation-zero specialization of the original pole predicate. There is no nonzero guard on e_r, s_r, p_r, q_r, or h_r.

The scalar block shares the point H. Define

$$r_\beta = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1).$$

It is nonzero for rational X: the only positive possible root not excluded by $\beta \geq 2$ would require $X^2 = 2399$. Let $Y_1, \ldots, Y_6$ be, in order, the two rational components of $f(H), f(S), z(S)/z(H)$, and set

$$(e_1, \ldots, e_6) = (1,1,1,1,32,32), \qquad y_i = Y_i / r_\beta^{e_i}, \qquad N_r(u,v) = u^2 + rv^2.$$

The scalar norm tree has no padding leaf. It is

$$\begin{aligned} n_0 &= N_{r_\beta}(1, y_1), \quad n_1 = N_{r_\beta}(y_2, y_3), \quad n_3 = N_{r_\beta}(y_5, y_6), \\ W_\beta &= N_{r_\beta}(N_{r_\beta}(n_0, n_1), N_{r_\beta}(y_4, n_3)). \end{aligned} \tag{43}$$

Use eight Boolean coordinates ϵ_-, ϵ_p, for $p \in \{2,3,5,7,11,13,17\}$, impose $\epsilon^2 - \epsilon = 0$, and set

$$c = (1-2\epsilon_-)\prod_p (1+(p-1)\epsilon_p), \qquad b_\beta = c r_\beta W_\beta^2, \qquad C_\beta = 17 b_\beta.$$

Require $b_\beta \neq 0$. Retain five rational witnesses $t_\beta, e_\beta, s_\beta, p_\beta, q_\beta$ and impose

$$(1 + b_\beta t_\beta^2) U_\beta = b_\beta (1 - b_\beta t_\beta^2), \qquad h_\beta = e_\beta (C_\beta - U_\beta^2), \tag{44}$$

$$p_\beta^2 = C_\beta (1 - e_\beta)(1 + 3e_\beta), \qquad C_\beta q_\beta^2 = (s_\beta^2 - C_\beta h_\beta)(s_\beta^2 + 3 C_\beta h_\beta). \tag{45}$$

Here U_β, h_β are arithmetic values. The preparation equation forces $1 + b_\beta t_\beta^2 \neq 0$: otherwise $b_\beta t_\beta^2 = -1$ but its right side is $2b_\beta \neq 0$.

Finally fix the integers

$$C = 3^{2^{26}}, \qquad \mathsf{S} = C^2 = 3^{2^{27}}, \qquad \kappa = \mathsf{S}^2 = 3^{2^{28}}. \tag{46}$$

The symbol S in the coefficient files means S; it is not the point S and is not a quantified coordinate. Write $f(S) = f_0 + f_1\sqrt{2}$, retain three private rational witnesses u, v, w, and impose

$$u^2 + 6v^2 - 2w^2 - 3(f_0^2 + f_0 f_1 - f_1^2)^2 + 2\mathsf{S}^2 = 0. \tag{47}$$

This completes the rational source. Every rational expression has the displayed guard or a domain implication already proved. Equations (42) and (45) retain their degenerate witness values.

Lemma 17.3 (The three-witness anchor)**.** *Equation* (47) *is equivalent to* $f_0/C, f_1/C \in \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}$. *For the nonzero integer label* a *of* $S = [a]P_*$ *it implies*

$$0 \neq a \in 2\,3^{2^{25}-1}\mathbb{Z}, \qquad |a| > (21 H_*)^{13/2} > (20 H_*)^4.$$

Its permitted labels contain arbitrarily divisible positive integers.

Proof. Put $R = \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}$. The quaternion algebra $(2,3)_\mathbb{Q}$ ramifies exactly at 2 and 3. The quadratic embedding criterion [89, Proposition 14.6.7] shows, for rational $t \neq 0$, that $1 - 2/(3t^2)$ is represented by $2x^2 + 3y^2 - 6z^2$ precisely when it is nonsquare in both $\mathbb{Q}_2$ and $\mathbb{Q}_3$. This last condition is exactly $t \in R$. Indeed at 2 a positive valuation of t gives odd valuation of the expression, valuation zero gives residue 3 modulo 8, and negative valuation gives a square in $1 + 8\mathbb{Z}_2$. At 3, nonnegative valuation of t gives odd negative valuation, while negative valuation gives a square in $1 + 3\mathbb{Z}_3$. The expression is nonzero because $t^2 = 2/3$ has no rational solution. Multiplying by $3t^2$ gives

$$t \in R \quad \Longleftrightarrow \quad \exists u, v, w \in \mathbb{Q} : \; u^2 + 6v^2 - 2w^2 - 3t^2 + 2 = 0.$$

The case $t = 0$ has witnesses $(0, 0, 1)$. A rational-square value of $1 - 2/(3t^2)$ is excluded by the division-algebra criterion as well: a nonzero trace-zero quaternion cannot square to a nonzero rational square in a division algebra.

The form $A^2 + AB - B^2$ has no nontrivial zero over $\mathbb{F}_2$ or $\mathbb{F}_3$. Scaling a pair therefore proves $v_p(a^2 + ab - b^2) = 2\min(v_p(a), v_p(b))$ for $p = 2, 3$. Apply the preceding three-witness formula to $(f_0^2 + f_0 f_1 - f_1^2)/C^2$ and multiply its witnesses by C^2. This gives exactly (47).

The local calculation in Section 14.2.2 applies with exponent 2^{26}: the reduction of P_* at 3 has order two and $v_3(z(2P_*)) = 1$. For even nonzero a it gives $v_3(f([a]P_*)) = 2 + 2v_3(a)$. At 2 the rational components of $f([a]P_*)$ are integral. The anchor is consequently equivalent to $0 \neq a \in 2\,3^{2^{25}-1}\mathbb{Z}$. Its minimum possible absolute value has binary logarithm greater than

$$1 + \tfrac{3}{2}(2^{25} - 1) = 50331647.5,$$

whereas $\log_2((21H_*)^{13/2}) < \frac{13}{2}(5+4783227) = 31091008$. The last comparison with $(20H_*)^4$ follows from $21^{13} > 20^8$ and $H_* \geq 1$. The subgroup description proves the last assertion. □

17.3 Local tests and the height implication

Lemma 17.4 (One place for every test)**.** *Let $R \in \mathbb{Q}^*$, $D_j \in K$, and positive integers m_j define*

$$W = \prod_j (1 + R(D_j/R^{m_j})^2), \quad T = \mathrm{Tr}(W^{-1}), \quad B = R/T^2,$$

with $W, T \neq 0$ and a preparation-zero pole certificate for B. If $p > 76176$ and $v_p(R)$ is odd, then $v_p(R) > 0$ and some $\mathfrak{p} \mid p$ has $v_\mathfrak{p}(D_j) \geq m_j$ for every j.

Proof. Set $\alpha = v_p(R)$. The nonconstant term in each nontrivial factor has odd valuation $\alpha + 2(v(D_j) - m_j\alpha)$; unramifiedness makes these valuations integral. The factor valuation is its minimum with zero. Thus $\lambda_\mathfrak{p} = v_\mathfrak{p}(W) \leq 0$. Put $\lambda = \max_{\mathfrak{p}|p} \lambda_\mathfrak{p}$. Integrality of trace gives $v_p(T) \geq -\lambda$. Hence

$$v_p(B) \leq \alpha + 2\lambda.$$

The left side is odd and, by Lemma 14.4, not negative. It is positive, so $\alpha > 0$ and $\lambda > -\alpha/2$. At a place attaining λ, each factor has valuation at least λ, because all factor valuations are nonpositive. For nonzero D_j this implies

$$v(D_j) > (m_j - 3/4)\alpha \geq m_j - 3/4.$$

Integrality gives $v(D_j) \geq m_j$; a zero D_j satisfies it as well. □

Lemma 17.5 (The two-anchor comparison)**.** *Write the integer labels of (A, S, T, Z) as (n, a, t, b). If the first fourteen tests in* (38) *have their specified orders at a place $\mathfrak{p} \mid p$, $p > 76176$, then*

$$L_i \in \mathcal{O}_{K_\mathfrak{p}}, \qquad L_i \equiv n_i \pmod{p^{k_i}}, \qquad (n_1, n_2, n_3, n_4) = (n, a, t, b).$$

This holds at split and inert primes.

Proof. Write $H = [h]P_*$ and $U_i = [u_i]P_*$. The order-one tests put H, U_i in the formal group at $\mathfrak{p}$ and V_0, V_i in the formal group at the conjugate place. Positive valuation of $f = 1/x$ gives formal membership on this good model. The guarded points give $h \neq 0, 4$. Their full logarithm ratios, after conjugating the second, are

$$r_i = u_i/h, \qquad r_i' = (u_i - 4n_i)/(h - 4).$$

These identities follow from the homomorphism property on equal formal multiples, and do not require P_* itself to be formal. At least one of $h, h-4$ is a p-unit. Its ratio is integral, and Lemma 17.2 gives the same valuation to its truncated ratio. The cross test makes the other truncated ratio integral too. Both relative errors are therefore absolute errors, and $L_i \equiv r_i \equiv r_i'$ modulo p^{k_i}. Multiplication by $h, h-4$ and subtraction give $4L_i \equiv 4n_i$. Divide by four. At an inert prime conjugation preserves the single local valuation, so the same argument applies. □

Lemma 17.6 (Height transfer)**.** *Let M_r be the odd-contact radical of t_r for the five contacts Θ, omitting primes at most 76176. If M_r^k divides a nonzero integer N for every r, then*

$$h(x(Q_i)) < 7H_*|N|^{24/k} \qquad (1 \le i \le 4).$$

At every prime counted in M_r, the source imposes all sixteen field tests with their specified orders at one place above that prime.

Proof. The fractional linear map $\phi(t) = 2 + 1/t$ carries Θ to $\infty, 0, 1, 2401, -343/32$. Its integral matrix has determinant -1. All primes dividing the relevant fixed coefficients and contact differences, including differences from 1, lie in $\{2, 3, 5, 7, 11, 37, 109, 439, 2399\}$. Thus outside the cutoff $M_r = M_5(\phi(t_r))$, and

$$h(t_r) \le h(\phi(t_r)) + \log 3 \le (H_* + \log 3)M_r^{24}.$$

At a prime of M_r, δ_r is a unit. Exactly one contact factor has odd positive valuation, while $\mathcal{A}_r^{-5}$ is a rational square. Therefore $v_p(R_r)$ is odd. Lemma 17.4 applied to $\widetilde{D}_{r,j}$ gives the asserted orders for D_j.

Set $T_0 = (H_* + \log 3)|N|^{24/k}$. The case $r = 0$ gives $h(\lambda) \le T_0$, and the other cases give $h(\mathcal{A}_r\lambda) \le T_0$. Hence $h(\mathcal{A}_r) \le 2T_0$. For the first component, $h(x_{i,0}) \le T_0$. For $x_{i,1} = a/b$ in lowest terms,

$$h(1 + x_{i,1}^2) = \log(a^2 + b^2) \ge 2h(x_{i,1}),$$

so $h(x_{i,1}) \le T_0/2$. Finally

$$h(x(Q_i)) \le h(x_{i,0}) + h(x_{i,1}) + \tfrac{3}{2}\log 2 < 7H_*|N|^{24/k}.$$

□

Lemma 17.7 (The two reduced group cardinalities)**.** *Let $p > 76176$ be prime and let $\pi \in \mathbb{Z}[i]$ have norm p, with $\pi \equiv 9 \pmod{16}$ and $\pi \equiv 1 + 3i \pmod 7$. The two reductions of E/K at the places above p have group cardinalities $N(\pi - 1)$ and $N(\pi + 1)$, in some order.*

Proof. The congruences give $p \equiv 1 \pmod 8$ and $p \equiv 3 \pmod 7$. Thus p splits in K and $(7/p) = -1$. The reductions of $d = 3 - \sqrt{2}$ have product 7 and opposite square classes. The two curves $y^2 = x^3 - d^2x$ are consequently opposite quadratic twists of $y^2 = x^3 - x$ and have opposite traces.

The Frobenius calculation already given in Section 14.4.1 applies: the order-four automorphism is defined over $\mathbb{F}_p$, its rational centralizer is $\mathbb{Q}(i)$, and integrality puts Frobenius in $\mathbb{Z}[i]$. Write it as $a+bi$, so $a^2+b^2 = p$ and the two cardinalities are $p+1-2a$ and $p+1+2a$. Full rational two-torsion makes both cardinalities divisible by four. Since $p \equiv 1 \pmod 4$, this forces a odd and b even. The real part of π is odd and its imaginary part is even. Unique factorization in $\mathbb{Z}[i]$ therefore makes $a + bi$ one of $\pm\pi, \pm\overline{\pi}$; the associates with factor i have the wrong parity. Hence the unordered cardinalities are exactly $N(\pi - 1), N(\pi + 1)$. □

Theorem 17.8 (Arithmetic interpretation of the source)**.** *With the inherited arithmetic results and Theorem I.1, the rational source of Section 17.2 has input projection exactly $\mathbb{Z}$.*

Soundness. Let $\mathcal{B} = \max(|n|, |a|, |t|, |b|)$ and $\Delta = F(a, t, b) \in \mathbb{Z}$. Polynomial substitution preserves congruences between integral local elements. Lemmas 17.5 and 17.6 therefore give $M_r^{52} \mid \Delta$ whenever $\Delta \neq 0$. The index-height estimate (17) yields

$$\mathcal{B}^2 - 6 < 7H_*|\Delta|^{24/52}, \qquad |\Delta| < 5\mathcal{B}^4,$$

the latter because $\mathcal{B} \geq |a| > 7$. Consequently

$$\mathcal{B}^{2/13} < (7 \cdot 5^{6/13} + 6)H_* < 21H_*.$$

For the last strict inequality, $5^6 7^{13} < 15^{13}$. This contradicts Lemma 17.3. Thus $\Delta = 0$, and Lemma 14.9, with $b \neq 0$, gives $\log \mathcal{B} > |a| - 3$.

Lemma 16.1 applies to the exact tree (43) and gives

$$h(\beta) \leq H_*|a|^{3/4} < |a|/20.$$

If $\beta \neq n$, put $N = |\operatorname{num}(\beta - n)|$ in lowest terms. The order-16 test gives $M_r^{16} \mid N$. Therefore

$$\mathcal{B}^2 < 7H_*N^{3/2} + 6, \qquad N \leq 2e^{h(\beta)}\mathcal{B},$$

and $\log \mathcal{B} < 2\log(32H_*) + 3h(\beta)$. The anchor also gives $\log(32H_*) < |a|/20$, so the right side is less than $|a|/4 < |a| - 3$, a contradiction. Hence $\beta = n \in \mathbb{Z}$. Since $X \in \mathbb{Q}$ and $X^2 = n - 2 \in \mathbb{Z}$, the reduced denominator of X is one. □

Completeness. Fix $X \in \mathbb{Z}$, set $n = X^2 + 2$, and choose the Boolean correction c so that $cr_\beta > 0$ is square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, as in the finite squareclass computation accompanying Lemma 14.4. Let $\mathcal{S}$ be the prime support of cr_β. The scalar completeness construction in Lemma 16.1 first chooses $h_0 > 0$ divisible by 64 with $[h_0]P_*$ sufficiently deep at these primes. It then chooses a positive label a sufficiently divisible relative to h_0 and satisfying the anchor. The normalized scalar leaves are integral at primes dividing r_β and divisible at primes dividing c but not r_β. Choose positive t, b with $F(a, t, b) = 0$ by Lemma 14.9. This fixes the four root points and all scales $\mathcal{A}_r$.

Temporarily put $H = [h]P_*$ and $U_i = [hn_i]P_*$. All comparison tests are rational functions of $[h]P_*$ with nonzero denominators at $h = h_0$. Their poles lie above finite sets under fixed nonzero multiplication and translation maps on the two real factors, as required by the real progression lemma. Continuity preserves the scalar conditions at every place over $\mathcal{S}$ and bounds below the valuations of all sixteen comparison tests at the places over $2, 3, 17$, throughout a progression

$$h \equiv h_0 \pmod{L}, \qquad 64 \mid L.$$

One obtains L by putting sufficiently small open formal subgroups inside the finitely many translated neighborhoods and taking a common multiple of their indices. Lemma 16.3 gives, independently of future progression moduli, real bounds for the tests attained by arbitrarily large positive h in each nonconstant progression. The direct polynomial F introduces no denominator and has the same boundedness property.

Choose a positive integer D_0 such that every $D_0^2\theta/\mathcal{A}_r$ is integral, and put

$$v = D_0^2 w^2, \qquad \lambda = u/v.$$

The contact positions are the integer forms $u - (D_0^2\theta/\mathcal{A}_r)w^2$. Delete repeated forms, but retain their incidences in the nine detectors. The five forms within one detector are distinct, and the zero contact is the common form u. Apply Theorem I.1 to these forms. Prescribe arbitrarily high

divisibility of w, a sufficiently large lower bound for u/w^2, and primes larger than 76176 and all prime factors of the fixed scales and contact differences. In its odd squarefree avoidance modulus include every odd prime factor of $3 \cdot 17L$ except 7. Writing this modulus as M, the prescribed Gaussian classes are

$$\pi_p \equiv 9 \pmod{16}, \qquad \pi_p \equiv 1 + 3i \pmod{7}, \qquad \pi_p \equiv i \pmod{M}.$$

Let the resulting Gaussian generators be π_p, and write $d_-(p) = N(\pi_p - 1)$ and $d_+(p) = N(\pi_p + 1)$ for the cardinalities of the two reduced elliptic groups, orienting the places by Lemma 17.7. These cardinalities annihilate the reductions of P_*; their exact point orders are not needed. They satisfy

$$v_2(d_-) = 6, \quad v_2(d_+) = 2, \quad \gcd(d_-(p), d_+(q)) = 4$$

for every pair of selected primes. No odd prime factor of L divides either order. For 7 this follows separately from $d_+ \equiv 6$ and $d_- \equiv 2 \pmod 7$. Set $M_- = \operatorname{lcm}_p d_-(p)$ and $M_+ = \operatorname{lcm}_p d_+(p)$. The congruences

$$h \equiv 0 \pmod{M_-}, \quad h \equiv 4 \pmod{M_+}, \quad h \equiv h_0 \pmod{L}$$

are compatible: their pairwise gcds are $4, 64, 4$, respectively. Choose a sufficiently large positive h in this progression attaining the real bounds. It avoids $0, 4, -4$, so all point and subtraction guards hold. At a selected prime, H, U_i are formal at the place of order d_-, while V_0, V_i are formal at its conjugate place. Their exact ratios are n_i. Lemma 17.2 gives all sixteen required orders, including the order-52 polynomial test because $F(a, t, b) = 0$.

We check that the nine preparation-zero poles can be filled. For each detector,

$$R_r = \frac{\prod_{\theta \in \Theta}(u - (\theta/\mathcal{A}_r)v)}{v^5 \delta_r^6}.$$

The denominator is a rational square. At the five distinct contact primes of this detector R_r has valuation one, and elsewhere its valuation is even. The primes do not divide v: otherwise two distinct forms in the same detector would give the same prime value, impossible since $w > 0$ and their coefficients are distinct. The lower bound for the primes excludes the fixed scale and difference denominators. At either place over a contact prime a factor of W_r has valuation zero or negative, and its inverse has residue one or zero, respectively. At the successful place every inverse factor has residue one. Consequently T_r has residue one or two and is a unit. It follows that B_r has positive odd valuation at its contact primes and even valuation elsewhere. Its reduced denominator is a square.

The prescribed Gaussian classes make every contact prime square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$. The last product formula therefore makes B_r square in each of those fields. The uniform real bounds and $\lambda \longrightarrow +\infty$ give

$$R_r \sim \mathcal{A}_r^{-6} \lambda^{-1}, \quad W_r = 1 + O(\lambda^{-3}), \quad T_r = 2 + O(\lambda^{-3}), \quad B_r \sim (4\mathcal{A}_r^6 \lambda)^{-1}.$$

In particular $0 < B_r < 17$. At a small prime $p = 2, 3, 17$, put $\nu_p = v_p(v)$ and $\eta_{p,r} = v_p(\mathcal{A}_r)$. The common form u is a selected prime greater than these small primes, so $v_p(u) = 0$ and $v_p(\lambda) = -\nu_p$. The fixed neighborhoods give $v(D_j) \geq -c_p$. For sufficiently large prescribed ν_p one has

$$v_p(R_r) = \nu_p - 6\eta_{p,r}, \qquad v(\widetilde{D}_{r,j}) \geq -c_p + (m_j + 1)(\nu_p - \eta_{p,r}).$$

The nonconstant factor term has valuation at least $3\nu_p + (10m_j - 8)\eta_{p,r} - 2c_p$, tending to infinity. Thus T_r is as close to 2 as needed, including at the ramified prime 2, and

$$v_p(B_r) = \nu_p - 6\eta_{p,r} - 2v_p(2).$$

Prescribing the divisibility of w before applying the prime theorem therefore ensures $v_2(B_r) \geq 3$, $v_3(B_r) \geq 1$, $v_{17}(B_r) \geq 2$. Lemma 16.2 supplies the four witnesses in every coordinate pole.

Finally the progression preserved all scalar leaf conditions. The constant leaf on the all-left path makes W_β a unit at every prime dividing r_β; the divisibility of the other leaves makes it a unit at primes dividing c but not r_β. Since $|r_\beta| > 1$, it is nonzero. Therefore $b_\beta = cr_\beta W_\beta^2$ is positive, has square reduced denominator, and is locally square at $2, 3, 17$. Lemma 14.4 supplies the scalar preparation and pole witnesses for this final H. The anchor already holds for S. All witnesses have now been supplied without changing the root points. □

17.4 The quadratic compilation in 962 variables

We compile the shared-point source above using different point charts, pole coordinates, and arithmetic circuits. Every change has explicit rational witness maps. The arithmetic source, including its simultaneous prime input, is unchanged. Throughout this subsection

$$K = \mathbb{Q}(\sqrt{2}), \qquad D = (3 - \sqrt{2})^2 = 11 - 6\sqrt{2}, \qquad \mathsf{S} = 3^{2^{27}}.$$

An equation in K means its two rational component equations. The fixed integer S is a coefficient, not a coordinate. An assignment of a quadratic expression to one rational coordinate adds one quadratic equation. Affine expressions require no coordinate.

Theorem 17.9 (The 962-variable quartic)**.** *The complete table specified below defines*

$$P_{962} \in \mathbb{Z}[X_0, \ldots, X_{961}]$$

with degree four, exactly 55409 *nonzero coordinate monomials, integer coefficient content one, and coefficient height*

$$H(P_{962}) = 64\mathsf{S}^2 = 64 \cdot 3^{2^{28}} < 2^{429496736}.$$

It is the sum of squares of 919 *quadratic residuals and has the same rational input projection as the shared-point source of Theorem 17.8. Consequently, under that theorem's arithmetic hypotheses, for every $x \in \mathbb{Q}$,*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists X_1, \ldots, X_{961} \in \mathbb{Q}, \quad P_{962}(x, X_1, \ldots, X_{961}) = 0.$$

The 962 *variables include the input and* 961 *existential witnesses.*

A quadratic chart for the nine points. For each point introduce $t, h \in K$ and set

$$u = t^2, \qquad X = h^2, \qquad u(X - u/4) = D, \qquad Y = h(u/2 - X). \tag{48}$$

The three assignments use six rational coordinates. The displayed constraint contributes two quadratic residuals. The four rational components of t, h replace the four halfpoint components.

Lemma 17.10 (Exact point-chart maps)**.** *The chart* (48) *describes precisely the finite doubles used in the shared-point source, with the same rational witnesses up to the following invertible changes. It preserves all required nonzero point-coordinate domains. The rational component X_0 of X is positive.*

Proof. For an old halfpoint (a, b) with $b^2 = a^3 - Da$ and its tangent equation, both a, b are nonzero. Indeed $b = 0$ would give $a = 0$ or $a^2 = D$, whereas the tangent numerator $3a^2 - D$ would be $-D$ or $2D$. Then $a = 0$ is excluded by the curve equation. Define

$$t = b/a, \qquad h = (a^2 + D)/(2b) = a/t - t/2.$$

The curve equation gives $t^2 = a - D/a$. Substitution gives $t^2(h^2 - t^2/4) = D$. The doubling formulas give exactly $X = h^2$ and $Y = h(t^2/2 - h^2)$.

Conversely, a chart solution has $t \neq 0$ because $D \neq 0$. Set

$$a = th + t^2/2, \qquad b = at, \qquad m = h + t.$$

If $a = 0$, then $h = -t/2$, contradicting the chart equation. Thus $b \neq 0$. Direct expansion gives

$$a^2 - t^2 a = D, \quad b^2 = a^3 - Da, \quad 2bm = 3a^2 - D, \quad m^2 - 2a = h^2,$$

and $m(a-h^2)-b = h(t^2/2-h^2)$. The substitutions are inverse, so they recover exactly the old finite double. Lemma 17.1 therefore supplies its nonzero abscissa and ordinate, and the same remains true for the unchanged guarded differences and subsequent doublings.

Both real embeddings of D are positive. Since $t \neq 0$, the chart equation implies $h^2 = t^2/4 + D/t^2 > 0$ under both embeddings. Hence X_0, the average of these two positive values, is positive. □

The five affine subtraction formulas keep their original inverses of the differences of abscissas. Neither a subtraction guard nor a point occurrence is deleted. Relative to the direct halfpoint doubling circuit, the chart saves two rational arithmetic coordinates for each of the nine points.

The logarithm polynomial. Retain the source's coefficients

$$a_0 = 1, \quad a_1 = -2/5, \quad a_j = a_{j-1}(4j - 5)/(4j + 1).$$

For $P(w) = \sum_{j=0}^{12} a_j w^j$ write

$$P(w) = a_{12} Q(w)^2 + R(w), \qquad \deg Q = 6, \quad Q \text{ monic}, \quad \deg R \leq 5. \tag{49}$$

This identity is constructed without any witness division. Starting with the leading coefficient of Q equal to one, match degrees eleven down to six successively. At each step the new coefficient is determined by division by $2a_{12} \neq 0$. The remainder then has degree at most five. After computing w^2, w^3, every term through w^6 is quadratic in $1, w, w^2, w^3$. Assigning Q and then $a_{12}Q^2+R$ therefore uses four field assignments, rather than five. The eight order-52 logarithms save sixteen rational coordinates. The order-16 logarithms keep their exact degree-three polynomials. No truncation order changes.

Usually the final logarithm is $\ell = zP(Df^2)$, with $f = 1/x$ and $z = -x/y$. For U_0 and V_0 only, omit the separate z coordinates and impose the direct quotient equations

$$y\ell = -xP(Df^2). \tag{50}$$

The polynomial output is an affine expression in assigned coordinates, so these are quadratic component equations. The ordinate is nonzero by Lemma 17.1; hence (50) is exactly the old logarithm. The source's logarithm-nonvanishing result and every logarithm-ratio domain remain unchanged. In the scalar block replace $z(H)\rho = z(S)$ by

$$x(H)\rho = -y(H)z(S). \tag{51}$$

Both $x(H), y(H)$ are nonzero, so this has the same unique value $\rho = z(S)/z(H)$.

Products, tests, and detector evaluation. For each of the three orders, compute the elementary symmetric coefficients of $\prod_j (1 + qD_j^2)$ by a balanced product tree. At an internal node with coefficient vectors a_i, b_j, assign each component of $\sum_{i+j=k} a_i b_j$ as one quadratic expression. This is the ordinary coefficient-convolution identity. It preserves every coefficient and every test, including its assigned order.

The degree-ten detector polynomial is evaluated in two blocks:

$$\sum_{j=0}^{10} c_j q^j = \sum_{j=0}^{4} c_j q^j + q^5 \left(\sum_{j=0}^{5} c_{j+5} q^j \right). \tag{52}$$

Compute q^2, q^3, q^4, q^5 using four scalar assignments, assign the two components of the parenthesized block, and assign the two final components directly. The low block is not assigned separately. Every final summand is quadratic in the available coordinates. This uses eight rational coordinates. The degree-two and degree-four polynomials use their direct power sums, with their two component sums assigned once. These formulas are identities over K.

The Lucas expression is regrouped as

$$t^2 - a(a-2)tb + ((a-2)b)^2 - 1 = t^2 - (at)C + C^2 - 1, \qquad C = (a-2)b.$$

Thus one field multiplication is removed without changing the tested polynomial. The fixed quintic is evaluated by its exact coefficients using t^2; it remains $F_5(t) = t(t + 1/2)(t + 1)(t - 1/2399)(t + 32/407)$.

Weight quotients and their domains. Fix a detector and write A for its scale, $\delta = A\lambda - 1$, and e for its weight. The scales are

$$1, \quad x(Q)_0^2, \quad (1 + x(Q)_1^2)^2 \qquad (Q = A, S, T, Z).$$

They are all positive by Lemma 17.10. Retain the inverse $\delta i = 1$, and replace the separate fifth-power product and inverse of F_5 by

$$F_5(A\lambda)e = (A\delta)^4 (A\delta). \tag{53}$$

Here $(A\delta)^4$ denotes the output of its retained power-assignment gates; the displayed product is quadratic in the available coordinates. The right side is nonzero, so this equation forces both $F_5(A\lambda)$ and e nonzero. In particular the removed inverse of F_5 is recovered uniquely. Define the low-order weight by

$$\delta q_1 = e i^2,$$

where i^2 is likewise a retained square-assignment output. This gives the unchanged $q_1 = e/\delta^3$. Retain the repeated-squaring values e^6, e^{24}, e^{96} and set

$$c = q_1 e^6, \qquad q_{16} = c e^{24}, \qquad q_{52} = c e^{96}.$$

These are the old $q_1 e^{30}$ and $q_1 e^{102}$, with one shared product.

If the detector product is $W = w_0 + w_1\sqrt{2}$, retain $w_0 j = 1$ and use

$$w_0 n = (w_0^2 - 2w_1^2)/2, \qquad ev = in, \qquad B = nv. \tag{54}$$

The trace guard is essential and is retained. It gives $W \neq 0$ in K, so $\mathrm{Norm}(W) \neq 0$ and $n = \mathrm{Norm}(W)/(2w_0) \neq 0$. Equations (53)–(54) therefore give

$$B = \frac{i}{e}\left(\frac{\mathrm{Norm}(W)}{2w_0}\right)^2 \neq 0,$$

the original detector input. The reverse map restores $R = i/e$, its old inverse intermediates, and n^2 by their unique rational values. No inverse is inferred from an unguarded zero-over-zero quotient.

The scalar norm and finite corrections. The scalar contact polynomial r and its inverse $ri = 1$ remain. Writing $f = f(H)_0$, $u = f(H)_1$, and $g = f(S)_0$, retain the leaf $\ell = ui$ and replace two norm-tree nodes by

$$rn_0 = r + f^2, \qquad rn_1 = u\ell + g^2.$$

with the existing leaf ui retained. Since $r \neq 0$, these equal $n_0 = 1 + r(fi)^2$ and $n_1 = (ui)^2 + r(gi)^2$. All other nodes and all orders in the scalar norm tree are unchanged.

Replace the eight Boolean correction choices and their seven product coordinates by $c_0, \ldots, c_7 \in \mathbb{Q}$ satisfying

$$c_0^2 = 1, \qquad (c_j - c_{j-1})(c_j - p_j c_{j-1}) = 0 \quad (1 \leq j \leq 7), \tag{55}$$

where $(p_1, \ldots, p_7) = (2, 3, 5, 7, 11, 13, 17)$. Induction shows $c_j \neq 0$ and $c_j/c_{j-1} \in \{1, p_j\}$. Thus c_7 is exactly an allowed signed squarefree correction. Conversely every original choice determines such a chain. Recover the original Booleans by

$$\epsilon_- = (1 - c_0)/2, \qquad \epsilon_j = \frac{c_j/c_{j-1} - 1}{p_j - 1}.$$

All denominators are nonzero. The scalar preparation coordinate, its defining quotient, and the explicit inverse of its pole input are retained.

The ten pole blocks. Write b for a pole input, $C = 17b$, and U for its prepared value. For the nine coordinate poles $U = b$. In the scalar pole U retains the source's preparation formula and its denominator domain. Every C is nonzero: this was proved above for coordinate detectors, and the scalar input has its retained inverse.

Replace the old witness e_0 by $z = Ce_0$. Introduce the five intermediates

$$P = p^2, \quad F = U^2, \quad H = z(C - F), \quad V = s^2, \quad Q = q^2,$$

and impose

$$CP = (C - z)(C + 3z), \qquad CQ = (V - H)(V + 3H). \tag{56}$$

Every equation is quadratic. Under $z = Ce_0$, the first residual is C times the old residual $p^2 - C(1 - e_0)(1 + 3e_0)$, and the second is unchanged because $H = Ce_0(C - U^2)$. The inverse map $e_0 = z/C$ is defined. This saves one arithmetic coordinate at each pole, without changing the pole predicate or any preparation condition.

Counts and the final polynomial. The initial allocation is

	coordinates	structural residuals
input	1	0
nine point charts	36	18
common slope	1	0
nine coordinate poles	36	18
scalar pole with preparation	5	2
private anchor	3	1
signed correction chain	8	8
total	90	47

The complete program has 904 arithmetic coordinates and their 904 defining equations. Hence the raw counts are 994 coordinates and 951 quadratic residuals.

The private anchor uses exactly the old affine change

$$u = \mathsf{S} + (U - 3V + 3W)/4, \quad v = \mathsf{S}/2 + V/4, \quad w = 3\mathsf{S}/2 + W/4.$$

Its determinant is $1/64$. If $T = f_0^2 + f_0 f_1 - f_1^2$, sixteen times the old anchor after substitution is

$$-48T^2 + U^2 - 6UV + 6UW + 15V^2 - 18VW + 7W^2 + 8\mathsf{S}U.$$

No anchor condition or witness is removed.

Each certified compiler elimination solves a row $aY + q = 0$ with $a \in \mathbb{Q}^*$ and q independent of Y, substitutes $Y = -q/a$ everywhere, and removes that row. It is accepted only if every remaining residual is still quadratic. The reverse map reinserts the unique displayed value of Y. There are exactly 32 such eliminations, no row additions, and no omission of an unused coordinate. The input is retained. Consequently

$$90 + 904 - 32 = 962, \qquad 47 + 904 - 32 = 919.$$

Normalize the remaining rows to primitive integer polynomials $q_1, \dots, q_{919}$ and define

$$P_{962}(X_0, \dots, X_{961}) = \sum_{\nu=1}^{919} q_\nu^2. \tag{57}$$

The complete coefficient table is `quartics/quartic-962/polynomial.json`, with SHA-256

`6b798de7d00ac3c7d725ca0a69d39f1eac1ddb02fee2756cb6323770b952af32`.

An entry with coefficient c, exponent `sPower` k, and coordinate list $[i_1, \dots, i_d]$ means $c\mathsf{S}^k X_{i_1} \cdots X_{i_d}$. Every entry is included and repeated indices give powers. This defines an ordinary integer polynomial.

Proof of Theorem 17.9. For a source solution apply the point-chart, correction-chain and pole substitutions, and evaluate all new assignment and quotient coordinates in dependency order. The identities and domain proofs above show that every structural residual vanishes. Conversely apply the inverse substitutions, restore every deleted quotient by its proved nonzero denominator, and evaluate the source assignments. Every source point, logarithm, test, detector input, scalar value and anchor value is recovered. These maps fix X_0. The affine anchor change, certified eliminations and nonzero row normalizations preserve the same projection. Over $\mathbb{Q}$, (57) vanishes exactly

when all 919 residuals vanish. This proves the projection statement, and Theorem 17.8 gives its conditional arithmetic interpretation.

Independent exact convolution gives 55409 distinct monomials. Of their coefficients, 55401 are S-independent integers, seven are $c\mathsf{S}$ with $0 < |c| \le 768$, and one is $64\mathsf{S}^2$. The largest absolute S-independent coefficient has bit length 1526. Thus no coefficient vanishes on specialization and $64\mathsf{S}^2$ is the exact height. It occurs at X_{79}^2, where $X_{79} = U$. The coefficient of X_0^4 is one, proving degree four and integer content one. Finally $\log_2(64\mathsf{S}^2) < 6 + (8/5)2^{28} < 429496736$ because $3^5 < 2^8$. □

The frozen package includes the builder, raw systems, all 904 gate definitions, the exact elimination and normalization certificate, compact coordinate map, final residuals, complete coefficient table, source-equivalence proofs and independent reviews. Its manifest has SHA-256

`36fca72cb0eb126193ffa43d890ff09d5e29cbf1274561d2632ed1e227163787`.

The independent checker verifies every gate, all 894 certificate operations, the complete sum-of-squares expansion, support, integrality, content and height. A fresh isolated reproduction gives the same seven deterministic artifacts byte for byte. These finite checks establish the stated algebraic facts; they do not prove any inherited arithmetic input.

18 Why a single defining polynomial must have degree at least four

The argument in this section comes from the agentic notes [6, section on unconditional lower bounds]. We give its specialization to $\mathbb{Q}$, including the treatment of singular cubics, cones and geometric reducibility. It is independent of the arithmetic inputs to our constructions.

For $F \in \mathbb{Q}[T, Y_1, \dots, Y_r]$, write

$$S_F = \{t \in \mathbb{Q} : \exists\, y \in \mathbb{Q}^r,\ F(t, y) = 0\}.$$

All degrees below are total degrees, including the input variable T.

Theorem 18.1 (Degree obstruction)**.** *If $S_F = \mathbb{Z}$, then* $\deg F \ge 4$.

We use two published results. The integral form of Hilbert's irreducibility theorem says that an irreducible polynomial $g(T, Y) \in \mathbb{Q}(T)[Y]$ admits infinitely many specializations $T = t \in \mathbb{Z}$ which preserve its degree in Y and are irreducible over $\mathbb{Q}$. Deleting finitely many values also avoids its coefficient poles and any prescribed nonzero polynomial in t. This integral formulation is recalled in the introduction of Müller [5]; merely obtaining specializations in $\mathbb{Q}$ would not suffice here. The second result is Kollár's theorem [4, Theorem 2]: a geometrically integral cubic hypersurface of dimension at least two over a perfect field is unirational over that field if it has a rational point and is not a cone. This is a special case of his statement for irreducible cubics. We apply it only over $\mathbb{Q}$, and remove every vertex direction before applying it. In particular, we do not assume that a cone over a plane cubic is unirational.

Lemma 18.2. *A nonconstant rational function $R \in \mathbb{Q}(U_1, \dots, U_m)$ cannot take only integer values at the rational points of a nonempty Zariski-open subset of $\mathbb{A}^m_{\mathbb{Q}}$ on which it is defined.*

Proof. Shrink the open subset to a nonempty principal open subset and choose a connected real open ball in it, avoiding the denominator of R. Rational points are dense in this ball. If their images belong to the closed set $\mathbb{Z}$, continuity puts the image of the entire ball in $\mathbb{Z}$. Connectedness then makes R constant on the ball. Its numerator minus that constant times its denominator vanishes on a real open set, hence is the zero polynomial. This contradicts nonconstancy. □

Consequently, if an integral variety $X/\mathbb{Q}$ is unirational and $f \in \mathbb{Q}(X)$ is nonconstant, the rational values of f cannot all be integers. Indeed, compose f with a dominant rational map from projective space to X, and restrict to a nonempty affine open subset of its domain. Dominance keeps this composite nonconstant, so Lemma 18.2 applies.

Lemma 18.3. *If* $\deg F \leq 2$ *and* $S_F \subseteq \mathbb{Z}$*, then* $|S_F| \leq 2$*.*

Proof. Otherwise choose rational zeros a, b, c with distinct input coordinates. If they lie on an affine line, the restriction of F to that line has three zeros and degree at most two, so vanishes identically. The input is nonconstant on the line and takes every rational value, a contradiction.

If the points are not collinear, let L be their rational affine plane. If $F|_L = 0$, a line in L with nonconstant input gives the same contradiction. If $F|_L$ factors over $\mathbb{Q}$ into affine linear polynomials, its zero set consists of at most two rational lines. One contains two of the three selected points and again has nonconstant input. A nonzero polynomial of degree at most one cannot vanish at three noncollinear points.

It remains to consider an irreducible quadratic on L. Its projective conic has a smooth rational point: a reduced irreducible quadratic in characteristic zero has at most one geometric singular point, while the three chosen affine points are distinct. A conic with a smooth rational point is rational. Explicitly, intersecting it with the pencil of lines through that point gives a rational parametrization by taking the second intersection point. The input function is nonconstant because it takes three different values. Composing it with this parametrization contradicts Lemma 18.2. □

Proof of Theorem 18.1. Suppose $S_F = \mathbb{Z}$ and $\deg F \leq 3$. The polynomial is neither zero nor a nonzero constant. Lemma 18.3 excludes degree at most two. It also excludes a reducible cubic: every nonconstant irreducible factor then has degree at most two, and the finite union of their projections cannot equal $\mathbb{Z}$. Thus F is irreducible of degree three.

Homogenize F to H on a rational vector space E, using X_0 as the homogenizing coordinate and X_1 for the input coordinate. Thus on $X_0 \neq 0$ the input is X_1/X_0. The homogeneous polynomial H is irreducible over $\mathbb{Q}$. If it is geometrically reducible, its geometric irreducible factors form a single Galois orbit with equal degrees. In characteristic zero they occur without repetition. Since the total degree is three, there are exactly three conjugate linear factors. Every rational zero lies on each of their hyperplanes. Take rational zeros normalized by $X_0 = 1$ with inputs zero and one. Their whole affine joining line lies on all three hyperplanes and its input takes every value in $\mathbb{Q}$, a contradiction. We may therefore assume H is geometrically integral.

Define the rational vertex subspace

$$W = \{v \in E : D_v H = 0 \text{ identically}\}, \qquad D_v H(x) = \frac{d}{ds} H(x + sv)\Big|_{s=0}.$$

These are linear conditions on v. In characteristic zero, $D_v H = 0$ is equivalent to $H(x+sv) = H(x)$ for all x, s. Thus H descends to a cubic $\overline{H}$ on E/W. After a basis of W is extended to one of E, this simply deletes the coordinates on which H does not depend. The quotient has no remaining vertex direction. This assertion holds over the algebraic closure as well, since the kernel of a matrix over $\mathbb{Q}$ commutes with extension of scalars. Hence its projective hypersurface is not a cone.

For $v \in W$ and a rational zero x with $x_0 = 1$, translation by sv gives rational zeros whose inputs, whenever defined, are

$$\frac{x_1 + sv_1}{1 + sv_0}.$$

Unless $v_1 = x_1 v_0$, this is a nonconstant fractional linear function of s, with cofinite image in $\mathbb{Q}$. Its image cannot be contained in $\mathbb{Z}$. Applying the resulting equality first at input zero and then at

input one gives $v_0 = v_1 = 0$ for every $v \in W$. Therefore X_0 and X_1 descend to independent linear forms on E/W.

Choose quotient coordinates beginning with these forms. The affine chart $X_0 = 1$ of $\overline{H} = 0$ has exactly the same input projection as $F = 0$. To see both inclusions, project rational points from E to E/W, and in the reverse direction use a rational linear section of that projection. This section preserves X_0 and X_1, and H is constant along each fiber.

If the quotient projective hypersurface has dimension at least two, Kollár's theorem applies. It is geometrically integral, has a rational point in $X_0 = 1$, is not a cone, and the ground field is perfect. It is therefore unirational. Its input function is nonconstant, since its rational values include all integers. The consequence of Lemma 18.2 then contradicts that its input projection is contained in $\mathbb{Z}$.

If the quotient has dimension one, its affine chart is defined by an irreducible polynomial $G(T, Y) \in \mathbb{Q}[T, Y]$. If its degree in Y is zero, its input projection is finite. If that degree is one, solving the linear equation for Y gives a rational point above every rational T outside a finite set, also impossible. If its degree in Y is at least two, Gauss's lemma makes it irreducible in $\mathbb{Q}(T)[Y]$. Integral Hilbert irreducibility supplies an integer t for which $G(t, Y)$ remains irreducible of that same degree. It has no rational root, contradicting $S_G = \mathbb{Z}$. Finally, a zero-dimensional projective quotient has only finitely many rational points and cannot have input projection $\mathbb{Z}$. These exhaust the possibilities and prove the theorem. □

Remark 18.4. This lower bound concerns a single polynomial equation and counts degree in every variable. It allows systems of quadratic equations; over $\mathbb{Q}$, their sum of squares is quartic. It puts no lower bound here on the number of variables in a degree-four definition.

A Dependency order in the arithmetic argument

The even coefficient bound uses stationary-base selection as well as integrality. The odd minimum uses a fixed nonzero witness and the prior genus-index lower bound. Support removal is applied only to the actual unramified specialization. The functional bound and determinant clearing are proved separately, and the same corrected coordinate is used at every address. These constructions are given in Sections 7–9 and Appendices E–G.

The conversion to ordinary integers uses the canonical model and total multiplication, followed by the exact computation relation. Only the shared-shift separation argument in Section 11 identifies the external rational variable with the ordinary value of its integer index. The literal substitution and size calculation then apply to the same expression P_*.

B Source identities and scope

The companion `SOURCE_MAP.json` records exact paths, hashes and locators, distinguishing mathematical and construction sources, artifacts, and version records. The base collection contains 83 sources; parity, construction and uniformity supplements are in separate addenda. The principal local sources are listed below; full paths are in the source map.

ID	Source basename	Use
S01	`ordinary-integer-polynomial.js.txt`	Outer construction
S02	`rational-circuit-translator.js.txt`	Substitution semantics
S03	`finite-polynomial-expressions.js.txt`	Fixed constructors

ID	Source basename	Use
S04	`variable-shift-polynomial-library.js.txt`	Close and Test
S05	`elliptic-generator-proof.txt`	Generator certificate
S06	`explicit-elliptic-data.txt`	Rank, torsion, codes
S07	`verify-elliptic-generator.js.txt`	Exact finite algorithm
S08	`six-contact-height-proof.txt`	Contacts and constants
S09	`z-in-q-height-independent-audit.txt`	Height proof
S10	`explicit-pole-predicate.txt`	Pole polynomial
S11	`z-in-q-pole-completeness-audit.txt`	Simultaneous descent
S12	`parity-frozen-4eaee875.tex`	Raw local matrices
S13	`multiplication-certificate-correctness-proof.txt`	Multiplication proof
S14	`twenty-two-prime-pattern-proof.txt`	Selected primes
S15	`z-in-q-variable-shift-bridge.txt`	Ordinary-value graph
S16	`z-in-q-ordinary-integer-assembly-proof.txt`	Outer composition
S17	`z-in-q-restricted-converse-assembly.txt`	RC assembly inputs
S18	`ring-limits-frozen-4eaee875.tex`	Finite models and limits
S19	`paired-functional-and-uniformity-proof.txt`	Paired functional
S20	`determinant-transfer-audit.txt`	Clearing and transfer
S21	`variable-shift-precision.py.txt`	Exact decision program
S22	`precision-register.js.txt`	Ideal IR and lowering
S23	`specialize-precision.js.txt`	Wrapper specialization
S24	`integer-halting-correctness-proof.txt`	Natural compiler
S25	`z-in-q-explicit-cutoff-bound.txt`	Literal cutoff
S26	`z-in-q-polynomial-size-bound-proof.txt`	Syntax/norm argument
S27	`z-in-q-numerical-polynomial-bounds.txt`	Tower comparison
S28	`determinant-switch-audit.txt`	Switches and outer rank
S29	`ring-determinants-frozen-4eaee875.tex`	Ring data and derivatives
S30	`preliminaries-frozen-4eaee875.tex`	Selmer coranks and lengths
S31	`fixed-certificate-constants.js.txt`	Exact least-multiple search
S32	`pointwise-frozen-4eaee875.tex`	Distinct point indices

In particular S30, S18, S29 and S32 are, respectively, Sections 2, 4, 5 and 9 of OpenAI's *A pointwise 2-converse for elliptic curves with rational two-torsion* [74], frozen at `openai/math` commit `4eaee87548250b2682affda7d05d947c05554bae`. The source filenames are provenance labels for those sections, not independent publications or independent arithmetic assumptions.

The supplement manifest also records the revised fixed-degree CM proof, the complete natural compiler proof, actual arithmetic and analytic producer arguments, finite-Artin and determinant-count clarifications, the fixed BFH witness application, exact substitution checks, and the new fixed-form, weighted-theta, operator and normalization derivations. For the height estimate, the derivation in Section 4 supersedes the historical arithmetic-height arguments in S08–S09 and the earlier CM supplements. Those historical alternatives are not additional premises. Primary theorem applications remain explicit in the mathematical text. The version records identify the source files used in the construction.

C Literal finite-constructor specification

The supplementary file `ordinary-integer-definition.json` is part of this specification. It is a finite directed acyclic description, not an expanded monomial list. The following rules determine one polynomial from its bytes. Diagnostic labels and intended-theorem strings add no equations.

The root is `integers`. Variable zero is the external input and variables one through 292890231904 are its rational witnesses. A nonnegative point reference r means $X_r, \dots, X_{r+4}$. References $-1, -2, -3$ respectively mean

$$(1,0,0,0,0), \quad (0, g_0, -145, g_1, g_2), \quad (0, g_0, -145, -g_1, -g_2).$$

Every model recipe conjoins $8g_0 - 1641 = 0$, $4g_1 + 15465 = 0$, $32g_2 - 87483 = 0$.

C.1 Scalar nodes and fixed expressions

Read nodes in their stored topological order. A `const` is its decimal integer; `var` is its local variable; `add` and `mul` are ordinary polynomial operations. For a `call` to a template with p parameters, substitute the specified node polynomials for variables $0, \dots, p-1$, and X_{w+j} for variable $p+j$, where w is the witness start. The value is the callee's entire root polynomial.

A `fixed_integer` evaluates a finite expression using literals, references to preceding definitions, addition, multiplication, nonnegative powers, maximum, factorial, the catalogue product or maximum, and the least formal multiple. The catalogue uses the first 22 primes in increasing order in each unit class. The searches are deterministic and terminate by Section 12. They produce constants, not quantified witnesses.

A `psi_component` uses the division-polynomial recurrence in Section 2, pair multiplication $(a,b)(c,d) = (ac + 2bd, ad + bc)$ and $A = (-11243, 7950)$. It returns component zero or one; every index is fixed. The positive fixed point codes use

$$x_n = x(P_0) - \frac{\psi_{n+1}(P_0)\psi_{n-1}(P_0)}{\psi_n(P_0)^2}, \qquad y_n = \frac{\psi_{2n}(P_0)}{2\psi_n(P_0)^4}.$$

Nontorsion makes these fixed denominators nonzero. Negative indices negate both ordinate components.

A `finite_guard_product` ranges over $(a,b,c) \in [-T,T]^3 \cap \mathbb{Z}^3$ with $ab \neq c$, with fixed T. Write the fifteen coordinates of the three codes as reduced A_j/B_j, $B_j > 0$, and put $D = \prod_j B_j$. Its factor is

$$\sum_{j=1}^{15} (DX_j - (D/B_j)A_j)^2.$$

Multiply these factors, with empty product one. This is an integer polynomial with precisely the finite false-triple zero set used above.

C.2 Recipe constraints and witness scope

A model-evaluation recipe sums squares of the generator equations, listed natural calls, scalar-constant calls, add/multiply/power-value calls, and the five coordinate differences of `rootEquality`. A source-variable row with a non-null natural witness start contributes exactly that natural call. Operation kinds zero, one and two select addition, multiplication and power value. Source-node and provenance fields add no equations.

A sum-of-squares composition sums squares of its callee polynomials; a product composition multiplies them. Parameter `[var,i]` means X_i. Each private block uses its recorded start and

length. The checked ranges are disjoint from parameters and other private blocks. These rules include globally conjoined fixed equations in active callees.

C.3 The saved templates and the mathematical predicates

The following dictionary identifies the relations encoded by the saved templates, in addition to the decoding rules above. Point arguments are canonical five-coordinate codes. Field values have their two rational components in the basis $(1, \sqrt{2})$; conjugation negates the second component. An inverse equation guards each displayed quotient. Conjunction is encoded by a sum of squares, and disjunction by a product, with the private witness blocks specified in the file. The identities are identities of zero sets over $\mathbb{Q}$; squaring a callee polynomial can change the literal polynomial without changing the relation it defines.

Saved template	Relation and constraints
`member`	Membership in the canonical code domain $D = 2E(K)$ of Section 3.
`raw_add`, `raw_z`, `raw_inverse_x`	The group-law relation on typed inputs, $z = -x/y$, and $w = 1/x$, respectively, with $z(O) = w(O) = 0$.
`log25`	The two-component equality to the displayed scaled degree-25 polynomial L in Section 6.
`pole`	$Q_{\mathrm{pole}}(B)^2$, where Q_{pole} is the eleven-witness polynomial of Section 5. The outer square comes from the singleton conjunction added by the rational builder.
`test_group`	The twelve field tests of Section 6, including $A_i - \sigma(A_i')$ for $i = 1, 2, 3$. Its four input points are $[c_*]P_0, [c_*]P_1, [c_*]P_2, [c_*]P_3$. It types the first four auxiliary points and obtains the second four by subtraction; its two denominator points and logarithms are nonzero.
`slope`	The normalized twelve-test products in six private groups, their traced inverses, and $\Phi(R/\Theta^2)$ from Section 6, with all contact and denominator guards.
`height_certificate`	The 27 slopes summing in triples to the nine homogeneous coordinate quantities in that section, together with membership of the three input points.
`multiply.scalar`, `multiply.guard`, `multiply`	The fixed $[c_*]$ graph; the inverse of the finite false-triple product; and their conjunction with the height certificate at the fixed ℓ_0.
`natural`	Four model squares, formed by multiplication, whose sum is the input code, as in Section 10.
`value.signed_encode`	$N = \iota(n)$ and a natural code $C = \iota(c)$ with $n = c/2$ for even c, and $n = -(c+1)/2$ for odd c. The two branches use $c = 2q$ and $c = 2q + 1$ with $q \geq 0$.
`value.abs`	$a = \lvert x \rvert$: a is a sum of four rational squares and $a^2 = x^2$.
`value.close_coordinates`	The strict coordinate window of Section 11, using the typed points A, L, J, $r = \mathrm{Tr}(z(J)/z(A))/2$ and $\epsilon = \lvert(z(L))_0\rvert + \lvert(z(L))_1\rvert > 0$, with $\lvert t - r\rvert < \epsilon$. This template alone does not assert the precision relation.

Saved template	Relation and constraints
`value.test`	Exactly Test(N, κ, t) of Section 11: $C_1 = [\kappa n]P_0$, $V_0 = H - \kappa P_0$, $V_1 = U - C_1$, the six displayed raw-z tests, their normalization by $F(t)$, and the single guarded pole input. These ratios use z, not L.
`value.exceptions`	The six branches $N = \iota(c)$ and $t = c$, $c \in \mathcal{C}$, with the constant points computed by the fixed group-law chains.
`scalar.integer_constant_`i	The fixed integer scalar graph requested by the translated precision computation.

The composition recipes make the remaining identifications explicit. `power_value` translates the source power template with its three input codes and output code; `precision` translates the five-natural-input precision polynomial, in the order (n, κ, a, ℓ, j) after signed encoding. The recipe `close` introduces A, L, J, forms those five signed encodings, conjoins `precision`, and calls `value.close_coordinates`. Thus it is the full Close. The recipe `main` introduces one shared κ and conjoins `close` and `value.test` at exactly that same code. The product recipe `value_union` joins this branch with `value.exceptions`; `value` also imposes `member` on N. Finally `integers` existentially introduces N and calls `value` with external coordinate $t = X_0$. These are precisely Val and its input projection. The ordered calls and private ranges are checked by `auditOrdinaryIntegerPolynomial` in `construction/ordinary-integer-polynomial.js`. `scripts/check_construction.py` regenerates the saved expression byte for byte. The dictionary supplies the mathematical identification; those finite checks verify the stored syntax and wiring, not the arithmetic theorems used to prove the predicates.

The following parameter and witness counts are extracted from the exact specification, without expanding its polynomial.

Template	Parameters	Witnesses	Root node
`member`	5	7	149
`raw_add`	15	6	308
`raw_z`	7	1	68
`raw_inverse_x`	7	0	54
`log25`	4	0	229
`pole`	1	11	106
`test_group`	44	162	515
`slope`	22	1,298	2681
`height_certificate`	36	35,106	260
`multiply.scalar`	10	1	136
`multiply.guard`	15	1	22
`multiply`	15	35,134	83
`natural`	5	140,604	79
`value.signed_encode`	10	140,654	138
`value.abs`	2	4	23
`value.close_coordinates`	16	53	271
`value.test`	11	35,282	775
`value.exceptions`	6	406	487
`scalar.integer_constant_0`	10	1	136

Template	Parameters	Witnesses	Root node
`scalar.integer_constant_1`	10	1	136
`scalar.integer_constant_2`	10	1	136
`scalar.integer_constant_3`	10	1	136
`scalar.integer_constant_4`	10	1	136

Recipe	Parameters	Witnesses	Total variables
`power_value`	20	17,786,698	17,786,718
`precision`	25	292,889,492,836	292,889,492,861
`close`	11	292,890,196,199	292,890,196,210
`main`	6	292,890,231,486	292,890,231,492
`value_union`	6	292,890,231,892	292,890,231,898
`value`	6	292,890,231,899	292,890,231,905
`integers`	1	292,890,231,904	292,890,231,905

The five scalar-constant templates use the values in `library.fixedConstants`. Large program literals are retained byte-for-byte in the supplementary specification. Replacing them by unspecified sufficiently large constants would change this definition.

D Primary theorem applications

Input	Exact application boundary
Selmer parity [49]	Rational two-isogeny, $p = 2$, full Selmer corank; no finite-Sha hypothesis.
Kato [55]	Fixed weight-two form and symbol, integral transfer and fixed-valuation smoothing; rational divisibility is Theorem 13.4(2), with oddness, rational irreducibility and the unipotent condition.
BFH and Hoffstein–Luo [56, 57]	Actual signed-prime twist with trivial character and arbitrary finite splitting set. Only the nonvanishing clause has the four-prime bound before the extra prime.
Waldspurger and Ramsey [63, 64]	Full good-Hecke eigenspace, same dyadic squareclass, and algebraic vectors by flat field extension. No arbitrary-ring base change in weight 3/2.
CM norms and heights [60, 59, 62]	Split level, conductors coprime to level and the new prime, no extra units, unaveraged sums and one fixed parametrization.
Duality [42, 44]	Actual derived Kummer conditions and their homotopies. Support change only after the coefficient representation is unramified at the removed place.
Forward finiteness [53, 54]	Only at known analytic nonvanishing/simple zeros or a known nontorsion Heegner trace.
Isogeny ratio [42, proof of I.7.3]	Finite Sha, dual sum/difference maps, canonical heights and pull-back measures; no BSD leading-term assumption.

Input	Exact application boundary
Height [23, 27, 28, 29, 30, 36]	Odd irreducible de Rham representation at five; explicit conductor and modular-Jacobian transfer. CM order class number bounded by the degree of the original Hom field.
Primes [38, 39, 40, 41]	Fixed finite-complexity system after fixing sieve cutoff, and the unconditional explicit progression estimate.

The source package includes the specification, natural precision polynomial and construction sources in `SUPPLEMENT_MANIFEST.json`. The larger proof corpus is identified by hashes and stable locators in `SOURCE_MAP.json`. A source identity is not a proof verdict.

E Fixed-form arithmetic and the even coefficient source

This appendix supplies the constructions used in (3.1)–(3.3). Throughout, $E = E_\ell$ is fixed, with the lattice, parametrization and period convention of Section 7. Constants may depend on these data and on the prescribed finite list of local types. They do not depend on the number or sizes of the later good conductor primes.

E.1 One retained class, including discriminant $4h$

All numbered primary references in this subsection are to Kato [55].

Fix a non-CM elliptic curve $E/\mathbf{Q}$ of conductor N, its weight-two newform f, $T = T_2E$, and $V = T \otimes \mathbf{Q}_2$. In the application $E = E_\ell$ is fixed first. Put $\Lambda = \mathbf{Z}_2[[\Gamma]] = \mathbf{Z}_2[[t]]$, where Γ is the real cyclotomic Galois group. Let $\kappa : G_{\mathbf{Q}} \to \Gamma \subset \Lambda^\times$ be the quotient map, and set $A = \Lambda[1/2]$. We use arithmetic Frobenius throughout. The action on the cyclotomic group-ring coefficient is the inverse deck action.

The precise output.

Proposition E.1 (One retained Kato system and its two comparisons)**.** *Use the fixed curve, lattice and cyclotomic convention just specified. Fix $e \geq \max\{3, v_2(N)\}$ large enough for the fixed initial local data; the smoothing choice is (K5). Let G be a finite elementary abelian two-group, let $\psi : G_{\mathbf{Q}} \to G$ be a continuous homomorphism, and let $\chi_x = \chi_0(x \circ \psi)$, for $x \in \widehat{G}$, be a finite family of quadratic characters with one fixed real sign and fixed local squareclasses at the primes dividing $2N$. Since the characters of G separate its elements, $\psi(c) = 1$ for complex conjugation c, and $\psi|_{G_{\mathbf{Q}_v}} = 1$ at every $v \mid 2N$. All varying conductor primes are good odd primes for E. Let F be the squarefree least common multiple of the odd parts of the primitive conductors. There is a single retained-smoothed class*

$$Z_F \in 2^{-c_E} H^1(G_{\mathbf{Q},S}, T \otimes \Lambda[G](\chi_0\psi\kappa^{-1})), \qquad S = \mathrm{prime}(2NF), \tag{K1}$$

where c_E is fixed before F and G. There is also one $F_E \in \Lambda$ with $F_E(0) \neq 0$, fixed before F, G, such that at every height-one prime $\mathfrak{p}$ of A,

$$\ell_{A_\mathfrak{p}} H^2(C_{S,x})_\mathfrak{p} \leq \ell_{A_\mathfrak{p}} \left(H^1(C_{S,x})/AF_E(Z_F)_x \right)_\mathfrak{p}. \tag{K2}$$

Here $C_{S,x}$ is the full rational global Iwasawa complex of the constituent. For an odd squarefree signed parameter h giving $\chi_x = \chi_h$, central reciprocity for this same class gives

$$v_2 \exp^*\big(F_E(Z_F)_x(0)\big) = v_2 L_*(h) + \sum_{q|F,\ q\nmid h} v_2 \# E^h(\mathbf{F}_q) + 2e + v_2 F_E(0) + \epsilon_h, \tag{K3}$$

when the value is nonzero. The numbers ϵ_h belong to a bounded set depending only on the fixed curve, period conventions and local types at two. A zero central L-value gives zero dual exponential. Formula (K3) does not assert that its remaining unit is one. No part of (K1)–(K3) compares integral lattices of varying newforms.

Proof.

The retained route and its comparison. All classes in this subsection retain the smoothing integers of (K5). The comparison used below is the exact identity (E.1), obtained by integral weighted corestriction and the norm relation of [55, Proposition 8.12]. We do not divide by the smoothing operators, or compare a chosen generator with a separately normalized zeta class for the varying newform. In particular the central factor is s_F of (K6), whose valuation is $2e$ for every F. The use of power-series zero finiteness in paragraph 7 establishes nonvanishing of the supplied Euler system in both generic sign components; it is not a comparison of two Euler systems.

1. One original lattice and one symbol combination. Kato (8.1.3), p. 180, constructs retained-smoothed classes

$$_{c,d_s}z_m(f,1,1,\xi,S_m) \in H^1(\mathbf{Z}[\zeta_m,1/2],L_f(1)), \qquad S_m = \mathrm{prime}(2mN), \tag{K4}$$

where $L_f = V_{\mathcal{O}_\lambda}(f)$ is the image lattice of the fixed form. For $\xi \in \mathrm{SL}_2(\mathbf{Z})$, section 8.9, p. 185, explicitly constructs the trace homomorphism into the integral induced lattice

$$\mathbf{Z}_2[\mathrm{Gal}(\mathbf{Q}(\zeta_m)/\mathbf{Q})] \otimes V_{2,\mathbf{Z}_2}(Y_1(N)).$$

Section 8.11 projects this lattice onto L_f. Auxiliary modular curve levels used in the trace do not change its target lattice. Choose a rational identification $L_f(1)\otimes\mathbf{Q}_2 = V$. One fixed 2^{c_E} sends its lattice into T.

By Kato 13.6–13.7, p. 227, the symbol span has a nonzero projection to each Betti sign line. The union of the two rational kernels does not contain the whole span. Choose an integral finite linear combination ξ with both projections nonzero, clearing its fixed denominators once. Norm relations and reciprocity are linear, so the same combination works at every m. Although (K4) is written with one symbol, it henceforth denotes this fixed combination.

2. Retain one smoothing choice for the whole family. Fix $e \geq \max\{3, v_2(N)\}$, enlarged only for fixed initial data, and take

$$c = d_s = 1 + 2^e\, 3 N_{\mathrm{odd}} F. \tag{K5}$$

These integers exceed one, are prime to $6NF$, are one modulo N and F, and are one modulo every dyadic quadratic conductor under consideration. Distinctness or primality of c, d_s is not a hypothesis of Kato's construction. The exponents (u,v) in (4.2.4), p. 143, are $(1,1)$ when $k = 2, r = r' = 1$. The retained-smoothed version of Theorem 6.6, p. 163, therefore has central scalar

$$s_F = (c^2-c)(d_s^2-d_s) = c(c-1)d_s(d_s-1), \qquad v_2(s_F) = 2e. \tag{K6}$$

No smoothing factor is divided out. The prime factors of cd_s exclude some Euler-system indices in Kato 13.1; they are not added to the inverted arithmetic support S, as is explicit in (8.1.3) and in the definition of H^q in section 8.2. In particular their number creates no local H^2 correction.

3. The integral j_* condition survives ramified transfer. For a number field K, let $X_K = \operatorname{Spec}\mathcal{O}_K[1/2]$ and $j_K : \operatorname{Spec} K \to X_K$. At finite coefficients $M = T/2^aT$, the low-degree Leray sequence gives an injection

$$H^1(X_K, j_{K,*}M) \longrightarrow H^1(K, M)$$

whose image is the kernel of all inertia restrictions outside two. Indeed the higher-direct-image stalk formula and Leray sequence [43, Tags 03Q9 and 03QA] identify the stalks of $R^1j_{K,*}M$ at a finite place with $H^1(I_v, M)$; the edge map is restriction to these stalks. The generic-point map j_K is affine, hence quasi-compact and quasi-separated, as required by the stalk formula. Kato's section 8.2, pp. 180–181, defines the integral groups by the inverse limit of these finite-coefficient groups, so this assertion applies to the classes (K4) before rationalization.

Let L/K be finite, with absolute Galois groups $H \subset G_K$. For a class $z \in H^1(H, M)$ and an inertia subgroup $I \subset G_K$, the Mackey formula is

$$\operatorname{res}_I^{G_K} \operatorname{cor}_H^{G_K} z = \sum_{g \in I\backslash G_K/H} \operatorname{cor}^I_{I\cap gHg^{-1}} \operatorname{res}^{gHg^{-1}}_{I\cap gHg^{-1}}(gz). \tag{K7}$$

The intersection in each summand is the upper inertia subgroup for the corresponding place of L. If all upper inertia restrictions vanish, every summand is zero. This does not require L/K to be unramified. It is a finite double-coset identity at each finite coefficient precision. Uniqueness under the Leray injection makes the extensions across X_K compatible in a. Thus the integral j_* condition survives corestriction, including at primes ramified in a quadratic twisting field.

4. Primitive twisting is an actual Euler system. Fix a constituent $\chi = \chi_h$ of primitive conductor $D = |\Delta_h| = 2^ad$, d odd, and a basis e_χ of $\mathbf{Z}_2(\chi)$. Put $\Sigma = \operatorname{prime}(2Ndcd_s)$. For every positive integer m with $(m, cd_s) = 1$, set

$$K_m = \mathbf{Q}(\zeta_m), \quad L_m = \mathbf{Q}(\zeta_{Dm}), \quad y_m = \operatorname{cor}_{L_m/K_m}(z_{Dm} \otimes e_\chi). \tag{K8}$$

The character is trivial over L_m, so tensoring is an integral map there. Formula (K7) places (K8) in the precise j_* group required by Kato 13.1 on its index set $\operatorname{prime}(m) \cap \Sigma = \{2\}$. The larger definition is permitted by (8.1.3), Sections 8.9–8.11 and Proposition 8.12: $S_{Dm} = \operatorname{prime}(2NDm)$ is disjoint from $\operatorname{prime}(cd_s)$, since $d \mid F$ and (K5) holds. Each class uses the original f, ξ, c, d_s. We will also use this same construction on the different index set required by [73, Section 0.1], as checked below.

For $m \mid m'$, transitivity of corestriction reduces the norm of (K8) to Proposition 8.12, p. 186. For a deck automorphism g, equivariance gives

$$\operatorname{cor}((gz) \otimes e_\chi) = \chi(g)^{-1} g \operatorname{cor}(z \otimes e_\chi).$$

Consequently the factor at a newly introduced good prime q is

$$E_q(\chi) = 1 - \frac{a_q\chi(q)}{q}\sigma_q^{-1} + \frac{1}{q}\sigma_q^{-2}. \tag{K9}$$

In Kato 13.1, pp. 224–225, the polynomial is $P_q(X) = \det(1 - \operatorname{Fr}_q X \mid T\chi)$ and the norm factor is $P_q(q^{-1}\sigma_q^{-1})$, exactly (K9). Repeated prime powers introduce no factor because Proposition 8.12 has a product over newly added support primes. Thus the subfamily on Kato 13.1's index set is an Euler system of the stated lattice, not a class inferred from proportionality of rank-one Iwasawa modules. The required norm relations on the 1999 index set are verified separately below.

For $m = 2^n$ with $n \geq a$, the field L_m can be replaced by $\mathbf{Q}(\zeta_{d2^n})$: apply repeated-two norm compatibility from exponent $n + a$ to n. This introduces no factor 2^a. The construction therefore covers discriminant $4h$ on the same cofinal tail. At the coefficient level it retains the full character $\chi_2\chi_{\mathrm{odd}}$. A dyadic character nontrivial on Γ changes its action by the integral automorphism $1+t \mapsto -(1+t)$, or $t \mapsto -2-t$. A dyadic sign character instead changes the finite real-sign component. Neither operation is a division by a sign projector.

5. One coefficient map produces every constituent. For n large enough to contain the dyadic characters, put $A_{F,n} = \mathrm{Gal}(\mathbf{Q}(\zeta_{F2^n})/\mathbf{Q})$. Apply Shapiro to z_{F2^n}, then the map on group bases

$$\Phi_n : \mathbf{Z}_2[A_{F,n}] \longrightarrow \mathbf{Z}_2[\Gamma_n \times G], \qquad [s] \longmapsto \chi_0(s)[\bar{s}][\psi(s)]. \tag{K10}$$

With inverse deck actions, both sides transform under $g \in G_{\mathbf{Q}}$ by the inverse of its group image. On the target the extra scalar is $\chi_0(g)$; since $\chi_0^{-1} = \chi_0$, (K10) is equivariant. This is an integral coefficient homomorphism. Shapiro restriction to an inertia subgroup is the direct sum of the upper restrictions; the same zero restrictions used in (K7) therefore remain zero after (K10). In particular the map acts on actual integral classes. The map in (K10) is defined on this finite group for the following reason. The constituent indexed by the trivial character of G is χ_0, and $x \circ \psi = \chi_x\chi_0$ for every $x \in \widehat{G}$. All these quadratic Dirichlet characters have conductor dividing $F2^n$ once n is large enough. If an element of $G_{\mathbf{Q}}$ acts trivially on $\mathbf{Q}(\zeta_{F2^n})$, every character of G is therefore one on its image under ψ. Character separation makes that image the identity. Thus both χ_0 and ψ factor through $A_{F,n}$. Explicitly, for $g \in G_{\mathbf{Q}}$,

$$\Phi_n([g^{-1}s]) = \chi_0(g)\,[\bar{g}]^{-1}[\psi(g)]^{-1}\Phi_n([s]).$$

This verifies equivariance on each integral group basis before evaluation at any character. The quotient to the real tower sends each basis element to its image; its matrix has entries zero and one. The transition maps and character evaluations are also maps of integral coefficient modules. None of these operations is the idempotent $|G|^{-1}\sum_g x(g)[g]$.

The maps Φ_n commute with the norm transition maps. Proposition 8.12 has no new prime when n increases, so their images define the single class Z_F in (K1). The finite-coefficient degree-zero groups are finite and satisfy Mittag–Leffler. The inverse-limit exact sequence [44, Corollary 4.1.3, p. 96] therefore has zero $\varprojlim^1 H^0$ term, so it gives this degree-one class in completed cochains. The quotient from the full cyclotomic tower to Γ is the integral quotient of group bases, equivalently corestriction to the real subfield. No factor $(1+\tau)/2$ occurs.

After evaluation at x, (K10) is the coefficient form of the unnormalized corestriction weighted by χ_x. Factor that corestriction through the primitive conductor field in (K8). Transitivity and Proposition 8.12 give the exact identity

$$(Z_F)_x = \prod_{q \mid F/d} E_q(\chi_x)\, y_{2^\infty}. \tag{E.1}$$

It is an identity of the given classes. The same smoothing choice appears on both sides. There is no group-order denominator and no comparison with a canonical zeta class of $f \otimes \chi_x$.

For clarity, at a constituent write $\chi_h = \chi_2\chi_{\mathrm{odd}}$ and $|\Delta_h| = 2^a d$. Before taking the cofinal limit, the corresponding scalar coefficient map is exactly

$$[s] \longmapsto \chi_2(s)\chi_{\mathrm{odd}}(s)[s|_{\Gamma_n}], \qquad n \geq \max\{a, 2\}. \tag{E.2}$$

Thus the dyadic character is part of the module action, including when the field discriminant is $4h$.

6. Central trace and period normalization. First use (E.1) to remove the unused tame primes. The central specialization of (K8) is the corestriction of $z_D \otimes e_\chi$ from $\mathbf{Q}(\zeta_D)$ to $\mathbf{Q}$, with omitted support prime$(2Nd)$. For $D = 1$ use the trivial character and the same omitted support. This follows from the same repeated-two norm identity even when $D = 4|h|$. One must do this before using a Gauss basis; the Gauss sum of an imprimitive character modulo a larger cyclotomic level can be zero.

Theorem 9.7, p. 189, identifies the dual exponential of z_D with the retained-smoothed de Rham zeta element. Dual exponential commutes with corestriction, whose de Rham map is field trace. Thus the coefficient of e_χ is the exact unnormalized sum

$$\sum_{g \in \mathrm{Gal}(\mathbf{Q}(\zeta_D)/\mathbf{Q})} \chi(g)\, g(\exp^* z_D). \tag{K12}$$

Theorem 6.6, p. 163, evaluates precisely this sum. In weight two, $r = r' = 1$, its Betti sign is $\epsilon = \chi(-1)$ and its power of $2\pi i$ is zero. If e_B^ϵ is a fixed rational generator of that Betti line and $\delta(f, 1, \xi)^\epsilon = b_\epsilon e_B^\epsilon$, then $b_\epsilon \in \mathbf{Q}^\times$ is fixed and

$$\mathrm{per}_f\big((\mathrm{K12})\big)^\epsilon = s_F L_{\mathrm{prime}(2Nd)}(f, \chi, 1)\, b_\epsilon e_B^\epsilon. \tag{K13}$$

Choose the original de Rham generator so that its Betti period on this line is $\Omega_E^\epsilon e_B^\epsilon$, absorbing a fixed rational comparison in b_ϵ. Formula (K13) says that the scalar multiplying e_χ in (K12) is $s_F b_\epsilon L_{\mathrm{prime}(2Nd)}(f, \chi, 1)/\Omega_E^\epsilon$. Its Galois character is χ.

To descend it, take the invariant basis $\sqrt{h}\, e_\chi$ of the quadratic character line. With the usual complex convention $\Omega_E^- = i|\Omega_E^-|$ and $\sqrt{h} = i\sqrt{|h|}$ for $h < 0$, division by $\sqrt{h}$ gives, up to the already fixed rational period choice, exactly

$$s_F b_\epsilon \frac{L_*(h)}{h} \prod_{p|2N} (\text{the omitted local factor at } p). \tag{K14}$$

The signed integer h is odd, so h^{-1} has valuation zero. This computation also explains the usual Gauss-basis description: for the primitive character, $\tau(\chi)^2 = \Delta_h$ and $\tau(\chi)e_\chi$ is invariant; replacing it by $\sqrt{h}e_\chi$ costs the fixed dyadic factor $\sqrt{\Delta_h/h}$. It does not cost $\varphi(D)$ or any power of the number of prime divisors of h. The discriminant-$4h$ case has ratio two in this last basis change, and the direct basis calculation (K14) already includes it. One can verify the normalization without choosing an idempotent. Write $\beta = \exp^* z_D$ in the original de Rham line and put $\beta_\chi = \sum_g \chi(g) g\beta$. Then $u\beta_\chi = \chi(u)\beta_\chi$ for every u in the finite cyclotomic group. Thus $\beta_\chi \otimes e_\chi$ is invariant, and its coordinate in the invariant basis $\sqrt{h}\, e_\chi$ is $\beta_\chi/\sqrt{h}$. Theorem 6.6 uses precisely β_χ, rather than $\beta_\chi/\varphi(D)$. Theorem 9.7 applies here since $1 \le r = r' = 1 \le k - 1 = 1$; it has no restriction that p be odd. The dual form f^* in Theorem 6.6 equals f, because the elliptic newform has rational coefficients.

In the real-period convention, $L_*(h) = \sqrt{|h|}L(E^h, 1)/|\Omega_E^\epsilon|$, up to the fixed rational period comparison already included in b_ϵ. For $h > 0$, $\Omega_E^+\sqrt{h} = |\Omega_E^+|\sqrt{|h|}$; for $h < 0$, $\Omega_E^-\sqrt{h} = -|\Omega_E^-|\sqrt{|h|}$. Consequently

$$\frac{L(E^h, 1)}{\Omega_E^\epsilon \sqrt{h}} = \frac{L_*(h)}{h}$$

with the same fixed period comparison. This proves the signed factor in (K14), not only its valuation. For a primitive quadratic character, $\tau(\chi)^2 = \Delta_h$ and $\tau(\chi)/\sqrt{h}$ equals a fixed choice of $\sqrt{\Delta_h/h}$. Changing from its Gauss basis to the displayed invariant basis multiplies the coordinate

by this ratio, which is one or two up to sign. In particular the discriminant-$4h$ case adds no factor depending on the number of odd conductor primes.

Identifying this generator with the chosen local generator for E^h at two has bounded valuation: odd squarefree h belongs to one of finitely many classes in $\mathbf{Q}_2^\times/\mathbf{Q}_2^{\times 2}$, and changing a representative by a unit square scales the quadratic twist differential by a two-adic unit. The local factors at $2N$ also form a fixed finite nonzero list. At a good active prime the local L-factor is one; at an unused good prime (K9) at $t=0$ is $\#E^h(\mathbf{F}_q)/q$. This proves (K3), including all its uniformity claims and the assertion about zero values.

7. Every hypothesis of Kato 13.4(2). Apply the theorem to the integral system (K8), after multiplying by the fixed lattice comparison if needed. Its hypotheses are:

- Each generic sign component is nonzero. For a finite-order cyclotomic character ρ, corestriction and the ρ-sum combine into the single $(\chi\rho)$-weighted sum on the original class. Theorems 6.6 and 9.7 apply with the same ξ, c, d_s. Their smoothing factors are nonzero, since $c, d_s > 1$ cannot be roots of unity. Theorem 13.5(2), p. 227, gives only finitely many zero L-values among characters supported on the fixed finite set prime$(2Nd)$. There are infinitely many ρ of each parity. A torsion Iwasawa class can have a nonzero specialization at only finitely many such characters, since a nonzero annihilating power series has only finitely many cyclotomic zeros. Hence both generic sign components are nonzero.
- The two real eigenspaces of $T\chi$ have rank one.
- The theorem uses inverse Frobenius in its purity condition. On $T\chi$ its good eigenvalues have complex absolute value $q^{-1/2}$, so its integer w is -1.
- $V\chi$ is irreducible; twisting preserves irreducibility of the non-CM rational Tate representation.
- The image over $\mathbf{Q}(\mu_{2^\infty})$ contains an open subgroup of $\mathrm{SL}_2(\mathbf{Z}_2)$. Intersecting with $\ker\chi$ keeps an open subgroup, which contains a nonidentity unipotent. It fixes a one-dimensional subspace and gives condition (v).

These are all rational hypotheses. The residual irreducibility, integral cokernel and $p \neq 2$ conditions of part (3) are not used. Theorem 13.4(1) gives torsion $\mathbf{H}^2(T\chi)$, and part (2) gives, at every height-one prime away from two,

$$\ell\mathbf{H}^2(T\chi)_{0,\mathfrak{p}} \le \ell(A_\mathfrak{p}/J_\mathfrak{p}), \qquad \mathbf{H}^2_0 = \ker(\mathbf{H}^2 \to \mathbf{H}^2_{\mathrm{loc}}). \tag{K15}$$

The finiteness and Euler characteristic (12.2.1)–(12.2.2), p. 220, give generic rank one in $\mathbf{H}^1$. Its localization away from two is torsion-free. Indeed a torsion element for a scalar prime would come from invariants of the corresponding finite-dimensional rational coefficient algebra at that prime; its coefficient action is abelian, whereas the open commutator subgroup on $V\chi$ has no fixed vectors. The coefficient exact sequence excludes these invariants. The scalar specialization need not be a finite-order character. Thus $\mathbf{H}^1_\mathfrak{p}$ is free of rank one. Localization commutes with Hom for this finitely presented module. Writing $y_{2^\infty} = ae$ in a local basis gives $J_\mathfrak{p} = (a)$. Formula (K15) is exactly the index bound for the supplied class, not for an arbitrarily rescaled generator.

The published prime-two Euler-system input. For $p = 2$ we use the rational, away-from-two case of [73, Theorem 0.8, pp. 317–318], with the stronger hypothesis in its condition 0.6(ii), to justify the application of [55, Theorem 13.4(2)]. For our non-CM elliptic representation the image of $G_{\mathbb{Q}^{\mathrm{ab}}}$ contains an open subgroup of $\mathrm{SL}_2(\mathbb{Z}_2)$, as stated in [73, Example 0.6.2, pp. 317–318], using

Serre's open-image theorem. A quadratic character χ is trivial on $G_{\mathbb{Q}^{\mathrm{ab}}}$. Thus the same subgroup acts on $V\chi$ and contains a matrix $\left(\begin{smallmatrix}1 & b\\ 0 & 1\end{smallmatrix}\right)$ with $b \neq 0$. Its fixed space over $\mathbb{Q}_2$ has dimension one. This supplies the unipotent in $G_{\mathbb{Q}^{\mathrm{ab}}}$, not merely in the cyclotomic subgroup. The latter subgroup also contains this open SL_2 subgroup, so $V\chi$ remains absolutely irreducible over $\mathbb{Q}(\mu_{2^\infty})$. Indeed, a line invariant under both upper and lower nonidentity unipotents cannot exist, even over an algebraic closure. No inference from irreducibility over $G_{\mathbb{Q}}$ alone is needed.

We check the index set and the supplied class in that theorem explicitly. Put $N_K = 2Nd$ and $N'_K = cd_s$. These integers are coprime by (K5), since $d \mid F$. Section 0.1 of [73] requires every index m with $N_K \mid m$ and $(m, N'_K) = 1$. The enlarged definition (K8) supplies all these classes. They originate in the integral j_* group over $\mathbf{Z}[\zeta_m, 1/2]$ and map to full cohomology over $\mathbf{Z}[\zeta_m, 1/N_K]$. The varying construction support S_{Dm} is not an instruction to invert all its primes in this last scheme. Apply the fixed lattice map 2^{c_E} from (K4) if needed, identically at every level.

For a prime $\ell \nmid N'_K$, if $\ell \mid m$ then $S_{Dm\ell} = S_{Dm}$, so Proposition 8.12 gives norm factor one. If $\ell \nmid m$, the condition $N_K \mid m$ implies $\ell \nmid 2Nd$, and hence $S_{Dm\ell} \setminus S_{Dm} = \{\ell\}$. The equivariance identity before (K9) gives the factor $P_\ell(\ell^{-1}\sigma_\ell^{-1}) = E_\ell(\chi)$, with $P_\ell(X) = 1 - a_\ell\chi(\ell)X + \ell X^2$. These are exactly the two cases of (0.1.1) in [73]. Smoothing primes are excluded by N'_K and need not be inverted.

For the class in Section 0.7 of that source take its parameter $d = 1$; this parameter is distinct from our odd conductor part d. For every $n \geq 1$,

$$S_{DN_K2^n} = S_{D2^n} = \mathrm{prime}(2Nd).$$

Proposition 8.12 therefore gives

$$\mathrm{cor}_{\mathbf{Q}(\zeta_{DN_K2^n})/\mathbf{Q}(\zeta_{D2^n})}\, z_{DN_K2^n} = z_{D2^n}.$$

Transitivity of corestriction, through the two intermediate fields in the common upper field, now yields

$$\mathrm{cor}_{\mathbf{Q}(\zeta_{N_K2^n})/\mathbf{Q}(\zeta_{2^n})}\, y_{N_K2^n} = \mathrm{cor}_{\mathbf{Q}(\zeta_{D2^n})/\mathbf{Q}(\zeta_{2^n})}(z_{D2^n} \otimes e_\chi) = y_{2^n}.$$

No Euler factor occurs because the supports are equal. No field degree occurs because this is corestriction of the actual upper class, not corestriction of a restriction. These equalities commute with transition maps, and $n \geq 1$ is a cofinal tail. Write ξ_K for the Section 0.7 class, to distinguish it from the modular-symbol combination ξ. It is exactly the full-cohomology image of y_{2^∞}, multiplied by 2^{c_E} if the fixed lattice map was needed. That factor is a unit at every height-one prime under consideration, so its coordinate ideal is precisely $J_{\mathfrak{p}}$. The preceding nonvanishing argument supplies a nontorsion class on each sign component, purity gives $w = -1$, and oddness gives condition $0.6(\mathrm{iv}_{\mathfrak{p}})$. At the height-one primes used here, 2 is invertible and the localized Iwasawa algebra is a DVR. Theorem 0.8 then bounds the coordinate ideal of this actual retained class. Its strong residual and integral-cokernel assumptions concern primes containing 2, which are not used here.

We also account for the difference between the two published cohomology conventions. Write C_j for the j_* Iwasawa complex of [55, Section 12.1], and C_S for full cochains on the open scheme with the odd support inverted, as in [73, Section 0.5]. Work in one rational sign component and localize at $\mathfrak{p}$. Their quotient is the direct sum of the odd local complexes (K16). Each differential has nonzero determinant, so its degree-one cohomology vanishes. Hence $H^1(C_j) \simeq H^1(C_S)$ and $H^2(C_j) \hookrightarrow H^2(C_S)$. At an odd prime q, the unramified local term has cohomological dimension one after inverting 2. The localization triangle identifies the degree-two cohomology of (K16) with full local $H^2(\mathbb{Q}_q, T\chi)$ in Iwasawa cohomology. The map from $H^2(C_S)$ is the usual local restriction.

Exactness therefore identifies $H^2(C_j)$ with the kernel of all those odd local restrictions. Intersecting with the kernel of restriction at 2 gives

$$\ker\bigl(H^2(C_j)\to H^2_{\mathrm{loc},2}\bigr)\simeq\ker\bigl(H^2(C_S)\to\bigoplus_{q\in S}H^2_{\mathrm{loc,q}}\bigr).$$

This is precisely the comparison of the subscript-zero groups in the two sources. The H^1 identification preserves the actual Euler-system class and its coordinate ideal J. Thus Theorem 0.8 gives (K15) in the convention used here. The argument for (K16) below is local and does not use (K15). We do not invoke the odd-prime-only Perrin–Riou argument or an unchecked version of Rubin's theorem at 2.

8. Paying for the full global complex. At an odd prime put $W_q=H^1(I_q,V\chi)$. The rational cone from j_* to full cochains is

$$[\,W_q\otimes A\xrightarrow{1-\sigma_q^{-1}\,\mathrm{Fr}_{W_q}}W_q\otimes A\,],\qquad\deg=(1,2).\tag{K16}$$

Its determinant D_q is nonzero, since the cyclotomic Frobenius exponent is nonzero. The map is injective even after localization at a prime dividing D_q; it is not being reduced modulo that prime. Thus the cone contributes only degree-two length $v_{\mathfrak{p}}(D_q)$. The long exact sequence shows that full H^2 increases by at most their sum and that H^1 is unchanged. This inequality does not require vanishing of the next j_* group.

At a good inactive prime, $W_q=V\chi(-1)$, so $D_q=E_q(\chi)$ with exactly the convention of (K9). At a good active prime, ramified quadratic inertia acts by -1; its rational cohomology vanishes and $D_q=1$. This includes every active base prime, regardless of how many there are. At an old bad prime retain $H^1(I_q,V\chi)$ itself; replacing it by $V^{I_q}\chi(-1)$ would in general be false.

Choose a fixed integral F_2 divisible over A by the characteristic elements of $\mathbf{H}^2_{\mathrm{loc}}(T\chi)$ for the finitely many allowed local types at two and finite sign components. Its central value can be chosen nonzero: derived local control, and the absence of local Iwasawa H^3, identify its rational central H^2 with $H^2(\mathbf{Q}_2,V\chi)=0$. The last equality follows from local duality and finiteness of local elliptic torsion. Hence the characteristic elements have no factor t. Kato (12.2.3), p. 220, gives the required finiteness and torsion of this local Iwasawa module.

At the finitely many old bad primes, choose stable lattices in the W_q and multiply all the determinants D_q for the finitely many local twisting types. Clear their fixed scalar denominators. Their central values are nonzero, since the cokernel of $1-\mathrm{Fr}_{W_q}$ is $H^2(\mathbf{Q}_q,V\chi)=0$ by the same local duality. Call this product F_{bad} and set $F_E=F_2F_{\mathrm{bad}}$.

The quotient $\mathbf{H}^2/\mathbf{H}^2_0$ injects into the local $\mathbf{H}^2_{\mathrm{loc}}$, so F_2 pays for it in (K15). Multiplying a class in a free rank-one module by a nonzero scalar adds that scalar's valuation to its index. Formula (E.1) already pays for every unused good determinant in (K16); F_{bad} pays for the remaining old ones. Adding these length inequalities proves (K2). Here is the length calculation with all contributions separated. Fix a height-one prime $\mathfrak{p}$ of A and write $R=A_{\mathfrak{p}}$. Let M be the localized degree-one module for the primitive j_* system, let $y=y_{2\infty}$, and write $\mathrm{ind}_R(y)=\ell_R(M/Ry)$. The preceding freeness argument gives $M\simeq R$. Put $Q_\chi=\prod_{q\mid F/d}E_q(\chi)$. All primes in F/d are good: the family has identical local characters at primes dividing N, so a bad prime cannot ramify in one constituent and be unramified in another. The cone (K16) has zero degree-one cohomology, hence identifies M

with the localized full degree-one module. Its long exact sequence and (K15) give

$$\begin{aligned}\ell_R H^2(C_{S,x})_{\mathfrak{p}} &\le \ell_R \mathbf{H}^2(T\chi)_{0,\mathfrak{p}} + \ell_R \mathbf{H}^2_{\mathrm{loc}}(T\chi)_{\mathfrak{p}} + \sum_{q\in S,\ q\neq 2} v_{\mathfrak{p}}(D_q)\\ &\le \mathrm{ind}_R(y) + v_{\mathfrak{p}}(F_2) + v_{\mathfrak{p}}(F_{\mathrm{bad}}) + v_{\mathfrak{p}}(Q_\chi)\\ &= \mathrm{ind}_R(F_E Q_\chi y) = \mathrm{ind}_R(F_E(Z_F)_x).\end{aligned}$$

The second line uses $D_q = 1$ at active good primes, and $D_q = E_q(\chi)$ at inactive good primes. It does not charge an extra fixed constant for each active prime.

For completeness, nonvanishing at the center of the fixed correction factors follows before multiplying over local types. Local derived control and $H^3_{\mathrm{Iw}} = 0$ identify $\mathbf{H}^2_{\mathrm{loc}}/t\mathbf{H}^2_{\mathrm{loc}}$ rationally with $H^2(\mathbf{Q}_2, V\chi)$. Local duality identifies the latter with the dual of $H^0(\mathbf{Q}_2, V\chi)$, which is zero because local elliptic torsion is finite. Nakayama at (t) therefore excludes a factor t from its characteristic element. At an old odd prime the cokernel of $1 - \mathrm{Fr}_{W_q}$ is $H^2(\mathbf{Q}_q, V\chi) = 0$, by the same duality. Its determinant at $t = 0$ is nonzero. Taking products over the finite list of local types, followed by one fixed power of two to clear scalar denominators, gives $F_E \in \Lambda$ with $F_E(0) \neq 0$. This operation is independent of the binary dimension and all later good conductor primes.

9. Uniformity and the boundary of this proposition. The only lattice denominator is fixed before the family. The smoothing integers may grow, but their central valuation is $2e$. The multiplier F_E depends only on the original local types. Transfer, coefficient push, real quotient and norm compatibility are integral. The proof never divides by a conductor degree, $|G|$, a class number, or a Fourier projector. It keeps one class through (K1), (K2), and (K3). □

We now use this same class in the positive complex. The preceding argument supplies its integral construction, its rational height-one divisibility and its exact central reciprocity normalization.

Proposition E.2 (One integral even scalar for the family)**.** *Retain Proposition E.1 and assume, as for the present curve, that $E[2](\mathbb{Q}) \simeq (\mathbb{Z}/2\mathbb{Z})^2$. Let C^+ be the fiber of the ordinary global cochains to the ordinary real cochains, with real degree zero retained. It has a compatible finite free model under the coefficient changes used here. For $B = \mathcal{O}[G]$, it satisfies $C^+ \otimes B \simeq B^2[-1]$. The fixed real invariant generator, the positive lift of the retained class, and the original determinant basis define the single scalar u^{ev} by (E.4), with*

$$u^{\mathrm{ev}} \in \Lambda[1/2][G], \qquad 2^{c_E} u^{\mathrm{ev}} \in \Lambda[G].$$

Here c_E may be enlarged for the fixed data but is independent of G and the later good conductor primes. The assertion requires the common real sign and local squareclasses at $2N$ in Proposition E.1.

Proof. Let C^+ be the fiber of ordinary global cochains to ordinary real cochains. Real H^0 is included. Over a finite Artin quotient, continuous cochains are filtered unions of finite free function modules, hence flat and therefore free. A residual splitting into finite cohomology and disks lifts by the nilpotent perturbation series to a finite free model with actual homotopies. Minimal models at successive quotients are related by invertible matrices: a homotopy equivalence reduces to an isomorphism of minimal residual complexes, and Nakayama applies. Lifting these matrices and comparison homotopies makes the models compatible. Their acyclic, termwise surjective comparison cones have acyclic inverse limit. Here is the compatibility argument. At successive finite coefficient quotients, reduction of a minimal finite free model is a minimal model of the reduced

cochains. Two such models are homotopy equivalent; reducing a homotopy inverse modulo the maximal ideal shows that the comparison is degreewise invertible. Lift these invertible matrices to the next quotient and change bases. The model differentials then reduce to the preceding differentials exactly. The comparison maps to the original cochains agree after reduction up to chain homotopy. Since reduction of continuous cochains is termwise surjective, lift that homotopy and adjust the upper comparison by its boundary. The comparison maps are now exactly compatible and remain quasi-isomorphisms.

Let $\mathcal{K}_n$ be their acyclic comparison cones. Their transition maps are termwise surjective. If s denotes the transition shift on the product, successive lifting gives the exact sequence of complexes

$$0 \longrightarrow \varprojlim_n \mathcal{K}_n \longrightarrow \prod_n \mathcal{K}_n \xrightarrow{1-s} \prod_n \mathcal{K}_n \longrightarrow 0.$$

Products of complexes of modules preserve acyclicity, so the long exact cohomology sequence makes $\varprojlim_n \mathcal{K}_n$ acyclic. The inverse-limit comparison is therefore a quasi-isomorphism from the bounded finite free limit model to the continuous coefficient complex. This also identifies its derived coefficient changes with the finite-quotient constructions. This proves the needed perfection and coefficient change.

The residual positive complex. Write $K_n = \mathbf{Q}(\zeta_{2^{n+2}})^+$, and let S_n consist of the places above two and a fixed finite set of odd rational primes. Put $r_n = 2^n$ and $s_n = \#S_n$. We first compute with the scalar coefficient $\mathbf{F}_2 = \mu_2$.

The ordinary two-class group of K_n is trivial. Indeed this holds for $K_0 = \mathbf{Q}$. Assuming it for K_{n-1}, let H be the Hilbert two-class field of K_n, with the real places split. The canonical extension H/K_{n-1} is Galois with a finite two-group G_H. Only the unique dyadic prime can ramify, and its inertia I has order two. The quotient of G_H^{ab} by the image of I gives an everywhere unramified abelian two-extension of K_{n-1} with real places split, hence is trivial by induction. Thus G_H^{ab} is cyclic, generated by an element of order at most two. The Frattini quotient of G_H has one generator, so G_H is cyclic and has order at most two. Since it contains K_n, we obtain $H = K_n$. The S_n-class group is a quotient, so its two-part also vanishes.

The signatures of units span all r_n real signs. For $n \geq 1$ put $m = n + 2$, $\zeta = \zeta_{2^m}$, and

$$u = 1 + \zeta + \zeta^{-1} = \zeta^{-1}\frac{\zeta^3 - 1}{\zeta - 1}.$$

This is a unit: the numerator and denominator generate the same dyadic prime ideal in the cyclotomic field. Its real conjugates are $1 + 2\cos(2\pi a/2^m)$ for odd $0 < a < 2^{m-1}$. The negative ones have $2^m/3 < a < 2^{m-1}$, and their number is

$$2^{m-2} - \left\lfloor \frac{2^{m-1}}{3} + \frac{1}{2} \right\rfloor,$$

an odd integer. Hence $N_{K_n/\mathbf{Q}}(u) = -1$. The signature vector of u has augmentation one in the local ring $\mathbf{F}_2[\mathrm{Gal}(K_n/\mathbf{Q})]$ and is therefore a unit. Its Galois translates span all signatures. At $n = 0$ use -1.

Kummer theory now gives $\dim H^1(G_{K_n,S_n}, \mathbf{F}_2) = r_n + s_n$, and its signature map to $\mathbf{F}_2^{r_n}$ is onto. The map in degree zero is the diagonal injection. The positive complex is the fiber of ordinary global cochains to the sum of ordinary real cochains, so its long exact sequence gives $\dim H^1(C_n^+) = r_n + s_n - 1$. The vanishing two-part of the S_n-class group identifies global H^2 with $\mathrm{Br}(\mathcal{O}_{K_n,S_n})[2]$. The local invariant description has $r_n + s_n$ coordinates with one sum relation.

Projection to the r_n real coordinates is onto, because $s_n \geq 1$, and has kernel of dimension $s_n - 1$. Together with signature surjectivity this gives

$$\dim H^1(C_n^+) = 2^n + s_n - 1, \qquad \dim H^2(C_n^+) = s_n - 1, \qquad H^i(C_n^+) = 0 \quad (i \neq 1, 2).$$

The last assertion uses the usual global-to-real comparison in degrees greater than two; degree zero vanishes by the diagonal injection.

For each fixed support, s_n is bounded. The dyadic prime is unique; each fixed odd prime has open decomposition group in the real cyclotomic $\mathbf{Z}_2$-extension, since its image in $\mathbf{Z}_2^\times/\{\pm 1\}$ has nonzero two-adic logarithm. Use now the finite free model and coefficient change already constructed. Its scalar reduction is a bounded finite free $\mathbf{F}_2[[t]]$-complex, and its finite cyclotomic specializations are derived quotients by t^{2^n}. Smith decomposition shows that torsion disks and their Tor terms contribute bounded dimensions, while a free cohomology generator contributes 2^n. The displayed dimensions therefore give generic rank one in degree one and zero in all other degrees.

For the present curve, $E[2]$ is the direct sum of two trivial $\mathbf{F}_2$-representations. Thus the residual positive complex has rank two in degree one only. The residue field of the local ring $B = \mathcal{O}[G]$ is $\mathbf{F}_2((t))$; the augmentation ideal of the finite two-group lies in its maximal ideal. A minimal finite free B-model has zero residual differential, so the just-computed dimensions force exactly two generators in degree one and no others. Consequently

$$C^+ \otimes B \simeq B^2[-1]. \tag{E.3}$$

This uses the local ring B, not semisimplicity of $\mathbf{F}_2[G]$.

The common real sign is needed here. The real cyclotomic character and ε are trivial on complex conjugation, so the real action is the fixed action on $T_2E \otimes \chi_0$, extended to the group ring. Its invariant module is the fixed free rank-one $\mathbb{Z}_2$-lattice extended to $\Lambda[G]$, and then to $\mathcal{O}[G]$. Thus a generator can be chosen before G varies. Real H^1 is killed by two over these coefficient rings, so the positive lift below costs the same single factor two for every binary dimension. Choose a generator of the fixed real invariant lattice and its connecting class a. Since real H^1 is killed by two, $2^{c_E+1}F_EZ_F$ has an integral positive lift b. For a basis θ of the original inverse determinant line, define

$$u^{\mathrm{ev}}\theta = 2^{-c_E}\, a \wedge b. \tag{E.4}$$

The wedge does not depend on the choice of the positive lift. Indeed, the difference of two lifts is in the image of real degree-zero cohomology by the cone exact sequence. That module is generated by the chosen real invariant generator, so the difference is a multiple of a and wedges to zero with a. After rationalization set $z = F_EZ_F$ and choose any lift $\widetilde{z}$ of z. Formula (E.4) then reads

$$u^{\mathrm{ev}}\theta = 2a \wedge \widetilde{z}.$$

Thus the factor 2^{c_E} used to make the lift integral cancels exactly. Increasing the fixed clearing exponent does not change the rational scalar. Character evaluation and derived central specialization preserve this equality and its cone maps. The class whose dual exponential occurs in (E.7) is consequently the same $F_E(Z_F)_h$ as in (K3), and the single remaining factor two is exactly the $+1$ in (E.7).

From the full cohomology bound to the same determinant scalar. Here is the rational divisibility step explicitly. Fix a character x of G, a height-one prime $\mathfrak{p}$ of A, and $R = A_{\mathfrak{p}}$. Put $z = F_E(Z_F)_x$, without changing its normalization. The full global degree-one module is free of rank one over R by the argument following (K15) and the injectivity of the maps in (K16). Write it as

Re, and write $z = \alpha e$. The higher real cohomology vanishes over R, since two is invertible. Global degree-zero cohomology is zero: an invariant would restrict to an invariant of the open nonabelian subgroup on the elliptic representation, on which the scalar coefficient character is trivial. Thus the positive-cone exact sequence gives

$$0 \longrightarrow Ra \longrightarrow H^1(C_x^+ \otimes R) \longrightarrow Re \longrightarrow 0, \qquad H^2(C_x^+ \otimes R) = H^2(C_{S,x} \otimes R).$$

The element a here is the image of the fixed real invariant generator in (E.4), and it is primitive in this positive degree-one module, because the quotient Re is free. Choose a lift $\widetilde{e}$ of e; then $(a, \widetilde{e})$ is a basis. The equality already proved from (E.4) becomes

$$u_x^{\mathrm{ev}} \theta_x = 2\alpha \, a \wedge \widetilde{e}.$$

Let $d_1, \dots, d_r$ be the nonzero Smith entries of a finite free model of $C_x^+ \otimes R$. All its cohomology outside degrees one and two is zero, its degree-one cohomology is free, and its degree-two cohomology is torsion. Its inverse determinant lattice, under the rational identification with $\bigwedge^2 H^1$, is therefore generated, up to an R-unit, by $(\prod_i d_i) a \wedge \widetilde{e}$. It follows that

$$\begin{aligned} v_{\mathfrak{p}}(u_x^{\mathrm{ev}}) &= v_{\mathfrak{p}}(\alpha) - \sum_i v_{\mathfrak{p}}(d_i) \\ &= \ell_R(H^1(C_{S,x} \otimes R)/Rz) - \ell_R H^2(C_{S,x} \otimes R) \;\geq\; 0, \end{aligned} \tag{E.4a}$$

where two is a unit and the inequality is exactly (K2). This uses the full H^2 corrected in (K16), rather than only Kato's $\mathbf{H}_0^2$.

The intersection of the height-one valuation rings of the normal domain A gives $u_x^{\mathrm{ev}} \in A$ for every x. Since $|G|$ is a power of two, the character idempotents belong to $A[G]$ and these memberships give $u^{\mathrm{ev}} \in A[G]$. This establishes membership only; it is not a bound for the denominator from character inversion. The uniform bound comes separately from (E.3): in its free rank-two $\mathcal{O}[G]$-module the integral vectors a, b of (E.4) have integral wedge, so $2^{c_E} u^{\mathrm{ev}} \in \mathcal{O}[G]$ with the fixed exponent already used for the original lattice. Coefficientwise intersection in the original group basis gives

$$A \cap \mathcal{O} = \Lambda, \qquad 2^{c_E} u^{\mathrm{ev}} \in \Lambda[G]. \tag{E.5}$$

Indeed an element of A has bounded powers of two as denominators in its power-series coefficients; membership in $\mathcal{O}$ makes their minimum valuation nonnegative. The coefficientwise intersection retains c_E, without a term $v_2|G|$. □

E.2 Central saturation and the numerical lower bound

Proposition E.3 (Central calibration of the even scalar)**.** *Use the scalar and determinant basis of Proposition E.2. At a corank-zero constituent E^h with finite* Ш$[2^\infty]$*, its nonzero central value satisfies*

$$v_2 u_h^{\mathrm{ev}}(0) = v_2 L_*(h) - 2\omega(h) - s(h) + \epsilon_h, \qquad |\epsilon_h| \leq C_{\mathrm{loc}},$$

with one constant for the fixed local types. At such a constituent $u_h^{\mathrm{ev}}(0) = 0$ if and only if $L_*(h) = 0$. Consequently every admissible h with $L_*(h) \neq 0$ satisfies $v_2 L_*(h) \geq 2\omega(h) + s(h) - c_E - C_{\mathrm{loc}}$: the forward rank-zero theorem in Lemma 5.5 supplies the preceding corank and finiteness hypotheses in this case. The central saturation corrections are those in (E.6)–(E.7); no primitivity after inverting t is assumed to survive central specialization.

Proof. At a corank-zero constituent put

$$g = \ell_{\mathbb{Z}_2} E^h(\mathbb{Q})[2^\infty], \quad g_v = \ell_{\mathbb{Z}_2} E^h(\mathbb{Q}_v)[2^\infty], \quad r_i = \ell_{\mathbb{Z}_2} H^i(\mathbf{R}, T_h) \quad (i = 1, 2).$$

Let i be the index length of the image of global H^1 in the free rank-one quotient $L_2 = H^1(\mathbb{Q}_2, T_h)/H^1_f(\mathbb{Q}_2, T_h)$. Orthogonal Kummer duality, with its real terms, gives

$$\ell H^2_{\mathrm{glob}} = s(h) - i + \sum_v g_v + r_2 - g. \tag{E.6}$$

Write $K = \ker(H^1_{\mathrm{glob}} \to H^1_{\mathbf{R}})$, let k be its torsion length and m its free index. If $P = H^1(C_h^+)$, p_0 is its torsion length and a_s the saturation index of the real image in P/P_{tors}, exactness gives

$$\ell \operatorname{coker}(H^1_{\mathrm{glob}} \to H^1_{\mathbf{R}}) = r_1 - g + k - m, \qquad k = p_0 + a_s.$$

For the numerical calculation suppose also $L_*(h) \neq 0$. The wedge in (E.4) adds $a_s + 1 + v_{\mathrm{free}}(z) - m$. Conversion to the inverse determinant adds $p_0 - \ell H^2(C_h^+)$. Substitution of these equalities cancels m and a_s exactly:

$$v_2 u_h^{\mathrm{ev}}(0) = v_2 \exp^* z - s(h) - \sum_v g_v + 2g + 1 - r_1 - e_2, \tag{E.7}$$

where e_2 is the valuation of $\exp^*$ on a generator of L_2. Primitivity of the real image after inverting t has not been specialized to $t = 0$.

Here are the real local convention and the exact length sequence used in (E.6). Put $T = T_2(E^h)$ and $D = E^h[2^\infty]$. Since $E^h[2]$ is rational, complex conjugation J satisfies $J \equiv 1 \pmod 2$. The integral projectors $(1 \pm J)/2$ give $T = T^+ \oplus T^-$. At the real place use the bounded Kummer complex

$$U_{\mathbf{R}} = \tau_{\leq 1} R\Gamma(\mathbf{R}, T) = [\, T^+ \oplus T^- \xrightarrow{J-1} T^- \,]$$

in degrees zero and one, with its map to ordinary real cochains. Modulo 2^n its degree-zero cohomology is $T^+/2^n \oplus (T^-/2^n)[2] = E^h(\mathbf{R})[2^n]$, and its degree-one image is the real Kummer line $T^-/2$. The latter can also be checked using a rectangular period lattice: dividing the nonidentity real component, represented by half the imaginary period, produces the minus-line cocycle. Thus this line is its exact annihilator under finite real Tate duality. The ordinary completion $E^h(\mathbf{R})^\wedge_2[-1]$ alone would lose the first summand of degree-zero cohomology.

At a finite place take the derived Kummer map from $E^h(\mathbb{Q}_v)^\wedge_2[-1]$. Its reduction has full local degree-zero cohomology and injective Kummer degree-one map. If C_{Sel} denotes the resulting ordinary global Selmer cone, these assertions identify

$$H^1(C_{\mathrm{Sel}} \otimes^{\mathbf{L}} \mathbb{Z}/2^n) = \mathrm{Sel}_{2^n}(E^h/\mathbb{Q}).$$

At corank zero the discrete Selmer group is finite, the Mordell–Weil rank is zero, and its order is that of $\mathrm{Sha}[2^\infty]$. The orthogonal-Kummer specialization of finite Poitou–Tate duality [44, 5.7.1.4–5.7.1.7], followed by the compact/discrete limits, gives

$$0 \longrightarrow L_2/J_2 \longrightarrow \mathrm{Sel}_{2^\infty}(E^h/\mathbb{Q})^\vee \longrightarrow H^2_{\mathrm{glob}} \longrightarrow \bigoplus_{v \in S_f} H^2(\mathbb{Q}_v, T) \oplus H^2(\mathbf{R}, T) \longrightarrow H^0(\mathbb{Q}, D)^\vee \longrightarrow 0,$$

where J_2 is the image of global H^1. There is no real degree-one quotient: the compact real Kummer image is all of $H^1(\mathbf{R}, T)$, and its discrete orthogonal Kummer group is zero. At finite $v \neq 2$ the

compact Kummer image is likewise all of local H^1, while at two its quotient is L_2. The real degree-two term is retained. The finite-level images satisfy Mittag–Leffler, and high-degree localization at the real place terminates the sequence.

This also makes H^2_{glob} finite. The cone sequence then makes $H^2(C_{\mathrm{Sel}})$ finite, and universal coefficients give

$$0 \to E^h(\mathbb{Q})/2^n \to \mathrm{Sel}_{2^n}(E^h/\mathbb{Q}) \to H^2(C_{\mathrm{Sel}})[2^n] \to 0.$$

Comparison with the Kummer sequence gives $\ell H^2(C_{\mathrm{Sel}}) = s(h)$, with no extra real torsion. Equivalently the cone sequence is

$$0 \to L_2/J_2 \to H^2(C_{\mathrm{Sel}}) \to H^2_{\mathrm{glob}} \to \bigoplus_{v \in S_f} H^2(\mathbb{Q}_v, T) \oplus H^2(\mathbf{R}, T) \to H^0(\mathbb{Q}, D)^\vee \to 0.$$

Its alternating lengths are exactly (E.6).

For the positive cone, $H^2_{\mathrm{glob}} \to H^2(\mathbf{R}, T)$ is onto: any real component's image in $H^0(\mathbb{Q}, D)^\vee$ can be canceled by a component at two, since restriction $H^0(\mathbb{Q}, D) \to H^0(\mathbb{Q}_2, D)$ is injective. Thus its exact sequences are

$$0 \to T^+ \xrightarrow{a} P \to K \to 0, \qquad 0 \to \mathrm{coker}(H^1_{\mathrm{glob}} \to H^1_{\mathbf{R}}) \to H^2(C_h^+) \to H^2_{\mathrm{glob}} \to H^2_{\mathbf{R}} \to 0.$$

If k and p_0 are the torsion lengths of K and P, m is the free index of K in global H^1, and a_s is the saturation index of the real line in P/P_{tors}, these sequences give

$$\ell\,\mathrm{coker}(H^1_{\mathrm{glob}} \to H^1_{\mathbf{R}}) = r_1 - g + k - m, \qquad k = p_0 + a_s.$$

The wedge $2^{-c_E} a \wedge b$ contributes $a_s + 1 + v_{\mathrm{free}}(z) - m$, and the inverse determinant comparison contributes $p_0 - \ell H^2(C_h^+)$. Substitution leaves $1 + v_{\mathrm{free}}(z) + i - s(h) - \sum_v g_v + 2g - r_1$. Finally $v_2 \exp^* z = v_{\mathrm{free}}(z) + i + e_2$, proving (E.7) without specializing a primitivity assertion from the punctured family. Apply (K3) to the retained class $z = F_E(Z_F)_h$ used in the wedge above. The primitive-conductor trace and the invariant $\sqrt{h}\, e_{\chi_h}$ basis have already been computed in paragraph 6, including discriminant $4h$. Absorb the fixed smoothing and old-support factors into ϵ_{rec}. Then

$$v_2 \exp^* z = v_2 L_*(h) + \sum_{q \text{ unused}} v_2 \# E^h(\mathbf{F}_q) + \epsilon_{\mathrm{rec}}, \qquad |\epsilon_{\mathrm{rec}}| \le C_E.$$

At unused good primes $g_q = v_2 \# E^h(\mathbf{F}_q)$, so these terms cancel. At an active good prime the inertia invariants in $E^h[2^\infty]$ are $E[2]$, of length two for the present curve. The remaining local types have fixed bounds, including global torsion by its injection at two. Thus

$$v_2 u_h^{\mathrm{ev}}(0) = v_2 L_*(h) - 2\omega(h) - s(h) + \epsilon_h, \qquad |\epsilon_h| \le C_{\mathrm{loc}}. \tag{E.8}$$

The central rational positive complex has only rank-two H^1, so this is specialization of (E.4). Corank-zero duality also shows $u_h^{\mathrm{ev}}(0) = 0$ if and only if $L_*(h) = 0$. At $L_*(h) \neq 0$, Lemma 5.5 supplies the corank-zero and finite-Sha hypotheses just used. Equations (E.5) and (E.8), already with $G = 1$, yield

$$v_2 L_*(h) \ge 2\omega(h) + s(h) - c_E - C_{\mathrm{loc}}. \tag{E.9}$$

No binary-family dimension is needed for this lower bound. □

E.3 The actual even form and its phase

Proposition E.4 (An actual even form with rational normalized phase)**.** *Fix a sign σ, unit squareclasses at a finite support S_0 containing $2N$, and a positive squarefree reference D_0 in that filter with $L_*(\sigma D_0) \neq 0$. For every fixed finite extra exclusion set B prime to D_0, there is an algebraic holomorphic cuspidal form of weight $3/2$, normalized by $a(D_0) = 1$, with the scalar characters ψ, η constructed below and $\psi/\eta = \chi_{-\sigma}$. It has the prescribed coefficient support. For every retained positive squarefree D in the original S_0 filter,*

$$a(D) = z_D r_D, \qquad z_D \in \mu_{2^\infty}, \quad r_D \in \mathbb{Q}, \qquad r_D^2 = L_*(\sigma D)/L_*(\sigma D_0).$$

It satisfies the all-index good-prime recurrence (E.12) outside $S_0 \cup B$. Nonzero retained squarefree coefficients have integral $v_2 a(D)$, and $v_2 a(D) - \omega(D)$ has a finite attained minimum. The form and the initial sieve are fixed before this minimum is chosen. Additional exclusions do not authorize a new minimum.

Proof. Fix a sign σ and unit squareclasses at a finite support S_0 containing two and every prime dividing N. Let $D_0 > 0$ be squarefree in these prescribed classes, with $L_*(\sigma D_0) \neq 0$. In (E.10)–(E.11), D ranges over positive squarefree parameters in these same unit squareclasses at every prime of S_0. Fix also a finite extra exclusion set B with $\gcd(D_0, \prod_{p \in B} p) = 1$.

A nonzero reference functional and its scalar local type. Use the global Shimura–Waldspurger packet with underlying PGL_2 representation $E^{(\sigma)}$ in the normalization of Waldspurger's Corollary 2. The nonzero central value at σD_0 supplies the member with a nonzero D_0-Whittaker functional. Equivalently, take the nonzero theta lift associated with $E^{(\sigma D_0)}$ and the additive character of frequency D_0; the change of additive character identifies its underlying packet with the preceding one. The nonvanishing and the local identification are the content of Waldspurger's global theta theorem and its Fourier-functional calculation [67, Theorem 1 and Proposition 1] and [63, Proposition 18 and its proof, pp. 466–467]. The positive real frequency selects the holomorphic member of lowest weight $3/2$. The lift is in the unary-orthogonal cuspidal spectrum.

Here is a local construction of a scalar type that retains the reference functional. At $p \in S_0$, write $\ell = \ell_{D_0}$ and choose v with $\ell(v) \neq 0$. Normalize Haar measure in

$$P_R v = \frac{1}{\mathrm{vol}(p^{-R}\mathbb{Z}_p)} \int_{p^{-R}\mathbb{Z}_p} e_p(-D_0 b)\pi(n(b))v \, db.$$

Then $\ell(P_R v) = \ell(v)$. Let $M(u)$ be the unit action normalized by the Weil multiplier. Explicitly, if the chosen torus lifts have cocycle $(u, v)_p$, their theta-cubed multiplier γ has two-power root-of-unity values and satisfies $\gamma(u)\gamma(v) = (u, v)_p \gamma(uv)$; put $M(u) = \gamma(u)^{-1}\pi(\widetilde{m}(u))$. It is therefore an action of $\mathbb{Z}_p^\times$ and $M(u)\pi(n(b))M(u)^{-1} = \pi(n(u^2 b))$. Changing u to its inverse changes none of the following support conditions. Direct substitution gives

$$\ell(M(u)P_R v) = \mathbf{1}_{u^2 \equiv 1 \pmod{p^R}} \ell(M(u)v).$$

The integral is a finite sum on this smooth vector.

The normalized $M(-1)$ acts by a scalar $\delta \in \{1, -1\}$. Choose a continuous unit character α of two-power order with $\alpha(-1) = \delta$. Such a character exists: for odd p, -1 lies in the two-primary factor of the finite cyclic group of residue-field units; for $p = 2$ use the factor $\{1, -1\}$ of $\mathbb{Z}_2^\times$. Take

R large enough that both v and α are constant on the small neighborhoods appearing above. The projection

$$w = \int_{\mathbb{Z}_p^\times} \alpha(u)^{-1} M(u) P_R v \, d^\times u$$

satisfies $M(u)w = \alpha(u)w$. Its ℓ-value is nonzero: the neighborhoods of 1 and -1 contribute the same nonzero value. For odd p the solution set is $1 + p^R\mathbb{Z}_p$ together with $-1 + p^R\mathbb{Z}_p$. For $p = 2$ and $R \geq 3$ it is

$$(1 + 2^{R-1}\mathbb{Z}_2) \ \cup \ (-1 + 2^{R-1}\mathbb{Z}_2).$$

Thus the four residue classes modulo 2^R form two neighborhoods; there are not four independently assignable phases. For completeness, the noncancellation in this projection is an exact calculation. Normalize multiplicative Haar measure to give $\mathbb{Z}_p^\times$ volume one, and put $K_R = 1 + p^R\mathbb{Z}_p$ for odd p, and $K_R = 1 + 2^{R-1}\mathbb{Z}_2$ for $p = 2$. Increase R until $M(k)v = v$ and $\alpha(k) = 1$ for $k \in K_R$. On K_R the integrand after applying ℓ is $\ell(v)$. On $-K_R$ it is $\alpha(-1)^{-1}\ell(M(-1)v) = \delta^{-1}\delta\ell(v)$. It vanishes elsewhere by the displayed additive projection formula. Consequently

$$\ell(w) = 2\,\mathrm{vol}(K_R)\ell(v) \neq 0.$$

This calculation takes place in characteristic zero; reduction at two is not involved. For the later support projection choose m so that the required unit squareclass is a union A of residue classes modulo p^m. Its operator is

$$\Pi_A = \frac{1}{p^m} \sum_{b \bmod p^m} \left(\sum_{a \in A} e_p(-ab/p^m) \right) \pi(n(b/p^m)).$$

On the Fourier functional of integral frequency D it acts as $\mathbf{1}_A(D)$. It therefore preserves the reference value and annihilates every excluded frequency. Conjugation by $M(u)$ permutes the summands and replaces A by $u^2A = A$, proving the asserted commutation. Its denominators are fixed before the form is normalized.

The vector $P_R v$, and then w, is fixed by upper unipotents $n(\mathbb{Z}_p)$, because D_0 is a unit. Smoothness makes w fixed by lower unipotents at sufficiently deep level. Together with the unit eigencharacter, the elementary decomposition of $K_0(p^m)$ gives a scalar K_0 type for sufficiently large m. The Weil normalization contributes only two-power roots of unity, so the resulting classical character has two-power order. The forced values at -1 give the global parity required for a weight-3/2 scalar character: the product of the finite central actions is the inverse of the holomorphic archimedean action, and the theta-cubed multiplier has the same product relation on rational $-I$. Thus the quotient character satisfies $\psi(-1) = 1$. At 2 one may, if desired, choose α of arbitrarily large two-power conductor before increasing R; this does not enlarge the prime support.

Apply the finite additive Fourier idempotent for the prescribed unit-squareclass subset modulo a sufficiently high power of p. This idempotent keeps the D_0 coefficient. Its support condition is stable under multiplication by unit squares, so it commutes with the unit action and retains a scalar type, at a possibly deeper level. At every prime outside S_0 take the spherical vector. The packet there is unramified; its standard squarefree coefficient c_p^0 is 1 at both valuations zero and one [63, Section I.3, pp. 377–378]. This includes the primes dividing D_0. Local Whittaker uniqueness and the tensor-product factorization show that the resulting global reference coefficient is still nonzero.

The exact full-space comparison. Let the resulting scalar character be ψ and put $\eta = \psi\chi_{-\sigma}$. Both have two-power order. Let V be the full good-Hecke eigenspace of this scalar character, at

the level just constructed. Keep all old multiplicities. In Waldspurger's notation its integral-weight eigenform and character are

$$\phi_0 = f_E \otimes \eta, \qquad \chi_0 = \psi\chi_{-1}, \qquad \phi_0 \otimes \chi_0^{-1} = f_E \otimes \chi_\sigma.$$

This also fixes the signs in the packet convention: Waldspurger's positive index D twists the last representation by χ_D. The Hecke eigenvalues in V are $\eta(p)a_p(E)$.

The nonzero space implies hypothesis (H1) by Waldspurger's Proposition 2 [63, Section I.2, p. 377]. In explicit terms, at a principal-series place its two inducing characters take value 1 at -1. Corollary 2 [63, pp. 483–484] applies to every $f \in V$ and any positive squarefree pair whose ratio is a square in every $\mathbb{Q}_p$ at a prime dividing the level. Our D, D_0 satisfy this condition because the level has prime support S_0. The final paragraph of that proof removes (H2) for this same-$\mathbb{Q}_2$-squareclass comparison. No assumption of non-supercuspidality at two is required.

The printed coefficient identity and its dyadic hypothesis. For clarity, write $\mathcal{L}_D = L(E^{(\sigma D)}, 1)$ and let N_{W} denote the level in Waldspurger's notation, not the conductor of E. In [63, Corollary 2, pp. 483–484] take

$$k = 3, \quad \chi = \psi, \quad \chi_0 = \psi\chi_{-1}, \quad \rho_0 = \rho_E \otimes \eta, \quad n_1 = D_0, \quad n_2 = D.$$

Then $\rho_0 \otimes \chi_0^{-1} = \rho_E \otimes \chi_\sigma$. The corollary, in its printed order, is

$$a_f(D_0)^2 \mathcal{L}_D \, \psi(D/D_0) D^{1/2} = a_f(D)^2 \mathcal{L}_{D_0} D_0^{1/2}.$$

The passage from the automorphic central argument $1/2$ to the classical argument 1 is the same for both terms. Dividing only by the nonzero reference value gives (E.10). In particular this is a literal square identity, with neither complex conjugation nor an unspecified absolute-value normalization.

Every prime of N_{W} belongs to S_0 at this point of the construction. The prescribed local classes make D/D_0 a square in $\mathbb{Q}_p^\times$ at each such prime, including $p = 2$. In the proof of the corollary, the local coefficient functions at these primes have equal values at D, D_0. Outside the level, the squarefree arguments have valuation zero or one and the unramified coefficient function has value one in both cases. The real coefficient contributes exactly the power $(D/D_0)^{(k-2)/4}$ before squaring. The quotient of epsilon factors is computed on pp. 483–484: its Hilbert-symbol product disappears by reciprocity and its remaining character is $\chi(D/D_0) = \psi(D/D_0)$. Thus these local terms do not leave an additional prime-dependent factor in the displayed identity.

The final paragraph on p. 484 explicitly explains the use of (H2): it is needed in that argument only to compare Fourier indices in different squareclasses of $\mathbb{Q}_2^\times$. Our pair is in one squareclass. Hence the corollary as stated requires (H1) here, but does not impose (H2), a nonsupercuspidal condition at two, or a new dyadic conductor restriction. The nonzero scalar eigenspace supplies (H1) by [63, Proposition 2, p. 377]. Moreover the proof on p. 483 extends its linear relation to every vector in the full eigenspace before squaring. The resulting identity therefore applies simultaneously to old multiplicities, and later to both Galois-conjugated and character-twisted vectors at a common level. At that later level the same squareclass test is repeated; its prime support is still S_0. Putting $n_1 = D_0, n_2 = D$ in the corollary, with weight $3/2$, gives

$$a_f(D)^2 = \psi(D/D_0) \, \frac{\sqrt{D}\, L(E^{(\sigma D)}, 1)}{\sqrt{D_0}\, L(E^{(\sigma D_0)}, 1)} a_f(D_0)^2. \tag{E.10}$$

The period cancels, so the quotient in this formula is the quotient of the specified L_*-values. No product over the primes dividing D or D_0 is missing: the unramified factors at squarefree valuations are exactly those checked above.

For clarity, (E.10) gives proportional coefficient functionals on the whole space. If $\ell_0(f) = a_f(D_0)$ and $\ell_D(f) = a_f(D)$, then $\ell_D^2 = C\ell_0^2$ as a polynomial identity on V. If $C = 0$, then $\ell_D = 0$. Otherwise, choose a square root c of C. The polynomial ring on the finite-dimensional vector space V is an integral domain, so $(\ell_D - c\ell_0)(\ell_D + c\ell_0) = 0$ implies $\ell_D = c\ell_0$ or $\ell_D = -c\ell_0$. The scalar is common to every vector. This argument does not assume that V is one-dimensional.

An algebraic vector and the Galois comparison. Ramsey's section model [64, Definition 3.1 and Theorem 4.2] realizes the half-integral-weight space as sections of the indicated divisor sheaf on $X_1(4N_{\mathrm{aux}})$. Here N_{aux} is supported on S_0 and includes the chosen local depths; it need not equal the conductor N of E. Over characteristic-zero fields, flat field extension commutes with these sections. Cuspidality, scalar character, support conditions and prescribed good-Hecke eigenvalues are algebraic linear conditions. Although infinitely many coefficients or primes can occur in their description, a finite subset cuts out the same subspace, by finite dimensionality.

One need not impose Petersson orthogonality as an additional transcendental linear condition. The elliptic good-Hecke eigenspace is automatically orthogonal to unary theta series: at a good prime, the unary eigenvalues have absolute value $p + 1$, whereas $|\eta(p)a_p(E)| \le 2\sqrt{p} < p + 1$. The unary space and its orthogonal complement are stable under the Hecke operators and their adjoints. Thus their elliptic eigenspaces are disjoint. The nonzero reference functional on the complex space is consequently nonzero on its algebraic model. Choose such an algebraic g and scale it to $a(D_0) = 1$. This uses flat field extension only, not Ramsey's arbitrary-ring base-change assertion in weight $3/2$; see [64, Remark 4.4]. Here the field-extension assertion can be checked without a cohomology-vanishing theorem. Work over a number field K containing the character values and the roots of unity specifying the cusps and the support idempotents. A finite affine cover of the modular curve computes global sections as the kernel of the difference of restriction maps to pairwise intersections. Tensoring with a field extension is flat and preserves that kernel. Thus the sections in Ramsey's Definition 3.1 commute with $K \subset \overline{\mathbb{Q}} \subset \mathbf{C}$, also in weight $3/2$. Impose vanishing at the cusps, the diamond character, the support idempotents, and the kernels of $T(p^2) - \eta(p)a_p(E)$ inside this finite-dimensional space. The span in its dual of all the defining linear equations has a finite basis. These conditions therefore commute with the same field extensions. Since the reference functional is nonzero after extension to $\mathbf{C}$, it is nonzero before extension; a vector in a K-basis has nonzero reference value and may be divided by that value. This also places every coefficient of the chosen form in one number field. No assertion about base change to a residue field, nor about vanishing of H^1 in weight $3/2$, is used.

Let $\tau \in \mathrm{Aut}(\overline{\mathbb{Q}}/\mathbb{Q})$. Choose an odd integer e describing its action on all values of η and ψ, and put $\nu = \eta^{(e-1)/2}$. Coefficientwise conjugation gives a form g^τ with character ψ^e and good eigenvalues $\eta(p)^e a_p(E)$. One can reduce this conjugation assertion to integral weight by multiplying by the ordinary theta series: it has rational coefficients, and its fourth power has trivial scalar character. The algebraic divisibility condition at the cusps is precisely the one in Ramsey's section model, so division after conjugation is again a half-integral-weight form. The coefficientwise twist $g \otimes \nu$ has character $\psi\nu^2 = \psi^e$ and exactly these good eigenvalues. The identities hold because ψ/η is quadratic. The twist is modular at a level still supported on S_0; this follows as well by writing it as the finite additive translation sum using the Gauss sum of the primitive character inducing ν.

Pass to a common such level. Both forms lie in its full good-Hecke eigenspace, so the proportionality of its D and D_0 functionals gives

$$\tau(a(D)) = \frac{\nu(D)a(D)}{\nu(D_0)a(D_0)} = \nu(D/D_0)a(D).$$

There is no vanishing reference denominator: D_0 is prime to S_0 and $a(D_0) = 1$. No rationality of

the central-value quotient has been assumed in obtaining this equality.

Write the Dirichlet character as a product of its finite prime-power factors η_p. For every p dividing its conductor choose $u_p \in \mathbb{Z}_p^\times$ with $u_p^2 = D/D_0$, and set

$$z_D = \prod_{p|\mathrm{cond}(\eta)} \eta_p(u_p).$$

Then z_D is a two-power root of unity, $z_D^2 = \eta(D/D_0)$, and

$$\tau(z_D) = z_D^e, \qquad \nu(D/D_0) = z_D^{e-1}.$$

Changing the selected odd representative of e has no effect on the last identity, because D/D_0 is a local square at the conductor. The preceding Galois identity now shows that $a(D)/z_D$ is fixed by every τ, hence lies in $\mathbb{Q}$. Different choices of the local square roots change z_D by a sign, which changes only the sign of this rational number.

Finally, $\psi/\eta = \chi_{-\sigma}$ is trivial on D/D_0: its conductor is supported at two and the ratio is a square there. Thus $\psi(D/D_0) = z_D^2$. Dividing (E.10) by z_D^2 gives

$$a(D) = z_D r_D, \qquad z_D \in \mu_{2^\infty}, \qquad r_D \in \mathbb{Q}, \qquad r_D^2 = \frac{L_*(\sigma D)}{L_*(\sigma D_0)}. \tag{E.11}$$

The argument also covers the zero coefficient case. In particular, every nonzero coefficient in (E.11) has integral normalized two-adic valuation.

Recurrence, later exclusions, and valuations. Only now impose the extra finite coprimality sieve B. The zero-or-retain Fourier sieve is a finite additive translation operator. It preserves cuspidality, keeps every coefficient whose index is prime to B, and commutes with good $T(p^2)$ for $p \notin S_1 = S_0 \cup B$. In particular it keeps $a(D_0) = 1$. The squarefree comparison (E.11) was proved before this enlargement, so it requires no extra squareclass hypotheses at the primes of B. After the sieve it is asserted at the retained indices. The standard good $T(p^2)$ formula, with $\psi^2 = \eta^2$, gives

$$a(p^2 n) = \{\eta(p) a_p(E) - \psi(p)(-n/p)\} a(n) - p\eta(p)^2 a(n/p^2), \qquad p \notin S_1. \tag{E.12}$$

For squarefree D, write $a(Dp^{2r}) = \eta(p)^r b_r$. The first multiplier is $a_p(E) - (\psi/\eta)(p)(-D/p)$; subsequent multipliers satisfy $b_{r+1} = a_p(E) b_r - p b_{r-1}$. All are integers, including when $p \mid D$. Iteration over the good primes gives the same assertion for every retained square shell after removal of the η phase. More explicitly, for $p \notin S_1$ set $A_{p,0}(D) = 1$, $A_{p,1}(D) = a_p(E) - (\psi/\eta)(p)(-D/p)$, and $A_{p,r+1}(D) = a_p(E) A_{p,r}(D) - p A_{p,r-1}(D)$ for $r \geq 1$. Then, for every retained index Dm^2 with D squarefree,

$$a(Dm^2) = z_D \eta(m) r_D \prod_{p^r \| m} A_{p,r}(D).$$

The formula includes $p \mid D$, when the initial Legendre symbol is zero. When applying the recurrence at one prime, the other prime-square factors do not alter that Legendre symbol. This proves the formula by induction on the prime factors of m, with $a(n/p^2) = 0$ when $p^2 \nmid n$. All factors $A_{p,r}(D)$ are integers. Thus every coefficient used by the extraction and shell operators of Proposition F.5 is a rational number times a two-power root of unity, including coefficients with non-squarefree index. In the chosen residue field such a root reduces to one. A rational coefficient of valuation zero reduces to the element 1 of its prime field $\mathbb{F}_2$. Consequently the fresh coefficient-one reading of that proposition detects precisely equality in the fixed minimum-depth bound. It uses the same g, D_0 and normalization throughout.

Combining (E.9) and (E.11) makes the values $v_2 a(D) - \omega(D)$ at nonzero retained squarefree coefficients a bounded-below subset of $\mathbb{Z}$. It is nonempty because D_0 is retained, and therefore has an attained finite minimum. The form and the initial finite sieve are fixed before this minimum is chosen. Restricting later fresh primes by further exclusions does not authorize choosing another minimum. □

F Actual odd theta forms and their operators

We now construct the forms in (3.8), with the same h_*, D_w, L and actual Kummer stages as in Section 7. Write $m = |h_*|$. An admissible D is positive, squarefree, prime to the fixed support and in the prescribed unit classes. Then $K_D = \mathbb{Q}(\sqrt{Dh_*})$ is imaginary, split at $2N$, has no extra units, and Dh_* is an odd fundamental discriminant. The conductor-one point is the unaveraged sum

$$P_D = \sum_{[\mathfrak{a}] \in \mathrm{Pic}(\mathcal{O}_{K_D})} \chi_{h_*}(\mathrm{N}\,\mathfrak{a})\, \pi(x_D^{[\mathfrak{a}]}). \tag{O.1}$$

It is distinct from the conductor-$|h|$ ring-class point used in Section 9.

F.1 A fixed witness and one compatible sequence

Proposition F.1 (One witness and a compatible detecting sequence)**.** *For the fixed $E = E_\ell$ under the full-corank-one hypothesis of Theorem 5.3, use the prepared positive unit filter containing* 1*. There exist a negative odd fundamental discriminant h_* with $L(E^{h_*}, 1) \neq 0$, a positive fundamental discriminant D_w in that filter with $L'(E^{D_w}, 1) \neq 0$, and a fixed integer L such that LP_D belongs to the D-twist rational line for every admissible D and $Q = LP_{D_w}$ is nontorsion. There is one sequence of actual stages with $M \to \infty$, good primes ℓ_0, and additive maps*

$$\lambda_M : E(\mathbb{F}_{\ell_0^2}) \longrightarrow \mathbb{Z}/2^M\mathbb{Z}, \qquad v_2\lambda_M(\overline{Q}) = c_w < M,$$

where c_w is fixed. At these stages Frobenius squared is trivial on $E[2^M]$, $(D_w/\ell_0) = 1$, $(h_*/\ell_0) = -1$, and λ_M has Frobenius sign $+1$. Any specified finite prime exclusion is permitted. Additional field prescriptions require a compatible common lift; no preservation of c_w under arbitrary later field prescriptions is asserted.

Proof. We use the positive unit-squareclass filter containing one, with its fixed support enlarged before choosing the witness. The rational two-isogeny parity theorem [49, Theorem 1.8] gives root number -1 for E under the full-corank-one assumption. Apply clause (ii) of [56, pp. 543–544] to its weight-two newform, with every old prime in the splitting set, to obtain a negative fundamental discriminant h_* with $L(E^{(h_*)}, 1) \neq 0$. Including two makes h_* odd and excludes the extra-unit fields. Now add the primes of h_* to that set and apply clause (i) to obtain a positive fundamental discriminant D_w with $L'(E^{(D_w)}, 1) \neq 0$. Splitting at every old prime puts D_w in the filter containing one and makes it prime to h_*. These choices precede every detecting prime and every theta precision.

The conductor-one genus character for $K_D = \mathbb{Q}(\sqrt{Dh_*})$ cuts out $F_D = \mathbb{Q}(\sqrt{D}, \sqrt{h_*})$. Thus the unaveraged sum P_D in (O.1) belongs to $E(F_D)$. Its rational character components are the D- and h_*-components. In the degenerate case $D = 1$ these are the trivial and h_* characters of the actual quadratic group. The h_*-component has rank zero by the forward rank-zero theorem. For D_w, the unaveraged height formula [59, Theorem 1.1] has Rankin derivative $L'(E^{(D_w)}, 1)L(E^{(h_*)}, 1) \neq 0$, so

P_{D_w} is nontorsion. Here the conductor is one, all primes of N split, and the extra-unit fields have been excluded.

We make the fixed multiplier explicit. Choose distinct good rational primes p_1, p_2 for E and put

$$B_j = \prod_{f=1}^{4} \#E(\mathbb{F}_{p_j^f}), \qquad B = B_1 B_2.$$

For every number field of degree at most four, prime-to-p_j torsion injects into a reduction group with residue degree at most four. The two primes therefore show that B kills its entire torsion subgroup. Put $B_{\mathrm{cusp}} = C_\pi$, where C_π annihilates the image of the cuspidal subgroup defined after (4.1). This includes every Atkin–Lehner translation $\pi(W\infty)$, which need not vanish merely because $\pi(\infty) = 0$. For $\Delta_D = \mathrm{Gal}(F_D/\mathbb{Q})$, the integral character operator

$$4e_D = \frac{4}{|\Delta_D|} \sum_{\gamma \in \Delta_D} \chi_D(\gamma)\gamma$$

is defined also when $D = 1$. The difference $4P_D - 4e_D P_D$ is torsion, since its only possible rational component is the rank-zero h_*-component. Hence $4BP_D = B(4e_D P_D)$ transforms exactly by χ_D. Choose L to be a common multiple of the previously required fixed multiplier and $4BB_{\mathrm{cusp}}$, for example their product. This retains every earlier descent and local-condition factor. Then LP_D is a rational point of $E^{(D)}$ under the standard twist identification, for every admissible D. In particular

$$Q = LP_{D_w} \in A(\mathbb{Q}), \qquad A = E^{(D_w)},$$

is nontorsion. No assertion that the original P_D is $E(K_D)$-rational is required.

Choose compatible division points Q_n with $2^n Q_n = Q$ and $2Q_{n+1} = Q_n$, and let $c_Q(g) = (gQ_n - Q_n)_n \in T_2 A$ be the Kummer cocycle. Write ρ_A for the Tate representation and $H = \ker \rho_A$. On H, this cocycle is a homomorphism. Its image is compact, and hence a closed $\mathbb{Z}_2$-submodule of $T_2 A$. Moreover

$$c_Q(ghg^{-1}) = \rho_A(g) c_Q(h) \qquad (h \in H),$$

so its rational span is invariant under the Tate image.

The curve A is non-CM. Open image [69] supplies both irreducibility of its rational Tate representation and a central scalar $z \neq 1$ in its Tate image. If $c_Q(H) = 0$, the rational cocycle descends to that image. Commuting z with g gives

$$(z-1)c_Q(g) = (g-1)c_Q(z).$$

Since $z - 1$ is invertible over $\mathbb{Q}_2$, this is a coboundary. It contradicts the Kummer injection $A(\mathbb{Q}) \otimes \mathbb{Q}_2 \hookrightarrow H^1(\mathbb{Q}, V_2 A)$ and nontorsion of Q. Thus the translation image is nonzero. Irreducibility makes its rational span all of $V_2 A$. It consequently contains $2^{a_0} T_2 A$ for some a_0.

Let $F_0/\mathbb{Q}$ be any fixed finite Galois auxiliary extension, including $\mathbb{Q}(\sqrt{D_w}, \sqrt{h_*})$ and any fixed ring-class data to be retained with the action of complex conjugation. Put $H_0 = H \cap G_{F_0}$. It has finite index d in H. For $h \in H$, $h^d \in H_0$ and $c_Q(h^d) = d\, c_Q(h)$. Hence $c_Q(H_0)$ still contains $2^a T_2 A$ for some fixed a. This argument uses no independence between F_0 and the Tate or division-point fields.

Fix actual complex conjugation τ. Since $D_w > 0$, the real twist identification identifies its matrix on $T_2 A$ with its matrix C on $T = T_2 E$. We have $C^2 = 1$ and $\det C = -1$. Its dual $+1$-eigenspace therefore contains a primitive integral row $b : T \to \mathbb{Z}_2$, with $bC = b$ and $b(T) = \mathbb{Z}_2$.

Choose $t_0 \in T$ with $b(t_0) = 1$, and choose $h \in H_0$ with $c_Q(h) = v = 2^a t_0$. Set $\sigma = h\tau$. If $d_\tau = c_Q(\tau)$, then $(1 + C)d_\tau = 0$, because $\tau^2 = 1$. Consequently

$$\rho_A(\sigma) = C, \qquad c_Q(\sigma^2) = (1 + C)(v + d_\tau) = (1 + C)v, \qquad b\, c_Q(\sigma^2) = 2^{a+1}.$$

Put $c_w = a + 1$. The element σ and this finite valuation are fixed before precision varies. Its restriction to F_0 is that of τ: it fixes $\sqrt{D_w}$ and negates $\sqrt{h_*}$. Since h fixes $\sqrt{D_w}$, the matrix of σ on T_2E is also C.

For each $M > c_w$, take the finite Galois extension generated by F_0, $E[2^M]$, $A[2^M]$, and all 2^M-division points of Q. Restrict this same actual σ to the extension. Finite Chebotarev supplies infinitely many good rational primes ℓ_0 in its conjugacy class. Choose a place realizing this representative, and exclude two, all bad primes, the support of $D_w h_*$ and any other finite set. Thus, with the torsion basis transported by this place,

$$\mathrm{Fr}_{\ell_0} \equiv C \pmod{2^M}, \qquad (D_w/\ell_0) = 1, \qquad (h_*/\ell_0) = -1.$$

The determinant also gives $\ell_0 \equiv -1 \pmod{2^M}$. In particular $\mathrm{Fr}_{\ell_0}^2$ acts trivially on $E[2^M]$.

Here is the finite-field map without a normalization ambiguity. Put $n = 2^M$ and write F for Frobenius on the reduction of E. For $P \in E(\mathbb{F}_{\ell_0^2})$, choose R with $nR = P$ and set

$$\kappa_M(P) = F^2R - R \in E[n] = T/nT.$$

Changing R by n-torsion does not change this value. The map is additive, with kernel $nE(\mathbb{F}_{\ell_0^2})$. It is onto: $F^2 - 1$ is a nonzero separable isogeny, so for $S \in E[n]$ one can solve $(F^2 - 1)R = S$ and take $P = nR$. Also $\kappa_M(FP) = F\kappa_M(P)$, by choosing FR as a division point. Therefore $\lambda_M = b\kappa_M$ has Frobenius sign $+1$.

Good reduction identifies the reduction of the global division cocycle with this exact map. The chosen place makes the twist identification defined over $\mathbb{F}_{\ell_0}$, since $(D_w/\ell_0) = 1$. Thus the fixed witness satisfies

$$\lambda_M : E(\mathbb{F}_{\ell_0^2}) \longrightarrow \mathbb{Z}/2^M\mathbb{Z}, \qquad v_2\lambda_M(\overline{Q}) = c_w < M. \tag{O.2}$$

Choosing increasing M and excluding previous primes gives one sequence of actual stages. Subsequent finite prime exclusions do not alter c_w. New field prescriptions require a compatible common lift and, if added after this choice, are not asserted to preserve this particular constant. □

F.2 Local functions and the theta sum

Proposition F.2 (The actual odd theta form and its recurrence)**.** *Fix an actual stage of Proposition F.1 with $M \geq 2$. Use $m = |h_*|$, the same L, the Eichler order of level N in the quaternion algebra ramified at ℓ_0, ∞, and the local Schwartz functions specified below, including (O.3). With $c_0 = |\,\mathrm{GL}_4(\mathbb{F}_3)| = 24\,261\,120$, the sum H_M in (O.4) is well defined modulo 2^M in the character ring. It gives an actual holomorphic form*

$$g_M = \sum_n a_M(n)q^n, \qquad a_M(n) = H_M(mn), \qquad g_M \in M_{3/2}(\Gamma_0(K), \psi),$$

with level supported only at the specified coefficient primes. The characters are exactly $\psi = \eta_B(m/\cdot)$ and $\eta = \eta_B\chi_{h_*}$, so $\psi/\eta = \chi_{-1}$. For every prime outside that support, the recurrence (O.6) holds at every integer index, with nonintegral indices interpreted as zero. Changing the lifts of P_M loses no precision. A specified further local coefficient sieve retains this conclusion outside the enlarged support.

Proof. In this appendix $M = M_\theta$; take $M \geq 2$. Retain the same $m = |h_*|$, L, and detecting prime ℓ_0 as in (O.2). Since $\mathrm{Fr}_{\ell_0}^2 = 1$ on $E[2^M]$, evaluation at arithmetic Frobenius squared identifies the finite-field Kummer quotient with $T_2E/2^MT_2E$. Equivariance and $bC = b$ imply $\lambda_M(\mathrm{Fr}_{\ell_0} P) = \lambda_M(P)$. The determinant gives $\ell_0 \equiv -1 \pmod 4$; hence $(m/\ell_0) = (-1/\ell_0)(h_*/\ell_0) = 1$.

The finite-field evaluation map. Write $k_0 = \mathbb{F}_{\ell_0^2}$ and let F denote its arithmetic Frobenius. For $P \in E(k_0)$, choose $Q_P \in E(\overline{k_0})$ with $2^MQ_P = P$. Then

$$\kappa_M(P) = FQ_P - Q_P \in E[2^M]$$

is independent of the choice of Q_P, since F is the identity on $E[2^M]$. This is an additive map. It kills $2^ME(k_0)$, and its kernel is exactly that subgroup: a zero value means that Q_P itself lies in $E(k_0)$. The finite group $E(k_0)/2^ME(k_0)$ has order $|E(k_0)[2^M]| = 2^{2M}$, as does $E[2^M]$, so the induced injection is an isomorphism. Composing κ_M with the fixed row b modulo 2^M gives λ_M. The arithmetic Frobenius over $\mathbb{F}_{\ell_0}$ commutes with F and acts on the torsion by C. Hence $\kappa_M(\mathrm{Fr}_{\ell_0} P) = C\kappa_M(P)$ and $bC = b$ proves the asserted Frobenius sign. For the fixed witness, specialization of a compatible division point identifies this evaluation with the cocycle value at σ^2, whose b-value is 2^{a+1}. Thus the visibility valuation is fixed as the precision grows.

By [82, Main Theorem 14.6.1], there is a quaternion algebra $B/\mathbb{Q}$ ramified exactly at ℓ_0 and infinity. Let R be an Eichler order of level N. Supersingular pairs are $X = B^\times\backslash\widehat{B}^\times/\widehat{R}^\times$; their automorphism groups are $R_z^\times$ for $R_z = B \cap z\widehat{R}z^{-1}$. This description allows every N prime to ℓ_0, including powers of two. One may choose each underlying supersingular curve over $\mathbb{F}_{\ell_0^2}$ with scalar Frobenius; its cyclic subgroups of order N are then stable. Thus all the corresponding coarse moduli points are defined over that field. The geometric and adelic descriptions are those of [61, Sections 1.1–1.2 and 2.5].

Define $P_M(z) = \lambda_M(L\pi_{\ell_0}(z))$. Here the specialization is legitimate because $\ell_0 \nmid N$: the modular curve has good reduction and the map to the good-reduction elliptic curve extends over the local base. The good Hecke correspondences also extend. Writing π as the Abel–Jacobi map based at infinity followed by the elliptic quotient gives, for $p \nmid N\ell_0$,

$$\sum_{z' \text{ a } p\text{-neighbor of } z} \pi_{\ell_0}(z') = a_p(E)\pi_{\ell_0}(z).$$

Indeed the quotient intertwines T_p with $[a_p(E)]$ and $T_p(\infty) = (p+1)\infty$. For an Atkin–Lehner involution W_{p^e} the analogous identity has eigenvalue w_{p^e} and the constant $\pi(W_{p^e}\infty)$, which L kills. Frobenius acts on the supersingular set as the right division-uniformizer operation. Consequently P_M has the asserted good Hecke and Atkin–Lehner eigenvalues modulo 2^M, and uniformizer eigenvalue $+1$. Choose arbitrary integral lifts of this finite function.

The finite Schwartz function. All local functions below are zero off their specified supports. Set $V = B^0$ and $Q_B(x) = \mathrm{Nrd}(x) = -x^2$. For $p \nmid 2Nm\ell_0$ use $1_{R_p^0}$. For odd $p^e\|N$, use the split integral unit support. The cyclic level quotient selects one of the two eigensummands and its root $r \in \mathbb{Z}_p^\times$. The projectors $(x+r)/(2r)$ and $(r-x)/(2r)$ are integral, so reduction and Hensel lifting show that the selected root modulo any fixed power of p is locally constant. Weight it by a unit character $\alpha_p(r)$ of finite 2-power order with

$$\alpha_p(-1) = w_{p^e}\chi_{h_*}(p)^e. \tag{O.3}$$

Such a character exists: at an odd prime the cyclic 2-primary part of $\mathbb{F}_p^\times$ admits a character taking -1 to -1, and the trivial character supplies the other parity. Conjugation by $R_p^\times$ preserves the

cyclic quotient and its selected summand. Multiplication of x by a unit u preserves the summands and replaces r by ur, giving multiplier $\alpha_p(u)$.

At two require $e_x = (1+x)/2 \in R_2$ and $x^2 = r^2 \in \mathbb{Z}_2^{\times 2}$. The eigenvalues $a = (1+r)/2$ and $a' = (1-r)/2$ of e_x differ by the unit r. Its projectors $(e_x - a')/r$ and $(a - e_x)/r$ are integral and remain distinct modulo two. If $2^e \| N$, the cyclic quotient is a free rank-one $\mathbb{Z}/2^e$-module. Of these complementary projectors exactly one acts as the identity on it, since the only idempotents of $\mathbb{Z}/2^e$ are zero and one. This defines the selected root without a choice of its parity. For every odd u,

$$e_{ux} = ue_x + (1-u)/2.$$

This affine transformation preserves the two projectors individually, so the selected root changes to ur, even when the two eigenvalue parities switch. The same root character rule therefore applies at two. A character with either prescribed parity exists, using the trivial character or the character modulo four. If $2 \nmid N$ there is no root character at two.

For $p \mid m$, require $x \in M_2(\mathbb{Z}_p)^0$ and $v_p Q_B(x) = 1$. The reduction of x is a nonzero nilpotent: zero reduction would force $p^2 \mid Q_B(x)$. All nonzero nilpotents are conjugate to $X_0 = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. The centralizer of X_0 in $\mathrm{GL}_2(\mathbb{F}_p)$ is $\{\left(\begin{smallmatrix} a & b \\ 0 & a \end{smallmatrix}\right) : a \neq 0\}$ and has square determinant. Thus

$$\epsilon_p(gX_0g^{-1}) = (\det g/p)$$

is well-defined. Conjugation by u multiplies it by $(\det u/p)$, and scalar multiplication by $u \in \mathbb{Z}_p^\times$ multiplies it by (u/p), as follows from conjugation of X_0 by $\mathrm{diag}(u, 1)$.

At ℓ_0 require $x \in R^0_{\ell_0}$ with nonzero reduction $\bar{x} = t\beta$ in the trace-zero line of $\mathbb{F}_{\ell_0^2}$, and use (t/ℓ_0). Order units act trivially on the residue field. A division uniformizer acts by Frobenius, which sends β to $-\beta$, giving multiplier -1. Unit scaling gives (u/ℓ_0). Moreover β^2 is a nonsquare in $\mathbb{F}_{\ell_0}$: otherwise $\beta = \pm a$ for $a \in \mathbb{F}_{\ell_0}$, contrary to its nonzero trace-zero condition. Both -1 and β^2 are nonsquares, so $Q_B(x) = -t^2\beta^2$ is a nonzero square. Since $(m/\ell_0) = 1$, extracted indices have $(D/\ell_0) = 1$.

These supports are compact and open, and all weights are locally constant. Additional fixed unit-squareclass conditions multiply them by locally constant characteristic functions. They are preserved by $x \mapsto ux$, which multiplies the norm by u^2. Let $\mathcal{O}$ be the valuation ring at a place above two of the field generated by the fixed root-character values at primes dividing N. All other weights are quadratic, so this field is independent of M and ℓ_0. With $f = \prod f_p$, the preceding calculations give

$$f(u^{-1}xu) = \chi_{h_*}(\mathrm{Nrd}\, u)^{-1} f(x) \quad (u \in \widehat{R}^\times), \qquad f(vx) = \alpha(v) f(x) \quad (v \in \widehat{\mathbb{Z}}^\times).$$

The ramified local quadratic character on units at $p \mid m$ is $(\cdot/p)$; all other order-unit character factors are trivial. The scaling character has

$$\alpha(-1) = \Big(\prod_{p^e \| N} w_{p^e} \chi_{h_*}(p)^e\Big)\Big(\prod_{p \mid m} (-1/p)\Big)(-1) = 1.$$

For the last equality, $m \equiv 3 \pmod 4$ and the first product is $w(E^{h_*}) = +1$. More explicitly, $w(E) = -\prod w_{p^e}$ and, since $(h_*, N) = 1$, $w(E^{h_*}) = w(E)\chi_{h_*}(-N)$. The latter twist formula also follows by multiplying the local unramified-twist factors $\chi_{h_*}(p)^{v_p N}$ at $p \mid N$ and the factors $(-1/p)$ at the newly ramified good primes.

Let η_B denote the classical character associated with α in the inverse-idele convention. The formula $\eta_B(d) = \prod_p \alpha_p(d)$ for positive integers d prime to the support is verified in the transformation calculation below.

Set $c_0 = |\operatorname{GL}_4(\mathbb{F}_3)| = 24\,261\,120$, so $v_2(c_0) = 9$, and define

$$H_M(n) = c_0 \sum_{z \in X} \frac{P_M(z)\chi_{h_*}(\operatorname{Nrd} z)}{|R_z^\times|} \sum_{x \in V(\mathbb{Q}),\ Q_B(x)=n} f(z^{-1}xz). \tag{O.4}$$

Left multiplication embeds each $R_z^\times$ into $\operatorname{GL}_4(\mathbb{Z})$; reduction modulo three is injective on finite subgroups. Indeed, a prime-order element $1 + 3^a A$ in its kernel, with $a \geq 1$ and A nonzero modulo three, has nonzero first term in its prime-th power modulo 3^{a+1}, or modulo 3^{a+2} when its order is three, a contradiction. Thus $|R_z^\times| \mid c_0$. The right-conjugation identity proves independence of the right representative. On the left, rational reduced norms are positive and the global quadratic idele character is trivial on them. The transformation law below shows that each summand is an actual holomorphic theta series with the same level and character; $c_0/|R_z^\times|$ is an integer. Consequently the coefficients lie in the fixed character ring $\mathcal{O}$. Changing any lift of P_M by 2^M changes every coefficient by an element of $2^M\mathcal{O}$. In particular there is no loss of precision from an automorphism denominator.

F.3 Transformation law and recurrence

Use the standard global additive character, self-dual measures, and the Gaussian $e^{-2\pi Q_B}$ at infinity. Weil's operator formulas [68, I.13] give multiplication by $\mathbf{e}_p(bQ_B)$ for $n(b)$, scaled pullback $f_p(x) \mapsto \gamma_{Q_B}(u) f_p(ux)$ for $m(u)$ with u a unit, and a constant times Fourier transform for the Weyl element. The upper unipotents $n(\mathbb{Z}_p)$ fix f_p, because Q_B is integral on its support. If L_p' is any lattice of translations fixing f_p, its Fourier transform has support in $(L_p')^\vee$. Choose A_p with $p^{A_p} Q_B((L_p')^\vee) \subseteq \mathbb{Z}_p$. Fourier conjugation then proves that $n^-(p^{A_p}\mathbb{Z}_p)$ fixes f_p. This includes $p = 2$. At two take $A_2 \geq 2$ and use the comparison with the ordinary theta multiplier given below; no splitting of the metaplectic cover over all diagonal units is asserted. For clarity, the local normalizations and the lower-unipotent argument are as follows. Take $\mathbf{e}_\infty(t) = e^{2\pi i t}$ and $\mathbf{e}_p(t) = e^{-2\pi i \{t\}_p}$, so that $\prod_v \mathbf{e}_v(r) = 1$ for $r \in \mathbb{Q}$. For $Q(x) = x^{\mathsf{T}} A x$ put $B_Q(x, y) = Q(x + y) - Q(x) - Q(y) = 2x^{\mathsf{T}} A y$ and set

$$\mathcal{F}_Q \phi(y) = \int \phi(x) \mathbf{e}_p(B_Q(x, y))\, dx_Q, \qquad dx_Q = |\det(2A)|_p^{1/2}\, dx.$$

Here dx is the coordinate product measure with $\operatorname{vol}(\mathbb{Z}_p^3) = 1$. At the real place use the same formula $dx_Q = |\det(2A)|_\infty^{1/2} dx$ with coordinate Lebesgue measure. Thus $\mathcal{F}_Q^2 \phi(x) = \phi(-x)$. If $\phi(x + \lambda) = \phi(x)$ for $\lambda \in L'$, translation in this integral shows that its Fourier transform vanishes unless $\mathbf{e}_p(B_Q(\lambda, y)) = 1$ for every $\lambda \in L'$. This is precisely the asserted support in $(L')^\vee$. Since this dual lattice is compact, an integer A_p with $p^{A_p} Q((L')^\vee) \subset \mathbb{Z}_p$ exists. The operator for $n^-(c)$ is a Fourier conjugate of multiplication by $\mathbf{e}_p(-cQ)$; consequently it fixes ϕ whenever $c \in p^{A_p}\mathbb{Z}_p$. A scalar used to normalize the Weyl operator cancels with its inverse in this conjugation.

All the specified local functions have such a translation lattice. At an odd level prime the unit-root condition and its character depend on a finite residue quotient. At two the square-unit condition is a congruence modulo eight; the integral projectors selecting the cyclic quotient, and the finite root character, again depend on a finite residue quotient. At a prime dividing m, valuation one and the nonzero-nilpotent sign depend on finite residue data. The division-prime function is already a residue function. Each support is contained in a lattice and is a finite union of cosets of a sufficiently small lattice. This proves the Schwartz–Bruhat and translation assertions for the actual weights.

For the reference $Q_0 = \sum X_i^2$, the dyadic dual of $\mathbb{Z}_2^3$ is $\frac{1}{2}\mathbb{Z}_2^3$ and $\operatorname{vol}_{dx_{Q_0}}(\mathbb{Z}_2^3) = 2^{-3/2}$. In particular its Fourier transform is $2^{-3/2} 1_{\frac{1}{2}\mathbb{Z}_2^3}$, not $1_{\mathbb{Z}_2^3}$. The lower-unipotent conclusion nevertheless

holds at depth two, because $4Q_0(\frac{1}{2}\mathbb{Z}_2^3) \subset \mathbb{Z}_2$. These positive measure factors affect neither the diagonal eigenvalue quotient nor the counting coefficients of the theta sum.

The diagonal factor can be computed directly. For $Q_B = \sum_{i=1}^3 a_i X_i^2$, the one-dimensional Weil-index identity

$$\frac{\gamma(ua_i)}{\gamma(a_i)} = \frac{\gamma(u)}{\gamma(1)}(u, a_i)_p$$

and multiplication under orthogonal sums give, relative to the ordinary theta multiplier cubed, the factor $(u, a_1a_2a_3)_p$. These identities are [68, II.25 and II.28]; the latter obtains the Hilbert symbol from the four-dimensional quaternion norm form. For $B = (-a, -b)$ the norm on B^0 is $aX^2 + bY^2 + abZ^2$, whose determinant is $(ab)^2$. There is therefore no additional determinant character for Q_B.

The normalized diagonal factor. The Weil-index identity used here can be derived without choosing Gauss-sum signs. The norm form of the quaternion algebra (u, a) is $\langle 1, -u, -a, ua\rangle$. By multiplicativity under orthogonal sums and $\gamma(-q) = \gamma(q)^{-1}$, its index is

$$\frac{\gamma(1)\gamma(ua)}{\gamma(u)\gamma(a)}.$$

For a split quaternion algebra the norm form is hyperbolic and its index is one; for the division algebra its index is minus one by [68, II.28, Proposition 4]. These are exactly the two values of $(u, a)_p$. Solving this equality gives

$$\frac{\gamma(ua)}{\gamma(a)} = \frac{\gamma(u)}{\gamma(1)}(u, a)_p.$$

Applying it to three diagonal entries gives the ordinary one-dimensional theta factor cubed times $(u, \det Q_B)_p$. This derivation includes $p = 2$. A change of diagonalization changes the determinant only by a square, so it changes no symbol. For Q_B/m the determinant becomes $m^{-3}\det Q_B$, whose squareclass is m. Thus the extra factor is precisely $(u, m)_p$. This establishes the determinant-character contribution in (O.5) with the same additive character as the original theta multiplier.

An operator construction of the common double cover. Here is a direct justification that the same lifts may be used in the comparison. It also fixes the signs of the diagonal operators. Work over $k = \mathbb{Q}_p$, including $p = 2$, or $k = \mathbb{R}$, with the additive character just specified. For a nondegenerate ternary Q write $\gamma_Q = \gamma(Q)$ and define

$$\begin{aligned} U_Q(b)\phi(x) &= \mathbf{e}_k(bQ(x))\phi(x), \\ D(a)\phi(x) &= |a|_k^{3/2}\phi(ax), \\ W_Q = \gamma_Q\mathcal{F}_Q, &\qquad M_Q(a) = r_Q(a)D(a), \qquad r_Q(a) = \frac{\gamma(aQ)}{\gamma(Q)}. \end{aligned}$$

The Weyl matrix in this convention is $w = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. The opposite Fourier-sign convention changes both compared representations together.

The Fourier formula of [68, I.14, Theorem 2], or its completion-of-the-square calculation, says for $b \neq 0$ that

$$\int_{k^3} \mathbf{e}_k\big(bQ(z) + B_Q(z, v)\big)\, dz_Q = |b|_k^{-3/2}\gamma(bQ)\mathbf{e}_k(-b^{-1}Q(v)).$$

The integral has the oscillatory meaning of that theorem. Applying it to a Schwartz function gives the operator identity

$$\mathcal{F}_Q U_Q(b) \mathcal{F}_Q = \gamma(bQ) U_Q(-b^{-1}) D(b^{-1}) \mathcal{F}_Q^{-1} U_Q(-b^{-1}).$$

Thus no choice of an individual square root of a Gauss sum is needed. The index identities proved above also give

$$\gamma(Q)^2 = (\det Q, -1)_k \gamma(1)^6, \qquad \gamma(1)^4 = (-1,-1)_k, \qquad r_Q(a) r_Q(b) = (a,b)_k r_Q(ab).$$

For the middle equality apply the quaternion-norm formula to $\langle 1,1,1,1\rangle$; the first follows by applying the same formula to $\langle 1, -a_i, 1, -a_i\rangle$ for each diagonal entry a_i of Q.

Consequently the displayed operators satisfy

$$\begin{gathered} U_Q(b) U_Q(c) = U_Q(b+c), \qquad M_Q(a) U_Q(b) M_Q(a)^{-1} = U_Q(a^2 b), \\ M_Q(a) M_Q(b) = (a,b)_k M_Q(ab), \qquad W_Q M_Q(a) W_Q^{-1} = M_Q(a^{-1}), \\ W_Q^2 = (-1,-1)_k M_Q(-1), \\ W_Q U_Q(b) W_Q = (-b,-1)_k U_Q(-b^{-1}) M_Q(-b^{-1}) W_Q U_Q(-b^{-1}) \quad (b \neq 0). \end{gathered}$$

For the last equality, the preceding Fourier identity leaves the scalar $\gamma(Q)^2 \gamma(bQ)^2$; the index-square formula reduces it to $(-b,-1)_k$. Every scalar in these relations is independent of the ternary form. Interpreting a scalar -1 as the central element ε gives one common double cover of $\mathrm{SL}_2(k)$. Indeed these relations reduce every word to one of $\varepsilon^e U(x) M(a)$ or $\varepsilon^e U(x) M(a) W U(y)$, with $e \in \{0,1\}$. The ordinary Bruhat decomposition makes their projections unique, apart from ε^e; the operators send ε to $-\mathrm{id}$, so the central kernel has exactly two elements. This realizes the rank-three pullback of Weil's metaplectic operators, rather than assuming a section over the diagonal units.

For $Q_0 = \langle 1,1,1\rangle$ these operators are the tensor products of the three ordinary one-dimensional theta operators. For either Q or Q_0, changing a lift multiplies its operator by the same -1. Hence the quotient of their eigenvalues is independent of the lift. At two the relation $M_Q(-1)^2 = (-1,-1)_2 \,\mathrm{id} = -\mathrm{id}$ explicitly retains the order-four lift of $-I$.

On a unit a the quotient of the two diagonal index factors is

$$\frac{r_Q(a)}{r_{Q_0}(a)} = (a, \det Q)_k.$$

Together with $f_p(ax) = \alpha_p(a) f_p(x)$, this is exactly the eigenvalue quotient $\alpha_p(a)(a, \det Q)_p$ used below. There is no remaining eighth-root factor at two.

The metaplectic comparison at two and the extracted character. We specify how the preceding index calculation fixes the multiplier, including its dyadic part. Let

$$Q_0 = X_1^2 + X_2^2 + X_3^2, \qquad \phi_{0,p} = 1_{\mathbb{Z}_p^3}, \qquad \theta(\tau) = \sum_{n \in \mathbb{Z}} e^{2\pi i n^2 \tau}.$$

Use the same additive character and metaplectic lifts for the rank-three Weil representations of Q_B and Q_0. The latter theta series is literally $\theta(\tau)^3$; this fixes the meaning of the ordinary theta multiplier cubed, including its eighth roots of unity. For either representation the nontrivial central element of the metaplectic cover acts by -1. Consequently the ratio of two eigenvalues computed using the same lift is independent of that lift. In particular a central factor in multiplying the three

lifts in the displayed triangular factorization cancels in this ratio. We compare eigenvalues, and do not choose a group section of the cover over $\Gamma_0(2^{A_2})$.

Here is the comparison on the three factors. Both upper unipotents act trivially on the indicated vectors. Choose A_p large enough that both lower unipotents do also. This is the Fourier-support argument above for f_p; for $\phi_{0,p}$ it requires no positive depth at an odd prime and depth two at two, since the bilinear dual of $\mathbb{Z}_2^3$ for Q_0 is $\frac{1}{2}\mathbb{Z}_2^3$ and $4Q_0(\frac{1}{2}\mathbb{Z}_2^3) \subset \mathbb{Z}_2$. The Weyl scalar cancels in conjugating an upper unipotent to a lower one, so it introduces no additional eigenvalue. On a diagonal unit u, the scaling part acts by $\alpha_p(u)$ on f_p and by one on $\phi_{0,p}$. The quotient of the Weil-index parts, by the preceding calculation, is

$$\frac{\gamma(uQ_B)/\gamma(Q_B)}{(\gamma(u)/\gamma(1))^3} = (u, \det Q_B)_p. \tag{O.5a}$$

If the inverse Weil-index convention is used for the diagonal operator, both numerator and denominator are inverted; the quotient is unchanged because the Hilbert symbol has order two. Thus this computation does not leave an undetermined inverse or a dyadic root of unity. These operator comparisons are precisely the scaled pullback, Fourier conjugation and scalar relations of [68, I.13–I.14 and II.25–II.28].

Here $\det Q$ means the determinant of the matrix in $Q(x) = x^{\mathsf{T}}Ax$. The associated bilinear Gram matrix is $2A$. If that convention is used instead, both Q and Q_0 acquire the same factor 2^3 in their determinants, which cancels in the comparison. In particular no extra squareclass of two is introduced at the dyadic place.

For a general ternary form Q in this argument, at each prescribed prime take $A_p \geq 1$ and increase A_p also to the conductor of $u \mapsto (u, \det Q)_p$. The eigenvalue relative to the ordinary theta multiplier, on a matrix whose upper-left entry is a, is then

$$\alpha_p(a)(a, \det Q)_p. \tag{O.5b}$$

Indeed the triangular factorization has just the two trivial unipotent factors and this diagonal factor, and its lift ambiguity cancels as explained above. This also verifies multiplication: the upper-left entry of a product is congruent to the product of the upper-left entries modulo p^{A_p}, so both local characters in (O.5b) multiply. The argument therefore applies to every matrix in the indicated congruence subgroup at a prescribed prime, not only to a choice of generators. At an unramified prime outside the support, the full spherical invariance established below treats matrices with nonunit upper-left entry as well.

For Q_B the determinant squareclass is one at every place, including two. Thus (O.5b) has no determinant factor. For the extracted form $Q'_B = Q_B/m$ it is $m^{-3} \equiv m$. At an odd prime $p \mid m$ and a unit u, and at two, respectively, the additional factors are explicitly

$$(u, m)_p = (u/p)^{v_p(m)}, \qquad (u, m)_2 = (-1)^{((u-1)/2)((m-1)/2)}.$$

Here m is odd and $m \equiv 3 \pmod 4$, so the second factor is $(-1)^{(u-1)/2}$. It is already trivial on $1 + 4\mathbb{Z}_2$. At every odd prime outside m the factor on units is one. There is consequently no new level prime, and there is no omitted dyadic character.

Finally take a positive integer d prime to the complete level. The finite operator used in the classical transformation is the one at γ^{-1}, with upper-left entry d. The root-character factor is therefore $\prod_p \alpha_p(d) = \eta_B(d)$. Hilbert reciprocity gives

$$\prod_{p \mid 2m} (d, m)_p = \prod_{q \mid d} (m/q)^{v_q(d)} = (m/d). \tag{O.5c}$$

Indeed the infinite symbol is one because $m, d > 0$, all other finite symbols are one, and at an odd $q \mid d$ the symbol is $(m/q)^{v_q(d)}$. This proves the displayed product without discarding its dyadic factor. Both sides extend to the same even Dirichlet character. Since $h_* = -m$, division by the genus character now gives

$$\frac{\eta_B(d)(m/d)}{\eta_B(d)\chi_{h_*}(d)} = (-1/d).$$

Thus the characters in (O.5), with the stated inverse-idele convention for η_B, are exactly the characters supplied by these local Weil operators.

Take A_p also at least the conductor exponent of α_p. For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with $c \in p^{A_p}\mathbb{Z}_p$ and a a unit, use

$$\gamma = n^-(c/a)m(a)n(b/a).$$

The scalar is the diagonal theta factor times $\alpha_p(a)$. In passing to the classical transformation law one applies the finite operator at γ^{-1}, whose upper-left entry is d. Thus the classical Dirichlet character is $\eta_B(d) = \prod_p \alpha_p(d)$, not its inverse. Its value at -1 is one, as required by the weight-$3/2$ theta multiplier.

There are no additional level primes hidden in this argument. Outside the prescribed support, p is odd, B_p is split, and $R_p^0 = \mathfrak{sl}_2(\mathbb{Z}_p)$ is self-dual for the bilinear form of $Q_B = -a^2 - bc$. Its determinant is a unit. Hence the Fourier transform of $1_{R_p^0}$ equals itself; the unramified Weil constant is one, as also follows from [68, II.27] by integrating over this self-dual lattice. Upper unipotents, diagonal units, and the Weyl element fix it. These generate $\mathrm{SL}_2(\mathbb{Z}_p)$, so the exponent there is zero. At the finitely many prescribed primes the preceding A_p suffice. Conjugation by any $z \in \widehat{B}^\times$ is an isometry of Q_B, commutes with the Weil representation, and preserves the self-dual measure. The same level and character therefore work for every summand indexed by z.

[68, III, Theorem 6] applies to this Schwartz–Bruhat function on $V(\mathbb{A})$ and proves invariance of its rational theta sum. The normalized archimedean expression is $\sum_x f(x)e^{2\pi i\tau Q_B(x)}$: the factor $y^{3/4}$ in the Weil action is removed in passing to the classical form. It has holomorphic weight $3/2$. At a rational cusp, the slash operation replaces the finite Schwartz function by its finite Weil transform. Its support meets $V(\mathbb{Q})$ in a finite union of rational lattice cosets, with bounded denominators. Positive definiteness gives nonnegative rational exponents of bounded denominator and normal convergence for $\Im\tau > 0$. This proves holomorphy at every cusp. Theorem 6 alone asserts automorphic invariance; this last argument supplies the holomorphy conclusion used here.

Extraction. The support at every $p \mid m$ has $v_pQ_B = 1$. All nonzero indices are integral and divisible by m. Repeat the preceding construction for $Q'_B = Q_B/m$ with the same finite weights and the Gaussian for Q'_B. This quadratic form is integral on the support. Away from m it differs by a unit, so the good-prime self-duality argument remains valid. Its determinant squareclass is $m^{-3} \equiv m$, giving the extra Dirichlet character $(m/\cdot)$. Hence

$$g_M(\tau) = \sum_n H_M(mn)q^n \in M_{3/2}(\Gamma_0(K), \psi).$$

Here K has only the stated prime support, and

$$\psi = \eta_B(m/\cdot), \qquad \eta = \eta_B\chi_{h_*}, \qquad \psi/\eta = \chi_{-1}. \tag{O.5}$$

This directly constructs the extracted form without a half-integral-weight extraction operator. The inverse in this classical passage can also be checked before any character notation is introduced. Put

$g_\tau = n(x)m(\sqrt{y})$ for $\tau = x + iy$, so $g_\tau i = \tau$. For $\gamma \in \mathrm{SL}_2(\mathbb{Q})$ write $\gamma g_\tau = g_{\gamma\tau} k_\infty$ with $k_\infty \in \mathrm{SO}_2(\mathbb{R})$. In the adelic group,

$$(g_{\gamma\tau}, 1_f) = \gamma\,(g_\tau, \gamma_f^{-1})(k_\infty^{-1}, 1_f).$$

This equality also holds with consistent metaplectic lifts. The global rational lifts are compatible for the two forms: [68, II.30, Proposition 5] gives $\prod_v \gamma_v(Q) = \prod_v \gamma_v(aQ) = 1$ for rational Q, a, and hence the local normalizing factors in W_Q and $M_Q(a)$ have product one. The product formula also gives $\prod_v |\det(2A)|_v^{1/2} = 1$, so the product of the stated self-dual measures is the measure for which the rational lattice has covolume one. Apply the rational invariance of [68, III.41, Theorem 6] to the displayed adelic identity. The archimedean Gaussian for either positive ternary form has the same weight $3/2$; therefore its factor from k_∞^{-1} is the same as that for θ^3. The relative finite factor is evaluated at γ_f^{-1}, whose upper-left entry is d. Thus the relative classical character is $\prod_p \alpha_p(d)(d, \det Q)_p$, over the prescribed primes. This proves the inverse-idele convention directly. It compares the transformation identities and does not divide by a value of a theta function.

For the extracted form there is likewise no measure or degree normalization in its coefficients. With exactly the same finite weights, replace only the quadratic form and archimedean Gaussian by Q_B/m and $e^{-2\pi Q_B/m}$. The summand indexed by x now has exponent $Q_B(x)/m$. The original support makes $Q_B(x)$ an integer with valuation one at every prime dividing m, so these exponents are nonnegative integers. Coefficient n is exactly the original sum over $Q_B(x) = mn$, namely $H_M(mn)$. The self-dual measure changes by $|m|_v^{-3/2}$ locally, but a theta coefficient is a discrete sum and contains no Haar measure; this change is used only in the Fourier operators. Outside the original support it is a unit rescaling of the same unimodular local form, and at primes dividing m the lower-unipotent depth can be increased within the existing support. The determinant changes by m^{-3}, whose squareclass is m, so (O.5c) supplies exactly $(m/\cdot)$. In particular no new level prime, factor of m, or dyadic root of unity is introduced by extraction.

All-index recurrence and its phases. Let p avoid the full support and put $s = \eta_B(p)$. Locally use $R_p = M_2(\mathbb{Z}_p)$ and $V_p = \mathfrak{sl}_2$. The $p+1$ neighbors are indexed by lines of $\mathbb{F}_p^2$. For integral x with nonzero reduction and $Q_B(x) = n$, the number of preserved lines is $1 + (-n/p)$, including the nilpotent case. If $x \in pR_p^0$, all $p+1$ lines are preserved; the excess over the preceding formula is p.

A nonintegral x in a neighbor lies in $p^{-1}R_p^0$. Write $x = y/p$ with y primitive and $Q_B(y) = p^2 n$. Then $\bar{y}$ is nonzero nilpotent. In a basis with $\bar{y} = X_0$, write $y = \left(\begin{smallmatrix} a & b \\ c & -a \end{smallmatrix}\right)$. Here $p \mid a, c$, b is a unit, and $p^2 \mid -a^2 - bc$ forces $p^2 \mid c$. Exactly the neighbor belonging to the kernel line makes y/p integral. The integral and nonintegral parts therefore exhaust the sum with no genericity assumption on n.

Scaling by p at all other places multiplies the weight by s. Thus the primitive nonintegral contribution is $s^{-1}H_M(p^2n) - H_M(n)$, and the integral contribution is $(1+(-n/p))H_M(n) + psH_M(n/p^2)$. Their sum is

$$s^{-1}H_M(p^2n) + (-n/p)H_M(n) + psH_M(n/p^2).$$

Here and below a coefficient at a nonintegral index is zero. For completeness, if N_{ij} counts neighbor kernels from i to j and $w_i = |\mathrm{Aut}(i)|$, counting actual isogenies and dualizing gives $N_{ij}w_j = N_{ji}w_i$, or $N_{ij}/w_i = N_{ji}/w_j$. This proves the self-adjointness with the mass $1/w_i$ appearing in H_M. A neighbor changes the norm character by $\chi_{h_*}(p)$. Moving the neighbor operator onto P_M and using its actual modular-parametrization identity consequently gives

$$H_M(p^2n) + s(-n/p)H_M(n) + ps^2H_M(n/p^2) = s\chi_{h_*}(p)a_p(E)H_M(n) \pmod{2^M\mathcal{O}}.$$

This computation explains both powers of s even when s is not quadratic.

Put $a_M(n) = H_M(mn)$ and $\eta = \eta_B \chi_{h_*}$. Since $(-mn/p) = (m/p)(-n/p)$ and $\eta(p)^2 = s^2$, one obtains

$$a_M(p^2 n) = \{\eta(p) a_p(E) - \psi(p)(-n/p)\} a_M(n) - p\eta(p)^2 a_M(n/p^2) \pmod{2^M \mathcal{O}}. \tag{O.6}$$

The proof covers $p \mid n$ and $p^2 \mid n$ explicitly. At zero the coefficients vanish by the local support, and negative indices vanish by positive definiteness. An added finite coprimality or unit-squareclass sieve is realized by the corresponding local characteristic function for Q'_B; the same modularity proof applies. At primes outside the enlarged support the three indices $p^2 n, n, n/p^2$ have the same retain-or-zero status. Hence these sieves preserve the displayed recurrence. □

F.4 The unaveraged point comparison

The actual weighted coefficient comparison.

Proposition F.3 (The unaveraged weighted theta comparison)**.** *Put $m = |h_*|$ and $F = \mathbf{Q}(\sqrt{Dh_*})$, where D is a positive squarefree parameter in the fixed odd filter. Assume $(D/\ell_0) = 1$ at the chosen actual stage, so that ℓ_0 is inert in F. Keep the local Schwartz functions, the actual function $P_M(z) = \lambda_M(L\pi_{\ell_0}(z))$, and the integer c_0 of (O.4). Then a 2-power root of unity $\zeta_M(D)$ satisfies*

$$H_M(mD) = c_0\, 2^{\omega(N)} \zeta_M(D) \lambda_M(\overline{LP_D}) \pmod{2^M}. \tag{O.7}$$

The root of unity belongs to the field generated by the fixed characters at the primes of N. No class number is divided out.

Proof. The filter makes Dh_* an odd fundamental discriminant, with F split at every prime dividing $2N$. Its unit group is $\{\pm 1\}$. The condition $Q_B(x) = mD$ is equivalent to $x^2 = Dh_*$. Together with $(1+x)/2 \in R_{z,2}$ and the other integrality conditions, this says exactly that

$$\mathcal{O}_F \longrightarrow R_z, \qquad \sqrt{Dh_*} \longmapsto x$$

is an optimal embedding. Optimality follows because the source order is already maximal. All nonzero local weights on such an embedding are roots of unity. The valuation-one condition at each prime of h_* holds because $(D, h_*) = 1$; at ℓ_0 the residue of x is nonzero. The fixed unit filters hold by admissibility.

First classify the local embeddings. Away from $N\ell_0$ there is one local orbit: an $\mathcal{O}_{F,p}$-stable lattice is locally a fractional ideal, hence principal, and this also covers ramified primes dividing Dh_*. At $p^e \| N$, the algebra is split. The two integral idempotents decompose both lattices of the Eichler lattice pair. Its quotient is cyclic of order p^e, so its elementary exponents lie wholly in one summand. The two possibilities give exactly two orientations. This applies at two using the integral spectral projectors specified in the construction. At ℓ_0, the maximal order of the division algebra has residue field $\mathbf{F}_{\ell_0^2}$. The two residue-field embeddings of the unramified quadratic order give the two orientations. Consequently there are $2^{\omega(N)+1}$ local orientation tuples.

Fix one tuple. Choose a rational embedding $\iota : F \hookrightarrow B$ and write $x_0 = \iota(\sqrt{Dh_*})$. Such an embedding exists because F is a field at the two ramified places of B, namely ℓ_0 and infinity [82, Proposition 14.6.7, $(v) \Rightarrow (iii)$]. Choose z_0 realizing the selected integral local embeddings. The adelic representatives with this tuple are exactly

$$\widehat{F}^\times z_0 \widehat{R}^\times.$$

Their quotient by the rational centralizer $F^\times$ is

$$F^\times\backslash\widehat{F}^\times/(\widehat{F}^\times\cap z_0\widehat{R}^\times z_0^{-1}) = F^\times\backslash\widehat{F}^\times/\widehat{\mathcal{O}}_F^\times = \mathrm{Pic}(\mathcal{O}_F).$$

Thus this is an actual class-group torsor, not an average or a quotient of a class-group orbit.

For an ideal-class representative $a\in\widehat{F}^\times$, the element a centralizes x_0, and therefore

$$f((az_0)^{-1}x_0(az_0)) = f(z_0^{-1}x_0z_0).$$

On the other hand,

$$\chi_{h_*}(\mathrm{Nrd}(az_0)) = \chi_{h_*}(\mathrm{N}_{F/\mathbf{Q}}\,a)\chi_{h_*}(\mathrm{Nrd}\,z_0).$$

The first factor is exactly the genus character in (O.1). To see that it is well-defined on the class group, for $D>1$ the biquadratic field $\mathbf{Q}(\sqrt{D},\sqrt{h_*})$ has discriminant $D\,h_*\,(Dh_*) = (Dh_*)^2$. Its relative discriminant over F is therefore one. The character obtained by restricting χ_{h_*} to G_F is unramified and, under class field theory, is $a\mapsto\chi_{h_*}(\mathrm{N}_{F/\mathbf{Q}}\,a)$. For $D=1$ it is the trivial character. Ideal representatives can be chosen prime to h_*, giving exactly the elementary norm formula in (O.1). Reversing the Artin convention has no effect because this character is quadratic.

Here is the precise specialization of the reduction theorem, including its orbit and orientation data. In the notation of [62, §§3.1–3.3], take

$$F_{\mathrm{CJ}} = \mathbb{Q},\quad K_{\mathrm{CJ}} = F = \mathbb{Q}(\sqrt{Dh_*}),\quad B_{\mathrm{CJ}} = M_2(\mathbb{Q}),\quad S=\{\ell_0\}.$$

The field F is imaginary because $D>0$ and $h_*<0$. It embeds into $M_2(\mathbb{Q})$ by its regular representation. The source algebra has no finite ramification, so H.1 holds and the integer in H.2 is $1+0+1=2$. Moreover $\ell_0\nmid Dh_*N$ and

$$(Dh_*/\ell_0) = (D/\ell_0)(h_*/\ell_0) = 1\cdot(-1) = -1$$

by the present support condition and (O.2). Thus H.3 holds in its inert, rather than ramified, case. The resulting target algebra is the definite algebra B ramified precisely at ℓ_0 and infinity.

Take a source Eichler order $R^{\mathrm{src}} = R'\cap R''$ of level N. In [62, §3.2] the source subgroup is $H = \widehat{R^{\mathrm{src}}}^\times$ and the target subgroup is $H_S = \widehat{R}^\times$. Indeed, away from ℓ_0 the local orders are transported by the fixed algebra isomorphisms. At ℓ_0 the source subgroup is $\mathrm{GL}_2(\mathbb{Z}_{\ell_0})$, whose determinant is $\mathbb{Z}_{\ell_0}^\times$; its inverse image under the target reduced norm is exactly the unit group of the maximal division order. The target level is the prime-to-S part of N, which is N itself. This construction allows arbitrary prime-power exponents in N, including its exponent at two.

Both fine conductors are one. To check this assertion using the definition in [62, §3.3], an embedding of the maximal order $\mathcal{O}_F$ into $R'\cap R''$ embeds it into each maximal order. The intersection of either maximal order with F is an order containing $\mathcal{O}_F$, hence is $\mathcal{O}_F$. Conversely, if both intersections are $\mathcal{O}_F$, their intersection with $R'\cap R''$ is $\mathcal{O}_F$. The same argument applies on the target. Thus $c'=c''=c=c_S=(1)$ in that theorem. Its parity condition at ℓ_0 is $0-0\equiv v_{\ell_0}(N)=0 \pmod 2$. The local lattice pair there has distance zero. Since F_{ℓ_0} is the unramified quadratic extension, its maximal order fixes a unique vertex of the local tree. Consequently

$$N_{\ell_0}(0,0,0)=1,\qquad |\,\mathrm{Gal}(F[c]/F[c_S])| = 1,$$

which also follows directly from [62, Lemma 2.1(ii)]. The multiplicity κ in [62, Theorem 3.1] is therefore exactly one.

It is essential to retain the sign fiber in that theorem. At ℓ_0 the quotient

$$F_{\ell_0}^\times\backslash B_{\ell_0}^\times/R_{\ell_0}^\times \simeq \mathbb{Z}/2\mathbb{Z}$$

is measured by the valuation of the reduced norm modulo two: norms from $F_{\ell_0}^\times$ have even valuation, whereas a division uniformizer has norm valuation one. By [62, §3.6] this sign is precisely the choice of residue-field embedding of the unramified quadratic order, with origin fixed by the chosen reference embedding. Theorem 3.1 gives a bijection onto one such sign, not onto both. Away from ℓ_0 its map on local orbit data is a bijection, so we may also restrict to the chosen orientations at the primes of N. The resulting map is an equivariant bijection of the source class-group torsor onto the target torsor for that full orientation tuple. Corollary 3.2 gives the same bijection after fixing any supersingular pair in the target.

Choose the reduction place compatibly with the place used in (O.2). Sections 4.9–4.10 of [62] identify the projection of this adelic map with geometric reduction of the corresponding enhanced elliptic curve on $X_0(N)$. Their local formula is the identity under the chosen identifications away from ℓ_0, and at ℓ_0 replaces a matrix by a division-uniformizer power whose exponent is the valuation of its determinant. Thus it is the same sign and the same reduction map as above. Choose z_0 to be the image of the actual reference Heegner point x_D in (O.1). Writing

$$\chi_F([a]) = \chi_{h_*}(\mathrm{N}_{F/\mathbb{Q}}\, a), \qquad z_a = [az_0],$$

equivariance gives the exact finite-sum identity

$$\sum_{[a]\in \mathrm{Pic}(\mathcal{O}_F)} \chi_F([a]) P_M(z_a) = \lambda_M\left(\overline{L \sum_{[a]\in \mathrm{Pic}(\mathcal{O}_F)} \chi_F([a])\pi(x_D^{[a]})}\right) = \lambda_M(\overline{LP_D}) \quad \text{in } \mathbb{Z}/2^M\mathbb{Z}.$$

Here the supersingular pairs and their parametrization values are defined over $\mathbb{F}_{\ell_0^2}$, so every term is in the domain of the same λ_M. Additivity, specialization, and multiplication by the fixed integer L commute with this finite sum. If the Artin action is written with inverses, reindexing $[a] \mapsto [a]^{-1}$ leaves it unchanged since χ_F is quadratic. Changing the reference point within its torsor multiplies the sum only by χ_F of that reference class.

The weights are constant on each embedding orbit used in the following count. Indeed a unit of the definite order R_z has positive integral reduced norm with integral inverse, so its norm is one. The conjugation law for f therefore gives weight one for its action. Along the class-group torsor the Schwartz factor is constant because a centralizes x_0, and the remaining norm factor is exactly $\chi_F([a])$. Hence the finite-sum identity just proved applies with the single phase $\chi_{h_*}(\mathrm{Nrd}\, z_0) f(z_0^{-1} x_0 z_0)$.

For each fixed z, the stabilizer of an embedding in $R_z^\times$ is $\mathcal{O}_F^\times = \{\pm 1\}$. Each embedding orbit thus has $|R_z^\times|/2$ elements. The denominator $|R_z^\times|$ in (O.4) leaves the factor $1/2$ per embedding class. The vectors x and $-x$ are not identified by this stabilizer: at ℓ_0 their residues define the two different embeddings of the residue field, whereas conjugation by an order unit acts trivially on that field. They already lie in the two different ℓ_0 orientation tuples counted above. The contribution of the reference orientation tuple is

$$\frac{c_0}{2}\, \zeta_M(D) \lambda_M(\overline{LP_D}), \qquad \zeta_M(D) = \chi_{h_*}(\mathrm{Nrd}\, z_0) f(z_0^{-1} x_0 z_0),$$

up to the harmless quadratic reference-class factor just described. All nonzero factors in this displayed phase are fixed-character values or signs. Thus it is a 2-power root of unity; its field does not grow with D, M, or ℓ_0.

It remains to account for the other orientations; they are not all in the image of the one sign component of the reduction map. At $p^e \| N$, use the local Atkin–Lehner normalizer. It reverses the selected root, so multiplies the Schwartz weight by

$$\alpha_p(-1) = w_{p^e} \chi_{h_*}(p)^e.$$

It multiplies the norm-weighted parametrization value by the same sign: its reduced norm has valuation e, and the actual parametrization transforms by w_{p^e} after L kills the fixed cuspidal translation. The two signs multiply to one. This normalizer has trivial components at the other primes, so it changes no other orientation or local weight.

At ℓ_0, a division-algebra uniformizer exchanges the two residue embeddings. It sends the trace-zero residue β to $-\beta$, multiplying its weight by $(-1/\ell_0) = -1$. Its reduced norm has valuation one and hence contributes $\chi_{h_*}(\ell_0) = -1$. Its action on supersingular pairs is Frobenius. Since λ_M has Frobenius sign $+1$ and the parametrization is defined over $\mathbf{Q}$, this introduces no further sign. Again the two factors multiply to one. The uniformizer operation used here may be described with arithmetic or geometric Frobenius: they have the same action on the supersingular pairs and their parametrization values over $\mathbb{F}_{\ell_0^2}$. The identity $\lambda_M(FP) = \lambda_M(P)$ from (O.2) consequently applies to the very operation that exchanges the residue embeddings. Its norm has valuation one, and χ_{h_*} is unramified at ℓ_0, so its norm-character value is exactly $\chi_{h_*}(\ell_0) = -1$, independently of the uniformizer's unit part. Likewise, at $p^e \| N$, the norm of the Atkin–Lehner normalizer has valuation e and χ_{h_*} is unramified at p. Its norm-character value is exactly $\chi_{h_*}(p)^e$. These observations fix the norm signs in both orientation changes.

It follows that every one of the $2^{\omega(N)+1}$ tuples has the same contribution as the reference tuple. Multiplying by their number gives precisely $c_0 2^{\omega(N)}$, as asserted. This counts individual embeddings with their stabilizer; a conjugate-pair formulation of the geometric correspondence must not be used to insert or remove another factor of two.

Finally, $c_0/|R_z^\times|$ is an integer for every z. Thus a change of the chosen integer lift of $P_M(z)$ by 2^M changes the actual coefficient by an element of $2^M\mathcal{O}$. Every preceding equality is consequently valid at the original retained precision. No denominator depending on the class number, the discriminant, or the stage has entered.

Precision in the orientation count. The factor $1/2$ in the orbit count does not require dividing a congruence modulo 2^M by two. Indeed, $-1 \in R_z^\times$ implies $2 \mid |R_z^\times|$, and the chosen integer c_0 is divisible by $|R_z^\times|$. Hence $c_0/2$ is an integer. Choose lifts of all $P_M(z)$ before summing. Each orbit of embeddings contributes $(c_0/|R_z^\times|)(|R_z^\times|/2)P_M(z) = (c_0/2)P_M(z)$, with its prescribed unit weight. Replacing any lift by a multiple of 2^M changes this expression by a multiple of 2^M. The comparison with the reduced point may therefore be made at precision M, separately for each orientation. Summing the $2^{\omega(N)+1}$ orientations gives the stated multiplier $c_0 2^{\omega(N)}$ without any loss of precision. In particular, write $c_\theta = 9 + \omega(N)$, the valuation of the fixed integer $c_0 2^{\omega(N)}$. At the fixed witness D_w, (O.2) and (O.7) imply

$$v_2 a_M(D_w) = c_\theta + c_w \qquad \text{whenever } M > c_\theta + c_w.$$

The phase is a unit and the congruence error has strictly larger valuation, so this is an equality, not merely a lower bound. Discarding the finitely many earlier stages is sufficient. Thus the visibility used in choosing the odd minimum retains a fixed witness and a fixed valuation; neither the class number nor the changing prime ℓ_0 introduces a new loss of precision.

□

Proposition F.4 (Visibility and the native genus index)**.** *Use the actual odd form and unaveraged point of Propositions F.2 and F.3. There is one constant C_{det}, independent of the actual stage and admissible D, for which (O.8) holds. Explicitly, for a known simple-zero D, $v_2 a_M(D) \geq \min\{M, j_{\mathrm{gen}}(D) - C_{\mathrm{det}}\}$. For every admissible D, the strict inequality $v_2 a_M(D) < M$ implies $\mathrm{ord}_{s=1} L(E^D, s) = 1$ and $j_{\mathrm{gen}}(D) \leq v_2 a_M(D) + C_{\mathrm{det}}$. If an admissible odd-sign D is not a simple*

zero, its coefficient vanishes modulo 2^M. No genus index of a torsion point is defined or used in this assertion.

Proof. For a known simple-zero D, write $P_D = m_D G + T$ in its native rank-one free lattice. The same L kills T. An integral finite-field Kummer map cannot decrease $v_2(m_D)$, so (O.7) gives the lower bound by the native index. The rational free line has index at most two in that native lattice, by conjugation and norm. Absorb this and the fixed initial convention in C_{det}. The first inequality below holds for a known simple-zero D; the visibility implication holds for every admissible D:

$$\begin{aligned} v_2 a_M(D) &\geq \min\{M, j_{\mathrm{gen}}(D) - C_{\mathrm{det}}\}, \\ v_2 a_M(D) < M \implies \operatorname{ord}_{s=1} L(E^D, s) &= 1, \quad j_{\mathrm{gen}}(D) \leq v_2 a_M(D) + C_{\mathrm{det}}. \end{aligned} \tag{O.8}$$

If an odd-sign D is not a simple zero, the height formula gives a torsion genus point, killed by L, and its coefficient is zero modulo 2^M. Combined with the first inequality on simple-zero parameters, this gives the all-D truncated bound used below, without assigning a rank-one lattice index to a torsion point. □

F.5 Integer divisions and actual trace operators

Proposition F.5 (Arithmetic realization of the coefficient rules)**.** *Use the rational full-two-torsion graph conventions of Definitions 7.4 and 7.5. For an even source of Proposition E.4, use its fixed minimum depth. For an odd source, first fix a finite operator budget $|U| \leq W$, take $R \geq 2$, and choose an actual stage satisfying the depth inequalities of Proposition G.3. Then the integer divisions (O.9)–(O.11) are valid at every Fourier index, with the stated truncation for the odd source. The fresh coefficient-one reading is $\overline{(t(\mathrm{Fr}_q)\theta H_U)[1]} = \mathcal{A}(Uq)$. On all tested nonempty configurations, these actual symbols depend only on their typed graph and satisfy the rules of Lemma 7.6: a pair difference contracts the endpoints and adds their types and every external incidence; whole-prime omission uses the same source, phase, depth and stage; and an isolated primary vertex of neutral type and neutral external incidences makes the symbol zero. Every contraction has nonempty output. Stationary character tests are carried along in the same product rule. This statement asserts neither values at untested odd-stage weights nor an identity at an empty contraction output.*

Proof. All new primes below avoid the full coefficient support S, including every late sieve and the current ℓ_0. Here S is the actual coefficient support. The coefficients of g vanish at all indices not prime to S. For the odd source this is forced at $2, N, h_*, \ell_0$ by the local Schwartz support and the extraction at $|h_*|$. At every additional coefficient-sieve prime it is imposed by an exact zero-or-retain Fourier sieve, which commutes with the good recurrence outside the enlarged support. This is the vanishing used in the condition $(m, S) = 1$ in the unary series below. Later stationary prime-selection tests may enrich the common label alphabet without resieving the intrinsic odd source; such tests do not enlarge this coefficient-support condition. Use the minimum-depth inequalities proved in Appendix G. The general bookkeeping assigns weight one to a split prime of $\mathbb{Q}(E[2])$ and one-half to a prime with transposition Frobenius, called simple. In our actual full-two-torsion family every prime is split, $w(U) = |U|$, and the depth δ is an integer. Thus only matching parity, meaning $e_U = 0$, is needed here; the displayed general formula also records where the unused simple-prime term would enter. For a finite set U, put

$$n_U = \lceil \delta + w(U) \rceil, \quad e_U = n_U - \delta - w(U), \qquad B_U = 2^{-n_U} \prod_{p \in U} (U_p - \eta(p) V_p) g,$$

where $U_p(\sum a(n) q^n) = \sum a(pn) q^n$ and $V_p g(\tau) = g(p\tau)$. These are operations on the actual form g. We verify the level at a new odd prime using the classical theta multiplier. Write the old level as

$4A$, with $p \nmid 4A$, and put $\beta = \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$. For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(4Ap)$ one has

$$\beta\gamma\beta^{-1} = \begin{pmatrix} a & pb \\ c/p & d \end{pmatrix} \in \Gamma_0(4A).$$

The square-root factor in the theta multiplier is unchanged on evaluating the latter matrix at $p\tau$. The quotient of its arithmetic factor by that of γ is $(c/p\,/d)/(c/d) = (p/d)$. Cubing the multiplier leaves this quadratic factor unchanged. Hence $V_p g$ has level $4Ap$ and character $\psi(p/\cdot)$.

At this level the normalized sum

$$U_{p^2} h(\tau) = \frac{1}{p^2} \sum_{r \bmod p^2} h((\tau + r)/p^2)$$

preserves the same weight and character. Indeed, put $\alpha_r = \left(\begin{smallmatrix} 1 & r \\ 0 & p^2 \end{smallmatrix}\right)$. For $\gamma \in \Gamma_0(4Ap)$ choose $r' \equiv (b + rd)/(a + rc) \pmod{p^2}$. Then

$$\alpha_r \gamma = \gamma'_r \alpha_{r'}, \qquad \gamma'_r = \begin{pmatrix} a + rc & (b + rd - (a + rc)r')/p^2 \\ p^2 c & d - cr' \end{pmatrix} \in \Gamma_0(4Ap).$$

The map $r \mapsto r'$ is a permutation, since $a + rc$ is a unit modulo p and the fractional linear map has determinant one. Writing $d' = d - cr'$, its nebentypus factor is unchanged since $d' \equiv d \pmod{4Ap}$. Its theta arithmetic factor is unchanged as well: $d' \equiv d \pmod{c}$, $d' \equiv d \pmod 4$, and $(p^2 c/d') = (c/d') = (c/d)$. These are the usual Kronecker-symbol identities for the theta multiplier on $\Gamma_0(4)$. Its square-root factor, evaluated at $(\tau + r')/p^2$, is $(c\tau + d)^{1/2}$. Thus the sum transforms with the same theta multiplier cubed and the same character. The formulas also hold for the triangular matrices, and holomorphy at the cusps is preserved by the finite rational translates. On Fourier expansions $U_{p^2} V_p = U_p$. Consequently both terms of $U_p - \eta(p) V_p$ have level exponent at most one at p and character $\psi(p/\cdot)$. Iteration introduces no higher exponent at previously added primes.

The same computation applies to the unary corrections. Start with the odd-character unary theta series at the old support, including its fixed coprimality sieve. Its J-series is its transform by V_J, for squarefree $J \subset U$. A prime $p \mid J$ therefore has exponent at most one and the extra character $(p/\cdot)$; a prime $p \nmid J$ is absent from its original level. Multiplication by the ordinary theta series introduces no odd level prime. This proves the precise level and oldform alternatives used in (O.12).

Write an index as Jrm^2 with $J \subset U$ and r squarefree, prime to U. Products of prime sets are understood here. The originating squarefree parameter in the coefficient of g is $D = (U \setminus J)r$, not the squarefree part Jr of the tested index. Remove the phase $\eta(p)^s$ in (O.6) and put

$$b_{-1} = 0, \quad b_0 = 1, \quad b_1 = a_p - \kappa, \quad b_{s+1} = a_p b_s - p b_{s-1}, \quad \kappa = (\psi/\eta)(p)(-D/p).$$

For $p \in U \setminus J$ the factor $b_s - b_{s-1}$ is odd; for $p \in J$ the factor $b_{s+1} - b_s$ is even. Consequently, for the even source,

$$v_2 B_U[Jrm^2] \geq w(r) + \tfrac{1}{2}\#\{p \in J : p \text{ is simple}\} - e_U. \tag{O.9}$$

The right side is an integer on the allowed filter and is nonnegative. In our full-two-torsion application all primes are split and $w = \omega$. For the odd source, fix an integer $R \geq 1$ and a weight bound W first. At one sufficiently advanced actual stage, simultaneously for all U with $w(U) \leq W$ and every Fourier index, (O.9) holds with its right side replaced by its minimum with R. Thus the

untruncated bound on a prescribed bounded range of squarefree weights follows by choosing R and then the stage. For the unary-removal and two-trace tests below, choose $R \geq 2$ after fixing the operator budget. For a split prime $a_p \equiv p+1 \pmod 4$. The recurrence gives the same residue on each divided shell,

$$(b_{s+1} - b_s)/2 \equiv c_p(D) := \{a_p - (\psi/\eta)(p)(-D/p) - 1\}/2 \pmod 2. \tag{O.10}$$

The two integral shell calculations. Here every prime is split in the rational two-torsion field, so $a_p \equiv p+1 \pmod 4$. For $p \mid D$ one has $\kappa = 0$. Reducing the recurrence modulo two gives $b_{s+1} = b_{s-1}$, with $b_0 = 1$ and $b_1 = 0$. Thus $b_s - b_{s-1}$ is odd for all $s \geq 0$, including $s = 0$ with $b_{-1} = 0$. For $p \nmid D$, one has $\kappa \in \{1, -1\}$, so every b_s is odd. Write $d_s = b_{s+1} - b_s$. The recurrence modulo four gives $d_s \equiv p d_{s-1} \pmod 4$ for $s \geq 1$. Each d_s is even, and multiplication of an even residue modulo four by an odd number fixes that residue. Consequently

$$d_s \equiv d_0 = a_p - \kappa - 1 \pmod 4 \qquad (s \geq 0).$$

This proves (O.10) without dividing a residue-field identity by two. For an index Jrm^2, each $p \in J$ therefore supplies an actual factor two, whereas the factors for $p \in U \setminus J$ are integral units modulo two. The squarefree input has $\omega(D) = |U| - |J| + \omega(r)$. Its depth bound plus these $|J|$ factors gives $\delta + \omega(D) + |J| = n_U + \omega(r)$ before division by 2^{n_U}. All removed character factors are units. For the odd source the same argument is a congruence modulo $2^M \mathcal{O}_{\mathrm{coef}}$ before that division. It therefore retains precision $M - n_U$ afterward; the subsequent division in (O.11) retains precision $M - n_U - 1$. Choose $i \in \{0, 1\}$ with $\eta\chi_{-4}^i$ odd and let

$$\Theta_J = \sum_{\substack{m>0 \\ (m,S)=1}} \eta(m)\chi_{-4}(m)^i m q^{Jm^2}.$$

Poisson summation gives this genuine unary weight-3/2 form. Its scalar character is $\eta\chi_{-4}^i(-J/\cdot)$ and every $p \mid J$ enters the level at exponent one. In matching parity the shell recurrence gives $B_U[Jm^2] - B_U[J]\eta(m) \in 2\mathcal{O}_{\mathrm{coef}}$. Thus

$$H_U = \frac{B_U - \sum_J B_U[J]\Theta_J}{2} \tag{O.11}$$

is integral; in opposite parity take $H_U = B_U$. This is division by the integer two, not reduction modulo the maximal ideal of a ramified coefficient ring. For odd-stage forms, the coarse bound and bounded-weight stabilization proved below pay every fixed denominator before choosing the stage.

Write $\mathfrak{m}_{\mathrm{coef}}$ for the maximal ideal of the chosen coefficient valuation ring $\mathcal{O}_{\mathrm{coef}}$. In the following trace argument a reduced expansion means coefficientwise reduction modulo $\mathfrak{m}_{\mathrm{coef}}$. The preceding divisions by integer powers of two have already been justified in $2^a\mathcal{O}_{\mathrm{coef}}$; reduction modulo the maximal ideal alone would not justify them. No saturation of the integral Fourier module is assumed, and no identification of its abstract quotient with all reduced expansions is needed. Multiply (O.11) and the undivided corrections by $\theta = 1 + 2\sum_{m>0} q^{m^2}$. They are actual weight-two forms. For a fixed stage and U, choose a finite complete coefficient-field extension containing all constituents and retain their old multiplicities and diamond translates in the finite-dimensional characteristic-zero space. Use modular Galois representations on cuspidal constituents and the two character summands on Eisenstein constituents. Holomorphy alone does not exclude the latter. Diamond translates remain integral: their correction coefficients have the form

$$\tfrac{1}{2}B_U[J]\{\psi_U(u) - \eta(u)\chi_{-4}(u)^i(-J/u)\},$$

whose character ratio is quadratic. Good Hecke operations preserve integrality by $(T_q f)[n] = f[qn] + q(\langle q\rangle f)[n/q]$. Finitely many Fourier coordinates inject this space into a finite coordinate space. The module generated by these translates is therefore finite over the coefficient DVR, and closed. Chebotarev and continuity make scalar trace and determinant operators limits of good Hecke and diamond operations, so they preserve this module. No integral eigenspace projector is used.

Reading an added vertex. For the actual full-two-torsion family, all primes are split, $\delta \in \mathbb{Z}$, and $n_U = \delta + |U|$. Put $F_U = \theta H_U$ and let $q \notin S \cup U$ be a fresh prime, where S includes every late sieve and the current moving theta prime. Write $u = \prod_{p\in U} p$ and let $\mathcal{A}(D)$ denote the normalized valuation symbol on the tested range: it is one exactly when $v_2 a(D) = \delta + \omega(D)$ on the filter. For an odd source, that range has a fixed weight bound chosen before the valid actual stage, as specified before (F.7). No finite-stage value at larger weights is asserted; the compatible limit tables used later are a separate object. Expanding the operators, every term containing a V_p has zero coefficient at q, and no unary index Jm^2 with $J \subset U$ equals q. Therefore

$$B_U[q] = 2^{-n_U} a(uq), \qquad H_U[q] = 2^{-n_U-1} a(uq).$$

At coefficient one the diamond term in the good Hecke formula vanishes. Since $F_U \equiv H_U \pmod{2\mathcal{O}_{\mathrm{coef}}}$,

$$\overline{(t(\mathrm{Fr}_q)F_U)[1]} = \overline{H_U[q]} = \overline{2^{-\delta-|U|-1}a(uq)} = \mathcal{A}(uq) \in \mathbb{F}_2.$$

Here residues lie in the coefficient residue field, with $\mathbb{F}_2$ its prime subfield. Every permitted phase is a two-power root of unity and reduces to one. For an odd source, fix the finite operator budget $|U| \le W$ and $R \ge 2$ first; the stage choice in (O.9) may be taken with $M_\theta \ge \max(0, \delta + W) + R$. Thus $M_\theta - n_U - 1 \ge 1$, so the normalized finite-stage error is still divisible by the integer two. The same reading holds with value zero off the fixed or moving coefficient support.

The diamond correction above is a sum of integral multiples of $\theta\Theta_J$; its fresh coefficient q is zero after reduction. It introduces neither an extra phase nor a denominator in this reading. All readings and subsequent omissions use this same g, η, ψ, δ and actual stage; only the set U changes.

For any fixed Fourier index n, Chebotarev approximation lets one test a finite product of trace operators using distinct fresh primes $q_1, \dots, q_a$ avoiding n and the level. The good Hecke formula then reads the coefficient at $nq_1 \cdots q_a$ of the original vector. Different indices may use different fresh primes; the conclusion is an identity of coefficientwise reduced expansions.

The determinant identity uses the actual diamond formula, not merely the reduction of its eigenvalues. For a good prime r, $d(\mathrm{Fr}_r) = r\langle r\rangle$. The formula above shows that $(d(\mathrm{Fr}_r) - 1)F_U$ reduces to a sum of integral multiples of the undivided corrections $\theta B_U[J]\Theta_J$: the scalar $r\psi_U(r) - 1$ lies in $\mathfrak{m}_{\mathrm{coef}}$. A fresh trace kills every such correction. Indeed, to test at n choose $q \nmid nJ$; then nq cannot have squarefree part J, and $\theta \equiv 1 \pmod{\mathfrak{m}_{\mathrm{coef}}}$. This remains true after further fresh traces. Continuity and the closed integral module therefore give, on the reduced trace orbit of F_U,

$$d(u)t(v) = t(v), \qquad t(u)t(v) = t(uv) + t(uv^{-1}), \qquad t(1) = 0.$$

The middle equality follows from the characteristic-zero matrix identity $\mathrm{tr}(A)\,\mathrm{tr}(B) = \mathrm{tr}(AB) + \det(B)\,\mathrm{tr}(AB^{-1})$, using the determinant identity just proved; the last follows from $\mathrm{tr}(1) = 2$.

Two fresh primes outside $S \cup U$ contribute two to the squarefree weight in (O.9). Thus $B_U[nq_1q_2]$ is divisible by four, no unary correction occurs there, and $H_U[nq_1q_2]$ remains divisible by two. Here $R \ge 2$ and the actual-stage budget were fixed before the test. It follows that $t(g)t(h)F_U = 0$ after reduction. Applying the middle identity to $t(gh)t(h)$ gives $t(gh^2)F_U = t(g)F_U$. Invariance

under all squares, and continuity, give factorization through the elementary quadratic quotient unramified outside $S \cup U$. Over $\mathbb{Q}$ this quotient is finite: its squareclasses are generated by -1, 2, and the primes in that support. This does not assert abelianity of the characteristic-zero Galois representations.

At $p \in U$, first keep the summands of F_U separately. The block θB_U and each unary block with $p \in J$ have level exponent at most one and primitive quadratic character at p. The unary blocks with $p \notin J$ are degeneracies of forms at a level omitting p. This distinction is preserved when retaining all old multiplicities. In particular, the latter blocks cannot acquire a Steinberg constituent merely by being viewed at a larger level.

Enlarge the characteristic-zero coefficient field to realize all the cuspidal newform constituents and the Eisenstein character pairs of these blocks. On a constituent and all its degeneracies, define $t(g)$ and $d(g)$ to be the scalar trace and determinant of its two-dimensional Galois representation, acting as the identity on its multiplicity space. Distinct characteristic-zero constituents with the same good eigenvalues have the same trace and determinant by Chebotarev, so these definitions agree on coinciding summands. In particular they give well-defined operators on their sum. No decomposition modulo two, and no residual irreducibility assertion, enters this construction.

For a cuspidal eigenform f, let Π_f denote its unitary adelic normalization: at a good prime q its Satake parameters α_q, β_q satisfy $a_q = \sqrt{q}(\alpha_q + \beta_q)$ and $\alpha_q \beta_q = \chi(q)$. Apply [66, §§0.3–0.7, Theorem A, pp. 409–411] to $\Pi_f^\vee$, and set $\rho_f = \sigma(\Pi_f^\vee)^\vee$. Here $F = \mathbb{Q}$, the real weight is 2, and the coefficient place lies above 2. The degree of F is odd, so the additional finite discrete-series-place hypothesis for even degree does not apply. The local prime p is odd. Carayol's reciprocity sends geometric Frobenius to a uniformizer, and his ω_p is $|\cdot|_p$. His normalized-induction formula is

$$\sigma(\mathrm{Ind}(\xi_1, \xi_2)) = \xi_1^{-1} \omega_p^{-1/2} \oplus \xi_2^{-1} \omega_p^{-1/2}.$$

Thus if $\Pi_{f,p} = \mathrm{Ind}(\mu_1, \mu_2)$, ρ_f on arithmetic Frobenius has eigenvalues $\sqrt{p}\,\mu_1(p)$, $\sqrt{p}\,\mu_2(p)$ whenever these characters are unramified. In particular its good polynomial is $X^2 - a_q X + q\chi(q)$. Both the automorphic contragredient and the dual on the Galois side are part of this convention.

For a block with ramified character, its local conductor is exactly one: it is at most the level exponent and at least the conductor of its determinant. Casselman's newvector theorem and conductor calculation [65, Theorem 1, pp. 302–306] show that a special representation has conductor one only for an unramified twist; a ramified twist has conductor at least two. A supercuspidal representation has conductor at least two. Consequently $\Pi_{f,p} = \mathrm{Ind}(\mu_1, \mu_2)$ with μ_1 unramified and μ_2 of conductor one, after interchanging the characters if necessary. Its newvector space is a line. On inertia the ramified character is the prescribed quadratic character.

We check the Fourier normalization of the operator on this line. Use normalized induction, so that

$$v\left(\begin{pmatrix} a & x \\ 0 & d \end{pmatrix} g\right) = \mu_1(a)\mu_2(d)|a/d|_p^{1/2} v(g).$$

Put $K_0(p) = \{g \in \mathrm{GL}_2(\mathbb{Z}_p) : g_{21} \in p\mathbb{Z}_p\}$. Use its character $k \mapsto \mu_2(k_{22})$. This is the lower-right-character convention. Casselman's upper-left-character convention is carried to it by $h_p = \left(\begin{smallmatrix} 0 & 1 \\ -p & 0 \end{smallmatrix}\right)$, which normalizes $K_0(p)$ and interchanges its diagonal entries under conjugation. We label the inducing characters so that μ_1 is unramified, using the interchange isomorphism for irreducible principal series. Normalize Haar measure by $\mathrm{vol}(K_0(p)) = 1$. The character-weighted double-coset operator for $K_0(p)\,\mathrm{diag}(p, 1)K_0(p)$, with value one at $\mathrm{diag}(p, 1)$, is

$$H_p = \sum_{b \in \mathbb{Z}/p\mathbb{Z}} \Pi_{f,p}\left(\begin{pmatrix} p & b \\ 0 & 1 \end{pmatrix}\right).$$

Take the newvector supported on $BK_0(p)$ with $v(1) = 1$. Evaluating at 1 gives

$$(H_p v)(1) = p\mu_1(p)p^{-1/2} = \sqrt{p}\,\mu_1(p).$$

This is the Fourier operator $U_p^{(2)}$. Indeed, in the Whittaker model use an additive character of conductor $\mathbb{Z}_p$. Writing $a(n) = \mathrm{diag}(n, 1)$ and $n(x) = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)$,

$$a(n)\begin{pmatrix} p & b \\ 0 & 1 \end{pmatrix} = n(nb)a(np) \quad (n \in \mathbb{Z}_p), \qquad (H_p W)(a(n)) = pW(a(np)).$$

Normalize the archimedean Whittaker function by $W_\infty(a(y)) = y\exp(-2\pi y)$ for $y > 0$. Comparison of its positive-n term with $yf(iy)$ gives the finite Whittaker coefficient a_n/n. The displayed equality therefore sends a_n to a_{pn}. Hence $U_p^{(2)} = \sqrt{p}\,\mu_1(p)$ on the new line, exactly the arithmetic-Frobenius eigenvalue on the unramified Galois line. This scalar is algebraic; the square roots only express the unitary normalization. Old multiplicities at primes other than p carry the same scalar. Casselman's oldspace dimension formula [65, Corollary, p. 306, and Theorem 4, p. 313] does not introduce an additional multiplicity at this conductor-one prime. For Eisenstein constituents, the two Galois characters and the divisor-sum Fourier coefficients give the same equality directly; Carayol's cuspidal theorem is not applied to them.

Choose an inertia element τ on which the quadratic character is -1, and a Frobenius lift ϕ. On a ramified block, a basis of the two character lines gives

$$\rho(\tau) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \rho(\phi) = \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}, \qquad U_p^{(2)} = \lambda, \quad d(\phi) = \lambda\mu.$$

For $\rho(g) = \left(\begin{smallmatrix} a & b \\ c & e \end{smallmatrix}\right)$, the two inertia differences are

$$t(g\tau) - t(g) = -2e, \qquad t(g\tau\phi) - t(g\phi) = -2e\mu.$$

It follows in characteristic zero that

$$U_p^{(2)}\{t(g\tau\phi) - t(g\phi)\} = d(\phi)\{t(g\tau) - t(g)\}. \tag{O.12}$$

On a block coming from level prime to p, the representation is unramified and both differences vanish. This proves the identity on its entire old multiplicity space even if $U_p^{(2)}$ is not scalar or semisimple on that space. It therefore proves (O.12) on the actual sum containing F_U.

These are identities on characteristic-zero forms before any reduction. To relate them to the integral trace module above, choose finitely many Fourier coordinates that inject its ambient space. Every generator of that module has integral coordinates; hence the module is a submodule of a finite free module over the complete coefficient DVR. It is finite and closed. Simultaneous Chebotarev approximation in the finitely many Galois representations makes $t(g)$ and $d(g)$ limits of good Hecke and diamond operators on this space. They preserve this integral module, so the identity can be evaluated on the integral F_U and then reduced. This does not require integral projectors onto constituents. In particular no division by the two in the displayed inertia differences is used.

Apply (O.12) at coefficient one. Put $V = U \setminus \{p\}$, $v = \prod_{r\in V} r$, so $n_U = n_V + 1$. For a fresh prime r set

$$C_p(vr) = \frac{a_p - (\psi/\eta)(p)(-vr/p) - 1}{2} \in \mathbb{Z}, \qquad c_p(vr) = C_p(vr) \bmod 2.$$

The two surviving operator terms at pr and the good recurrence give

$$H_U[pr] \equiv \eta(p) C_p(vr)\, 2^{-n_V-1} a(vr) \pmod{2\mathcal{O}_{\mathrm{coef}}}.$$

For the even source this is an equality before reduction. For the odd source the error is in $2^{M_\theta - n_U - 1}\mathcal{O}_{\mathrm{coef}}$, which is contained in $2\mathcal{O}_{\mathrm{coef}}$ at our chosen stage. Choose fresh q, q_τ, r_0, r_1 representing respectively $g, g\tau, g\phi, g\tau\phi$ in one sufficiently precise joint quotient for the U and V forms and all retained characters. The omitted form uses the same g, η, ψ, δ and stage, and its level omits the entire rational prime p. Thus the two V symbols at vr_0, vr_1 agree. Their Legendre signs at p are opposite, so $c_p(vr_0) + c_p(vr_1) = 1$ in $\mathbb{F}_2$. The coefficient-one reading and the reduced trace identity now give

$$\mathcal{A}_U(Uq_\tau) + \mathcal{A}_U(Uq) = \mathcal{A}_V(Vr_0) \qquad \text{in } \mathbb{F}_2.$$

Here the subscript records the operator set, not a new coefficient source or minimum. The factor $\eta(p)$ reduces to one; no factor two is cancelled in (O.12). For every retained quadratic character χ unramified at p, the representative of the omitted vertex satisfies

$$\chi(g\phi) = \chi(g)\chi(\phi) = \chi(\mathrm{Fr}_q)\chi(\mathrm{Fr}_p).$$

For a fixed-support character this is the product of the two vertex labels. For the character of a third vertex $r \in V$, it adds the two bits (r/q) and (r/p) in $\mathbb{F}_2$. Reciprocity then gives the reverse bits from the same endpoint types. All additional stationary tests were included in the joint quotient, so they obey this formula as well. This proves the full product-label omission rule, including every external incidence.

A fresh vertex with trivial label in the elementary extension for $S \cup U$ has the same reduced reading as $t(1)$, hence gives zero. An isolated primary vertex has precisely this fixed label and these zero incidence bits. A contracted cluster of primary vertices has the same property when its external incidences are neutral. The result is still a nonempty configuration, including the one-vertex case; no contraction deletes the last vertex.

Finally the fixed-U factorization gives dependence on the entire finite typed graph. For two disjoint prime realizations of the same graph, choose a fresh bridge copy with the prescribed types and matching incidences to both realizations. CRT and Dirichlet provide the primes successively, and reciprocity determines their reverse bits. Replace the vertices of the first copy by bridge vertices one at a time, then the bridge vertices by those of the second copy. At each step the other vertices form the same fixed U, and the changed vertex has the same elementary Frobenius data for $S \cup U$; its symbol therefore agrees. If the original realizations overlap, first replace one by a fresh copy avoiding both original prime sets, with incidences matching that first realization. A vertex and its own replacement are never tested together, so their mutual bit can be chosen freely. The resulting two disjoint realizations admit the preceding bridge argument without conflicting prescriptions. Only graphs of the original bounded size are evaluated; extra bridge incidences are prime-selection conditions. Thus these are rules for the actual arithmetic graph symbols, at the already chosen valid stage. The simultaneous finite graph budget and stabilization are treated below. □

G Nonvanishing, normalization and the final family

The uses of the coefficient and address propositions have the following order. First apply only their even-source case, whose depth is already fixed by Proposition E.4. This gives the rough companion and the genus lower bound. These in turn establish the odd minimum and its finite-stage inequalities. Only then apply the odd case of Proposition F.5 and the two-symbol address

construction. In particular the proof of the rough companion does not use the later odd minimum. We give the choice order underlying (3.3)–(3.9). This also separates the two valuation arguments: the even upper transfer uses the common Kato scalar, while the final contradiction uses the common Heegner coordinate of Section 9.

G.1 An absolute bound of five new primes

The theorem of Hoffstein–Luo [57, p. 436] supplies fundamental discriminants with nonzero central value and at most four prime factors. Its finite splitting set is arbitrary. The bound four is absolute; no uniform size threshold for those discriminants is needed.

Proposition G.1 (Nonvanishing with five primes and finite local prescriptions)**.** *Let $A/\mathbb{Q}$ be a fixed modular elliptic curve for this application, let S contain $2, 3$ and all its bad primes, and let B be a finite prime exclusion set. In every nonempty signed unit-squareclass filter at S of even root number, there is a signed odd squarefree parameter u, prime to $S \cup B$, with $L(A^u, 1) \neq 0$ and $\omega(u) \leq 5$. Only the intermediate parameter supplied by Hoffstein–Luo is required to be fundamental; the final field discriminant may be $4u$. The number five is independent of the prescribed finite set. No uniform bound on the sizes of the selected primes is asserted.*

Proof. Let $A/\mathbb{Q}$ be any fixed modular elliptic curve for this application, S contain two, three and all its bad primes, and B be any finite exclusion set. Fix a nonempty signed unit filter of even root number. By CRT and Dirichlet choose one signed prime q_0 in its classes, outside $S \cup B$. Apply the theorem to the actual curve A^{q_0}, putting

$$S' = S \cup B \cup \{|q_0|\} \cup \{p : p \mid N_{A^{q_0}}\}$$

in its splitting set. The theorem gives a fundamental d with $\chi_d(p) = 1$ for all $p \in S'$ and at most four prime factors. The functional equation forces $d > 0$, since A^{q_0} has even root number and every conductor prime splits. Splitting at two gives $d \equiv 1 \pmod 8$. Thus $q_0 d$ has all original unit classes, is prime to B, is squarefree, and has at most five prime factors. If $q_0 d \equiv 3$ or $7 \pmod 8$, its field discriminant is $4q_0 d$; only d was required to be fundamental in the theorem. This is compatible with the coefficient construction (E.2). □

Here h is restricted to the prepared positive odd-sign unit filter containing one, as in (3.3), so $h \equiv 1 \pmod 8$. The multiset of dummy labels and its size are fixed before h; its realizing primes are chosen after h and avoid its support.

For the rough companion the curve itself varies: write a for the bounded dummy product and take $A = E^{ha}$. Choose q_0 so $k_0 = aq_0$ is negative, 1 modulo eight and a square at every prime of $2Nh$, with the required exclusions. Then k_0 is odd fundamental, prime to N_{E^h}, and

$$w(E^{hk_0}) = w(E^h)\chi_{k_0}(-N_{E^h}) = +1.$$

This proves the sign directly for the actual application. Apply Hoffstein–Luo to E^{hk_0} with every conductor, old, h, dummy and forbidden prime in its splitting set. The resulting $k' = k_0 d$ adds at most five new primes. The constants and sizes in the analytic theorem may depend on this varying curve and the whole finite set; the bound five does not.

G.2 The even upper transfer

Proposition G.2 (A uniformly bounded partitionable companion)**.** *Fix E and the finitely many prepared even sources, local types and unit seeds used for (3.3). There are constants A, C_E, fixed*

before h and before any additional finite prime exclusion set, with the following property. Let h be positive squarefree in the prepared odd-sign unit filter containing 1, prime to its fixed support; in particular $h \equiv 1 \pmod 8$. There is a negative odd fundamental discriminant k', prime to $2Nh$ and the additional exclusions, square at every place dividing $2Nh$, for which hk' is partitionable in the sense of Definition 7.8, $L_*(hk') \neq 0$, and

$$\omega(k') \le A, \qquad v_2 L_*(hk') \le 2\omega(h) + 2\omega(k') + s(hk') + C_E.$$

Here A is the fixed dummy-label count plus five. Dummy types and their number are fixed first, but their distinct prime realizations are selected after h and the exclusions. The proof uses the one-symbol stationary-address conclusion of Lemma 7.11, whose proof is given in Proposition G.6.

Proof. For one fixed even source write (E.11) as

$$v_2 L_*(\sigma D) = 2v_2 a(D) + \kappa_e.$$

The lower bound (E.9) and a nonzero reference coefficient imply that the nonempty set of integers $v_2 a(D) - \omega(D)$ has a finite attained minimum δ_e. A unit of the normalized symbol means precisely equality to this minimum. It therefore gives

$$L_*(\sigma D) \neq 0, \qquad v_2 L_*(\sigma D) = 2\omega(D) + C_{e,\mathrm{unit}}, \quad C_{e,\mathrm{unit}} = 2\delta_e + \kappa_e. \tag{F.1}$$

This constant belongs to the fixed source, not to a curve later introduced solely to obtain nonvanishing.

For the negative types used in the rough companion, the unsigned index is 7 modulo eight. Its total $(-1/D)$ label is nonzero, so every unit seed is nonempty. A seed of least positive size has no dependence on internal edge bits: a nonzero edge derivative would be a smaller nonempty unit. Its labels therefore suffice. If a list with the same total label contains the first $r-1$ seed anchors and a nonempty remainder, contract that remainder into the last anchor. Its total label is forced to be the remaining seed label. This proves partitionability regardless of how many old h-primes are in the remainder.

For a partitionable d already known to have Selmer corank zero, the one-symbol address construction keeps every prime of d stationary and gives arbitrarily large cubes d_x, $d_0 = d$, with unit symbol at each $x \neq 0$. The scalar in (E.5) is common to that cube. By (E.8) and (F.1),

$$v_2 u_x^{\mathrm{ev}}(0) \le B_e := C_{e,\mathrm{unit}} + C_{\mathrm{loc}} \quad (x \neq 0).$$

Choose $M - c_E > B_e$. The constant coefficient of $2^{c_E} u^{\mathrm{ev}} \in \Lambda[G_b]$ has sign-value bits of total Boolean degree at most $2^{M-1} - 1$. For a larger cube dimension, the Boolean comparison in Section 9 gives $x \neq 0$ with $v_2(u_x^{\mathrm{ev}}(0) - u_0^{\mathrm{ev}}(0)) \ge M - c_E$. If the base were zero or had valuation greater than B_e, this would contradict the upper bound at x. Therefore

$$v_2 L_*(d) - 2\omega(d) - s(d) \le C_{e,\mathrm{unit}} + 2C_{\mathrm{loc}}. \tag{F.2}$$

The required dimension may depend on d; this bound does not. It precedes the odd minimum and uses no Heegner transfer.

Prepare these sources and nonempty seeds for every total type at the fixed support. There are finitely many. A bounded supply of dummy anchors and reserves works for all of them. For each h realize that supply by distinct primes avoiding the entire fixed support and h. Put every dummy and every other already fixed forbidden prime in the exclusion set of the preceding nonvanishing construction. It gives k' with $\omega(k') \le A$, where A is the fixed dummy count plus five. For the

total type that actually occurs, hk' is partitionable. Nonvanishing gives corank zero by the forward theorem, so (F.2) applies. Taking the maximum over the finitely many prepared constants proves

$$v_2 L_*(hk') \le 2\omega(h) + 2\omega(k') + s(hk') + C_E. \tag{F.3}$$

No new even minimum is chosen after h or its exclusion set. Those excluded primes are not added as relaxed places in a Selmer complex. □

G.3 The odd minimum is chosen after its lower bound

Proposition G.3 (The odd minimum and every finite operator budget)**.** *Fix the original odd filter, the detecting sequence of Proposition F.1, and one nonprincipal ultrafilter on its stages. The arithmetic lower bound (3.5), the native comparisons (3.4), and Proposition G.2 give (F.4)–(F.5) before any odd minimum is chosen. The bounded-weight visible depths therefore have an attained finite integer minimum δ_o. For every fixed weight cutoff, the simultaneous inequality (F.6) holds on a set in this same ultrafilter. For every finite operator budget and $R \ge 2$, one may choose a stage where all the integer divisions and residual errors required in Proposition F.5 are valid at every Fourier index. On a fixed squarefree-output range the unit symbol defined before (F.7) is exhaustive, and its value* 1 *implies (F.7), with the one constant $C_o = \delta_o + C_{\mathrm{det}}$. No minimum is changed on omitting a vertex or adding stationary tests, and no single finite stage is claimed to serve every unbounded weight range.*

Proof. The arithmetic result (3.5) and the two native-lattice height comparisons (3.4) give, using (F.3),

$$2j_{\mathrm{gen}}(h) \ge 2\omega(h) + s(h) - C_{\mathrm{gen}}, \qquad C_{\mathrm{gen}} = 2A + C_R + C_E - v_2 L_*(h_*) + C_H. \tag{F.4}$$

The variable $s(hk')$ cancels. This estimate precedes the theta prime, precision, odd minimum, final field and binary dimension.

Combining (F.4) with (O.8) gives on all admissible D

$$v_2 a_M(D) \ge \min\{M, \omega(D) - C_{\mathrm{coarse}}\}, \qquad C_{\mathrm{coarse}} = C_{\mathrm{gen}}/2 + C_{\mathrm{det}}. \tag{F.5}$$

At visible coefficients (O.8) first gives a simple zero, allowing (F.4); at the other odd-sign parameters the coefficient is zero modulo 2^M. This avoids a circular use of the future minimum.

Fix one nonprincipal ultrafilter on the actual stages with $M = M_\theta \to \infty$. Consider the integers r for which a bounded-weight sequence of visible D_M has $v_2 a_M(D_M) - \omega(D_M) = r$ on a set in the ultrafilter. The fixed witness D_w and (O.2),(O.7) make this set nonempty. Equation (F.5) bounds it below. Let its attained minimum be δ_o. For every fixed W, on a set in the same ultrafilter,

$$v_2 a_M(D) \ge \omega(D) + \delta_o \quad \text{for every visible } D,\ \omega(D) \le W. \tag{F.6}$$

Otherwise choose a violating D_M at each bad stage. One of the finitely many integers between $-C_{\mathrm{coarse}}$ and $\delta_o - 1$ would occur on an ultrafilter-large set, a contradiction. Invisible coefficients satisfy the required truncated inequality once $M > W + \delta_o$.

For a fixed finite operator budget $|U| \le W_U$, put $N_0 = \max(0, \delta_o + W_U)$ and fix $R \ge 2$. Choose an integer $T \ge W_U + 1$ with $T - C_{\mathrm{coarse}} \ge N_0 + R$, then choose a stage in the ultrafilter-large set where (F.6) holds through weight T, with $M \ge N_0 + R$ and $M > T + \delta_o$. For the index Jrm^2 put $D = (U \setminus J)r$. The integral shell recurrence has an error in $2^M\mathcal{O}$. If $\omega(D) > T$, (F.5) gives

$$v_2\big(2^{n_U} B_U[Jrm^2]\big) \ge N_0 + R.$$

If $\omega(D) \le T$, (F.6) for visible coefficients and $v_2 a_M(D) \ge M$ for invisible ones instead give

$$v_2(2^{n_U} B_U[Jrm^2]) \ge \min\{M, n_U + \omega(r)\}.$$

Consequently $v_2 B_U[Jrm^2] \ge \min\{R, \omega(r)\}$ for every Fourier index and every U in the budget at this one stage. At $r = 1$ the normalized originating coefficient $2^{|J|-n_U} a_M(U \setminus J)$ is integral, so (O.10) proves the unary congruence modulo the integer two. The additional division in (O.11) leaves error valuation at least $R - 1 \ge 1$. In particular $M - n_U - 1 \ge 1$ for every reading and first-shell omission in the budget. It does not select one stage for infinitely many budgets.

More precisely, fix a squarefree-output cutoff L. On a set of stages in the same ultrafilter, (F.6) holds through L and $M_\theta > \delta_o + L$. Define $\mathcal{A}_{M_\theta,L}(D)$ only for positive squarefree D with $\omega(D) \le L$: it is zero off the full coefficient filter, one when $v_2 a_{M_\theta}(D) = \delta_o + \omega(D)$ on that filter, and zero otherwise. This is exhaustive on the tested range: visible coefficients satisfy (F.6), while invisible ones have valuation at least $M_\theta > \delta_o + \omega(D)$. In particular a unit in this range is visible and satisfies

$$\operatorname{ord}_{s=1} L(E^D, s) = 1, \qquad j_{\mathrm{gen}}(D) \le \omega(D) + C_o, \quad C_o = \delta_o + C_{\mathrm{det}}. \tag{F.7}$$

For the preceding operator budget take $L \ge W_U + 1$ and include its valid-stage requirements; all fresh-vertex and omission outputs then lie in this range. The constant C_o is independent of L and of the stage. No value is assigned by this definition at unbounded weights of one finite stage. No minimum is changed when a vertex is omitted or a stationary condition is later added. □

G.4 Nonempty rational seed and a common finite stage

Proposition G.4 (Nonempty seeds after fixing the complementary twist)**.** *Choose the negative even unit k with the fresh ramification and exclusions specified in Section 7, and fix $K = \mathbb{Q}(\sqrt{k})$ before the address dimension. Then E/K has full two-primary Selmer corank one. Transport the even coefficient source to E^k, normalize at index 1, and use its positive square-unit filter. On that filter its coefficients have $v_2 a_{\mathrm{new}}(D) \ge \omega(D)$, and the source has a nonempty K-split unit seed. If the intrinsic odd limit symbol is a unit at the empty configuration, E already has a simple zero. Otherwise it too has a nonempty unit seed. In this remaining case both seeds can be read through the actual common alphabet, whose projections onto the two intrinsic alphabets are separately surjective. No surjectivity onto their product is required.*

Proof. Choose the final negative even unit k as in Section 7, with fresh ramification outside the old odd label and torsion fields. Fix $K = \mathbb{Q}(\sqrt{k})$ before the binary dimension. The forward theorem gives corank zero for E^k; the sum-and-difference isogeny, on compact Selmer modules tensored with $\mathbb{Q}_2$, gives full corank one for E/K. The intrinsic odd symbol omits the primes of k. Later ℓ_0 avoids them. Fresh ramification implies that every old odd label still lifts to a K-split label. Additional stationary tests can have relations; use their actual common alphabet and its separate surjective projections.

Transport to the fixed complementary twist. Put $d_k = |k|$, so that $k = -d_k < 0$. The old source has sign -1; its Fourier coefficient at the positive index d tests E^{-d}. Choose the new source for E^k with sign $+1$, normalize its coefficient at 1 to be 1, and impose a support containing the old support and all primes dividing k. For the following comparison D is positive and squarefree, is prime to that support, and lies in the square unit classes of 1 required by the new filter. In particular D is a square in every old support completion, so $d_k D$ and d_k lie in the same old filter.

Waldspurger comparison on the two sources gives

$$\begin{aligned} v_2 a_{\mathrm{new}}(D) &= v_2 a_{\mathrm{old}}(d_k D) - v_2 a_{\mathrm{old}}(d_k) \\ &\geq \omega(D). \end{aligned}$$

The equality is a valuation equality; the two sources may have different root-of-unity phases. The normalized central-value ratios agree, since both are

$$\sqrt{D}\,\frac{L(E^{kD},1)}{L(E^{k},1)}.$$

Fixed period constants cancel in each ratio. The old unit condition at d_k, and $\omega(d_k D) = \omega(d_k) + \omega(D)$, prove the inequality. The signed twist is kD throughout; no Fourier coefficient at a negative index is used. The final family already satisfies these unit classes. In the nonempty-seed argument the fresh prime p' is chosen to match all fixed labels of p, so pp' satisfies the same filter, including its dyadic condition.

For a unit of this normalized source equality holds, so $v_2 L_*(h_x k) \leq 2\omega(h_x) + C_{e,\mathrm{final}}$ with one fixed period-normalization constant. Its K-split seed must be nonempty. Suppose otherwise. The integral coefficient valuations then improve the preceding lower bound to $\omega(D) + 1$ on every nonempty product of allowed K-split primes. For a fresh good K-split prime p outside the support, apply the trace identity to $f = \theta H_{\{p\}}$:

$$U_p^2 t(\phi) = U_p^3 + d(\phi) U_p, \qquad (t(\phi) f)[p^2] = f[p^3] + (d(\phi) f)[p].$$

Here and in this calculation all congruences are modulo the ideal $2\mathcal{O}_{\mathrm{coef}}$ before passage to the residue field. Write $a_p = a_p(E^{(k)})$. We check every theta convolution term in the three coefficients. Put

$$B = B_{\{p\}} = \frac{(U_p - \eta(p) V_p) g}{2}, \qquad b_1 = B[1] = \frac{a(p)}{2}, \quad b_p = B[p],$$
$$H = H_{\{p\}} = \frac{B - b_1 \Theta_1 - b_p \Theta_{\{p\}}}{2}.$$

The inherited depth is zero, and H is integral by (O.11). Also $H[0] = 0$, since the cuspidal source and the unary series have zero constant term. The ordinary lower bound gives $b_1 \in \mathcal{O}_{\mathrm{coef}}$. Under the assumed absence of a nonempty K-split unit seed one has the stronger assertion

$$b_1 \in 2\mathcal{O}_{\mathrm{coef}}. \tag{G.4a}$$

Indeed, if p lies in the coefficient filter, the improved bound is $v_2 a(p) \geq 2$; otherwise the actual filter gives $a(p) = 0$. No condition that p itself belong to that filter is imposed.

Let $\delta_B, \delta_1, \delta_p$ be the scalars by which $d(\phi)$ acts on the full weight-two blocks $\theta B, \theta\Theta_1, \theta\Theta_{\{p\}}$, respectively. They are integral units. The determinant is the cyclotomic character times the weight-two nebentypus on every cuspidal constituent and on the Eisenstein character pair, independently of old multiplicity. For this positive-twist source $\psi/\eta = \chi_{-1}$ and $i = 0$. The half-integral-weight characters of B and $\Theta_{\{p\}}$ are therefore both

$$\psi(p/\cdot) = \eta(-p/\cdot).$$

Multiplication by the same θ preserves this equality. Consequently $\delta_B = \delta_p$ for the actual Frobenius lift ϕ, including its action on the ramified character. The ratio of the characters of $\theta\Theta_{\{p\}}$ and $\theta\Theta_1$ is quadratic. Their cyclotomic determinant factors are identical, so $\delta_p/\delta_1 \in \{1, -1\}$ and

$\delta_p - \delta_1 \in 2\mathcal{O}_{\mathrm{coef}}$. We must not simply equate these two scalars. On the actual characteristic-zero module the exact identity is

$$d(\phi)f = \delta_p f + \frac{(\delta_p - \delta_1)b_1}{2}\,\theta\Theta_1.$$

Now $H[p] = 0$, since $B[p] = b_p$, $\Theta_{\{p\}}[p] = 1$ and $\Theta_1[p] = 0$. Integrality of H implies $f[p] \in 2\mathcal{O}_{\mathrm{coef}}$. Also $\theta\Theta_1[p] = 2\sum_{0<j^2<p} \Theta_1[p - j^2]$ is even. The displayed determinant identity therefore already proves $(d(\phi)f)[p] \in 2\mathcal{O}_{\mathrm{coef}}$ using only the ordinary integrality of b_1. The following direct expansion checks the individual cross terms under the no-seed assumption in force here.

For $n_j = p - j^2 > 0$, one has $p \nmid n_j$ and $\Theta_{\{p\}}[n_j] = 0$. We claim that

$$B[n_j] \in 2\mathcal{O}_{\mathrm{coef}}. \tag{G.4b}$$

If n_j meets the old support, the coprimality sieve makes this coefficient zero. In particular $2 \in S$, so every odd j gives an even n_j and vanishes in this way. Otherwise write $n_j = rm^2$, with r squarefree. Since $p \nmid n_j$, the V_p term is zero and $B[n_j] = a(prm^2)/2$. The integral good-prime shell recurrences express $a(prm^2)$ as an integral multiple, up to unit character factors, of $a(pr)$. If $r > 1$, the ordinary coefficient lower bound gives $v_2 a(pr) \geq \omega(pr) = 1 + \omega(r) \geq 2$, or the coefficient is zero off the filter. This uses no assertion that the primes dividing r split in K. If $r = 1$, the same shell recurrences express $B[m^2]$ as an integral multiple of b_1, and (G.4a) applies. This proves (G.4b). Also every coefficient of $b_1\Theta_1$ is even by (G.4a).

Now expand $\theta = 1 + 2\sum_{j>0} q^{j^2}$ before discarding any term. Linearity of the determinant operator on the three actual blocks gives the exact expression

$$\begin{aligned}(d(\phi)f)[p] = {} & \frac{\delta_B B[p] - \delta_p b_p}{2} \\ & + \sum_{0<j^2<p} (\delta_B B[p - j^2] - \delta_1 b_1 \Theta_1[p - j^2] - \delta_p b_p \Theta_{\{p\}}[p - j^2]).\end{aligned} \tag{G.4c}$$

There is no constant-theta $\Theta_1[p]$ term since p is not a square, and $\Theta_{\{p\}}[p] = 1$. The first line is zero because $B[p] = b_p$ and $\delta_B = \delta_p$. Every summand in the second line is even by (G.4a)–(G.4b) and the vanishing of its last term. In particular representations $p = j^2 + m^2$, which can occur when $p \equiv 1 \pmod 4$, are covered by the case $r = 1$ and are not omitted. Thus $(d(\phi)f)[p] \in 2\mathcal{O}_{\mathrm{coef}}$.

The other two readings have no determinant operator. Since H is integral, for every positive integer N,

$$f[N] = H[N] + 2\sum_{0<j^2\leq N} H[N - j^2] \equiv H[N] \pmod{2\mathcal{O}_{\mathrm{coef}}}. \tag{G.4d}$$

This includes $N = p^3$ and every term in its theta convolution. Choose a fresh K-split prime p' matching ϕ in a sufficiently precise finite Galois quotient for the retained integral module, and matching all fixed rational characters. The good Hecke formula at $p' \neq p$ identifies $(t(\phi)f)[p^2]$ modulo two with $f[p^2p']$, hence with $H[p^2p']$ by (G.4d). Neither unary series has support at p^2p', and the recurrence at $p \mid pp'$ has $\kappa = 0$. Consequently

$$H[p^2p'] = \frac{a(p^3p') - \eta(p)a(pp')}{4} = \frac{\eta(p)(a_p - 1)a(pp')}{4}.$$

The product pp' lies in the square filter even if p alone does not. Its two primes split in K, so the no-seed assumption gives $v_2 a(pp') \geq 3$. This last coefficient is therefore even. The trace identity and (G.4c)–(G.4d) now imply $H[p^3] \in 2\mathcal{O}_{\mathrm{coef}}$. The following computation uses precisely this unary-subtracted coefficient.

The coefficient behind the mod-eight contradiction. Write $a_p = a_p(E^k)$. The transported source tests positive twists of E^k. Thus $\psi/\eta = \chi_{-1}$ and, at the squarefree index 1, $\kappa = (\psi/\eta)(p)(-1/p) = 1$. Also η is odd, so the unary correction uses $i = 0$. With $a(1) = 1$ the recurrence gives

$$a(p^2) = \eta(p)(a_p - 1), \qquad a(p^4) = \eta(p)^2\{a_p(a_p - 1) - p\}.$$

The inherited depth is zero. Hence

$$B_{\{p\}}[p] = \frac{\eta(p)(a_p - 2)}{2}, \qquad B_{\{p\}}[p^3] = \frac{\eta(p)^2((a_p - 1)^2 - p)}{2}.$$

Only $\Theta_{\{p\}}$ contributes at p^3, with coefficient $\eta(p)p$. Therefore the actual integer division gives

$$H_{\{p\}}[p^3] = \frac{\eta(p)^2(a_p - 1)(a_p - p - 1)}{4}.$$

This expression is integral because full rational two-torsion gives $a_p \equiv p + 1 \pmod 4$. The positive-twist character convention is essential in this computation. Under the assumed absence of a nonempty unit seed, the trace identity makes this coefficient zero in the residue field. Since its phase is a unit and $a_p - 1$ is odd, it follows that $a_p \equiv p + 1 \pmod 8$. The contradiction is obtained only after this conditional deduction. Choose p also with $(k/p) = 1$, $p \equiv 1 \pmod 4$ and $(\ell/p) = -1$, outside the finite support. These prescriptions are compatible: K has a ramified prime outside the ramification of $\mathbf{Q}(i, \sqrt{\ell})$, so $K \cap \mathbf{Q}(i, \sqrt{\ell}) = \mathbf{Q}$. The corresponding quadratic residue classes are therefore nonempty and Dirichlet's theorem supplies infinitely many such primes. Since k is a square modulo p, the reductions of $E^{(k)}$ and E_ℓ are isomorphic. A half of a nonzero two-torsion point with root e_i on a split cubic requires both differences $e_i - e_j$ to be squares. For the roots $0, \ell, -3\ell$, one of these required differences is, respectively, $-\ell, \ell, -4\ell$. All three are nonsquares under the chosen conditions. Thus none of the three nonzero points of order two has a half over $\mathbf{F}_p$. If the two-primary subgroup had order greater than four, its structure as a finite abelian group with exactly four elements killed by two would give an element of order four, and hence such a half. Its order is consequently exactly four. This gives $v_2(p + 1 - a_p) = 2$, contradicting the congruence modulo eight above. This proves the required nonempty even seed. If the intrinsic odd symbol already has a unit at the empty configuration, its detection formula gives a simple zero for E and the converse is finished. Otherwise take a least nonempty odd unit seed. Thus the remaining two-symbol address argument has two nonempty seeds. □

Proposition G.5 (Compatible finite graph tables on one ultrafilter)**.** *Use δ_o and the ultrafilter of Proposition G.3. First fix the abstract type of the bounded elementary label groups and their projection data on a set in this same ultrafilter. For every finite cutoff W, the valid odd stages give finitely many possible symbol tables on the compatible typed graphs with at most $W+1$ vertices. One table occurs on a set in that ultrafilter. These selected tables agree on their common restrictions, and their union defines the intrinsic odd symbol on all compatible finite typed graphs. It retains its bounded-weight witness and the nonempty-output coefficient rules. After fixing k, its stationary tests may be included in a common alphabet with separately surjective projections on the same ultrafilter, without changing δ_o. Every specified finite collection of tests has a common actual stage. No stage common to all cutoffs is selected.*

Proof. Here the finite graph argument has a definite arithmetic order. The moving theta prime contributes only a bounded number of quadratic label coordinates. The root characters at the old

level were fixed before M_θ; growing Kummer precision introduces no growing-order character on the variable primes. Thus the elementary label groups, their projections and their finite addition tables have bounded size. Choose their abstract type on the same ultrafilter as δ_o. For each fixed weight cutoff W, restrict first to the ultrafilter-large set of stages where the preceding budget and precision conditions hold for every $|U| \le W$, with $R = 2$. On those stages the coefficient reading, factorization, contraction and isolation rules hold on all configurations of at most $W + 1$ vertices, with the stated nonempty-output restriction. They therefore define actual symbol truth tables on the compatible typed graph data. There are finitely many such tables; select the one occurring on a set in the same ultrafilter. Two selected tables agree on their common restriction by taking the finite intersection of their valid-stage sets. Their union defines the odd limit symbol on compatible finite typed graphs of every size; it is not a total valuation symbol at any one finite stage. Every seed or contraction identity uses only finitely many such graphs and is inherited at a cutoff containing all of them. The bounded-weight witness remains a witness in these tables. No stage common to all cutoffs is selected. After fixing k, include its tests and choose the common projection data on the same filter, without redefining the intrinsic odd minimum. Stabilize the finite alphabet and its two projections in this way. □

Proposition G.6 (Stationary and simultaneous address constructions)**.** *Use the definitions and realization freedoms in Definitions 7.4, 7.5, 7.10 and Lemma 7.7, and the coefficient rules of Proposition F.5. Fix $b \ge 1$. Either take one symbol on a nonempty set of always active terminals with a nonzero polynomial in their pair bits, or take one or two symbols on terminals with nonzero linear activations, including a terminal of every nonzero activation, such that each symbol separately has a nonzero terminal polynomial at every $x \ne 0$. In the latter case the symbols may read different compatible projections of the common alphabet and fixed stationary external data. Require the active data to satisfy each prescribed filter at every nonzero address. Starting from fixed compatible typed terminal and stationary data, one can adjoin finitely many primary auxiliaries with nonzero linear activations and choose only incidences involving them so that all the symbols equal 1 at every $x \ne 0$. All original data stay fixed, no auxiliary is present at zero, and the required neutral external incidences and splitting conditions are retained. Only nonempty configurations and nonempty contraction outputs are used. A nonzero iterated difference, hence a partitionability certificate, suffices for the required nonzero terminal polynomial. For the two final symbols the zero-total packets below supply these terminal hypotheses using only separate projection surjectivity. For the even stationary application, the original prime representatives may be fixed at the outset. For a stabilized odd symbol, first fix the abstract data and the finite budget, then choose an actual stage carrying them, and only then choose prime representatives. Previously fixed representatives may be retained when compatible with the full type at that stage; arbitrary earlier primes are not asserted to satisfy a later moving condition.*

Proof.

Total labels of the final seeds. Both final symbols have projected total label zero. The intrinsic odd filter contains 1: its fixed unit squareclasses, positive sign, and moving condition $(D/\ell_0) = 1$ are those of 1. The even source for E^k is likewise normalized in the unit filter containing 1. Every nonzero coefficient, and hence every unit seed, lies in its respective filter. Additional stationary tests at primes dividing k refine the common alphabet but are forgotten by the intrinsic odd projection. They do not add conditions to that odd symbol.

Take repeated packets with total zero in the common elementary label group, including sufficiently many copies of every label. For either projection separately, keep the first $r - 1$ anchors of

its unit seed and contract the nonempty remainder into the last anchor. Since the total projected label is zero, the remainder has the last seed label. This argument uses surjectivity of each projection separately. It does not require that a fixed old odd seed admit lifts whose total is zero in the common alphabet. The whole packet already has that total, and contractions preserve it. Thus the final twists satisfy the stationary tests as well as both coefficient filters. Concretely, choose a packet with an even number of copies of every common label, and enough copies to provide the anchors of either seed under its respective projection. Its total label is zero. For every nonzero activation λ, take a disjoint copy of this whole packet and give all its vertices activation λ. At every $x \neq 0$ there are exactly 2^{b-1} active packets. The active list is therefore nonempty, has total common label zero, and contains enough anchors for each projected seed separately. Its terminal polynomial for each symbol is nonzero by the contraction certificate, regardless of the initially fixed bits. It also contains a terminal of each required activation. This constructs the terminal input of the address lemma explicitly. No simultaneous lifting of a prescribed pair of seed labels is needed. Every added network vertex is primary, so the total common label remains zero at each address.

Network operators and their degree. We spell out the simultaneous argument, including the product over all addresses. Write ∂_e for the finite difference in a terminal pair bit e. Products of these operators commute and $\partial_e^2 = 0$. The contraction rule identifies a forest product with its contracted graph. A cycle product is zero: after a spanning path is contracted its last edge is internal and changes neither a type nor any external incidence. No contraction removes the last vertex.

A network of primary auxiliary vertices acts on any of the symbols by an operator in the ∂_e. To see this directly, expand in the bits incident to the auxiliary vertices. Contract the chosen forest edges. A component with no terminal becomes an isolated primary vertex, and its contribution vanishes by primary isolation. A component meeting several terminals gives their forest difference. If Q is the grounded auxiliary Laplacian and c_a is its attachment column at terminal a, the degree-zero and degree-one terms are

$$\det(Q)\,1 + \sum_{a<b} c_a^{\mathsf{t}} \operatorname{adj}(Q) c_b\, \partial_{ab}.$$

All other terms have difference degree at least two. The displayed coefficients are the rooted-forest minors, by Cauchy–Binet on the incidence matrix. They are polynomial identities, so apply to singular Q as well. Distinct networks have no edges between their auxiliary vertices. Their operators multiply: expanding both forests gives their union, and unions containing a terminal cycle give zero by the preceding observation. This also shows that the operator is the same for both projected symbols.

Fix a target $x_* \neq 0$. Choose a basis $\lambda_1, \dots, \lambda_b$ of the dual address space with $\lambda_i(x_*) = 1$ for all i. Such a basis is obtained from one functional taking value one by adding it to a basis of the space of functionals vanishing at x_*. Group i has three primary vertices, each activated by λ_i. In its physical three-dimensional coordinate space let $e_i = (1,1,1)$ and choose u_i, v_i spanning its orthogonal complement. The vector e_i has norm one. Prescribe the symmetric matrix Q by its bilinear matrix I_b on the e_i and

$$H(T,B) = \begin{pmatrix} 0 & T \\ T^{\mathsf{t}} & B \end{pmatrix}$$

on the remaining coordinates. In physical coordinates each $h_i = Qe_i = e_i$ is supported on its own group. Attach this column to a terminal activated by λ_i, which exists by hypothesis. For an active

set J the sum of these active attachment columns is $Q_J\mathbf{1}_J$. Thus the prescribed principal matrix Q_J really is the grounded graph Laplacian on that address, including its diagonal entries.

For a selected terminal pair a, b active at x_*, let their activations be α, β. Attach an extra column c, dual to u_1, to both. If $\alpha \neq \beta$, attach the same column to a terminal of activation $\alpha + \beta$; it is absent at x_*. The extra columns sum to zero at every address since $\alpha + \beta + (\alpha + \beta) = 0$. If $\alpha = \beta$, the two columns already cancel. Hence they do not alter Q_J.

For the singular network set $T = I + P$, where P is the cyclic shift, and give B one diagonal entry one. On every proper group subset, the restricted cyclic shift is a union of paths, so T_J is invertible and $\det Q_J = \det(T_J)^2 = 1$. On all groups, the kernels of T and T^{t} are generated by the all-one vector. The one nonzero diagonal entry of B forces the lower component of a vector in $\ker H$ to vanish. Thus $\ker Q$ is precisely the line generated by $r = \sum_i u_i$, and $\operatorname{adj}(Q) = rr^{\mathsf{t}}$. The anchor columns pair to zero with r, while $c^{\mathsf{t}}r = 1$. Only a, b receive active extra columns at x_*. Consequently its network operator $S_{x_*,e}$, for $e = ab$, has the expansions

$$S_{x_*,e}|_{x_*} = \partial_e + \text{terms of degree at least two}, \qquad S_{x_*,e}|_x = 1 + \text{terms of positive degree} \quad (x \neq x_*).$$

For $b = 1$, $T = 0$ and $B = (1)$ give exactly the same calculation.

For a varying network set $T = I + L$, with L the consecutive lower shift, and $B_\varepsilon = \varepsilon E_{bb}$ for $\varepsilon \in \{0, 1\}$. Every T_J is invertible and both operators $V^{\varepsilon}_{x_*,e}|_x$ have constant coefficient one. The upper-left block of H^{-1} is $T^{-\mathsf{t}}B_\varepsilon T^{-1}$. Its change on the selected columns is $((T_J^{-1})_{b1})^2$, interpreted as zero when an endpoint is absent. This is one exactly when the entire path is present, which means $x = x_*$. Anchor columns have zero change. Therefore

$$\begin{aligned}(V^0_{x_*,e} + V^1_{x_*,e})|_{x_*} &= \partial_e + \text{terms of degree at least two},\\ (V^0_{x_*,e} + V^1_{x_*,e})|_x &= \text{terms of degree at least two} \quad (x \neq x_*).\end{aligned}$$

One choice for both symbols at every nonzero address. For each $x \neq 0$ let $P_{1,x}, P_{2,x}$ be the two nonzero terminal polynomials, of degrees $d_{1,x}, d_{2,x}$, and let $H_{s,x}$ be their highest homogeneous parts. Work in the ordinary polynomial ring in the terminal pair variables for this address. Since it is a domain, $H_{1,x}H_{2,x} \neq 0$. Choose a coefficient-one monomial

$$M_x = \prod_e X_e^{n_{x,e}}, \qquad n_{x,e} \in \{0, 1, 2\}, \qquad \sum_e n_{x,e} = d_{1,x} + d_{2,x}.$$

For $n_{x,e} = 2$ install one singular network $S_{x,e}$; it acts on both symbol factors. For $n_{x,e} = 1$ install one varying network $V^{\varepsilon_{x,e}}_{x,e}$. Use disjoint auxiliary sets for all these networks. Let $\Gamma_y(\varepsilon)$ be the resulting graph at address y, and consider the finite sum in $\mathbf{F}_2$

$$\sum_{\varepsilon} \prod_{y \neq 0} \prod_{s=1}^{2} A_s(\Gamma_y(\varepsilon)). \tag{G.1a}$$

The total degree available to all terminal polynomials is $D = \sum_{y \neq 0}(d_{1,y} + d_{2,y})$.

Each singular network at x has zero constant at x, so its action costs at least one difference in each of the two factors there, hence at least two in (G.1a). For a varying network, summing over its ε cancels the all-constant term. Its first-degree terms are independent of ε away from its target, and their difference at the target is ∂_e. Thus every surviving term costs at least one difference, and a cost-one term acts at its target on exactly one of the two factors. The sum of all these minimum costs is exactly D. Any higher-degree action, or any additional action at a wrong address, exceeds D and annihilates at least one terminal polynomial. Hence only the minimum-cost terms survive.

At address x, a variable with exponent two is therefore differentiated once in each factor; a variable with exponent one is differentiated in exactly one factor, summed over the two possibilities. The resulting scalar is precisely $[M_x](H_{1,x}H_{2,x}) = 1$. Lower-degree parts cannot contribute. Taking the product over all $x \neq 0$ proves that (G.1a) equals one. At least one choice of the finitely many ε consequently makes every factor in that product one. This is one graph for both symbols at every nonzero address. Boolean reduction of $H_{1,x}H_{2,x}$ before selecting M_x would destroy this exponent-two accounting and is not used. Here is the stationary and one-symbol specialization, so that the application in Proposition G.2 uses the same proved construction. If all terminals are always active, attach each column $h_i = Qe_i$ to any one fixed terminal. Since h_i is supported on its own group, their restricted sum is $Q_J\mathbf{1}_J$ at every address. The two additional columns c at a selected terminal pair cancel directly; no balancing terminal of activation $\alpha+\beta$ is needed. The same singular program therefore isolates that pair difference at its target address. For one symbol, choose a coefficient-one top monomial of each nonzero P_x and put a separate singular program at x for each variable in that monomial. Its own programs already spend $\deg P_x$ contraction degrees at x; any extra degree kills the term. Thus exactly the selected top coefficient survives, and it is one. A constant nonzero P_x needs no program. This proves the stationary one-symbol case as well as the moving one-symbol case. In all cases the primary realization of Lemma 7.7 changes only incidences with the newly adjoined vertices. These are finite constructions for each b, with no assertion of one infinite prime array.

□

Proposition G.7 (One actual stage for the final family)**.** *In the nonempty-seed case of Proposition G.4, fix the final k, K and the stabilized common alphabet. For every sufficiently large b, the address construction is realized at one actual theta stage by a finite set Q of distinct good odd primes split in K, avoiding the fixed support and the specified finite exclusions. With nonzero activations λ_q it gives exactly the family in (3.9),*

$$h_x = \prod_{q\in Q} (q^*)^{\lambda_q(x)}, \qquad h_0 = 1, \qquad q^* = (-1)^{(q-1)/2}q.$$

All fixed local types are retained, and for every $x \neq 0$,

$$\operatorname{ord}_{s=1} L(E^{h_x}, s) = 1, \quad L_*(h_x k) \neq 0, \qquad j_{\mathrm{gen}}(h_x) \leq \omega(h_x) + C_o, \quad v_2 L_*(h_x k) \leq 2\omega(h_x) + C_e.$$

The constants precede b. The finite graph budget and then the actual theta precision are chosen after b. The theta prime remains a coefficient prime and exclusion, not a relaxed Selmer prime.

Proof. For each chosen b, this construction has a finite budget W_b. Enlarge it to bound every graph product and squarefree output in its finite auxiliary tests, as well as every operator set. Contractions only decrease the number of vertices. Thus every final h_x lies in the tested range of (F.7); this enlargement changes no fixed constant. Only now take one actual theta stage in the finite intersection of the selected table sets through $W_b + 1$, the valid-stage set for the full budget $|U| \leq W_b$ with $R = 2$, and the required precision and finite exclusion sets. In particular (F.6) holds through the threshold T used for that budget. Realize its prescribed forward residue bits by distinct primes using CRT and Dirichlet; quadratic reciprocity supplies the reverse bits. There is no uniform bound on W_b, no intersection over all budgets, and no product-surjectivity assumption on the alphabets. The resulting one family satisfies (3.9), with constants fixed before b. The theta prime ℓ_0 is a coefficient prime and an exclusion, not an active relaxed Selmer prime or a moving arithmetic prime r_i.

Finite intersection and the upper bound. The required stage conditions form a finite list once the address program is fixed. Let $\mathcal{U}$ be the chosen ultrafilter and let $A_1, \dots, A_t \in \mathcal{U}$ be the selected truth-table sets used by that program, including all intermediate graphs. Let $B \in \mathcal{U}$ impose (F.6) through the threshold T for its operator budget and impose $M \geq \max(0, \delta_o + W_b) + 2$ and $M > T + \delta_o$. Finite prime exclusions are included in B. Then $B \cap \bigcap_{i=1}^{t} A_i \in \mathcal{U}$, hence is nonempty. Choose a stage there before realizing the prime labels. The coefficient rules hold for every realization at this stage, not merely for a previously selected finite list of Fourier indices. The address construction makes each nonzero address a unit of both sources. For the odd source this says exactly

$$v_2 a_M(h_x) = \delta_o + \omega(h_x) < M.$$

The strict inequality follows from the tested-range precision condition. Applying (O.8) gives the simple zero and $j_{\mathrm{gen}}(h_x) \leq \omega(h_x) + \delta_o + C_{\mathrm{det}}$. The normalized even source gives $v_2 L_*(h_x k) \leq 2\omega(h_x) + C_{e,\mathrm{final}}$. Thus one may take $C_o = \delta_o + C_{\mathrm{det}}$ and $C_e = C_{e,\mathrm{final}}$, both fixed before b. Only the number of tests, the threshold T, and the chosen stage depend on the address dimension. □

G.5 One coordinate in the final contradiction

The family in Proposition G.7 is now used with the same coordinate of Lemma 8.11 and its central calibration in Proposition 8.10. The denominator and coefficient comparisons are those of Propositions 9.8 and 9.9; the zero-base assertion is Lemma 8.12. For this fixed K, (3.4), (F.7), and the even bound just proved give

$$2j_{\mathrm{rc}}(h_x, k) \leq 4\omega(h_x) + C_J, \qquad C_J = 2C_o + C_{e,\mathrm{final}} - v_2 L_*(h_*) + C_{H,\mathrm{final}}$$

at every $x \neq 0$. The finite support comparison uses the same class, duality map, bases and coefficient stages. Its integral correction

$$\eta = \prod_q (1 + v_q(1 + g_q)), \qquad l_q/e_q^2 = 1 + 2v_q, \qquad \widetilde{u} = \eta u_{\mathrm{rel}}$$

is formed before characters. The actual integral determinant triangle gives $\widetilde{u}_x = \epsilon_x u_{\mathrm{prim},x}$ with $\epsilon_x \in \Lambda^\times$, rather than independently chosen units in $\mathcal{O}^\times$. Therefore the central identity is

$$\begin{aligned} v_2 \widetilde{u}_x(0) = {} & 2j_{\mathrm{rc}}(h_x, k) - 4\omega(h_x) - s(h_x) - s(h_x k) \\ & - \sum_{p \mid k} t_p + \epsilon'_x, \qquad 0 \leq t_p \leq 2, \quad |\epsilon'_x| \leq C_{\mathrm{cal}}. \end{aligned} \tag{F.8}$$

The nonnegative finite Sha terms give one upper bound $v_2 \widetilde{u}_x(0) \leq B$ for all $x \neq 0$, independent of b. At the unknown base no finiteness of Sha is assumed. The retained central class and corank-one argument instead give $\widetilde{u}_0(0) = 0$ if the base derivative vanishes.

The fixed-field construction of Section 9 has

$$\begin{gathered} c_{\mathrm{det}} = a + 4(h_K + 2c_E + 2t_K), \quad L_T(M) = 16(d_K + 5)M, \\ f_x \widetilde{u}_x = d_x, \qquad c = v_2 f_0(0) \leq c_{\mathrm{det}}. \end{gathered}$$

Choose $M > \max(c_{\mathrm{det}}, B + 2c_{\mathrm{det}}, 0)$, then $b > (L_T(M) + 2)(2^M - 1)$ large enough for the address construction, then its finite graph budget and actual theta stage M_θ. The residual comparison tests coefficients of f from $-L_T(M)$ through zero and only the constant coefficient of d. It gives $x \neq 0$ with

$$v_2(\widetilde{u}_x(0) - \widetilde{u}_0(0)) \geq M - 2c > B.$$

The estimate uses both $v_{\mathcal{O}}(f_x) \le c$ and $\widetilde{u}_x \in \Lambda[1/2]$; neither a bound on the whole numerator nor a finite-Artin inverse of t is asserted. This contradicts (F.8) under the supposed base zero. The Rankin factorization and $L(E^k, 1) \neq 0$ give the asserted simple zero of E.

This completes the proof of the restricted converse, Theorem 5.3, from the displayed classical inputs and the arithmetic, coefficient and determinant constructions. The conversion from that converse to pole witnesses, the integer model, the natural compiler and finally the ordinary rational value is the separate argument in the main text.

H De Rham comparison and finite descent in the height argument

Lemma H.1. *Let $K/\mathbb{Q}_\ell$ and F/K be finite extensions, let A/F be an abelian surface, and let $C_\ell/\mathbb{Q}_\ell$ be a finite coefficient field. Suppose a continuous representation $r : G_K \longrightarrow \mathrm{GL}_2(C_\ell)$ satisfies*

$$V_\ell(A) \otimes_{\mathbb{Q}_\ell} C_\ell \simeq r|_{G_F} \oplus r|_{G_F}.$$

Then r is de Rham. At every coefficient embedding its Hodge–Tate weights are $0, -1$, with $\mathbb{Q}_\ell(1)$ assigned weight -1.

Proof. The smooth proper variety A has a smooth proper rigid analytification. Apply [15, Corollary 1.8], with the corrections in [16]. The passage from algebraic to analytic cohomology uses [15, Theorems 9.1 and 9.3]: Theorem 9.1 identifies coherent cohomology, and hence the filtered de Rham hypercohomology, while Theorem 9.3 identifies the $\mathbb{Z}/\ell^n\mathbb{Z}$-cohomology, compatibly in n. Its invertibility condition holds because F has characteristic zero. Taking inverse limits and then inverting ℓ identifies the algebraic and analytic ℓ-adic cohomology, compatibly with the action of G_F. Thus $H^1_{\mathrm{et}}(A_{\overline{F}}, \mathbb{Q}_\ell)$ is de Rham.

The canonical, Galois-equivariant identification [18, Part I, Theorem 12.1(a)]

$$V_\ell(A) \simeq H^1_{\mathrm{et}}(A_{\overline{F}}, \mathbb{Q}_\ell)^\vee$$

gives the same property for the Tate module. De Rham representations are closed under duals and subquotients [17, Theorem 5.2.1 and Proposition 6.3.3]. Extension of the coefficient field is, on underlying $\mathbb{Q}_\ell$-representations, a finite direct sum. The displayed decomposition therefore makes $r|_{G_F}$ de Rham.

Here finite-extension descent requires no semistable descent assertion. To see it explicitly, pass to a finite Galois extension F'/K containing F, and let W be the underlying $\mathbb{Q}_\ell$-representation of r. The vector space

$$D_{F'} = (B_{\mathrm{dR}} \otimes_{\mathbb{Q}_\ell} W)^{G_{F'}}$$

has its natural semilinear $\mathrm{Gal}(F'/K)$-action, and its invariants are $D_K = (B_{\mathrm{dR}} \otimes_{\mathbb{Q}_\ell} W)^{G_K}$. Finite Galois descent gives $F' \otimes_K D_K \simeq D_{F'}$, also on each filtration step. Since $W|_{G_{F'}}$ is de Rham, $\dim_K D_K = \dim_{F'} D_{F'} = \dim_{\mathbb{Q}_\ell} W$. This proves the assertion over K; it is the finite-extension case of [17, Proposition 6.3.8].

The two Hodge numbers of $H^1_{\mathrm{dR}}(A/F)$ are $2, 2$. Consequently the Tate module has weights $0, 0, -1, -1$. At each coefficient embedding, its two isomorphic summands have the same weight multiset, so each summand has weights $0, -1$. The filtration-compatible descent preserves these weights. □

The Hodge–Tate comparison in [34, Chapter III, Theorem 4.1] alone would not prove the de Rham assertion. The preceding argument uses the de Rham comparison itself and does not infer de Rhamness from Hodge–Tate weights.

I Simultaneous primes with coprime adjacent norms

This appendix gives the prime-pattern argument needed for the construction. Its strategy is due to Sun [7]. We supply the local calculations, the two ranges of the sieve, the normalization of the ray-class densities, and the order in which the parameters are chosen. For the analytic input we prove the special case for fixed ray class fields of $\mathbb{Q}(i)$ directly. The ideal-norm method follows Frei and Tinková [10]; no theorem from that work is assumed. The proof uses the published fixed-field prime estimate of Mitsui, the one-parameter equidistribution and factorization results of Green–Tao, and published ordinary-prime transference results. The field and its ray modulus are fixed before every asymptotic limit. The exact source versions checked are identified in the bibliography and the accompanying theorem-application inventory.

Write $\mathrm{N}(z) = z\bar{z}$ for $z \in \mathbb{Z}[i]$, and put $d_-(z) = \mathrm{N}(z-1)$ and $d_+(z) = \mathrm{N}(z+1)$. An Artin or Frobenius element in this appendix is arithmetic Frobenius: on residue fields it sends x to $x^{\mathrm{N}\mathfrak{p}}$.

Theorem I.1 (The simultaneous prime input)**.** *Let $c_1, \dots, c_k$ be distinct integers, where $k \geq 2$. Let $q \geq 1$, let M be an odd squarefree positive integer with $7 \nmid M$, and let $H, Y > 0$. There are positive integers u, w such that*

$$q \mid w, \qquad u/w^2 > H, \qquad p_j = u + c_j w^2 > Y \quad (1 \leq j \leq k),$$

where every p_j is prime. Each p_j has a Gaussian prime generator π_j of norm p_j such that

$$\pi_j \equiv 9 \pmod{16}, \qquad \pi_j \equiv 1 + 3i \pmod{7}, \qquad \pi_j \equiv i \pmod{M}.$$

No odd prime divides both $d_-(\pi_j)$ and $d_+(\pi_j)$. For $j \neq h$, no odd prime divides both $d_\epsilon(\pi_j)$ and $d_\eta(\pi_h)$, for any $\epsilon, \eta \in \{-,+\}$. Moreover,

$$v_2(d_-(\pi_j)) = 6, \quad v_2(d_+(\pi_j)) = 2, \quad \gcd(d_-(\pi_j), d_+(\pi_h)) = 4 \quad \text{for all } j, h,$$

and all these norms are coprime to $7M$.

The quantifiers concern fixed data. No estimate uniform in q, M, k or the c_j is asserted. We first prove the result for $q = 1$. Replacing the coefficients by $q^2 c_j$ and then replacing the parameter v by $w = qv$ proves the general case, since the ratio before this substitution can be made greater than $q^2 H$.

I.1 Local counts and the medium primes

Fix a nonzero integer κ and write

$$Q_\kappa = a^2 + b^2 - c^2 - d^2 - \kappa w^2, \quad G_\epsilon = (a+\epsilon)^2 + b^2, \quad H_\eta = (c+\eta)^2 + d^2, \qquad \epsilon, \eta \in \{1, -1\}.$$

We use $\chi(\ell) = (-1/\ell)$ for odd primes ℓ.

Lemma I.2 (Finite-field counts)**.** *If $\ell \nmid 2\kappa$, then $Q_\kappa = 0$ has $\ell^4 - 1$ nonzero solutions over $\mathbb{F}_\ell$. At most $4\ell^2$ of its solutions satisfy $G_\epsilon = H_\eta = 0$. If $\chi(\ell) = -1$, the latter set has exactly one element.*

Let $\gamma_1, \dots, \gamma_k$ be distinct integers, including 0 and $-\kappa$, and put $L_j = a^2 + b^2 + \gamma_j w^2$. Outside a fixed set of primes containing the divisors of $2\kappa \prod_{j<h}(\gamma_j - \gamma_h)$, the proportion $g(\ell)$ of nonzero zeros of Q_κ on which some L_j vanishes is

$$g(\ell) = \frac{(\ell-1)(\ell-\chi(\ell))[2\nu_0 + (k-2)(\ell-\chi(\ell))] + \nu_0^2 - 1}{\ell^4 - 1}, \qquad \nu_0 = \ell + (\ell-1)\chi(\ell). \tag{58}$$

Consequently $g(\ell) = (k + 2\chi(\ell))/\ell + O_k(\ell^{-2})$.

Proof. The norm x^2+y^2 has ν_0 representations of zero and $\ell-\chi(\ell)$ representations of every nonzero element. This follows either by factoring over $\mathbb{F}_\ell$ when $\chi=1$, or by the norm map from $\mathbb{F}_{\ell^2}$ when $\chi=-1$. Summing these fibres for Q_κ gives ℓ^4 solutions including the origin. In the split case, each of the equations $G_\epsilon=0$ and $H_\eta=0$ is a union of two lines. On each of the four pairs of lines the remaining equation is $-2\epsilon a+2\eta c-\kappa w^2=0$, leaving at most ℓ^2 choices. In the inert case these equations force $(a,b,c,d)=(-\epsilon,0,-\eta,0)$, and then $w=0$.

For $w\neq 0$, the conditions $L_j=0$ are disjoint. For the two distinguished coefficients one norm is zero and the other is nonzero, giving $2\nu_0(\ell-\chi)$ choices. Each other coefficient gives $(\ell-\chi)^2$ choices. There are $\ell-1$ possible nonzero w. For $w=0$, both norms are zero, giving ν_0^2-1 nonzero vectors. This proves (58). In particular, when $k=2$ and $\chi=-1$, its value is $2/(\ell^2+1)$, not zero. □

Lemma I.3 (Primitive congruence count)**.** *Let ψ be a smooth nonnegative compactly supported function away from the origin, and let m be squarefree and prime to 2κ. For each $\ell\mid m$ prescribe an arbitrary subset Ω_ℓ of the nonzero zeros of Q_κ over $\mathbb{F}_\ell$. The weighted number of primitive integral vectors on $Q_\kappa=0$ with these reductions equals*

$$C_\kappa\sigma_\infty(\psi)B^3\prod_{\ell\mid m}\frac{|\Omega_\ell|}{\ell^4-1}+O_{\kappa,\psi,\varepsilon}(B^{5/2+\varepsilon}m^{5/2+\varepsilon}). \tag{59}$$

The constant C_κ is positive and independent of m and the subsets.

Proof. Apply Browning–Loughran [8], Theorem 4.1, to Q_κ in five variables. It is isotropic, since $(1,0,1,0,0)$ is a zero, and nonsingular, since $\kappa\neq 0$. Its discriminant has no prime divisor outside 2κ. The theorem permits $n=5$ and arbitrary nonzero residue subsets; its error is uniform in the modulus and in these subsets. Impose primitivity by $\sum_{h\mid(a,b,c,d,w)}\mu(h)$. If $(h,m)>1$ there are no permitted vectors. Otherwise the subsets become $h^{-1}\Omega_\ell$, of the same cardinalities, and the scale becomes B/h. The sum of the error terms is convergent. Extending the main-term sum to infinity costs $O(B)$. At a good prime the unrestricted local density is

$$\sigma_\ell=\frac{1-\ell^{-4}}{1-\ell^{-3}}.$$

Indeed, nonsingular primitive zeros lift with four free coordinates, and the vectors divisible by ℓ contribute the geometric factor ℓ^{-3}. The primitive density is therefore $1-\ell^{-4}$. This turns the factor $|\Omega_\ell|/\ell^4$ of the unprimitive count into $|\Omega_\ell|/(\ell^4-1)$ in (59). The remaining finitely many local factors are positive because the form has a nonsingular rational zero. Empty subsets give the assertion directly. □

In using this count, the two norms lie between fixed positive multiples of B^2, and $|w|\le\tau B$, where $0<\tau\le\tau_0$. A smooth majorant can be chosen with $\sigma_\infty(\psi)\ll\tau$ uniformly in τ: integrate first in the two radial norm variables, on which the derivative of their difference is bounded away from zero. The constants in error terms may depend on derivatives of this majorant, and hence on τ.

Lemma I.4 (Medium-prime bound)**.** *For fixed coefficients and sufficiently large fixed L, the contribution of prime patterns with a common adjacent-norm divisor $L<\ell\le B^{1/100}$ is at most*

$$\frac{C\tau B^3}{(\log B)^k L\log L}+O_\tau(B^{27/10}). \tag{60}$$

The count is unweighted by logarithms, and C is independent of L,τ. The bound covers both distinct positions and the two signs at one position.

Proof. For two positions, use Q_κ with their fixed nonzero coefficient difference and impose $G_\epsilon = H_\eta = 0 \pmod{\ell}$. For squarefree d supported on good primes other than ℓ, impose also that some L_j vanishes at every prime dividing d. The preceding lemmas give a sieve sequence

$$A_{\ell,d} = X_\ell g(d) + r_{\ell,d}, \qquad X_\ell \ll \tau B^3/\ell^2, \qquad |r_{\ell,d}| \ll_{\tau,\varepsilon} B^{5/2+\varepsilon}(\ell d)^{5/2+\varepsilon}.$$

Enlarge the fixed exceptional set so that $0 < g(p) < 1$ outside it. Set $h(p) = g(p)/(1-g(p))$ and extend h multiplicatively to squarefree integers. The elementary Selberg upper sieve has denominator

$$\mathcal{G}_\ell(z) = \sum_{\substack{d<z \text{ squarefree} \\ (d,\ell\prod_{p\in S}p)=1}} h(d) \gg (\log z)^k.$$

Here is the dimension calculation. Williams [14], Theorem 1 with fixed modulus 4 and residues $1, 3$, gives Mertens' estimates in the two reduced classes. Taking logarithms gives $\sum_{p\le y} g(p) = k\log\log y + O(1)$ and $\sum_{p\le y} g(p)\log p \ll \log y$. For the latter bound only the elementary Chebyshev estimate is needed: $\prod_{n<p\le 2n} p \mid \binom{2n}{n} < 4^n$ gives $\vartheta(x) \ll x$ by dyadic summation, and partial summation gives $\sum_{p\le y}(\log p)/p \ll \log y$. In the Euler product $\prod_{p\le y}(1+h(p))$, normalize the weights $h(d)$ to a probability distribution. Its expected value of $\log d$ is $\sum_{p\le y} g(p)\log p$. Taking y to be a sufficiently small fixed power of z leaves at least half of the mass on $d < z$, by Markov's inequality. Removing one prime ℓ changes the estimate by a bounded factor.

For completeness the sieve weights may be taken as

$$\lambda_d = \frac{\mu(d)}{\mathcal{G}_\ell(z)} \prod_{p\mid d}(1-g(p))^{-1} \sum_{\substack{r<z/d \ (r,d)=1 \ r \text{ squarefree} \\ (r,\ell\prod_{p\in S}p)=1}} h(r).$$

They have $\lambda_1 = 1$, $|\lambda_d| \ll_\varepsilon d^\varepsilon$, and direct expansion gives $\sum_{d,e}\lambda_d\lambda_e g([d,e]) = 1/\mathcal{G}_\ell(z)$. Thus the number with no prime divisor below z in any L_j is at most

$$\frac{C\tau B^3}{\ell^2(\log z)^k} + O_{\tau,\varepsilon}(B^{5/2+\varepsilon}\ell^{5/2+\varepsilon}z^{7+4\varepsilon}).$$

All the prime values under consideration exceed z. Put $z = B^{1/100}$ and sum over $L < \ell \le B^{1/100}$. The main term uses $\sum_{\ell>L}\ell^{-2} \ll 1/(L\log L)$, another consequence of the same Chebyshev bound; for example $\varepsilon = 1/1000$ makes the error smaller than $B^{27/10}$.

At one position, an odd common divisor of $(a-1)^2+b^2$ and $(a+1)^2+b^2$ forces $a = 0$ and $b^2 = -1 \pmod{\ell}$. There are at most 2ℓ such triples (a, b, w) over $\mathbb{F}_\ell$. For $w \neq 0$ the zero sets of the k prime forms are disjoint, and the ordinary three-dimensional local sieve density is $(k+\chi(\ell))/\ell + O_k(\ell^{-2})$. Counting residue classes in a smooth box gives the corresponding three-dimensional congruence estimate. The same Selberg weights and the same, weaker error in (60) apply. □

I.2 The large-prime tail

We use Lemmas 2.1 and 4.1 of Browning–Heath-Brown [9]. The adjacent-norm polynomials are inhomogeneous, so we give the fixed-coordinate argument rather than invoke their homogeneous quadric-sieve theorem directly. Their Lemma 2.1 says that coprime polynomials in $\mathbb{Z}[X_1, \dots, X_m]$ of bounded degree and coefficient height at most H have at most

$$O\left(\frac{V\log(VH)}{L\log L} + \frac{V\log(VH)}{B_{\min}}\right) \tag{61}$$

integer points in a box of side parameters $B_i \geq 2$ at which a prime greater than $L \geq 2$ divides every polynomial; $V = \prod B_i$. "Coprime" here excludes a common constant prime factor as well as a common nonconstant factor. Their Lemma 4.1 covers the zeros modulo a squarefree q of a rank-m quadratic form, with $m \geq 3$, by at most $(3m)^{\omega(q)} q^{m-2}$ lattices of determinant q^{m-1}. The number with largest Euclidean successive minimum exceeding T is

$$O_m((3m)^{\omega(q)} q^{2m-3} T^{-m}). \tag{62}$$

Every prime dividing q must be odd and preserve the rank of the form. The lattices in that lemma contain $q\mathbb{Z}^m$; in particular their largest successive minima are at most q.

Lemma I.5 (Large common divisors on the quadric)**.** *Fix the implied constant in the box of size $O(B)$. For fixed $\kappa \neq 0$ and $\epsilon, \eta \in \{1, -1\}$, uniformly for $B, L \geq 2$, the number of integral vectors in this box satisfying $Q_\kappa = 0$ and $\ell \mid G_\epsilon, H_\eta$ for some prime $\ell > L$ is*

$$O_{\kappa,\alpha}\left(\frac{B^{3+\alpha}}{L \log L} + B^{11/4+\alpha}\right) \qquad (\alpha > 0). \tag{63}$$

Proof. Sign changes of a, c reduce to $\epsilon = \eta = 1$. Put $x_0 = a + c$, $x_1 = a - c$, $x = b - d$, $z = b + d$, and $R = \kappa w^2 - xz$. Then $x_0 x_1 = R$. Parity restrictions can be discarded in an upper bound. For a fixed nonzero value of either x_0 or x_1, the triple (x, z, w) determines the other coordinate uniquely whenever it gives an integral solution. We shall use this injection when discarding divisibility conditions.

First suppose that $\ell \mid x_0, x_1$. Then $\ell^2 \mid R$. For each fixed integer $v = O(B^2)$, the equation $R = v$ has $O_{\kappa,\alpha}(B^{1+\alpha})$ triples: for each w, count the factorizations $xz = \kappa w^2 - v$ by the divisor bound; the zero right-hand side occurs for at most two w and contributes $O(B)$ each. If $R \neq 0$, it has $O_\alpha(B^\alpha)$ factorizations $x_0 x_1 = R$. Since $0 < |h| \ll B^2/\ell^2$ and $\ell \ll B$, summation gives

$$O_{\kappa,\alpha}\left(B^{3+2\alpha} \sum_{\ell > L} \ell^{-2}\right) = O_{\kappa,\alpha}(B^{3+2\alpha}/(L \log L)).$$

The same bound for triples with $R = 0$, followed by the $O(B)$ choices with $x_0 x_1 = 0$, gives $O_{\kappa,\alpha}(B^{2+\alpha})$. We may discard all these $R = 0$ solutions before proceeding.

In the remaining cases choose $t = x_1$ or $t = x_0$ such that $\ell \nmid 2t$. Take $t = x_1$ first. Here $G_1 - H_1 = \kappa w^2 + 2t$ and $4t^2 G_1 = K$. Thus the two norm congruences are equivalent to divisibility of

$$P = \kappa w^2 + 2t, \qquad K = (\kappa w^2 - xz + t^2 + 2t)^2 + t^2(x + z)^2.$$

The identity

$$K - (t^2 + x^2)(t^2 + z^2) = P(P + 2t^2 - 2xz)$$

shows that it suffices to sieve the coprime pair $P, (t^2 + x^2)(t^2 + z^2)$. They have no common nonconstant factor over $\mathbb{Q}$, because the first depends nontrivially only on w, whereas the second is independent of w. Their contents have no common prime, since the second has coefficient 1 on $x^2 z^2$. For $t = x_0$ we still have $4t^2 G_1 = K$, whereas $t(G_1 - H_1) = \kappa(t + 2)w^2 - 2xz$. The pair is therefore

$$P' = \kappa(t + 2)w^2 - 2xz, \qquad K.$$

If $t \neq -2$, P' is an irreducible ternary quadratic form of rank three and K has nonzero constant term $t^2(t + 2)^2$; homogeneity rules out divisibility. If $t = -2$, neither x nor z divides K. The

coefficient 1 of x^2z^2 in K excludes every common content prime. In both charts the fixed value t is nonzero.

In either chart, $t \mid R$. Put

$$S_R = \{p : p \mid 2\kappa\}, \qquad q = \prod_{p \mid t \ p \notin S_R} p.$$

The matrix of R has determinant $-\kappa/4$, so R has rank three at every prime in this product, and those primes are odd. Also $q \ll B$. For a fixed q, every eligible nonzero $t = O(B)$ is supported on $S_R \cup \{p : p \mid q\}$. For every $\delta > 0$, Rankin's bound gives

$$\#\{t : q(t) = q,\ 0 < |t| \ll B\} \ll B^\delta \prod_{p \in S_R \text{ or } p \mid q} (1 - p^{-\delta})^{-1} \ll_{\kappa,\delta} B^\delta q^\delta \ll_{\kappa,\delta} B^{2\delta}.$$

For the middle inequality, remove the finitely many primes with $p^{-\delta} > 1/2$ and bound the remaining factors by $2^{\omega(q)} \ll_\delta q^\delta$. Renaming the small exponent gives $O_{\kappa,\alpha}(B^\alpha)$ choices of t.

Apply (62) with $m = 3$ to $q \mid R$. If $\lambda_1 \le \lambda_2 \le \lambda_3$ are the successive minima, then $\lambda_3 \le q \ll B$ and $\lambda_1\lambda_2\lambda_3 \asymp q^2$. A basis comparable to these minima therefore gives $O(\prod_i (1 + B/\lambda_i)) = O(B^3/q^2)$ points in the box. The lattices with $\lambda_3 > T$ contribute at most

$$B^\alpha \sum_{q \ll B} O\left(9^{\omega(q)} q^3 T^{-3} \frac{B^3}{q^2}\right) \ll_{\kappa,\alpha} B^{5+2\alpha} T^{-3}.$$

For the other lattices choose a basis with lengths comparable to the successive minima. The inverse basis matrix bounds the new coordinates by $O(B/\lambda_i)$, so the resulting rectangular box has volume $O(B^3/q^2)$ and minimum side parameter $\gg B/T$. Enlarge its side parameters to at least 2, as required in (61); since $\lambda_i \ll B$, this changes volume only by a fixed factor. The substitution into the polynomial pair is invertible over $\mathbb{Q}$. For $\ell \nmid 2t$ its determinant, of absolute value q^2, is invertible modulo ℓ too. Thus no new common content prime outside $2t$ appears. Remove common content supported on $2t$, which cannot affect the primes being counted. Coefficient heights are powers of B with fixed exponents. Formula (61) bounds this lattice's contribution by

$$O\left(\frac{B^3 \log B}{q^2 L \log L} + \frac{B^2 T \log B}{q^2}\right).$$

There are at most $9^{\omega(q)} q$ lattices. Since $9^{\omega(q)} \ll_\alpha q^\alpha$, summing over $q \ll B$ and the $O_{\kappa,\alpha}(B^\alpha)$ values of t gives $O_{\kappa,\alpha}(B^{3+3\alpha}/(L \log L) + B^{2+3\alpha} T)$. Choose $T = B^{3/4}$ and rename the small exponent to obtain (63). □

Proposition I.6 (Weighted obstruction tail)**.** *For fixed distinct c_j there are $L_* \ge 2$ and $\tau_0, C > 0$ such that, for every fixed $0 < \tau \le \tau_0$, uniformly for $L \ge L_*$, the logarithmically weighted number of pairs*

$$N/2 \le u \le N, \qquad 1 \le w \le \tau\sqrt{N}, \qquad p_j = u + c_j w^2 \text{ all prime},$$

having any of the odd common divisors forbidden in Theorem I.1 greater than L, is at most

$$\frac{C\tau N^{3/2}}{L \log L} + O_\tau(N^{3/2 - 1/800}). \tag{64}$$

The weight is $\prod_j \log p_j$, and the bound allows every Gaussian generator, so it also holds for any prescribed choice of generators.

Proof. Take $B = \sqrt{N}$ and $L_0 = B^{1/100}$, and choose τ_0 so that every prime value in the displayed region is between fixed positive multiples of B^2. Thus $\prod_j \log p_j \ll (\log B)^k$. Choose L_* above the fixed exceptional primes in the medium-prime bound. When $L \le L_0$, that bound gives

$$\frac{C\tau B^3}{L \log L} + O_\tau(B^{27/10}(\log B)^k)$$

for $L < \ell \le L_0$. Its proof is uniform in this interval of L: the remainder sum is bounded by the sum over all $\ell \le L_0$. When $L > L_0$ this part is empty.

For distinct positions, apply Lemma I.5 at L_0 with $\alpha = 1/1000$. After multiplication by the weight, the bound is

$$O\left(\left(\frac{B^{3+1/1000}}{B^{1/100}\log L_0} + B^{11/4+1/1000}\right)(\log B)^k\right) = O(B^{3-1/200}).$$

This bounds all the remaining primes, also when $L > L_0$. At one position, an odd divisor of both adjacent norms divides a and b^2+1. The coprime pair a, b^2+1 in variables (a, b, w), inserted in (61) at L_0, gives

$$O\left(\left(\frac{B^3 \log B}{L_0 \log L_0} + B^2 \log B\right)(\log B)^k\right) = O(B^{3-1/200}).$$

No factor τ is needed in these power-saving errors.

For a fixed pair of positions, the vector (a, b, c, d, w) determines $u = a^2 + b^2 - c_j w^2$; at one position (a, b, w) does so. Thus the vector counts bound the pattern counts. There are finitely many pairs of positions and signs. Each split rational prime has eight Gaussian elements of that norm, so counting all generator choices changes the bound by at most a fixed factor. Every vector used in the medium-prime count is primitive, since a common integer divisor greater than one would have its square divide a prime norm. All logarithmic factors in the remainders are absorbed for sufficiently large B, with thresholds depending only on the fixed data and τ. Weakening the error to $B^{3-1/400} = N^{3/2-1/800}$ proves the claim. □

I.3 Gaussian ray weights and their progression main terms

Let $E = \mathbb{Q}(i)$ and let H be a fixed positive integer divisible by 16. Let K/E be the ray class field for $H\mathbb{Z}[i]$. This field is Galois over $\mathbb{Q}$, since complex conjugation preserves both the modulus and its principal ray subgroup. Put

$$A = \mathrm{Gal}(K/E) \simeq (\mathbb{Z}[i]/H\mathbb{Z}[i])^\times/\{1, -1, i, -i\}, \qquad G = \mathrm{Gal}(K/\mathbb{Q}).$$

We use arithmetic Frobenius throughout. Let $C \subseteq A$ be invariant under complex conjugation, so that C is a union of conjugacy classes in G. Define $\Lambda'_C(p) = \log p$ for unramified rational primes whose Frobenius lies in C, and put $\Lambda'_C(n) = 0$ otherwise. In particular this function is supported on rational primes which split in E.

For ideals coprime to H let $[\mathfrak{a}] \in A$ denote their ray class, and set

$$T_C(n) = \sum_{\substack{\mathrm{N}\mathfrak{a}=n \\ (\mathfrak{a},H)=1}} \Lambda_E(\mathfrak{a}) 1_C([\mathfrak{a}]).$$

Here $\Lambda_E(\mathfrak{p}^j) = \log \mathrm{N}\mathfrak{p}$ and it is zero on all other ideals. Every unramified split rational prime has two conjugate prime ideals of norm p, so

$$T_C(p) = 2\Lambda'_C(p) \qquad (p \nmid 2H). \tag{65}$$

The remaining difference between T_C and $2\Lambda'_C$ is supported on rational prime powers of exponent at least two and the finitely many excluded primes. Its total absolute value up to X is $O_H(\sqrt{X}\log^2(2X))$. This bound follows by summing the at most two prime ideals above each rational prime and their powers; no density theorem is used in this comparison.

For $\chi \in \widehat{A}$ put

$$c_C(\chi) = |A|^{-1} \sum_{a \in C} \overline{\chi(a)}, \qquad T_\chi(n) = \sum_{\mathrm{N}\mathfrak{a}=n} \Lambda_E(\mathfrak{a})\chi(\mathfrak{a}),$$

where each character is extended by zero on ideals not coprime to H. Finite Fourier inversion gives the exact identity

$$T_C(n) = \sum_{\chi \in \widehat{A}} c_C(\chi) T_\chi(n). \tag{66}$$

The field and all these characters are fixed before X tends to infinity.

Lemma I.7 (Fixed-field Hecke Siegel–Walfisz estimate)**.** *Let $E = \mathbb{Q}(i)$, let χ be a fixed finite-order ray character of E with finite modulus $\mathfrak{m}$, and let $B > 0$. There is $c = c(\chi, B) > 0$ such that, uniformly for $X \geq 3$, $1 \leq Y \leq X$, positive integers $q \leq (\log X)^B$, and Dirichlet characters ρ modulo q,*

$$\sum_{\mathrm{N}\mathfrak{a} \leq Y} \Lambda_E(\mathfrak{a})\chi(\mathfrak{a})\rho(\mathrm{N}\mathfrak{a}) = r_{\chi,\rho} Y + O_{\chi,B}\big(X \exp(-c\sqrt{\log X})\big). \tag{67}$$

Here $\Lambda_E(\mathfrak{p}^j) = \log \mathrm{N}\mathfrak{p}$ for a prime ideal $\mathfrak{p}$ and $j \geq 1$, and it is zero otherwise; characters are extended by zero on ideals not coprime to their moduli. The number $r_{\chi,\rho}$ is one if the primitive Hecke character inducing $\chi(\rho \circ \mathrm{N})$ is trivial, and zero otherwise. There is also $c_B > 0$ such that, with the same ranges of X, Y, q, ρ,

$$\sum_{n \leq Y} \Lambda(n)\rho(n) = 1_{\rho \text{ principal}} Y + O_B\big(X \exp(-c_B\sqrt{\log X})\big). \tag{68}$$

In particular, either error is $O(X(\log X)^{-D})$ for every fixed $D > 0$, with the corresponding dependence on D allowed.

Proof. We use the published fixed-field estimate of Mitsui [1, §1, Lemma 15, p. 12]. In his notation, for a fixed number field K, a fixed $A > 0$, and any ray character ϑ whose finite modulus has norm at most $(\log x)^A$, it states

$$\sum_{\mathrm{N}\mathfrak{p} \leq x} \vartheta(\mathfrak{p}) \log \mathrm{N}\mathfrak{p} = E(\vartheta)x + O_{K,A}\big(x \exp(-c_{K,A}\sqrt{\log x})\big), \tag{69}$$

where $E(\vartheta) = 1$ for the principal character and is zero otherwise. The sum is over prime ideals of every residue degree. Mitsui defines these characters and $E(\vartheta)$ on p. 2, includes imprimitive characters, and states the uniformity in the character and its modulus immediately before Lemma 14 on p. 11. Infinite primes may occur in the ray modulus; only its finite part enters the norm bound.

For completeness, the elementary passage from Mitsui's proved smoothed estimate (Lemma 14, pp. 11–12) to (69) also applies to complex characters. Put

$$P(x, \vartheta) = \sum_{\mathrm{N}\mathfrak{p} \leq x} \vartheta(\mathfrak{p}) \log \mathrm{N}\mathfrak{p} \log \frac{x}{\mathrm{N}\mathfrak{p}}.$$

Lemma 14 gives $P(x, \vartheta) = E(\vartheta)x + O_{K,A}(xe^{-c_1\sqrt{\log x}})$. Subtract its values at xe^h and x, where $h = e^{-c_1\sqrt{\log x}/2}$, and divide by h. The resulting left side is the prime sum in (69) plus boundary terms of absolute value at most

$$\sum_{x < \mathrm{N}\mathfrak{p} \leq xe^h} \log \mathrm{N}\mathfrak{p} \ll_K (xh + 1)\log x.$$

Indeed, for each integer norm there are at most $[K:\mathbb{Q}]$ prime ideals. The main term is $E(\vartheta)x(e^h-1)/h = E(\vartheta)x + O(xh)$, and the error divided by h is $O_{K,A}(xe^{-c_1\sqrt{\log x}/2})$. Reducing the positive exponent constant proves (69).

Prime powers of exponent at least two contribute $O_K(\sqrt{x}\log^2(2x))$: sum over exponents $j \geq 2$ and use the elementary bound $O_K(x^{1/j})$ for the number of prime ideals of norm at most $x^{1/j}$. Consequently (69) holds with the prime sum replaced by $\sum_{\mathrm{N}\mathfrak{a}\leq x}\Lambda_K(\mathfrak{a})\vartheta(\mathfrak{a})$.

The character $\rho\circ\mathrm{N}$ on E has ray modulus (q): if $\alpha\equiv 1 \pmod{q\mathbb{Z}[i]}$, then $\mathrm{N}\alpha\equiv 1 \pmod q$. Thus $\chi(\rho\circ\mathrm{N})$ has a modulus dividing $\mathfrak{m}(q)$, whose norm is $\mathrm{N}\mathfrak{m}\,q^2$. For $\sqrt{X}\leq Y\leq X$, the bounds $q\leq(\log X)^B$ and $\log Y\geq\frac{1}{2}\log X$ imply

$$\mathrm{N}\mathfrak{m}\,q^2 \leq (\log Y)^{2B+1}$$

for all sufficiently large X, with the threshold depending only on $\mathfrak{m}$ and B. Apply the preceding estimate with $K=E$, $x=Y$, and $A=2B+1$. Its error is at most the error in (67), after reducing c. For $1\leq Y<\sqrt{X}$, the elementary bound $\sum_{\mathrm{N}\mathfrak{a}\leq Y}\Lambda_E(\mathfrak{a}) \ll_E Y\log^2(2Y)$ suffices, and the remaining bounded values of X are absorbed in the implied constant.

The principal character at the chosen modulus is exactly the character induced by the trivial primitive character, giving the stated $r_{\chi,\rho}$. Equivalently one can work at the primitive conductor and remove the extra Euler factors: the corresponding prime-power towers contribute only $O_\chi(\log(q+2)\log(Y+2))$, which is absorbed in the error.

For (68), take $K=\mathbb{Q}$ and use the ray character attached to ρ, allowing the real infinite place for odd characters. Its finite modulus norm is q. The same proof uses $A=B+1$ for $Y\geq\sqrt{X}$ and the elementary bound for smaller Y. Finally, $e^{-c\sqrt{\log X}} \ll_{c,D} (\log X)^{-D}$ for every fixed $D>0$. □

The preceding lemma gives the following consequence, including $Y=0$ by the empty-sum convention. For every $B,D>0$, every fixed finite-order ray character χ of E, and every Dirichlet character ρ of modulus $q\leq(\log X)^B$,

$$\sum_{\mathrm{N}\mathfrak{a}\leq Y}\Lambda_E(\mathfrak{a})\chi(\mathfrak{a})\rho(\mathrm{N}\mathfrak{a}) = r_{\chi,\rho}Y + O_{H,B,D}(X(\log X)^{-D}) \quad (0\leq Y\leq X), \tag{70}$$

where $r_{\chi,\rho}=1$ if the induced Hecke character $\chi(\rho\circ\mathrm{N})$ is trivial and is zero otherwise. The omitted Euler factors only change the sum by a polylogarithmic quantity. The constant may be ineffective. We require uniformity in ρ, q, Y, but not in H.

Lemma I.8 (Progression density from the fixed-field input)**.** *For $(a,q)=1$ define $J_q = K\cap\mathbb{Q}(\zeta_q)$ and let $\sigma_a(\zeta_q)=\zeta_q^a$. For fixed $B,D>0$, uniformly for $q\leq(\log X)^B$ and $0\leq Y\leq X$,*

$$\sum_{\substack{n\leq Y\\ n\equiv a \pmod q}}\Lambda'_C(n) = \frac{\#\{g\in C : g|_{J_q}=\sigma_a|_{J_q}\}}{[K\mathbb{Q}(\zeta_q):\mathbb{Q}]}Y + O_{H,B,D}(X(\log X)^{-D}).$$

Proof. Orthogonality of Dirichlet characters and (66) show that the coefficient of Y for T_C is

$$\frac{1}{\varphi(q)}\sum_{\chi\in\widehat{A}}c_C(\chi)\sum_{\rho \bmod q}\overline{\rho(a)}r_{\chi,\rho}.$$

The finite sum over χ has size depending only on H, while the average over ρ cancels its own factor $\varphi(q)$ in the error.

To evaluate this coefficient without a convention about inverse Artin symbols, put $L = K\mathbb{Q}(\zeta_q)$ and $B_q = \mathrm{Gal}(L/E)$. Restriction identifies B_q with

$$\{(g, s) \in A \times \mathrm{Gal}(\mathbb{Q}(\zeta_q)/\mathbb{Q}) : g|_{J_q} = s|_{J_q}\}.$$

The characters $\chi(g)\rho(s)$ restrict to characters of B_q. Their average on B_q is one precisely when their pullback by Artin reciprocity is the trivial Hecke character; otherwise it is zero. This is exactly $r_{\chi,\rho}$. Expanding $1_C(g)$ and $1_{s=\sigma_a}$ by Fourier inversion therefore identifies the displayed coefficient with the proportion of elements of B_q satisfying $g \in C$ and $s = \sigma_a$. Its numerator is the numerator in the lemma and its denominator is $|B_q| = [L : E]$. Dividing by two using (65) changes the denominator to $[L : \mathbb{Q}]$. The prime-power error is smaller than the stated error. This proves the formula. □

Choose a fixed integer W divisible by the conductor of K^{ab}. It is in particular divisible by 4. For a unit $b \pmod W$ put

$$\mu_C(b) = \frac{\#\{g \in C : g|_{K^{\mathrm{ab}}} = \sigma_b|_{K^{\mathrm{ab}}}\}}{[K : K^{\mathrm{ab}}]},$$
$$f_{C,b,W}(n) = \frac{\varphi(W)}{W}\Lambda'_C(Wn + b),$$
$$\lambda_{b,W}(n) = \frac{\varphi(W)}{W}\Lambda'(Wn + b).$$

The prime-supported ordinary von Mangoldt function is denoted by Λ'. Replacing it by the usual Λ costs only the same negligible prime-power error. The centered weight is

$$h_{C,b,W} = f_{C,b,W} - \mu_C(b)\lambda_{b,W}. \tag{71}$$

Here $0 \le \mu_C(b) \le 1$, so $|h_{C,b,W}(n)| \le \lambda_{b,W}(n) = O_W(\log(2N))$ on $[N]$.

Corollary I.9 (Centered sums on progressions)**.** *For every $D > 0$,*

$$\sum_{n \in P} h_{C,b,W}(n) = O_{H,W,D}(N(\log N)^{-D})$$

uniformly over arithmetic progressions P contained in $[N]$.

Proof. Write P as a residue class modulo q intersected with an interval. If $q > (\log N)^{D+3}$, its cardinality is at most $N/(\log N)^{D+3} + 1$, and the pointwise logarithmic bound proves the claim directly. Otherwise the condition $n \in P$ becomes one residue class $a \pmod{Wq}$ and an interval for $Wn + b$.

If $(a, Wq) > 1$, every prime in this residue class divides Wq. There are at most $O(\log(Wq))$ such primes, with total logarithmic weight polylogarithmic in N, which is negligible here. Suppose instead that $(a, Wq) = 1$. Since $\mathbb{Q}(\zeta_{Wq})$ is abelian and already contains K^{ab}, one has $K \cap \mathbb{Q}(\zeta_{Wq}) = K^{\mathrm{ab}}$. Also $a \equiv b \pmod W$, so the lemma's main coefficient is

$$\frac{\mu_C(b)}{\varphi(Wq)}.$$

For the ordinary prime weight, the Siegel–Walfisz main coefficient is $1/\varphi(Wq)$. The two terms of (71) therefore have equal main terms. Apply the progression lemma at both interval endpoints, with an error exponent larger than D and with $X = O_W(N)$, and subtract. This proves the claimed bound. All moduli used are at most a fixed power of $\log N$, and W is fixed throughout this argument. □

I.4 Transfer from Gaussian ideal norms to polynomial nilsequences

Throughout this subsection $E = \mathbb{Q}(i)$, $\mathcal{O}_E = \mathbb{Z}[i]$, and H_0 is a fixed positive integer. A ray character χ modulo $H_0\mathcal{O}_E$ is extended by zero to ideals not coprime to $H_0\mathcal{O}_E$. The trivial character on all ideals is included by taking $H_0 = 1$. We use the conventions of [2]: G is connected, simply connected and nilpotent, Γ is a discrete cocompact subgroup, and the filtration consists of closed connected rational subgroups. Lipschitz norms include the supremum norm.

Lemma I.10 (Gaussian norm transfer)**.** *Fix positive integers m, d. There are $\kappa, \sigma > 0$, depending only on m, d, and constants $C, X_0, \delta_0 > 0$, depending on H_0, m, d, with the following property. Suppose*

$$X \geq X_0, \qquad X^{-\sigma} \leq \delta \leq \delta_0, \qquad 1 \leq Q \leq \delta^{-\kappa}, \qquad 1 \leq q \leq \delta^{-\kappa}, \quad q \in \mathbb{Z}.$$

Let G/Γ have dimension m, a filtration $G_\bullet$ of degree at most d, and a Q-rational adapted Mal'cev basis. Suppose $g \in \mathrm{poly}(\mathbb{Z}, G_\bullet)$ is totally δ-equidistributed on $[X]$. If $F : G/\Gamma \to \mathbb{C}$ is Lipschitz and $\int F = 0$, then for every $0 \leq A < B \leq X$ and every $b \in \mathbb{Z}$,

$$\left| \sum_{\substack{A < \mathrm{N}\mathfrak{a} \leq B \\ \mathrm{N}\mathfrak{a} \equiv b \pmod q}} \chi(\mathfrak{a}) F(g(\mathrm{N}\mathfrak{a})\Gamma) \right| \leq C X \delta^{\kappa} \|F\|_{\mathrm{Lip}}. \tag{72}$$

There is no lower bound on $B - A$ or on the number of progression terms. Inserting $\log \mathrm{N}\mathfrak{a}$ on the left costs at most an additional factor $O(\log(2X))$.

We prove the lemma using only the one-parameter quantitative Leibman theorem [2, Theorem 2.9], together with elementary polynomial and lattice arguments. No multiparameter equidistribution theorem is used.

Lemma I.11 (Quadratic coefficient extrapolation)**.** *Let $X \geq 1$, $P(t) = a_0 + a_1 t + a_2 t^2 \in \mathbb{Z}[t]$, $a_2 \neq 0$, and suppose*

$$|a_i| \leq B_0 X^{1-i/2}, \qquad L \geq X^{1/2}/B_0, \qquad B_0 \geq 2.$$

For every scalar polynomial $p : \mathbb{Z} \to \mathbb{R}/\mathbb{Z}$ of degree at most d, there is an integer $1 \leq k \leq C_d B_0^{C_d}$ such that

$$\|kp\|_{C^\infty[X]} \leq C_d B_0^{C_d} \|p \circ P\|_{C^\infty[L]}.$$

Here, for $p(n) = \sum_j \alpha_j \binom{n}{j}$, we use $\|p\|_{C^\infty[X]} = \max_{1 \leq j \leq d} X^j \|\alpha_j\|_{\mathbb{R}/\mathbb{Z}}$.

Proof. Write $\binom{P(t)}{j} = \sum_\ell s_{j,\ell} \binom{t}{\ell}$. These coefficients are integers since the left side is integer-valued. Its degree is $2j$, and comparison of leading coefficients gives

$$s_{j,2j} = \frac{(2j)!}{j!} a_2^j \neq 0, \qquad |s_{j,\ell}| \ll_d B_0^j X^{j-\ell/2}.$$

For the bound, expand powers of P and change between the monomial and binomial bases. The change-of-basis matrices have fixed rational coefficients. Descending induction in ℓ shows that every term contributing to $s_{j,\ell}$ has weighted input degree at least ℓ, which gives the displayed estimate.

If $p = \sum_j \alpha_j \binom{n}{j}$ and $p \circ P = \sum_\ell \beta_\ell \binom{t}{\ell}$, then

$$\beta_{2j} = \sum_{h=j}^{d} s_{h,2j} \alpha_h.$$

Put $k_j = \prod_{h=j}^{d} s_{h,2h}$. Descending induction in j gives

$$X^j \|k_j \alpha_j\|_{\mathbb{R}/\mathbb{Z}} \ll_d B_0^{O_d(1)} \|p \circ P\|_{C^\infty[L]}.$$

Indeed the β_{2j} term is bounded using $X^j L^{-2j} \le B_0^{2j}$, and a higher coefficient is multiplied by $s_{h,2j} = O_d(B_0^h X^{h-j})$, canceling its X^{-h} error. All multipliers used to clear higher rows are integers of size $B_0^{O_d(1)}$. Multiplying to the common nonzero integer $|k_1|$ proves the assertion. The constant coefficient α_0 is not used. □

Lemma I.12 (Small horizontal character obstruction)**.** *There are $c_0, \sigma_0 > 0$, depending only on m, d, such that the following holds for sufficiently large X and sufficiently small $\delta \ge X^{-\sigma_0}$. Suppose the basis is T-rational and a nontrivial horizontal character η satisfies*

$$|\eta| \le T, \qquad \|\eta \circ g\|_{C^\infty[X]} \le T, \qquad T \le \delta^{-c_0}.$$

Then g is not totally δ-equidistributed on $[X]$.

Proof. Choose representatives of the nonconstant binomial coefficients of $\eta \circ g$ satisfying $|\alpha_j| \le TX^{-j}$. On the initial interval of length $\lfloor X/(100dT) \rfloor$ its values lie in an arc of length less than $1/10$, translated by its constant coefficient. Choose a Lipschitz function on $\mathbb{R}/\mathbb{Z}$ which equals one on that arc, has mean zero and bounded Lipschitz norm. Its composition with η has Lipschitz norm at most $T^{C(m,d)}$, by the Mal'cev metric estimates in [2, Appendix A]. Its Haar mean is zero: a nontrivial horizontal character maps Haar measure onto Haar measure on the circle. Choose c_0 with $C(m,d)c_0 < 1/4$ and $c_0 < 1/4$. The initial interval is longer than δX, while the discrepancy one exceeds $\delta T^{C(m,d)}$. Taking σ_0 sufficiently small absorbs the floors. This contradicts total equidistribution. □

Proof of Lemma I.10. Fix a constant C_* which dominates the exponents in Lemma I.11, the degree-$2d$ case of [2, Theorem 2.9], and the preceding obstruction. Choose

$$0 < \kappa < \min\{(1000C_*^2)^{-1}, c_0/(12C_*)\}, \qquad 0 < \sigma < \min\{(1000C_*^2)^{-1}, \sigma_0\}, \qquad 2\kappa\sigma < 1/4.$$

These choices depend only on m, d. Decrease δ_0, depending on H_0, m, d, whenever a fixed multiplicative constant must be absorbed.

We first record a pullback consequence. Suppose P and L satisfy Lemma I.11 with $B_0 \le \delta^{-5\kappa}$. Then $(g(P(t))\Gamma)_{t\in[L]}$ is δ^κ-equidistributed. Otherwise the quantitative Leibman theorem gives a nontrivial η with

$$|\eta|, \ \|\eta \circ g \circ P\|_{C^\infty[L]} \ll \delta^{-C_*\kappa}.$$

The basis is $\delta^{-\kappa}$-rational as required. The composed sequence has degree at most $2d$ for the stretched filtration $\widetilde{G}_j = G_{\lceil j/2 \rceil}$, with the same adapted basis. This follows from the elementary one-parameter coordinate characterization in [2, Lemma 6.7]. Extrapolation gives a nonzero integer multiple $k\eta$ with

$$|k\eta|, \ \|k\eta \circ g\|_{C^\infty[X]} \ll \delta^{-6C_*\kappa},$$

with C_* chosen sufficiently large initially. It remains nontrivial. The choice $6C_*\kappa < c_0/2$ and a reduction of δ_0 make this contradict Lemma I.12. Only ordinary equidistribution of this pullback is needed.

Every nonzero Gaussian ideal has exactly four generators. The function $z \mapsto \chi((z))$, extended by zero when z is not coprime to H_0, is periodic modulo $H_0\mathbb{Z}[i]$ and invariant under units. Thus the sum in (72) is exactly

$$\frac{1}{4} \sum_{\substack{z=x+iy\neq 0 \\ A<x^2+y^2\le B \\ x^2+y^2\equiv b \pmod q}} \chi((z))F(g(x^2+y^2)\Gamma).$$

The factor $1/4$ also applies to ideals with generators on an axis; we have introduced no angular fundamental domain.

Set $\rho = \delta^\kappa$ and $M = \lfloor \rho\sqrt{X}/H_0 \rfloor$. On each horizontal row $y \in \mathbb{Z}$, and in each residue class of x modulo H_0, partition the points into consecutive blocks of M points with step H_0. The character is constant on a block. The intersection of the annulus with a horizontal row consists of at most two intervals, so at most four blocks per residue class meet an endpoint. There are $O(\sqrt{X})$ relevant rows. Consequently the boundary blocks contain

$$O(H_0\sqrt{X}\, M) = O(\rho X)$$

points. Estimate their contribution trivially. Excluding $z = 0$, if necessary, costs one more bounded summand, absorbed by ρX.

Every remaining contributing block lies wholly in the annulus. Split its index modulo q. The norm congruence is constant on each resulting class. Such a class has the form

$$x = x_0 + H_0 q n, \quad 1 \le n \le L, \qquad P(n) = (x_0 + H_0 q n)^2 + y^2, \qquad L \ge \frac{\rho\sqrt{X}}{4H_0 q}.$$

The hypotheses on δ and σ ensure $M \ge 2q$ and absorb the floors for large X. A harmless endpoint adjustment gives $|x_0| \le 2\sqrt{X}$. The coefficients of P satisfy

$$|a_0| \ll X, \qquad |a_1| \ll H_0 q\sqrt{X}, \qquad a_2 = H_0^2 q^2.$$

Since H_0 is fixed and $q \le \delta^{-\kappa}$, these bounds and the lower bound for L fit the pullback consequence with $B_0 \le \delta^{-5\kappa}$ after decreasing δ_0. Hence the sum on the class is $O(L\delta^\kappa \|F\|_{\mathrm{Lip}})$. The classes partition the interior points, of which there are $O(X)$; this elementary bound follows from the enclosing square. Summing and adding the boundary contribution proves (72).

The number of row endpoints is independent of $B - A$, explaining why no lower interval length is required. Finally apply Abel summation to the uniform bound with variable upper endpoint to insert $\log \mathrm{N}\mathfrak{a}$ at cost $O(\log(2X))$. □

Remark I.13. For Type II, apply the lemma in dimension $2m$ to $(g_1(n), g_2(n))$ and $F_1 \otimes \overline{F_2}$, when this entire product orbit is totally equidistributed. Mean-zero factors ensure that the product test function has mean zero; they do not imply equidistribution of the product orbit. Use the smaller exponents for dimensions m and $2m$. In dimension zero mean-zero test functions vanish, so the assertion is immediate.

The constraints $rn \equiv b \pmod q$ and $r'n \equiv b \pmod q$ either are inconsistent or specify one residue class modulo a divisor of q. Intersecting their interval restrictions only changes A, B. Thus all Type I and Type II truncations are covered by the lemma. If $\delta = (\log X)^{-B_1}$ and $Q, q \le (\log X)^{C_1}$, any desired saving $X(\log X)^{-A_1}$ follows once $B_1\kappa > \max\{C_1, A_1\}$, with H_0 fixed.

I.5 Gaussian ideal Vaughan decomposition and Type I/II estimates

We give the ideal decomposition and the subsequent integer-parameter recombination in full. The architecture is that of [39, Section 3], with the Gaussian ideal-norm adaptation attributed to Frei–Tinkova. The analytic inputs used here are the separately proved Lemma I.10 and the published one-parameter results [2, Theorem 2.9 and Lemma 4.5]. In particular, no theorem of Frei–Tinkova is an assumed input.

Fix the Gaussian ray modulus H_0. All ideals in this subsection are nonzero integral ideals of $\mathbb{Z}[i]$. Write $\mathrm{N}\mathfrak{a}$ for the absolute ideal norm, μ_E for ideal Mobius, and $\Lambda_E(\mathfrak{p}^j) = \log \mathrm{N}\mathfrak{p}$ for $j \geq 1$, with Λ_E zero on other ideals. A ray character χ is extended by zero on ideals not coprime to H_0. It is completely multiplicative and satisfies $|\chi| \leq 1$. We use a Lipschitz norm that includes the supremum norm. All nilmanifolds have connected, simply connected nilpotent covering group, a cocompact lattice, and an adapted rational Mal'cev basis, with the filtration and polynomial-sequence conventions of [2].

Lemma I.14 (Ideal multiplicities and moments)**.** *Let $a_E(n) = \#\{\mathfrak{a} : \mathrm{N}\mathfrak{a} = n\}$. For $X \geq 1$,*

$$a_E(n) \leq \tau(n), \tag{73}$$

$$\sum_{n \leq X} a_E(n)^2 \leq X(1+\log X)^3, \qquad \sum_{n \leq X} \frac{a_E(n)^2}{n} \leq (1+\log X)^4. \tag{74}$$

Moreover

$$A_E(X) := \#\{\mathfrak{a} : \mathrm{N}\mathfrak{a} \leq X\} = \frac{\pi}{4} X + O(\sqrt{X}+1). \tag{75}$$

In particular $A_E(X) \ll X$, and the number of ideals of norm in $(B/2, B]$ is comparable to B for all sufficiently large B.

Proof. At a split rational prime p one has $a_E(p^k) = k+1$; at an inert prime it is one for even k and zero for odd k; and $a_E(2^k) = 1$. Multiplicativity proves (73). Prime by prime, $(k+1)^2 \leq \binom{k+3}{3}$, so $\tau(n)^2 \leq d_4(n)$. Summing over ordered quadruples with product at most X, and summing the last coordinate first, bounds $\sum_{n \leq X} d_4(n)$ by $X(1+\log X)^3$. Bounding each of the four harmonic sums separately gives $\sum_{n \leq X} d_4(n)/n \leq (1+\log X)^4$. Every ideal has exactly four Gaussian generators. Counting lattice points in the disk of radius $\sqrt{X}$, and dividing by four, proves (75): the unit squares meeting its boundary lie between circles of radii $\sqrt{X} - \sqrt{2}$ and $\sqrt{X} + \sqrt{2}$ and number $O(\sqrt{X}+1)$. □

Lemma I.15 (Exact ideal Vaughan identity)**.** *Let $N \geq 8$, $U = V = N^{1/3}$, $L = \log(2N)$, and let $f : \mathbb{N} \longrightarrow \mathbb{C}$ satisfy $|f| \leq 1$ and vanish outside $(N/2, N]$. Then*

$$\sum_{\mathfrak{a}} \Lambda_E(\mathfrak{a}) \chi(\mathfrak{a}) f(\mathrm{N}\mathfrak{a}) = T_0 - T_1 + T_2, \tag{76}$$

where

$$T_0 = \sum_{\mathrm{N}\mathfrak{d} \leq U} \mu_E(\mathfrak{d}) \chi(\mathfrak{d}) \sum_{\mathfrak{b}} \log(\mathrm{N}\mathfrak{b}) \chi(\mathfrak{b}) f(\mathrm{N}\mathfrak{d}\,\mathrm{N}\mathfrak{b}),$$

$$T_1 = \sum_{\mathrm{N}\mathfrak{d} \leq UV} \alpha_{\mathfrak{d}} \chi(\mathfrak{d}) \sum_{\mathfrak{b}} \chi(\mathfrak{b}) f(\mathrm{N}\mathfrak{d}\,\mathrm{N}\mathfrak{b}),$$

$$T_2 = \sum_{V < \mathrm{N}\mathfrak{d} \leq N/U} \beta_{\mathfrak{d}} \chi(\mathfrak{d}) \sum_{\mathrm{N}\mathfrak{b} > U} \mu_E(\mathfrak{b}) \chi(\mathfrak{b}) f(\mathrm{N}\mathfrak{d}\,\mathrm{N}\mathfrak{b}),$$

with

$$\alpha_{\mathfrak{d}} = \sum_{\substack{\mathfrak{bc}=\mathfrak{d} \\ \mathrm{N}\mathfrak{b}\le U,\ \mathrm{N}\mathfrak{c}\le V}} \mu_E(\mathfrak{b})\Lambda_E(\mathfrak{c}), \qquad\qquad \beta_{\mathfrak{d}} = \sum_{\substack{\mathfrak{c}\mid\mathfrak{d} \\ \mathrm{N}\mathfrak{c}>V}} \Lambda_E(\mathfrak{c}).$$

Both coefficients have absolute value at most L.

Proof. Convolution on ideals satisfies $\mu_E * 1 = \mathbf{1}_{\mathfrak{a}=(1)}$, $1 * \Lambda_E = \log \mathrm{N}$, and $\mu_E * \log \mathrm{N} = \Lambda_E$, as one checks on prime powers. Splitting μ_E at U and Λ_E at V gives the exact identity

$$\Lambda_E = \mu_{\le U} * \log \mathrm{N} + \Lambda_{\le V} - \mu_{\le U} * 1 * \Lambda_{\le V} + \mu_{>U} * 1 * \Lambda_{>V}. \tag{77}$$

The second term vanishes on the support of f. Complete multiplicativity of χ, including its zero extension, permits separation of each convolution factor and yields (76) with exactly the stated cutoffs. Finally $\sum_{\mathfrak{c}\mid\mathfrak{d}} \Lambda_E(\mathfrak{c}) = \log \mathrm{N}\mathfrak{d}$ and every summand is nonnegative. This proves the coefficient bounds, with no discarded endpoint or error term. □

Proposition I.16 (Quantitative extraction)**.** *There are absolute constants $c_0 > 0$ and N_0 such that the following holds. Under Lemma I.15, suppose $N \ge N_0$, $0 < \epsilon \le 1$, and*

$$\left| \sum_{\mathfrak{a}} \Lambda_E(\mathfrak{a})\chi(\mathfrak{a}) f(\mathrm{N}\mathfrak{a}) \right| \ge \epsilon N L. \tag{78}$$

Put $\theta = c_0 \epsilon^4 / L^{24}$. At least one of the following holds.

1. *Some dyadic $R \le 2N^{1/3}$ has at least θR integers $r \in (R/2, R]$ satisfying*

$$\left| \sum_{\mathfrak{b}} \log(\mathrm{N}\mathfrak{b})\chi(\mathfrak{b}) f(r\mathrm{N}\mathfrak{b}) \right| \ge \theta N / r.$$

2. *Some dyadic $R \le 2N^{2/3}$ has at least θR integers $r \in (R/2, R]$ satisfying the same inequality without the logarithm.*

3. *There are dyadic R, B with*

$$N^{1/3} < R \le 2N^{2/3}, \qquad N/2 < RB < 4N, \tag{79}$$

and at least θR^2 pairs $r, r' \in (R/2, R]$ satisfying

$$\left| \sum_{B/2 < \mathrm{N}\mathfrak{b} \le B} f(r\mathrm{N}\mathfrak{b}) \overline{f(r'\mathrm{N}\mathfrak{b})} \right| \ge \theta B. \tag{80}$$

Proof. One of $|T_i|$ is at least $\epsilon NL/3$. For T_0 put

$$G(r) = \frac{r}{NL} \left| \sum_{\mathfrak{b}} \log(\mathrm{N}\mathfrak{b})\chi(\mathfrak{b}) f(r\mathrm{N}\mathfrak{b}) \right|;$$

for T_1 use r/N times the absolute unweighted inner sum. In either case Lemma I.14 gives $G \ll 1$ and

$$\epsilon \ll \sum_{r \le N^e} \frac{a_E(r) G(r)}{r}, \qquad e = 1/3 \text{ or } 2/3.$$

Cauchy–Schwarz and (74) imply $\sum_{r\le N^e} G(r)^2/r \gg \epsilon^2/L^4$. There are $O(L)$ dyadic cells, so one satisfies

$$\sum_{R/2<r\le R} G(r)^2 \gg R\epsilon^2/L^5.$$

Since $G \ll 1$, at least $cR\epsilon^2/L^5$ terms have $G(r) \ge c\epsilon/L^{5/2}$. Both the count and the claimed amplitudes follow after decreasing c_0.

For T_2, split both ideal norms into dyadic cells. There are $O(L^2)$ rectangles, and a nonzero cell satisfies (79) and $B \ge N^{1/3}/4$. A rectangle contributes at least $c\epsilon N/L$; we use the weaker bound $c\epsilon N/L^2$. Grouping $\mathfrak{d}$ by $r = \mathrm{N}\mathfrak{d}$ writes this contribution as

$$T = \sum_{R/2<r\le R} A_r \sum_{\mathfrak{b}\in\mathcal{B}} c_{\mathfrak{b}} f(r\mathrm{N}\mathfrak{b}), \qquad \mathcal{B} = \{\mathfrak{b} : B/2 < \mathrm{N}\mathfrak{b} \le B\}.$$

Here $|A_r| \le La_E(r)$ and $|c_{\mathfrak{b}}| \le 1$; the restriction $\mathrm{N}\mathfrak{b} > U$ belongs to $c_{\mathfrak{b}}$. All product cutoffs remain in f, so the domain is rectangular. By (74), $\sum_r |A_r|^2 \ll RL^5$. Thus, with $F_{r,\mathfrak{b}} = f(r\mathrm{N}\mathfrak{b})$,

$$T_* := \sum_r \left|\sum_{\mathfrak{b}\in\mathcal{B}} c_{\mathfrak{b}} F_{r,\mathfrak{b}}\right|^2 \gg \epsilon^2 RB^2/L^9. \tag{81}$$

Put $M = |\mathcal{B}| \asymp B$. Expansion and Cauchy–Schwarz in the ordered pair $(\mathfrak{b}, \mathfrak{b}')$ give

$$\begin{aligned} T_*^2 &= \left|\sum_{\mathfrak{b},\mathfrak{b}'} c_{\mathfrak{b}}\overline{c_{\mathfrak{b}'}} \sum_r F_{r,\mathfrak{b}}\overline{F_{r,\mathfrak{b}'}}\right|^2 \\ &\le M^2 \sum_{\mathfrak{b},\mathfrak{b}'} \left|\sum_r F_{r,\mathfrak{b}}\overline{F_{r,\mathfrak{b}'}}\right|^2 = M^2 \sum_{r,r'} \left|\sum_{\mathfrak{b}} F_{r,\mathfrak{b}}\overline{F_{r',\mathfrak{b}}}\right|^2. \end{aligned} \tag{82}$$

Consequently

$$\mathbb{E}_{r,r'} \left|\mathbb{E}_{\mathfrak{b}\in\mathcal{B}} F_{r,\mathfrak{b}}\overline{F_{r',\mathfrak{b}}}\right|^2 \gg \epsilon^4/L^{18}.$$

Every summand is at most one. Thresholding gives (80) and its stated count, with the weaker common parameter $\theta = c_0\epsilon^4/L^{24}$. The character and Mobius coefficients have disappeared. All conjugates in (82) are necessary. □

Progressions and interval restrictions. In the application,

$$f(n) = 1_{N/2<n\le N}\, 1_{u<n\le v}\, 1_{n\equiv b \pmod q} F(g(n)\Gamma).$$

For Type I the congruence $rn \equiv b \pmod q$ is either insoluble or one class modulo $q/\gcd(r,q)$. In Type II its intersection with $r'n \equiv b \pmod q$ is either empty or one class modulo

$$\operatorname{lcm}(q/\gcd(r,q), q/\gcd(r',q)),$$

which divides q. The remaining restrictions are interval intersections. Lemma I.10 covers them without a lower bound on their lengths. On the product nilmanifold the function is $F(x)\overline{F(y)}$, whose Haar integral is $|\int F|^2 = 0$. After normalizing $\|F\|_{\mathrm{Lip}} = 1$, its Lipschitz norm for the sum metric is at most two. Thus all mean-zero, modulus and interval hypotheses for product norm transfer are satisfied.

Lemma I.17 (Integer-parameter recombination)**.** *For each m, d there are $C_{\mathrm{rec}}, K_{\mathrm{rec}} \geq 2$ with the following property. Let $0 < \eta < \eta_0$, $N \geq \eta^{-K_{\mathrm{rec}}}$, and let g have degree at most d on an m-dimensional nilmanifold with an η^{-1}-rational adapted basis. Each of the following implies that $(g(n)\Gamma)_{n\leq N}$ is not totally $\eta^{C_{\mathrm{rec}}}$-equidistributed.*

1. *For at least ηR values $r \in (R/2, R]$, where $R \leq 2N^{2/3}$, the orbit $(g(rn)\Gamma)_{n\leq \lceil N/r\rceil}$ is not totally η-equidistributed.*
2. *For at least ηR^2 pairs $r, r' \in (R/2, R]$, where $N^{1/3}/4 \leq R \leq 2N^{2/3}$ and $N/2 < RB < 4N$, the product orbit $(g(rn), g(r'n))_{n\leq B}$ is not totally η-equidistributed.*

Proof. All constants in this proof depend only on m, d. Failure of total equidistribution gives a progression $a+q[\ell] \subseteq [M]$ with $\ell \geq \eta M$, $q \leq 2/\eta$, and $|a| \leq 2M$ on which ordinary η-equidistribution fails. Apply [2, Theorem 2.9], in dimension m or $2m$. A nonzero horizontal character has magnitude and restricted smoothness norm at most η^{-C_1}, after shrinking η_0 to absorb fixed factors.

Here is the required removal of a possibly large shift. Write the scalar phase $p(a + qt)$ in monomials. Multiplication by $d!$ clears conversion from its binomial coefficients. Inverting $t = (n - a)/q$ and multiplying by q^d clears every remaining denominator. The error in the coefficient of n^i is bounded by

$$C_d\eta^{-C_1}\sum_{j=i}^{d} q^{d-j}|a|^{j-i}\ell^{-j} \leq \eta^{-C_2}M^{-i}, \qquad i \geq 1. \tag{83}$$

The integer parts disappear modulo one. Thus a nonzero integer multiple of the character has magnitude and $C^\infty[M]$ norm at most η^{-C_2}. This argument does not treat $a = O(M)$ as a bounded-height rational parameter. There are at most η^{-C_3} possible resulting characters, since their lattice has rank at most $2m$. Pigeonholing therefore fixes one character on a set of density at least η^{C_4}.

In Type I write $\psi(g(n)) = \sum_{j=0}^{d} \beta_j n^j$ modulo one. For the selected set $S \subseteq (R/2, R]$, coefficient conversion gives

$$\|d!\beta_j r^j\|_{\mathbb{R}/\mathbb{Z}} \leq \eta^{-C_5}(R/N)^j, \qquad 1 \leq j \leq d. \tag{84}$$

If $R \leq \eta^{-C_6}$ for a sufficiently large fixed C_6, choose one $r \in S$ and multiply the character by $d!r^d$. Multiplication of the jth bound by r^{d-j} gives magnitude and $C^\infty[N]$ norm bounded by a fixed power of η^{-1}. This treats every small dyadic scale, including $R = 1$.

Otherwise apply [2, Lemma 4.5] to $r \mapsto d!\beta_j r^j$ on $[\lceil R\rceil]$. The selected set has density at least a fixed power of η, and its values lie in an arc of length at most $2\eta^{-C_5}(R/N)^j$. Taking K_{rec} large, the bound $R \leq 2N^{2/3}$ makes this length at most half the density. The choice of C_6 also places the recurrence length beyond the fixed inverse density powers. The published recurrence lemma supplies nonzero integers t_j, $|t_j| \leq \eta^{-C_7}$, such that

$$\|t_j(d!)^2\beta_j\|_{\mathbb{R}/\mathbb{Z}} \leq \eta^{-C_8}N^{-j}. \tag{85}$$

The extra factorial clears the leading-coefficient conversion. Taking the product of the t_j and the fixed factorials gives a single nonzero horizontal character of magnitude and $C^\infty[N]$ norm at most η^{-C_9}.

For Type II the pigeonholed character is $\psi_1 \oplus \psi_2$, not both zero. Write the two scalar phases with coefficients β_j, β'_j. The information replacing (84) is

$$\|d!(\beta_j r^j + \beta'_j r'^j)\|_{\mathbb{R}/\mathbb{Z}} \leq \eta^{-C_5}(R/N)^j. \tag{86}$$

Interchange the two coordinates if necessary so that $\psi_1 \neq 0$. Fix an r' with at least $\eta^{C_4}R$ partners. Equation (86) puts $d!\beta_j r^j$ in an arc centered at $-d!\beta'_j r'^j$. The same published recurrence lemma

permits this arbitrary center and gives (85) for ψ_1. Its large-length requirement follows from $R \ge N^{1/3}/4$ and a sufficiently large K_{rec}. There is no substitution of $(r-r_0)^j$ for $r^j - r_0^j$.

Finally let Ψ denote the nonzero character just obtained. Its binomial expansion shows that for $1 \le n \le c\eta^{2C_9+4}N$ the phase $\Psi(g(n))$ stays within $1/10$ of $\Psi(g(0))$. Choose a mean-zero Lipschitz circle function equal to one on that arc and compose it with Ψ. Its mean is zero because a nonzero horizontal character pushes Haar measure to circle Haar measure. Its Lipschitz norm is bounded by a fixed power of η^{-1} by [2, Appendix A]. Choose C_{rec} larger than the length and Lipschitz exponents, and then enlarge K_{rec} so the initial interval is nonempty. Its average one contradicts total $\eta^{C_{\mathrm{rec}}}$-equidistribution. This proves both assertions. All exponent constants are finite combinations of the published one-parameter constants and degree-d polynomial algebra. □

Proposition I.18 (Power-form Gaussian ideal minor-arc bound)**.** *For each m, d there is $c_{\mathrm{G}} > 0$ with the following property. Fix H_0 and a finite-order Gaussian ray character χ of modulus H_0, extended by zero. Let G/Γ have dimension m, filtration degree d, and a Q-rational adapted basis, with $Q \ge 2$. Suppose $(g(n)\Gamma)_{n\le N}$ is totally δ-equidistributed, where $0 < \delta < 1$. If F is Lipschitz and $\int F = 0$, then for every interval progression $P \subseteq [N]$ of common difference at most Q,*

$$\left| \sum_{\mathrm{N}\mathfrak{a}\le N} \Lambda_E(\mathfrak{a})\chi(\mathfrak{a})1_P(\mathrm{N}\mathfrak{a})F(g(\mathrm{N}\mathfrak{a})\Gamma) \right| \ll_{H_0,m,d} QN\log(2N)^7\delta^{c_{\mathrm{G}}}\|F\|_{\mathrm{Lip}}. \tag{87}$$

No lower bound on the length of P or on δ is required.

Proof. Normalize $\|F\|_{\mathrm{Lip}} = 1$. Take the smaller positive exponents κ, σ in Lemma I.10 for dimensions m and $2m$, and decrease κ to at most one. Partial summation gives the same transfer bound with an extra factor $\log(2M)$ for a $\log \mathrm{N}\mathfrak{b}$ weight: the unweighted lemma holds for every partial interval, and $\int_1^M dt/t \le \log(2M)$. Let $C_{\mathrm{rec}}, K_{\mathrm{rec}}$ be as in Lemma I.17, and choose

$$a_* = \frac{1}{4C_{\mathrm{rec}}}, \qquad c_* = \frac{a_*\kappa}{32}, \qquad s_* = \min\left\{\frac{1}{100}, \frac{1}{4a_*K_{\mathrm{rec}}}, \frac{\sigma}{12a_*}\right\}. \tag{88}$$

First assume $N^{-s_*} \le \delta \le \delta_0$, with the fixed cutoff δ_0 sufficiently small, and restrict the sum to $(N/2, N]$. We claim that its absolute value is

$$\ll QNL^7\delta^{c_*}, \qquad L = \log(2N). \tag{89}$$

The elementary bound is $O(NL)$. Thus the claim is immediate when $QL^6\delta^{c_*} \ge 1$, and in the other case $Q \le \delta^{-c_*}$ and $L \le \delta^{-c_*/6}$. If (89) fails with a sufficiently large implied constant, Proposition I.16 applies with ϵ a fixed large multiple of $QL^6\delta^{c_*}$; if this exceeds the permitted range, the elementary bound already proves the claim. The extracted density and normalized amplitude are at least a fixed positive constant times δ^{4c_*}, hence at least δ^{5c_*} after decreasing δ_0.

Set $\eta = \delta^{a_*}$. If an extracted orbit were totally η-equidistributed, Lemma I.10, with the progression analysis above, would bound the extracted sum by

$$O(M\delta^{a_*\kappa}L^e), \qquad e = 0 \text{ or } 1.$$

Here $M = \lceil N/r \rceil \ge N^{1/3}/2$ in Type I and $M = B \ge N^{1/3}/4$ in Type II. The rationality and modulus hypotheses hold because $c_* < a_*\kappa$. The lower-parameter hypothesis holds because $a_*s_* \le \sigma/12$; increasing the fixed threshold for N absorbs the constants two and four. Since $a_*\kappa = 32c_*$, the displayed bound is smaller than $M\delta^{5c_*}$, even for $e = 1$. This is a contradiction. Thus every extracted orbit fails total η-equidistribution. Its density is at least η, since $a_* > 5c_*$. Also $N \ge \eta^{-K_{\mathrm{rec}}}$ by

$a_* s_* K_{\mathrm{rec}} \le 1/4$, and $Q \le \eta^{-1}$. Lemma I.17 now contradicts total δ-equidistribution, because $\eta^{C_{\mathrm{rec}}} = \delta^{1/4} > \delta$. This proves (89). Fixed values $\delta \ge \delta_0$ are covered by the elementary bound with a larger constant.

We spell out the passage through all scales. Use exact real endpoints $M_j = 2^{-j}N$ and retain shells with $M_j \ge 2\sqrt{\delta}N$. The discarded norms are at most $2\sqrt{\delta}N$ and their total absolute weight is $O(\sqrt{\delta}NL)$ by Lemma I.14. There is no rounding of ideal-norm cutoffs. On $[\lfloor M_j \rfloor]$ the orbit is totally $2\sqrt{\delta}$-equidistributed: every progression of the requisite length has at least δN terms for sufficiently large N, and the original discrepancy bound is stronger. For $\delta \ge N^{-s_*/4}$ the lower-parameter condition for the shell bound holds on every retained scale, since $M_j \ge N^{1-s_*/8}$ up to fixed constants. The progression is merely intersected with the shell; its common difference remains at most Q. Summing the shell estimates and using $\sum_j M_j \le 2N$ gives

$$O(QNL^7\delta^{c_*/2}). \tag{90}$$

For $1/N < \delta < N^{-s_*/4}$ replace δ by $\delta^{s_*/8} \ge N^{-s_*/8}$. Total equidistribution at the original parameter implies it at this larger one. The exponent in (90) becomes $c_* s_*/16$. For $\delta \le 1/N$, singleton progressions imply $|F(g(n)\Gamma)| \le \delta$ at each integer n, so the elementary $O(NL)$ weight bound suffices. All remaining bounded N are absorbed by the fixed constant. Thus one may take

$$c_{\mathrm{G}} = c_* s_*/16 > 0. \tag{91}$$

Its dependence is only on m, d and the corresponding norm-transfer exponents; H_0 enters constants and thresholds, not this exponent. □

Corollary I.19 (Ordinary prime term)**.** *Under the nilmanifold hypotheses of Proposition I.18, there is $c_{\mathbb{Q}} > 0$, depending only on m, d, such that*

$$\left| \sum_{n \le N} \Lambda'(n) 1_P(n) F(g(n)\Gamma) \right| \ll_{m,d} QN \log(2N)^7 \delta^{c_{\mathbb{Q}}} \|F\|_{\mathrm{Lip}}, \tag{92}$$

where $\Lambda'(p) = \log p$ on rational primes and is zero otherwise. The same bound holds for ordinary von Mangoldt Λ.

Proof. For positive integers the convolution identities $\mu * 1 = 1_{n=1}$, $1 * \Lambda = \log$, and $\mu * \log = \Lambda$ give (77) with exactly the displayed norm cutoffs replaced by integer cutoffs. Its coefficients are bounded by $\sum_{c|d} \Lambda(c) = \log d$. Integer multiplicity is one, so both moment bounds in Lemma I.14 remain valid. The two Cauchy–Schwarz calculations (81) and (82) apply to the resulting finite integer sums and give Proposition I.16 with the same conservative parameter $c_0\epsilon^4/L^{24}$. Lemma I.17 concerns only integer dilation parameters and is unchanged.

The unweighted replacement for norm transfer follows directly from total equidistribution. If an interval progression $J \subseteq [X]$ has $|J| \ge \zeta X$, then

$$\left| \sum_{n \in J} F(g(n)\Gamma) \right| \le |J|\zeta \|F\|_{\mathrm{Lip}} \le X\zeta \|F\|_{\mathrm{Lip}}.$$

If $|J| < \zeta X$, its cardinality and the supremum norm give the same bound. This treats every modulus and every truncated interval. Partial summation supplies the logarithmic weight; the identical long-progression/short-progression dichotomy treats the mean-zero product function $F \otimes \overline{F}$ on the product nilmanifold. Thus the exponent choices and all shell steps in the proof of Proposition I.18

apply, with any sufficiently small fixed positive transfer exponents, to give (92) for Λ. No extension of a Mobius theorem to Λ is assumed.

Finally proper prime powers contribute at most $O(\sqrt{N}\log(2N)^2\|F\|_{\mathrm{Lip}})$, by counting at most $N^{1/k}$ possible bases for each $2 \le k \le \log_2 N$. Shrink $c_{\mathbb{Q}}$ below $1/4$. If $\delta > 1/N$, this error is absorbed because $N\delta^{c_{\mathbb{Q}}} \ge N^{1-c_{\mathbb{Q}}}$. If $\delta \le 1/N$, singleton equidistribution and $\sum_{n\le N}\Lambda'(n) \le N\log(2N)$ prove the bound directly. This gives the assertion for Λ' as well. □

I.6 Fixed-modulus affine interpolation

Fix integers $W \ge 1$ and $0 \le b < W$. The modulus W remains fixed as the size parameter tends to infinity. We first justify the affine interpolation needed to apply ideal-norm estimates to a W-tricked sequence.

Lemma I.20 (Linear coefficient extrapolation)**.** *Let $X \ge 1$ and let $p : \mathbb{Z} \to \mathbb{R}/\mathbb{Z}$ have degree at most d. Suppose $P(t) = u + vt$ with $u, v \in \mathbb{Z}$, $v \ne 0$, $|u| \le CX$, $|v| \le E$, and $L \ge X/E$, where $E \ge 2$. There is an integer $1 \le k \le C_d E^{C_d}$ such that*

$$\|kp\|_{C^\infty[X]} \le C_d(C+E)^{C_d}\|p \circ P\|_{C^\infty[L]}.$$

Proof. Expand $\binom{u+vt}{j}$ in the binomial basis in t. All coefficients are integers, the diagonal coefficient is v^j, and the coefficient in row ℓ is $O_d((C+E)^j X^{j-\ell})$. Starting in degree d, clear the nonzero diagonal entries successively. Their product is a nonzero integer bounded by $E^{O_d(1)}$. At degree j use $X^j/L^j \le E^j$; a higher-row error is multiplied by $O_d((C+E)^h X^{h-j})$, canceling its X^{-h} factor. Multiplication to the common product of diagonal entries proves the claim, exactly as in Lemma I.11. □

Lemma I.21 (Affine interpolation preserves quantitative distribution)**.** *Fix dimension and degree bounds m, d. There are $a, \sigma > 0$, depending only on m, d, and thresholds depending on W, m, d, with the following property. Suppose a nilmanifold and its rational filtration satisfy the Green–Tao conventions, its adapted basis is Q-rational, and $g \in \mathrm{poly}(\mathbb{Z}, G_\bullet)$. Suppose also that*

$$N^{-\sigma} \le \delta \le \delta_0(W, m, d), \qquad Q \le \delta^{-a},$$

and that g is totally δ-equidistributed on $[N]$. There is $\widetilde{g} \in \mathrm{poly}(\mathbb{Z}, G_\bullet)$ satisfying

$$\widetilde{g}(Wn + b) = g(n) \quad (n \in \mathbb{Z})$$

which is totally δ^a-equidistributed on $[WN + b]$.

Proof. Use the one-parameter coordinate characterization in [2, Lemma 6.7]. Each second-kind Mal'cev coordinate of $g(n)$ is a polynomial, with the prescribed degree bound from the filtration. Evaluate those same polynomials at real arguments and apply the inverse coordinate map, obtaining $g_{\mathbb{R}}(t)$. Set

$$\widetilde{g}(t) = g_{\mathbb{R}}((t-b)/W).$$

Affine composition preserves the coordinate degree bounds, so the same characterization shows that this is a polynomial sequence for the original filtration. It gives the stated identity on integer n. Equivalently, one may use the Taylor product with real powers; those powers exist in a connected simply connected nilpotent group.

Put $X = WN + b$ and $\epsilon = \delta^a$. Suppose the asserted total ϵ-equidistribution fails. Then on a progression

$$u + v[L] \subseteq [X], \qquad L \ge \epsilon X, \quad u, v \in \mathbb{Z}, \quad 1 \le v \le 2/\epsilon, \quad |u| \le 2X,$$

the sequence is not ϵ-equidistributed. By the one-parameter quantitative Leibman theorem [2, Theorem 2.9], there is a nontrivial horizontal character η satisfying

$$|\eta| \ll \epsilon^{-C}, \qquad \|\eta \circ \widetilde{g} \circ (u + vt)\|_{C^\infty[L]} \ll \epsilon^{-C}.$$

The rationality hypothesis holds since $Q \le \epsilon^{-1}$. Lemma I.20 gives a positive integer $k \ll \epsilon^{-C}$ such that

$$\|k\eta \circ \widetilde{g}\|_{C^\infty[X]} \ll \epsilon^{-C}.$$

Forward composition with the fixed integer polynomial $Wn + b$ gives

$$\|k\eta \circ g\|_{C^\infty[N]} \ll_{W,d} \epsilon^{-C}.$$

For this last bound expand $\binom{Wn+b}{j}$ in the binomial basis; its matrix is integral, and X is comparable to WN. The character $k\eta$ remains nontrivial.

Let C_{aff} bound all exponents in these estimates, including the character size. Fix explicitly

$$0 < a < \min\{1, c_0/(2C_{\mathrm{aff}})\}, \qquad 0 < \sigma < \min\{\sigma_0, 1/(2a)\},$$

where c_0, σ_0 are from Lemma I.12. Decrease δ_0 to absorb fixed W-dependent constants. The resulting character bounds contradict that lemma and total δ-equidistribution of g. The inequality $a\sigma < 1/2$ ensures $\epsilon X \gg N^{1/2}$, so all progression parametrizations and floor estimates used above hold for large N. This proves the assertion on the entire interval $[WN + b]$, including progressions outside the residue class b modulo W. □

The Gaussian input in the following lemma is furnished by Proposition I.18; its rational counterpart is Corollary I.19.

Lemma I.22 (W-tricked minor-arc interface)**.** *For $E = \mathbb{Q}(i)$ and a fixed ray character χ, put*

$$T_\chi(t) = \sum_{\mathrm{N}\mathfrak{a}=t} \Lambda_E(\mathfrak{a})\chi(\mathfrak{a}).$$

Assume the mean-zero ideal estimate

$$\left| \sum_{\substack{\mathrm{N}\mathfrak{a}\le X \\ \mathrm{N}\mathfrak{a}\in R}} \Lambda_E(\mathfrak{a})\chi(\mathfrak{a})F_0(h(\mathrm{N}\mathfrak{a})\Gamma) \right| \ll Q_1 X \log(2X)^7 \eta^{c_E} \|F_0\|_{\mathrm{Lip}} \tag{93}$$

for a positive c_E, whenever h is totally η-equidistributed, the basis is Q_1-rational, and R is any interval progression of modulus at most Q_1. Then, under the hypotheses of Lemma I.21, for every Lipschitz F_0 with Haar mean zero and every interval progression $P \subseteq [N]$ of modulus $q \le Q$,

$$\left| \sum_{n\in P} T_\chi(Wn + b)F_0(g(n)\Gamma) \right| \ll_{H_0,W,m,d} QN \log(2N)^7 \delta^{ac_E} \|F_0\|_{\mathrm{Lip}}. \tag{94}$$

The same implication holds for $E = \mathbb{Q}$ and the corresponding ordinary von Mangoldt estimate.

Proof. Take $X = WN + b$ and the sequence $\widetilde{g}$ constructed above. The conditions $t = Wn + b$ and $n \in P$ specify an interval progression R of modulus Wq. Apply (93) with $Q_1 = \max\{Q, Wq, 2\} \ll_W Q$ and $\eta = \delta^a$. The ideal sum then equals the left side of (94) exactly, while $X \ll_W N$ and $\log(2X) \ll_W \log(2N)$. No assertion about fractional-argument distribution has been assumed: it was proved in Lemma I.21. □

Corollary I.23 (Centered ray-prime weights)**.** *Suppose a conjugation-invariant fixed Gaussian ray condition has the checked finite-character expansion*

$$\Lambda'_C(t) = \frac{1}{2}\sum_{\chi} c_\chi T_\chi(t) + e(t), \qquad \sum_{t\le X} |e(t)| \ll_{H_0} \sqrt{X}\log(2X)^2.$$

Assume both the Gaussian estimate (93) *and its ordinary rational counterpart, with positive exponents $c_E, c_{\mathbb{Q}}$. Let G/Γ and g satisfy the hypotheses of Lemma I.21, and let F_0 be Lipschitz with Haar mean zero. Then any fixed linear combination of the W-tricked ray-prime weight and the W-tricked ordinary-prime weight obeys, for every interval progression $P \subseteq [N]$ of modulus at most Q,*

$$\left|\sum_{n\in P} w_N(n) F_0(g(n)\Gamma)\right| \ll QN\log(2N)^7\delta^{c_W}\|F_0\|_{\mathrm{Lip}}, \qquad c_W = a\min\{c_E, c_{\mathbb{Q}}\} > 0,$$

for mean-zero F_0, in a regime $\delta \ge N^{-\sigma_1}$ and $Q \le \delta^{-a}$ with fixed $\sigma_1 > 0$. Constants may depend on H_0, W and the fixed linear combination. In particular this applies to the proposed centering

$$w_N(n) = \frac{\varphi(W)}{W}\Lambda'_C(Wn+b) - \mu_C(b)\lambda_{b,W}(n),$$

once the normalization of $\lambda_{b,W}$, the scalar $\mu_C(b)$ and the displayed character expansion have been fixed and checked.

Proof. Apply Lemma I.22 to the finitely many characters and to the rational von Mangoldt function. The Gaussian factor $1/2$ accounts for the two conjugate prime ideals above an eligible split rational prime. Replacing ordinary von Mangoldt by its prime-only version also has a proper-prime-power error $O_W(\sqrt{N}\log(2N)^2)$. Thus the stated estimate holds with the additional error $O_{H_0,W}(\sqrt{N}\log(2N)^2\|F_0\|_\infty)$. Choose $\sigma_1 \le \sigma$ so small that $\sigma_1 c_W < 1/4$. Then $N\delta^{c_W} \ge N^{3/4}$ throughout the claimed regime, and the square-root errors are absorbed. All normalization factors are fixed constants because W is fixed. This proves the minor-arc estimate but does not assert the centered progression estimate (95), which must be proved separately from the arithmetic main terms. □

Remark I.24. Fix the arithmetic moduli first, then the positive exponents in this subsection. They do not depend on the later factorization exponent B. Proposition I.25 chooses B large enough to satisfy the resulting upper complexity condition. Its output M is polylogarithmic in N, so its equidistribution parameter also obeys $\delta \ge N^{-\sigma_1}$ for sufficiently large N. The affine interpolation argument is motivated by the corresponding step of Frei–Tinkova, but uses no theorem from that work as an input.

I.7 Quantitative factorization for centered weights

We isolate the analytic implication needed after the arithmetic progression main terms have been established. Throughout, $L = \log(2N)$, all nilpotent groups are connected and simply connected, and all filtrations are by closed connected rational subgroups. Constants may depend on fixed arithmetic moduli and on the fixed dimension and degree bounds.

Proposition I.25 (Factorization implication)**.** *Fix dimension and degree bounds m, d. Let $w_N : [N] \to \mathbb{C}$ satisfy the following three hypotheses.*

1. *For a fixed $K_w \ge 0$, $|w_N(n)| \ll L^{K_w}$.*

2. *For every $A_0, C_0 > 0$, uniformly over all interval progressions $P \subseteq [N]$ of common difference at most L^{C_0},*

$$\left|\sum_{n\in P} w_N(n)\right| \ll_{A_0,C_0} NL^{-A_0}. \tag{95}$$

No lower bound on $|P|$ is imposed here.

3. *There are fixed $c > 0$, $C_1, K_1 \geq 0$ such that if a nilmanifold of dimension at most m and filtration degree at most d has a Q-rational adapted Mal'cev basis, h is a polynomial sequence totally δ-equidistributed on $[N]$, and F_0 is Lipschitz with Haar mean zero, then*

$$\left|\sum_{n\in P} w_N(n)F_0(h(n)\Gamma)\right| \ll N\delta^c Q^{C_1} L^{K_1}\|F_0\|_{\mathrm{Lip}} \tag{96}$$

for every interval progression $P \subseteq [N]$ of modulus at most Q and length at least N/Q. It is sufficient for this hypothesis to hold only when

$$\delta \geq N^{-\sigma}, \qquad Q \leq \delta^{-\kappa} \tag{97}$$

for fixed positive σ, κ. An estimate allowing arbitrary progression lengths is stronger than required.

Then, for every $A, C_{\mathrm{init}} > 0$, uniformly over initial nilmanifolds of dimension at most m and degree at most d with an $L^{C_{\mathrm{init}}}$-rational adapted basis, all polynomial sequences g, and all Lipschitz F,

$$\left|\sum_{n\leq N} w_N(n)F(g(n)\Gamma)\right| \ll_{A,C_{\mathrm{init}}} NL^{-A}\|F\|_{\mathrm{Lip}}. \tag{98}$$

Proof. Normalize $\|F\|_{\mathrm{Lip}} = 1$ and put

$$D = A + K_w + 3, \qquad M_0 = L^{\max\{1,C_{\mathrm{init}}\}}.$$

Leave a factorization exponent B unspecified for the moment. By [2, Theorem 1.19], there is an integer M with

$$M_0 \leq M \ll_{B,m,d} M_0^{E(B,m,d)}$$

and a factorization $g = \varepsilon g'\gamma$ into polynomial sequences. Here ε is (M,N)-smooth, $\gamma(n)\Gamma$ has period $q \leq M$, and g' is totally M^{-B}-equidistributed in a rational subgroup quotient. The new adapted basis consists of M-rational combinations of the initial one.

For each residue j modulo q, choose a reduced rational representative γ_j of $\gamma(j)\Gamma$. Split that residue class into $\lceil ML^D\rceil$ consecutive blocks of nearly equal cardinality. For sufficiently large N after B is fixed, the resulting progressions $P_{j,k}$ satisfy

$$\#\{P_{j,k}\} \ll M^2L^D, \qquad |P_{j,k}| \gg \frac{N}{M^2L^D}, \qquad \mathrm{diam}(P_{j,k}) \ll \frac{N}{ML^D}. \tag{99}$$

Their common difference is q. Indeed each original residue class has cardinality comparable to N/q, and rounding the balanced block sizes changes these estimates only by an absolute factor, since all powers of M are polylogarithmic in N once B is fixed.

Choose $n_{j,k} \in P_{j,k}$ and put $a_{j,k} = \varepsilon(n_{j,k})$. The right-invariant ambient metric and smoothness imply

$$\begin{aligned}&|F(\varepsilon(n)g'(n)\gamma_j\Gamma) - F(a_{j,k}g'(n)\gamma_j\Gamma)|\\ &\qquad \leq d(\varepsilon(n),\varepsilon(n_{j,k})) \ll L^{-D} \qquad (n \in P_{j,k}).\end{aligned}$$

There is no factor depending on the coordinates of $g'(n)$ in this estimate: right invariance removes it. The total weighted freezing error is therefore $O(NL^{K_w-D}) = O(NL^{-A-3})$.

We use the quantitative conjugation claim proved in [39, p. 547], whose inputs are the rational-subgroup, rational-element and metric comparisons of [2, Appendix A]. There are fixed $a > 0$, $C_2 \geq 1$, depending only on m, d, such that, with

$$H_j = \gamma_j^{-1} G' \gamma_j, \quad \Lambda_j = H_j \cap \Gamma, \quad h_j(n) = \gamma_j^{-1} g'(n) \gamma_j,$$

the quotient H_j/Λ_j has an adapted basis of rationality at most M^{C_2}, h_j is totally M^{-aB+C_2}-equidistributed, and

$$F_{j,k}(x\Lambda_j) = F(a_{j,k}\gamma_j x \Gamma) \quad \text{satisfies} \quad \|F_{j,k}\|_{\mathrm{Lip}} \leq M^{C_2}, \qquad \|F_{j,k}\|_\infty \leq 1.$$

We have enlarged C_2 to absorb fixed multiplicative constants. These exponents do not depend on B. In applying this published claim, one must account for the commensurable lattice resulting from rational conjugation; conjugation is not asserted to be an isometry of the original quotients. Its finite-index and metric losses are included in a, C_2.

Write $F_{j,k} = m_{j,k} + F^0_{j,k}$ by subtracting its Haar mean on H_j/Λ_j. Then $|m_{j,k}| \leq 1$, $\int F^0_{j,k} = 0$, and $\|F^0_{j,k}\|_{\mathrm{Lip}} \leq 2M^{C_2}$. This subtraction is needed even if the initial function F happened to have Haar mean zero. For a dimension-zero factor the centered function simply vanishes.

Put

$$U = C_2 + D + 5, \qquad Q_* = M^U, \qquad b_* = aB - C_2.$$

Since $M \geq L$, Q_* dominates the rationality, cell modulus and $N/|P_{j,k}|$. Applying (96) to all centered cell functions and using (99) gives a total of at most

$$CNL^{K_1} M^{-b_* c + C_1 U + C_2 + 2 + D}. \tag{100}$$

Every cell is counted here. In particular, the bound N in the minor-arc hypothesis has not been replaced by an unsupported cell cardinality.

We now fix B sufficiently large that

$$b_* > 1, \qquad b_* c \geq C_1 U + C_2 + 2 + D + A + K_1 + 4, \qquad b_* \kappa \geq U + 1 \tag{101}$$

when the restricted regime (97) is used. The last condition is omitted for an unrestricted minor-arc estimate. These inequalities involve only fixed exponents and not $E(B, m, d)$. They make (100) $O(NL^{-A-4})$. They also ensure $Q_* \leq (M^{-b_*})^{-\kappa}$. The other regime condition, $M^{-b_*} \geq N^{-\sigma}$, holds for all sufficiently large N: after fixing B the upper bound on M is a fixed power of $\log N$.

Only at this point use that upper bound, writing $M \leq L^{E_*}$ after enlarging the fixed exponent E_*. There are at most $O(L^{2E_*+D})$ cells of modulus at most L^{E_*}. Apply (95) with

$$C_0 = E_* + 1, \qquad A_0 = A + 2E_* + D + 4.$$

Summing the constant terms $m_{j,k} \sum_{n \in P_{j,k}} w_N(n)$ gives $O(NL^{-A-4})$. Combined with the centered estimates and freezing error, this proves (98). □

Remark I.26 (Order of parameter choices). The order in the proof is

$$(A, K_w, C_{\mathrm{init}}, c, C_1, K_1, a, C_2, \kappa, \sigma) \longrightarrow (D, U) \longrightarrow B \longrightarrow E_* \longrightarrow (A_0, C_0) \longrightarrow N_0.$$

Thus there is no requirement that B dominate the factorization upper exponent $E(B, m, d)$. The centered progression estimate allows every logarithmic saving, so its exponent can be chosen after E_* is known. The minor-arc input must have a fixed power form such as (96); unspecified polylogarithmic specializations alone would not justify this argument.

Corollary I.27 (Fixed-modulus centered nilsequence estimate)**.** *Fix the Gaussian ray field and a conjugation-invariant union C of ray classes as above. Let W be a fixed multiple of the conductor of $K^{\mathrm{ab}}/\mathbb{Q}$, and let b be a unit residue modulo W. For every $A, C_{\mathrm{init}} > 0$ and fixed dimension and degree bounds, uniformly over the nilmanifolds and polynomial sequences in Proposition I.25 with an $(\log(2N))^{C_{\mathrm{init}}}$-rational adapted basis,*

$$\left|\sum_{n\le N} h_{C,b,W}(n)F(g(n)\Gamma)\right| \ll_{H,W,A,C_{\mathrm{init}},m,d} N(\log(2N))^{-A}\|F\|_{\mathrm{Lip}}.$$

The constants are uniform in b and in the polynomial orbit. No uniformity as W varies is asserted.

Proof. We check the three hypotheses of Proposition I.25. The pointwise inequality $|h_{C,b,W}| \le \lambda_{b,W}$ gives its first hypothesis with $K_w = 1$. Corollary I.9 gives its second hypothesis for every A_0, C_0, in fact for every interval progression.

For the third hypothesis, take $H_0 = H$ in Corollary I.23. Its character expansion is (66), with the factor $1/2$ and the prime-power error supplied by (65) and the bound following that equation. Its two assumed minor-arc estimates are Proposition I.18 and Corollary I.19. Apply that corollary to the fixed linear combination defining $h_{C,b,W}$ in (71). It gives (96) with $c = c_W$, $C_1 = 1$ and $K_1 = 7$ in the regime $\delta \ge N^{-\sigma_1}$, $Q \le \delta^{-a}$. The affine-interpolation result initially restricts δ to a fixed sufficiently small interval. For larger δ, the bound $|h_{C,b,W}| \ll_W \log(2N)$ gives (96) directly after enlarging its fixed constant. For a dimension-zero quotient the mean-zero test function vanishes. For all other dimensions and degrees within the fixed bounds, take the minimum of the finitely many positive exponents c_W, a, σ_1 and the maximum of the fixed constants. Thus the restricted regime (97) holds with fixed positive κ, σ, uniformly throughout the required class.

Proposition I.25 now proves the displayed estimate. There are only finitely many unit residues modulo the fixed W, so their constants can be bounded by a single constant. In particular, division by N and the relabeling $W = V$, $N = L$ give the fixed-complexity estimate (102). □

I.8 From Gaussian nilsequence estimates to the polynomial count

Fix the Gaussian ray field $K/\mathbb{Q}(i)$ and the conjugation-invariant ray classes used in the preceding argument. Let D be divisible by the conductor of $K^{\mathrm{ab}}/\mathbb{Q}$. For each relevant union C of classes, let Λ'_C be its rational prime weight and let $\mu_C(b)$ be the conditional density in (108). In particular $0 \le \mu_C(b) \le 1$. We use the following conclusion of the preceding Gaussian norm, progression and factorization arguments, as assembled in Corollary I.27: for every fixed multiple V of D, every unit $b \pmod V$, every fixed bounded-complexity nilsequence $F(g(n)\Gamma)$, and every $B > 0$,

$$\begin{aligned} |\mathbb{E}_{n\le L} h_{C,V,b}(n)F(g(n)\Gamma)| &\ll_{V,B,\mathrm{complexity}} (\log L)^{-B}, \\ h_{C,V,b}(n) &= \frac{\varphi(V)}{V}\big(\Lambda'_C(Vn+b) - \mu_C(b)\Lambda'(Vn+b)\big). \end{aligned} \tag{102}$$

The bound is uniform in the polynomial orbit and functions with the stated complexity bound. The field, its ray modulus and V are fixed in this statement. We do not assume any uniform rate as V varies.

Published transference inputs. We use Green–Tao, *Linear equations in primes* [38], Proposition 6.4 and Appendix D for the ordinary prime majorant, Proposition 7.1 and Appendix C for its generalized von Neumann bound, and Lemma 4.4 with its subsequent normal-form reduction. These are statements about the ordinary prime majorant and arbitrary functions dominated by it;

they do not assert an Artin-prime theorem. We also use Dodos–Kanellopoulos [11], Theorem 5.1, in the following form, checked in arXiv:1610.00487v4, pp. 12–13. The final typeset statement has not been compared. For target interval norm $U^s[L]$, $s \geq 2$, its hypothesis on a nonnegative majorant on a prime cyclic group of size $L' \in [CL, 2CL]$, $C \geq 20$, is

$$\|\nu - 1\|_{U^{2s}(\mathbb{Z}/L'\mathbb{Z})} \leq \eta. \tag{103}$$

For a real function $|f| \leq \nu$ with $\|f\|_{U^s[L]} \geq \delta$, the conclusion is a correlation bounded below with a bounded Lipschitz linear nilsequence on an $(s-1)$-step nilmanifold. Its nilmanifold, Lipschitz bound and positive correlation threshold depend only on s, C, δ. The interval norm is the normalized zero-extension norm. In particular a target U^{u+1} requires majorant norm U^{2u+2}.

An auxiliary cutoff, distinct from the polynomial cutoff. Put

$$V_z = D \prod_{\substack{p \leq z \\ p \nmid D}} p.$$

Here z is an auxiliary cutoff used only to prove the next linear replacement statement. It is not the cutoff v defining W, A in (111). For every fixed integer $s \geq 2$, one can choose $z = z(L) \to \infty$ sufficiently slowly that

$$\max_{C,\ (b,V_z)=1} \|h_{C,V_z,b}\|_{U^s[L]} = o(1). \tag{104}$$

The maximum is over the fixed finite list of ray-class unions and the finitely many unit residues for the current V_z.

Here are all normalization and limit details. First choose the complexity of the ordinary prime majorant larger than both the requirements of the intended generalized von Neumann bound and $\max(4^s, 4s)$. Choose its ambient constant $C \geq 20$ large enough for both published majorant and generalized von Neumann statements. If $\lambda_{V,b}(n) = (\varphi(V)/V)\Lambda'(Vn + b)$, then $|h_{C,V,b}| \leq \lambda_{V,b}$. The majorant simultaneously dominates a fixed number of such ordinary weights on $[L^{3/5}, L]$, with a multiplicative constant depending only on the chosen complexities. Consequently there is $\alpha > 0$, depending only on the fixed complexities and the fixed factor in the modulus, such that $\alpha 1_{[L^{3/5},L]} h_{C,V_z,b}$ is dominated in absolute value by ν on $[L]$. The same α works for all cutoffs and unit residues. In applying the inverse theorem to a lower bound δ for the unscaled norm, use the lower bound $\alpha\delta/2$ for this truncated, scaled function, once the truncation error is at most $\delta/2$.

For sufficiently large z, $V_z = E \prod_{p \leq z} p$ with E fixed. The ordinary divisor-sum construction works with this V_z: at primes dividing E, the unit residue excludes every divisor term containing that prime, and at all other primes its local factors are unchanged. The same observation applies to the correlation upper bounds in Appendix D. Thus the stated pseudorandomness conditions remain valid, uniformly in unit residues, with constants allowed to depend on E. Expand the centered $(2s)$-cube of $\nu - 1$. It contains at most 4^s distinct forms, in $2s + 1$ variables, with coefficients zero or one. Every subproduct has mean $1 + o(1)$ by the linear-forms condition; inclusion–exclusion gives (103) with a left side tending to zero. This estimate is on the full cyclic group.

The removed initial interval contributes $o(1)$ to the interval U^s norm. Indeed the weights are $O(\log L)$ if $V_z = L^{o(1)}$, and a cube with its initial vertex in this interval has relative count $O(L^{-2/5})$. The 2^sth power of the norm of the removed function is at most $O((\log L)^{2^s} L^{-2/5})$. Its correlation with a bounded nilsequence is also $o(1)$.

If (104) fails by some fixed $\delta > 0$, Theorem 5.1 gives a fixed-complexity nilsequence correlation bounded away from zero, contradicting (102). There is no illicit varying-modulus use of that input.

For each integer z and each of the finitely many residues modulo V_z, first take a size threshold for the required nilsequence estimates. Diagonalize these thresholds also over $\delta = 1, 1/2, 1/3, \dots$ and over the finite list of norm orders needed. Impose in addition $z(L) \le \log\log\log L$ and all size thresholds of the majorant. At stage j, choose the cutoff z_j large enough for the limiting majorant errors to be below half the inverse-theorem tolerances needed for the first j tests; only then choose their size threshold L_j. The resulting increasing step function tends to infinity and proves the assertion for all sufficiently large L. The fixed smooth metric in the inverse theorem is comparable to a fixed Mal'cev metric, and a linear orbit with arbitrary base point is a polynomial orbit, so the nilsequence input applies to exactly the supplied correlations. When this construction is used at the refined scale $L \asymp N/V_{z_j}$ below, choose the physical threshold N_j so large that every constant multiple of N/V_{z_j} used by the normal-form reduction is at least L_j for $N \ge N_j$. Impose the same growth constraints there. Thus the cutoff is constant on each physical-scale stage; no implicit equation $z = z(N/V_z)$ is being assumed.

Lemma I.28 (Fixed-system linear replacement)**.** *Let $\Psi = (\psi_1, \dots, \psi_t) : \mathbb{Z}^d \to \mathbb{Z}^t$ have fixed nonzero integer linear parts of finite complexity. Allow its constant terms to vary with N, subject to $\|\Psi\|_N \le L_0$ for a fixed L_0. Uniformly over convex regions $\mathcal{K} \subseteq [-N, N]^d$ with $\Psi(\mathcal{K}) \subseteq [1, L_0 N]^t$,*

$$\sum_{\mathbf{n} \in \mathcal{K} \cap \mathbb{Z}^d} \left(\prod_{i=1}^{t} \Lambda'_{C_i}(\psi_i(\mathbf{n})) - \prod_{i=1}^{t} \mu_{C_i}(\psi_i(\mathbf{n}) \bmod D) \Lambda'(\psi_i(\mathbf{n})) \right) = o(N^d). \tag{105}$$

The same conclusion holds for a mixed product in which any selected factors have already been replaced by their ordinary-prime models.

Proof. We can discard points where some $\psi_i \le N^{9/10}$. For a nonconstant fixed linear form their number is $O_{\Psi, L_0}(N^{d-1}(N^{9/10}+1))$; the product of weights is $O((\log N)^t)$. This is $o(N^d)$, uniformly in the constant terms. Apply the auxiliary V_z-trick to the variables: $\mathbf{n} = V_z \mathbf{m} + \mathbf{r}$. If one output residue is not a unit modulo V_z, its prime weight vanishes on the remaining range, since the output is larger than every prime dividing V_z. For an admissible residue vector write

$$\psi_i(V_z \mathbf{m} + \mathbf{r}) = V_z \widetilde{\psi}_{i,\mathbf{r}}(\mathbf{m}) + b_{i,\mathbf{r}}, \qquad 1 \le b_{i,\mathbf{r}} \le V_z.$$

The new linear coefficients are the original ones. Its normalized constant terms remain bounded at scale N/V_z. Because $D \mid V_z$, the factor μ_{C_i} is constant on this residue vector.

Telescope the difference of the two products. After normalization by $(\varphi(V_z)/V_z)^t$, each term has one factor h_{C_i, V_z, b_i}; all other factors are nonnegative and at most λ_{V_z, b_j}. Introduce a common ordinary-prime majorant for these at most t weights and truncate as above. Multiply every factor by a common fixed $\alpha_0 > 0$ so that each has absolute value at most that majorant, and apply the normal-form extension of Green–Tao Lemma 4.4 and Proposition 7.1. Here α_0 depends only on the fixed system, the chosen complexities and the fixed conductor, not on z or the residue vector. The small factor has the required norm by (104). Dividing the resulting estimate by α_0^t changes only a fixed constant. The normal-form extension adds only a fixed number of variables with fixed coefficients; its convex-region reduction is the one following Lemma 4.4, and divides out their box volume. The error on each residue vector is therefore $o((N/V_z)^d)$, uniformly in the vector and convex region. The discarded initial segment stays outside the retained output range since $V_z = N^{o(1)}$.

One can make this reduction explicit. For each pair $i \ne j$ choose an integer vector v_{ij} with $\dot{\psi}_j(v_{ij}) = 0$ and $\dot{\psi}_i(v_{ij}) \ne 0$; fixed nonproportional linear parts ensure such vectors exist with fixed size. Replace $\mathbf{m}$ by $\mathbf{x} + \sum_{i \ne j} y_{ij} v_{ij}$ and allow each new y_{ij} to range over an integer interval of

length comparable to N/V_z. The block y_{ij} for fixed i certifies normal form. The region defined by the original convex constraint on this sum and the new box is convex. The integer change $(\mathbf{x},\mathbf{y}) \mapsto (\mathbf{x} + \sum y_{ij} v_{ij}, \mathbf{y})$ is bijective, so its sum is exactly the original sum times the number of points in the new box. Division by that number gives the asserted error. If there is only one form, choose an integer vector v with $\dot{\psi}_1(v) \neq 0$ and replace $\mathbf{m}$ by $\mathbf{x} + yv$. The same box and bijection argument gives the reduction in this case.

No growing factor is lost when summing residue vectors. If $\mathcal{R}_z$ is the set of admissible vectors, then

$$\frac{|\mathcal{R}_z|}{V_z^d} \left(\frac{V_z}{\varphi(V_z)} \right)^t = \prod_{p \mid V_z} \beta_p(\Psi) = O_{\Psi, L_0}(1). \tag{106}$$

Exclude the finitely many primes dividing a chosen nonzero coefficient of each linear part, and a chosen nonzero two-by-two minor for each pair of linear parts. At every remaining prime each output hyperplane has probability p^{-1}, and any two meet with probability p^{-2}. Hence $\beta_p = 1 + O_t(p^{-2})$, uniformly in the constant terms. This also covers $t = 1$, when there are no pair minors and the good-prime factor is exactly one. The finitely many remaining factors are bounded by $(p/(p-1))^t$. The Chinese remainder theorem proves the equality, also for the fixed prime powers dividing D. Multiplying the per-residue error by this bounded mass gives $o(N^d)$. The proof works unchanged for the mixed products. All auxiliary thresholds can be chosen uniformly over bounded constant terms and convex regions, because the published majorant and generalized von Neumann conclusions have this uniformity. □

The exact local cube needed in the appendix. Now fix the appendix cutoff v, and put W, A, n, R as in its proof. This is a new limiting argument; the auxiliary cutoff in the proof of Lemma I.28 has already been eliminated from that lemma's statement. Let

$$f_{C,b}(x) = \frac{\varphi(W)}{W} \Lambda'_C(Wx + b) 1_{R \le x \le n}, \qquad g_{C,b} = f_{C,b} - \mu_C(b).$$

For fixed $a \ge 2$ and $C_1 < C_2 < C_3$, uniformly over $q \le A^{C_1}$ and $nA^{-C_3} \le J \le nA^{-C_2}$,

$$\|g_{C,b}\|_{U^a_{q[J]}}^{2^a} \ll_{a,C_1,C_2,C_3,\rho,D} \frac{1}{\log v} + o_v(1). \tag{107}$$

The norm uses the difference multiset $q(h - h')$, $h, h' \in [J]$, on $\mathbb{Z}/n\mathbb{Z}$.

To prove this, remove the cutoff $x < R$ at cost $O((\log n)^{2^a} R/n) = o_v(1)$. Expand the centered cube and partition its domain according to every wrap-around integer m_ω. The relevant affine forms are precisely

$$W \left(x + q \sum_{r=1}^{a} \omega_r (h_r - h'_r) - m_\omega n \right) + b, \qquad \omega \in \{0,1\}^a.$$

They have pairwise nonproportional linear parts. Each region is convex; there are $O_a(1)$ regions because $qJ/n < 1$. For fixed v, the number of possible q is finite and J/n stays in a fixed positive compact interval. Apply Lemma I.28 to every subproduct and region. Its error is $o_v(n^{2a+1})$ uniformly in J, and division by nJ^{2a} gives $o_v(1)$. Since every output is b modulo D, all its conditional densities are the same $\mu_C(b)$. Thus the centered Artin cube equals $\mu_C(b)^{2^a}$ times the centered ordinary-prime cube, plus $o_v(1)$.

The ordinary cube is evaluated exactly as in Tao–Ziegler [12], Proposition 6, published pages 39–40, using the ordinary-prime linear correlation theorem, for example the published formulation in

Bienvenu [3], Theorem 1.1, pages 79–80. The factors at $p \mid W$ cancel the normalization. For $p \nmid Wq$, pairwise independent gradients give local factor $1 + O_a(p^{-2})$, uniformly in the wrap constants. For $p \mid q$, $p > v$, the factor is $1 + O_a(p^{-1})$. Since $\log q = O_{C_1,\rho,D}(v)$, there are $O(v/\log v)$ such prime divisors. Their product is $1 + O(1/\log v)$, and the other large-prime factors give $1 + O(1/v)$. Centering cancels the common main term. Prime powers, if the ordinary theorem is stated for Λ, contribute $o_v(1)$ by their $O(\sqrt{n}\log^2 n)$ total weight and the fixed linear-fiber bound. This proves (107), and hence supplies (111) without an Artin linear-correlations theorem.

Completion by the published polynomial transfer. The remainder of the proof of Proposition I.29, starting with its arbitrary-function dense-model step, now applies unchanged to these fixed Gaussian ray fields. Specifically use the published Tao–Ziegler [12] Theorem 11 and Corollary 3, then Theorem 9 and Theorem 13. The model for an arbitrary $f \le \lambda_{W,b}$ is used; one does not assert that an Artin weight has model one. Equation (107) shows that its bounded model, minus $\mu_C(b)$, has small global norms. The model is bounded by a fixed constant depending on the chosen sieve exponent. Divide its difference from $\mu_C(b)$ by a fixed bound for that difference before applying Theorem 9, whose hypothesis is a bound by one. After restoring this fixed factor, concatenation and model approximation give the required local norm bounds. None of these scaling constants depends on v.

Keep the original parameter order: local box order and coefficient bound first, then $B_0 > C+2$, then the global order a, and finally small ρ. Only then let v increase. The quantities $M_0 = \lfloor T/W \rfloor$, $M_1 = \lfloor M_0/A \rfloor$ and $J = \lfloor A^{-2B_0 a} M_1^2 \rfloor$ satisfy the size hypotheses recorded in the appendix when $T/\sqrt{N} \ge 4\sqrt{W}/A$. For each fixed v the estimates are uniform for T in that allowed range. The final assertion $T = o(\sqrt{N})$ is imposed only along the subsequent diagonal $v = v(N) \to \infty$, not while v is frozen.

After $u = Wx + b$, $w = Wm + c$, the polynomials are $c_j W m^2 + 2c_j cm + (b + c_j c^2 - b_j)/W$, with distinct leading coefficients. The bounded-model transfer kills every term containing a centered factor. Output-boundary strips have width $O(T^2) = o(N)$. For completeness, dominate them first by strips of width ϵN. Here is the needed localization of the ordinary majorant estimate. This argument also covers negative coefficients c_j. For the true integer count use an enlarged period $n_* = \lceil BN/W \rceil$, where $B > 2 + \max_j |c_j|$ is fixed. We may assume $T \le \sqrt{N}$. Nonpositive outputs have prime weight zero, and every positive output is at most $(1 + \max_j |c_j|)N$. Discard also tuples for which some positive output is at most $W\lceil n_*^\rho \rceil + W$. For each w there are $O_k(Wn_*^\rho + W)$ such u, so their total weight is $O_k((Wn_*^\rho + W)T(\log N)^k) = o_v(NT)$, uniformly in the allowed T range. All remaining true outputs have their ordinary prime majorants on $[n_*]$ without wrap-around. Use the same localization calculation on the original period n to bound the artificial wrapped terms of the cyclic count. The parameters n_* and n differ by a fixed factor, so both applications satisfy the same polynomial-forms hypotheses. In the following calculation n denotes either chosen ambient period. Write $\nu_0(x)$ for the majorant factor with constant shift if there is one, and set $\nu_0 = 1$ otherwise. There is at most one such factor. Put $F(x) = \mathbb{E}_m \prod_i \nu_i(x + P_i(m))$, omitting that factor. Tao–Ziegler, Theorem 15 in Appendix A (published page 57), applied to $\nu_0 F$, $\nu_0 F^2$ and ν_0, gives

$$\mathbb{E}_x \nu_0(x)(F(x) - 1)^2 = o_{v\to\infty}(1) + o_v(1).$$

In the square use independent variables m, m'; the differences between all its nonconstant shift polynomials are nonconstant, including those from distinct copies. Weighted Cauchy–Schwarz now shows, for every interval I in the x variable,

$$\mathbb{E}_x 1_I(x)\nu_0(x)F(x) = \mathbb{E}_x 1_I(x)\nu_0(x) + o_{v\to\infty}(1) + o_v(1).$$

For fixed relative interval length the one-dimensional divisor-sum mean estimate gives $\mathbb{E}_x 1_I \nu_0 = |I|/n + o_{v\to\infty}(1) + o_v(1)$; it follows either directly from the ordinary majorant construction or from its linear-forms estimate on that fixed interval. The normalized prime weights are bounded by fixed multiples of their sieve majorants, with constants depending on the fixed sieve exponent. Thus a strip of width ϵN has total weight $[O(\epsilon) + o_{v\to\infty}(1) + o_{v,\epsilon}(1)]NT$, with that fixed domination factor included in the implied constants. The errors here have the same iterated order as the rest of the polynomial transfer. Then take $\epsilon \downarrow 0$ as described below. This avoids a spurious factor $(\log N)^k$ from a pointwise bound. The same positive majorant treats interval endpoints.

Choose increasing cutoffs v_j for which the global/local errors and $4\sqrt{W(v_j)}/A(v_j)$ tend to zero, and then choose size thresholds valid for all the finitely many residues, step values and transfer parameters at that cutoff. At stage j impose the boundary estimates for every $\epsilon \in \{1, 1/2, \dots, 1/j\}$, preserving all earlier tests. Let $v(N) = v_j$ between successive thresholds and take $\omega(N) \geq 4\sqrt{W(v(N))}/A(v(N))$, tending to zero sufficiently slowly. This gives the uniform assertion for $\omega(N)\sqrt{N} \leq T = o(\sqrt{N})$. Indeed, for an arbitrary such T, first fix k; eventually $T^2/N < 1/k$ and the estimate for the strip of width N/k applies. Take the upper limit as $N \to \infty$ and then $k \to \infty$. There is no requirement that $T^2/N < 1/j(N)$ along the chosen stages. The main term is precisely (110); the admissible residue mass is bounded, so summation does not amplify errors. The $u \in [N/2, N]$ version follows by using that convex interval directly, or by subtraction with thresholds enlarged for $N/2$.

This bridge uses the separately established Gaussian nilsequence input and published ordinary-prime transference. It uses no statement or estimate from Frei–Tinková as a theorem.

I.9 Fixed Artin conditions in quadratic patterns

Here the field is fixed throughout the limiting argument. Let K be the ray class field of $\mathbb{Q}(i)$ for $H\mathbb{Z}[i]$, where $16 \mid H$. Put $G = \mathrm{Gal}(K/\mathbb{Q})$, and let $C_1, \dots, C_k$ be conjugation-invariant subsets of $\mathrm{Gal}(K/\mathbb{Q}(i))$, regarded as conjugacy-invariant subsets of G. Let D be divisible by the conductor of K^{ab}. For a unit $b \pmod D$, define

$$\mu_j(b) = \frac{\#\{g \in C_j : g|_{K^{\mathrm{ab}}} = \sigma_b|_{K^{\mathrm{ab}}}\}}{[K : K^{\mathrm{ab}}]}, \qquad \sigma_b(\zeta_D) = \zeta_D^b, \tag{108}$$

and set $\mu_j(b) = 0$ for nonunits. Define $\Lambda'_j(p) = \log p$ for unramified primes with arithmetic Frobenius in C_j, and zero otherwise.

Proposition I.29 (Fixed-field polynomial count)**.** *There is a function $\omega(N) \to 0$, depending on the fixed data, such that whenever $\omega(N)\sqrt{N} \leq T = o(\sqrt{N})$,*

$$\sum_{u\leq N} \sum_{1\leq w\leq T} \prod_{j=1}^{k} \Lambda'_j(u + c_j w^2) = \mathfrak{S}\, NT + o(NT), \tag{109}$$

where

$$\mathfrak{S} = \left(\frac{D}{\varphi(D)}\right)^k \mathbb{E}_{u,w \bmod D} \prod_j \mu_j(u + c_j w^2) \prod_{p\nmid D} \beta_p, \quad \beta_p = \mathbb{E}_{u,w \bmod p} \prod_j \frac{p}{p-1} 1_{p\nmid u + c_j w^2}. \tag{110}$$

The same statement on $N/2 \leq u \leq N$ has main term $\mathfrak{S}NT/2$.

Proof. The Gaussian linear replacement of Lemma I.28 and its local-cube consequence (107) supply the analytic input. We record the normalization and polynomial-transfer parameters here.

Choose a small positive ρ and first keep v fixed. Put

$$W = D\prod_{\substack{p\le v\\ p\nmid D}} p, \quad A = W^{1/\rho}, \quad n = \lfloor N/W\rfloor, \quad R = n^{\rho},$$

and, for unit $b \pmod W$, put

$$f_{j,b}(x) = \frac{\varphi(W)}{W}\Lambda'_j(Wx+b)1_{R\le x\le n}, \qquad g_{j,b} = f_{j,b} - \mu_j(b).$$

View these functions on $\mathbb{Z}/n\mathbb{Z}$, with representatives $1,\dots,n$. If $a\ge 2$ and $C_1 < C_2 < C_3$ are fixed, (107) gives

$$\|g_{j,b}\|^{2^a}_{U^a_{q[J]}} \ll_a (\log v)^{-1} + o_v(1), \quad q\le A^{C_1}, \quad nA^{-C_3}\le J\le nA^{-C_2}. \tag{111}$$

The implicit constant may also depend on the other fixed complexities and the fixed ray field. The norm, wrap-around regions, conditional densities, and order of the auxiliary cutoff are all treated explicitly in Section I.8.

We now use the arbitrary-function dense-model theorem of Tao–Ziegler [12]. Their published Theorem 11 and Corollary 3 give a single bounded model accurate in both global Gowers norms and the local polynomial box norms. Their Theorem 9 is the bounded-function concatenation statement, and their Theorem 13 bounds the polynomial average by one of those local norms. All majorants here are the ordinary prime majorants, since $0\le\Lambda'_j\le\Lambda'$. The fixed factor D in W is allowed in their divisor-sum construction: for sufficiently large v, $W = E\prod_{p\le v}p$ with a fixed E whose prime factors are at most v. A unit residue makes every small-prime divisibility condition identically false, including at the extra prime powers. At the remaining primes the divisor-sum local factors are unchanged. Polynomial coefficient bounds absorb the fixed factor E. Thus the polynomial-forms estimates for the majorant have the same $1+o(1)$ conclusion.

The following choice records the size hypotheses of these results. First choose their local box order d' and a coefficient exponent C; choose an integer $B_0 > C+2$, then a sufficiently large global order a, and finally a sufficiently small ρ. Set $\rho_* = \rho/B_0$ and $\mathcal{A} = A^{B_0}$. Put $M_0 = \lfloor T/W\rfloor$, $M_1 = \lfloor M_0/A\rfloor$. For $T/\sqrt{N}\ge 4\sqrt{W}/A$ one has

$$A^{-1}\sqrt{n}\le M_0\le\sqrt{n}, \qquad A^{-3}\sqrt{n}\le M_1\le\sqrt{n}.$$

The model at cutoff n^{ρ_*} dominates the function already truncated at n^{ρ}. Corollary 3 supplies approximation for the degree-one local step polynomials, whose coefficients are $O(A^C)$, and simultaneously for $q\le A^{B_0a}$ with $J = \lfloor A^{-2B_0a}M_1^2\rfloor$. In particular

$$nA^{-2B_0a-7}\le J\le nA^{-2B_0a}.$$

This is within (111). Let $f'_{j,b}$ be the model supplied by Corollary 3 at cutoff n^{ρ_*}. Its bound is $0\le f'_{j,b}\le B_*-1$, where $B_*\ge 2$ depends on the fixed ray field, complexities and on ρ_*, but not on v,n,b or the polynomials within the stated coefficient bounds. This is the bounded model of Tao–Ziegler's Proposition 4: its construction first rescales the prime weight by a fixed positive constant and then divides the model by that constant. In particular, it need not be bounded by one. Since $0\le\mu_j(b)\le 1$, the function

$$F_{j,b} = \frac{f'_{j,b}-\mu_j(b)}{B_*}$$

is bounded in magnitude by one. The triangle inequality, the global part of Corollary 3, and (111) show that its required global norms tend to zero under the diagonal limit below. Apply Theorem 9

to $F_{j,b}$ with parameters $d = d_0 = 2$, $M = M_1$, $A = \mathcal{A}$, and global order a. Its smallness threshold is independent of $\mathcal{A}$; B_* has already been fixed before v is chosen. Multiplication by B_* and the local part of Corollary 3 therefore show that the local norm of $g_{j,b}$ tends to zero. Theorem 13, with the original parameter A and coarse scale M_0, makes every polynomial average containing a centered factor $o(1)$. These are the reasons for choosing the complexities before v.

For the final W-trick write $u = Wx + b$ and $w = Wm + c$. If b_j is the residue of $b + c_j c^2$ modulo W, the resulting integer polynomial is

$$c_j W m^2 + 2c_j c m + (b + c_j c^2 - b_j)/W.$$

Its leading coefficient differences are nonzero, and its coefficients are $O(W)$. Expand the product into its means and centered factors and apply the preceding conclusion. End intervals of length $O(T^2) = o(N)$ are controlled by the weighted second-moment localization and the fixed-strip argument in Section I.8. Summing residue classes does not amplify the error: their normalized admissible mass is $\prod_{p|W} \beta_p = O(1)$. Indeed, outside the fixed coefficient-difference primes,

$$\beta_p = \left(\frac{p}{p-1}\right)^k \frac{(p-1)(p-k+1)}{p^2} = 1 + O_k(p^{-2}).$$

This also proves convergence of the product in (110). Let N tend to infinity for each fixed v and then let v tend to infinity sufficiently slowly. A diagonal choice supplies $\omega(N)$ and yields (109). Subtract the corresponding formula with upper limit $N/2$, enlarging ω if necessary, to obtain the last assertion. □

Remark I.30 (Published numbering). The labels in the published 2018 paper differ from those in arXiv 1603.07817v1. The relevant dictionary is

arXiv	Theorem 1.4	Theorem 2.5	Theorem 3.1	Corollary 3.9	Theorem 4.1
published	Theorem 4	Theorem 9	Theorem 11	Corollary 3	Theorem 13

The published locations are respectively pages $4, 21, 28, 38, 41$. The preprint's equations (3.5) and (4.1) are published equations (36) and (57). The dense model used above is the theorem's assertion for an arbitrary function bounded by the prime weight. The special model 1 for the ordinary prime weight is not an assertion about Artin weights.

I.10 The ray-class lower bound

For ray class fields we use the published existence and reciprocity statements in Kopp–Lagarias [13], Theorems 1.1, 1.3 and 7.2, and the modulus exact sequence of Theorem 6.5, specialized to the maximal order $\mathbb{Z}[i]$. There are no real places. This replaces an appeal to unpublished course notes. In this principal ideal domain the ray group for the finite modulus $H\mathbb{Z}[i]$ is

$$(\mathbb{Z}[i]/H\mathbb{Z}[i])^\times/\{1, -1, i, -i\}. \tag{112}$$

It also follows directly by sending a residue r to the ideal class of (r): surjectivity is principality, and the kernel consists exactly of unit residues. The Artin map sends this ideal class to arithmetic Frobenius. Conjugation acts by $r \mapsto \bar{r}$.

Choose a fixed L_1 larger than all coefficient-difference primes and larger than 11. Let H_0 be divisible by $16 \cdot 7M$ and by all odd primes at most L_1. Prescribe a unit residue $r \pmod{H_0}$ which is 9 modulo 16, $1 + 3i$ modulo 7, and i modulo every other odd prime dividing H_0. Compatible

higher powers can be chosen arbitrarily. The associated rational-prime Frobenius condition is the two-element conjugacy class $C_0 = \{h_r, h_{\bar{r}}\}$ in the Galois group G_0 of the ray field K_0 over $\mathbb{Q}$. The two elements are distinct: an equality of their ray classes would require $r = \epsilon\bar{r} \pmod{H_0}$ for a Gaussian unit ϵ; modulo 16 this forces $\epsilon = 1$, contradicted modulo 7. The field is Galois over $\mathbb{Q}$, because conjugation preserves the modulus and the defining ray subgroup.

Let D_0 be the conductor of K_0^{ab} and choose a unit $b_0 \pmod{D_0}$ inducing the restriction of h_r. The value in (108) is then $\mu_0 = |C_0|/|G_0'| > 0$. For $L \geq L_1$ let $T_L = \prod_{L_1 < \ell \leq L,\, \ell \nmid H_0} \ell$ and let K_L be the ray field for $H_0 T_L \mathbb{Z}[i]$. For each new prime write

$$B_\ell = (\mathbb{Z}[i]/\ell\mathbb{Z}[i])^\times, \qquad N_\ell = \ker(B_\ell \xrightarrow{\mathrm{N}} \mathbb{F}_\ell^\times).$$

The norm map is onto and $|N_\ell| = \ell - \chi(\ell)$.

Lemma I.31 (Abelian fibres and their surviving proportion)**.** *Put $N_L = \prod_{\ell \mid T_L} N_\ell$ and $G_L = \mathrm{Gal}(K_L/\mathbb{Q})$. Then*

$$N_L \subseteq G_L', \qquad G_L'/N_L \simeq G_0', \qquad K_L^{\mathrm{ab}} = K_0^{\mathrm{ab}}\mathbb{Q}(\zeta_{T_L}).$$

Let C_L consist of those elements above C_0 whose normalized Gaussian residue z satisfies $\mathrm{N}(z-1)\mathrm{N}(z+1) \not\equiv 0 \pmod{\ell}$ at every $\ell \mid T_L$. For the density μ_L of this set,

$$\mu_L(b) \geq \mu_0 \prod_{\ell \mid T_L} \left(1 - \frac{4}{\ell - 1}\right) \qquad (b \equiv b_0 \pmod{D_0},\ (b, D_0 T_L) = 1). \tag{113}$$

Proof. In the CRT presentation of (112), the unit group embeds in the H_0 factor, because it already embeds modulo 16. Once the representative above r or $\bar{r}$ is normalized, the kernel of reduction to the base ray group is exactly $\prod B_\ell$. There is no further quotient by the four units.

The map $z \mapsto z/\bar{z}$ from B_ℓ onto N_ℓ is surjective. For split primes this is $(x, y) \mapsto (x/y, y/x)$. For inert primes it is $z \mapsto z^{1-\ell}$ in the cyclic group $\mathbb{F}_{\ell^2}^\times$. These maps are commutators with complex conjugation, so $N_L \subseteq G_L'$. The norm on each new residue factor gives the cyclotomic action $\zeta_\ell \mapsto \zeta_\ell^{\mathrm{N}(z)}$. This is also seen by applying arithmetic Frobenius to ζ_ℓ; it is compatible with the ray map by reciprocity. The reduction map gives $K_0 \subseteq K_L^{N_L}$. The cyclotomic action just computed is trivial on N_L, so also $\mathbb{Q}(\zeta_{T_L}) \subseteq K_L^{N_L}$. The fields K_0 and $\mathbb{Q}(\zeta_{T_L})$ are disjoint over $\mathbb{Q}$. Indeed, K_0 is unramified at every prime dividing T_L, whereas the inertia groups at these primes generate the Galois group of $\mathbb{Q}(\zeta_{T_L})/\mathbb{Q}$. Thus their intersection is trivial. Moreover, the kernel computation above gives

$$[K_L^{N_L} : \mathbb{Q}] = [K_0 : \mathbb{Q}] \prod_{\ell \mid T_L} \frac{|B_\ell|}{|N_\ell|} = [K_0 : \mathbb{Q}]\varphi(T_L).$$

This is the degree of $K_0\mathbb{Q}(\zeta_{T_L})$, proving

$$K_L^{N_L} = K_0\mathbb{Q}(\zeta_{T_L}), \qquad G_L/N_L \simeq G_0 \times \prod_{\ell \mid T_L} \mathbb{F}_\ell^\times.$$

Since $N_L \subseteq G_L'$, taking commutators in this quotient gives $G_L'/N_L \simeq G_0'$ and $K_L^{\mathrm{ab}} = K_0^{\mathrm{ab}}\mathbb{Q}(\zeta_{T_L})$.

Fix a nonzero norm $s \in \mathbb{F}_\ell^\times$. For $z = a + bi$ in that norm fibre, $\mathrm{N}(z \pm 1) = s + 1 \pm 2a$. For each sign, vanishing fixes a and leaves at most two values of b. Thus at most four of the $\ell - \chi(\ell)$ elements are excluded. The conditions are invariant under conjugation, so C_L is a union of rational conjugacy classes. In every compatible abelian fibre its numerator in (108) is at least $|C_0| \prod(\ell - \chi(\ell) - 4)$, and its denominator is $|G_0'| \prod(\ell - \chi(\ell))$. Their quotient proves (113). □

Proposition I.32 (Finite-cutoff lower bound)**.** *There is a constant $c > 0$ independent of L such that the singular series in Proposition I.29 for the fixed field K_L and the set C_L at every position satisfies*

$$\mathfrak{S}_L \geq c(\log L)^{-4k}.$$

Proof. At the base modulus retain just $u = b_0$, $w = 0 \pmod{D_0}$. At each new prime retain all pairs on which the k norm values are units. Apply (113) separately to each position. The norm-fibre denominators cancel as shown in the previous lemma; a factor inverse to the degree of K_L is not left in the bound. The normalized proportion of the retained pairs at each new prime is $\beta_\ell = 1 + O_k(\ell^{-2})$. Each is positive, since $w = 0$, $u \neq 0$ is allowed. Their product, together with the tail Euler product, is bounded below by a fixed positive number. Williams' theorem with modulus 1, or its product for the two reduced classes modulo 4, gives

$$\prod_{\ell \mid T_L} (1 - 4/(\ell - 1))^k \gg (\log L)^{-4k}.$$

The base modulus contributes another fixed positive factor. □

I.11 The final order of limits

Proof of Theorem I.1. Fix the coefficients, M, and the base modulus first. Choose one fixed $L \geq \max(L_1, L_*)$ so large that

$$\frac{C}{L \log L} < \frac{c}{32(\log L)^{4k}},$$

where C, c are from the obstruction and lower-bound propositions. Now fix K_L, C_L. Only after this choice do we use the asymptotic polynomial theorem or let its small-prime cutoff tend to infinity.

Choose numbers $\tau_j \downarrow 0$, all at most τ_0. Choose successively N_j so large that for $N \geq N_j$ $\omega(N) \leq \tau_j/2$, that $\tau_j\sqrt{N} \geq 2$, and that the error term in (64) is less than $c\tau_j N^{3/2}/(32(\log L)^{4k})$. This is possible for each fixed τ_j. For $N_j \leq N < N_{j+1}$ use $\tau(N) = \tau_j$ and define $T = \lfloor \tau(N)\sqrt{N} \rfloor$. Then $T = o(\sqrt{N})$ and $T \geq \tau\sqrt{N}/2$. Apply Proposition I.29 to this piecewise choice of T. For all sufficiently large N its lower count is at least $cNT/(4(\log L)^{4k})$ and exceeds the full obstruction tail. Hence a pattern remains with no forbidden odd common divisor greater than L. The ray conditions exclude every odd divisor at most L from each individual adjacent norm. At 7 the adjacent norms of $1+3i$ are 2 and 6; at every other odd base prime the adjacent norms of i are 2. Thus no forbidden odd common divisor remains at all.

The residue modulo 16 gives, on writing $a = 9 + 16A$ and $b = 16B$,

$$(a+1)^2 + b^2 \equiv 4 \pmod 8, \qquad (a-1)^2 + b^2 \equiv 64 \pmod{128}.$$

These yield the exact two-adic valuations and hence the cross-sign gcd 4. The residue modulo M gives coprimality with M. Finally $u \geq N/2$ and $w \leq \tau\sqrt{N}$ imply $u/w^2 \geq 1/(2\tau^2) \to \infty$. For sufficiently small τ_0, all $u + c_j w^2$ are between fixed positive multiples of N, so they exceed Y when N is large. The dilation at the start imposes $q \mid w$. □

J The odd-product quartics in 523 and 540 variables

The following construction incorporates the November 7 contribution of Gabriel Conant, developed with ChatGPT. We give its arithmetic argument, including the connection to the results proved above. Variable counts include the input. The older quartics remain as source and proof baselines.

Theorem J.1 (The Version 9 quartics)**.** *There are specified integer polynomials P_{523} and P_{540}, of total degree four in respectively 523 and 540 variables, such that*

$$q \in \mathbb{Z} \iff \exists v \in \mathbb{Q}^{522} : P_{523}(q, v) = 0 \iff \exists w \in \mathbb{Q}^{539} : P_{540}(q, w) = 0.$$

The first has 35,163 nonzero monomials and largest coefficient magnitude $64 \cdot 3^{2^{27}}$. Its coefficient table uses the fixed integer $S_0 = 3^{2^{26}}$. The second has 35,213 nonzero monomials and largest coefficient magnitude $3^{2048}+1005072314631333 < 2^{3247}$. Its complete decimal table occupies 708,848 bytes. Both have content one.

The proof is the composition of the source equivalences, local and global arguments, and exact polynomial certificates below. The intermediate 554-variable construction is included to separate the odd-product argument from the additional formal-depth refinement. It is not a further claimed improvement over the two polynomials in the theorem.

J.1 Inherited inputs and the exact changes of convention

Use the normalized curve and generator of Section 17.1, with $P = P_*$, not a new generator. The coefficient basis is $(1, \sqrt{2})$ and conjugation sends $a + b\sqrt{2}$ to $a - b\sqrt{2}$. At unramified finite places $v(p) = 1$; at ramified places the integral valuation has value group $\mathbb{Z}$. Heights are absolute logarithmic Weil heights, with the local degree weights divided by $[K : \mathbb{Q}] = 2$. Throughout this section set $\mathcal{H} = H_{19}$ of Theorem 15.2.

The source is the system in Section 17.2, with the following explicit changes. Replace the comparison targets $[4]P_*, [4]Q_i$ by P_*, Q_i; replace the orders 16 and 52 by 12 and 40 before the further refinement below. Retain the sixteen-test product, the same five contacts, the scalar norm tree with orders $(1, 1, 1, 1, 32, 32)$, the Boolean correction and the preparation-zero pole equations. Change β to $2X^2+3$; use the point $S-P$ only in the scalar and physical-anchor blocks; and change the fixed coefficient S to $S_0 = 3^{2^{26}}$. The point-chart replacement, the three new scales, and the final order-one/order-forty test list are given below. All unchanged quotient guards and permitted zero values of pole witnesses are retained. This description, the displayed equations below, and the pinned rational straight-line program specify the source without a choice of arithmetic witnesses.

Input	Exact result used and translation
Elliptic labels and domains	Section 17.1, Lemma 17.1, and (17): $2E(K) = \mathbb{Z}P_*$, nonzero coordinate domains, and the same lower bound $h(x([n]P_*)) \geq n^2 - 6$.
Height of rational contacts	Theorem 15.2, with $\mathcal{H} < 2^{2093966}$, applied through the fractional linear map and contact calculation of Lemma 17.6. Only that calculation is reused; its exponent 24 is replaced by 19.
Pole obstruction and realization	Lemmas 14.4, 14.6, and 17.4. These include the restricted-converse dependency proved in the earlier arithmetic sections; it has not been discarded.
Scalar and anchor	Lemma 16.1, with its proof applied to $2X^2 + 3$ and $S - P$, and Lemma 17.3 with the smaller exponent calculated just below.
Local comparisons	Lemma 17.2 and the recurrence of Lemma 15.4; orders and target changes are justified below, not inferred by renaming indices.
Global choices	Theorem I.1, Lemma 17.7, Proposition I.29, Section I.11, and Lemma 16.3. All fixed data precede the asymptotic prime choice.
Final contradictions	Lemma 14.9 and the explicit new height inequalities below.

The published results behind these inputs retain the versions and convention qualifications in the theorem-application inventory. The new contribution does not supply an assumed replacement for the restricted converse, the height theorem, or the Gaussian prime argument.

Lemma J.2 (Direct targets and smaller anchor)**.** *For the changed source above, the local comparison and completeness arguments remain valid with direct targets. The scalar anchor on $R = [a-1]P_*$ implies*

$$0 \neq a - 1 \in 2\,3^{2^{24}-1}\mathbb{Z}, \qquad \log_2 |a-1| > 25165823.5, \qquad h(\beta) \leq \mathcal{H}|a-1|^{19/32}.$$

It permits arbitrarily divisible positive scalar indices $a - 1$.

Proof. For direct targets set $V_0 = H - P_*$ and $V_i = U_i - Q_i$, with $H = [h]P_*$, $U_i = [u_i]P_*$ and $Q_i = [n_i]P_*$. The quotient guards give finite nonidentity points. At a consumed prime the full logarithm ratios are u_i/h and $(u_i - n_i)/(h-1)$. One of $h, h-1$ is a unit. As in Lemma 17.5, the cross congruence first makes both approximate ratios integral, and then makes their relative errors absolute. Subtraction of the two integral congruences gives $L_i \equiv n_i$, without division by four. The logarithm recurrence and denominator bound in Lemma 15.4 give the same estimates at orders 12 and 40. The test list below strengthens the input order to 40 in the final refinement.

For completeness, the reduced elliptic groups have all four rational two-torsion points, since $p > 76176$ and the three roots $0, \pm d$ are distinct. Thus their doubling images have order $d_\pm(p)/4$. The point P_* is itself a double, so these integers annihilate its reductions. The prime theorem and Lemma 17.7 therefore give, on setting $M_\pm = \operatorname{lcm}_p(d_\pm(p)/4)$,

$$\gcd(M_-, M_+) = 1, \quad \gcd(M_-, L) = 16, \quad \gcd(M_+, L) = 1.$$

Here $64 \mid L$, the exact two-parts are 2^6 and 2^2 before division by four, and all odd prime divisors of L were avoided. Consequently $h \equiv 0 \pmod{M_-}$, $h \equiv 1 \pmod{M_+}$ and $h \equiv h_0 \pmod{L}$ are compatible when $64 \mid h_0$. The real-progression argument supplies large positive choices and avoids the finite forbidden point and quotient loci. The later prime-separation lemma justifies adding the factor Π^{19}.

For the anchor, repeat the proof of Lemma 17.3 with $C = 3^{2^{25}}$ and $\mathsf{S} = C^2 = S_0$. Its quaternion criterion and anisotropic binary form are unchanged. Thus the two rational coefficients of $f(R)/C$ belong to $\mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}$. The exact local calculation of Section 14.2.2 now gives $a - 1 \in 2\,3^{2^{24}-1}\mathbb{Z} \setminus \{0\}$, and

$$\log_2 |a-1| \geq 1 + (2^{24} - 1)\log_2 3 > 1 + \tfrac{3}{2}(2^{24} - 1) = 25165823.5.$$

For the scalar proof the new label is at least three, its five-contact product never vanishes, and all its negative valuations at the consumed odd primes are even. These are exactly the properties of X^2+2 used in Lemma 16.1; they hold for $2X^2 + 3$ by the calculation below. That proof therefore gives $M_5(\beta)^{32} \mid a - 1$. Applying Theorem 15.2 gives the asserted exponent 19/32. Its finite correction and continuity arguments apply to the same norm tree with the scalar point R; choosing $a - 1$ sufficiently divisible meets the anchor and any further prescribed divisibility. □

J.2 An exact replacement of the point chart

Put $K = \mathbb{Q}(\sqrt{2})$, $\delta = 3 - \sqrt{2}$ and $D = \delta^2 = 11 - 6\sqrt{2}$. The old chart, with letters in K, is

$$u = t^2, \qquad x = h^2, \qquad u(x - u/4) = D, \qquad y = h(u/2 - x). \tag{114}$$

The new chart is

$$x = h^2, \qquad j^2 = x^2 - D, \qquad y = hj. \tag{115}$$

An equation over K means its two rational component equations. The last expression in (115) need not be assigned its own coordinates.

Lemma J.3. *If $U, V \in K$, $U \neq 0$, and $V^2 = U^3 + 4DU$, then U is a square in K.*

Proof. The ring $\mathcal{O} = \mathbb{Z}[\sqrt{2}]$ is a PID: rounding both coefficients of an element of K to nearest integers leaves a remainder of absolute norm at most $1/2$, giving a Euclidean algorithm. Its units are $\pm(1+\sqrt{2})^n$, and every totally positive unit is a square. For completeness, multiply a positive unit by a power of $1+\sqrt{2}$ to place its first real embedding in $[1, 1+\sqrt{2})$. Its other embedding is the positive or negative reciprocal. Writing the unit as $a + b\sqrt{2}$ then forces $a, b \in \{0, 1\}$, and the norm condition and excluded upper endpoint leave only the unit 1. The two signs of the fundamental unit under the real embeddings give the assertion about totally positive units.

Let $m = v_{\mathfrak{p}}(U)$, with integral normalization. Away from $(\sqrt{2})$ and (δ) the two terms U^3 and $4DU$ have valuations $3m$ and m. If m is odd, their unequal minimum is odd, a contradiction. At $(\sqrt{2})$ these valuations are $3m$ and $m + 4$, with equality only at the even value $m = 2$. Again m cannot be odd. At (δ) they are $3m$ and $m + 2$, with equality only at $m = 1$. In that exceptional case write $U = \delta u$ with u a local unit. Then

$$V^2 = \delta^3 u(u^2 + 4).$$

The residue field is $\mathbf{F}_7$, in which -4 is a nonsquare. Thus the right side has valuation three, also impossible. Every finite valuation of U is even.

Under each real embedding, $D > 0$ and $U(U^2 + 4D) = V^2$, so $U > 0$. By the PID property, $U = \gamma^2\eta$ for $\gamma \in K^*$ and a unit η. This unit is totally positive, hence a square. □

Proposition J.4. *The charts* (114) *and* (115) *have exactly the same projected pairs $(x, y) \in K^2$. Every solution of the new chart has $h, j, x, y \neq 0$, and x is positive under both real embeddings.*

Proof. Given (114), put $j = t^2/2 - h^2$. Then

$$j^2 = h^4 - t^2h^2 + t^4/4 = h^4 - D,$$

and $y = hj$. Conversely, given (115), set $U = 2(h^2 + j)$ and $V = 2hU$. The identity $j^2 = h^4 - D$ gives

$$U^2 - 4h^2U + 4D = 0,$$

hence $V^2 = U^3 + 4DU$. Also $U \neq 0$, since $U = 0$ would imply $D = 0$. By Lemma J.3, choose $t \in K$ with $t^2 = U$. It follows that

$$t^2(h^2 - t^2/4) = D, \qquad h(t^2/2 - h^2) = hj,$$

as required.

If $h = 0$, then $j^2 = -D$, contradicting either real embedding. If $j = 0$, then $h^4 = D$, so $h^2 = \pm\delta$. Taking norms yields $\mathrm{N}(h)^2 = 7$, impossible for $\mathrm{N}(h) \in \mathbb{Q}$. The remaining assertions follow immediately. □

The proof is specific to this field and this value of D. It does not use the generally false assertion that square abscissa characterizes doubles on every elliptic curve. It also avoids adding a Mordell–Weil rank or generator hypothesis to the chart replacement.

J.3 Using the chart without ordinate variables

All nine point modules use the new chart. It has six rational coordinates: the two components of each of h, j, x, subject to four quadratic component equations. The old module had ten coordinates, including the components of t, h, u, x, y. Thus the change saves four coordinates per point, provided each downstream ordinate is consumed as the expression hj.

For the difference of chart points Q, R, retain the existing inverse equations for $x_Q - x_R$ and define the slope by

$$(x_Q - x_R)m = h_Q j_Q + h_R j_R.$$

Its numerator and defining equation are quadratic in the available coordinates. Define the difference abscissa by the existing square of m, and its ordinate by

$$y_{Q-R} = m(x_Q - x_{Q-R}) - h_Q j_Q.$$

This again is a quadratic assignment. The retained inverse ensures that the slope has exactly its old value.

For a chart point,

$$z = -x/y = -h/j.$$

Thus the local parameter is defined by $jz = -h$. When a logarithm value is combined with its parameter, replace $y\ell = -xG$ by $j\ell = -hG$, where G is the unchanged truncated logarithm polynomial. At the scalar ratio replace

$$x_H \rho = -y_H z_S \qquad \text{by} \qquad h_H \rho = -j_H z_S.$$

All cancellations are reversible by Proposition J.4. Difference-point denominators and all other guards remain unchanged. The fixed target has

$$h = -1 + \tfrac{7}{4}\sqrt{2}, \qquad j = \tfrac{39}{8} - \tfrac{9}{2}\sqrt{2};$$

these give exactly its previous abscissa and ordinate.

Together with Proposition J.4, these identities give rational witness maps in both directions. To reconstruct the old chart, choose the square root supplied by Lemma J.3; then evaluate all old assignments. The choice of sign does not alter (x, y) or any downstream value.

J.4 Shorter weight circuits

For each of nine detectors, write e for its nonzero weight and q for its order-one weight. The required outputs are $q_{12} = qe^{22}$ and $q_{40} = qe^{78}$. The old construction uses seven pure powers of e and two final quotients. Instead construct

$$a_1 = e^{-1}, \quad a_3 = a_1^2/e, \quad a_7 = a_3^2/e, \quad a_{14} = a_7^2, \quad a_{29} = a_{14}^2/e, \quad a_{55} = a_{29}^2/a_3,$$

and impose

$$a_{29} q_{12} = q a_7, \qquad a_{55} q_{40} = q_{12} e.$$

These are six power coordinates and two output coordinates. They give the same outputs and save one coordinate per detector.

The nonzero condition is inherited, rather than presumed from a quotient notation. In the source, w_0 has a retained inverse and

$$w_0 n = (w_0^2 - 2w_1^2)/2, \qquad (e\Delta)b = n^2.$$

Since $w_0 \neq 0$ and 2 is a nonsquare in $\mathbb{Q}$, the first equation gives $n \neq 0$. The second gives $e\Delta \neq 0$. Therefore every Laurent denominator above is nonzero. Conversely all Laurent coordinates are uniquely defined for each source solution. The changes introduce no new zero branches and exclude no source solution.

J.5 Scalar preparation with one fewer coordinate

The scalar pole uses a nonzero input b and a preparation witness t, with

$$U = \frac{b(1 - bt^2)}{1 + bt^2}.$$

The old circuit stores $b^{-1}, t_2 = t^2, bt_2, U$. Replace these four coordinates by t_2, T, V, with

$$t_2 = t^2, \qquad b(T - t_2) = 1, \qquad TV = 2,$$

and use the affine expression $U = V - b$. The equation $b(T - t_2) = 1$ forces $b \neq 0$, and $TV = 2$ forces $T \neq 0$. Consequently $T = t^2 + b^{-1}$, $1 + bt^2 = bT \neq 0$, and

$$U = 2/T - b = \frac{b(1 - bt^2)}{1 + bt^2}.$$

Conversely every old solution extends uniquely by these values of T, V. Thus the modification saves one coordinate and preserves all downstream quantities. It also supplies the nonzero condition formerly imposed by the separate coordinate b^{-1}.

J.6 An exact product-height identity

Put

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2 = 11 - 6\sqrt{2}, \qquad E : y^2 = x^3 - Dx,$$

and put

$$R_0 = (1 + \sqrt{2}, 2\sqrt{2}), \qquad P = 2R_0 = \left(\frac{57 - 28\sqrt{2}}{8}, \frac{-660 + 417\sqrt{2}}{32} \right).$$

The model calculation in Section 14 proves that $P = P_*$ has infinite order and $2E(K) = \mathbb{Z}P_*$; this is the same normalized curve and generator, not the earlier model of Proposition 3.1. Write h for the absolute logarithmic Weil height. The following lemma uses the displayed coordinates and doubling identity, together with the infinite order of P.

Lemma J.5. *For odd integers $n_1, \dots, n_r$,*

$$\mathrm{h}\left(\prod_{i=1}^{r} x([n_i]P) \right) = \sum_{i=1}^{r} \mathrm{h}(x([n_i]P)).$$

Proof. Let $T_0 = (0, 0)$. At a finite place v with $v(D) = 0$, use the finite quotient $E(K_v)/E_1(K_v)$, where E_1 denotes the formal neighborhood of infinity for this integral equation. Here we use the integral formal group construction of [48, Chapter IV, Sections 1–3]. Its parameter is $z = -x/y$, and $x = z^{-2}(1 + O(z))$, $y = -z^{-3}(1 + O(z))$ identify this neighborhood with the maximal ideal. It is an open subgroup; the proper curve $E(K_v)$ is compact, so its quotient is finite. This argument does not assume that the equation is minimal at the place above two. For a finite nonidentity point Q, membership in E_1 is equivalent to $v(x(Q)) < 0$. Since

$$x(Q + T_0) = -D/x(Q),$$

positive valuation of $x(Q)$ is equivalent to $Q+T_0 \in E_1$. The image of T_0 in this quotient has order two.

If an odd multiple of P belongs to E_1, the image of P has odd order. None of its multiples can then equal the image of T_0, so all $x([n_i]P)$ have nonpositive valuation. If no odd multiple belongs to E_1, all these abscissae have nonnegative valuation. Thus the valuations have a common weak sign. This argument includes the place above two.

The only place with $v(D) > 0$ is $\mathfrak{d} = (3-\sqrt{2})$ above seven. The displayed integral model is minimal there, since $v_{\mathfrak{d}}(-D) = 2 < 4$. Its reduction is $y^2 = x^3$ over $\mathbb{F}_7$. Substituting $\sqrt{2} = 3$ shows that P reduces to $(1,6)$, a nonsingular point. By [48, Proposition VII.2.1, pp. 188–190], all multiples therefore belong to the subgroup $E_0(K_{\mathfrak{d}})$ of points with nonsingular reduction. A positive valuation of x would force a positive valuation of y, and hence reduction to the singular point $(0,0)$. It cannot occur. The valuations again have a common weak sign.

At either real embedding, each $[n_i]P$ is a double of a non-torsion point. The duplication formula makes its abscissa a nonzero square. Its ordinate is nonzero, so the equation of E gives $x([n_i]P) > d$ at that embedding. Both real values $3 \pm \sqrt{2}$ of d exceed one.

At a finite place, numbers whose valuations have the same weak sign satisfy

$$\log \max\left(1, \left|\prod_i x([n_i]P)\right|_v\right) = \sum_i \log \max(1, |x([n_i]P)|_v).$$

At both real places the same equality holds because all the factors exceed one. Summing with normalized local weights proves the assertion. □

The product here is an element of K. It is not its norm down to $\mathbb{Q}$. Norms can cancel a pole at one split place against a zero at the other, whereas the proof above works separately at every place.

Corollary J.6. *Suppose the four root points $Q_i = [n_i]P$ have odd indices, and write*

$$C = \prod_{i=1}^{4} x(Q_i) = C_0 + C_1\sqrt{2}.$$

The original nine height-detector scales may be replaced by

$$1, \qquad C_0^2, \qquad (1+C_1^2)^2.$$

The root-height estimate retains the original constant $7\mathcal{H}$, where $\mathcal{H}$ is the exponent-19 height constant of the source.

Proof. Both real values of C are positive, so $C_0 > 0$. All three scales are positive nonzero rational squares. If a nonzero integer N is divisible by the kth powers of the three contact radicals, the unchanged detector estimate gives

$$\mathrm{h}(\lambda),\ \mathrm{h}(C_0^2\lambda),\ \mathrm{h}((1+C_1^2)^2\lambda) \le T := (\mathcal{H} + \log 3)|N|^{19/k}.$$

The elementary rational height identities yield

$$\mathrm{h}(C_0) \le T, \qquad \mathrm{h}(C_1) \le T/2.$$

Indeed, height subadditivity first bounds the heights of the two scales by $2T$; also $\mathrm{h}(C_0^2) = 2\,\mathrm{h}(C_0)$ and $\mathrm{h}((1+C_1^2)^2) \ge 4\,\mathrm{h}(C_1)$. Hence

$$\mathrm{h}(C) \le \mathrm{h}(C_0) + \mathrm{h}(C_1) + \tfrac{3}{2}\log 2 \le \tfrac{3}{2}T + \tfrac{3}{2}\log 2 < 7\mathcal{H}|N|^{19/k}.$$

Here $\mathcal{H} \geq 1$. Lemma J.5 bounds each root height by $\mathrm{h}(C)$. In particular, the inherited lower bound, with $B = \max_i |n_i|$, gives

$$B^2 - 6 < 7\mathcal{H}|N|^{19/k}. \tag{116}$$

The old completeness proof uses its scales only after the roots have been fixed, and requires fixed positive rational squares. Thus its prime selection and pole construction apply to this shorter list. □

J.7 Enforcing odd indices by quadratic equations

Put $\epsilon = 1 + \sqrt{2}$. The existing chart for a point Q has K-coordinates

$$x = h^2, \qquad j^2 = x^2 - D, \qquad Q = (x, hj).$$

To this chart add $t, r \in K$ and the equations

$$t^2 = 2(x + j), \qquad th + x + j = \epsilon r^2. \tag{117}$$

These are four rational quadratic equations in four additional rational coordinates. No extra inverse witnesses are needed.

Proposition J.7. *The projected points of the augmented chart are exactly $P + 4E(K)$, that is, the odd multiples of P.*

Proof. Use the squareclass homomorphism

$$\alpha : E(K) \longrightarrow K^*/K^{*2}, \qquad \alpha(O) = 1, \quad \alpha(T_0) = [-D] = [-1], \quad \alpha((a, b)) = [a] \quad (a \neq 0).$$

For clarity, its homomorphism property follows from the line-intersection identity

$$x(R + S)x(R)x(S) = \left(\frac{x(R)y(S) - x(S)y(R)}{x(R) - x(S)}\right)^2$$

when the displayed abscissae are nonzero and distinct. Doubling follows from $x(2R) = ((x(R)^2 + D)/(2y(R)))^2$; addition by T_0 follows from $x(R + T_0) = -D/x(R)$. Inverse pairs and identity summands are immediate. If $R + S = T_0$, then $x(R)x(S) = -D$. These cases cover the exceptional pairs.

Since $2E(K) = \mathbb{Z}P$ and $2R_0 = P$, every point of $E(K)$ has a unique expression $mR_0 + T$, where $m \in \mathbb{Z}$ and $T \in E[2](K)$. The two-torsion is generated by T_0 and $T_d = (d, 0)$, so

$$\alpha(mR_0 + aT_0 + bT_d) = [\epsilon]^m[-1]^a[d]^b.$$

The classes $[\epsilon], [-1], [d]$ are independent. A square $\epsilon^c(-1)^a d^b$, with $a, b, c \in \{0, 1\}$, would have rational square norm $(-1)^c 7^b$. This forces $c = b = 0$, and the real embeddings then force $a = 0$. Consequently

$$\alpha(R) = [\epsilon] \quad \Longleftrightarrow \quad R \in R_0 + 2E(K). \tag{118}$$

Given the augmented chart, set $u = t^2 = 2(x + j)$ and

$$a = th + u/2, \qquad b = at.$$

Expansion gives

$$u(x - u/4) = D, \qquad a^2 - ua = D, \qquad b^2 = a^3 - Da.$$

In particular t, a, b are nonzero. The tangent slope at $R = (a, b)$ is $h + t$; duplication gives $2R = (x, hj) = Q$. Equation (117) says $a = \epsilon r^2$, so $\alpha(R) = [\epsilon]$. By (118), $Q \in P + 4E(K)$.

Conversely, for $Q = (2m + 1)P$, take $R = (2m + 1)R_0 = (a, b)$. Then $a, b \neq 0$, and $a = \epsilon r^2$ for some $r \in K$. Set

$$t = b/a, \qquad h = (a^2 + D)/(2b), \qquad x = h^2, \qquad j = t^2/2 - x.$$

The same duplication identities give $Q = (x, hj)$, $j^2 = x^2 - D$, and both equations (117). □

Only the four root points receive this augmentation. The shared point and the four comparison points retain their old charts: their indices need not be odd. The total cost is sixteen coordinates and sixteen quadratic residuals. The projected-point formulation matters, since the added equations can select a sign representation of the old h, j chart.

J.8 Odd labels, scalar bounds, and completeness

Write the root points as

$$A = [n]P, \qquad S = [a]P, \qquad T = [t]P, \qquad Z = [b]P.$$

Here S denotes a point; the fixed coefficient remains S_0. Replace the old input expression by

$$\beta = 2X^2 + 3.$$

For rational X, integrality of β is equivalent to integrality of X: if $X = u/v$ in lowest terms, then $\beta \in \mathbb{Z}$ forces $v^2 \mid 2$, and hence $v = 1$. For integral X, β is odd.

In the scalar norm block and the physical anchor, replace S by

$$R = S - P = [a - 1]P.$$

The comparison and product-height blocks still use S. The retained subtraction guard excludes $a = \pm 1$, which is harmless for completeness. The original scalar contact product is

$$\beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1).$$

It never vanishes for rational X: $\beta \geq 3$, and $\beta = 2401$ would give $X^2 = 1199$, which is not a rational square. At an odd prime, a negative valuation of β is twice that of X, as required by the scalar argument.

J.8.1 Soundness

Put $c = |a - 1|$. Applying the scalar and anchor lemmas to R gives

$$\mathrm{h}(\beta) \leq \mathcal{H} c^{19/32}, \qquad \log_2 c > 25{,}165{,}823.5, \qquad 1 \leq \mathcal{H} < 2^{2{,}093{,}966}.$$

The anchor is still $S_0 = 3^{2^{26}}$. These explicit bounds imply

$$c > (80\mathcal{H})^{10}, \qquad |a| \geq c - 1 \geq c/2 > (40\mathcal{H})^{10},$$

and

$$\mathrm{h}(\beta) < c/80 \leq |a|/40, \qquad \log(40\mathcal{H}) < c/80 \leq |a|/40. \tag{119}$$

For example $10(7 + 2{,}093{,}966) < 25{,}165{,}823.5$, and $80 < 2^7$. The first estimate in (119) follows from $c^{13/32} > 80\mathcal{H}$; the logarithmic estimate is weaker still.

The local comparison lemmas retain their original conclusions. Let

$$F = t^2 - a(a-2)tb + (a-2)^2b^2 - 1, \qquad B = \max(|n|, |a|, |t|, |b|).$$

For $B > 4$, the triangle inequality gives $|F| \le 2B^4 + 6B^3 + 5B^2 + 1 < 5B^4$. If $F \neq 0$, the three radicals divide it to order 40, so (116) gives

$$B^{1/10} < 7\mathcal{H}\, 5^{19/40} + 6 < 24\mathcal{H}.$$

This contradicts the anchor. Thus $F = 0$. Since the point Z is finite and nonidentity, $b \neq 0$. Lemma 14.9 now gives $\log B > |a| - 3$.

If $n \neq \beta$, the nonzero numerator N of $n - \beta$, in reduced form, satisfies $|N| \le 2e^{\mathrm{h}(\beta)}B$, and the radicals divide it to order 12. Equation (116) and $B > 4$ give

$$B^{5/12} < 40\mathcal{H}e^{(19/12)\,\mathrm{h}(\beta)},$$

so

$$\log B < \frac{12}{5}\log(40\mathcal{H}) + \frac{19}{5}\,\mathrm{h}(\beta) < \frac{31}{200}|a| < |a| - 3.$$

This is the other contradiction. Therefore $n = \beta \in \mathbb{Z}$, and $X \in \mathbb{Z}$.

J.8.2 Completeness

Fix $X \in \mathbb{Z}$, and set $n = \beta$. Perform the old scalar preparation: choose its finite correction and then a positive shared index h_0 divisible by 64 and sufficiently deep at the fixed small primes. Choose a positive scalar index s sufficiently divisible to satisfy the scalar leaves and physical anchor for $[s]P$. We may require $6 \mid s$ and take s arbitrarily large. Set $a = s + 1$. Then a is odd, $a \equiv 1 \pmod 3$, and $S - P = [s]P$ is exactly the prepared scalar point.

Define

$$U_0 = 0, \qquad U_1 = 1, \qquad U_{j+1} = aU_j - U_{j-1}, \qquad k = 2(a-2),$$

and take $b = U_k/(a-2)$ and $t = U_{k+1}$. Modulo $a - 2$, induction gives $U_j \equiv j$, so $b \in \mathbb{Z}$. Modulo two the sequence has period $0, 1, 1$; since $k \equiv 1 \pmod 3$, both U_k and U_{k+1} are odd. The positive odd denominator $a - 2$ shows that b is odd. The Lucas identity gives

$$t^2 - a(a-2)tb + (a-2)^2b^2 = 1.$$

Thus all four root indices n, a, t, b are positive and odd. Proposition J.7 supplies their chart witnesses.

Fix these roots and the three scales of Corollary J.6. Local continuity of the scalar and comparison functions at the prepared shared point gives the same fixed modulus L, with $64 \mid L$. Only then choose the contact primes by the inherited simultaneous-prime theorem. The direct-target construction chooses a common index h satisfying the primary and opposite reduction-order congruences and $h \equiv h_0 \pmod L$, with the required real bounds. Its comparison points are $H = [h]P$ and $U_i = [hn_i]P$. Their shifted points have indices $h - 1$ and $(h-1)n_i$.

All quotient-domain arguments remain valid. The product scales are fixed positive squares, so the original pole witnesses apply unchanged. The scalar block is preserved by the congruence modulo L. This gives a rational solution for every integral X and completes the arithmetic proof of the primary construction.

J.9 Replacing five logarithms by asymmetric formal depth

This section is used only for the 523-variable construction. Let

$$U_0 = H = [h]P, \quad U_i = [u_i]P, \quad V_0 = H - P, \quad V_i = U_i - Q_i,$$

where $Q_i = [n_i]P$ are the four roots. Put $f = 1/x$ and $z = -x/y$. Keep the alternate truncated logarithms $\ell_{40}(V_i) = z(V_i)G_9(Df(V_i)^2)$, but replace every primary truncated logarithm by $z(U_i)$. Thus

$$L_i = z(U_i)/z(H), \qquad L_i' = \overline{\ell_{40}(V_i)/\ell_{40}(V_0)}.$$

The bar denotes conjugation in K. The sixteen tests now have the following orders:

1	$f(U_1), \dots, f(U_4),\ \overline{f(V_0)}, \dots, \overline{f(V_4)}$
40	$f(H),\ L_1 - L_1', \dots, L_4 - L_4',\ L_1 - \beta,$ the direct Lucas residual.

The scalar tests retain their original order 32. The stronger beta tests imply all former order-12 conclusions.

J.9.1 Local soundness

At a consumed unramified place above $p > 76176$, a formal point Q with $r = v(z(Q)) \geq 1$ satisfies

$$\log_E(Q)/z(Q) \in 1 + p^{4r}\mathcal{O}. \tag{120}$$

This is the same formal-logarithm expansion used in the source, starting its tail at the first nonconstant term. Namely $v(f(Q)) = 2r$, and for its coefficients a_j the factorial estimate is $v(a_j) \geq -(4j+1)/(p-1)$. If $4j + 1 < p$, the coefficient is integral. For the remaining $j \geq 19044$, the displayed bound still makes the jth term have valuation at least $4r$. Both $z(Q)$ and its full logarithm have valuation r.

The new test on $f(H)$ gives $v(z(H)) \geq 20$. The order-one tests put the other points on the formal branch. The full logarithm ratios at the primary and conjugate places are

$$r_i = u_i/h, \qquad r_i' = (u_i - n_i)/(h-1).$$

At least one of $h, h-1$ is a p-unit, so at least one of these ratios is integral. Even before knowing the depth of U_i, the weak unit estimate (120) and the alternate truncation estimate show that L_i, L_i' have the same valuations as r_i, r_i'. The order-40 cross test transfers integrality to both sides. Now L_i integral implies $v(z(U_i)) \geq v(z(H)) \geq 20$. The primary logarithmic corrections are therefore 1 modulo p^{80}. The alternate corrections are 1 modulo p^{40}. Since the ratios are integral, these relative estimates become absolute:

$$L_i \equiv r_i \equiv r_i' \pmod{p^{40}}.$$

Multiplying by h and $h-1$ and subtracting gives $L_i \equiv n_i \pmod{p^{40}}$. At an inert prime conjugation preserves the local valuation, so the same proof applies. The height contradictions of Section J.8 remain unchanged.

J.9.2 A stronger choice within the existing prime-pattern argument

For completeness, the common index must acquire extra primary divisibility. The following observation ensures compatibility with the opposite reduction orders.

Lemma J.8. *The contact primes in the source completeness argument may be chosen so that no selected prime divides any opposite reduced group order.*

Proof. After merging repeated contact forms and fixing all scale and congruence data, the prime-pattern proof uses distinct forms $p_j = u + c_j w^2$, with fixed coefficients c_j. Integer dilations and ray congruences change only fixed constants. It counts $u \in [N/2, N]$ and $1 \le w \le T$, where $T = \lfloor \tau(N)\sqrt{N} \rfloor$ and $\tau(N) \to 0$ can be chosen arbitrarily slowly in the final diagonal argument. Its weighted count, after the prescribed obstruction removal, retains a fixed positive proportion of NT. These are the fixed-field weighted count in Proposition I.29 and the final diagonal argument of Section I.11, including its obstruction removal.

If there are k forms, impose also

$$\frac{\tau(N)N^{1/6}}{(\log N)^k} \longrightarrow \infty.$$

This is compatible with the same diagonal choice: for each fixed positive value of τ, its threshold can be added to the existing finite list of thresholds. The strip $w < N^{1/3}$ has at most $O(N^{4/3})$ integer pairs and weight at most $O((\log N)^k)$ per pair. Its total weight is $o(NT)$, so an admissible pattern remains outside it.

For these patterns $w^2 \ge N^{2/3}$ and $w^2 = o(N)$. All p_j are asymptotic to the same u. Hasse's bound [48, Theorem V.1.1, p. 138] gives $d_+(p_j) = p_j + O(\sqrt{N})$ for the opposite reduced group order. Every ratio $d_+(p_j)/p_i$ is eventually strictly between $1/2$ and 2. Hence $p_i \mid d_+(p_j)$ would force equality. If $i \ne j$, this contradicts

$$|p_i - p_j| \ge c_* N^{2/3}, \qquad c_* := \min_{i \ne j} |c_i - c_j| > 0.$$

If $i = j$, equality is impossible because $4 \mid d_+(p_i)$ whereas p_i is odd. □

J.9.3 Completeness with deeper primary points

Keep the choice order in Section J.8: the roots, scales and continuity modulus L precede the prime selection. Choose primes using Lemma J.8, beyond all prime factors of L and the fixed scale data. Put

$$M_\pm = \operatorname*{lcm}_p(d_\pm(p)/4), \qquad \Pi = \prod_p p.$$

The direct-target prime theorem gives

$$\gcd(M_-, M_+) = 1, \quad \gcd(M_-, L) = 16, \quad \gcd(M_+, L) = 1.$$

Lemma J.8 gives $\gcd(\Pi, M_+) = 1$, and the cutoff gives $\gcd(\Pi, L) = 1$. Thus

$$h \equiv 0 \pmod{\Pi^{19} M_-}, \qquad h \equiv 1 \pmod{M_+}, \qquad h \equiv h_0 \pmod{L}$$

are compatible, since $64 \mid h_0$. The inherited real-progression argument supplies a sufficiently large h in this progression with the same real bounds.

The point P is globally a double, so multiplication by the primary doubling-image order first makes it formal, with z-valuation at least one. Multiplication by p^{19} raises this valuation by nineteen, using the formal logarithm isometry of [48, Theorem IV.6.4, p. 132]; these completions are unramified, $v(p) = 1$, and $1 > 1/(p-1)$. Consequently $H = [h]P$ and $U_i = [hn_i]P$ have primary depth at least twenty, and their parameter ratios equal n_i to order eighty. At the opposite places,

$V_0 = [h-1]P$ and $V_i = [(h-1)n_i]P$ are formal and their retained logarithm ratios equal n_i to order forty.

The pole-completeness estimates still apply to these fixed test orders. In the source notation their large-real-parameter decay is λ^{-3} for each order m, and the small-prime lower bound is $3\nu+(10m-8)\eta-2c$, which still tends to infinity with the preparatory divisibility. Scalar conditions and the anchor are preserved by $h \equiv h_0 \pmod{L}$. This proves completeness for the optional refinement.

J.10 Polynomial construction and exact verification

The generators apply three changes to the earlier point-chart source: the four odd-coset charts, the product and three scales, and the shifted scalar point with $\beta = 2X^2 + 3$. The depth variant first changes the logarithms and test orders as in Section J.9. Each field equation is represented by two rational component equations. Intermediate products are assigned coordinates as needed to keep every residual quadratic.

The full-log source has 577 raw coordinates; the pinned upstream compiler reduces this to 561. Seven further certified eliminations give 554. The depth source has 540 raw coordinates; compilation gives 532 and nine certified eliminations give 523. Every elimination is justified by exact polynomial identities and reversible substitutions preserving the input. Clearing residual denominators and taking the sum of their squares gives the displayed integer quartics. Over $\mathbb{Q}$, the sum vanishes exactly when every residual does.

The constant-chain versions add $c_0, \dots, c_{16}$ with

$$c_0 = 3^{1024}, \qquad c_{i+1} = c_i^2 \quad (0 \le i < 16).$$

These witnesses have unique rational values and $c_{16} = S_0$. Substitution for S_0 leaves every residual quadratic. Direct expansion gives the 3,247-bit maximum coefficient bound.

The accompanying archive contains the complete polynomial tables, residual systems, coordinate maps, elimination certificates, pinned source generators, arithmetic notes and independent checkers. Each final package has a standard-library-only `verify_all.py`. It replays all eliminations, verifies the unique constant lift, independently expands the sums of squares, and compares every coefficient. Both full packages have passed fresh independent runs. A separate source replay rebuilds the two compiled precursors and checks the match to their frozen residual systems.

These finite checks certify the counts, degrees and stated algebraic transformations. The arithmetic assertions in Sections J.6–J.9 received separate adversarial AI review; they have not received independent human peer review or a complete formal proof check. The inherited arithmetic inputs are the results of this manuscript identified in Section J.1; no theorem from the contribution archive is assumed as an additional arithmetic input.

Proof of Theorem J.1. The chart identities and Lemma J.2 identify the exact source used by the generators with the arithmetic system described here. Proposition J.7 enforces the odd root indices, so Corollary J.6 supplies the required height bound. Section J.9.1 verifies the local implications for the final test orders. Section J.8.1 then forces the input to be an integer. Conversely, for an integer input, Sections J.8.2 and J.9.3, using Lemma J.8, construct witnesses for every source equation and retained guard. The certified substitutions preserve this projection in both directions. The sum of squares vanishes over $\mathbb{Q}$ precisely when all its residuals vanish. Finally the 17-coordinate constant chain has exactly one rational solution and ends at S_0, so it preserves the same projection. This proves the statements for both polynomials. □

The complete tables are in the files

quartics/quartic-523-symbolic-coefficients.json
quartics/quartic-540-fully-expanded.txt.

Their SHA-256 digests, respectively, are

6a3685980de4512bdc7f3a781d0c0d4b32cce5829ec83f03f318264931bdbbe3
83e7a31e11f9ad2edc2f67935e50bff437d87ef4b77d37b6a4391e8117770680.

The reproduction command is pythonscripts/check_odd_product.py; the complete source, certificates and repaired, non-mutating verifiers are in quartics/quartic-523-540/.

References

[1] T. Mitsui, *Generalized Prime Number Theorem*, Japanese Journal of Mathematics **26** (1956), 1–42. https://doi.org/10.4099/jjm1924.26.0_1.

[2] B. Green and T. Tao, *The quantitative behaviour of polynomial orbits on nilmanifolds*, Annals of Mathematics **175** (2012), 465–540. https://doi.org/10.4007/annals.2012.175.2.2.

[3] P.-Y. Bienvenu, *A higher-dimensional Siegel–Walfisz theorem*, Acta Arithmetica **179** (2017), 79–100. https://doi.org/10.4064/aa8600-10-2016.

[4] J. Kollár, *Unirationality of cubic hypersurfaces*, J. Inst. Math. Jussieu **1** (2002), no. 3, 467–476. doi:10.1017/S1474748002000117.

[5] P. Müller, *Finiteness results for Hilbert's irreducibility theorem*, Ann. Inst. Fourier **52** (2002), no. 4, 983–1015. doi:10.5802/aif.1907.

[6] Agentic notes, *Conditional polynomial constructions, lower bounds, and number-field obstructions for rational Diophantine integrality*, working version 3, 6 November 2026. Public source at Git revision f4d1b5e6c23d4adeb3ae1f986ab0bf87884bdff6. Public source at the pinned revision.

[7] Z.-W. Sun, *86 unknowns for Hilbert's tenth problem over the rationals*, arXiv:2610.20691v2, 2026. Theorem 4.1, Sections 4–6 and Appendix A. https://arxiv.org/abs/2610.20691v2.

[8] T. Browning and D. Loughran, *Sieving rational points on varieties*, Trans. Amer. Math. Soc. **371** (2019), 5757–5785. https://doi.org/10.1090/tran/7514. Theorem 4.1 was checked in the institutional accepted manuscript and arXiv:1705.01999v2; the publisher's final typeset text was not accessible in this audit.

[9] T. D. Browning and D. R. Heath-Brown, *The geometric sieve for quadrics*, Forum Math. **33** (2021), no. 1, 147–165. https://doi.org/10.1515/forum-2020-0074. Lemmas 2.1 and 4.1 were checked in the Oxford accepted manuscript dated 14 September 2020 and arXiv:2003.09593v1; the publisher's final typeset text was not accessible in this audit.

[10] C. Frei and M. Tinková, with an appendix by C. Frei, J. König and M. Tinková, *Linear constellations in primes with arithmetic restrictions*, arXiv:2408.17441v3, 5 November 2025. https://arxiv.org/abs/2408.17441v3. Theorem 1.4. Listed as *Mathematika*, to appear, on https://www.math.tugraz.at/~frei/, checked 6 November 2026. No final published version was verified.

[11] P. Dodos and V. Kanellopoulos, *Uniformity norms, their weaker versions, and applications*, Acta Arith. **203** (2022), no. 3, 251–270. https://doi.org/10.4064/aa210728-27-2. Theorem 5.1 was checked in arXiv:1610.00487v4, 9 June 2022, pp. 12–13; the final typeset statement was not compared.

[12] T. Tao and T. Ziegler, *Polynomial patterns in the primes*, Forum Math. Pi **6** (2018), e1, 60 pp. https://doi.org/10.1017/fmp.2017.3. Published Theorems 9, 11, 13 and Corollary 3.

[13] G. S. Kopp and J. C. Lagarias, *Ray class groups and ray class fields for orders of number fields*, Essential Number Theory **4** (2025), no. 1, 1–65. https://doi.org/10.2140/ent.2025.4.1. Theorems 1.1, 1.3, 6.5, 7.2 and Remark 7.3.

[14] K. S. Williams, *Mertens' theorem for arithmetic progressions*, J. Number Theory **6** (1974), 353–359, Theorem 1, p. 356. https://doi.org/10.1016/0022-314X(74)90032-8.

[15] P. Scholze, *p-adic Hodge theory for rigid-analytic varieties*, Forum of Mathematics, Pi 1 (2013), e1, 1–77, Corollary 1.8 and Theorems 9.1, 9.3. https://doi.org/10.1017/fmp.2013.1.

[16] P. Scholze, *p-adic Hodge theory for rigid-analytic varieties—corrigendum*, Forum of Mathematics, Pi 4 (2016), e6, 1–4. https://doi.org/10.1017/fmp.2016.4.

[17] O. Brinon and B. Conrad, *CMI summer school notes on p-adic Hodge theory*, preliminary version, June 24, 2009, Theorem 5.2.1 and Propositions 6.3.3, 6.3.8. https://math.stanford.edu/~conrad/papers/notes.pdf.

[18] J. S. Milne, *Abelian varieties*, version 2.00 (2008), Part I, Theorem 12.1(a). https://www.jmilne.org/math/CourseNotes/AV.pdf.

[19] S. Baba and H. Granath, *Genus 2 curves with quaternionic multiplication*, 2005 preprint, Sections 2.1, 3 and Proposition 10. https://webdoc.sub.gwdg.de/ebook/serien/e/mpi_mathematik/2005/18.pdf.

[20] K. Buzzard, *Integral models of certain Shimura curves*, 1996, corrected 2002, Theorem 2.1 and Corollary 2.3. https://www.ma.imperial.ac.uk/~buzzard/maths/research/papers/shimura.pdf.

[21] Y.-H. Lin and Y. Yang, *Quaternionic loci in Siegel's modular threefold*, arXiv:1807.00466v1, 2018, equations (2), (14). https://arxiv.org/abs/1807.00466v1.

[22] L. Berger, *Représentations p-adiques et équations différentielles*, Inventiones Mathematicae 148 (2002), 219–284, p-adic monodromy theorem. https://doi.org/10.1007/s002220100202.

[23] L. Pan, *The Fontaine–Mazur conjecture in the residually reducible case*, arXiv:1901.07166v2, Theorem 1.0.4, the version used here. Published in Journal of the American Mathematical Society 35 (2022), 1031–1169. https://arxiv.org/abs/1901.07166v2.

[24] B. Conrad, *Lifting global representations with local properties*, author preprint, December 11, 2011, Appendix B, pp. 32–33. https://math.stanford.edu/~conrad/papers/locchar.pdf.

[25] W.-C. W. Li, *Newforms and functional equations*, Mathematische Annalen 212 (1975), 285–315, p. 296.

[26] G. Shimura, *On the factors of the jacobian variety of a modular function field*, Journal of the Mathematical Society of Japan 25 (1973), 523–544, Theorem 1. https://doi.org/10.2969/jmsj/02530523.

[27] A. Javanpeykar, with an appendix by P. Bruin, *Polynomial bounds for Arakelov invariants of Belyi curves*, Algebra & Number Theory 8 (2014), 89–140, Theorem 1.1.1 and Section 2.3. https://doi.org/10.2140/ant.2014.8.89.

[28] É. Gaudron and G. Rémond, *Polarisations et isogénies*, Duke Mathematical Journal 163 (2014), 2057–2108, Theorem 1.4 and Section 9. https://doi.org/10.1215/00127094-2782528.

[29] G. Fowler, *Some uniform effective results on André–Oort for sums of powers in* $\mathbb{C}^n$, Mathematical Proceedings of the Cambridge Philosophical Society 180 (2026), 607–641, Proposition 2.7. https://doi.org/10.1017/S0305004125101825.

[30] A. Barquero-Sanchez, L. Cadwallader, O. Cannon, T. Genao and R. Masri, *Faltings heights of CM elliptic curves and special Gamma values*, Research in Number Theory 3 (2017), article 13, Section 6, equation (6.2). https://doi.org/10.1007/s40993-017-0077-7.

[31] A. Silverberg and Yu. G. Zarhin, *Semistable reduction and torsion subgroups of abelian varieties*, Annales de l'Institut Fourier 45 (1995), 403–420, Theorem 3.5. https://www.numdam.org/article/AIF_1995__45_2_403_0.pdf.

[32] J.-P. Serre, *Local class field theory*, in J. W. S. Cassels and A. Fröhlich (eds.), *Algebraic Number Theory*, Academic Press, 1967, Chapter VI, 128–161, Sections 4.1–4.3.

[33] G. Faltings, *Endlichkeitssätze für abelsche Varietäten über Zahlkörpern*, Inventiones Mathematicae 73 (1983), 349–366; erratum, 75 (1984), 381.

[34] G. Faltings, *p-adic Hodge theory*, Journal of the American Mathematical Society 1 (1988), 255–299.

[35] K. Lauter, M. Naehrig and T. Yang, *Hilbert theta series and invariants of genus 2 curves*, Journal of Number Theory, 2015, Section 2.2, doi:10.1016/j.jnt.2015.02.020. https://people.math.wisc.edu/~tonghaiyang/2015-LNYFinal-JNT.pdf.

[36] F. Pazuki, *Theta height and Faltings height*, Bulletin de la Société Mathématique de France, 2012, Sections 2.2–2.3 and Corollary 1.3. https://www.numdam.org/item/10.24033/bsmf.2623.pdf.

[37] B. Poonen and M. Stoll, *The Cassels–Tate pairing on polarized abelian varieties*, Annals of Mathematics 150 (1999), 1109–1149. https://math.mit.edu/~poonen/papers/sha.pdf.

[38] B. Green and T. Tao, *Linear equations in primes*, Annals of Mathematics 171 (2010), 1753–1850. https://annals.math.princeton.edu/2010/171-3/p08.

[39] B. Green and T. Tao, *The Möbius function is strongly orthogonal to nilsequences*, Annals of Mathematics 175 (2012). https://annals.math.princeton.edu/2012/175-2/p03.

[40] B. Green, T. Tao and T. Ziegler, *An inverse theorem for the Gowers* $U^{s+1}[N]$*-norm*, arXiv:1009.3998v5, 2026 corrected version. https://arxiv.org/abs/1009.3998v5; author erratum: https://people.maths.ox.ac.uk/greenbj/papers/GI(s)-erratum.pdf.

[41] M. A. Bennett, G. Martin, K. O'Bryant and A. Rechnitzer, *Explicit bounds for primes in arithmetic progressions*, Illinois Journal of Mathematics 62 (2018), 427–532, arXiv:1802.00085v3, Theorem 1.2. https://arxiv.org/abs/1802.00085v3.

[42] J. S. Milne, *Arithmetic duality theorems*, second edition, 2006, Chapter I. https://www.jmilne.org/math/Books/ADTnot.pdf.

[43] The Stacks Project, *Stalks of higher direct images* and *The Leray spectral sequence*, Theorem 59.53.1, Tag 03Q9, and Proposition 59.54.2, Section Tag 03QA, HTML snapshots of November 5, 2026. https://stacks.math.columbia.edu/tag/03Q9; https://stacks.math.columbia.edu/tag/03QA.

[44] J. Nekovář, *Selmer complexes*, Astérisque 310, 2006, Proposition 5.2.4, Theorems 6.3.4 and 7.8.8, and Sections 5.7 and 6.4. https://www.numdam.org/item/AST_2006__310__1_0/.

[45] Archive of Formal Proofs, *DPRM_Theorem*, Alpha_Sequence, Exponential_Relation, and Exponentiation, directory thys/DPRM_Theorem/Diophantine/, AFP development git mirror revision b4f3b892d0ae25cea27412ba4007cdb63e44b0eb (May 10, 2026). The three source files match this revision byte for byte. https://isa-afp.org/entries/DPRM_Theorem.html.

[46] B. Edixhoven, *Néron models and tame ramification*, Compositio Mathematica 81 (1992), 291–306, Theorem 5.3 and Remark 5.4.1. https://www.numdam.org/item/CM_1992__81_3_291_0/.

[47] A. V. Sutherland, *Isogeny kernels and division polynomials*, MIT 18.783, Fall 2025, Lecture 5, September 18, 2025, §§5.5–5.6. https://ocw.mit.edu/courses/18-783-elliptic-curves-fall-2025/mit18_783_f25_lec05.pdf.

[48] J. H. Silverman, *The arithmetic of elliptic curves*, second edition, Springer, 2009, Chapters IV and VII.

[49] T. Dokchitser and V. Dokchitser, *Root numbers and parity of ranks of elliptic curves*, Journal für die reine und angewandte Mathematik 658 (2011), 39–64, Theorem 1.8. https://arxiv.org/abs/0906.1815v1.

[50] C. Breuil, B. Conrad, F. Diamond and R. Taylor, *On the modularity of elliptic curves over $\mathbb{Q}$: wild 3-adic exercises*, author manuscript, Theorem A and Introduction, conditions (1)–(6). https://virtualmath1.stanford.edu/~conrad/papers/tswfinal.pdf. Published in Journal of the American Mathematical Society 14 (2001), 843–939.

[51] J. S. Milne, *Modular functions and modular forms*, v1.31, March 22, 2017, Remark 11.18 and Theorem 11.19. https://www.jmilne.org/math/CourseNotes/MF.pdf.

[52] R. Elkik, *Le théorème de Manin–Drinfeld*, Astérisque 183 (1990), 59–67, Theorem, p. 59. https://www.numdam.org/item/AST_1990__183__59_0.pdf.

[53] B. Gross and D. Zagier, *Heegner points and derivatives of L-series*, Inventiones Mathematicae 84 (1986), 225–320.

[54] V. A. Kolyvagin, *On the Mordell–Weil group and the Shafarevich–Tate group of modular elliptic curves*, MPIM preprint MPI/90-69 (1990), Theorems 1–5, pp. 4–5. https://archive.mpim-bonn.mpg.de/id/eprint/325/1/preprint_1990_69.pdf.

[55] K. Kato, *p-adic Hodge theory and values of zeta functions of modular forms*, Astérisque 295 (2004), 117–290. https://www.numdam.org/article/AST_2004__295__117_0.pdf.

[56] D. Bump, S. Friedberg and J. Hoffstein, *Nonvanishing theorems for L-functions of modular forms and their derivatives*, Inventiones Mathematicae 102 (1990), 543–618. https://wstein.org/papers/bib/bump-friedberg-hoffstein-nonvanishing.pdf.

[57] J. Hoffstein and W. Luo, *Nonvanishing of L-series and the combinatorial sieve*, Mathematical Research Letters 4 (1997), 435–444. https://intlpress.com/site/pub/files/_fulltext/journals/mrl/1997/0004/0003/MRL-1997-0004-0003-a012.pdf.

[58] J. Desjardins, *Root number of twists of an elliptic curve*, Journal de Théorie des Nombres de Bordeaux 32 (2020), 73–101. https://jtnb.centre-mersenne.org/item/10.5802/jtnb.1112.pdf.

[59] L. Cai, J. Shu and Y. Tian, *Explicit Gross–Zagier and Waldspurger formulae*, Algebra & Number Theory 8 (2014). https://arxiv.org/abs/1408.1733.

[60] L. Cai, Y. Chen and Y. Liu, *Heegner points on modular curves*, Transactions of the American Mathematical Society 370 (2018), 3721–3743. https://archive.ymsc.tsinghua.edu.cn/pacm_download/632/12220-Heegner_points_on_modular_curves.pdf.

[61] D. Jetchev and B. Kane, *Equidistribution of Heegner Points and Ternary Quadratic Forms*, arXiv:0908.3905 (2009), Sections 1.1–1.2 and 2.5. https://arxiv.org/abs/0908.3905.

[62] C. Cornut and D. Jetchev, *Liftings of reduction maps for quaternion algebras*, 2010 preprint. https://webusers.imj-prg.fr/~christophe.cornut/papers/liftings.pdf.

[63] J.-L. Waldspurger, *Sur les coefficients de Fourier des formes modulaires de poids demi-entier*, Journal de Mathématiques Pures et Appliquées 60 (1981), 375–484, Proposition 18 and Corollary 2.

[64] N. Ramsey, *Geometric and p-adic modular forms of half-integral weight*, Annales de l'Institut Fourier 56 (2006), 599–624. https://www.numdam.org/item/10.5802/aif.2195.pdf.

[65] W. Casselman, *On some results of Atkin and Lehner*, Mathematische Annalen 201 (1973), 301–314, Theorem 1 and its proof, pp. 302–306, and Theorem 4, p. 313.

[66] H. Carayol, *Sur les représentations ℓ-adiques associées aux formes modulaires de Hilbert*, Annales Scientifiques de l'École Normale Supérieure 19 (1986), 409–468, Theorem A. https://www.numdam.org/item/ASENS_1986_4_19_3_409_0/.

[67] J.-L. Waldspurger, *Correspondances de Shimura*, Proceedings of the International Congress of Mathematicians, 1983, 525–531. https://www.imj-prg.fr/wp-content/uploads/2020/prix/waldspurger1983.pdf.

[68] A. Weil, *Sur certains groupes d'opérateurs unitaires*, Acta Mathematica 111 (1964), 143–211, I.13, II.25, II.28, III, Theorem 6. https://webhomes.maths.ed.ac.uk/~v1ranick/papers/weil2.pdf.

[69] J.-P. Serre, *Propriétés galoisiennes des points d'ordre fini des courbes elliptiques*, Inventiones Mathematicae 15 (1972), 259–331.

[70] B. Mazur, *The topology of rational points*, Experimental Mathematics 1 (1992), 35–45. https://doi.org/10.1080/10586458.1992.10504244.

[71] J. Robinson, *Definability and decision problems in arithmetic*, Journal of Symbolic Logic 14 (1949), 98–114, Theorem 3.1. https://doi.org/10.2307/2266510.

[72] B. Poonen, *Characterizing integers among rational numbers with a universal-existential formula*, American Journal of Mathematics 131 (2009), 675–682, Theorems 3.1 and 4.1. Author version with the April 2019 correction to Lemma 2.3. https://math.mit.edu/~poonen/papers/ae.pdf.

[73] K. Kato, *Euler systems, Iwasawa theory, and Selmer groups*, Kodai Mathematical Journal 22 (1999), 313–372, Sections 0.1 and 0.4–0.7, Example 0.6.2 and Theorem 0.8, pp. 314–318. https://doi.org/10.2996/kmj/1138044090.

[74] OpenAI, *A pointwise 2-converse for elliptic curves with rational two-torsion*, preprint, 7 October 2026, Theorem 1.1; source revision 4eaee87548250b2682affda7d05d947c05554bae. Pinned source PDF.

[75] OpenAI, *Goldfeld's analytic density conjecture and the 2-converse for elliptic curves*, preprint, 7 October 2026, Theorem 1.1; same frozen source revision. Pinned source PDF.

[76] B. Mazur, *Speculations about the topology of rational points: an up-date*, Astérisque 228 (1995), 165–181, Conjecture 3, p. 165. https://www.numdam.org/article/AST_1995__228__165_0.pdf.

[77] Yu. V. Matiyasevich, *Diophantine representation of enumerable predicates*, Mathematics of the USSR-Izvestiya 5 (1971), 1–28, Basic Theorem, p. 3; integer-witness conversion, p. 2. https://doi.org/10.1070/IM1971v005n01ABEH001004.

[78] OpenAI, *Hilbert's tenth problem over the rational numbers*, OpenAI Math Release preprint, 6 October 2026. https://github.com/openai/math/blob/main/preprints/Hilberts-tenth-problem-over-the-rational-numbers-October-6-2026/main.pdf.

[79] B. Howard, *The Heegner point Kolyvagin system*, Compositio Mathematica 140 (2004), 1439–1472; arXiv:1202.6340v1.

[80] S. Lang, *Algebraic groups over finite fields*, American Journal of Mathematics 78 (1956), 555–563.

[81] K. S. Williams, *Mertens' theorem for arithmetic progressions*, J. Number Theory **6** (1974), 353–359, Theorem 1. https://people.math.carleton.ca/~williams/papers/pdf/057.pdf.

[82] J. Voight, *Quaternion algebras*, corrected online edition v1.0.7u, August 5, 2026, Main Theorem 14.6.1 and Proposition 14.6.7, pp. 223–224. https://jvoight.github.io/quat-book.pdf.

[83] J. Koenigsmann, *Defining $\mathbb{Z}$ in $\mathbb{Q}$*, Annals of Mathematics **183** (2016), 73–93. doi:10.4007/annals.2016.183.1.2.

[84] J. González and V. Rotger, *Non-elliptic Shimura curves of genus one*, Journal of the Mathematical Society of Japan **58** (2006), 927–948, Theorem 4.2 and Table 2, pp. 938–939; Proposition 5.2, pp. 940–941. https://doi.org/10.2969/jmsj/1179759530.

[85] X. Yuan, *Explicit Kodaira–Spencer maps over Shimura curves*, Acta Mathematica Sinica, Chinese Series **67** (2024), 227–249. https://doi.org/10.12386/A20220154. We use the English translation dated April 30, 2024, identified by the author as a translation of the published Chinese article: https://faculty.bicmr.pku.edu.cn/~yxy/preprints/explicit_KS.pdf. The author identifies it at https://faculty.bicmr.pku.edu.cn/~yxy/publications.html. The relevant locators in that translation are Theorem 1.1, p. 3; metric definitions, pp. 2–3; Sections 2.1–2.2 and Lemma 2.1, pp. 4–10; and the analytic and metric calculations in Sections 4.2–4.3, pp. 20–26.

[86] P. Deligne, *Variétés de Shimura: interprétation modulaire, et techniques de construction de modèles canoniques*, in *Automorphic Forms, Representations and L-Functions*, Proceedings of Symposia in Pure Mathematics **33**, Part 2 (1979), 247–289, Sections 2.2.5 and 2.3.1. English translation by J. S. Milne, corrected September 6, 2020, p. 29: https://www.jmilne.org/math/Documents/DeligneSV.pdf.

[87] É. Gaudron and G. Rémond, *Nouveaux théorèmes d'isogénie*, Mémoires de la Société Mathématique de France **176** (2023), vi+123 pp., Section 1.5 and Theorem 1.9(1), pp. 7–8. https://doi.org/10.24033/msmf.484.

[88] R. von Känel, *The effective Shafarevich conjecture for abelian varieties of* GL_2*-type*, Forum of Mathematics, Sigma **9** (2021), e39, 1–29, equations (2.2) and (5.9), pp. 5 and 13. https://doi.org/10.1017/fms.2021.29.

[89] J. Voight, *Quaternion algebras*, Graduate Texts in Mathematics **288**, Springer, 2021, Chapter 43, equations (43.1.2)–(43.1.3), p. 800, and Main Theorem 43.6.14, p. 819; proof pp. 822–823. https://doi.org/10.1007/978-3-030-56694-4_43.

[90] F. Calegari, *Reciprocity in the Langlands program since Fermat's Last Theorem*, in *Proceedings of the International Congress of Mathematicians 2022*, Vol. 2, EMS Press, 2023, 610–651, Section 7, p. 624; definition of geometric representations, pp. 611–612. https://doi.org/10.4171/ICM2022/101.

[91] F. Diamond and J. Im, *Modular forms and modular curves*, in Seminar on Fermat's Last Theorem, CMS Conference Proceedings 17 (1995), 39–133, §12.5.

[92] M. H. Baker, E. González-Jiménez, J. González and B. Poonen, *Finiteness results for modular curves of genus at least 2*, American Journal of Mathematics 127 (2005), 1325–1387, §§3.1, 3.2.1.

[93] N. Mascot, *Moduli-friendly Eisenstein series over the p-adics and the computation of modular Galois representations*, Research in Number Theory 8 (2022), article 48, doi:10.1007/s40993-022-00329-6, §2.1, (2.1.8).

[94] T. Pheidas, *An effort to prove that the existential theory of* $\mathbb{Q}$ *is undecidable*, in *Hilbert's Tenth Problem: Relations with Arithmetic and Algebraic Geometry*, J. Denef, L. Lipshitz, T. Pheidas and J. Van Geel, eds., Contemporary Mathematics **270**, American Mathematical Society, Providence, RI, 2000, 237–252. https://doi.org/10.1090/conm/270/04376.

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